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Claus Sorensen

From Classical *L*-Functions to Modern Reciprocity Laws

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For
Ben, Eric, and Tania
with love

Preface

This book has its origins in lecture notes for a variety of courses I taught over the past 20 or so years. The material has thus been thoroughly tested on students at Princeton University (2006–2013) and the University of California San Diego (from 2013 onward). The impetus for polishing my notes, and assembling them in book form, was the COVID-19 pandemic where all instruction was moved online. With the extra time on my hands, I thought it would be useful to have typeset notes accompany my video lectures. In the wake of the pandemic, I then continued to expand on the notes intermittently.

The target audience is first and second year graduate students with an interest in number theory. However, a considerable portion of the manuscript (such as the entire Chap. 1 and Appendix A) should also appeal to advanced undergraduate students. It is my hope that the book will also benefit researchers, from other areas of mathematics, harboring a desire to quickly grasp the basic theory of L-functions.

The overarching goal of the book is to smoothen the transition from the standard graduate algebraic number theory sequence to more advanced coursework, at the forefront of modern arithmetic. Hence, the prerequisite for reading the book, in its entirety, is a good understanding of number fields and local fields (at the level of Neukirch's book [Neu99] for example).

The starting point is the Riemann zeta function and its analytic properties. From there, we move on to Dirichlet L-functions and Dedekind zeta functions, before discussing the general case of Hecke L-functions encompassing all these types of functions. We adopt Tate's perspective (in contrast to [Neu99] for instance) which he introduced in his 1950 Ph.D. thesis [Tat50], completed under the supervision of Emil Artin at Princeton University. Tate's approach is an extremely elegant application of harmonic analysis for number fields and local fields. It greatly simplifies, and conceptualizes, Hecke's original technique which led to formidable technical computations. Perhaps more importantly, Tate's work paved the way for the development of automorphic L-functions in the 1970s, and to some extent his work provided a blueprint for establishing their fundamental analytic properties. Nowadays, the theory of automorphic L-functions is the bread and butter of the celebrated Langlands program, which among other things predicts a deep connection to L-functions arising from Galois

representations. Chapter 2 of this book contains a detailed discussion of Tate's thesis, including the necessary background material, examples, and exercises. All calculations are fleshed out. Hopefully, the exposition will serve as a useful supplement, for the novice, to some of the existing go-to sources, such as [Bum97] and [RV99] (and Tate's thesis itself of course).

Another goal of the book is to give a freestanding proof of the meromorphic continuation of Artin L-functions, by appealing to Tate's thesis rather than Hecke's theory of Grossencharacters (unlike [Neu99]). This is done in Chap. 4, which also covers the all-important Brauer induction theorem from representation theory. This meromorphicity is an influential cornerstone of the subject: It spawned the key Artin conjecture (concerning the poles) and, in turn, the Langlands reciprocity conjecture (relating Artin L-functions to automorphic L-functions).

In broad strokes, the Langlands correspondence foresees a link between Galois representations of a geometric nature, and automorphic forms. This correspondence is mediated by an equality of L-functions. The upshot of this, is it allows us to transfer the relatively easily verifiable analytic properties on the "automorphic side" to the "Galois side" where the analogous properties are still widely unkown in general. In fact this works both ways: If the totality of all L-functions on the Galois side have all the desired analytic properties (analytic continuation, a functional equation, etc.) there are "converse theorems" which let us identify these L-functions of "geometric origin" with automorphic L-functions.

Wiles's proof of Fermat's Last Theorem established a very important special case of the Langlands correspondence, previously known as the Taniyama–Shimura–Weil conjecture, which was formulated in the 1960s. They conjectured that every rational elliptic curve "comes from" a modular form. In the 1950s, Eichler and Shimura associated a rational elliptic curve to every modular form (of weight two, with rational Fourier coefficients). What is now often referred to as the "modularity theorem" asserts that indeed every rational elliptic curve arises from a modular form via the Eichler–Shimura construction. This implies that the Hasse–Weil L-function, which encodes the number of points on the curve over finite fields, has analytic continuation to the whole complex s-plane. In particular it is defined at $s = 1$, although initially it is defined only to the right of $\frac{3}{2}$. We note that the Birch and Swinnerton-Dyer conjecture, which is one of the seven Millennium Prize Problems, states that the order of vanishing at $s = 1$ equals the rank of the group of rational points on the curve. At the time of writing, the BSD conjecture remains one of the biggest mysteries of number theory, despite astounding progress in recent years.

Uninitiated readers may wonder what modularity has to do with Diophantine equations. Fermat's Last Theorem has a long and very interesting history, which we will not dwell on here. Fermat speculated (in 1637) that the equation $a^n + b^n = c^n$ has no integer solutions (a, b, c) when $n > 2$—other than the trivial ones, where one of the integers is allowed to be zero. The trick, explored in the 1980s, is to consider the so-called Frey curve $y^2 = x(x - a^n)(x + b^n)$ attached to a hypothetical solution to Fermat's equation. Using monumental work of Ribet, this curve can be shown to *not* be modular.

By passing to the so-called ℓ-adic Tate module, Eichler–Shimura theory produces a two-dimensional representation of the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, having coefficients in the field of ℓ-adic numbers. If the input is a modular form f (of weight two, with rational Fourier coefficients) the pth Fourier coefficient of f determines the eigenvalues of a pth Frobenius element of the Galois group, and vice versa. Here p is a large enough prime number. We say the Galois representation is associated with f, or attached to f. In fact one can push this construction further and remove the rationality assumption on the Fourier coefficients of f (which is still of weight two, for now). In this generality, one considers the Tate module of a certain abelian variety associated with f (a quotient of the Jacobian of a modular curve) as opposed to an elliptic curve.

Galois representations take the spotlight in the general narrative of Wiles's proof, through deformation theory and "modularity lifting" theorems. Roughly speaking, you start with a two-dimensional representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with coefficients in the finite field F_ℓ. Suppose it comes from a modular form, as discussed in the previous paragraph, after reducing the coefficients modulo ℓ. The objective is then to show that *all* "lifts" (to the ℓ-adic integers $\mathbb{Z}_\ell$, say) of the representation necessarily arise from modular forms as well.

For a modular form f of weight $k \geq 2$ there is still a way to attach an ℓ-adic Galois representation to it, by utilizing the arithmetic geometry of modular curves. This construction was carried out by Deligne in the early 1970s. Rather than working with the more hands-on Tate modules, as in the $k = 2$ case, one draws on somewhat intricate algebro-geometric methods such as étale cohomology.

The weight one case stands out in several respects. This is the only case where one can expect to attach a *complex* Galois representation, and a fortiori produce an Artin L-function which should agree with the L-function of the modular form. However, in stark contrast to the $k \geq 2$ case, this Galois representation does not appear naturally in the cohomology of a variety. The by-now-standard construction of Deligne and Serre reduces the $k = 1$ case to the $k \geq 2$ case. The basic idea is to multiply a weight one form f by a suitable Eisenstein series $E_\kappa \equiv 1$ modulo ℓ, in order to increase the weight. Here E_κ is a weight 2κ modular form, so the product fE_κ is of weight $2\kappa + 1$ and $fE_\kappa \equiv f$ modulo ℓ. Essentially by invoking Deligne's construction, applied to a "relative" of fE_κ, one manufactures the sought-after complex Galois representation.

Reciprocally, in the other direction there is the prominent Langlands–Tunnell theorem. The tables are turned, and the setup is now an *odd* two-dimensional complex representation of $\mathrm{Ga}\ell(\overline{\mathbb{Q}}/\mathbb{Q})$. If the image of the representation in $\mathrm{PGL}_2(\mathbb{C})$ is solvable, then the theorem guarantees the representation is associated with a weight one form (and consequently the Artin L-function extends to an entire function if the representation is irreducible). The oddness condition is satisfied by all Galois representations attached to modular forms, and is therefore necessary. By definition it means that complex conjugation acts by a linear operator with determinant minus one. The solvability assumption, however, was eventually removed by Khare and Wintenberger, around 2010, as a result of their proof of the Serre modularity conjecture.

Chapter 6 of the book contains a concise discussion of Eichler–Shimura theory, Deligne's construction, the Langlands–Tunnell theorem, and it culminates in an exposition of the proof of the Deligne–Serre theorem. A good amount of background material, which I believe may come in handy to a beginner, is interwoven into the text. For example, the preceding Chap. 5 provides a rudimentary introduction to modular forms and the automorphic point of view. It is my hope that bringing together this material in one place will prove to be profitable for someone starting out in the field.

The main text is meant to be instructional, and not encyclopedic. Partly for this reason, I have deliberately adopted a more conversational tone in many places throughout the text, in lieu of the typical research monograph style. The manuscript is interspersed with a multitude of exercises of varying difficulty. Some are routine calculations, whereas others are intended to solidify the reader's understanding of some of the more theoretical aspects. In the appendix we review a selection of topics most readers will have encountered in a standard undergraduate number theory class. This serves mainly as a handy reference, but in fact, I have used the appendix (plus Sect. 1.4) as notes for various undergraduate courses in basic arithmetic.

It brings me joy to take this opportunity to thank my mentors, mentees, colleagues, collaborators, and students over the years. I will refrain from listing any names but you know who you are. One way or the other, you have all helped shape me as a mathematician.

Despite my best efforts and attention to detail, surely inaccuracies, mistakes, and typos have slipped through the cracks. Please send comments, corrections, and constructive feedback to the email address csorensen@ucsd.edu.

La Jolla, CA, USA Claus Sorensen
July 2025

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Chapter 1
Classical Zeta Functions and L-Functions



Euler solved the so-called Basel problem in his twenties by evaluating sums like $\sum_{n=1}^{\infty} \frac{1}{n^2}$ in the mid 1730s, but it was Riemann who saw the importance of studying the sums $\sum_{n=1}^{\infty} \frac{1}{n^s}$ as a function $\zeta(s)$ of the *complex* variable s. In his pioneering 1859 article [Rie59], Riemann brought to light a powerful relationship between the zeros of the zeta function and the distribution of prime numbers. The connection can be traced back to the fact that the zeta function can also be represented by an infinite product $\prod_p \frac{1}{1-p^{-s}}$ with p ranging over all the prime numbers. For example, the mere fact that $\zeta(s)$ has a pole at $s = 1$ immediately implies Euclid's theorem that there are infinitely many primes.

Actually the zeta function does not have *any* zeros in the right half-plane $\mathrm{Re}(s) > 1$, where it is initially defined by a convergent infinite series. However, by Riemann's insight, $\zeta(s)$ extends uniquely to a meromorphic function on the whole s-plane. For instance it turns out that $\zeta(0) = -\frac{1}{2}$, although the series $\sum_{n=1}^{\infty} \frac{1}{n^0}$ of course diverges. When we talk about the zeros of the zeta function, we are usually interested in those lying in the critical strip $0 < \mathrm{Re}(s) < 1$. Nevertheless, the ζ-function has a set of 'trivial' zeros $\{-2, -4, -6, \ldots\}$ stemming from the functional equation, which relates $\zeta(s)$ to $\zeta(1-s)$. On the other hand, when we talk about the 'distribution of primes' we refer to the asymptotic behavior of the prime-counting function $\pi(x)$, defined to be the number of primes less than or equal to some real number x. The celebrated prime number theorem asserts that $\pi(x) \sim \frac{x}{\ln x}$ as x goes to infinity. In truth the 'offset' logarithmic integral $\mathrm{Li}(x) = \int_2^x \frac{dt}{\ln t}$ gives more precise asymptotics. In any case, the prime number theorem has an equivalent formulation solely in terms of the zeta function, namely that $\zeta(s) \neq 0$ on the line $\mathrm{Re}(s) = 1$.

Having a zero-free region often enables one to move the contour in various integrals and thereby find bounds for the error term $\pi(x) - \mathrm{Li}(x)$. The all-important *Riemann hypothesis*, one of the Millennium Prize Problems specified by the Clay Mathematics Institute in 2000, is widely regarded as one of the biggest unsolved problems in mathematics–perhaps *the* biggest. In its simplicity, it states that all the

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zeros of $\zeta(s)$ in the critical strip have real part $\frac{1}{2}$. This would give the optimal zero-free region. The implications of the hypothesis for prime-counting is a bound for $\pi(x) - \mathrm{Li}(x)$ of the order of $\sqrt{x}\ln x$.

In this Chapter we first introduce the Riemann zeta function from scratch, and establish its basic analytic properties. The proof of the meromorphic continuation, and the functional equation, follows a general strategy which we will employ in later chapters as well–in more complicated situations. To give the gist of the game plan, one 'completes' the zeta function by including a factor $\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$. The Γ-function, which interpolates factorials, has completely well-understood analytic properties. The resulting completed zeta function $\xi(s) = \pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\zeta(s)$ happens to have a nice integral representation as a 'Mellin transform' of a θ-function. Using basic Fourier analysis on the real line, to wit the Poisson summation formula, one proves a key transformation property of the θ-function with respect to the substitution $t \mapsto \frac{1}{t}$. Altogether this gives the meromorphic continuation of $\xi(s)$ to the entire s-plane, the location and order of its poles along with their residues, and its invariance under the point reflection $s \mapsto 1 - s$ fixing the 'center' of the critical strip $\frac{1}{2}$.

We also discuss a different approach to analytically continuing $\zeta(s)$ from Riemann's original paper. This method has the benefit of generating formulae for the special values $\zeta(1 - r)$ on the spot, where r is a positive integer. Surprisingly these values are *rational* numbers, indeed they are essentially given by the ubiquitous Bernoulli numbers. For $r = 1$ we recover the formula $\zeta(0) = -\frac{1}{2}$ mentioned earlier. Via the functional equation this solves the Basel problem: More generally it allows us to evaluate the sums $\zeta(2r)$ explicitly, for all r, in terms of Bernoulli numbers and other simple constants. For instance we get Euler's solution $\zeta(2) = \frac{\pi^2}{6}$.

The theory of L-functions got underway in Dirichlet's article [Dir37]. His motivation was to show the existence of infinitely many primes in any arithmetic progression $a + N\mathbb{Z}$, assuming a is relatively prime to N of course. He showed not only that, but the stronger result that the primes are equidistributed among the $\phi(N)$ invertible residue classes modulo N. In other words primes are unbiased: Half of all (odd) primes belong to $1 + 4\mathbb{Z}$ and the other half belong to $3 + 4\mathbb{Z}$. Similarly with 4 replaced by N.

Dirichlet proved his theorem by encoding a character χ of $(\mathbb{Z}/N\mathbb{Z})^\times$ in a variant of the zeta function, namely $L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$. This function enjoys many of the properties of $\zeta(s)$. When χ is primitive, i.e. it does not factor through $(\mathbb{Z}/M\mathbb{Z})^\times$ for a smaller $M|N$, there is a functional equation relating $L(s, \chi)$ to $L(1 - s, \overline{\chi})$. It is a bit more involved though. First of all, the Γ-factors in the completed version of the L-function depend on the sign $\chi(-1)$. Moreover, the functional equation entails a phase factor $W(\chi)$ which is basically a normalized Gauss sum. We note that Gauss introduced these trigonometric sums in the early 19th century, in part to prove quadratic reciprocity, cubic reciprocity, and quartic reciprocity. So their arithmetic significance is amply documented.

Going back to Dirichlet's theorem on primes in arithmetic progressions, more or less formal manipulations reduce the theorem to the assertion that $L(1, \chi) \neq 0$. We

use this non-vanishing claim at $s = 1$ as a segue to Dedekind zeta functions, which justify the claim in a natural way.

Dedekind generalized the Riemann zeta function to number fields $K/\mathbb{Q}$ by introducing the sum $\zeta_K(s) = \sum_{I \neq 0} \frac{1}{N(I)^s}$, where I now runs over the nonzero ideals of the ring of integers $\mathcal{O}_K$, and $N(I)$ denotes the cardinality of the quotient ring $\mathcal{O}_K/I$. For $K = \mathbb{Q}$ this of course boils down to $\zeta(s)$. Interestingly, Dedekind launched his theory in an appendix to a number theory textbook by Dirichlet published in 1863. This book, titled Vorlesungen über Zahlentheorie, was based on Dirichlet's lectures at the University of Göttingen. After Dirichlet's death, Dedekind apparently continued editing the book and added various appendices, among those one on $\zeta_K(s)$.

The completed version $\xi_K(s)$ now includes a bunch of Γ-factors, one for each 'infinite place' of K: Each real field embedding $K \hookrightarrow \mathbb{R}$ contributes a factor $\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$, whereas a pair of complex conjugate non-real embeddings $K \hookrightarrow \mathbb{C}$ gives rise to a factor $2(2\pi)^{-s}\Gamma(s)$. (The discriminant D_K also appears in the definition.) The major result is that $\xi_K(s)$ can be extended to a meromorphic function on all of $\mathbb{C}$, which is invariant under $s \mapsto 1 - s$. What's more, $s = 1$ and $s = 0$ are the only poles. Both are simple, and with residues $\pm\frac{2^r hR}{w}$ respectively. The striking factors here are the class number h and the regulator R. The residue $\operatorname{res}_{s=1}\zeta_K(s)$ of the actual Dedekind zeta function is given by a similar formula known as the analytic class number formula, which involves just about every arithmetic invariant assigned to K. As an obvious by-product this residue is always a *positive* constant. This simple observation is instrumental in proving that $L(1, \chi)$ is nonzero.

When K is a subfield of a cyclotomic field $\mathbb{Q}(\zeta_N)$, in other words when $K/\mathbb{Q}$ has an abelian Galois group, one can express $\zeta_K(s)$ as a product of Dirichlet L-functions $L(s, \chi)$. This is an analytic reformulation of how prime numbers split in K (ramification indices, inertia degrees, and so on). For example, when $N > 2$ is a prime and K is the quadratic subfield of $\mathbb{Q}(\zeta_N)$, the Dedekind zeta function factors as $\zeta(s)$ times the L-function of the Legendre symbol $\left(\frac{\bullet}{N}\right)$. This is *equivalent* to the quadratic reciprocity law. At the other extreme, when $K = \mathbb{Q}(\zeta_N)$ is a cyclotomic field, the Dedekind zeta function factors as a product of Dirichlet L-functions $L(s, \chi)$ with χ varying over all primitive characters of conductor dividing N. Thus one factors $\operatorname{res}_{s=1}\zeta_K(s)$ as a product of special values $\prod_{\chi \neq 1} L(1, \chi)$ and concludes that $L(1, \chi) \neq 0$.

The factorization formula for $\zeta_K(s)$ in the case of a *non*-abelian Galois extension $K/\mathbb{Q}$ will play a very important role in later chapters. In general one has to replace the Dirichlet L-functions with Artin L-functions attached to the irreducible representations of $\operatorname{Gal}(K/\mathbb{Q})$.

Sections 1.2 and 1.3 of this Chapter examine the general theory of Dirichlet L-functions and Dedekind zeta functions. Section 1.4 is more elementary in nature and can be skipped on a first reading. It covers related background material from a typical first undergraduate course in number theory (asymptotics of arithmetic functions, Mertens constants, Dirichlet series, and more). One highlight is Landau's Lemma, which essentially says the abscissa of convergence of a Dirichlet series with non-negative coefficients is a pole. Landau's result will play a major role later when we

discuss the Rankin-Selberg method and the irreducibility of Galois representations in Chaps. 5 and 6.

1.1 The Riemann Zeta Function

1.1.1 Convergence and First Remarks

We begin by recalling the mother of all zeta functions.

Definition 1.1 For $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$ the Riemann zeta function $\zeta(s)$ is defined as the sum of the infinite series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}.$$

(Note that $n^{-s} = e^{-s\ln(n)}$ has absolute value $n^{-\mathrm{Re}(s)}$ and therefore the series is absolutely convergent in the open right half-plane $\mathrm{Re}(s) > 1$.)

It is easy to see the series $\sum_{n=1}^{\infty} n^{-s}$ is uniformly convergent in the region $\mathrm{Re}(s) \geq 1 + \epsilon$ for any choice of $\epsilon > 0$. In particular it converges uniformly in all compact subsets of $\{s \in \mathbb{C} : \mathrm{Re}(s) > 1\}$ and by complex analysis $\zeta(s)$ defines a holomorphic function on the domain of convergence.

One of our immediate goals is to analytically continue the Riemann zeta function to a holomorphic function on $\mathbb{C} - \{1\}$ with a simple pole at $s = 1$. Note that naively plugging in $s = 1$ gives the harmonic series $\sum_{n=1}^{\infty} n^{-1} = \infty$.

Here are a few famous special values of the Riemann zeta function which go back to Euler:

$$\zeta(2) = \frac{\pi^2}{6} \qquad \zeta(4) = \frac{\pi^4}{90} \qquad \zeta(6) = \frac{\pi^6}{945} \quad \ldots \quad \zeta(12) = \frac{691 \cdot \pi^{12}}{638512875} \quad \ldots.$$

In general $\zeta(2n)\pi^{-2n}$ is a rational number expressed explicitly in terms of the Bernoulli number B_{2n}. More about that later. The odd zeta values $\zeta(2n+1)$ are far more mysterious, and they supposedly play an important role even in physics (look up antiferromagnetic spin chains). There is also a connection to the orders of the stable homotopy groups of spheres. The fact that $\zeta(3) = 1.202...$ is irrational is a big result of Apéry from 1978. Not much is known beyond that, but it is expected that the numbers $\zeta(2n+1)$ are algebraically independent.

Example 1.2 As a first attempt at meromorphic continuation we briefly mention the following simple-minded idea. We introduce an alternating variant of $\zeta(s)$ as follows.

$$D(s) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s} = 1 - 2^{-s} + 3^{-s} - 4^{-s} + \cdots.$$

One can show (and we will eventually, using the partial summation formula) that $D(s)$ is conditionally convergent for $\mathrm{Re}(s) > 0$, uniformly in compact subsets. The trick is that

$$\zeta(s) - D(s) = 2^{1-s}\zeta(s) \Longrightarrow \zeta(s) = \frac{D(s)}{1-2^{1-s}}$$

for $\mathrm{Re}(s) > 1$. The key point is $D(s)$ makes sense in the larger region $\mathrm{Re}(s) > 0$ and we can use the above formula to extend $\zeta(s)$ into the critical strip where $0 < \mathrm{Re}(s) < 1$. The main drawback of this approach are the zeroes of $1 - 2^{1-s}$ which lead to potential poles at $s = 1 + \frac{2\pi i}{\ln(2)}m$ with $m \in \mathbb{Z}$.

Exercise 1.3 In Example 1.2 show that $1 + \frac{2\pi i}{\ln(2)}m$ is not a pole for $m \neq 0$ by considering

$$\widetilde{D}(s) = 1 + \frac{1}{2^s} - \frac{2}{3^s} + \frac{1}{4^s} + \frac{1}{5^s} - \frac{2}{6^s} + \cdots = (1 - 3^{1-s})\zeta(s).$$

Exercise 1.4 In the previous example, use the observation $D(1) = \ln(2)$ to show $\zeta(s)$ has a simple pole at $s = 1$ with residue one. That is $\zeta(s) = \frac{1}{s-1} + O(1)$ as $s \to 1$.

Exercise 1.5 Prove that $\zeta(s) \neq 0$ for real inputs $s \in (0, 1)$.

If you read these notes you have surely heard about the Riemann hypothesis, which is one of the most important unsolved problems in mathematics (one of the Clay Mathematics Institute's Millennium Prize Problems).

Conjecture 1.6 (The Riemann hypothesis) *All zeroes of $\zeta(s)$ in the critical strip $0 < Re(s) < 1$ have real part $Re(s) = \frac{1}{2}$.*

Once we extend $\zeta(s)$ to $\mathbb{C} - \{1\}$ we will see that in fact $\zeta(s)$ does have zeroes outside the critical strip, namely the so-called trivial zeroes

$$\{-2, -4, -6, \ldots\}.$$

Curiously $\zeta(0) = -\frac{1}{2}$ which is a special case of a more general formula relating the special values $\zeta(-n)$ to the Bernoulli number B_{n+1} for $n = 0, 1, 2, \ldots$. We will return to that later on in this course.

1.1.2 The Euler Product

Riemann introduced $\zeta(s)$ with the asymptotics of the prime-counting function $\pi(x)$ in mind. It turns out the location of the ζ-zeroes and the distribution of primes are intricately linked. The connection comes from the following expression for $\zeta(s)$ as an infinite product.

Proposition 1.7 *For $Re(s) > 1$ the Riemann zeta function is given by the product*

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

(Here the product ranges over all prime numbers p.)

Proof For a fixed p we have the geometric sum $\frac{1}{1-p^{-s}} = \sum_{r=0}^{\infty} p^{-rs}$ for $\mathrm{Re}(s) > 0$. Taking the product over all $p \leq x$ gives

$$\prod_{p \leq x} \frac{1}{1 - p^{-s}} = \sum_{n \in N_x} n^{-s}$$

where N_x is the set of positive integers with all prime factors $\leq x$. Therefore, for $\mathrm{Re}(s) > 1$, we have

$$\zeta(s) - \prod_{p \leq x} \frac{1}{1 - p^{-s}} = \sum_{n \notin N_x} n^{-s}.$$

Now note that $n \notin N_x$ has a prime factor $p > x$. In particular $n \geq p > x$ which gives the estimate

$$|\sum_{n \notin N_x} n^{-s}| \leq \sum_{n > x} n^{-\mathrm{Re}(s)} = O(x^{1-\mathrm{Re}(s)})$$

as $x \to \infty$. □

Exercise 1.8 For $\mathrm{Re}(s) > 1$ we define $\ln \zeta(s)$ by the infinite series

$$\ln \zeta(s) = \sum_p \ln\big(\frac{1}{1 - p^{-s}}\big) = \sum_p \sum_{r=1}^{\infty} \frac{p^{-rs}}{r}.$$

(Recall the Taylor expansion $\ln\big(\frac{1}{1-z}\big) = \sum_{r=1}^{\infty} \frac{z^r}{r}$ valid for $|z| < 1$.)

(a) Check that the series defining $\ln \zeta(s)$ is indeed absolutely convergent for $\mathrm{Re}(s) > 1$, uniformly in $\mathrm{Re}(s) \geq 1 + \epsilon$ for any $\epsilon > 0$. Conclude $\ln \zeta(s)$ is holomorphic in the domain of convergence.
(b) Explain why $\zeta(s) = \exp\big(\ln \zeta(s)\big)$ and deduce that $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$.
(c) Show that $\mathrm{Re}\big(\ln \zeta(s)\big) = \ln |\zeta(s)|$ in the region $\mathrm{Re}(s) > 1$.

Asymptotic behavior of $\zeta(s)$ as $s \to 1^+$ (approaching $s = 1$ along the real axis from the right): Let $s > 1$. Write

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \int_1^{\infty} t^{-s} dt + \sum_{n=1}^{\infty} \int_n^{n+1} (n^{-s} - t^{-s}) dt = \frac{1}{s-1} + \sum_{n=1}^{\infty} \int_n^{n+1} (n^{-s} - t^{-s}) dt.$$

Here, by the fundamental theorem of calculus,

$$0 < n^{-s} - t^{-s} = \int_n^t sx^{-s-1}dx < \int_n^{n+1} sx^{-s-1}dx < sn^{-s-1} < \frac{s}{n^2}$$

for $n < t < n+1$. Therefore

$$0 < \zeta(s) - \frac{1}{s-1} < s\zeta(2).$$

In particular $\zeta(s) - \frac{1}{s-1}$ remains bounded as $s \to 1^+$. This of course implies the Euler product cannot be finite, in other words it gives an analytic proof of the existence of infinitely many primes: $\pi(x) \to \infty$ as $x \to \infty$.

Exercise 1.9 Show that $\ln \zeta(s) = \ln\left(\frac{1}{s-1}\right) + O(s-1)$ as $s \to 1^+$.

1.1.3 Partial Summation

This is a fundamental tool in analytic number theory. The setup is as follows. Suppose we are given some sequence of complex numbers $a_1, a_2, a_3, \ldots$. The associated sum function is $A(x) = \sum_{n \le x} a_n$. This is a step function $A : \mathbb{R} \to \mathbb{C}$ and clearly $A(x) = 0$ for $x < 1$.

Lemma 1.10 *For any function $f : [1, \infty) \to \mathbb{C}$ we have the identity below for all $x \ge 1$,*

$$\sum_{n \le x} a_n f(n) = A([x])f([x]) + \sum_{n=1}^{[x]-1} A(n)(f(n) - f(n+1))$$

Proof Let $x \ge 1$. We work out the left-hand side by inserting $a_n = A(n) - A(n-1)$,

$$\begin{aligned}\sum_{n \le x} a_n f(n) &= A(1)f(1) + \sum_{n=2}^{[x]} (A(n) - A(n-1))f(n) \\ &= A(1)(f(1) - f(2)) + A(2)(f(2) - f(3)) + \cdots + A([x])f([x]) \\ &= A([x])f([x]) + \sum_{n=1}^{[x]-1} A(n)(f(n) - f(n+1))\end{aligned}$$

as desired. □

When f is $\mathcal{C}^1$ we obtain the neat formula below.

Proposition 1.11 (The partial summation formula) *Adopt the notation above and assume f is continuously differentiable. Then for all $x \ge 1$,*

$$\sum_{n \le x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)dt.$$

Proof By our assumption that f is $\mathcal{C}^1$ the fundamental theorem of calculus tells us

$$A(n)(f(n) - f(n+1)) = -\int_n^{n+1} A(t)f'(t)dt.$$

(Note that $A(t) = A(n)$ for $t \in [n, n+1)$.) Summing up these identities for $n = 1, \ldots, [x] - 1$ yields

$$\sum_{n=1}^{[x]-1} A(n)(f(n) - f(n+1)) = -\int_1^{[x]} A(t)f'(t)dt.$$

Finally split the integral as $\int_1^x = \int_1^{[x]} + \int_{[x]}^x$ and observe $\int_{[x]}^x A(t)f'(t)dt = A(x)f(x) - A([x])f([x])$. □

Example 1.12 Take all $a_n = 1$ and $f(t) = \frac{1}{t}$. In this case $A(x) = [x]$ and the partial summation formula implies

$$\begin{aligned}
\sum_{n \le x} \frac{1}{n} &= \frac{[x]}{x} + \int_1^x \frac{[t]}{t^2} dt \\
&= \frac{[x]}{x} + \int_1^x \frac{t}{t^2} dt + \int_1^x \frac{[t]-t}{t^2} dt \\
&= 1 - \frac{\{x\}}{x} + \ln(x) - \int_1^x \frac{\{t\}}{t^2} dt \\
&= 1 - \frac{\{x\}}{x} + \ln(x) - \int_1^\infty \frac{\{t\}}{t^2} dt + \int_x^\infty \frac{\{t\}}{t^2} dt \\
&= \ln(x) + \big(1 - \int_1^\infty \frac{\{t\}}{t^2} dt\big) + O(1/x) \\
&= \ln(x) + \gamma + O(1/x)
\end{aligned}$$

where we have introduced the notation $\{t\} = t - [t] \in [0, 1)$ for the fractional part of a real number t. The number $\gamma = 1 - \int_1^\infty \frac{\{t\}}{t^2} dt = 0.577\ldots$ is known as Euler's constant, or the Euler–Mascheroni constant. The irrationality of γ remains a big open question in mathematics.

Next we apply the partial summation formula to analytically continue $\zeta(s)$ across the critical strip. Take all $a_n = 1$ as in the previous example, but now let $f(t) = t^{-s}$ for some fixed $s \in \mathbb{C}$. Then

$$\begin{aligned}\sum_{n\le x} n^{-s} &= [x]x^{-s} + s\int_1^x [t]t^{-s-1}dt \\ &= x^{1-s} - \{x\}x^{-s} + s\int_1^x tt^{-s-1}dt - s\int_1^x \{t\}t^{-s-1}dt \\ &= x^{1-s} + \frac{s}{1-s}(x^{1-s}-1) - s\int_1^x \{t\}t^{-s-1}dt + O(x^{-\mathrm{Re}(s)}).\end{aligned}$$

When $\mathrm{Re}(s) > 1$ both sides converge as $x \to \infty$ and in the limit we get the relation

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{t\}t^{-s-1}dt$$

The expression on the right defines a meromorphic function in $\mathrm{Re}(s) > 0$ with a simple pole at $s = 1$. This approach even gives us the constant term in the Laurent expansion:

Corollary 1.13 $\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$ *as* $s \to 1$.

Proof Letting $s \to 1$ in the above formula for $\zeta(s)$ gives the limit

$$\zeta(s) - \frac{1}{s-1} = 1 - s\int_1^\infty \{t\}t^{-s-1}dt \quad\longrightarrow\quad 1 - \int_1^\infty \{t\}t^{-2}dt = \gamma$$

as desired. □

Exercise 1.14 Return to Example 1.2 and show the series defining $D(s)$ is convergent for $\mathrm{Re}(s) > 0$.

1.1.4 Non-vanishing for Re(s) = 1

The Riemann hypothesis predicts all non-trivial zeroes of $\zeta(s)$ have real part $\frac{1}{2}$. Our next result shows that at least there are no zeroes on the boundary $\mathrm{Re}(s) = 1$ of the domain of convergence. This is already enough to prove the prime number theorem $\pi(x) \sim \frac{x}{\ln(x)}$.

Theorem 1.15 $Re(s) = 1 \Longrightarrow \zeta(s) \neq 0$.

Proof The argument is based on the following miraculous inequality $\forall\theta \in \mathbb{R}$. By the double-angle formula for cosine

$$\begin{aligned}3 + 4\cos(\theta) + \cos(2\theta) &= 3 + 4\cos(\theta) + 2\cos(\theta)^2 - 1 \\ &= 2(1 + 2\cos(\theta) + \cos(\theta)^2) \\ &= 2(1+\cos(\theta))^2 \\ &\geq 0.\end{aligned}$$

In Exercise 1.8 we introduced

$$\ln \zeta(s) = \sum_p \ln\Big(\frac{1}{1-p^{-s}}\Big) = \sum_p \sum_{r=1}^{\infty} \frac{p^{-rs}}{r}$$

for $\mathrm{Re}(s) > 1$. Taking absolute values on both sides of $\zeta(s) = \exp\big(\ln \zeta(s)\big)$ we find that

$$\ln|\zeta(s)| = \mathrm{Re}\big(\ln \zeta(s)\big) = \sum_p \sum_{r=1}^{\infty} \frac{p^{-r\sigma}}{r} \cos(r\tau \ln p)$$

where $s = \sigma + i\tau$ with $\sigma, \tau \in \mathbb{R}$ and $\sigma > 1$. We fix a $\tau \in \mathbb{R}$ and use this formula for $s \in \{1+\epsilon, 1+\epsilon+i\tau, 1+\epsilon+2i\tau\}$ and eventually let $\epsilon \to 0^+$. With the preliminary observation in mind the trick is to look at the combination below:

$$3\mathrm{Re}\big(\ln \zeta(1+\epsilon)\big) + 4\mathrm{Re}\big(\ln \zeta(1+\epsilon+i\tau)\big) + \mathrm{Re}\big(\ln \zeta(1+\epsilon+2i\tau)\big) =$$

$$\sum_p \sum_{r=1}^{\infty} \frac{p^{-r(1+\epsilon)}}{r}\big(3 + 4\cos(r\tau \ln p) + \cos(2r\tau \ln p)\big) \geq 0.$$

Exponentiating both sides gives us the inequality $\forall \epsilon > 0$:

$$|\zeta(1+\epsilon)|^3 \cdot |\zeta(1+\epsilon+i\tau)|^4 \cdot |\zeta(1+\epsilon+2i\tau)| \geq 1. \tag{1.1}$$

For the sake of contradiction suppose $\zeta(1+i\tau) = 0$ for some $\tau \in \mathbb{R}^\times$. From the Taylor expansion we find that $\zeta(1+\epsilon+i\tau) = O(\epsilon)$ as $\epsilon \to 0$. Moreover, we know $\zeta(1+\epsilon) = \epsilon^{-1} + O(1)$ since $s = 1$ is a simple pole with residue 1. Finally $\zeta(1+\epsilon+2i\tau) = O(1)$ as $1+2i\tau$ is not a pole. Altogether

$$\zeta(1+\epsilon)^3 \cdot \zeta(1+\epsilon+i\tau)^4 \cdot \zeta(1+\epsilon+2i\tau) = O(\epsilon^{-3}) \cdot O(\epsilon^4) \cdot O(1) = O(\epsilon)$$

In particular $\zeta(1+\epsilon)^3 \cdot \zeta(1+\epsilon+i\tau)^4 \cdot \zeta(1+\epsilon+2i\tau) \to 0$ as $\epsilon \to 0^+$. This contradicts the inequality (1.1). All these products have absolute value at least one. □

Theorem 1.15 is in fact equivalent to the prime number theorem. We will not go into too much detail, but here is a rough outline: The Riemann zeta function has logarithmic derivative

$$\frac{\zeta'(s)}{\zeta(s)} = -\sum_{n=1}^{\infty} \Lambda(n) n^{-s}$$

where Λ is the von Mangoldt function (thus $\Lambda(n) = \ln p$ if $n = p^r$ with $r > 0$ and $\Lambda(n) = 0$ otherwise). Why? Find $\zeta'(s)$ by term-wise differentiation and note that $\ln n = \sum_{d|n} \Lambda(d)$ for all $n \in \mathbb{N}$. Now we know $-\frac{\zeta'(s)}{\zeta(s)}$ has no poles on $\mathrm{Re}(s) = 1$

other than $s = 1$ and is given by a Dirichlet series with non-negative coefficients. By so-called Tauberian results in complex analysis ('Wiener-Ikehara') we deduce that

$$\psi(x) := \sum_{n \le x} \Lambda(n) \sim x$$

as $x \to \infty$. This is known as the Chebyshev ψ-function. Completely elementary, but non-trivial, arguments show that $\pi(x) \sim \frac{\psi(x)}{\ln(x)}$ as $x \to \infty$, and altogether this gives the prime number theorem. We elaborate on the last claim in Sect. A.3 of the appendix, see Theorem A.22 and Corollary A.23 in particular.

The Riemann hypothesis in some sense implies the optimal estimate for $\pi(x)$. It is well-known that the logarithmic integral $\mathrm{li}(x) = \int_0^\infty \frac{dt}{\ln t}$ gives a better approximation to $\pi(x)$ than $\frac{x}{\ln x}$. Because of the singularity at $t = 1$ this integral is really a principal value, i.e. the limit of $\int_0^{1-\epsilon} + \int_{1+\epsilon}^\infty$ as $\epsilon \to 0^+$. One can show that the error term $\pi(x) - \mathrm{li}(x)$ has order of magnitude $O(x^b \ln x)$ where $b \in [\frac{1}{2}, 1]$ is any upper bound for the real parts $\mathrm{Re}(\rho)$ of all the zeroes ρ. Thus the Riemann hypothesis implies a bound $O(x^{\frac{1}{2}} \ln x)$. In fact there is a more precise version, apparently due to Schoenfeld, which says that

$$\text{Riemann hypothesis} \quad \Longrightarrow \quad |\pi(x) - \mathrm{li}(x)| < \frac{1}{8\pi} x^{\frac{1}{2}} \ln x, \qquad \forall x \ge 2657.$$

An expert has informed me that one even expects a bound $O(x^{\frac{1}{2}} \ln \ln x)$ by taking the imaginary parts $\mathrm{Im}(\rho)$ into account.

Establishing the bound $O(x^b \ln x)$ again make use of the ψ-function. One can strengthen the previous discussion and in fact give an explicit formula for $\psi(x)$ in terms of the ζ-zeroes:

$$\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} - \ln(2\pi).$$

Here ρ runs over all zeroes of $\zeta(s)$ (including the trivial ones which add up to a contribution $\frac{1}{2}\ln(1 - \frac{1}{x^2})$).

How do we know $\zeta(s)$ actually does have zeroes in the critical strip? The argument principle tells us a good deal more. The contour integral $\frac{1}{2\pi i} \oint \frac{\zeta'(s)}{\zeta(s)} ds$ equals # zeroes − # poles within the contour (counting multiplicity). By integrating around a tall rectangle inside the critical strip one can show that $N(T) \sim \frac{T}{2\pi} \ln\left(\frac{T}{2\pi}\right)$ as $T \to \infty$ where $N(T)$ is the number of ζ-zeroes ρ with $0 < \mathrm{Im}(\rho) < T$. See [Tit86, Theorem 9.3, p. 212] for example.

1.1.5 The Gamma Function

Eventually we will complete L-functions by throwing in Euler factors at the infinite primes built out of the gamma function $\Gamma(s)$. Doing so makes the functional equation more transparent as we will see. In this section we briefly review the properties of $\Gamma(s)$ we will use, for the most part without proofs.

Definition 1.16 $\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}dt$ for $\mathrm{Re}(s) > 0$.

The integral converges absolutely and uniformly in compact subsets of $\mathrm{Re}(s) > 0$ (convergence issues arise from $t = 0$). Thus Γ is holomorphic in this right half-plane, and easily extends to a meromorphic function on the whole complex plane using the next "recursive" relation.

Lemma 1.17 $\Gamma(s+1) = s\Gamma(s)$ *and* $\Gamma(1) = 1$.

Proof This is partial integration. Just integrate $\frac{d}{dt}\left[(-e^{-t})t^s\right] = e^{-t}t^s + (-e^{-t})st^{s-1}$ from 0 to ∞. □

In particular $\Gamma(n+1) = n!$ for $n \in \mathbb{Z}_{\geq 0}$. In this sense the Γ-function interpolates factorials.

Exercise 1.18 Use the Gaussian integral $\int_{-\infty}^{+\infty} e^{-t^2}dt = \sqrt{\pi}$ to show $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

We move on to locating the poles of the Γ-function and finding their residues.

Proposition 1.19 *The Γ-function admits meromorphic continuation to $\mathbb{C}$. The poles are precisely the non-positive integers $\{0, -1, -2, -3, \ldots\}$. They are all simple poles, with residues given by the formula*

$$res_{s=-n}\Gamma(s) = \frac{(-1)^n}{n!}$$

for $n \in \mathbb{Z}_{\geq 0}$.

Proof For $N \in \mathbb{Z}_{\geq 0}$ let $P(N)$ be the following statement: The Γ-function admits meromorphic continuation to $\mathrm{Re}(s) > -N$ with simple poles $\{0, -1, -2, \ldots, -N+1\}$ whose residues are given by the formula in the Proposition. $P(0)$ is true for trivial reasons. We show $P(N) \Rightarrow P(N+1)$. The relation $\Gamma(s) = \frac{\Gamma(s+1)}{s}$ extends Γ to a meromorphic function on $\mathrm{Re}(s) > -(N+1)$. The only poles are $\{0, -1, -2, \ldots, -N\}$. They are simple with the stated residues by the induction hypothesis combined with the observation

$$\mathrm{res}_{s=-N}\Gamma(s) = \lim_{s\to -N}(s+N)\Gamma(s) = \lim_{s\to -N}(s+1+N-1)\frac{\Gamma(s+1)}{s} = -\frac{1}{N}\cdot \mathrm{res}_{s=-N+1}\Gamma(s).$$

This proves the Proposition. □

What about the zeroes? Easy. The Γ-function does not have any zeroes, as we will show momentarily. First we recall without proof two fundamental identities satisfied by $\Gamma(s)$.

Proposition 1.20 *The Γ-function satisfies the following two functional equations.*

(1) ("Euler's reflection formula")

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}.$$

(2) ("Legendre's duplication formula")

$$\Gamma(s)\Gamma(s+\frac{1}{2}) = 2^{1-2s}\sqrt{\pi}\Gamma(2s).$$

Proof Omit. Look up properties of the beta function $B(s,s') = \int_0^1 t^{s-1}(1-t)^{s'-1}dt = \frac{\Gamma(s)\Gamma(s')}{\Gamma(s+s')}$. □

The reflection formula derives its name from the geometric nature of the transformation $s \mapsto 1-s$ of the complex plane. This is the point reflection over $\frac{1}{2}$; the "center" of the critical strip. Presumably the duplication formula gets its name from the fact that it involves s and its double $2s$.

Exercise 1.21 Use the formulas in Proposition 1.20 to find the special values $\Gamma(\frac{1}{2}+n)$ for all $n \in \mathbb{Z}$. (`Hint`: Use (2) to deal with the case $n \geq 0$. Then use (1) when $n \leq 0$.)

Corollary 1.22 *The Γ-function has no zeroes. In other words $\frac{1}{\Gamma(s)}$ is an entire function with simple zeroes at the non-positive integers.*

Proof By the reflection formula $\Gamma(s)\Gamma(1-s)\sin(\pi s) = \pi \neq 0$ for all s. Suppose $\Gamma(s) = 0$. Since sine is entire $1-s$ must be a pole of Γ, that is $1-s = -n$ for some $n \in \mathbb{Z}_{\geq 0}$. This implies $\sin(\pi s) = \sin(\pi(n+1)) = 0$. Therefore $1-s = -n$ must be a pole of Γ of order at least two. This is a contradiction, the Γ-function only has simple poles. □

The poles of $\Gamma(s)$ account for the trivial zeroes of $\zeta(s)$ (once we prove the latter has analytic continuation). We will soon show the *completed* Riemann zeta function $\xi(s) = \pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ has meromorphic continuation to $\mathbb{C}$ and satisfies the functional equation

$$\xi(s) = \xi(1-s).$$

The poles are $s = 0, 1$ and both are simple. More geometrically $\xi(s)$ is invariant under point reflection in $\frac{1}{2}$. We often interpret the extra factor $\pi^{-s/2}\Gamma(\frac{s}{2})$ as an Euler factor pertaining to the infinite prime of $\mathbb{Q}$. Tate's thesis solidifies this interpretation.

For now we will assume these analytic properties of $\xi(s)$ and explain why $\zeta(s)$ has simple zeroes at $\{-2, -4, -6, \ldots\}$. So let $s = -2n$ for some positive integer n. Then

$$\xi(s) = \xi(1-s) = \pi^{-(\frac{1}{2}+n)}\Gamma(\frac{1}{2}+n)\zeta(1+2n) \neq 0$$

since Γ has no zeroes and obviously $\zeta(1+2n) > 0$. On the other hand Γ has a simple pole at $-n$ and to make up for that we must have $\zeta(-2n) = 0$ as claimed.

Note that the previous argument breaks down for $n = 0$ since $\zeta(1) = \infty$. Also, what goes wrong at the *odd* negative integers?

Exercise 1.23 Modify the argument and show $\zeta(0) = -\frac{1}{2}$. (`Hint`: Let $s \to 0$ in $s\xi(s) = s\xi(1-s)$ and use $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.)

1.1.6 Fourier Analysis on the Real Line

In this section we will briefly review parts of Fourier theory for $\mathbb{R}$ needed to establish the analytic continuation of $\zeta(s)$ to all of $\mathbb{C}$.

First recall the Fourier transformation of an integrable function f is

$$\widehat{f}(x) = \int_{-\infty}^{+\infty} f(t)e^{-2\pi ixt}dt.$$

We will only be taking Fourier transforms of *Schwartz* functions f. Those are $\mathcal{C}^\infty$-functions ("smooth") with rapid decay, which means

$$\|f\|_{m,n} = \sup_{t\in\mathbb{R}} |t^m f^{(n)}(t)| < \infty$$

for all $m, n \in \mathbb{Z}_{\geq 0}$. Loosely speaking this says all higher order derivatives of f exist and decay faster than any polynomial. We denote the vector space of all Schwartz function by $\mathcal{S}(\mathbb{R})$.

Here is a fundamental example which we will make frequent use of throughout the course.

Example 1.24 Let $f(t) = e^{-t^2}$ be the Gaussian function, which clearly belongs to $\mathcal{S}(\mathbb{R})$. We compute its Fourier transform.

$$\begin{aligned}\widehat{f}(x) &= \int_{-\infty}^{+\infty} e^{-t^2}e^{-2\pi ixt}dt \\ &= \int_{-\infty}^{+\infty} e^{-\pi^2x^2}e^{-(t+\pi ix)^2}dt\end{aligned}$$

$$
\begin{aligned}
&= e^{-\pi^2 x^2} \cdot \int_{-\infty}^{+\infty} e^{-(t+\pi i x)^2} dt \\
&\overset{(\dagger)}{=} e^{-\pi^2 x^2} \cdot \int_{-\infty}^{+\infty} e^{-t^2} dt \\
&= \sqrt{\pi} e^{-\pi^2 x^2}.
\end{aligned}
$$

In step (†) we have shifted the contour integration of e^{-s^2} from $\mathrm{Im}(s) = \pi x$ to the real axis (we leave it to reader to make this precise by integrating around a rectangle and checking the vertical contributions vanish in the limit).

From the above calculation one easily deduces the more general result that $f_a(t) = f(at) = e^{-a^2 t^2}$ has Fourier transformation $\widehat{f_a}(x) = \frac{1}{a}\widehat{f}(\frac{x}{a}) = \frac{\sqrt{\pi}}{a} e^{-\pi^2 (x/a)^2}$ for all $a > 0$. In particular $e^{-\pi t^2}$ is its own Fourier transform.

The next basic result says that $\widehat{(\cdot)}$ is involutive on $\mathcal{S}(\mathbb{R})$ up to a sign.

Theorem 1.25 (Fourier inversion) *Fourier transformation defines an automorphism* $\mathcal{S}(\mathbb{R}) \xrightarrow{\widehat{(\cdot)}} \mathcal{S}(\mathbb{R})$. *More precisely, one can recover* $f \in \mathcal{S}(\mathbb{R})$ *from* $\widehat{f}$ *via the Fourier inversion formula:*

$$
f(x) = \hat{\hat{f}}(-x) = \int_{-\infty}^{+\infty} \widehat{f}(t) e^{+2\pi i x t} dt.
$$

Proof Omitted. This is a standard result from real analysis, see [Rud87, Chap. 9] for example. □

The key result in this section is the following summation formula due to Poisson, which is crucial for establishing the functional equation for the Riemann zeta function.

Theorem 1.26 (The Poisson summation formula) *For any* $f \in \mathcal{S}(\mathbb{R})$ *with Fourier transform* $\widehat{f}$ *we have the equality of sums*

$$
\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \widehat{f}(n).
$$

(Both sides are absolutely convergent.)

Proof We symmetrize f by considering the sum of all its $\mathbb{Z}$-translates $F(x) = \sum_{n \in \mathbb{Z}} f(x+n)$. Convergence is guaranteed by the rapid decay of the Schwartz function. By construction F is a periodic $\mathcal{C}^\infty$-function, and may be viewed as a function on the circle $\mathbb{R}/\mathbb{Z}$. As such it has a Fourier series expansion

$$
F(x) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n x} \qquad a_n = \int_0^1 F(t) e^{-2\pi i n t} dt.
$$

Note that $F(0) = \sum_{n\in\mathbb{Z}} a_n$. On the other hand $F(0) = \sum_{n\in\mathbb{Z}} f(n)$ by definition of F. It suffices to show $a_n = \widehat{f}(n)$. Inserting the definition of F we unfold the formula for the Fourier coefficient

$$\begin{aligned} a_n &= \int_0^1 \sum_{m\in\mathbb{Z}} f(t+m)e^{-2\pi int}dt \\ &= \int_0^1 \sum_{m\in\mathbb{Z}} f(t+m)e^{-2\pi in(t+m)}dt \\ &= \int_{-\infty}^{+\infty} f(t)e^{-2\pi int}dt \\ &= \widehat{f}(n). \end{aligned}$$

This proves the summation formula. □

Exercise 1.27 Generalize Poisson's summation formula by showing the identity

$$\sum_{n\in\mathbb{Z}} f(an) = \frac{1}{|a|}\sum_{n\in\mathbb{Z}} \widehat{f}(a^{-1}n)$$

for all $f \in \mathcal{S}(\mathbb{R})$ and $a \in \mathbb{R}^\times$. (This is an archimedean analogue of the Riemann-Roch formula for global fields.)

Later we will develop Fourier theory for local fields in general. Not just for the archimedean local fields $\mathbb{R}$ and $\mathbb{C}$, but for finite extensions of $\mathbb{Q}_p$ as well. This is at the core of Tate's thesis.

1.1.7 Theta Series

We define a holomorphic function $\theta(z)$ on the complex upper half-plane $\mathcal{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$ by the infinite sum

$$\theta(z) = \sum_{n\in\mathbb{Z}} e^{\pi in^2 z}.$$

It is handy to also introduce $\omega(z) = \sum_{n=1}^{\infty} e^{\pi in^2 z}$. Of course $\theta(z) = 1 + 2\omega(z)$ so there is no substantial difference between the two.

Exercise 1.28 Check that $\theta(2z)^2 = \sum_{n=0}^{\infty} r_2(n)e^{2\pi inz}$ where $r_2(n)$ is the sum of squares function, that is the number of pairs $(a, b) \in \mathbb{Z}^2$ for which $n = a^2 + b^2$. (What we are about to discuss essentially shows $\theta(2z)^2$ is a weight one modular form for $\Gamma_0(4)$. Feel free to ignore this comment.)

The θ-function satisfies the following key transformation property under the linear fractional transformation $z \mapsto -\frac{1}{z}$ defined by $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$.

Theorem 1.29 *$\theta(\frac{i}{t}) = \sqrt{t}\theta(it)$ for all $t > 0$.*

Proof Fix $t > 0$. Observe that $\theta(it) = \sum_{n\in\mathbb{Z}} f(n)$ where $f(x) = e^{-\pi t x^2}$. As we know from Example 1.24 with $a = \sqrt{\pi t}$ the latter has Fourier transform $\widehat{f}(x) = \frac{1}{\sqrt{t}} e^{-\pi x^2/t}$. Hence the right-hand side of the Poisson summation formula becomes

$$\sum_{n\in\mathbb{Z}} \widehat{f}(n) = \frac{1}{\sqrt{t}} \sum_{n\in\mathbb{Z}} e^{-\pi n^2/t} = \frac{1}{\sqrt{t}} \theta(\frac{i}{t}).$$

This verifies the claim. □

What we will actually use is the following reformulation in terms of the ω-function:

Corollary 1.30 *$\omega(\frac{i}{t}) = \frac{1}{2}(\sqrt{t} - 1) + \sqrt{t}\omega(it)$ for all $t > 0$.*

Proof This is completely formal and left as an exercise to the reader. □

Exercise 1.31 Explain why $\omega(it) \sim e^{-\pi t}$ as $t \to \infty$. Deduce that $\omega(it) \sim \frac{1}{2\sqrt{t}}$ as $t \to 0^+$.

1.1.8 Analytic Continuation and Functional Equation

Finally we get to the main result about $\zeta(s)$.

Theorem 1.32 *The completed Riemann zeta function $\xi(s) = \pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ has meromorphic continuation to $\mathbb{C}$ and satisfies the functional equation $\xi(s) = \xi(1 - s)$. The only poles are $s = 0$ and $s = 1$. They are both simple with residues $res_{s=0}\xi(s) = -1$ and $res_{s=1}\xi(s) = +1$ respectively.*

Proof Suppose $\mathrm{Re}(s) > 1$. We first identify $\xi(s)$ with the so-called Mellin transform of $\omega(it)$.

$$\begin{aligned} \int_0^\infty \omega(it) t^{\frac{s}{2}-1} dt &= \int_0^\infty \sum_{n=1}^\infty e^{-\pi n^2 t} t^{\frac{s}{2}-1} dt \\ &= \sum_{n=1}^\infty \int_0^\infty e^{-\pi n^2 t} t^{\frac{s}{2}-1} dt \\ &= \sum_{n=1}^\infty \int_0^\infty e^{-x} (\frac{x}{\pi n^2})^{\frac{s}{2}-1} \frac{dx}{\pi n^2} \end{aligned}$$

$$
\begin{aligned}
&= \sum_{n=1}^{\infty} \pi^{-s/2} n^{-s} \int_0^{\infty} e^{-x} x^{\frac{s}{2}-1} dx \\
&= \pi^{-s/2} \Gamma(\frac{s}{2}) \zeta(s) \\
&= \xi(s).
\end{aligned}
$$

The Mellin transform $\int_0^{\infty} \omega(it) t^{\frac{s}{2}-1} dt$ only converges absolutely for $\mathrm{Re}(s) > 1$ since $\omega(it) = O(1/\sqrt{t})$ as $t \to 0^+$. To get around this problem we split the integral as $\int_0^{\infty} = \int_0^1 + \int_1^{\infty}$. Thus let

$$
I(s) = \int_1^{\infty} \omega(it) t^{\frac{s}{2}-1} dt, \qquad J(s) = \int_0^1 \omega(it) t^{\frac{s}{2}-1} dt.
$$

The first integral defining $I(s)$ converges for all $s \in \mathbb{C}$ since $\omega(it) = O(e^{-\pi t})$ as $t \to \infty$, and gives an entire function. We use the transformation property of ω to express $J(s)$ in terms of $I(1-s)$. Assume $\mathrm{Re}(s) > 1$ to ensure $J(s)$ converges. Then

$$
\begin{aligned}
J(s) &= \int_0^1 \omega(it) t^{\frac{s}{2}} \frac{dt}{t} \\
&= \int_1^{\infty} \omega(\frac{i}{x}) x^{-\frac{s}{2}} \frac{dx}{x} \\
&= \int_1^{\infty} \frac{1}{2}(\sqrt{x} - 1) x^{-\frac{s}{2}} \frac{dx}{x} + \int_1^{\infty} \sqrt{x} \omega(ix) x^{-\frac{s}{2}} \frac{dx}{x} \\
&= \frac{1}{2} \int_1^{\infty} x^{-\frac{s+1}{2}} dx - \frac{1}{2} \int_1^{\infty} x^{-\frac{s}{2}-1} dx + \int_1^{\infty} \omega(ix) x^{\frac{1-s}{2}-1} dx \\
&= \frac{1}{s-1} - \frac{1}{s} + I(1-s).
\end{aligned}
$$

Altogether this gives the following expression for $\xi(s)$.

$$
\xi(s) = \frac{1}{s-1} - \frac{1}{s} + I(s) + I(1-s).
$$

The right-hand side is visibly holomorphic on $\mathbb{C} - \{0, 1\}$ with simple poles $s = 0$ and $s = 1$ having residues -1 and $+1$ respectively. Furthermore it is clearly invariant under the point reflection $s \mapsto 1 - s$. □

The above proof gives a template for how we will approach these questions in a more general setting. Eventually we will associate a completed L-function $\Lambda(s, \chi)$ to a Hecke character χ of a number field. One of our noblest goals is to show it has analytic continuation and a functional equation relating $\Lambda(s, \chi)$ to $\Lambda(1-s, \bar{\chi})$. In

general they will not be equal. The functional equation involves the root number of χ which is of fundamental importance in number theory. Here is the general strategy:

1. Express $\Lambda(s, \chi)$ as the Mellin transform of some theta function θ_χ.
2. Establish a transformation property of θ_χ using Poisson summation.
3. Split the Mellin transform into two integrals $I_\chi(s) + J_\chi(s)$ with $I_\chi(s)$ entire.
4. Use the 2nd step to relate $J_\chi(s)$ and $I_{\bar{\chi}}(1-s)$ when $\mathrm{Re}(s)$ is large enough.

1.1.9 Special Values and Bernoulli Numbers

There is a small subtlety with Bernoulli numbers which is often overlooked. We will define two sequences of rational numbers B_n^+ and B_n^- where $n = 0, 1, 2, \ldots$. However, they only differ for $n = 1$. The defining equations are

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n^-}{n!} x^n$$

and

$$\frac{x}{1 - e^{-x}} = \sum_{n=0}^{\infty} \frac{B_n^+}{n!} x^n.$$

Both power series converge for $|x| < 2\pi$. Clearly $B_0^\pm = 1$ and one easily checks by differentiating n times that $B_n^\pm \in \mathbb{Q}$. Also, changing the sign of x in both power series immediately shows that $B_n^+ = (-1)^n B_n^-$. One checks that $B_1^- = -\frac{1}{2}$ and therefore $B_1^+ = \frac{1}{2}$. (Just expand e^x as a Taylor series and look at the leading terms; for instance $\frac{1}{x}\left(\frac{x}{e^x-1} - 1\right) \to -\frac{1}{2}$ as x goes to zero). In fact these are the only differing terms of the two sequences since $B_n^\pm = 0$ for odd $n \geq 3$ as follows from the next exercise.

Exercise 1.33 Verify that the function $\frac{x}{e^x-1} - 1 + \frac{1}{2}x = \sum_{n=2}^{\infty} \frac{B_n^-}{n!} x^n$ is even.

Here is a table containing the first few terms of the sequences $B_n^\pm$.

n	0	1	2	3	4	5	6	7	8	9	10	11	12
$B_n^\pm$	1	$\pm\frac{1}{2}$	$\frac{1}{6}$	0	$-\frac{1}{30}$	0	$\frac{1}{42}$	0	$-\frac{1}{30}$	0	$\frac{5}{66}$	0	$-\frac{691}{2730}$

The big result of this section is beneath.

Theorem 1.34 *For all positive integers n we have $\zeta(1-n) = -\frac{B_n^+}{n}$.*

We will give the proof below, as a result of yet another way to analytically continue $\zeta(s)$, but we pause to point out a few consequences. For $n = 1$ we of course recover

our earlier observation $\zeta(0) = -\frac{1}{2}$. For varying odd $n > 1$ we get the trivial zeroes $\{-2, -4, -6, \ldots\}$. For even $n > 1$ we get new information such as $\zeta(-1) = -\frac{1}{12}$.

Corollary 1.35 *For all integers $n \geq 0$ we have the special value formula*

$$\zeta(2n) = (-1)^{n+1} \frac{B_{2n}(2\pi)^{2n}}{2(2n)!}.$$

(Here we let $B_{2n} = B_{2n}^{+} = B_{2n}^{-}$.) In particular the sign of B_{2n} equals $(-1)^{n+1}$.

Proof This follows from the functional equation in conjunction with the formula

$$\Gamma(\frac{1}{2} - n) = (-1)^n 2^{2n} \sqrt{\pi} \frac{n!}{(2n)!}.$$

(Apply the duplication formula plus the residue formula for the Γ-function. We leave the details to the reader.) Indeed

$$\begin{aligned}
\zeta(2n) &= \frac{\pi^{n-\frac{1}{2}} \Gamma(\frac{1}{2} - n)\zeta(1-2n)}{\pi^{-n}\Gamma(n)} \\
&= \frac{\pi^{2n-\frac{1}{2}}}{(n-1)!} \cdot (-1)^n 2^{2n} \sqrt{\pi} \frac{n!}{(2n)!} \cdot (-1) \frac{B_{2n}}{2n} \\
&= (-1)^{n+1} \frac{B_{2n}(2\pi)^{2n}}{2(2n)!}
\end{aligned}$$

as desired. The first step uses the functional equation. In the second step we invoked Theorem 1.34. □

We return to the proof of Theorem 1.34. As mentioned this will be a byproduct of an alternative way of extending $\zeta(s)$ to $\mathbb{C}$, which is based on the following observation:

Lemma 1.36 $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$ *for $Re(s) > 1$.*

Proof For any fixed $n \geq 1$ we have

$$\int_0^\infty x^{s-1} e^{-nx} dx = \int_0^\infty (\frac{t}{n})^{s-1} e^{-t} \frac{dt}{n} = n^{-s} \int_0^\infty t^{s-1} e^{-t} dt = n^{-s} \Gamma(s).$$

Since we are assuming $\mathrm{Re}(s) > 1$ we may add all these and get

$$\zeta(s)\Gamma(s) = \sum_{n=1}^{\infty} \int_0^\infty x^{s-1} e^{-nx} dx = \int_0^\infty (\sum_{n=1}^{\infty} e^{-nx}) x^{s-1} dx = \int_0^\infty \frac{e^{-x}}{1 - e^{-x}} x^{s-1} dx = \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$$

as in the Lemma. □

The integral $\int_0^\infty \frac{x^{s-1}}{e^x-1}dx$ only converges for $\mathrm{Re}(s) > 1$ (the integrand has absolute value roughly $x^{\mathrm{Re}(s)-2}$ for $x \to 0^+$). The basic idea is to instead integrate the function $\frac{z^{s-1}}{e^z-1}$ around an infinitesimally narrow "keyhole" contour C composed of the following: Fix a radius ρ in the range $0 < \rho < 2\pi$ (to avoid poles).

- C_1 moves "above" the real axis from ∞ leftwards to ρ.
- C_2 goes counterclockwise around the radius ρ circle centered at 0.
- C_3 moves "below" the real axis from ρ back to ∞.

Here a picture is worth a thousand words, the reader is encouraged to draw one. We first have to make sense of z^{s-1} as a function of $z \in \mathbb{C} - [0, \infty)$. By definition $z^{s-1} = e^{(s-1)\mathrm{Log}(z)}$ where $\mathrm{Log} : \mathbb{C} - [0, \infty) \to \mathbb{C}$ is the branch of the logarithm given by

$$\mathrm{Log}(z) = \ln|z| + i\theta$$

if $z = |z|e^{i\theta}$ is nonzero with $0 \le \theta < 2\pi$.

The key point is that the contour C circumvents 0 and therefore $\oint_C \frac{z^{s-1}}{e^z-1}dz$ makes sense for all $s \in \mathbb{C}$ and this contour integral defines an entire function. In particular the next identity extends $\zeta(s)$ to a meromorphic function on $\mathbb{C}$ (with a simple pole at $s = 1$):

Proposition 1.37 $\zeta(s) = e^{-\pi i s}\Gamma(1-s)\frac{1}{2\pi i}\oint_C \frac{z^{s-1}}{e^z-1}dz$ *for all* $s \neq 1$.

Proof (*Sketch*) Fix an $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$ and let $h(z) = \frac{z^{s-1}}{e^z-1}$ for simplicity. First note that

$$\oint_{C_1} h(z)dz = -\int_\rho^\infty \frac{x^{s-1}}{e^x-1}dx$$

and

$$\oint_{C_3} h(z)dz = e^{2\pi i s}\int_\rho^\infty \frac{x^{s-1}}{e^x-1}dx.$$

(Note that $\theta = 0$ on C_1 but $\theta = 2\pi$ on C_3. Thus $\mathrm{Log}(z) = \ln|z| + 2\pi i$ for $z \in C_3$ which gives the extra factor $e^{2\pi i(s-1)} = e^{2\pi i s}$ in front of the second integral.)

We next argue why the circular contribution $\oint_{C_2} h(z)dz$ goes to 0 as $\rho \to 0^+$. Suppose $z = \rho e^{i\theta}$ where $0 \le \theta < 2\pi$ and we write $s = \sigma + i\tau$ with $\sigma > 1$. First of all

$$|z^{s-1}| = |e^{(s-1)(\ln\rho + i\theta)}| = e^{(\sigma-1)\ln\rho - \tau\theta} = \rho^{\sigma-1}e^{-\tau\theta} \le \rho^{\sigma-1}e^{2\pi|\tau|}.$$

The inequality comes from multiplying $\theta < 2\pi$ by $|\tau|$. Moreover, there is a constant $A > 0$ such that

$$|z| \le 1 \Longrightarrow |e^z - 1| \ge A|z|.$$

(Why? The function $\frac{1}{z}(e^z - 1) = 1 + O(z)$ is continuous and therefore bounded and without zeroes on the closed unit disc. Let $A > 0$ be its minimum.) Now

$$\oint_{C_2} h(z)dz = \int_0^{2\pi} h(\rho e^{it})i\rho e^{it}dt$$

and

$$|h(\rho e^{it})i\rho e^{it}| \leq \rho \cdot \frac{1}{A\rho} \cdot \rho^{\sigma-1}e^{2\pi|\tau|} = \frac{1}{A} \cdot \rho^{\sigma-1}e^{2\pi|\tau|}$$

for $\rho \leq 1$. This shows $\oint_{C_2} h(z)dz$ vanishes in the limit as $\rho \to 0^+$ since $\sigma > 1$.

By Lemma 1.36 we arrive at the following relation after letting $\rho \to 0^+$.

$$\oint_C h(z)dz = (e^{2\pi is} - 1)\Gamma(s)\zeta(s),$$

still assuming $\sigma > 1$. By rewriting this equation using the reflection formula

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)} = \frac{2\pi i}{e^{\pi is} - e^{-\pi is}} = \frac{2\pi i \cdot e^{\pi is}}{e^{2\pi is} - 1}$$

we infer the formula in the Proposition. □

Remark 1.38 The reason we call the previous proof a "sketch" is one should really let C_1 start at some $T + i\epsilon$ and let C_3 end at $T - i\epsilon$, and then let $T \to \infty$ and $\epsilon \to 0^+$. The unconvinced reader is invited to carry this out.

This way of analytically continuing $\zeta(s)$ is from Riemann's original paper. Without much effort it gives the special value formula $\zeta(1-n) = -\frac{B_n^+}{n}$ for $n \in \mathbb{N}$.

Proof (*Theorem* 1.34) Specialize to $s \in \mathbb{Z}$ in the previous proof. Since $e^{2\pi is} = 1$ we find that $\oint_{C_1} h(z)dz + \oint_{C_3} h(z)dz = 0$. In other words $\frac{1}{2\pi i}\oint_C h(z)dz = \frac{1}{2\pi i}\oint_{C_2} h(z)dz = \mathrm{res}_{z=0}(\frac{z^{-n}}{e^z-1})$ if we write $s = 1 - n$. We assume $n \in \mathbb{N}$. By definition of B_r^- we have the Laurent series expansion

$$\frac{z^{-n}}{e^z - 1} = \sum_{r=0}^{\infty} \frac{B_r^-}{r!} z^{r-n-1}$$

for $0 < |z| < 2\pi$. We read off its residue at $z = 0$ as the coefficient of z^{-1}. Taking $r = n$ we get

$$\mathrm{res}_{z=0}(\frac{z^{-n}}{e^z - 1}) = \frac{B_n^-}{n!}.$$

On the other hand, from the formula expressing $\zeta(s)$ in terms of $\frac{1}{2\pi i}\oint_C h(z)dz$ we get

$$\zeta(1-n) = e^{-\pi i(1-n)}\Gamma(n)\frac{1}{2\pi i}\oint_C h(z)dz = (-1)^{n-1}(n-1)!\frac{B_n^-}{n!} = -\frac{B_n^+}{n}.$$

Done. □

Exercise 1.39 What does this proof tell us for $n = 0$ and integers $n < 0$?

Exercise 1.40 Let n be a positive integer. Then $-2n$ is one of the trivial zeroes of $\zeta(s)$–which are all simple. Find an explicit formula for the leading coefficient

$$\zeta'(-2n) = \lim_{s\longrightarrow -2n}(s+2n)^{-1}\zeta(s)$$

in terms of the odd zeta value $\zeta(2n+1)$. What do you get for $n = 1$?

Remark 1.41 The Bernoulli numbers have numerous fascinating arithmetic properties. We mention a few. A prime $p > 2$ is said to be *regular* if $p \nmid h_{\mathbb{Q}(\zeta_p)}$ (the class number of the pth cyclotomic field). Kummer's criterion states that p is regular if and only if p does not divide the numerators of $B_2, B_4, \ldots, B_{p-3}$. A relatively simple argument, based on the case where $\mathbb{Z}[\zeta_p]$ is a UFD, shows that Fermat's Last Theorem holds for regular exponents p, that is $x^p + y^p + z^p \Longrightarrow xyz = 0$ for integers x, y, z. It is an open problem whether there are infinitely many regular primes, although the majority (60.65%) should be. Curiously Jensen proved in 1915 that there are infinitely many irregular primes (37, 59, 67, ...). Let us also point out that 691 is the first irregular prime appearing in the numerator of a Bernoulli number, namely $B_{12} = -\frac{691}{2730}$.

There is also the connection to p-adic L-functions. The Kummer congruences say that

$$m \equiv n \pmod{p^{r-1}(p-1)} \Longrightarrow (1-p^{m-1})\frac{B_m}{m} \equiv (1-p^{n-1})\frac{B_n}{n} \pmod{p^r}.$$

Here $r > 0$ and $m, n \not\equiv 0 \pmod{p-1}$ are even and positive. Moreover, for $a, b \in \mathbb{Q}$ we write $a \equiv b \pmod{p^r}$ if $v_p(a-b) \geq r$. The Kummer congruences amount to

$$\zeta^*(1-m) \equiv \zeta^*(1-n) \pmod{p^r}$$

where $\zeta(s) = (1-p^{-s})^{-1}\zeta^*(s)$. In other words $\zeta^*(s)$ is the Riemann zeta function with the pth Euler factor removed. As a result one can p-adically interpolate the values $\zeta^*(1-n)$ for positive $n \equiv 0 \pmod{p-1}$ by a p-adically continuous (even analytic) $\mathbb{C}_p$-valued function $\zeta_p(s)$ on $\mathbb{Z}_p$. This is called the Kubota-Leopoldt p-adic zeta function.

1.2 Dirichlet L-Functions

1.2.1 Density

Fix an $N \geq 1$ and an $a \in \mathbb{Z}$ coprime to N. A famous result of Dirichlet says that there are infinitely many primes in the arithmetic progression $a, a+N, a+2N, \ldots$. If the primes are uniformly distributed among the $\phi(N)$ possible residue classes modulo N one would expect a portion $\frac{1}{\phi(N)}$ of the primes to lie in the residue class $a + N\mathbb{Z}$. To make this precise we introduce the notion of density.

The idea behind the definition is the following observation. For $\mathrm{Re}(s) > 1$ we have

$$\ln \zeta(s) = \sum_p p^{-s} + \sum_p \sum_{r=2}^{\infty} \frac{p^{-rs}}{r}$$

and the double-sum converges absolutely for $\mathrm{Re}(s) > \frac{1}{2}$ (the inner sum starts from $r = 2$, so $\sum_p \sum_{r=2}^{\infty} \frac{p^{-r\sigma}}{r}$ is bounded by $\sum_p \sum_{r=2}^{\infty} \frac{p^{-r\sigma}}{2} = \frac{1}{2} \sum_p \frac{1}{p^\sigma(p^\sigma - 1)}$ which is finite for $\sigma > \frac{1}{2}$); in particular it is holomorphic in that region and $\ln \zeta(s) = \sum_p p^{-s} + O(1)$ as $s \to 1$. On the other hand

$$\ln \zeta(s) = \ln(\frac{1}{s-1}) + O(s-1)$$

as $s \to 1^+$. Altogether $\sum_p p^{-s} \sim \ln(\frac{1}{s-1})$ as $s \to 1^+$.

Definition 1.42 A set of primes A is said to have analytic density $\delta \in (0, 1]$ if

$$\sum_{p \in A} p^{-s} \sim \delta \cdot \ln(\frac{1}{s-1})$$

as $s \to 1^+$. (Such a set A with density $\delta > 0$ is obviously infinite.)

Remark 1.43 There is also the notion of *natural* density which is $\lim_{x \to \infty} \frac{1}{\pi(x)} \#\{p \in A : p \leq x\}$ when it exists. If so, the analytic density exists and the two densities are the same. However, the converse is not true. There are sets A with an analytic density but no natural density, such as $A =$ {primes with first digit 1}. Also, there are sets without an analytic density.

We can now formulate Dirichlet's theorem.

Theorem 1.44 (Dirichlet) *Let $N \geq 1$ and suppose $a \in \mathbb{Z}$ is coprime to N. Then the set of all primes $p \equiv a \,(\mathrm{mod}\, N)$ has analytic density $\frac{1}{\phi(N)}$. In other words $\sum_{p \equiv a \,(\mathrm{mod}\, N)} p^{-s} \sim \frac{1}{\phi(N)} \cdot \ln(\frac{1}{s-1})$ as $s \to 1^+$.*

The proof takes us to Dirichlet characters and Dirichlet L-functions.

1.2.2 Dirichlet Characters

A group homomorphism $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is called a Dirichlet character modulo N. Note that χ is necessarily unitary; its values are roots of unity. We often extend χ to an arithmetic function $\chi : \mathbb{N} \longrightarrow \mathbb{C}$ by declaring that $\chi(n) = 0$ when $\mathrm{GCD}(n, N) > 1$. This extension is strongly multiplicative in the sense that $\chi(mn) = \chi(m)\chi(n)$ for all $m, n \in \mathbb{N}$.

Example 1.45 Let $p > 2$ be a prime. The Legendre symbol $\left(\frac{\bullet}{p}\right) : (\mathbb{Z}/p\mathbb{Z})^\times \longrightarrow \{\pm 1\}$ is a Dirichlet character modulo p. By definition $\left(\frac{n}{p}\right) = +1$ precisely when the congruence $x^2 \equiv n \bmod p$ has solutions.

Example 1.46 Take $p = 7$. Any Dirichlet character modulo 7 necessarily takes values in the group of sixth roots of unity $\{\pm 1, \pm\omega, \pm\bar{\omega}\}$, where $\omega = e^{2\pi i/3} = \frac{1}{2}(-1 + i\sqrt{3})$. One easily checks that 3 is primitive modulo 7, i.e. the residue class of 3 generates $(\mathbb{Z}/7\mathbb{Z})^\times$. Indeed, working mod 7, we successively find that

$$3^1 \equiv 3, \quad 3^2 \equiv 2, \quad 3^3 \equiv 6, \quad 3^4 \equiv 4, \quad 3^5 \equiv 5, \quad 3^6 \equiv 1. \tag{1.2}$$

Thus a Dirichlet character modulo 7 is uniquely determined by where it sends 3. Let χ be the one with $\chi(3) = -\omega$. From (1.2) one reads off all the other values of χ, which we collect in the table below.

n	1	2	3	4	5	6
$\chi(n)$	1	$\bar{\omega}$	$-\omega$	ω	$-\bar{\omega}$	-1

For example, $\chi(2) = \chi(3)^2 = (-\omega)^2 = \bar{\omega}$. The powers of χ give all six Dirichlet characters modulo 7. On the list $\{1, \chi, \chi^2, \chi^3, \chi^4, \chi^5\}$ we identify $\left(\frac{\bullet}{7}\right)$ with the unique (non-trivial) quadratic character χ^3.

Example 1.47 ($N = 4$) The unique non-trivial Dirichlet character $\chi_4 : (\mathbb{Z}/4\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ modulo 4 is

$$\chi_4(n) = \begin{cases} +1 & \text{if } n \equiv 1 \\ -1 & \text{if } n \equiv 3. \end{cases}$$

(Obviously $\chi_4(n) = 0$ for even $n > 0$.)

We attach an L-function to any character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ given the Dirichlet series

$$L(s, \chi) = \sum_{n=1}^{\infty} \chi(n) n^{-s}.$$

This is absolutely convergent for $\mathrm{Re}(s) > 1$ and in fact conditionally convergent for $\mathrm{Re}(s) > 0$ provided $\chi \neq 1_N$. (By the orthogonality relations for $(\mathbb{Z}/N\mathbb{Z})^\times$ the

summatory function $\sum_{n\le x}\chi(n)$ is bounded. Now use the partial summation formula to control $\sum_{n\le x}\chi(n)n^{-s}$ as $x\to\infty$.)

Exercise 1.48 Show that the L-function of χ has an Euler product for $\mathrm{Re}(s)>1$:

$$L(s,\chi)=\prod_p \frac{1}{1-\chi(p)p^{-s}}$$

where the product ranges over all primes $p\nmid N$. Note that $L(s,1_N)$ is the Riemann zeta function with all Euler factors at primes dividing N removed. (Here and elsewhere 1_N denotes the trivial character of $(\mathbb{Z}/N\mathbb{Z})^\times$ which sends all residue classes to one.)

When $\chi\neq 1_N$ we now know $L(1,\chi)=\sum_{n=1}^\infty \chi(n)n^{-1}$ is convergent. In this section we will reduce Dirichlet's theorem to the non-vanishing of these special values.

Theorem 1.49 *$L(1,\chi)\neq 0$ for all Dirichlet characters χ.*

Example 1.50 Here is the proof of 1.49 for $N=4$. This is just the Leibniz formula

$$L(1,\chi_4)=\sum_{n=1}^\infty \chi_4(n)n^{-1}=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots=\arctan(1)=\frac{\pi}{4}\neq 0.$$

The proof of Theorem 1.49 will be postponed for a while. Eventually we will exhibit $L(s,\chi)$ as a factor of the Dedekind zeta function for $\mathbb{Q}(\zeta_N)$ and this will conclude the proof. For now our goal is to show the implication

$$\text{Theorem 1.51}\implies\text{Theorem 1.45.}$$

The idea is to relate the entire collection of L-functions $L(s,\chi)$ with χ varying mod N to the collection of shifted zeta functions $\zeta(s,\frac{a}{N})$ where $\mathrm{GCD}(a,N)=1$.

Definition 1.51 The Hurwitz zeta function with parameter $q>0$ is defined for $\mathrm{Re}(s)>1$ by the absolutely convergent series

$$\zeta(s,q)=\sum_{n=0}^\infty (q+n)^{-s}.$$

Note that $\zeta(s,1)=\zeta(s)$ and $\zeta(s,q)=\zeta(s,q+1)+q^{-s}$.

One can show by "keyhole" contour integration that $\zeta(s,q)$ has meromorphic continuation to $\mathbb{C}$ with a simple pole at $s=1$, but we will not use that. Here we only care about parameters $q=\frac{a}{N}$.

Lemma 1.52 *Fix an $1 \leq a < N$ coprime to N. Then for $Re(s) > 1$ we have the relation*

$$\zeta(s, \frac{a}{N}) = \frac{N^s}{\phi(N)} \sum_{\chi} \bar{\chi}(a) L(s, \chi).$$

Here χ runs over all characters of $(\mathbb{Z}/N\mathbb{Z})^\times$.

Proof Let $a, n > 0$ be coprime to N. By the orthogonality relations $\sum_\chi \bar{\chi}(a)\chi(n) = 0$ unless $a \equiv n \pmod N$, in which case the sum is $\phi(N)$. Therefore

$$\begin{aligned}\sum_{\chi} \bar{\chi}(a) L(s, \chi) &= \sum_{n=1}^{\infty} \frac{1}{n^s} \sum_{\chi} \bar{\chi}(a)\chi(n) \\ &= \phi(N) \sum\nolimits_{0<n\equiv a \pmod N} \frac{1}{n^s} \\ &= \phi(N) \sum_{m=0}^{\infty} (a + mN)^{-s} \\ &= \phi(N) N^{-s} \zeta(s, \frac{a}{N}).\end{aligned}$$

This proves the Lemma. □

Exercise 1.53 Suppose $N > 1$ and $\mathrm{Re}(s) > 1$. Show that $L(s, \chi) = \frac{1}{N^s} \sum_{a=1}^{N-1} \chi(a)\zeta(s, \frac{a}{N})$ for all χ.

The preceding proof can be adapted to show the following for $\mathrm{Re}(s) > 1$:

$$\sum_{p \in A} p^{-s} = \frac{1}{\phi(N)} \sum_{\chi} \bar{\chi}(a) f_\chi(s)$$

where A is the set of primes $p \equiv a \pmod N$ and

$$f_\chi(s) = \sum_{p \nmid N} \frac{\chi(p)}{p^s}.$$

To find the density δ of A we need to understand the asymptotics of $f_\chi(s)$ as $s \to 1^+$. Once we prove the next Lemma we are done.

Lemma 1.54 *Let χ be a Dirichlet character modulo N and assume $L(1, \chi) \neq 0$ (this assumption will be shown to be superfluous, cf. Theorem 1.49).*

(1) If $\chi = 1_N$ then $f_\chi(s) \sim \ln(\frac{1}{s-1})$ as $s \to 1^+$.
(2) If $\chi \neq 1_N$ then $f_\chi(s) = O(1)$ as $s \to 1^+$.

Proof Part (1) is immediate since the finite sum $\sum_{p|N} p^{-s}$ stays bounded near $s = 1$ and we already observed that $\sum_p p^{-s} \sim \ln(\frac{1}{s-1})$ as $s \to 1^+$.

For part (2) we need the non-vanishing assumption $L(1, \chi) \neq 0$. For $\mathrm{Re}(s) > 1$ introduce

$$\ln L(s, \chi) = \sum_p \sum_{r=1}^{\infty} \frac{\chi(p)^r p^{-rs}}{r} = f_\chi(s) + \sum_p \sum_{r=2}^{\infty} \frac{\chi(p)^r p^{-rs}}{r}.$$

The rightmost double-sum converges absolutely for $\mathrm{Re}(s) > \frac{1}{2}$ and is therefore bounded near $s = 1$. It suffices to control $\ln L(s, \chi)$. By the Euler product $L(s, \chi) = \exp\big(\ln L(s, \chi)\big)$.

Now recall the following fact from complex analysis: If Ω is a simply connected domain (for instance an open ball) and $h : \Omega \to \mathbb{C}$ is a holomorphic function without zeroes in Ω, then there exists a holomorphic function $\ell : \Omega \to \mathbb{C}$ such that $h(s) = \exp \ell(s)$ for all $s \in \Omega$. See part (h) of [Rud87, Theorem 13.11, p. 274] for example.

We apply this fact to $L(s, \chi)$ and a small enough open ball Ω centered at $s = 1$. We shrink the radius if necessary to avoid zeroes of $L(s, \chi)$. (This is where we use the non-vanishing hypothesis.) This allows us to write $L(s, \chi) = \exp \ell(s)$ for some holomorphic $\ell : \Omega \to \mathbb{C}$. Clearly $\ell(s) - \ln L(s, \chi) \in 2\pi i \mathbb{Z}$ and by continuity this constant C must be independent of $s \in \Omega$. Thus $\ln L(s, \chi) = \ell(s) - C$ is manifestly bounded on Ω. □

This shows the implication Theorem 1.51 $\Longrightarrow$ Theorem 1.45.

1.2.3 Primitivity

To get a nice functional equation relating the values $L(s, \chi)$ and $L(1 - s, \bar{\chi})$ we must impose the condition that χ is primitive mod N. Roughly this means χ genuinely belongs to $(\mathbb{Z}/N\mathbb{Z})^\times$ in the sense that it does not arise from a character of smaller level dividing N. For instance 1_N is imprimitive unless $N = 1$. When N is a prime all $\chi \neq 1_N$ are primitive. For example, the Legendre symbol is a primitive Dirichlet character.

We are about to give the precise definition, but first we introduce some notation: For any divisor $M|N$ there is a natural surjection of rings $\mathbb{Z}/N\mathbb{Z} \twoheadrightarrow \mathbb{Z}/M\mathbb{Z}$. Passing to units gives a surjective homomorphism of groups $(\mathbb{Z}/N\mathbb{Z})^\times \twoheadrightarrow (\mathbb{Z}/M\mathbb{Z})^\times$ which we label $\mathrm{pr}_{N,M}$.

Exercise 1.55 Prove that $\mathrm{pr}_{N,M}$ is indeed surjective. (`Hint`: By the Chinese remainder theorem you may assume M and N are powers of the same prime.)

Thus any character of $(\mathbb{Z}/M\mathbb{Z})^\times$ can be inflated to $(\mathbb{Z}/N\mathbb{Z})^\times$ by composition with $\mathrm{pr}_{N,M}$ (and the inflation determines the original character mod M by the surjectivity of $\mathrm{pr}_{N,M}$).

Definition 1.56 A character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is primitive if it does not factor through $\mathrm{pr}_{N,M}$ for any positive divisor $M < N$. We summarize this in the diagram

$$\begin{array}{ccc} (\mathbb{Z}/N\mathbb{Z})^\times & \xrightarrow{\mathrm{pr}_{N,M}} & (\mathbb{Z}/M\mathbb{Z})^\times \\ & \searrow^{\chi} & \downarrow ? \\ & & \mathbb{C}^\times. \end{array}$$

For an arbitrary character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ define its conductor N_χ to be the GCD of all $M|N$ for which χ factors through $\mathrm{pr}_{N,M}$. Thus χ is primitive modulo N if and only if $N_\chi = N$.

Exercise 1.57 Given χ mod N show there is a unique primitive character χ^{Prim} mod N_χ such that $\chi = \chi^{\mathrm{Prim}} \circ \mathrm{pr}_{N,N_\chi}$. In other words, χ factor uniquely through a primitive character modulo N_χ. (`Hint`: Again use the Chinese remainder theorem to reduce to the case where N is a prime power.)

Exercise 1.58 In continuation of the previous exercise, show that the L-functions of χ and χ^{Prim} are related as follows.

$$L(s, \chi^{\mathrm{Prim}}) = L(s, \chi) \cdot \prod_{p|N \text{ and } p\nmid N_\chi} \frac{1}{1 - \chi^{\mathrm{Prim}}(p)p^{-s}}.$$

Suppose $\chi \neq 1_N$. Deduce that $L(1, \chi) \neq 0$ is equivalent to the non-vanishing of $L(1, \chi^{\mathrm{Prim}})$. (So when proving Theorem 1.49 we may assume χ is primitive.)

The next exercise requires knowledge of the Möbius function μ, whose definition and basic properties are reviewed in the appendix (see Example A.11 and Theorem A.15).

Exercise 1.59 Use the Möbius inversion formula (Theorem A.15 in the appendix) to show that the number of primitive Dirichlet characters modulo N is given by the explicit formula

$$\mathrm{Prim}(N) = \sum_{M|N} \mu(M)\phi(\frac{N}{M}).$$

For prime powers p^r this reduces to $\mathrm{Prim}(p^r) = p^r - 2p^{r-1} + p^{r-2}$ when $r \geq 2$. What is $\mathrm{Prim}(p)$?

Exercise 1.60 Show by hand that there are exactly two primitive characters χ modulo 8, one even ($\chi(-1) = 1$) and one odd ($\chi(-1) = -1$). Tabulate their values on the four residue classes $\{[1], [3], [5], [7]\}$.

1.2.4 Gauss Sums

These are central invariants of a Dirichlet character which appear in the functional equation for Dirichlet L-functions.

Definition 1.61 Let $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ be a character. Then for any $m \in \mathbb{Z}$ we let

$$\tau(\chi, m) = \sum_{r=0}^{N-1} \chi(r) e^{2\pi i r m/N}.$$

For $m = 1$ we just write $\tau(\chi) = \tau(\chi, 1)$.

Note that $\tau(\chi, m)$ lies in the ring of integers of some cyclotomic field containing $e^{2\pi i/N}$ and all the values of χ. It turns out there is quite a bit of cancellation in the Gauss sum. We are interested in its magnitude $|\tau(\chi, m)|$. By the triangle inequality we get the trivial bound N. When χ is primitive this has a drastic improvement:

Lemma 1.62 *Suppose $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is a primitive Dirichlet character. Then:*

(1) $\tau(\chi, m) = \overline{\chi(m)}\tau(\chi)$ *for all* $m \in \mathbb{Z}$.
(2) $|\tau(\chi)| = \sqrt{N}$.

Proof First focus on part (1). When $\mathrm{GCD}(m, N) = 1$ this is an easy change of variables:

$$\tau(\chi, m) = \sum_{r \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(r) e^{2\pi i r m/N} = \sum_{r' \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(r'm') e^{2\pi i r'/N} = \chi(m')\tau(\chi, 1) = \overline{\chi(m)}\tau(\chi)$$

where $mm' \equiv 1 \pmod N$. The more difficult case $\mathrm{GCD}(m, N) > 1$ requires the primitivity assumption. Let $M = \frac{N}{\mathrm{GCD}(m,N)} < N$. Since χ is primitive it does not factor through $(\mathbb{Z}/M\mathbb{Z})^\times$. That is, there exists a residue class $[a] \in \ker(\mathrm{pr}_{N,M})$ for which $\chi(a) \neq 1$. This a is coprime to N and $a \equiv 1 \pmod M$. However, of course $a \not\equiv 1 \pmod N$ since $\chi(a) \neq 1$. Now scale $\tau(\chi, m)$ by $\chi(a)$ and observe that

$$\chi(a)\tau(\chi, m) = \sum_{r=0}^{N-1} \chi(ar) e^{2\pi i r m/N} \stackrel{?}{=\!=} \sum_{r=0}^{N-1} \chi(ar) e^{2\pi i a r m/N} = \sum_{r'=0}^{N-1} \chi(r') e^{2\pi i r' m/N} = \tau(\chi, m).$$

Here the second step is the remark that $\frac{rm}{N} - \frac{arm}{N} = \frac{rm}{N}(1-a) = \frac{rm}{N}Mx = \frac{rm}{\mathrm{GCD}(m,N)}x$ is an integer, where we have written $1 - a = Mx$ for some $x \in \mathbb{Z}$. We conclude that $\tau(\chi, m) = 0$. Note that also $\chi(m) = 0$ since $\mathrm{GCD}(m, N) > 1$. This proves part (1).

The proof of part (2) now goes as follows.

$$\begin{aligned}
|\tau(\chi)|^2 &= \tau(\chi)\overline{\tau(\chi)} \\
&= \tau(\chi)\sum_{r=0}^{N-1}\overline{\chi(r)}e^{-2\pi ir/N} \\
&\overset{(1)}{=} \sum_{r=0}^{N-1}\tau(\chi,r)e^{-2\pi ir/N} \\
&= \sum_{r=0}^{N-1}\sum_{s=0}^{N-1}\chi(s)e^{2\pi irs/N}e^{-2\pi ir/N} \\
&= \sum_{r=0}^{N-1}\sum_{s=0}^{N-1}\chi(s)e^{2\pi ir(s-1)/N} \\
&= \sum_{s=0}^{N-1}\chi(s)\Big(\sum_{r=0}^{N-1}e^{2\pi ir(s-1)/N}\Big) \\
&= \chi(1)N \\
&= N.
\end{aligned}$$

The parenthetical sum is a geometric sum in $e^{2\pi i(s-1)/N}$ which is zero unless $s \equiv 1 \pmod{N}$. □

Another invariant of χ is its parity $\mathrm{e} = \mathrm{e}(\chi) \in \{0, 1\}$ with the defining property $\chi(-1) = (-1)^{\mathrm{e}}$. We say χ is even if $\mathrm{e} = 0$ and odd if $\mathrm{e} = 1$.

Example 1.63 Let $p > 2$ be a prime. Since $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$ the Legendre symbol $\left(\frac{\bullet}{p}\right)$ has parity

$$\mathrm{e} = \begin{cases} 0 & \text{if } p \equiv 1 \pmod 4 \\ 1 & \text{if } p \equiv 3 \pmod 4. \end{cases}$$

More succinctly $\mathrm{e} \equiv \frac{p-1}{2} \pmod 2$.

These invariants are combined into what is called the root number of χ.

Definition 1.64 The root number of a primitive Dirichlet character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is

$$W(\chi) = \frac{\tau(\chi)}{i^{\mathrm{e}}\sqrt{N}}.$$

(Note that $|W(\chi)| = 1$ so up to the factor i^{e} one can think of $W(\chi)$ as the phase of the Gauss sum $\tau(\chi)$.)

The mysterious factor i for odd χ has a natural "archimedean" explanation in Tate's thesis, as we will see. As extra justification the factor $i^{\mathfrak{e}}$ guarantees $W(\cdot)$ commutes with complex conjugation:

Exercise 1.65 Let χ be a primitive Dirichlet character. Check that $\tau(\bar{\chi}) = (-1)^{\mathfrak{e}}\overline{\tau(\chi)}$ and deduce the first equality $W(\bar{\chi}) = \overline{W(\chi)} = W(\chi)^{-1}$.

Exercise 1.66 Consider the Legendre symbol $\chi = \left(\frac{\bullet}{p}\right)$ for a prime $p > 2$. Use what we proved in this section to show $\tau(\chi)^2 = p^* = (-1)^{\frac{p-1}{2}}p$. Deduce $W(\chi) = \pm 1$. (In fact we always have $W(\chi) = 1$ as we will show later in Example 1.113 as a result of the refinement $\tau(\chi) = i^{\mathfrak{e}}\sqrt{p}$.)

1.2.5 Analytic Continuation and Functional Equation

Throughout this section we fix a primitive Dirichlet character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$. When $N = 1$ the L-function is just $\zeta(s)$ so we assume $N > 1$, and in particular $\chi \neq 1_N$. As for the Riemann zeta function, the functional equation for $L(s, \chi)$ is best stated in terms of the completed L-function

$$\widehat{L}(s, \chi) = \left(\frac{\pi}{N}\right)^{-\frac{s+\mathfrak{e}}{2}} \Gamma\left(\frac{s+\mathfrak{e}}{2}\right) L(s, \chi).$$

(Note that the Γ-factor involves the parity e.) Here the big result is:

Theorem 1.67 *Let χ be a primitive Dirichlet character modulo $N > 1$. Then the completed L-function $\widehat{L}(s, \chi)$ has analytic continuation to en entire function on $\mathbb{C}$, and satisfies the functional equation*

$$\widehat{L}(1 - s, \bar{\chi}) = W(\chi)\widehat{L}(s, \chi).$$

One of the ultimate goals of Tate's thesis is to give a slick way of generalizing this result to number fields, using harmonic analysis on the adeles.

Remark 1.68 As a word of caution, beware that the functional equation is occasionally misstated as being $\widehat{L}(s, \chi) = W(\chi)\widehat{L}(1 - s, \bar{\chi})$ in several otherwise trustworthy sources. This is incorrect (unless of course the root number equals ± 1). We will prove Theorem 1.67 in great detail below as a special case of Tate's Theorem. The data base `dlmf.nist.gov` gives the correct functional equation.

Remark 1.69 There is a slight variant of $\widehat{L}(s, \chi)$ independent of the conductor N which we call the automorphic L-function:

$$L^{\text{aut}}(s, \chi) = \pi^{-\frac{s+\mathfrak{e}}{2}} \Gamma\left(\frac{s+\mathfrak{e}}{2}\right) L(s, \chi) = N^{-\frac{s+\mathfrak{e}}{2}} \widehat{L}(s, \chi).$$

In terms of the latter, the functional equation may be recast as

$$L^{\text{aut}}(s, \chi) = \epsilon(s, \chi) L^{\text{aut}}(1 - s, \bar{\chi})$$

where the ϵ-factor $\epsilon(s, \chi) = W(\bar{\chi}) N^{\frac{1}{2}-s} = i^{-\mathrm{e}} \tau(\bar{\chi}) N^{-s}$ is what depends on the conductor N. Note that the root number of the complex conjugate $\bar{\chi}$ is what appears in the ϵ-factor–not $W(\chi)$.

One of the most novel aspects of Tate's thesis is how to make sense of $\epsilon(s, \chi)$ for number fields, and show how it factors into local ϵ-factors from each place. This factorization is not something you see in the classical theory.

Exercise 1.70 Keep a primitive $\chi \bmod N > 1$. Show that $L(s, \chi)$ has the following trivial zeroes.

- $\{0, -2, -4, -6, \ldots\}$ if $\mathrm{e} = 0$;
- $\{-1, -3, -5, \ldots\}$ if $\mathrm{e} = 1$.

Are they all simple? At $s = 0$ you may need to use Theorem 1.49.

Exercise 1.71 Suppose χ is in addition quadratic (that is $\{\pm 1\}$-valued). Prove that $W(\chi) = (-1)^{\nu}$ where $\nu = \text{ord}_{s=\frac{1}{2}} L(s, \chi)$ is the order of vanishing at the center of the critical strip.

We dive into the proof of Theorem 1.67. Our general strategy is to first find an integral representation of $\widehat{L}(s, \chi)$ as the Mellin transform of some θ-function $\theta_\chi(z)$ with a nice transformation property under $z \mapsto -\frac{1}{z}$. Here it is.

$$\theta_\chi(z) = \sum_{n \in \mathbb{Z}} \chi(n) n^{\mathrm{e}} e^{\pi i n^2 z / N}$$

for $z \in \mathcal{H}$. Note that $\chi(-n)(-n)^{\mathrm{e}} = \chi(n) n^{\mathrm{e}}$ and $\chi(0) = 0$. Therefore $\theta_\chi(z) = 2 \sum_{n=1}^{\infty} \chi(n) n^{\mathrm{e}} e^{\pi i n^2 z / N}$.

Lemma 1.72 $\widehat{L}(s, \chi) = \frac{1}{2} \int_0^\infty \theta_\chi(it) t^{\frac{s+\mathrm{e}}{2}} \frac{dt}{t}$ *for* $Re(s) > 1$.

Proof This is a pleasant formal computation. We strongly encourage the reader to do this as an exercise. The steps involved are completely analogous to what we did for the Riemann zeta function. □

We content ourselves with just quoting the all-important transformation property of $\theta_\chi(z)$.

Proposition 1.73 *Let χ be a primitive Dirichlet character. Then for all $t > 0$ we have the transformation formula*

$$\theta_\chi\left(\frac{i}{t}\right) = W(\bar{\chi}) t^{e+\frac{1}{2}} \theta_{\bar{\chi}}(it).$$

Proof Omitted. □

Since we will eventually give another proof of Theorem 1.67 we will just admit this formula. We emphasize that its proof heavily uses the primitivity of χ. One starts off by expressing $\theta_\chi(z)$ in terms of

$$\theta_N(a,b,z) = \sum_{t\in N\mathbb{Z}} \exp\left[\pi i(a+t)^2 z + 2\pi i b t\right]$$

and $\theta_N^1(a,b,z) = \frac{1}{2\pi i z}\frac{\partial}{\partial a}\theta_N(a,b,z)$ for $a,b\in\mathbb{R}$. By the Poisson summation formula,

$$\theta_N(a,b,\frac{i}{t}) = e^{-2\pi i a b}\frac{1}{N}\sqrt{t}\,\theta_{\frac{1}{N}}(-b,a,it).$$

Differentiation gives the transformation property of $\theta_N^1(a,b,z)$. Weighing these series by $\chi(a)$ and summing over a introduces Gauss sums and proves Proposition 1.73. We skip the details.

Proof (*Theorem* 1.67) Our next step is to split the integral representing $\widehat{L}(s,\chi)$. Thus write

$$2\widehat{L}(s,\chi) = \int_0^1 \theta_\chi(it)t^{\frac{s+\mathfrak{e}}{2}}\frac{dt}{t} + \int_1^\infty \theta_\chi(it)t^{\frac{s+\mathfrak{e}}{2}}\frac{dt}{t} = J_\chi(s) + I_\chi(s).$$

Since $\theta_\chi(it)\sim 2e^{-\pi t/N}$ as $t\to\infty$, obviously $I_\chi(s)$ defines an entire function. Finally, for $\mathrm{Re}(s)\gg 0$ we have

$$\begin{aligned}
J_\chi(s) &= \int_0^1 \theta_\chi(it)t^{\frac{s+\mathfrak{e}}{2}}\frac{dt}{t}\\
&= \int_1^\infty \theta_\chi(\frac{i}{x})x^{-\frac{s+\mathfrak{e}}{2}}\frac{dx}{x}\\
&= W(\bar\chi)\int_1^\infty x^{\mathfrak{e}+\frac{1}{2}}\theta_{\bar\chi}(ix)x^{-\frac{s+\mathfrak{e}}{2}}\frac{dx}{x}\\
&= W(\bar\chi)\int_1^\infty \theta_{\bar\chi}(ix)x^{\frac{1-s+\mathfrak{e}}{2}}\frac{dx}{x}\\
&= W(\bar\chi)I_{\bar\chi}(1-s).
\end{aligned}$$

This shows $2\widehat{L}(s,\chi) = I_\chi(s) + W(\bar\chi)I_{\bar\chi}(1-s)$ extends to an entire function. The functional equation follows immediately by substituting $\chi\rightsquigarrow\bar\chi$ and $s\rightsquigarrow 1-s$ on the right-hand side, and using $W(\bar\chi) = \overline{W(\chi)} = W(\chi)^{-1}$. □

1.2.6 Generalized Bernoulli Numbers and L-Values

Let χ be a primitive Dirichlet character modulo $N > 1$. We associate a sequence of algebraic numbers $B_{n,\chi} \in \mathbb{Q}(\zeta_{\phi(N)})$ as $n = 0, 1, 2, \ldots$ given by the power series expansion

$$F_\chi(z) = \sum_{a=1}^{N} \chi(a) \frac{ze^{az}}{e^{Nz} - 1} = \sum_{n=0}^{\infty} \frac{B_{n,\chi}}{n!} z^n$$

for $|z| < \frac{2\pi}{N}$. Near zero $\frac{ze^{az}}{e^{Nz}-1}$ is roughly $\frac{1}{N}(1 + az)$. Therefore, by orthogonality $B_{0,\chi} = \frac{1}{N} \sum_{a=1}^{N} \chi(a) = 0$ since $\chi \neq 1_N$, and perhaps more importantly

$$B_{1,\chi} = \frac{1}{N} \sum_{a=1}^{N} \chi(a)a.$$

One easily verifies that $F_\chi(-z) = (-1)^{\mathrm{e}} F_\chi(z)$ by the substitution $a \rightsquigarrow N - a$. Consequently

$$B_{n,\chi} \neq 0 \Longrightarrow n \equiv \mathrm{e} \pmod 2.$$

In other words $B_{n,\chi}$ vanishes when n and χ have opposite parities.

For the Dirichlet L-function $L(s, \chi)$ we have the following special L-values.

Theorem 1.74 *For all positive integers n we have $L(1 - n, \chi) = -\frac{B_{n,\chi}}{n}$.*

Proof We will skip the details. After expressing $L(s, \chi)$ in terms of $\zeta(s, \frac{a}{N})$ as $a = 1, \ldots, N$ the proof is completely analogous to the keyhole integration argument for $\zeta(s)$. □

This of course also explains the trivial zeroes of $L(s, \chi)$ mentioned earlier: The non-positive even integers when $\mathrm{e} = 0$, the negative odd integers when $\mathrm{e} = 1$.

As for $\zeta(s)$ the functional equation yields the values at positive integers $n \equiv \mathrm{e} \pmod 2$.

Corollary 1.75 *For positive $n \equiv e \pmod 2$ we have the special value formula*

$$L(n, \chi) = (-1)^{\frac{n-e}{2}+1} \frac{B_{n,\bar{\chi}} (\frac{2\pi}{N})^n}{2 \cdot n!} i^{-e} \tau(\bar{\chi}).$$

Proof We leave this as an exercise for the reader. □

Exercise 1.76 Deduce the above Corollary 1.75 from Theorem 1.74 and Theorem 1.67.

We record the special case $n = 1$ separately.

Corollary 1.77 *For odd χ the value at $s = 1$ is given by the formula*

$$L(1, \chi) = \frac{\pi i}{N}\tau(\bar{\chi})B_{1,\bar{\chi}} = \frac{\pi i}{N^2}\tau(\bar{\chi})\sum_{a=1}^{N}\overline{\chi(a)}a.$$

Proof This is immediate. □

Although this formula is very explicit, it is not at all clear that $L(1, \chi) \neq 0$.

Example 1.78 Let χ_4 be the non-trivial character modulo $N = 4$. It is odd. In this case $\tau(\chi_4) = 2i$ and $B_{1,\chi_4} = -\frac{1}{2}$. In this case $L(1, \chi_4) = \frac{\pi i}{4}2i(-\frac{1}{2}) = \frac{\pi}{4}$; consistent with the Leibniz formula.

For even χ there is also an explicit formula for $L(1, \chi)$ of a different nature. We quote it here without proof:

$$L(1, \chi) = -\frac{\tau(\chi)}{N}\sum_{a=1}^{N}\overline{\chi(a)}\ln|1 - \zeta_N^a|.$$

(The sum runs over positive $a < N$ coprime to N.) We encourage the reader to try this as an exercise, or consult [Was97, p. 37].

1.3 The Dedekind Zeta Function

1.3.1 Definition and Basic Properties

Throughout these notes we fix a number field $K/\mathbb{Q}$ of degree $d = [K : \mathbb{Q}]$. We let $\mathcal{O}_K$ denote its Dedekind domain of algebraic integers. Recall from basic algebraic number theory that $N(I) = |\mathcal{O}_K/I|$ is finite for nonzero ideals $I \subset \mathcal{O}_K$ and $I \mapsto N(I)$ is multiplicative. When $I = \mathfrak{p}$ is a nonzero prime ideal (equivalently a maximal ideal) dividing the rational prime p, the norm is $N(\mathfrak{p}) = p^f$ where $f = f(\mathfrak{p}|p)$ is the degree of the residue field $\mathcal{O}_K/\mathfrak{p}$ over $\mathbb{F}_p$ ("inertia degree"). The ramification index $e = e(\mathfrak{p}|p)$ is the exponent of $\mathfrak{p}$ in the factorization of $p\mathcal{O}_K$ into maximal ideals.

Definition 1.79 The Dedekind zeta function of the number field K is defined for $\mathrm{Re}(s) > 1$ by the infinite series

$$\zeta_K(s) = \sum_{I \neq 0} N(I)^{-s}.$$

Here the sum ranges over all nonzero ideals $I \subset \mathcal{O}_K$.

For $K = \mathbb{Q}$ this is obviously just $\zeta(s)$ since $I = n\mathbb{Z}$ with $n > 0$. We will address convergence shortly. Since $\mathcal{O}_K$ is a Dedekind domain, nonzero ideals factor uniquely into nonzero prime ideals. Thus at least formally $\zeta_K(s)$ has an Euler product

$$\zeta_K(s) = \prod_{\mathfrak{p}} \frac{1}{1 - N(\mathfrak{p})^{-s}}$$

with $\mathfrak{p}$ ranging over all maximal ideals of $\mathcal{O}_K$. The series defining $\zeta_K(s)$ is absolutely convergent for $\mathrm{Re}(s) > 1$, uniformly in $\mathrm{Re}(s) \geq 1 + \epsilon$ for all $\epsilon > 0$. The Euler product is convergent for $\mathrm{Re}(s) > 1$. To show both claims it suffices to show

$$\ln \zeta_K(s) = \sum_{\mathfrak{p}} \sum_{r=1}^{\infty} \frac{1}{r} N(\mathfrak{p})^{-rs}$$

converges absolutely for $\mathrm{Re}(s) > 1$, and uniformly in $\mathrm{Re}(s) \geq 1 + \epsilon$. Let $\sigma = \mathrm{Re}(s) > 1$. Then

$$\begin{aligned}
\sum_{\mathfrak{p}} \sum_{r=1}^{\infty} \frac{1}{r} N(\mathfrak{p})^{-r\sigma} &= \sum_{p} \sum_{\mathfrak{p}|p} \sum_{r=1}^{\infty} \frac{1}{r} N(\mathfrak{p})^{-r\sigma} \\
&= \sum_{p} \sum_{\mathfrak{p}|p} \sum_{r=1}^{\infty} \frac{1}{r} p^{-rf(\mathfrak{p}|p)\sigma} \\
&\leq \sum_{p} \sum_{\mathfrak{p}|p} \sum_{r=1}^{\infty} \frac{1}{r} p^{-r\sigma} \\
&\leq d \sum_{p} \sum_{r=1}^{\infty} \frac{1}{r} p^{-r\sigma} < \infty.
\end{aligned}$$

In the third step we used $f(\mathfrak{p}|p) \geq 1$, in the fourth we used that there are at most $d = [K : \mathbb{Q}]$ distinct primes $\mathfrak{p}|p$. Indeed $\sum_{\mathfrak{p}|p} e(\mathfrak{p}|p) f(\mathfrak{p}|p) = d$ as we know from algebraic number theory.

Note that $\zeta_K(s)$ is given by a Dirichlet series for $\mathrm{Re}(s) > 1$. Namely $\zeta_K(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ where

$$a_n = \#\{I \subset \mathcal{O}_K : N(I) = n\}.$$

Eventually we will prove that $\zeta_K(s)$ has meromorphic continuation to $\mathbb{C}$ with a simple pole at $s = 1$, whose residue $\mathrm{res}_{s=1}\zeta_K(s) = \mathcal{M}_K$ carries a lot of arithmetic information about K (its class number h_K, discriminant D_K, regulator R_K, and more).

Once we bag this result, Tauberian theorems à la Wiener-Ikehara immediately give the distribution of ideals:

$$\sum_{n \leq x} a_n = \#\{I \subset \mathcal{O}_K : N(I) \leq x\} \sim \mathcal{M}_K \cdot x$$

as $x \to \infty$.

1.3.2 Normalized Gamma Factors

Before defining the completed Dedekind zeta function $\xi_K(s)$ we will introduce some notation which will remain in force throughout this lecture series.

Definition 1.80 We define the $\mathbb{R}$ and $\mathbb{C}$ normalizations of the Γ-function as follows.

- $L_{\mathbb{R}}(s) = \pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$;
- $L_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$.

Clearly $L_{\mathbb{R}}(s)$ and $L_{\mathbb{C}}(s)$ have no zeroes, and we know the location of their poles. Note that $L_{\mathbb{R}}(1) = 1$ and $L_{\mathbb{C}}(1) = \pi^{-1}$.

The factor 2 in the definition of $L_{\mathbb{C}}(s)$ may seem a bit silly. We include it at least for a couple of reasons. Firstly it ensures $L_{\mathbb{R}}(s)$ and $L_{\mathbb{C}}(s)$ have the same asymptotics $\sim \frac{2}{s}$ as $s \to 0$. Secondly it simplifies the duplication formula below, which gives a clean factorization of $L_{\mathbb{C}}(s)$ as a product of $L_{\mathbb{R}}(s)$ and its shift $L_{\mathbb{R}}(s+1)$. This will simplify our calculations later on.

Proposition 1.81 *The normalized Γ-functions $L_{\mathbb{R}}(s)$ and $L_{\mathbb{C}}(s)$ satisfy the functional equations:*

(1) ("Real reflection formula")

$$L_{\mathbb{R}}(1+s)L_{\mathbb{R}}(1-s) = \frac{1}{\cos(\frac{\pi s}{2})}.$$

(2) ("Complex reflection formula")

$$L_{\mathbb{C}}(s)L_{\mathbb{C}}(1-s) = \frac{2}{\sin(\pi s)}.$$

(3) ("Duplication formula")

$$L_{\mathbb{R}}(1+s)L_{\mathbb{R}}(s) = L_{\mathbb{C}}(s).$$

Proof Left as an exercise for the reader. □

Exercise 1.82 Deduce formulas (1) through (3) in Proposition 1.81 from Proposition 1.20.

1.3.3 Euler Factors at the Infinite Places

We first recall some standard notation. The r_1 real places of K correspond to the real embeddings $K \hookrightarrow \mathbb{R}$. The r_2 complex places of K correspond to pairs of complex conjugate non-real embeddings $K \hookrightarrow \mathbb{C}$. We let $r = r_1 + r_2$ denote the number of infinite places of K. Let us also point out that $d = r_1 + 2r_2$.

We complete the Dedekind zeta function by including a factor $L_{\mathbb{R}}(s)$ at each real place, and a factor $L_{\mathbb{C}}(s)$ at each complex place. Moreover, we include a factor $|D_K|^{\frac{s}{2}}$ to simplify the functional equation.

Definition 1.83 The completed Dedekind zeta function $\xi_K(s)$ is defined as

$$\xi_K(s) = |D_K|^{\frac{s}{2}} L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2} \zeta_K(s).$$

One of our main results will be its analytic continuations and functional equation:

Theorem 1.84 *The completed Dedekind zeta function $\xi_K(s)$ admits meromorphic continuation to $\mathbb{C}$ with only two simple poles at $s = 0$ and $s = 1$. Their residues are given by*

- $res_{s=0}\xi_K(s) = -\frac{2^r hR}{w}$
- $res_{s=1}\xi_K(s) = +\frac{2^r hR}{w}$.

(Here $h = h_K = |Cl_K|$ is the class number, $R = R_K$ the regulator, and $w = w_K = |\mathcal{O}_{K,tor}^{\times}|$ is the number of roots of unity in K.) Furthermore, it is invariant under point reflection over $\frac{1}{2}$:

$$\xi_K(s) = \xi_K(1 - s).$$

This has a classical proof along the same lines as for $\zeta(s)$ and $L(s, \chi)$ using Poisson summation in $\mathbb{R}^d$ and so on, but it is rather technical and a notational challenge. We find the approach in Tate's thesis far more transparent and elegant. We will delve into that before long.

For now we mention the following famous explicit formula for the residue $\mathcal{M}_K$.

Corollary 1.85 (The analytic class number formula) *The Dedekind zeta function $\zeta_K(s)$ has meromorphic continuation to $\mathbb{C}$ with $s = 1$ as its only pole. It is simple with residue*

$$res_{s=1}\zeta_K(s) = \mathcal{M}_K = \frac{2^{r_1}(2\pi)^{r_2}}{w\sqrt{|D_K|}} hR.$$

Proof By Theorem 1.84 we know that $(s-1)\xi_K(s) \to \frac{2^r hR}{w}$ as $s \to 1$. On the other hand

$$(s-1)\xi_K(s) = |D_K|^{\frac{s}{2}} L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2} (s-1)\zeta_K(s) \longrightarrow |D_K|^{\frac{1}{2}} \cdot 1^{r_1} \cdot \pi^{-r_2} \cdot \mathcal{M}_K.$$

Comparison gives the formula for $\mathcal{M}_K$. (The case $s \to 0$ will be dealt with momentarily.) □

At $s = 0$ the behavior of $\zeta_K(s)$ is summarized below.

Corollary 1.86 *$\zeta_K(s) \sim -\frac{hR}{w} s^{r-1}$ as $s \to 0$. In particular $ord_{s=0}\zeta_K(s) = r - 1 = rank(\mathcal{O}_K^\times)$.*

Proof From Theorem 1.84 we have $s\xi_K(s) \to -\frac{2^r hR}{w}$ as $s \to 0$. Expand $\zeta_K(s) = bs^a + \cdots$ where $a \in \mathbb{Z}$ is the analytic rank and $b \in \mathbb{C}^\times$ is the leading coefficient. In this notation

$$s\xi_K(s) = |D_K|^{\frac{s}{2}} L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2} s\zeta_K(s) \sim 1 \cdot (\frac{2}{s})^{r_1} \cdot (\frac{2}{s})^{r_2} \cdot bs^{a+1} = 2^r bs^{a+1-r}$$

as $s \to 0$. By comparison we must have $a + 1 - r = 0$ and $2^r b = -\frac{2^r hR}{w}$. Equivalently $a = r - 1$ and $b = -\frac{hR}{w}$ as claimed. Finally $\mathrm{rank}(\mathcal{O}_K^\times) = r - 1$ by the Dirichlet unit theorem. □

Exercise 1.87 What does Corollary 1.86 tell us for $K = \mathbb{Q}$?

Remark 1.88 One of the most prominent open problems in mathematics is to find an analogue of the previous Corollary for elliptic curves E/K. The celebrated Birch-Swinnerton-Dyer (BSD) conjecture predicts

$$\mathrm{ord}_{s=1} L(s, E) \stackrel{?}{=} \mathrm{rank} E(K).$$

Here the Hasse-Weil L-function $L(s, E)$ is a certain degree two Euler product encoding the number of points of E over finite fields. Even its analytic continuation is not known for all number fields K. For $K = \mathbb{Q}$ the analytic continuation was a salient byproduct of Wiles's proof of Fermat's Last Theorem, but despite great progress in recent years the BSD conjecture remains open even for the rationals.

There is also a conjectural formula for the leading Taylor coefficient at $s = 1$ in terms of arithmetic invariants associated with E. For instance the order of the so-called Tate-Shafarevich group $\mathrm{Sha}(E)$ plays the role of the class number. However, $\mathrm{Sha}(E)$ is not even known to be finite in general.

In terms of

$$\zeta_K^{\mathrm{aut}}(s) = L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2} \zeta_K(s) = |D_K|^{-\frac{s}{2}} \xi_K(s)$$

the functional equation takes the form

$$\zeta_K^{\text{aut}}(s) = |D_K|^{\frac{1}{2}-s}\zeta_K^{\text{aut}}(1-s).$$

This is the version we will ultimately prove.

What are the trivial zeroes of $\zeta_K(s)$? We know what happens at $s = 0$, but what is the order of vanishing at the negative integers?

Corollary 1.89 *For $n \in \mathbb{N}$ we have* $ord_{s=-n}\zeta_K(s) = \begin{cases} r_2 & \text{if } n \text{ is odd} \\ r_1 + r_2 & \text{if } n \text{ is even.} \end{cases}$

Proof Fix an $n \in \mathbb{N}$. Since $\xi_K(1+n) \neq 0$ the functional equation lets us conclude that

$$\text{ord}_{s=-n}\zeta_K(s) + r_1\text{ord}_{s=-n}L_{\mathbb{R}}(s) + r_2\text{ord}_{s=-n}L_{\mathbb{C}}(s) = 0.$$

When n is odd $-\frac{n}{2}$ is not a pole of $\Gamma(s)$ so the middle term above is zero, and the last term is $-r_2$ regardless of the parity of n. On the other hand, when n is even $-\frac{n}{2}$ is a simple pole of $\Gamma(s)$ so the middle term is $-r_1$. □

Example 1.90 When K is totally real ($r_2 = 0$) the zeta function $\zeta_K(s)$ has trivial zeroes at $\{-2, -4, -6, \ldots\}$, all of multiplicity $d = [K : \mathbb{Q}]$. When $K \neq \mathbb{Q}$ also $s = 0$ is a zero, but of multiplicity $d - 1$. In the totally real case there is a famous theorem of Siegel saying that $\zeta_K(-n) \in \mathbb{Q}^\times$ for $n \in \{1, 3, 5, \ldots\}$, thereby generalizing Theorem 1.34 for $K = \mathbb{Q}$.

At the other extreme, when K is totally imaginary ($r_1 = 0$) the zeta function $\zeta_K(s)$ has trivial zeroes at $\{-1, -2, -3, \ldots\}$, all of multiplicity $\frac{d}{2}$. When K is not quadratic imaginary also $s = 0$ is a zero, but with multiplicity $\frac{d}{2} - 1$.

Exercise 1.91 For $n \in \mathbb{N}$ let $b_n \in \mathbb{C}^\times$ denote the leading coefficient of $\zeta_K(s)$ at $s = -n$. In other words $\zeta_K(s) \sim b_n(s+n)^{a_n}$ as $s \to -n$ where $a_n = \text{ord}_{s=-n}\zeta_K(s)$. Find an explicit formula expressing b_n in terms of $\zeta_K(1+n)$ and other elementary factors. (A conjecture of Lichtenbaum predicts a formula for b_{odd} in terms of the algebraic K-theory of $\mathcal{O}_K$.)

We include one more application which is a special case of Chebotarev's density theorem, to be proved later after we discuss Artin L-functions (see Theorem 4.57). For this we assume $K/\mathbb{Q}$ is a finite Galois extension. Algebraic number theory tells us $\text{Gal}(K/\mathbb{Q})$ acts transitively on the set of $\mathfrak{p}$ dividing a given p. In particular $e(\mathfrak{p}|p)$ and $f(\mathfrak{p}|p)$ only depend on p. We say p splits completely in K if $e(\mathfrak{p}|p) = f(\mathfrak{p}|p) = 1$ for all $\mathfrak{p}|p$. In other words $p\mathcal{O}_K = \mathfrak{p}_1 \cdots \mathfrak{p}_d$ where the $\mathfrak{p}_i \subset \mathcal{O}_K$ are distinct maximal ideals. Let $\text{Spl}_{K/\mathbb{Q}}$ denote the set of primes p which split completely in K.

Exercise 1.92 Describe the set of primes $\text{Spl}_{K/\mathbb{Q}}$ by congruences for $K = \mathbb{Q}(\sqrt{5})$ and $K = \mathbb{Q}(\zeta_N)$.

Corollary 1.93 *The set $Spl_{K/\mathbb{Q}}$ has analytic density $\delta = \frac{1}{[K:\mathbb{Q}]}$.*

Proof We know $s = 1$ is a simple pole of $\zeta_K(s)$ by Corollary 1.85. In particular $\ln \zeta_K(s) \sim \ln\left(\frac{1}{s-1}\right)$ as $s \to 1^+$. On the other hand we may write

$$\begin{aligned}\ln \zeta_K(s) &= \sum\nolimits_{p\in \mathrm{Spl}_{K/\mathbb{Q}}} \sum_{\mathfrak{p}|p} \sum_{r=1}^{\infty} \frac{1}{r} N(\mathfrak{p})^{-rs} + \sum\nolimits_{p\notin \mathrm{Spl}_{K/\mathbb{Q}}} \sum_{\mathfrak{p}|p} \sum_{r=1}^{\infty} \frac{1}{r} N(\mathfrak{p})^{-rs} \\ &= d \cdot \sum\nolimits_{p\in \mathrm{Spl}_{K/\mathbb{Q}}} \sum_{r=1}^{\infty} \frac{1}{r} p^{-rs} + \sum\nolimits_{p\notin \mathrm{Spl}_{K/\mathbb{Q}}} \sum_{\mathfrak{p}|p} \sum_{r=1}^{\infty} \frac{1}{r} N(\mathfrak{p})^{-rs} \\ &= d \cdot \sum\nolimits_{p\in \mathrm{Spl}_{K/\mathbb{Q}}} p^{-s} + \mathcal{E}(s)\end{aligned}$$

where the error term $\mathcal{E}(s)$ remains bounded as $s \to 1^+$, the reason being that $f(\mathfrak{p}|p) \geq 2$ for all unramified $p \notin \mathrm{Spl}_{K/\mathbb{Q}}$ (so the triple sum and $\sum_{p\in \mathrm{Spl}_{K/\mathbb{Q}}} \sum_{r=2}^{\infty} \frac{1}{r} p^{-rs}$ both converge absolutely for $\mathrm{Re}(s) > \frac{1}{2}$). Altogether this shows

$$\sum\nolimits_{p\in \mathrm{Spl}_{K/\mathbb{Q}}} p^{-s} \sim \frac{1}{d} \cdot \ln\left(\frac{1}{s-1}\right)$$

as $s \to 1^+$. In other words $\frac{1}{d}$ is the analytic density of $\mathrm{Spl}_{K/\mathbb{Q}}$. □

An important consequence is that $\mathrm{Spl}_{K/\mathbb{Q}}$ fully determines the Galois extension $K/\mathbb{Q}$, see [Neu99, Corollary 13.10, p. 548]. Thus the finite Galois extensions of $\mathbb{Q}$ are classified by data intrinsic to the base field $\mathbb{Q}$, namely certain infinite subsets of prime numbers. Describing $\mathrm{Spl}_{K/\mathbb{Q}}$ when $\mathrm{Gal}(K/\mathbb{Q})$ is non-abelian is a central problem in number theory, which is closely tied to the Langlands program. In the abelian case one can deduce from class field theory that $\mathrm{Spl}_{K/\mathbb{Q}}$ is always given by a collection of congruences.

Exercise 1.94 Show that the logarithmic derivative $\frac{\zeta_K'(s)}{\zeta_K(s)}$ has a simple pole at $s = 1$ with residue -1.

Later below we will show a general non-vanishing result for Hecke L-functions (Theorem 3.29). In particular $\zeta_K(s)$ has no zeroes on the line $\mathrm{Re}(s) = 1$. Admitting this, one can emulate the discussion after Theorem 1.15 and deduce Landau's prime ideal theorem: The number of prime ideals $\mathfrak{p} \subset \mathcal{O}_K$ of norm $N(\mathfrak{p}) \leq x$ is asymptotically $\frac{x}{\ln x}$ as $x \to \infty$.

1.3.4 An Analytic Reformulation of Quadratic Reciprocity

In this section $N > 2$ denotes a prime number, and $N^* = (-1)^{\frac{N-1}{2}} N$. Recall that $\tau(\chi_N)^2 = N^*$ where $\chi_N = (\frac{\bullet}{N})$ is the Legendre symbol. In particular this shows $\mathbb{Q}(\sqrt{N^*})$ is the quadratic subfield of $\mathbb{Q}(\zeta_N)$.

Here we will achieve the following factorization of $\zeta_{\mathbb{Q}(\sqrt{N^*})}(s)$ which is equivalent to quadratic reciprocity.

Theorem 1.95 *Let $N > 2$ be a prime number, and $\chi_N = (\frac{\bullet}{N})$ the Legendre symbol. Then*

$$\zeta_{\mathbb{Q}(\sqrt{N^*})}(s) = \zeta(s)L(s, \chi_N).$$

Before we initiate the proof we pause to introduce some notation.

Definition 1.96 For a number field $K/\mathbb{Q}$ and a prime number p we introduce the finite Euler product

$$\zeta_{K,p}(s) = \prod_{\mathfrak{p}|p} \frac{1}{1 - N(\mathfrak{p})^{-s}}.$$

Obviously $\zeta_K(s) = \prod_p \zeta_{K,p}(s)$ for $\mathrm{Re}(s) > 1$.

We will compute these p-local Euler products for $K = \mathbb{Q}(\sqrt{N^*})$. Note that $D_K = N^* \equiv 1 \pmod 4$ and $\mathcal{O}_K = \mathbb{Z}[\frac{1}{2}(1 + \sqrt{N^*})]$. Elementary number theory gives us the splitting of p in $\mathbb{Q}(\sqrt{N^*})$. At first let us assume that $p > 2$.

(a) p is ramified if and only if $p = N$;
(b) $p \neq N$ is split if and only if $(\frac{p}{N}) = +1$;
(c) $p \neq N$ is inert if and only if $(\frac{p}{N}) = -1$.

By quadratic reciprocity $(\frac{p}{N}) = (\frac{N^*}{p})$ for $p \neq N$. The corresponding p-local Euler products are listed below.

(a) For $p = N$,

$$\zeta_{\mathbb{Q}(\sqrt{N^*}),p}(s) = \frac{1}{1 - p^{-s}}.$$

(b) For split $p \neq N$,

$$\zeta_{\mathbb{Q}(\sqrt{N^*}),p}(s) = (\frac{1}{1 - p^{-s}})^2.$$

(c) For inert $p \neq N$,

$$\zeta_{\mathbb{Q}(\sqrt{N^*}),p}(s) = \frac{1}{1 - p^{-2s}}.$$

Remark 1.97 The case $p = 2$ has to be treated separately. The congruence $N^* \equiv 1 \pmod 4$ always holds. The residue class of N^* modulo 8 determines whether 2 splits or not:

- 2 splits if and only if $N^* \equiv 1 \pmod 8$. In this case $\zeta_{\mathbb{Q}(\sqrt{N^*}),2}(s) = (\frac{1}{1-2^{-s}})^2$.
- 2 is inert if and only if $N^* \equiv 5 \pmod 8$. In this case $\zeta_{\mathbb{Q}(\sqrt{N^*}),2}(s) = \frac{1}{1-2^{-2s}}$.

Note that $N^* \equiv 1 \pmod 8$ precisely when $(\frac{2}{N}) = (-1)^{\frac{N^2-1}{8}} = +1$, as is easily checked.

Proof (Theorem 1.95) Putting all this together shows

$$\frac{\zeta_{\mathbb{Q}(\sqrt{N^*})}(s)}{\zeta(s)} = \prod\nolimits_{p\neq N:(\frac{p}{N})=+1} \frac{1}{1-p^{-s}} \cdot \prod\nolimits_{p\neq N:(\frac{p}{N})=-1} \frac{1}{1+p^{-s}} = L(s,\chi_N)$$

for $\mathrm{Re}(s) > 1$. By analytic continuation the final identity holds on all of $\mathbb{C}$. □

1.3.5 The Factorization Formula in the Abelian Case

Now let $N \geq 1$ be arbitrary (not necessarily a prime) and consider the cyclotomic field $\mathbb{Q}(\zeta_N)$. We set out to establish a factorization formula for $\zeta_{\mathbb{Q}(\zeta_N)}(s)$ analogous to the one in Theorem 1.95. It involves all Dirichlet L-functions $L(s,\chi)$ where χ is primitive of conductor $N_\chi | N$.

We start off by recalling some basic facts from algebraic number theory about how p splits in $\mathbb{Q}(\zeta_N)$. To warm up, let us first assume p is unramified. Equivalently $p \nmid N$. In this case

$$(p) = p\mathbb{Z}[\zeta_N] = \mathfrak{p}_1 \cdots \mathfrak{p}_t$$

for distinct maximal ideals $\mathfrak{p}_i \subset \mathbb{Z}[\zeta_N]$, all having the same inertia degree

$$f(\mathfrak{p}_i|p) = f = \mathrm{ord}_N(p) = \min\{m > 0 : p^m \equiv 1 \pmod N\}.$$

(Thus f is the order of $[p]$ in $(\mathbb{Z}/N\mathbb{Z})^\times$.) We necessarily have $t = \frac{\phi(N)}{f}$. In particular p splits completely precisely when $p \equiv 1 \pmod N$. In this case we clearly have

$$\zeta_{\mathbb{Q}(\zeta_N),p}(s) = \left(\frac{1}{1-p^{-fs}}\right)^{\frac{\phi(N)}{f}}.$$

In the ramified case $p|N$ this formula only requires minor modifications: Write $N = p^a M$ where $p \nmid M$ and $a > 0$. Now $(p) = (\mathfrak{p}_1 \cdots \mathfrak{p}_t)^{\phi(p^a)}$ where each $\mathfrak{p}_i$ has inertia degree $f = \mathrm{ord}_M(p)$. In this notation

$$\zeta_{\mathbb{Q}(\zeta_N),p}(s) = \left(\frac{1}{1-p^{-fs}}\right)^{\frac{\phi(M)}{f}}.$$

The key is to express these p-local Euler products in terms of Dirichlet L-factors $L_p(s,\chi)$.

Lemma 1.98 *Suppose $N \geq 1$ and let p be an arbitrary prime. Write $N = p^a M$ where $p \nmid M$ and $a \geq 0$. Then*

$$\zeta_{\mathbb{Q}(\zeta_N),p}(s) = \prod_{\chi} L_p(s,\chi)$$

where χ runs over all characters of $(\mathbb{Z}/M\mathbb{Z})^\times$.

Proof Throughout this proof let $G = (\mathbb{Z}/M\mathbb{Z})^\times$ and $H = \langle [p] \rangle \leq G$. Note that $|H| = \mathrm{ord}_M(p) = f$ in the above notation, and $[G:H] = \frac{\phi(M)}{f}$. By $\widehat{G}$ we mean the dual group of all characters $\chi : G \longrightarrow \mathbb{C}^\times$. Similarly for $\widehat{H}$. The restriction map $\widehat{G} \longrightarrow \widehat{H}$ is known to be surjective (any complex character of a finite index subgroup of an abelian group G extends to a character of G). Starting with a character $\psi : H \longrightarrow \mathbb{C}^\times$ we look at the fiber of the restriction map, that is the set of all extensions

$$\widehat{G}_\psi = \{\chi \in \widehat{G} : \chi|_H = \psi\}.$$

As $\psi \in \widehat{H}$ varies these fibers $\widehat{G}_\psi$ form a partition of $\widehat{G}$, and each fiber has size $[G:H]$. Indeed, lifts of ψ differ by a character of G/H, so choosing a lift gives a bijection between $\widehat{G}_\psi$ and the dual group $\widehat{G/H}$. The latter is non-canonically isomorphic to G/H. Having made these observations we infer that

$$\begin{aligned}
\prod_{\chi} L_p(s,\chi) &= \prod_{\psi\in\widehat{H}} \prod_{\chi\in\widehat{G}_\psi} \frac{1}{1-\chi(p)p^{-s}} \\
&= \prod_{\psi\in\widehat{H}} \left(\frac{1}{1-\psi(p)p^{-s}}\right)^{|\widehat{G}_\psi|} \\
&= \prod_{\psi\in\widehat{H}} \left(\frac{1}{1-\psi(p)p^{-s}}\right)^{\frac{\phi(M)}{f}}.
\end{aligned}$$

In remains to make the following remarks. Pick a generator ψ_0 for the cyclic group $\widehat{H} \simeq H$ of cardinality f. Then $\gamma = \psi_0(p)$ must be a primitive fth root of unity. Therefore

$$\prod_{\psi\in\widehat{H}} (1-\psi(p)X) = \prod_{i=0}^{f-1} (1-\psi_0^i(p)X) = \prod_{i=0}^{f-1} (1-\gamma^i X) = 1 - X^f$$

Evaluating at $X = p^{-s}$ gives the desired result. □

Inflation $\chi \mapsto \chi \circ \mathrm{pr}_{N,M}$ gives a bijection between the characters of $(\mathbb{Z}/M\mathbb{Z})^\times$ and the set of characters $\eta : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ with conductor dividing M. Equivalently those η mod N with conductor prime-to-p. Note that if $\eta = \chi \circ \mathrm{pr}_{N,M}$ we have

$$L_p(s,\chi) = L_p(s,\eta^{\mathrm{Prim}}).$$

As a result we may rewrite the product in Lemma 1.98 as $\prod_\chi L_p(s,\chi)$ where we rename χ and now let it range over all *primitive* Dirichlet characters with conductor $N_\chi | N$ but $p \nmid N_\chi$.

This calculation essentially gives the main result for $\mathbb{Q}(\zeta_N)$.

Theorem 1.99 *The Dedekind zeta function of $\mathbb{Q}(\zeta_N)$ has a factorization into Dirichlet L-functions*

$$\zeta_{\mathbb{Q}(\zeta_N)}(s) = \prod_\chi L(s,\chi)$$

where χ runs over all primitive Dirichlet characters of conductor dividing N.

Proof For $\mathrm{Re}(s) > 1$ we have, by interchanging the products over p and χ, that

$$\begin{aligned}\zeta_{\mathbb{Q}(\zeta_N)}(s) &= \prod_p \zeta_{\mathbb{Q}(\zeta_N),p}(s)\\ &= \prod_p \prod_{\chi: N_\chi | N \text{ and } p\nmid N_\chi} L_p(s,\chi)\\ &= \prod_{\chi: N_\chi | N} \prod_{p\nmid N_\chi} L_p(s,\chi)\\ &= \prod_{\chi: N_\chi | N} L(s,\chi).\end{aligned}$$

Here χ denotes a primitive character whose conductor divides N. This proves the claim. □

The factorization formula generalizes to any abelian extension $K/\mathbb{Q}$, by which we mean a finite Galois extension with abelian Galois group $\mathrm{Gal}(K/\mathbb{Q})$. By a central result from class field theory ("Kronecker-Weber") any such K is a subfield of a cyclotomic field. We will instead fix an $N \geq 1$ once and for all, and take a subfield $K \subset \mathbb{Q}(\zeta_N)$. Such a $K/\mathbb{Q}$ is obviously an abelian extension:

$$(\mathbb{Z}/N\mathbb{Z})^\times \xrightarrow{\sim} \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \twoheadrightarrow \mathrm{Gal}(K/\mathbb{Q}).$$

We say that a character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is coupled with K if it factors through the natural projection map $(\mathbb{Z}/N\mathbb{Z})^\times \twoheadrightarrow \mathrm{Gal}(K/\mathbb{Q})$. Such a χ may be viewed as a Galois character $\mathrm{Gal}(K/\mathbb{Q}) \longrightarrow \mathbb{C}^\times$.

Example 1.100 If $N > 2$ is a prime the Legendre symbol $\chi_N = (\frac{\bullet}{N})$ is the only non-trivial character coupled with $\mathbb{Q}(\sqrt{N^*})$.

Example 1.101 For arbitrary $N \geq 1$ consider the maximal totally real subfield $\mathbb{Q}(\zeta_N)^+ = \mathbb{Q}(\cos\frac{2\pi}{N})$ inside $\mathbb{Q}(\zeta_N)$. It is of degree $d = r_1 = \frac{\phi(N)}{2}$ over $\mathbb{Q}$. The residue class $[-1]$ induces complex conjugation on $\mathbb{Q}(\zeta_N)$, and therefore the restriction map defines an isomorphism

$$\frac{(\mathbb{Z}/N\mathbb{Z})^\times}{\{\pm 1\}} \xrightarrow{\sim} \mathrm{Gal}(\mathbb{Q}(\zeta_N)^+/\mathbb{Q}).$$

It follows that $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is coupled with $\mathbb{Q}(\zeta_N)^+$ exactly when χ is even ($e = 0$).

With a little extra effort one can extend the factorization formula to all abelian extensions $K/\mathbb{Q}$. We record the result without proof for now. Later below, in Corollary 4.11, we will extend the factorization formula to all *non*-abelian Galois extensions $K/\mathbb{Q}$. In this case the factors $L(s, \chi)$ are replaced by *Artin* L-functions.

Theorem 1.102 *Let $K \subset \mathbb{Q}(\zeta_N)$ be a subfield for some $N \geq 1$. Then*

$$\zeta_K(s) = \prod_\chi L(s, \chi)$$

where χ runs over all primitive Dirichlet characters of conductor $N_\chi | N$ whose inflation $\chi \circ pr_{N,N_\chi}$ is coupled with K.

Proof Omitted, but see Corollary 4.11. □

Exercise 1.103 Check that $\zeta_8 + \zeta_8^{-1} = \sqrt{2}$ and deduce $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}(\zeta_8)^+$. Conclude that $\zeta_{\mathbb{Q}(\sqrt{2})}(s) = \zeta(s)L(s, \chi_8)$ where $\chi_8 : (\mathbb{Z}/8\mathbb{Z})^\times \to \mathbb{C}^\times$ is the non-trivial even character. (Note that $(\mathbb{Z}/8\mathbb{Z})^\times$ is isomorphic to Klein's four-group V_4.) What is $\chi_8([3])$? Why is χ_8 primitive?

Exercise 1.104 Check that $\zeta_8^3 + \zeta_8 = \sqrt{-2}$ and infer that $\mathbb{Q}(\sqrt{-2}) \subset \mathbb{Q}(\zeta_8)$ is the fixed field of $\langle [3] \rangle$. Conclude that $\zeta_{\mathbb{Q}(\sqrt{-2})}(s) = \zeta(s)L(s, \psi_8)$ where $\psi_8 : (\mathbb{Z}/8\mathbb{Z})^\times \to \mathbb{C}^\times$ is the non-trivial character satisfying $\psi_8([3]) = 1$. Why is ψ_8 primitive?

It follows immediately that the ratio $\frac{\zeta_K(s)}{\zeta(s)} = \prod_{\chi \neq 1} L(s, \chi)$ is an entire function. Of course the simple poles at $s = 1$ of $\zeta_K(s)$ and $\zeta(s)$ cancel out; the point is that $\zeta(s) = 0 \Rightarrow \zeta_K(s) = 0$. This should be true in general:

Conjecture 1.105 (Dedekind) $\frac{\zeta_K(s)}{\zeta(s)}$ *is entire for any number field $K/\mathbb{Q}$.*

This is known for Galois extensions $K/\mathbb{Q}$ ("Aramata-Brauer") and we will prove this later in Corollary 4.20. More generally one can replace $\mathbb{Q}$ by any base number field F and consider extensions K/F but we stick to $F = \mathbb{Q}$ for simplicity.

As we mentioned above, the factorization formula vastly generalizes to non-abelian Galois extensions $K/\mathbb{Q}$, but instead of L-functions of Galois characters one has Artin L-functions $L(s, \rho)$ of complex Galois representations $\rho : \mathrm{Gal}(K/\mathbb{Q}) \longrightarrow \mathrm{Aut}_{\mathbb{C}}(V)$ on the right-hand side. We will not define $L(s, \rho)$ yet, but just signal that it is given by an Euler product of degree $n = \dim_{\mathbb{C}}(V)$. The theory of Artin L-functions is the topic of Chap. 4. A famous conjecture of Artin asserts that $L(s, \rho)$ is entire for irreducible $\rho \neq 1$ and this implies Dedekind's conjecture. Artin's conjecture is

one of the driving forces behind the Langlands program. The philosophy is that one can somehow match ρ with what is called an automorphic representation π of GL_n in such a way that $L(s, \rho) = L(s, \pi)$. The L-functions $L(s, \pi)$ are known to have many of the desired analytic properties.

The analytic class number formula immediately implies the following:

Corollary 1.106 *Suppose $K \subset \mathbb{Q}(\zeta_N)$ is a subfield for some $N \geq 1$. Then we have the relation*

$$\prod_{\chi \neq 1} L(1, \chi) = \mathcal{M}_K = \frac{2^{r_1}(2\pi)^{r_2}}{w\sqrt{|D_K|}} hR.$$

where χ runs over primitive Dirichlet characters of conductor dividing N whose inflation to $(\mathbb{Z}/N\mathbb{Z})^\times$ is coupled with K.

Proof Let $s \to 1$ in the factorization formula $\frac{(s-1)\zeta_K(s)}{(s-1)\zeta(s)} = \prod_{\chi \neq 1} L(s, \chi)$ and appeal to Corollary 1.85. □

The cyclotomic case $K = \mathbb{Q}(\zeta_N)$ finally proves Theorem 1.49 (and therefore also Theorem 1.44): Indeed $\mathcal{M}_K > 0$ so in particular all the factors $L(1, \chi)$ must be nonzero!

We have explicit expressions for $L(1, \chi)$ so in principle Corollary 1.106 gives an explicit formula for hR. However, the individual factors h and R are extremely hard to compute for a general number field K. To embellish the computation of class numbers let us mention the following renowned open problem from Disquisitiones Arithmeticae (published in 1801):

Conjecture 1.107 (Gauss) *There are infinitely many real quadratic fields with class number one.*

The Cohen–Lenstra heuristics give sharper predictions about the structure of class groups of quadratic fields. For real quadratic fields they suggest that 75.45% of the fields $\mathbb{Q}(\sqrt{p})$ have class number one.

1.3.6 The Conductor-Discriminant Formula

By combining all functional equations mentioned thus far we will derive the following beautiful (and computationally useful) formula for D_K for abelian extensions $K/\mathbb{Q}$. In Chap. 4 we will generalize the formula to non-abelian extensions; see Example 4.49. In general the factors N_χ are replaced by Artin conductors of irreducible representations of the Galois group.

Theorem 1.108 (Conductor-discriminant formula) *Consider a subfield $K \subset \mathbb{Q}(\zeta_N)$. Then*

$$D_K = (-1)^{r_2} \cdot \prod_{\chi} N_\chi$$

where χ ranges over primitive Dirichlet characters with conductor $N_\chi | N$ coupled with K.

In general $\operatorname{sign}(D_K) = (-1)^{r_2}$ for *any* number field $K/\mathbb{Q}$ ("Brill's Theorem"). This is relatively easy to see. Work with the definition $D_K = \det\left(\sigma_i(b_j)\right)^2$ where $\sigma_i : K \hookrightarrow \mathbb{C}$ and $\{b_j\}$ is a $\mathbb{Z}$-basis for $\mathcal{O}_K$, and perform row operations on the matrix $\left(\sigma_i(b_j)\right)$. Splitting its complex entries into real and imaginary parts you get a factor i out from r_2 of the rows.

Here is the idea for an imaginary quadratic field K: In this case $\sigma_2 = \overline{\sigma_1}$. Write $\sigma_1(b_1) = x + iy$ and $\sigma_1(b_2) = u + iv$. Then, using row operations, we find that

$$\begin{pmatrix} \sigma_1(b_1) & \sigma_1(b_2) \\ \sigma_2(b_1) & \sigma_2(b_2) \end{pmatrix} = \begin{pmatrix} x+iy & u+iv \\ x-iy & u-iv \end{pmatrix} \sim \begin{pmatrix} x+iy & u+iv \\ 2x & 2u \end{pmatrix} \sim \begin{pmatrix} x+iy & u+iv \\ x & u \end{pmatrix} \sim \begin{pmatrix} iy & iv \\ x & u \end{pmatrix} \sim \begin{pmatrix} y & v \\ x & u \end{pmatrix} = \begin{pmatrix} \operatorname{Im}(\sigma_1(b_1)) & \operatorname{Im}(\sigma_1(b_2)) \\ \operatorname{Re}(\sigma_1(b_1)) & \operatorname{Re}(\sigma_1(b_2)) \end{pmatrix}.$$

We scaled by $\frac{1}{2}$ and $\frac{1}{i}$ in the second and fourth $\sim$ respectively. This scales the determinant accordingly. Thus

$$\det\begin{pmatrix} \sigma_1(b_1) & \sigma_1(b_2) \\ \sigma_2(b_1) & \sigma_2(b_2) \end{pmatrix} = 2i \cdot \det\begin{pmatrix} \operatorname{Im}(\sigma_1(b_1)) & \operatorname{Im}(\sigma_1(b_2)) \\ \operatorname{Re}(\sigma_1(b_1)) & \operatorname{Re}(\sigma_1(b_2)) \end{pmatrix}.$$

Finally, taking the square shows that $D_K < 0$ in this case.

Thus what we really have to show in order to prove Theorem 1.108 is the formula $|D_K| = \prod_\chi N_\chi$. Before doing that we will give two quick examples showing the power of our new tool.

Example 1.109 Consider $\mathbb{Q}(\zeta_p)$ for a prime $p > 2$. Here $\operatorname{sign}(D_{\mathbb{Q}(\zeta_p)}) = (-1)^{\frac{p-1}{2}}$ since $r_2 = \frac{d}{2} = \frac{p-1}{2}$. All non-trivial characters $\chi : (\mathbb{Z}/p\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ are primitive so $\operatorname{Prim}(p) = p - 2$. They all have conductor p and therefore

$$D_{\mathbb{Q}(\zeta_p)} = (-1)^{\frac{p-1}{2}} \cdot p^{p-2}.$$

This expression should be familiar from basic algebraic number theory where it is usually shown by a brute force determinant computation.

Example 1.110 Now let us take a look at $\mathbb{Q}(\zeta_p)^+ = \mathbb{Q}(\cos\frac{2\pi}{p})$ for a prime $p > 2$. This is totally real ($r_2 = 0$) so its discriminant is positive. Recall that $\chi : (\mathbb{Z}/p\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ is coupled with $\mathbb{Q}(\zeta_p)^+$ if and only if χ is even. The quotient $\frac{(\mathbb{Z}/p\mathbb{Z})^\times}{\{\pm 1\}}$ has precisely $\frac{p-1}{2} - 1 = \frac{p-3}{2}$ non-trivial characters. Therefore

$$D_{\mathbb{Q}(\zeta_p)^+} = p^{\frac{p-3}{2}}.$$

This formula is more tricky to show by simple methods.

We merge into the proof of Theorem 1.108. First observe that $K \subset \mathbb{Q}(\zeta_N)$ is necessarily Galois over $\mathbb{Q}$ since $\operatorname{Gal}(\mathbb{Q}(\zeta_N)/K)$ is obviously normal in $(\mathbb{Z}/N\mathbb{Z})^\times$. Hence $\operatorname{Gal}(K/\mathbb{Q})$ acts transitively on the set of embeddings $K \hookrightarrow \mathbb{C}$. If K admits

one real embedding all embeddings must therefore be real. In other words either K is totally real ($r_2 = 0$) or totally imaginary ($r_1 = 0$). The proof divides into those two cases.

Proof (*Theorem 1.108 in the totally real case*) Let us first focus on the simplest case where K is totally real. All the characters $\chi : (\mathbb{Z}/N_\chi\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ coupled with K must be even (e = 0) which simplifies the functional equation for $L(s, \chi)$:

$$(\frac{\pi}{N_\chi})^{-\frac{s}{2}}\Gamma(\frac{s}{2})L(s,\chi) = W(\bar{\chi})(\frac{\pi}{N_\chi})^{-\frac{1-s}{2}}\Gamma(\frac{1-s}{2})L(1-s,\bar{\chi}).$$

Take the product over all such χ (there are r_1 of them) and invoke the factorization formula for $\zeta_K(s)$. This gives the relation

$$\mathcal{N}^{\frac{s}{2}}L_{\mathbb{R}}(s)^{r_1}\zeta_K(s) = \mathcal{W}\mathcal{N}^{\frac{1-s}{2}}L_{\mathbb{R}}(1-s)^{r_1}\zeta_K(1-s)$$

where we have introduced the two quantities $\mathcal{N} = \prod_\chi N_\chi$ and $\mathcal{W} = \prod_\chi W(\chi)$ for the purpose of this proof only. Note that $\chi \mapsto \bar{\chi}$ permutes the characters coupled with K. On the other hand $\zeta_K(s)$ satisfies its own functional equation

$$|D_K|^{\frac{s}{2}}L_{\mathbb{R}}(s)^{r_1}\zeta_K(s) = |D_K|^{\frac{1-s}{2}}L_{\mathbb{R}}(1-s)^{r_1}\zeta_K(1-s).$$

Dividing one by another yields the identity

$$\frac{\mathcal{N}^{\frac{s}{2}}}{|D_K|^{\frac{s}{2}}} = \mathcal{W}\frac{\mathcal{N}^{\frac{1-s}{2}}}{|D_K|^{\frac{1-s}{2}}} \Longrightarrow (\frac{\mathcal{N}}{|D_K|})^{s-\frac{1}{2}} = \mathcal{W}.$$

Inserting $s = \frac{1}{2}$ first shows $\mathcal{W} = 1$, and then plugging in $s = 0$ gives $\frac{\mathcal{N}}{|D_K|} = 1$ as claimed. □

Exercise 1.111 Supply the proof of Theorem 1.108 for a totally imaginary field K. (`Hints`: Why must there be an odd χ coupled with K? Use this to show #{odd χ} = #{even χ} = r_2. Finally make use of the duplication formula $L_{\mathbb{C}}(s) = L_{\mathbb{R}}(s)L_{\mathbb{R}}(s+1)$.)

As a byproduct of the previous proof (and exercise) we get for free that $\prod_\chi W(\chi) = 1$. In terms of Gauss sums this may be restated as follows.

Corollary 1.112 *Let $K \subset \mathbb{Q}(\zeta_N)$ be a subfield. Then*

$$\prod_\chi \tau(\chi) = i^{r_2}\sqrt{|D_K|} = \begin{cases} \sqrt{|D_K|} & \text{if } K \text{ is totally real } (r_2 = 0) \\ i^{\frac{d}{2}}\sqrt{|D_K|} & \text{if } K \text{ is totally imaginary } (r_1 = 0) \end{cases}$$

with χ in the usual range of characters coupled with K.

Proof This follows immediately from the very definition of $W(\chi)$ plus Theorem 1.108 and its proof. □

Example 1.113 When $K = \mathbb{Q}(\sqrt{p^*})$ for a prime $p > 2$ only the Legendre symbol $\chi_p = (\frac{\bullet}{p})$ is coupled with K. Here $D_K = p^*$ since we always have $p^* \equiv 1 \pmod 4$ so in this case we conclude that

$$\tau(\chi_p) = i^{\mathrm{e}}\sqrt{p}$$

where $\mathrm{e} = 0, 1$ according to whether $p \equiv 1, 3 \pmod 4$. In other words $W(\chi_p) = +1$ as announced earlier in exercise 1.66. This is a refinement of the property $\tau(\chi_p)^2 = p^*$.

Exercise 1.114 Verify the formula $\tau(\chi_p) = i^{\mathrm{e}}\sqrt{p}$ explicitly by hand for $p = 3$. (`Hint`: Write the Gauss sum $\tau(\chi_3)$ in terms of $\omega = e^{2\pi i/3} = \frac{1}{2}(-1 + i\sqrt{3})$ and $\omega^2 = \bar{\omega}$.)

1.3.7 Some Discriminant Calculations

As a neat application of Theorem 1.108 we will evaluate $D_{\mathbb{Q}(\zeta_{p^r})}$ for any prime $p > 2$ and $r \geq 1$. We already did $r = 1$ in Example 1.109. Here $r_2 = \frac{1}{2}\phi(p^r) = \frac{1}{2}p^{r-1}(p-1)$ has the same parity as $\frac{p-1}{2}$ so $\mathrm{sign}(D_{\mathbb{Q}(\zeta_{p^r})}) = (-1)^{\frac{p-1}{2}}$ is independent of r.

Corollary 1.115 *$D_{\mathbb{Q}(\zeta_{p^r})} = (-1)^{\frac{p-1}{2}} \cdot p^E$ where $E = p^{r-1}(r(p-1)-1)$.*

Proof From the Möbius inversion formula argument in Exercise 1.59 we know that $\mathrm{Prim}(p) = p - 2$ and $\mathrm{Prim}(p^s) = p^{s-2}(p-1)^2$ for $s \geq 2$. Consequently

$$|D_{\mathbb{Q}(\zeta_{p^r})}| = \prod_{s=0}^{r} \prod_{\chi : N_\chi = p^s} p^s = p^E$$

where

$$E = \sum_{s=0}^{r} s\mathrm{Prim}(p^s) = p - 2 + (p-1)^2 \frac{1}{p} \sum_{s=2}^{r} s p^{s-1}.$$

Differentiating the geometric sum $\sum_{s=0}^{r} X^s = \frac{X^{r+1}-1}{X-1}$ gives the formula

$$(X-1)^2 \sum_{s=1}^{r} s X^{s-1} = rX^{r+1} - (r+1)X^r + 1.$$

Inserting $X = p$ allows us to evaluate the component $(p-1)^2 \sum_{s=2}^{r} sp^{s-1}$ of the above expression for E:

$$(p-1)^2 \sum_{s=2}^{r} sp^{s-1} = -(p-1)^2 + (p-1)^2 \sum_{s=1}^{r} sp^{s-1} = rp^{r+1} - (r+1)p^r - p^2 + 2p.$$

This finally gives us the desired formula

$$\begin{aligned} E &= p - 2 + \frac{1}{p}\big(rp^{r+1} - (r+1)p^r - p^2 + 2p\big) \\ &= p - 2 + \big(rp^r - (r+1)p^{r-1} - p + 2\big) \\ &= rp^r - (r+1)p^{r-1} \\ &= p^{r-1}\big(rp - r - 1\big) \\ &= p^{r-1}\big(r(p-1) - 1\big). \end{aligned}$$

Done. □

See [Was97, Proposition 2.1, p. 9] for another derivation of this formula for $D_{\mathbb{Q}(\zeta_{p^r})}$ based on Vandermonde determinants. The reader may want to derive a similar formula for $p = 2$.

Exercise 1.116 Mimic the above calculation and show that $D_{\mathbb{Q}(\zeta_{p^r})^+} = p^{\frac{1}{2}\left(rp^r-(r+1)p^{r-1}-1\right)}$.

1.4 Related Topics

The material in this section deviates slightly from the main theme of the book, and it is more elementary in nature. Most readers will likely have seen this in an undergraduate number theory class. We recommend skipping this section on a first read and referring back to it as necessary. The only result we will use repeatedly later is Landau's Lemma 1.135, so the reader may want to at least understand that statement if they have not seen it before.

1.4.1 Asymptotics of Arithmetic Functions

Let $f : \mathbb{N} \longrightarrow \mathbb{C}$ be an arithmetic function. In other words, as discussed in Appendix A.2, a sequence $a_1, a_2, \ldots$ of complex numbers $a_n = f(n)$. In this section we are interested in the asymptotic behavior of a_n as $n \to \infty$. For instance we will show that $\frac{\phi(n)}{n}$ is on average $\frac{6}{\pi^2} = 0.60793...$ as $n \to \infty$. We will make use of the Möbius transformation

$$(f * \mathbf{1})(n) = \sum_{d|n} f(d).$$

We start with the following general basic inequality:

Lemma 1.117 $|\frac{1}{x}\sum_{n\le x}(f*\mathbf{1})(n) - \sum_{d\le x}\frac{f(d)}{d}| \le \frac{1}{x}\sum_{d\le x}|f(d)|$.

Proof Note that for all $x \ge 1$ we have

$$\begin{aligned}\sum_{n\le x}(f*\mathbf{1})(n) &= \sum_{n\le x}\sum_{d|n} f(d)\\ &= \sum_{d\le x}\sum\nolimits_{n=dm\le x} f(d)\\ &= \sum_{d\le x} f(d)\Big[\frac{x}{d}\Big].\end{aligned}$$

Therefore

$$|\sum_{n\le x}(f*\mathbf{1})(n) - \sum_{d\le x} f(d)\frac{x}{d}| \le \sum_{d\le x}|f(d)|$$

as claimed. □

The idea is, when f is small enough, that $\lim_{x\to\infty}\frac{1}{x}\sum_{d\le x}|f(d)| = 0$ and $\sum_{d=1}^{\infty}\frac{f(d)}{d}$ is the asymptotic mean value of $f*\mathbf{1}$. Here is the example alluded to earlier in this section:

Proposition 1.118 $\frac{1}{x}\sum_{n\le x}\frac{\phi(n)}{n} = \frac{6}{\pi^2} + O(\frac{\ln x}{x})$.

Proof We apply Lemma 1.117 to the function $f(n) = \frac{\mu(n)}{n}$ where μ is the Möbius function, see Example A.11. In this case

$$(f*\mathbf{1})(n) = \sum_{d|n}\frac{\mu(d)}{d} = \frac{1}{n}\sum_{d|n}\mu(d)\frac{n}{d} = \frac{\phi(n)}{n}$$

by the Möbius inversion formula (Theorem A.15) for ϕ. Indeed ϕ has Möbius transformation

$$(\phi*\mathbf{1})(n) = \sum_{d|n}\phi(d) = n.$$

In this situation, since $|\mu(d)| \le 1$, the inequality in Lemma 1.117 yields

$$|\frac{1}{x}\sum_{n\le x}\frac{\phi(n)}{n} - \sum_{d\le x}\frac{\mu(d)}{d^2}| \le \frac{1}{x}\sum_{d\le x}\frac{1}{d}.$$

Recall that $\sum_{d\le x}\frac{1}{d} = \ln x + \gamma + O(\frac{1}{x})$, so the right-hand side of the above inequality is $O(\frac{\ln x}{x})$. On the other hand

$$\sum_{d\le x}\frac{\mu(d)}{d^2}=\sum_{d=1}^{\infty}\frac{\mu(d)}{d^2}+\sum_{d>x}\frac{\mu(d)}{d^2}=\frac{1}{\zeta(2)}+O(\frac{1}{x})$$

by Example A.11. Now insert the special value $\zeta(2)=\frac{\pi^2}{6}$. □

Exercise 1.119 Show that $\lim_{n\to\infty}\frac{\phi(n)}{n}$ does not exist. (`Hint`: Take n to be a prime power.)

Exercise 1.120 Consider the divisor sum function σ_1. Apply Lemma 1.117 to $f(n)=\frac{1}{n}$ and prove that

$$\frac{1}{x}\sum_{n\le x}\frac{\sigma_1(n)}{n}=\frac{\pi^2}{6}+O(\frac{\ln x}{x}).$$

Check that $\lim_{n\to\infty}\frac{\sigma_1(n)}{n}$ does not exist.

There is an important variant of Lemma 1.117 which, among other things, gives the average order of the ϕ-function itself.

Lemma 1.121 $|\frac{1}{x}\sum_{n\le x}n(f*\mathbf{1})(n)-\frac{1}{2}\sum_{d\le x}f(d)(\frac{x}{d}+1)|\le\sum_{d\le x}|f(d)|$.

Proof For $x\ge 1$ we have

$$\begin{aligned}\sum_{n\le x}n(f*\mathbf{1})(n)&=\sum_{n\le x}\sum_{d|n}nf(d)\\&=\sum_{d\le x}\sum\nolimits_{n=dm\le x}dmf(d)\\&=\sum_{d\le x}df(d)\cdot\Big(\sum_{m=1}^{[x/d]}m\Big)\\&=\sum_{d\le x}df(d)\frac{1}{2}[\frac{x}{d}]([\frac{x}{d}]+1).\end{aligned}$$

Consequently, using the estimate

$$0\le\frac{1}{2}t(t+1)-\frac{1}{2}[t]([t]+1)<\frac{1}{2}t(t+1)-\frac{1}{2}(t-1)t=t$$

with $t=\frac{x}{d}$, we obtain

$$|\sum_{n\le x}n(f*\mathbf{1})(n)-\sum_{d\le x}df(d)\frac{1}{2}\frac{x}{d}(\frac{x}{d}+1)|\le\sum_{d\le x}d|f(d)|\frac{x}{d}=x\sum_{d\le x}|f(d)|.$$

Dividing by x gives the desired result. □

As an immediate application this gives the asymptotic behavior of ϕ on average:

Proposition 1.122 $\frac{1}{x}\sum_{n\le x}\phi(n) = \frac{3}{\pi^2}x + O(\ln x)$.

Proof Taking $f(n) = \frac{\mu(n)}{n}$ in Lemma 1.121 yields

$$|\frac{1}{x}\sum_{n\le x}\phi(n) - \frac{1}{2}\sum_{d\le x}\frac{\mu(d)}{d}(\frac{x}{d}+1)| \le \sum_{d\le x}\frac{1}{d}.$$

We again note that $\sum_{d\le x}\frac{1}{d} = \ln x + \gamma + O(\frac{1}{x})$. Furthermore

$$\frac{1}{2}\sum_{d\le x}\frac{\mu(d)}{d}(\frac{x}{d}+1) = \frac{1}{2}x\sum_{d\le x}\frac{\mu(d)}{d^2} + \frac{1}{2}\sum_{d\le x}\frac{\mu(d)}{d}.$$

The second term $\frac{1}{2}\sum_{d\le x}\frac{\mu(d)}{d}$ is clearly also $O(\ln x)$. Since $\sum_{d\le x}\frac{\mu(d)}{d^2} = \frac{1}{\zeta(2)} + O(\frac{1}{x})$ we are done. □

Here is an amusing interpretation of this result: For a fixed x start with the square $(0, x] \times (0, x]$ of area x^2. In this square plot all pairs of positive integers (m, n) for which $\mathrm{GCD}(m, n) = 1$. This set of points is symmetric about the diagonal through the origin, and $(1, 1)$ is the only point on the diagonal. Thus the total number of such points in the square is

$$\#\{(m,n) : 1 \le m, n \le x \text{ coprime}\} = 1 + 2\sum_{2\le n\le x}\phi(n) = \frac{6}{\pi^2}x^2 + O(x\ln x).$$

In conclusion, this number of points is $\sim \frac{6}{\pi^2}\cdot$ (the area of the square) as $x \to \infty$. In this sense, the probability that a randomly chosen pair of positive integers are coprime is $\frac{6}{\pi^2} = 0.60793....$

Exercise 1.123 Prove the identity $|\mu(n)| = \sum_{d:d^2|n}\mu(d)$ for all $n \in \mathbb{N}$. (`Hint`: Write $n = m^2 s$ with s square-free. Then $d^2|n$ if and only if $d|m$.) Deduce that

$$\sum_{n\le x}|\mu(n)| = \frac{6}{\pi^2}x + O(\sqrt{x}).$$

What is the probability that a randomly selected natural number is square-free?

Another prominent arithmetic function is $\sigma_s(n) = \sum_{d|n} d^s$ (see Example A.14). For $s = 0$ this specializes to the divisor function $\sigma_0(n)$ $(= d(n) =$ the number of positive divisors of $n)$. For $s = 1$ we get the divisor sum function $\sigma_1(n)$ $(=$ the sum of all positive divisors of $n)$ which appeared in Exercise 1.120 above. The next result gives the average asymptotics of σ_0 and σ_1.

Proposition 1.124 *As $x \to \infty$ we have*

(1) $\sum_{n\le x} \sigma_0(n) = x\ln(x) + (2\gamma - 1)x + O(\sqrt{x})$;
(2) $\frac{1}{x}\sum_{n\le x} \sigma_1(n) = \frac{\pi^2}{12}x + O(\ln x)$.

Proof Note that $\sigma_0 = \mathbf{1} * \mathbf{1}$. Reading the proof of Lemma 1.117 with $f = \mathbf{1}$ shows that $\sum_{n\le x} \sigma_0(n) = \sum_{d\le x}[\frac{x}{d}] = \#\{(d,m) : dm \le x\}$ is the number of lattice points under the hyperbola $uv = x$ in the first quadrant of the (u, v)-plane. We divide this region into three domains defined by projecting the point $(\sqrt{x}, \sqrt{x})$ on the hyperbola to the u-axis and the v-axis, and then we count the lattice points in each of those domains. More precisely, we let

- $A_x = \#\{(d,m) : dm \le x,\ d \le \sqrt{x}\}$;
- $B_x = \#\{(d,m) : dm \le x,\ m \le \sqrt{x}\}$;
- $C_x = \#\{(d,m) : d \le \sqrt{x},\ m \le \sqrt{x}\}$;

The C_x lattice points in the square $[0, \sqrt{x}] \times [0, \sqrt{x}]$ contribute to both A_x and B_x. Due to this overcount we have $\sum_{n\le x} \sigma_0(n) = A_x + B_x - C_x$. Clearly $C_x = [\sqrt{x}]^2 = x + O(\sqrt{x})$. On the other hand, by symmetry,

$$\begin{aligned} B_x = A_x &= \sum_{d\le\sqrt{x}} [\frac{x}{d}] \\ &= \sum_{d\le\sqrt{x}} \frac{x}{d} + O(\sqrt{x}) \\ &= x\big(\ln\sqrt{x} + \gamma + O(\frac{1}{\sqrt{x}})\big) + O(\sqrt{x}) \\ &= \frac{1}{2}x\ln(x) + \gamma x + O(\sqrt{x}). \end{aligned}$$

Altogether this gives the formula in part (1).

For part (2) we observe that $\sigma_1 = \mathrm{Id} * \mathbf{1}$. Taking $f = \mathrm{Id}$ in the proof of Lemma 1.117 gives the identity

$$\begin{aligned} \sum_{n\le x} \sigma_1(n) &= \sum_{d\le x} d[\frac{x}{d}] \\ &= \sum\nolimits_{(d,m):dm\le x} d \\ &= \sum_{m\le x} \sum_{d=1}^{[x/m]} d \\ &= \sum_{m\le x} \frac{1}{2}[\frac{x}{m}]([\frac{x}{m}] + 1) \end{aligned}$$

As in the proof of Lemma 1.121 this results in the estimate

$$|\sum_{n\le x}\sigma_1(n)-\sum_{m\le x}\frac{1}{2}\frac{x}{m}(\frac{x}{m}+1)|\le\sum_{m\le x}\frac{x}{m}.$$

The right-hand side is $x\ln(x)+\gamma x+O(1)$. Moreover, the main term is

$$\sum_{m\le x}\frac{1}{2}\frac{x}{m}(\frac{x}{m}+1)=\frac{1}{2}x^2\sum_{m\le x}\frac{1}{m^2}+\frac{1}{2}x\sum_{m\le x}\frac{1}{m}=\frac{\pi^2}{12}x^2+\frac{1}{2}x\ln(x)+O(x).$$

Combining these observations gives (2). □

Exercise 1.125 Let $\sigma_0^+(n)$ denote the number of even positive divisors $d|n$. Similarly, let $\sigma_0^-(n)$ be the number of odd positive divisors d of n.

(1) Check that $n\mapsto(-1)^n$ has Möbius transformation $n\mapsto\sigma_0^+(n)-\sigma_0^-(n)$.
(2) Prove that $\frac{1}{x}\sum_{n\le x}\left(\sigma_0^+(n)-\sigma_0^-(n)\right)=O(1)$ and deduce

$$\frac{1}{x}\sum_{n\le x}\sigma_0^{\pm}(n)=\frac{1}{2}\ln(x)+O(1).$$

(3) Sharpen these asymptotics by noting that $\sigma_0^+(n)=\begin{cases}\sigma_0(\frac{n}{2}) & \text{if } n \text{ is even,}\\ 0 & \text{if } n \text{ is odd.}\end{cases}$

– Conclude:

$$\frac{1}{x}\sum_{n\le x}\sigma_0^{\pm}(n)=\frac{1}{2}\ln x+\frac{1}{2}(2\gamma-1\mp\ln 2)+O(\frac{1}{\sqrt{x}}).$$

(4) Show $\lim_{x\to\infty}\frac{1}{x}\sum_{n\le x}\left(\sigma_0^+(n)-\sigma_0^-(n)\right)$ exists and find it.

1.4.2 Mertens' Constants and Theorems

One of the goals of this section is to give a rather precise description of the asymptotic behavior of the truncated prime-harmonic series $\sum_{p\le x}\frac{1}{p}$ (where p ranges over primes). We start with the following observations. We remind the reader that the von Mangoldt function Λ is defined by letting $\Lambda(n)=\ln(p)$ if $n=p^r>1$ is a prime-power, and $\Lambda(n)=0$ otherwise. (See Sect. A.3 for more details.)

Lemma 1.126 *As $x\to\infty$ we have*

(1) $\sum_{n\le x}\ln(n)=x\ln(x)+O(x)$;
(2) $\sum_{n\le x}\frac{\Lambda(n)}{n}=\ln(x)+O(1)$;
(3) $\sum_{p\le x}\frac{\ln(p)}{p}=\ln(x)+O(1)$.

Proof Part (1) is an immediate application of the partial summation formula (Proposition 1.11) with all $a_n = 1$ and $f(t) = \ln(t)$:

$$\sum_{n\le x} \ln(n) = [x]\ln(x) - \int_1^x \frac{[t]}{t}dt = x\ln(x) - \{x\}\ln(x) - \int_1^x dt + \int_1^x \frac{\{t\}}{t}dt = x\ln(x) + O(x).$$

Recall that $\{x\} = x - [x] \in [0, 1)$ denotes the fractional part of x.

For the proof of part (2) we replace x by a positive integer N and write the sum in part (1) as

$$\begin{aligned}\sum_{n\le N} \ln(n) = \ln(N!) = \sum_{p\le N} v_p(N!)\ln(p) = \sum_{p\le N}\sum_{r=1}^{\infty}\big[\frac{N}{p^r}\big]\ln(p) &= \sum_{n\le N}\big[\frac{N}{n}\big]\Lambda(n)\\ &= \sum_{n\le N}\frac{N}{n}\Lambda(n) + O\big(\sum_{n\le N}\Lambda(n)\big)\\ &= N\sum_{n\le N}\frac{\Lambda(n)}{n} + O(\psi(N))\end{aligned}$$

In the third step we used de Polignac's formula for the p-adic valuation of factorials (verified in the proof of Proposition A.21 for example). Finally, note that $\psi(x) = O(x)$ by Proposition A.21 (or the prime number theorem). Here $\psi(x) := \sum_{n\le x}\Lambda(n)$ is the Chebyshev function.

To show part (3) now simply note that

$$0 \le \sum_{n\le x}\frac{\Lambda(n)}{n} - \sum_{p\le x}\frac{\ln(p)}{p} = \sum_{r=2}^{\infty}\sum_{p^r\le x}\frac{\ln(p)}{p^r} < \sum_p \ln(p)\Big(\frac{1}{p^2} + \frac{1}{p^3} + \cdots\Big) = \sum_p \frac{\ln(p)}{p(p-1)} < \sum_{n=2}^{\infty}\frac{\ln(n)}{n(n-1)}$$

and check the last sum converges (by comparing it to the integral $\int_1^\infty \frac{\ln(t)}{t^2}dt$ for instance). □

Part (3) can be further improved to $\sum_{p\le x}\frac{\ln(p)}{p} = \ln(x) + B_3 + o(1)$ for a constant $B_3 = 1.33258....$

Theorem 1.127 *There is a constant $B_1 = 0.261496...$ (the "Meissel–Mertens constant") such that*

$$\sum_{p\le x}\frac{1}{p} = \ln\ln x + B_1 + O\big(\frac{1}{\ln x}\big).$$

Proof Apply the partial summation formula (1.11) to the function $f(t) = \frac{1}{\ln t}$ and the sequence $(a_n)_{n\in\mathbb{N}}$ with $a_p = \frac{\ln p}{p}$ for primes p, and $a_n = 0$ for composite n. For these data the sum-function A is

$$A(x) = \sum_{p\le x}\frac{\ln p}{p} = \ln(x) + O(1)$$

by the above Lemma 1.126. Note that $A(x) = 0$ for $x < 2$. For the purpose of this proof let $\tau(x) := A(x) - \ln(x) = O(1)$ denote the error term. The partial summation formula amounts to

$$\sum_{p \le x} \frac{1}{p} = \frac{A(x)}{\ln x} + \int_2^x \frac{A(t)}{t \ln(t)^2} dt = 1 + \frac{\tau(x)}{\ln x} + \int_2^x \frac{dt}{t \ln(t)} + \int_2^x \frac{\tau(t)}{t \ln(t)^2} dt$$

Note that $\int_2^x \frac{dt}{t \ln(t)} = \ln \ln x - \ln \ln 2$. It remains to observe that (with the substitution $u = \ln t$)

$$\int_x^\infty \frac{dt}{t \ln(t)^2} = \int_{\ln x}^\infty \frac{du}{u^2} = \frac{1}{\ln x}.$$

Consequently $\int_2^x \frac{\tau(t)}{t \ln(t)^2} dt = \int_2^\infty \frac{\tau(t)}{t \ln(t)^2} dt + O\big(\frac{1}{\ln x}\big)$, which shows the Theorem holds with

$$B_1 = 1 - \ln \ln 2 + \int_2^\infty \frac{\tau(t)}{t \ln(t)^2} dt.$$

This finishes the proof. □

Note the analogy with the Euler constant γ which satisfies $\sum_{n \le x} \frac{1}{n} = \ln x + \gamma + O(\frac{1}{x})$.

Exercise 1.128 The ω-function is defined by letting $\omega(n)$ denote the number of primes dividing n, *not* counting multiplicity. For example $\omega(60) = \omega(2^2 \cdot 3 \cdot 5) = 3$.

(1) Check that $\omega(mn) = \omega(m) + \omega(n)$ provided $\mathrm{GCD}(m, n) = 1$.
(2) Deduce that $n \mapsto (-1)^{\omega(n)}$ is a multiplicative arithmetic function. Then show that

$$(-1)^{\omega(n)} = \sum_{d|n} \mu(d) \sigma_0(d).$$

(`Hint`: The right-hand side defines a multiplicative function. Evaluate it on prime-powers.)
(3) Prove that

$$\frac{1}{x} \sum_{n \le x} \omega(n) = \ln \ln x + B_1 + O\big(\frac{1}{\ln x}\big).$$

(`Hint`: Apply Lemma 1.117 to the characteristic function on the set of primes. Its Möbius transformation is ω. Now appeal to Theorem 1.127 and the results in Sect A.3.)

In contrast, $\Omega(n)$ denotes the number of primes dividing n, *counting* multiplicity. For example $\Omega(60) = \omega(2^2 \cdot 3 \cdot 5) = 4$. One easily checks that $\Omega(mn) = \Omega(m) + \Omega(n)$ for all m and n, regardless of whether they are coprime or not. Thus $\lambda(n) := (-1)^{\Omega(n)}$ is a strongly multiplicative function, which goes by the name the Liouville λ-function.

Exercise 1.129 Compute the Möbius transformation of λ. Prove that

$$(\lambda * \mathbf{1})(n) = \begin{cases} 1 & \text{if } n \text{ is a square } \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

(Hint: Both sides define multiplicative functions. It therefore suffices to take $n = p^r$.)

The average asymptotics of Ω involves the constant $B_2 := B_1 + \sum_p \frac{1}{p(p-1)} = 1.034653....$

Proposition 1.130 $\frac{1}{x}\sum_{n\leq x} \Omega(n) = \ln\ln x + B_2 + O\left(\frac{\ln x}{x}\right)$.

Proof Apply Lemma 1.117 to the characteristic function f on the set of prime powers > 1. The number of prime powers dividing n is exactly $\Omega(n)$. (For example, p^r has r prime powers $\{p, p^2, \ldots, p^r\}$ dividing it.) In other words $f * \mathbf{1} = \Omega$. Lemma 1.117 gives us the inequality

$$\left|\frac{1}{x}\sum_{n\leq x}\Omega(n) - \sum_{p^r\leq x}\frac{1}{p^r}\right| \leq \frac{1}{x}\sum_{p^r\leq x} 1 = \frac{1}{x}\left[\frac{\ln x}{\ln p}\right].$$

We know the asymptotics of $\sum_{p\leq x}\frac{1}{p}$ from Theorem 1.127. We are left with understanding the nature of $\sum_{r=2}^{\infty}\sum_{p^r\leq x}\frac{1}{p^r}$. However, this clearly converges as $x \to \infty$:

$$\sum_{r=2}^{\infty}\sum_p \frac{1}{p^r} = \sum_p \frac{1}{p^2}\left(1 + \frac{1}{p} + \frac{1}{p^2} + \cdots\right) = \sum_p \frac{1}{p(p-1)} = B_2 - B_1.$$

This gives the result. □

There is an elegant explicit formula for the Meissel–Mertens constant, which we will not prove here:

$$B_1 = \gamma + \sum_p \left(\ln\left(1 - \frac{1}{p}\right) + \frac{1}{p}\right). \tag{1.3}$$

For a proof, which is quite technical, we refer the reader to [HW79, Theorem 428, p. 351]. It uses the fact that Euler's constant can be written as

$$\gamma = -\Gamma'(1) = -\int_0^{\infty} e^{-t}\ln(t)dt.$$

See [NZM91, Exercise 28, p. 401] for a sketch of an elementary proof of this. As a minimum, let us at least check the sum over p converges:

$$0 < -\ln\left(1 - \frac{1}{p}\right) - \frac{1}{p} = -\frac{1}{p} + \sum_{r=1}^{\infty}\frac{p^{-r}}{r} = \frac{1}{2p^2} + \frac{1}{3p^3} + \frac{1}{4p^4} + \cdots < \frac{1}{2p^2} + \frac{1}{2p^3} + \frac{1}{2p^4} + \cdots = \frac{1}{2p(p-1)}.$$

Clearly $\sum_p \frac{1}{2p(p-1)}$ converges and therefore so does $\sum_p \left(\ln\left(1-\frac{1}{p}\right) + \frac{1}{p} \right)$. The sum is a negative number, which by (1.3) turns out to be $B_1 - \gamma = -0.315719....$

Theorem 1.131 (Mertens) $\prod_{p\leq x}\left(1-\frac{1}{p}\right) \sim \frac{e^{-\gamma}}{\ln(x)}$.

Proof By taking logarithms, what we have to show is equivalent to:

$$\sum_{p\leq x} \ln\left(1-\frac{1}{p}\right) + \gamma + \ln\ln x \longrightarrow 0$$

as $x \to \infty$. In other words that

$$\sum_{p\leq x}\left(\ln\left(1-\frac{1}{p}\right) + \frac{1}{p} \right) + \gamma - \sum_{p\leq x}\frac{1}{p} + \ln\ln x \longrightarrow 0.$$

This is a direct consequence of (1.3) and Theorem 1.127. □

Without assuming (1.3) a similar argument shows $\prod_{p\leq x}\left(1-\frac{1}{p}\right) \sim \frac{C}{\ln(x)}$ for a constant $C > 0$. As an application of Theorem 1.131 we return to the fluctuation of $\frac{\phi(n)}{n}$ in the interval $[0, 1]$:

Corollary 1.132 $\limsup_{n\to\infty} \frac{\phi(n)}{n} = 1$ *and* $\liminf_{n\to\infty} \frac{\phi(n)}{n} = 0$.

Proof In fact $\sup_{m\geq n} \frac{\phi(m)}{m} = 1$ and $\inf_{m\geq n} \frac{\phi(m)}{m} = 0$, for all n. For the supremum just take $m = p$ to a prime $\geq n$ and note that $\frac{\phi(p)}{p} = 1 - \frac{1}{p}$ is arbitrarily close to 1 for large enough p. For the infimum take $m = N!$ and observe that $\frac{\phi(N!)}{N!} = \prod_{p|N!}(1-\frac{1}{p}) = \prod_{p\leq N}(1-\frac{1}{p}) \sim \frac{e^{-\gamma}}{\ln(N)}$. In particular $\frac{\phi(N!)}{N!} \to 0$ as $N \to \infty$, and we are done. □

To wrap up this section we briefly discuss the Mertens function $M(x) = \sum_{n\leq x} \mu(n)$. The trivial estimate is of course $|M(x)| \leq \sum_{n\leq x} |\mu(n)| \leq [x] \leq x$, just using that $\mu(n) \in \{0, \pm 1\}$. Mertens conjectured (in 1897) that $|M(x)| \leq \sqrt{x}$, but this was disproved in 1985 by Odlyzko and te Riele.

Exercise 1.133 Assume $M(x) = O(x^{\frac{1}{2}+\epsilon})$ for all $\epsilon > 0$.

(1) Show that $\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$ converges for $\mathrm{Re}(s) > \frac{1}{2}$, using the partial summation formula.
(2) Deduce the Riemann hypothesis. (`Hint`: Consider the poles of $\frac{1}{\zeta(s)}$ and use the functional equation.)

Conversely, the Riemann hypothesis can be shown to imply the estimate $M(x) = O(x^{\frac{1}{2}+\epsilon})$ for all $\epsilon > 0$.

1.4.3 More About Dirichlet Series

As in Sect. A.2 we let $a_n = f(n)$ be the sequence defined by an arithmetic function $f : \mathbb{N} \longrightarrow \mathbb{C}$. We encode this sequence in a Dirichlet series $\sum_{n=1}^{\infty} a_n n^{-s}$ whose sum we denote by $D(s, f)$ when it exists.

Let $A(x) = \sum_{n \leq x} a_n$. By the partial summation formula (1.11) we have the identity

$$\sum_{n \leq x} a_n n^{-s} = A(x)x^{-s} + s \int_1^x A(t)t^{-s-1}dt.$$

In particular, if $A(x) = O(x^{\alpha})$–for some $\alpha \in \mathbb{R}$–then the series $\sum_{n=1}^{\infty} a_n n^{-s}$ converges for $\mathrm{Re}(s) > \alpha$ (not necessarily absolutely). A minor modification of this argument shows the convergence is uniform in $\mathrm{Re}(s) \geq \alpha + \epsilon$ (for all $\epsilon > 0$) and therefore $D(s, f)$ is a holomorphic function in $\mathrm{Re}(s) > \alpha$.

If all $a_n \geq 0$ there is a partial converse to this observation. Suppose $\sum_{n=1}^{\infty} a_n n^{-\alpha}$ is convergent and $\alpha \geq 0$. Then the estimates

$$A(x) = \sum_{n \leq x} a_n \leq \sum_{n \leq x} a_n (\frac{x}{n})^{\alpha} \leq \big(\sum_{n=1}^{\infty} a_n n^{-\alpha}\big) x^{\alpha}$$

show that $A(x) = O(x^{\alpha})$.

In complex analysis a holomorphic function determines its Taylor coefficients. For instance, suppose $\sum_{n=0}^{\infty} c_n z^n = 0$ for all z in a disk at the origin, then $c_n = 0$ for all n. This is sometimes referred to as the identity theorem. An analogous result holds for Dirichlet series.

Proposition 1.134 *Suppose $\sum_{n=1}^{\infty} a_n n^{-s}$ is absolutely convergent and $D(s, f) = 0$ for $Re(s)$ sufficiently large. Then $a_n = 0$ for all $n \in \mathbb{N}$. In other words, if f and g are arithmetic functions, then*

$$D(s, f) = D(s, g) \text{ for } Re(s) \gg 0 \Longrightarrow f = g.$$

Proof Let $N \geq 1$ and assume $a_n = 0$ for all $n < N$. For $N = 1$ this is vacuous. We will deduce that $a_N = 0$. Pick a real number $\alpha \geq 0$ such that $\sum_{n=1}^{\infty} |a_n| n^{-\alpha} < \infty$. We consider the sequence of functions $f_n(x) := a_n (\frac{N}{n})^x$ on $\mathbb{R}$. Of course $f_n(x) = 0$ unless $n \geq N$ by hypothesis. We observe that

- $|f_n(x)| \leq |a_n| (\frac{N}{n})^{\alpha}$ for all $x \in [\alpha, \infty)$;
- $\lim_{x \to \infty} f_n(x) = \begin{cases} a_N & \text{if } n = N \\ 0 & \text{if } n \neq N \end{cases}$.

In particular all the functions f_n are uniformly bounded by the constant $M := \sum_{n=1}^{\infty} |a_n|(\frac{N}{n})^{\alpha}$ on $[\alpha, \infty)$. The Weierstrass M-test from real analysis shows that, as a result of these properties, we have

$$\sum_{n=1}^{\infty} \lim_{x\to\infty} f_n(x) = \lim_{x\to\infty} \sum_{n=1}^{\infty} f_n(x).$$

This amounts to $a_N = \lim_{x\to\infty} \sum_{n=1}^{\infty} a_n(\frac{N}{n})^x = \lim_{x\to\infty} N^x D(x, f) = 0$. We are done by induction. □

As an application $\frac{1}{\zeta'(s)}$ is *not* given by a Dirichlet series in any right half-plane (in contrast to $\frac{1}{\zeta(s)}$ and $\frac{\zeta'(s)}{\zeta(s)}$ for example). To see this suppose $\frac{1}{\zeta'(s)} = D(s, g)$ for some arithmetic function g and $\mathrm{Re}(s)$ sufficiently large. By term-wise differentiation we have $\zeta'(s) = D(s, f)$ where $f(n) = -\ln(n)$. Thus

$$D(s, f * g) = D(s, f) \cdot D(s, g) = 1 = D(s, 1_{\mathcal{A}}).$$

See A.2 for more details on the convolution of arithmetic functions. Therefore, by Proposition 1.134 we must have $f * g = 1_{\mathcal{A}}$. For $n = 1$ this yields the contradiction $1 = 1_{\mathcal{A}}(1) = (f * g)(1) = f(1)g(1) = 0$.

Here is another important basic fact from complex analysis: Suppose h is a holomorphic function on an open subset $\Omega \subset \mathbb{C}$. Pick a point $z_0 \in \Omega$. Then the Taylor series $\sum_{n=0}^{\infty} \frac{h^{(n)}(z_0)}{n!}(z - z_0)^n$ converges to $h(z)$ for all z in the largest open disc $B(z_0, r) \subset \Omega$. By convention we allow $r = \infty$ if $\Omega = \mathbb{C}$. See (the proof of) [Rud87, Theorem 10.16] for instance. In other words, if a power series $\sum_{n=0}^{\infty} c_n z^n$ has radius of convergence ϱ and its sum extends to a meromorphic function on a slightly larger disc $B(0, \varrho + \epsilon)$, then there must be a pole on the circle $\partial B(0, \varrho)$.

This translates into an analogous result for Dirichlet series with non-negative coefficients. This central observation goes by the name Landau's method, or Landau's Lemma:

Lemma 1.135 (Landau) *Assume $a_n \geq 0$ for all n. Suppose $\sum_{n=1}^{\infty} a_n n^{-s}$ is absolutely convergent for $Re(s) > t$, uniformly in compact subsets. If the sum has analytic continuation to an open neighborhood of t, then there is a $t' < t$ such that $\sum_{n=1}^{\infty} a_n n^{-s}$ converges absolutely for $Re(s) > t'$.*

Proof By replacing a_n with $a_n n^{-t}$ we may assume $t = 0$. Let $D(s) := \sum_{n=1}^{\infty} a_n n^{-s}$ for $\mathrm{Re}(s) > 0$, and suppose $D(s)$ extends to a holomorphic function in the open disc $B(0, R)$. Once and for all we select an $r < \frac{R}{2}$. For $\sigma > 0$ we have

$$\begin{aligned} D(\sigma) &= \sum_{n=1}^{\infty} a_n n^{-r} n^{r-\sigma} \\ &= \sum_{n=1}^{\infty} a_n n^{-r} e^{(r-\sigma)\ln(n)} \\ &= \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} a_n n^{-r} \frac{1}{m!} (r-\sigma)^m \ln(n)^m \\ &\overset{(ac)}{=} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} a_n n^{-r} \frac{1}{m!} (r-\sigma)^m \ln(n)^m \\ &= \sum_{m=0}^{\infty} \left(\sum_{n=1}^{\infty} a_n n^{-r} \ln(n)^m \right) \frac{1}{m!} (r-\sigma)^m \\ &= \sum_{m=0}^{\infty} c_m(r) \frac{1}{m!} (r-\sigma)^m, \end{aligned}$$

where we have introduced $c_m(r) := \sum_{n=1}^{\infty} a_n n^{-r} \ln(n)^m$. In the fourth step (ac) we used the assumption that $a_n \geq 0$ in order to interchange the order of summation. The power series $\sum_{m=0}^{\infty} c_m(r) \frac{1}{m!} (r-s)^m$ represents $D(s)$ for $s > 0$, and in particular in the disc $|s-r| < r$. However, since $D(s)$ is holomorphic in $B(0, R)$, the power series represents $D(s)$ in a slightly larger disc $|s-r| < r + \epsilon$. (We encourage the reader the draw a picture of the inscribed circles.) Revisiting the above calculation we deduce that $D(\sigma) < \infty$ for $\sigma > -\epsilon$. We conclude that $D(s) = \sum_{n=1}^{\infty} a_n n^{-s}$ converges absolutely for $\mathrm{Re}(s) > -\epsilon$. □

In the setup of Lemma 1.135 and its proof, let

$$t^\sharp := \inf\{t \in \mathbb{R} : \sum_{n=1}^{\infty} a_n n^{-t} < \infty\}$$

be the abscissa of convergence for the given Dirichlet series. If $t^\sharp \neq -\infty$, and $D(s)$ extends to a meromorphic function in an open neighborhood of $t^\sharp$, then Lemma 1.135 asserts that $t^\sharp$ is necessarily a pole of $D(s)$. Loosely speaking, for a Dirichlet series with non-negative coefficients there is a pole where the real axis meets the boundary of the right half-plane of convergence.

Chapter 2
Tate's Thesis



Hecke characters are analogues of Dirichlet characters for number fields. Letting the field K and the character χ vary gives a family of L-functions $L(s, \chi)$ subsuming both Dirichlet L-functions ($K = \mathbb{Q}$) and Dedekind zeta functions ($\chi = 1$). In his 1950 Princeton PhD thesis [Tat50], completed under Emil Artin, John Tate found an extremely elegant way of deriving the analytic properties of all the $L(s, \chi)$ collectively. Tate's point of view, which ultimately relies on harmonic analysis on local fields, dramatically simplifies Hecke's original computational heavy method. On top of that it gives a beautiful uniform explanation for the shape of the Euler factors, such as $\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$ and $\frac{1}{1-p^{-s}}$, at *all* places (archimedean as well as non-archimedean)–in perfect accordance with the Hasse principle that all completions of K should be treated on an equal footing. Another important point is that Tate's method is amenable to generalization: To some extent it gives a roadmap to defining an even more general collection of L-functions, where χ is replaced by a so-called automorphic representation of the general linear group GL_n. Tate's thesis deals with the GL_1 case. This collection of L-functions for GL_n (and other algebraic groups such as symplectic groups and orthogonal groups) is of momentous importance in the Langlands program–a sweeping area of modern-day number theory.

In Sect. 2.1 we set the stage by first introducing adeles and ideles. The terminology here was introduced by Chevalley: Supposedly *idele* (in French idèle) is short for ideal element and *adele* stands for additive idele. The adelic situation is reminiscent of the the circle group, to the degree that $\mathbb{Z}$ is a discrete subgroup of $\mathbb{R}$ with a compact quotient $\mathbb{R}/\mathbb{Z}$. Correspondingly, K sits as a discrete subgroup of the ring of adeles $\mathbb{A}_K$ and the quotient $\mathbb{A}_K/K$ is compact. This is essentially a reformulation of Minkowski's observation that $\mathcal{O}_K$ is a full lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$. Also, into the bargain, $\mathbb{A}_K$ is locally compact. This essential attribute is what makes it possible to integrate function on $\mathbb{A}_K$ and latch onto abstract Fourier analysis. The group of ideles $\mathbb{I}_K$ contains $K^\times$ as a discrete subgroup but the quotient is no longer compact–only nearly (it is isomorphic to $\mathbb{R}$ times a compact group). These properties are tied up with the techniques going into the proof of Dirichlet's unit theorem on the finite generation of $\mathcal{O}_K^\times$ (such as the notion of regulators, etc.). Thus several of the major

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milestones in basic algebraic number theory can be recast elegantly in the language of adeles and ideles.

The quotient $C_K = \mathbb{I}_K / K^\times$ is called the idele class group, and it is a central player in Tate's thesis as well as class field theory. Unlike the *ideal* class group Cl_K, the idele class group is infinite (even non-compact). However, they are intimately linked via a natural surjection $C_K \twoheadrightarrow \mathrm{Cl}_K$. Moreover, various generalizations of Cl_K (such as ray class groups, the narrow class group, etc.) have simple descriptions as quotients of C_K.

In Sect. 2.2 we discuss Hecke characters, which are no more than continuous group homomorphisms $\chi : C_K \to \mathbb{C}^\times$. In other words, we are considering characters χ of $\mathbb{I}_K$ which are trivial on $K^\times$. (This last condition is what makes χ an 'automorphic' representation of GL_1.) Because $\mathbb{I}_K$ is a restricted direct product of the multiplicative groups $K_v^\times$, with v ranging over all places of K, any character χ of $\mathbb{I}_K$ factors as a restricted tensor product of local components $\chi = \otimes_v \chi_v$. Here χ_v is a character of $K_v^\times$ which is *unramified* for all but at most finitely many v, in the sense that χ_v is trivial on the units $\mathcal{O}_v^\times$. When χ_v is unramified it is completely determined by a single complex number $\chi_v(\varpi_v)$, where ϖ_v can be any uniformizer in K_v. We encode this parameter in an Euler factor $\frac{1}{1-\chi_v(\varpi_v)q_v^{-s}}$ with q_v denoting the size of the residue field at v. On the other hand, when χ_v is ramified we declare the Euler factor at v to be 1. The product of all these Euler factors *is* the Hecke L-function $L(s, \chi)$. Initially it is defined in some right half-plane $\mathrm{Re}(s) \gg 0$ and the big accomplishment of Hecke and Tate, in this setting, was to extend it to a meromorphic function on all of $\mathbb{C}$ (and in the process establish a functional equation).

As in the case of a primitive Dirichlet character, the functional equation for $L(s, \chi)$ involves a factor $\epsilon(s, \chi)$ formed out of the discriminant of K, the conductor of χ, and the root number $\epsilon(\frac{1}{2}, \chi)$. One thing that is not transparent in the more classical theory à la Hecke, is the fact that this ϵ-factor decomposes as a finite product $\epsilon(s, \chi) = \prod_v \epsilon(s, \chi_v, \psi_v)$ where the ψ_v are auxiliary additive characters, and v runs over all places of K. Generally speaking the local factors $\epsilon(s, \chi_v, \psi_v)$ are 'interesting' at the ramified places and 'trivial' (or easily computable) at the unramified places–which is just the opposite for the L-function. To orient the reader, the factors $\epsilon(s, \chi_v, \psi_v)$ are local variants of Gauss sums when χ_v is ramified. Their computation takes up a large part of Sect. 2.4.

Section 2.3 summarizes the prerequisites from harmonic analysis on a locally compact abelian group G, such as the dual group $\widehat{G}$, Pontryagin duality, Haar measures, Fourier transforms, and the Poisson summation formula for a discrete and cocompact subgroup $\Gamma \subset G$. Readers who are already familiar with this material can safely skip this section and refer back to it as needed. In the local theory G will be either the additive group K_v or the multiplicative group $K_v^\times$. When v is non-archimedean these groups are locally profinite, meaning they admit an open subgroup which is profinite in the induced topology. For example, K_v contains $\mathcal{O}_v = \varprojlim_n \mathcal{O}_v/\mathfrak{p}_v^n$. For a locally *profinite* abelian group one can construct a Haar measure with bare hands by reducing to the case of a finite group. The bottom line is the measures we will eventually consider are not at all sophisticated.

Section 2.4 in some sense marks the beginning of our discussion of Tate's thesis. In this section we zero in on a place v and work exclusively over the local field K_v. The local theory revolves around certain *zeta* integrals $Z(f_v, \chi_v)$, where f_v is a nice (Schwartz) function on K_v and χ_v is a character of $K_v^\times$. The nomenclature stems from the fact that, for a natural choice of f_v, the zeta integral gives the Euler factor of χ_v. This is the 'uniform explanation' alluded to earlier in the first paragraph. The complex variable s comes in by varying χ_v in a family, i.e. by introducing $\chi_{v,s} = \chi_v |\cdot|_v^s$. In this fashion $Z(f_v, \chi_{v,s})$ defines a function for $\mathrm{Re}(s) \gg 0$. Tate's main local theorem first of all gives the meromorphic continuation of $Z(f_v, \chi_{v,s})$ to the whole s-plane. Secondly, it shows that the ratio $\frac{Z(f_v, \chi_{v,s})}{L(s, \chi_v)}$ is an entire function satisfying a functional equation relating it to $\frac{Z(\widehat{f_v}, \chi_{v,s}^\vee)}{L(1-s, \overline{\chi}_v)}$ with the epsilon factor $\epsilon(s, \chi_v, \psi_v)$ appearing. (See Theorem 2.92 for the precise statement.) The observation that $\epsilon(s, \chi_v, \psi_v)$ is always nonzero, as a result of their explicit calculation, is an essential ingredient in the proof of the local theorem.

In Sect. 2.5 we prove Tate's main global theorem (Theorem 2.101) which demonstrates the analytic features of the global zeta integrals $Z(f, \chi_s)$, where f is now a function on $\mathbb{A}_K$ and χ is a Hecke character (and $\chi_s = \chi|\cdot|^s$ with $|\cdot|$ denoting the idele norm). As in the local story, these zeta integrals have meromorphic continuation to $\mathbb{C}$, and they are entire functions unless χ is essentially the trivial character. When $\chi = 1$ the residues at $s = 0$ and $s = 1$ are basically given by $\mp$ the (suitably normalized) volume of the norm-one subgroup C_K^1 of the idele class group. The proof adheres to the general scheme from Chap. 1 (finding an integral representation as a Mellin transform, using Poisson summation for $K \subset \mathbb{A}_K$ to find a transformation property under $t \mapsto \frac{1}{t}$, and then splitting the integral as $\int_0^1$ plus $\int_1^\infty$.)

The global zeta integral $Z(f, \chi_s)$ is very closely related to the Hecke L-function. In fact, for any f there is a finite set of places S such that $Z(f, \chi_s)$ factors as the entire function $\prod_{v \in S} \frac{Z(f_v, \chi_{v,s})}{L(s, \chi_v)}$ from the local theory times (the automorphic version of) $L(s, \chi)$. With this in the bag it is now relatively straightforward to deduce the desired analytic properties of Hecke L-functions (see Theorem 2.103).

When $\chi = 1$ we deduce the analytic properties of the Dedekind zeta function $\zeta_K(s)$ as a special case. The analytic class number formula follows without much trouble by evaluating the volume of C_K^1.

2.1 Adeles and Ideles

2.1.1 Preliminary Motivating Remarks

We assemble our zeta and L-functions in a flowchart:

$$\begin{array}{ccc} \text{Riemann zeta function} & \longrightarrow & \text{Dirichlet } L\text{-functions} \\ \downarrow & & \downarrow \\ \text{Dedekind zeta function} & \longrightarrow & \text{Hecke } L\text{-functions.} \end{array}$$

In the top row the field is $K = \mathbb{Q}$ and in the first column $\chi = 1$. Tate's thesis gives a method to show analytic continuation for all these L-functions simultaneously, and the method is amenable to generalization to other groups such as GL_n. Tate's thesis for GL_1 deals with the Hecke L-functions in the diagram (which we have not defined yet).

We raise our ambitions and allow a general number field $K/\mathbb{Q}$ and a general character χ simultaneously. It is not obvious what should play the role of Dirichlet characters for $K \neq \mathbb{Q}$. If we emulate the definitions of $\zeta_K(s)$ and Dirichlet L-functions, $L(s, \chi)$ should be something like $\sum_{I \neq 0} \chi(I) N(I)^{-s}$ where I runs over nonzero ideals of $\mathcal{O}_K$. This suggests χ ought to be a character of the group of nonzero fractional ideals $\mathrm{Id}_K = \bigoplus_{\mathfrak{p}} \mathbb{Z}\mathfrak{p}$. This seems promising. For any such unitary character $\chi : \mathrm{Id}_K \to \mathbb{C}^\times$ we have an Euler product for $\mathrm{Re}(s) > 1$,

$$L(s, \chi) = \sum_{I \neq 0} \chi(I) N(I)^{-s} = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p}) N(\mathfrak{p})^{-s}}.$$

However, since Id_K is free on the set of maximal ideals we can assign a random value $\chi(\mathfrak{p}) \in \mathbb{C}^\times$ to any $\mathfrak{p}$. This is way too untamed and $L(s, \chi)$ does not have good analytic properties in this generality. The solution to the problem is to impose a "global" condition on χ which guarantees $L(s, \chi)$ has analytic continuation. Perhaps the first thing that comes to mind is to only consider χ which factor through the ideal class group Cl_K; in other words such that $\chi((a)) = 1$ for all $a \in K^\times$. This actually works! The bad news is this is far too restrictive. When $K = \mathbb{Q}$ you would only get the trivial character, and for a general $K \neq \mathbb{Q}$ most interesting characters happen to have infinite order.

The way out of this conundrum is to beef up Cl_K and consider an infinite variant $C_K \twoheadrightarrow \mathrm{Cl}_K$ called the idele class group. It has the shape $C_K = \mathbb{I}_K / K^\times$ where $\mathbb{I}_K \twoheadrightarrow \mathrm{Id}_K$ is the so-called idele group (which we are about to define) containing $K^\times$ as a discrete subgroup. Although C_K is not finite, it has a natural topology and is "almost" compact.

Once we have the precise definitions in place, a Hecke character will be a character $\chi : C_K \to \mathbb{C}^\times$ and there is a natural way to attach a Hecke L-function $L(s, \chi) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p}) N(\mathfrak{p})^{-s}}$ to it. One of our supreme goals is to analytically continue $L(s, \chi)$ to all of $\mathbb{C}$ and establish its functional equation. The latter involves the ϵ-factor $\epsilon(s, \chi)$ which encodes $|D_K|$ and the conductor of χ, along with the root number $\epsilon(\frac{1}{2}, \chi)$. This will put all our classical examples under one roof.

2.1.2 The Ring of Adeles

A place of K is an equivalence class of non-trivial absolute values. Ostrowski's Theorem gives a complete description of the places of K. Let V denote the set of places, and decompose it as $V = V_\infty \cup V_f$ where V_∞ is the archimedean part

("infinite places") and V_f is the non-archimedean part ("finite places"). Places in V_∞ arise from embeddings $K \hookrightarrow \mathbb{C}$ whereas places in V_f arise from maximal ideals $\mathfrak{p} \subset \mathcal{O}_K$. The set of infinite places further decomposes as $V_\infty = V_\mathbb{R} \cup V_\mathbb{C}$ where the "real" places in $V_\mathbb{R}$ correspond to the r_1 embeddings $\sigma : K \hookrightarrow \mathbb{R}$ and the "complex" places in $V_\mathbb{C}$ correspond to the r_2 pairs of complex conjugate non-real embeddings $\tau \neq \bar{\tau} : K \hookrightarrow \mathbb{C}$.

Each place $v \in V$ has a canonical representative called the normalized absolute value $|\cdot|_v$ (although this terminology should be taken with a grain of salt when $v \in V_\mathbb{C}$):

- If $v \in V_\mathbb{R}$ corresponds to $\sigma : K \hookrightarrow \mathbb{R}$ we let $|x|_v = |\sigma(x)|_\mathbb{R}$;
- If $v \in V_\mathbb{C}$ corresponds to the pair $\tau \neq \bar{\tau} : K \hookrightarrow \mathbb{C}$ we let $|x|_v = |\tau(x)|^2_\mathbb{C}$;
- If $v \in V_f$ corresponds to $\mathfrak{p} \subset \mathcal{O}_K$ we let $|x|_v = N(\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}(x)}$.

Here $\mathrm{ord}_\mathfrak{p}(x)$ denotes the exponent of $\mathfrak{p}$ in the factorization of $x\mathcal{O}_K$ into $\mathbb{Z}$-powers of maximal ideals. Also, $|\cdot|_\mathbb{R}$ and $|\cdot|_\mathbb{C}$ denote the usual absolute values on $\mathbb{R}$ and $\mathbb{C}$ respectively; $|z|_\mathbb{C} = \sqrt{z\bar{z}}$ and $|t|_\mathbb{R} = \mathrm{sign}(t)t$ to be precise.

Remark 2.1 Because of the square $|x|_v = |\tau(x)|^2_\mathbb{C}$ is technically *not* an absolute value (it fails to satisfy the triangle inequality). This will not cause any problems, but it is of course something to be cautious about.

With these normalizations we have:

Proposition 2.2 (The Artin product formula) $\prod_{v \in V} |x|_v = 1$ *for all* $x \in K^\times$.

Proof For $K = \mathbb{Q}$ this is easy to check for $x = \ell$ being a prime number and for $x = \pm 1$. Since the set $\{\pm\ell : \ell \text{ prime}\}$ generates the multiplicative group $\mathbb{Q}^\times$ the result follows for the rationals. The general case follows by observing that

$$\prod_{v \in V} |x|_v = \prod_{p \in V_\mathbb{Q}} \prod_{v|p} |x|_v = \prod_{p \in V_\mathbb{Q}} |N_{K/\mathbb{Q}}(x)|_p = 1.$$

This reduction step uses our normalizations; for example $|N_{\mathbb{C}/\mathbb{R}}(z)|_\mathbb{R} = |z|^2_\mathbb{C}$ at the complex places. □

At each place $v \in V$ we choose a completion $K \hookrightarrow K_v$ relative to $|\cdot|_v$. Thus $K_v \simeq \mathbb{R}$ when $v \in V_\mathbb{R}$ and $K_v \simeq \mathbb{C}$ for $v \in V_\mathbb{C}$. For $v \in V_f$ the completion is a finite extension $K_v/\mathbb{Q}_p$ for some p and we adopt the following notation throughout:

- $\mathcal{O}_v = \{x \in K_v : |x|_v \leq 1\}$ is the discrete valuation ring;
- $\mathfrak{p}_v = \{x \in K_v : |x|_v < 1\}$ is its maximal ideal;
- $k_v = \mathcal{O}_v/\mathfrak{p}_v \simeq \mathbb{F}_{q_v}$ is its residue field;
- ϖ_v is a choice of generator for $\mathfrak{p}_v$ ("uniformizer").

One word of caution: The normalized absolute value $|\cdot|_v$ on K_v is *not* the one extending $|\cdot|_p$ on $\mathbb{Q}_p$ (unless of course $K_v = \mathbb{Q}_p$). Indeed $|p|_v = p^{-[K_v:\mathbb{Q}_p]}$ whereas

$|p|_p = p^{-1}$. Later when we define measures on K_v the normalization of $|\cdot|_v$ will appear naturally.

Recall that K_v is locally compact. (A topological space is locally compact if every point has an open neighborhood contained in a compact subset.) More precisely $\mathcal{O}_v$ is compact and open for $v \in V_f$.

The Hasse principle says the following. Suppose $Q(x_1, \ldots, x_N)$ is a quadratic form over K. Then the equation $Q(x_1, \ldots, x_N) = 0$ has a solution over K if and only if it has a solution over every completion K_v. This suggests all completions of K should be treated on an equal footing in global questions. This leads us to look for a K-algebra built out of all the completions K_v. A first attempt at this would be to just take the direct product $\prod_{v\in V} K_v$. However, this is *not* locally compact in the product topology so doing harmonic analysis on $\prod_{v\in V} K_v$ is hopeless.

Exercise 2.3 Explain why $\prod_{v\in V} K_v$ is not locally compact in the product topology. (`Hint`: A basis of open sets consists of boxes $\prod_{v\in S} U_v \times \prod_{v\notin S} K_v$ where $S \subset V$ is a finite set parametrizing open sets $U_v \subset K_v$. Project onto K_v for some $v \notin S$.)

To fix this we consider instead a certain subring of $\prod_{v\in V} K_v$ known as the restricted direct product.

Definition 2.4 The ring of adeles $\mathbb{A}_K$ is the set of tuples $(x_v)_{v\in V}$ of elements $x_v \in K_v$ such that $x_v \in \mathcal{O}_v$ for all but at most finitely many $v \in V_f$. In other words the restricted direct product

$$\mathbb{A}_K = \prod_{v\in V}{}' K_v = \{x = (x_v)_{v\in V} \in \prod_{v\in V} K_v : x_v \in \mathcal{O}_v \text{ for all but finitely many } v\}.$$

The adeles $\mathbb{A}_K$ form a K-algebra under componentwise addition and multiplication, and via the diagonal embedding $K \hookrightarrow \mathbb{A}_K$ sending $x \mapsto (x, x, x, \ldots)$. Note that $\mathrm{ord}_{\mathfrak{p}}(x) = 0$ for all but finitely many $\mathfrak{p}$ so this map indeed lands in $\mathbb{A}_K$.

Example 2.5 For $K = \mathbb{Q}$ let $\mathbb{A} = \mathbb{A}_{\mathbb{Q}}$. Elements of $\mathbb{A} \subset \mathbb{R} \times \mathbb{Q}_2 \times \cdots \times \mathbb{Q}_p \times \cdots$ are infinite tuples

$$(x_\infty, x_2, \ldots, x_p, \ldots)$$

where $x_\infty \in \mathbb{R}$ and $x_p \in \mathbb{Q}_p$ for all p, under the constraint that $x_p \in \mathbb{Z}_p$ for all large enough primes.

Exercise 2.6 Show that $\mathbb{A}_K \simeq K \otimes_{\mathbb{Q}} \mathbb{A}$. (`Hint`: Use the isomorphism $K \otimes_{\mathbb{Q}} \mathbb{Q}_p \simeq \prod_{v|p} K_v$.)

Next we promote $\mathbb{A}_K$ to a topological ring. By a box in $\mathbb{A}_K$ we mean a subset of the following form. Let $S \subset V$ be a finite set of places such that $S \supset V_\infty$. Let $\{U_v\}_{v\in S}$ be a collection of open sets $U_v \subset K_v$. The box defined by this data is

$$U = \prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v.$$

A finite intersection of boxes is a box. The open sets of $\mathbb{A}_K$ are declared to be arbitrary unions of boxes, and this turns $\mathbb{A}_K$ into a topological K-algebra (addition and multiplication are continuous maps $\mathbb{A}_K \times \mathbb{A}_K \longrightarrow \mathbb{A}_K$).

Exercise 2.7 Explain why the inclusion $\mathbb{A}_K \hookrightarrow \prod_{v\in V} K_v$ is continuous but not a topological embedding. In other words the topology on $\mathbb{A}_K$ is stronger than the induced topology.

Exercise 2.8 Check $\mathbb{A}_K$ is locally compact (`Hint`: Use the compactness of $\mathcal{O}_v$ and appeal to Tychonoff's Theorem.)

We introduce a couple of important subsets of $\mathbb{A}_K$.

- $K_\infty = K \otimes_{\mathbb{Q}} \mathbb{R} \simeq \prod_{v\in V_\infty} K_v \simeq \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ ("Minkowski space");
- $\widehat{\mathcal{O}}_K = \mathcal{O}_K \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}} \simeq \prod_{v\in V_f} \mathcal{O}_v$ where $\widehat{\mathbb{Z}} = \varprojlim_N \mathbb{Z}/N\mathbb{Z} \simeq \prod_p \mathbb{Z}_p$.

Note that $K_\infty \times \widehat{\mathcal{O}}_K$ is the box consisting of $(x_v)_{v\in V}$ where $x_v \in \mathcal{O}_v$ for all finite places v.

Lemma 2.9 (The approximation lemma) *In the above notation we have the identities:*

(1) $\mathbb{A}_K = K + (K_\infty \times \widehat{\mathcal{O}}_K)$;
(2) $\mathcal{O}_K = K \cap (K_\infty \times \widehat{\mathcal{O}}_K)$.

Proof We start with the second half. An element $x \in K$ (viewed in $\mathbb{A}_K$) lies in $K_\infty \times \widehat{\mathcal{O}}_K$ precisely when $\mathrm{ord}_{\mathfrak{p}}(x) \geq 0$ for all $\mathfrak{p} \subset \mathcal{O}_K$. Equivalently $x\mathcal{O}_K = \prod_{\mathfrak{p}} \mathfrak{p}^{\mathrm{ord}_{\mathfrak{p}}(x)} \subset \mathcal{O}_K$, that is $x \in \mathcal{O}_K$.

In the first half we need to approximate a random $x \in \mathbb{A}_K$ by an element from K. Pick an $m \in \mathbb{N}$ such that $mx \in K_\infty \times \widehat{\mathcal{O}}_K$ (clear denominators at the finitely many v where $x_v \notin \mathcal{O}_v$). Let Σ denote the finite set of $v \in V_f$ for which $|m|_v < 1$. For each $v \in \Sigma$ pick a $y_v \in \mathcal{O}_K$ such that $|y_v - mx_v| \leq |m|_v$. By the Chinese remainder theorem for $\mathcal{O}_K$ there exists a $y \in \mathcal{O}_K$ such that

$$|y - y_v|_v \leq |m|_v$$

for all $v \in \Sigma$. By the strong triangle inequality $|y - mx_v|_v \leq |m|_v$ for all $v \in \Sigma$. Consequently $\frac{y}{m} \in K$ and $|\frac{y}{m} - x_v|_v \leq 1$ for all $v \in \Sigma$. Finally, for finite $v \notin \Sigma$ we have $|m|_v = 1$ and therefore in this case we still have $|\frac{y}{m} - x_v|_v = |y - mx_v|_v \leq 1$ since $mx_v \in \mathcal{O}_v$ even for $v \notin \Sigma$. In conclusion $\frac{y}{m} - x \in K_\infty \times \widehat{\mathcal{O}}_K$. □

Our main application will be the following adelic formulation of Minkowski's geometry of numbers. Consider the diagonal embedding into Minkowski space $\mathcal{O}_K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$. Basic algebraic number theory ([Neu99, Chap. 1, Sect. 5]) tells us the image of $\mathcal{O}_K$ is a lattice and $(\mathbb{R}^{r_1} \times \mathbb{C}^{r_2})/\mathcal{O}_K$ is compact with volume $2^{-r_2}\sqrt{|D_K|}$ relative to the quotient of the Lebesgue measure. We will get back to measures in more detail later in Sect. 2.3.2.

Theorem 2.10 *The image of $K \hookrightarrow \mathbb{A}_K$ is discrete and $\mathbb{A}_K/K$ is compact in the quotient topology.*

Proof For the discreteness first pick an open neighborhood $0 \in \Omega \subset K_\infty$ such that $\Omega \cap \mathcal{O}_K = \{0\}$. This is possible because the lattice $\mathcal{O}_K \hookrightarrow K_\infty$ is discrete. Then $\Omega \times \widehat{\mathcal{O}}_K$ is open in $\mathbb{A}_K$ and

$$K \cap (\Omega \times \widehat{\mathcal{O}}_K) = \mathcal{O}_K \cap \Omega = \{0\}$$

by part (2) of Lemma 2.9. That means 0 is the only point of K in a small open neighborhood of 0. By translation to other points of K this shows K is a discrete subring of $\mathbb{A}_K$.

For the compactness choose a closed parallelotope $F \subset K_\infty$ such that $K_\infty = F + \mathcal{O}_K$. Consider the composition

$$F \times \widehat{\mathcal{O}}_K \xrightarrow{i} (K_\infty \times \widehat{\mathcal{O}}_K)/\mathcal{O}_K \xrightarrow{\text{pr}} \mathbb{A}_K/K.$$

The map labelled pr is surjective by part (1) of Lemma 2.9. The map labelled i is surjective since $K_\infty = F + \mathcal{O}_K$. Since both maps are continuous and $F \times \widehat{\mathcal{O}}_K$ is compact (by Tychonoff and Heine-Borel) so is $\mathbb{A}_K/K$. □

2.1.3 The Group of Ideles

The group of ideles is just the units $\mathbb{A}_K^\times$ as an abstract group, but the topology on the ideles is not the topology induced from $\mathbb{A}_K$–it is stronger.

Definition 2.11 The group of ideles $\mathbb{I}_K$ is the set of tuples $(x_v)_{v \in V}$ of elements $x_v \in K_v^\times$ such that $x_v \in \mathcal{O}_v^\times$ for all but at most finitely many $v \in V_f$. In other words the restricted direct product

$$\mathbb{I}_K = \prod_{v \in V}{}' K_v^\times = \{x = (x_v)_{v \in V} \in \prod_{v \in V} K_v^\times : x_v \in \mathcal{O}_v^\times \text{ for all but finitely many } v\}.$$

The topology of $\mathbb{I}_K$ again has a basis consisting of boxes $U = \prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v^\times$ where $U_v \subset K_v^\times$ are open sets indexed by a finite set of places $S \supset V_\infty$. As before $\mathbb{I}_K$ is locally compact.

Exercise 2.12 Explain why the inclusion $\mathbb{I}_K \hookrightarrow \mathbb{A}_K$ is continuous but not a topological embedding. In other words the topology on $\mathbb{I}_K$ is stronger than the induced topology. (`Hint`: $K_\infty^\times \times \widehat{\mathcal{O}}_K^\times$ is idelic open but not open in the induced topology. Why not?)

Exercise 2.13 Verify that $\mathbb{I}_K$ is a dense subset of $\mathbb{A}_K$ and therefore neither a closed nor open subset.

Exercise 2.14 Prove that the map $\mathbb{I}_K \hookrightarrow \mathbb{A}_K \times \mathbb{A}_K$ given by $x \mapsto (x, x^{-1})$ is a topological embedding, meaning the product topology on $\mathbb{A}_K \times \mathbb{A}_K$ induces the idelic topology on $\mathbb{I}_K$.

Let us move on to ideles versus ideals. The principal ideal map $K^\times \to \mathrm{Id}_K$ sending $x \mapsto (x) = x\mathcal{O}_K$ extends naturally to the idele group:

$$\mathbb{I}_K \longrightarrow \mathrm{Id}_K, \qquad x \mapsto (x) = \prod_{v \in V_f} \mathfrak{p}_v^{\mathrm{ord}_v(x_v)}.$$

(Here we adopt the hopefully obvious notation, confusing $\mathfrak{p}_v$ with a prime ideal of $\mathcal{O}_K$. Furthermore $x_v \sim \varpi_v^{\mathrm{ord}_v(x_v)}$.) Note that the product $\prod_{v \in V_f}$ is finite since $\mathrm{ord}_v(x_v) = 0$ for all but finitely many v. It is fairly immediate that the map $\mathbb{I}_K \to \mathrm{Id}_K$ is surjective with kernel $K_\infty^\times \times \widehat{\mathcal{O}}_K^\times$.

Exercise 2.15 Show that the diagonal embedding $K^\times \hookrightarrow \mathbb{I}_K$ identifies $K^\times$ with a discrete subgroup of $\mathbb{I}_K$. (`Hint`: Choose an open neighborhood $1 \in \Omega \subset \mathbb{A}_K$ such that $\Omega \cap K = \{1\}$ and consider $\Omega \cap \mathbb{I}_K$.)

Definition 2.16 The idele class group of K is the quotient $C_K = \mathbb{I}_K / K^\times$ endowed with the quotient topology.

The map $\mathbb{I}_K \twoheadrightarrow \mathrm{Id}_K$ induces a map $C_K \twoheadrightarrow \mathrm{Cl}_K$ which fits in the following four-term exact sequence

$$0 \longrightarrow \mathcal{O}_K^\times \longrightarrow K_\infty^\times \times \widehat{\mathcal{O}}_K^\times \longrightarrow C_K \longrightarrow \mathrm{Cl}_K \longrightarrow 0.$$

(Why? An idele $x \in \mathbb{I}_K$ defines a principal ideal if and only if there is a $y \in K^\times$ such that $\mathrm{ord}_v(y^{-1}x_v) = 0$ for all $v \in V_f$. That is, such that $y^{-1}x \in K_\infty^\times \times \widehat{\mathcal{O}}_K^\times$. It is trivial to check $K^\times \cap (K_\infty^\times \times \widehat{\mathcal{O}}_K^\times) = \mathcal{O}_K^\times$ just by unwinding the definitions.)

This sequence intertwines the local and global units with the idele and ideal class groups, and it will play a fundamental role later on. For now we record the isomorphism

$$\mathbb{I}_K / K^\times K_\infty^\times \widehat{\mathcal{O}}_K^\times \xrightarrow{\sim} \mathrm{Cl}_K.$$

In particular $\mathbb{I}_K = K^\times K_\infty^\times \widehat{\mathcal{O}}_K^\times$ when $h_K = 1$. For $K = \mathbb{Q}$ we can do slightly better:

Proposition 2.17 *The multiplication map defines an isomorphism of topological groups*

$$\mathbb{Q}^\times \times \mathbb{R}_{>0}^\times \times \widehat{\mathbb{Z}}^\times \xrightarrow{\sim} \mathbb{I} = \mathbb{I}_\mathbb{Q}.$$

This induces an isomorphism $\mathbb{R}_{>0}^\times \times \widehat{\mathbb{Z}}^\times \xrightarrow{\sim} C_\mathbb{Q}$.

Proof We know the multiplication map is (a continuous homomorphism and) surjective since $h_{\mathbb{Q}} = 1$. Suppose the triple (r, t, u) satisfies $rtu = 1$. Then $r^{-1} \in \mathbb{Q}^\times \cap (\mathbb{R}^\times_{>0} \times \widehat{\mathbb{Z}}^\times) = \{1\}$. Therefore $r = 1$ and hence $t^{-1} = u \in \mathbb{R}^\times_{>0} \cap \widehat{\mathbb{Z}}^\times = \{1\}$ so that also $t = u = 1$. We leave it to the reader to write down an explicit inverse and check continuity. □

This will enter when we inflate a Dirichlet character to a character of $C_{\mathbb{Q}}$. Note that $\widehat{\mathbb{Z}}^\times = \varprojlim_N (\mathbb{Z}/N\mathbb{Z})^\times$.

Definition 2.18 The idele norm $|\cdot| : \mathbb{I}_K \to \mathbb{R}^\times_{>0}$ is defined as $|x| = \prod_{v \in V} |x_v|_v$. (This is a finite product since $|x_v|_v = 1$ for all but finitely many v.)

By the Artin product formula $|x| = 1$ for $x \in K^\times$ and therefore the idele norm factors through a norm on the idele class group $|\cdot| : C_K \to \mathbb{R}^\times_{>0}$. This is surjective so C_K is not compact. Our next goal is to show the norm one subgroup of C_K *is* compact. We already know this for $K = \mathbb{Q}$. Indeed $C^1_{\mathbb{Q}} \simeq \widehat{\mathbb{Z}}^\times$ in the notation introduced below, so $C^1_{\mathbb{Q}}$ is not only compact but profinite.

Definition 2.19 We let $\mathbb{I}^1_K = \{x \in \mathbb{I}_K : |x| = 1\}$ and $C^1_K = \{x \in C_K : |x| = 1\} = \mathbb{I}^1_K / K^\times$ denote the norm one subgroups with their induced topologies.

Before we proceed we notice that there are non-canonical isomorphisms $\mathbb{I}_K \simeq \mathbb{I}^1_K \times \mathbb{R}^\times_{>0}$ and $C_K \simeq C^1_K \times \mathbb{R}^\times_{>0}$ for any number field K. For instance, the short exact sequence

$$0 \longrightarrow \mathbb{I}^1_K \longrightarrow \mathbb{I}_K \longrightarrow \mathbb{R}^\times_{>0} \longrightarrow 0$$

has a splitting $s : \mathbb{R}^\times_{>0} \to \mathbb{I}_K$ obtained by choosing a $v \in V_\infty$ and letting $s(t) = (t, 1, 1 \ldots)$ with t in the vth slot if $v \in V_{\mathbb{R}}$. If v is complex let $s(t) = (\sqrt{t}, 1, 1 \ldots)$. Once we show C^1_K is compact we can conclude C_K is of the form (compact group) $\times$ (real line) as a topological group.

In preparation for the proof that C^1_K is compact we recall some of the ingredients going into the proof of the Dirichlet unit theorem. First and foremost there is the regulator homomorphism

$$\mathrm{reg} : K^\times_\infty = (\mathbb{R}^\times)^{r_1} \times (\mathbb{C}^\times)^{r_2} \longrightarrow \mathbb{R}^{r_1 + r_2}$$

which sends $(x_1, \ldots, z_1, \ldots) \mapsto (\ln |x_1|_{\mathbb{R}}, \ldots, \ln |z_1|^2_{\mathbb{C}}, \ldots)$. It has compact kernel $T = \{\pm 1\}^{r_1} \times (\mathbb{C}^1)^{r_2}$. Composing reg with the diagonal embedding $\mathcal{O}^\times_K \hookrightarrow K^\times_\infty$ identifies $\mathcal{O}^\times_K / \mu_K$ with a full lattice $L \subset H$ inside the hyperplane $H = \ker(\mathrm{sum}) \subset \mathbb{R}^{r_1 + r_2}$. The suitably normalized volume of the compact quotient H/L is by definition the regulator R_K.

Theorem 2.20 *The image of $K^\times \hookrightarrow \mathbb{I}^1_K$ is a discrete cocompact subgroup; that is C^1_K is compact.*

Proof We start with the following variant of the four-term exact sequence mentioned above:

$$0 \longrightarrow \mathcal{O}_K^\times \longrightarrow K_\infty^1 \times \widehat{\mathcal{O}}_K^\times \longrightarrow C_K^1 \longrightarrow \mathrm{Cl}_K \longrightarrow 0.$$

(The map $C_K^1 \to \mathrm{Cl}_K$ is surjective because $\mathbb{I}_K \simeq \mathbb{I}_K^1 \times \mathbb{R}_{>0}^\times$ and $(s(t)x) = (x)$ for $t \in \mathbb{R}_{>0}^\times$.) It suffices to show $\frac{K_\infty^1 \times \widehat{\mathcal{O}}_K^\times}{\mathcal{O}_K^\times}$ is compact since it has index h_K in C_K^1. Applying the snake lemma to the diagram (whose vertical arrows are inclusions)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mu_K & \longrightarrow & \mathcal{O}_K^\times & \longrightarrow & L & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & T \times \widehat{\mathcal{O}}_K^\times & \longrightarrow & K_\infty^1 \times \widehat{\mathcal{O}}_K^\times & \xrightarrow{\text{reg}} & H & \longrightarrow & 0 \end{array}$$

we arrive at a short exact sequence

$$0 \longrightarrow \frac{T \times \widehat{\mathcal{O}}_K^\times}{\mu_K} \longrightarrow \frac{K_\infty^1 \times \widehat{\mathcal{O}}_K^\times}{\mathcal{O}_K^\times} \longrightarrow \frac{H}{L} \longrightarrow 0.$$

As the two end terms are compact so is the middle term. The discreteness was taken care of in Exercise 2.15 and follows straightforwardly from the discreteness of K in $\mathbb{A}_K$. □

Towards the end of the previous proof we used the following observation:

Exercise 2.21 Let G be a locally compact topological group. Suppose $S \leq G$ is a compact subgroup for which G/S is also compact. Show that G itself must be compact. (`Hint`: Consider an open covering $G = \bigcup_{i \in I} U_i$ where each U_i is contained in a compact subset. Then project to G/S.)

Down the line we will revisit this argument and relate the volume of C_K^1 to the constant $\mathcal{M}_K$. Up to the factor $\sqrt{|D_K|}$ they are the same, with the appropriate normalization of measures of course. See Theorem 2.107.

2.2 Characters

2.2.1 Hecke Characters of a Number Field

Here is the generalization of Dirichlet characters to an arbitrary number field K.

Definition 2.22 A Hecke character of K is a continuous homomorphism $\chi : C_K \to \mathbb{C}^\times$. In other words a continuous homomorphism $\chi : \mathbb{I}_K \to \mathbb{C}^\times$ satisfying the automorphy condition $\chi(K^\times) = 1$.

We are not assuming χ is unitary. Some authors call such non-unitary χ quasi-characters, but we will just call them characters. There is really no big difference. One can twist any χ by $|\cdot|^s$ for suitable $s \in \mathbb{C}$ to make it unitary:

Proposition 2.23 *Any Hecke character $\chi : C_K \to \mathbb{C}^\times$ factors (non-uniquely) as $\chi = \chi_0|\cdot|^s$ for some $s \in \mathbb{C}$ and a unitary Hecke character χ_0.*

Proof Fix an isomorphism $C_K \simeq C_K^1 \times \mathbb{R}_{>0}^\times$ once and for all. We know $\chi|_{C_K^1}$ is unitary since C_K^1 is compact. Extend $\chi|_{C_K^1}$ to a unitary character $\chi_0 : C_K \to \mathbb{C}^\times$ by letting $\chi_0 = 1$ on $\mathbb{R}_{>0}^\times$ for instance. Then $\chi\chi_0^{-1}$ is trivial on C_K^1 and therefore factors through the idele norm $C_K/C_K^1 \xrightarrow{\sim} \mathbb{R}_{>0}^\times$. All continuous characters $\mathbb{R}_{>0}^\times \to \mathbb{C}^\times$ have the form $t \mapsto t^s$ for some $s \in \mathbb{C}$. Therefore $\chi\chi_0^{-1} = |\cdot|^s$. Done. □

Note that neither χ_0 nor s is uniquely determined by χ. One can always modify the factorization as $\chi = \chi_0|\cdot|^{-i\tau}|\cdot|^{s+i\tau}$ for any $\tau \in \mathbb{R}$, but that is the only alteration one can make. In particular $a_\chi = \mathrm{Re}(s)$ is completely determined by χ; we call it the exponent of χ. In this notation $a_\chi = 0$ means χ is unitary.

Remark 2.24 For $K = \mathbb{Q}$ one can boost the factorization $\chi = \chi_0|\cdot|^s$ and assume χ_0 is of finite order, which forces uniqueness. Why? We observed $C_\mathbb{Q}^1 \simeq \widehat{\mathbb{Z}}^\times$ is profinite, so by the no small subgroups argument below any continuous character $C_\mathbb{Q}^1 \to \mathbb{C}^\times$ factors through a finite quotient of $C_\mathbb{Q}^1$. Going back to the proof of Proposition 2.23 the restriction $\chi|_{C_\mathbb{Q}^1}$ therefore has finite order, and so does the extension χ_0.

The finite order characters $\chi_0 : C_\mathbb{Q} \to \mathbb{C}^\times$ correspond to Dirichlet characters $(\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$ via the projections

$$C_\mathbb{Q} \simeq C_\mathbb{Q}^1 \times \mathbb{R}_{>0}^\times \twoheadrightarrow C_\mathbb{Q}^1 \simeq \widehat{\mathbb{Z}}^\times \twoheadrightarrow (\mathbb{Z}/N\mathbb{Z})^\times.$$

More about that later.

We made use of the following extremely useful fact:

Lemma 2.25 ("No small subgroups argument") *Let G be a profinite group. Then any continuous homomorphism $\chi : G \to \mathbb{C}^\times$ factors through a finite quotient of G.*

Proof This is based on the observation that $\mathbb{C}^\times$ has no small subgroups, by which we mean there is an open neighborhood $1 \in \Omega \subset \mathbb{C}^\times$ not containing any subgroups of $\mathbb{C}^\times$ other than $\{1\}$. (Why? Suppose $z \in \Omega - \{1\}$. If Ω is small enough some powers of z will end up outside Ω.) On the other hand G has a neighborhood basis at the identity consisting of open normal subgroups $N \lhd G$ of finite index (that is what it means to be profinite). By continuity of χ we may choose an N small enough to ensure $N \subset \chi^{-1}(\Omega)$. Then $\chi(N)$ is a subgroup of $\mathbb{C}^\times$ entirely contained in Ω and therefore $\chi(N) = \{1\}$. In other words χ factor through the finite quotient G/N. □

For a moment we will ignore the automorphy condition $\chi(K^\times) = 1$ and describe a canonical factorization of any continuous homomorphism $\chi : \mathbb{I}_K \to \mathbb{C}^\times$ into local characters.

At each $v \in V$ we have the natural embedding $K_v^\times \hookrightarrow \mathbb{I}_K$ into the vth slot. Given χ we restrict it to $K_v^\times$ and generate a collection of continuous characters $\chi_v : K_v^\times \to \mathbb{C}^\times$ called the local components of χ.

Lemma 2.26 *We have $\chi_v(\mathcal{O}_v^\times) = \{1\}$ for all but at most finitely many $v \in V_f$.*

Proof Pick an $\Omega \subset \mathbb{C}^\times$ as in the proof of Lemma 2.25. By continuity of χ its inverse image $\chi^{-1}(\Omega)$ contains an open neighborhood of $1 \in \mathbb{I}_K$ which we may take to be a box $U = \prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v^\times$. Let $v \notin S$. Then $\chi_v(\mathcal{O}_v^\times)$ is a subgroup of $\mathbb{C}^\times$ entirely contained in Ω and therefore trivial. □

When $\chi_v(\mathcal{O}_v^\times) = \{1\}$ we say χ_v is unramified (or χ is unramified at v). For those χ_v, the complex parameter $\chi_v(\varpi_v) \in \mathbb{C}^\times$ is independent of the choice of uniformizer ϖ_v and it completely determines χ_v.

Since χ is unramified at all but finitely many $v \in V_f$ the product $\chi(x) = \prod_{v\in V} \chi_v(x_v)$ is finite for all $x \in \mathbb{I}_K$. We occasionally write this symbolically as an infinite tensor product $\chi = \otimes_{v\in V}\chi_v$.

Conversely, starting from a collection of continuous characters $\chi_v : K_v^\times \to \mathbb{C}^\times$ satisfying the condition that almost all the χ_v are unramified, we associate the character $\chi : \mathbb{I}_K \to \mathbb{C}^\times$ defined by $\chi = \otimes_{v\in V}\chi_v$. Of course there is no reason for this χ to be a Hecke character! The automorphy condition on χ imposes a subtle tension between its local components χ_v.

Notation 2.27 The infinity type of $\chi : \mathbb{I}_K \to \mathbb{C}^\times$ is the character of $K_\infty^\times$ given by $\chi_\infty = \otimes_{v\in V_\infty}\chi_v$.

Exercise 2.28 Show that there is a bijection between characters $\mathrm{Cl}_K \to \mathbb{C}^\times$ and Hecke characters $\chi : C_K \to \mathbb{C}^\times$ for which (1) $\chi_\infty = 1$; and (2) χ is unramified at every $v \in V_f$.

2.2.2 Parametrization of Local Characters

In this section we will drop the subscript v and let K denote a local field of characteristic 0, with normalized absolute value $|\cdot|$. If K is non-archimedean let $\mathcal{O}$ be its valuation ring, $\mathfrak{p} = (\varpi)$ the maximal ideal thereof, and $k = \mathcal{O}/\mathfrak{p} \simeq \mathbb{F}_q$ the residue field.

Our goal here is to describe all the continuous characters $\chi : K^\times \to \mathbb{C}^\times$. We first assume K is archimedean.

Proposition 2.29 *When $K = \mathbb{R}$ or $K = \mathbb{C}$ the local characters of $K^\times$ have the following form:*

(1) The real case: Every continuous character $\chi : \mathbb{R}^\times \to \mathbb{C}^\times$ can be expressed uniquely as

$$\chi(x) = \chi_{e,s}(x) = sign(x)^e |x|^s$$

where $e \in \{0, 1\}$ and $s \in \mathbb{C}$.

(2) *The complex case: Every continuous character* $\chi : \mathbb{C}^\times \to \mathbb{C}^\times$ *can be expressed uniquely as*

$$\chi(x) = \chi_{n,s}(x) = \Big(\frac{x}{|x|_{\mathbb{C}}}\Big)^n |x|^s$$

where $n \in \mathbb{Z}$ *and* $s \in \mathbb{C}$. *(We emphasize that* $|x| = x\bar{x} = |x|_{\mathbb{C}}^2$ *here denotes the normalized "absolute value" on* $\mathbb{C}$.*)*

Proof We already remarked that the continuous characters $\mathbb{R}_{>0}^\times \to \mathbb{C}^\times$ have the form $t \mapsto t^s$ for some $s \in \mathbb{C}$. Now use the isomorphism $\mathbb{R}^\times \simeq \{\pm 1\} \times \mathbb{R}_{>0}^\times$ sending $x \mapsto (\mathrm{sign}(x), |x|)$. This settles the real case.

The complex case is similar. We identify $\mathbb{C}^\times \simeq \mathbb{C}^1 \times \mathbb{R}_{>0}^\times$ via $x \mapsto (\frac{x}{|x|_{\mathbb{C}}}, |x|)$. Characters of the circle $\mathbb{C}^1 \simeq \mathbb{R}/\mathbb{Z}$ all have the form $x \mapsto x^n$ for some $n \in \mathbb{Z}$. □

Remark 2.30 In this parametrization $\chi_{n,s}|_{\mathbb{R}^\times} = \chi_{\mathrm{e},2s}$ where $\mathrm{e} \in \{0, 1\}$ gives the parity of n.

Now assume $K/\mathbb{Q}_p$ is a finite extension and let $\chi : K^\times \to \mathbb{C}^\times$ be a continuous character. By Lemma 2.25 continuity amounts to $\chi(1 + \mathfrak{p}^r) = \{1\}$ for $r \gg 0$. In other words $\chi|_{\mathcal{O}^\times}$ factors through the finite quotient $\mathcal{O}^\times/1 + \mathfrak{p}^r \simeq (\mathcal{O}/\mathfrak{p}^r)^\times$. The conductor of χ is the smallest $r \geq 0$ for which this happens:

$$\mathrm{cond}(\chi) = \min\{r \geq 0 : \chi|_{1+\mathfrak{p}^r} = 1\}.$$

By convention $\mathrm{cond}(\chi) = 0$ if and only if $\chi|_{\mathcal{O}^\times} = 1$; that is when χ is unramified. When χ is ramified $\mathrm{cond}(\chi) > 0$. Sometimes we will also refer to the ideal $\mathfrak{p}^{\mathrm{cond}(\chi)} \subset \mathcal{O}$ as the conductor of χ.

Here is an analogue of Proposition 2.23 in the non-archimedean case:

Proposition 2.31 *For non-archimedean fields* K *any continuous character* $\chi : K^\times \to \mathbb{C}^\times$ *factors (non-uniquely) as* $\chi = \chi_0 |\cdot|^s$ *for some* $s \in \mathbb{C}$ *and a finite order character* χ_0. *One can arrange that* $\chi_0(\varpi) = 1$ *for a given choice of uniformizer* ϖ.

Proof Note that $K^\times = \varpi^{\mathbb{Z}} \times \mathcal{O}^\times$. Pick an $s \in \mathbb{C}$ such that $\chi(\varpi) = |\varpi|^s = q^{-s}$. Such an s is only unique modulo $\frac{2\pi i}{\ln(q)}\mathbb{Z}$. The twist $\chi_0 = \chi|\cdot|^{-s}$ satisfies $\chi_0(\varpi) = 1$ and the restriction $\chi_0|_{\mathcal{O}^\times} = \chi|_{\mathcal{O}^\times}$ factors through $(\mathcal{O}/\mathfrak{p}^r)^\times$ for large enough r. Thus χ_0 has finite order. □

Exercise 2.32 Deduce from Proposition 2.31 that any unramified character of $K^\times$ has the form $|\cdot|^s$ for some $s \in \mathbb{C}$ well-determined modulo $\frac{2\pi i}{\ln(q)}\mathbb{Z}$.

We return to our initial hypothesis and let K denote an arbitrary local field of characteristic 0. We will now attach an L-factor $L(\chi) \in \mathbb{C}^\times$ to a continuous character $\chi : K^\times \to \mathbb{C}^\times$. This is done by hand so to speak, the justification will come later: Tate's thesis gives a uniform way of "explaining" the shape of these local L-factors.

Note that the exponent $a_\chi = \mathrm{Re}(s)$ makes sense for any local character $\chi : K^\times \to \mathbb{C}^\times$.

Definition 2.33 The local L-factors are defined as follows.

- The real case.
$$L(\chi_{\mathrm{e},s}) = L_{\mathbb{R}}(s+\mathrm{e}) = \pi^{-\frac{s+\mathrm{e}}{2}}\Gamma(\frac{s+\mathrm{e}}{2}).$$
- The complex case.
$$L(\chi_{n,s}) = L_{\mathbb{C}}(s+\frac{|n|}{2}) = 2(2\pi)^{-(s+\frac{|n|}{2})}\Gamma(s+\frac{|n|}{2}).$$
- The non-archimedean case.
$$L(\chi) = \frac{1}{1-\chi(\varpi)}$$
when χ is unramifed and non-trivial. Otherwise, when $\chi \neq 1$ is ramified we declare that $L(\chi) = 1$.

We leave $L(1)$ undefined in the non-archimedean case. The same with $L(\chi_{\mathrm{e},s})$ and $L(\chi_{n,s})$ when $\frac{s+\mathrm{e}}{2}$ respectively $s+\frac{|n|}{2}$ is a pole of the Γ-function. We will only consider $L(\chi)$ for local characters χ with exponent $a_\chi > 0$ so this will not be an issue.

Remark 2.34 The shifted dual of χ is the character $\chi^\vee(x) = \chi(x)^{-1}|x|$. Observe that $a_{\chi^\vee} = 1 - a_\chi$ and $(\chi|\cdot|^s)^\vee = \chi^{-1}|\cdot|^{1-s}$ for all $s \in \mathbb{C}$.

2.2.3 Definition of Hecke L-Functions

We return to the global setting, and K again denotes a number field with completions K_v et cetera. Finally we can define the L-function attached to a Hecke character $\chi : C_K \to \mathbb{C}^\times$. Factor the character into local components $\chi = \otimes_{v\in V}\chi_v$. Note that we necessarily have $a_\chi = a_{\chi_v}$ for all $v \in V$. That is, all local components must have the same exponent.

Definition 2.35 The Hecke L-function of $\chi : C_K \to \mathbb{C}^\times$ is defined by the following Euler product over all $v \in V_f$ for which χ_v is unramified:

$$L(s,\chi) = \prod_{v\in V_f} L(\chi_v|\cdot|_v^s) = \prod_{\text{unram } v} \frac{1}{1-\chi_v(\varpi_v)q_v^{-s}}.$$

This is absolutely convergent for $\mathrm{Re}(s) > 1$ provided χ is unitary. (The proof for $\zeta_K(s)$ works almost verbatim.)

We will not define a completed L-function, but only the variant $L^{\mathrm{aut}}(s,\chi)$ which takes the local L-factors at the infinite places into account:

$$L^{\mathrm{aut}}(s,\chi) = \prod_{v\in V} L(\chi_v|\cdot|_v^s) = \left(\prod_{v\in V_\infty} L(\chi_v|\cdot|_v^s)\right)\cdot L(s,\chi).$$

For $K=\mathbb{Q}$ this essentially recovers our definition for Dirichlet characters (when viewed as characters of $C_{\mathbb{Q}}$) and in general $L^{\mathrm{aut}}(s,1)=\zeta_K^{\mathrm{aut}}(s)$.

Some of the decisive motivation behind Tate's thesis is to show the analytic continuation of $L^{\mathrm{aut}}(s,\chi)$ along with an $s\mapsto 1-s$ functional equation. The latter involves an exponential factor $\epsilon(s,\chi)$ yet to be defined, which involves $|D_K|$ and the norm of the conductor $\prod_{v\in V_f}\mathfrak{p}_v^{\mathrm{cond}(\chi_v)}\subset\mathcal{O}_K$.

2.2.4 Additive Characters of a Local Field

The Fourier transformation integral involves the additive characters $e^{-2\pi i a(\cdot)}$ of $\mathbb{R}$ for varying a. To do Fourier analysis on all completions of a number field we need to first understand the additive characters. We go back to assuming K is a local field of characteristic 0, and adapt the notation from Sect. 2.2.2 (apologies for the back-and-forth). In this section we will describe all unitary characters of the additive group $(K,+)$; in other words all continuous homomorphisms $\psi: K\to\mathbb{C}^1$. In each case there is a standard choice $\psi_{\mathrm{std}}\neq 1$ and all other ψ are of the form $\psi_a(x)=\psi_{\mathrm{std}}(ax)$ as $a\in K$ varies.

- The real case. Here $\psi_{\mathrm{std}}:\mathbb{R}\longrightarrow\mathbb{C}^1$ is defined by
$$\psi_{\mathrm{std}}(x)=e^{-2\pi i x}.$$
- The complex case. Here $\psi_{\mathrm{std}}:\mathbb{C}\longrightarrow\mathbb{C}^1$ is defined by composing the above with $\mathrm{Tr}_{\mathbb{C}/\mathbb{R}}$,
$$\psi_{\mathrm{std}}(z)=e^{-2\pi i(z+\bar{z})}=e^{-4\pi i\mathrm{Re}(z)}.$$
- The non-archimedean case. Here $\psi_{\mathrm{std}}:K\longrightarrow\mathbb{C}^1$ is defined as $\psi_{\mathbb{Q}_p}\circ\mathrm{Tr}_{K/\mathbb{Q}_p}$ where $\psi_{\mathbb{Q}_p}$ is the composition
$$\mathbb{Q}_p\longrightarrow\mathbb{Q}_p/\mathbb{Z}_p\xrightarrow{\sim}(\mathbb{Q}/\mathbb{Z})[p^\infty]=\mathbb{Z}[\tfrac{1}{p}]/\mathbb{Z}\xrightarrow{e^{+2\pi i(\cdot)}}\mathbb{C}^1.$$
More explicitly $\psi_{\mathbb{Q}_p}$ takes a p-adic number $x=\sum_{i\gg-\infty}a_i p^i$ to $\psi_{\mathbb{Q}_p}(x)=e^{+2\pi i(\sum_{i<0}a_i p^i)}$.

Note the intentional change of signs in the exponential function. (This will be important later when we combine these into a character of the adeles, to guarantee the adelic standard character is trivial on the number field.)

Clearly $\mathrm{im}(\psi_{\mathbb{Q}_p}) = \mu_{p^\infty}(\mathbb{C})$ and $\ker(\psi_{\mathbb{Q}_p}) = \mathbb{Z}_p$. For a general $K/\mathbb{Q}_p$ we still have $\mathrm{im}(\psi_{\mathrm{std}}) = \mu_{p^\infty}(\mathbb{C})$, but $\ker(\psi_{\mathrm{std}}) \supset \mathcal{O}$ could be a strict inclusion; for instance $\psi_{\mathrm{std}}(\frac{1}{p}) = 1$ if p divides $[K : \mathbb{Q}_p]$.

Definition 2.36 Suppose K is non-archimedean and let $\psi \neq 1 : K \to \mathbb{C}^1$ be a non-trivial continuous additive character. The conductor $\mathrm{cond}(\psi)$ is the smallest $r \in \mathbb{Z}$ for which $\psi(\mathfrak{p}^r) = \{1\}$. In other words the integer r for which $\psi|_{\mathfrak{p}^r} = 1$ but $\psi|_{\mathfrak{p}^{r-1}} \neq 1$. We sometimes abuse terminology and also refer to the fractional ideal $\mathfrak{p}^{\mathrm{cond}(\psi)} \subset K$ as the conductor of ψ. We say ψ is unramified if $\mathrm{cond}(\psi) = 0$.

For example, since $\psi_{\mathrm{std}}(\mathfrak{p}^r) = \{1\}$ if and only if $\mathrm{Tr}_{K/\mathbb{Q}_p}(\mathfrak{p}^r) \subset \mathbb{Z}_p$ we find the conductor of the standard character

$$\mathfrak{p}^{\mathrm{cond}(\psi_{\mathrm{std}})} = \{x \in K : \mathrm{Tr}_{K/\mathbb{Q}_p}(x\mathcal{O}) \subset \mathbb{Z}_p\} = \partial^{-1}_{K/\mathbb{Q}_p}$$

is the inverse different. (The different is by definition its inverse ideal $\partial_{K/\mathbb{Q}_p} \subset \mathcal{O}$.)

Exercise 2.37 Consider $\psi_a(x) = \psi_{\mathrm{std}}(ax)$ for some $a \in K^\times$. Show that $\mathfrak{p}^{\mathrm{cond}(\psi_a)} = a^{-1}\partial^{-1}_{K/\mathbb{Q}_p}$. Conclude that any non-archimedean field $K/\mathbb{Q}_p$ admits an unramified non-trivial character $K \to \mathbb{C}^\times$.

Remark 2.38 Without proof we mention a few basic facts about how $\partial_{K/\mathbb{Q}_p}$ captures ramification in the extension $K/\mathbb{Q}_p$. Let $e = e(K/\mathbb{Q}_p)$ denote the ramification index–so $p \sim \varpi^e$. Then $\mathfrak{p}^{e-1}|\partial_{K/\mathbb{Q}_p}$. This is easy to see:

$$\mathrm{Tr}_{K/\mathbb{Q}_p}(\varpi^{1-e}\mathcal{O}) = \mathrm{Tr}_{K/\mathbb{Q}_p}(\frac{\varpi}{p}\mathcal{O}) = \frac{1}{p}\mathrm{Tr}_{K/\mathbb{Q}_p}(\mathfrak{p}) \subset \frac{1}{p} \cdot p\mathbb{Z}_p = \mathbb{Z}_p.$$

There are more precise bounds on the differential exponent involving $v_p(e)$, from which one can deduce:

- $K/\mathbb{Q}_p$ is ramified ($e > 1$) exactly when $\mathfrak{p}|\partial_{K/\mathbb{Q}_p}$;
- $K/\mathbb{Q}_p$ is tamely ramified ($p \nmid e$) if and only if $\mathfrak{p}^{e-1} = \partial_{K/\mathbb{Q}_p}$.

Furthermore $N(\partial_{K/\mathbb{Q}_p}) = |D_K|$ where $D_K = \det\left(\mathrm{Tr}_{K/\mathbb{Q}_p}(\alpha_i\alpha_j)\right)$ for some $\mathbb{Z}_p$-basis $\{\alpha_i\}$ for $\mathcal{O}$ is the local discriminant. There are analogous results for number fields. See [Fr67, Theorems 1 and 2, p. 21] for example.

To allow more flexibility we now start with an arbitrary non-trivial additive character $\psi \neq 1 : K \to \mathbb{C}^1$ (and K denotes a local field of characteristic 0–possibly archimedean). For $a \in K$ we let $\psi_a(x) = \psi(ax)$. Note that $\psi_a = 1 \Leftrightarrow a = 0$. Also $\psi_{a+a'} = \psi_a \cdot \psi_{a'}$. Thus $a \mapsto \psi_a$ defines an injective homomorphism $K \longrightarrow \widehat{K}$ into the dual group $\widehat{K} = \mathrm{Hom}_{\mathrm{cont}}(K, \mathbb{C}^1)$.

Proposition 2.39 *Sending $a \mapsto \psi_a$ gives an isomorphism $K \xrightarrow{\sim} \widehat{K}$ (of abstract groups).*

Proof This is well-known and easy to check for $\mathbb{R}$ and $\mathbb{C}$ so we will assume K is non-archimedean. For simplicity we will also assume our choice ψ has conductor $\mathcal{O}$ (we let the reader reduce the general case to the unramified case, essentially using Exercise 2.37). For a fixed $n \geq 1$ consider the isomorphism

$$\mathcal{O}/\mathfrak{p}^n \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cont}}(\mathfrak{p}^{-n}/\mathcal{O}, \mathbb{C}^1) \qquad a \mapsto \psi_a|_{\mathfrak{p}^{-n}}.$$

(It is injective because $\mathrm{cond}(\psi) = \mathcal{O}$ and therefore surjective since the target and source both have size q^n.) The maps are compatible when we vary n, and passing to the inverse limit gives an isomorphism

$$\mathcal{O} = \varprojlim_n \mathcal{O}/\mathfrak{p}^n \xrightarrow{\sim} \varprojlim_n \mathrm{Hom}_{\mathrm{cont}}(\mathfrak{p}^{-n}/\mathcal{O}, \mathbb{C}^1) = \mathrm{Hom}_{\mathrm{cont}}(\bigcup_{n\geq 1} \mathfrak{p}^{-n}/\mathcal{O}, \mathbb{C}^1) = \mathrm{Hom}_{\mathrm{cont}}(K/\mathcal{O}, \mathbb{C}^1).$$

From this the surjectivity of $K \longrightarrow \widehat{K}$ follows easily: If $\mu \in \widehat{K}$ has conductor $\mathrm{cond}(\mu) = r$ then μ_{ϖ^r} factors through $K/\mathcal{O}$ and is therefore of the form ψ_a for some $a \in \mathcal{O}$ by the above. Thus $\mu = \psi_{a\varpi^{-r}}$. □

This says the choice of a $\psi \neq 1$ gives a self-duality on $(K, +)$. Later we will topologize $\widehat{K}$ and then the map $K \xrightarrow{\sim} \widehat{K}$ becomes an isomorphism of topological groups.

2.2.5 Additive Characters of the Adeles

In the same vein we now parametrize all the continuous unitary characters $\psi : \mathbb{A}_K \to \mathbb{C}^1$ for a number field K.

First we define the adelic standard character. For each place $v \in V$ let $\psi_{\mathrm{std},v} : K_v \to \mathbb{C}^1$ be the local standard character. Note that $\psi_{\mathrm{std},v}$ is unramified precisely when $K_v/\mathbb{Q}_p$ is unramified. In particular $\psi_{\mathrm{std},v}(\mathcal{O}_v) = \{1\}$ for all but finitely many $v \in V_f$ and therefore the product $\psi_{\mathrm{std}}(x) = \prod_{v\in V} \psi_{\mathrm{std},v}(x_v)$ is finite for all $x \in \mathbb{A}_K$. We will sometimes express this more succinctly as $\psi_{\mathrm{std}} = \otimes_{v\in V}\psi_{\mathrm{std},v}$. Continuity of ψ_{std} is clear.

When $K = \mathbb{Q}$ we also write $\psi_{\mathbb{Q}} = \psi_{\mathrm{std}}$ for the standard character $\mathbb{A} \to \mathbb{C}^1$. Many questions about ψ_{std} for a general $K/\mathbb{Q}$ can be reduced to $\psi_{\mathbb{Q}}$ by the relation $\psi_{\mathrm{std}} = \psi_{\mathbb{Q}} \circ \mathrm{Tr}_{\mathbb{A}_K/\mathbb{A}}$ where

$$\mathrm{Tr}_{\mathbb{A}_K/\mathbb{A}} : \mathbb{A}_K = K \otimes_{\mathbb{Q}} \mathbb{A} \longrightarrow \mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{A} = \mathbb{A}$$

is the adelic trace $\mathrm{Tr}_{\mathbb{A}_K/\mathbb{A}} = \mathrm{Tr}_{K/\mathbb{Q}} \otimes 1$–for instance the fact that $\psi_{\mathrm{std}}|_K = 1$ ("automorphic"):

Lemma 2.40 *The character ψ_{std} factors through the quotient $\mathbb{A}_K/K$. That is $\psi_{std}(K) = \{1\}$.*

Proof Since $\mathrm{Tr}_{\mathbb{A}_K/\mathbb{A}}(K) \subset \mathbb{Q}$ it suffices to prove $\psi_{\mathbb{Q}}$ factors through $\mathbb{A}/\mathbb{Q}$. Decompose the torsion module $\mathbb{Q}/\mathbb{Z}$ into p-primary parts

$$\mathbb{Q}/\mathbb{Z} = \bigoplus_{p<\infty} (\mathbb{Q}/\mathbb{Z})[p^\infty].$$

Let $r \in \mathbb{Q}$ and find $r_p \in \mathbb{Z}[\frac{1}{p}]$ such that $r \equiv \sum_p r_p$ modulo $\mathbb{Z}$ (almost all r_p lie in $\mathbb{Z}$). Then

$$\prod_{p<\infty} \psi_{\mathbb{Q}_p}(r) = \prod_{p<\infty} e^{+2\pi i r_p} = e^{+2\pi i r} = \psi_\infty(r)^{-1}.$$

In other words $\psi_{\mathbb{Q}}(r) = 1$. □

Choose any locally non-trivial continuous character $\psi : \mathbb{A}_K \to \mathbb{C}^1$ (not necessarily automorphic). By locally non-trivial we mean that $\psi_v \neq 1$ for all $v \in V$. An element $a \in \mathbb{A}_K$ defines an additive character $\psi_a(x) = \psi(ax)$. As for local fields this choice ψ gives a self-duality for $\mathbb{A}_K$ (which will later be promoted to an isomorphism $\mathbb{A}_K \xrightarrow{\sim} \widehat{\mathbb{A}_K}$ of topological groups).

Proposition 2.41 *Sending $a \mapsto \psi_a$ gives an isomorphism $\mathbb{A}_K \xrightarrow{\sim} \widehat{\mathbb{A}_K}$ (of abstract groups).*

Proof This follows from the local analogue and its proof. Any $\mu \in \widehat{\mathbb{A}_K}$ factors as $\mu = \otimes_{v\in V} \mu_v$ where $\mu_v = \psi_{v,a_v}$ for some $a_v \in K_v$. Moreover $\mathcal{O}_v \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cont}}(K_v/\mathcal{O}_v, \mathbb{C}^1)$ and therefore $a_v \in \mathcal{O}_v$ for all but finitely many v (where μ is unramified). Hence $a = (a_v)_{v\in V}$ is an adele and $\mu = \psi_a$. □

The automorphic part of $\widehat{\mathbb{A}_K}$ corresponds to K. Here is a more precise formulation:

Proposition 2.42 *Fix a non-trivial character $\psi : \mathbb{A}_K/K \to \mathbb{C}^1$. Then ψ is locally non-trivial ($\psi_v \neq 1$ for all $v \in V$) and the map $a \mapsto \psi_a$ restricts to an isomorphism $K \xrightarrow{\sim} \widehat{\mathbb{A}_K/K}$ (of discrete groups).*

Proof Once we show $a \mapsto \psi_a$ gives an isomorphism $K \xrightarrow{\sim} \widehat{\mathbb{A}_K/K}$ the rest follows from the diagram

$$\begin{array}{ccc} K & \xrightarrow{\sim} & \widehat{\mathbb{A}_K/K} \\ \downarrow & & \downarrow \\ K_v & \xrightarrow{\sim} & \widehat{K_v}. \end{array}$$

Noting that $\mathbb{A}_K/K \simeq K \otimes_{\mathbb{Q}} (\mathbb{A}/\mathbb{Q}) \simeq (\mathbb{A}/\mathbb{Q})^{\oplus d}$ it suffices to show the isomorphism $\mathbb{Q} \xrightarrow{\sim} \widehat{\mathbb{A}/\mathbb{Q}}$ for the choice $\psi_{\mathbb{Q}}$. Injectivity is clear. By the Approximation Lemma 2.9, there is an isomorphism

$$\frac{\mathbb{R} \times \widehat{\mathbb{Z}}}{\mathbb{Z}} \xrightarrow{\sim} \mathbb{A}/\mathbb{Q}.$$

We see that $\mu \in \widehat{\mathbb{A}/\mathbb{Q}}$ corresponds to a pair of unitary characters $\mu_\infty : \mathbb{R} \to \mathbb{C}^1$ and $\mu_f : \widehat{\mathbb{Z}} \to \mathbb{C}^1$ which are compatible on $\mathbb{Z}$, in the sense that $\mu_\infty|_\mathbb{Z} \cdot \mu_f|_\mathbb{Z} = 1$. Since $\widehat{\mathbb{Z}}$ is profinite μ_f has finite order. Consequently $\mu_\infty|_\mathbb{Z}$ must have finite order and therefore $\mu_\infty = e^{-2\pi i a(\cdot)}$ for some $a \in \mathbb{Q}$. We claim $\mu = \psi_{\mathbb{Q},a}$. Indeed $(\mu^{-1}\psi_{\mathbb{Q},a})_\infty = 1$ and therefore $(\mu^{-1}\psi_{\mathbb{Q},a})_f = 1$ first on $\mathbb{Z}$, and then by continuity on all of $\widehat{\mathbb{Z}}$. □

2.3 A Crash Course in Harmonic Analysis

2.3.1 Pontryagin Duality

For now let G denote an arbitrary topological group. We consider the dual group $\widehat{G} = \mathrm{Hom}_{\mathrm{cont}}(G, \mathbb{C}^1)$ of all continuous homomorphisms $\chi : G \to \mathbb{C}^1$. The group structure is given by $(\chi\chi')(g) = \chi(g)\chi'(g)$, and $\widehat{G}$ is visibly always abelian. Note that $\widehat{G}$ may very well be trivial in this generality (this happens for A_n when $n \geq 5$).

We first turn $\widehat{G}$ into a topological group by endowing it with the so-called compact-open topology. The latter is generated by subsets

$$W(C, U) = \{\chi \in \widehat{G} : \chi(C) \subset U\}$$

where $C \subset G$ is compact and $U \subset \mathbb{C}^1$ is open. This means the open subsets of $\widehat{G}$ are unions of finite intersections of subsets of the form $W(C, U)$. We say these sets $W(C, U)$ form a subbase for the topology. Convergence of $\chi_i \to \chi$ in $\widehat{G}$ amounts to uniform convergence on compact subsets. For this reason the compact-open topology is also known as the topology of compact convergence.

Remark 2.43 There are too many hats! Depending on the context $\widehat{(\cdot)}$ could mean completion or Pontryagin dual. An example of conflicting notation is $\widehat{\mathbb{Z}}$. We always take this to mean the profinite completion of $\mathbb{Z}$, not the Pontryagin dual of $\mathbb{Z}$ (which is the circle group $\mathbb{R}/\mathbb{Z}$).

Exercise 2.44 Verify that $\widehat{G}$ indeed becomes a topological group this way. (Check that multiplication $\widehat{G} \times \widehat{G} \longrightarrow \widehat{G}$ and inversion $\widehat{G} \longrightarrow \widehat{G}$ are continuous maps.)

Exercise 2.45 Recall that a topological space is discrete if all subsets are open. Prove the following:

(a) G discrete $\Longrightarrow \widehat{G}$ compact;
(b) G compact $\Longrightarrow \widehat{G}$ discrete.

(`Hints`: In (a) note that compact $C \subset G$ are finite so $\widehat{G} \hookrightarrow \prod_{g \in G} \mathbb{C}^1$ is a topological embedding. For part (b) note that $W(G, U) = \{1\}$ for small enough open neighborhoods $1 \in U \subset \mathbb{C}^1$, cf. Lemma 2.25.)

Exercise 2.46 Improve Proposition 2.39 by showing $a \mapsto \psi_a$ gives a topological isomorphism $K \xrightarrow{\sim} \widehat{K}$. (Hint: Without loss of generality $\mathrm{cond}(\psi) = 0$. To show the bijection is continuous and open check that $\mathfrak{p}^{-n} \xrightarrow{\sim} \widehat{K/\mathfrak{p}^n}$ for all $n \in \mathbb{Z}$. Moreover $\widehat{K/\mathfrak{p}^n} = W(\mathfrak{p}^n, U)$ is open in $\widehat{K}$ for small enough $U \ni 1$.)

From now on G will be assumed to be abelian, and locally compact. (Our main examples are K_v, $K_v^\times$, $\mathbb{A}_K$, $\mathbb{I}_K$.)

Theorem 2.47 (Pontryagin duality) *Let G be a locally compact abelian group. Then $\widehat{G}$ is also locally compact (and obviously abelian). Moreover, the natural biduality map*

$$\begin{aligned} G &\longrightarrow \widehat{\widehat{G}} \\ g &\longmapsto \left[\chi \mapsto \chi(g)\right] \end{aligned}$$

is an isomorphism of topological groups.

Proof This is a rather long and technical proof which we omit. It occupies a significant portion of [RV99, Chap. 3]. Even the fact that $\widehat{G}$ is locally compact is non-trivial, see part (v) of [RV99, Proposition 3.2]. □

Exercise 2.48 Prove the previous Theorem for profinite abelian groups G. (Hint: Write $G = \varprojlim G/N$. By Proposition 2.25 we get $\widehat{G} = \varinjlim \widehat{G/N}$. Reduce to the case of a finite abelian group.)

As a quick application we apply Pontryagin duality to the compact group $\mathbb{A}_K/K$. As we have seen, a choice of $\psi \neq 1 : \mathbb{A}_K/K \to \mathbb{C}^1$ gives an isomorphism $K \xrightarrow{\sim} \widehat{\mathbb{A}_K/K}$ (between discrete groups). Passing to duals yields an isomorphism (between compact groups)

$$\mathbb{A}_K/K \xrightarrow{\sim} \widehat{\widehat{\mathbb{A}_K/K}} \xrightarrow{\sim} \widehat{K}.$$

In particular $\mathbb{A}/\mathbb{Q}$ can be identified with the Pontryagin dual of $(\mathbb{Q}, +)$.

2.3.2 *Haar Measures*

Fix a locally compact abelian group G. We want to do Fourier analysis on G. In order to define the Fourier transformation we need a sensible way of integrating functions on G. For $G = \mathbb{R}$ one integrates with respect to the Lebesgue measure, which is translation-invariant. For a general G it turns out there is a natural analogue of the Lebesgue measure, called the Haar measure μ, which is again translation invariant–in the sense that $\mu(gA) = \mu(A)$ for all $g \in G$ and measurable sets $A \subset G$. Imposing additional regularity hypotheses on μ characterizes it uniquely up to a positive constant.

We will first review bits and pieces from measure theory. A good reference is the first two Chapters of [Rud87]. All our measures will be defined on Borel sets (those in the smallest σ-algebra containing all open sets); let $\mathcal{B}$ denote the σ-algebra of Borel sets. Recall that a σ-algebra of sets is closed under complements and countable unions and intersections.

A Borel measure on G is a function $\mu : \mathcal{B} \longrightarrow [0, \infty]$ which is countably additive. This means

$$\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$$

for any sequence $(A_n)_{n\in\mathbb{N}}$ of mutually disjoint sets $A_n \in \mathcal{B}$. (One also assumes $\mu(\varnothing) = 0$.) Integrating relative to μ defines a functional $L_\mu : \mathcal{C}_c(G) \longrightarrow \mathbb{C}$. Here $\mathcal{C}_c(G)$ denotes the space of compactly supported continuous functions $f : G \to \mathbb{C}$ and we adopt the notation $L_\mu(f) = \int_G f(x)d\mu(x)$. The functional L_μ is "positive" in the sense that $f \geq 0 \Rightarrow L_\mu(f) \geq 0$. Conversely, the Riesz representation theorem says that any positive functional $L : \mathcal{C}_c(G) \longrightarrow \mathbb{C}$ is of the form $L = L_\mu$ for a suitable measure μ. Moreover, μ is uniquely determined by L if one imposes the following conditions:

(1) The measure μ is "outer regular" on all $A \in \mathcal{B}$, which means

$$\mu(A) = \inf\{\mu(U) \colon U \supset A \text{ open}\}.$$

(2) The measure μ is "inner regular" on all open A, which means

$$\mu(A) = \sup\{\mu(C) \colon C \subset A \text{ compact}\}.$$

(3) $\mu(C) < \infty$ for all compact subsets $C \subset G$.

Everything we have stated so far in fact holds for any locally compact space. We now invoke the group structure.

Definition 2.49 Let G be a locally compact abelian group. A Haar measure on G is a nonzero Borel measure $\mu : \mathcal{B} \longrightarrow [0, \infty]$ satisfying conditions (1)–(3) above which is furthermore translation-invariant. That is $\mu(gA) = \mu(A)$ for all $g \in G$ and $A \in \mathcal{B}$.

We assume G is abelian since that suffices for our purposes. For non-abelian locally compact groups one can still make sense of left/right Haar measures; they do not coincide in general.

The big Theorem here is Haar's existence and uniqueness result:

Theorem 2.50 *Any locally compact abelian group G admits a Haar measure. It is uniquely determined up to a positive constant.*

Proof Skipped. We refer the interested reader to [RV99, Sect. 1.2]. □

This is one of those Theorems which is very easy to use, but a real pain to prove. The proof goes via the Riesz representation theorem, so one actually constructs a G-invariant positive functional $L : \mathcal{C}_c(G) \longrightarrow \mathbb{C}$. Here G acts on $\mathcal{C}_c(G)$ by $(gf)(x) = f(g^{-1}x)$ and by L being G-invariant we mean $L(gf) = L(f)$. Note that $g\mathrm{Ch}_A = \mathrm{Ch}_{gA}$ so this amounts to $\mu(gA) = \mu(A)$. Here and elsewhere in this book, Ch_A denotes the characteristic function of A, i.e.

$$\mathrm{Ch}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Since a Haar measure is only defined up to a positive constant, the normalization of measures will soon become a recurring theme.

Example 2.51 As a toy example consider a finite abelian group G. In this case it is often convenient to normalize μ such that $\mu(G) = 1$ ("probability measure"). Thus $\mu(A) = \frac{|A|}{|G|}$ for any subset A. The corresponding functional is given by averaging– that is $L(f) = \frac{1}{|G|}\sum_{x\in G} f(x)$ for all $f \in \mathcal{C}(G)$.

Exercise 2.52 Let G be a Hausdorff topological group and consider the counting measure $\mu_{\mathrm{ct}}(A) = |A|$. First check μ_{ct} is translation-invariant. Then show μ_{ct} is outer regular if and only if G is discrete.

For $\mathbb{R}$ and $\mathbb{C}$ the Lebesgue measure is a Haar measure. When v is non-archimedean K_v and $K_v^\times$ are both locally profinite groups, and a Haar measure can be constructed quite explicitly by hand as we will now sketch. (A topological group G is locally profinite if it has an open subgroup which is profinite in the induced topology. Think of the examples $\mathcal{O}_v \subset K_v$ and $\mathcal{O}_v^\times \subset K_v^\times$.)

First we tackle the profinite case, and for technical reasons we assume G has a *countable* neighborhood basis at the identity consisting of open normal subgroups of finite index, say $N_1 > N_2 > \cdots > N_r > \cdots$. (For instance take $N_r = \mathfrak{p}_v^r$ in the case $G = \mathcal{O}_v$.) This assumption guarantees G is metrizable; a metric is given by $d(x, y) = 2^{-\max\{r>0:\, x^{-1}y\in N_r\}}$. This will allow us to use the Heine-Cantor Theorem on uniform continuity of continuous functions on a compact metric space.

We now start building a G-invariant functional $L : \mathcal{C}(G) \to \mathbb{C}$. From the previous example we have the averaging functionals $L_r : \mathcal{C}(G/N_r) \to \mathbb{C}$. Because of our "probability" normalization these L_r are compatible and glue to a functional $L : \mathcal{C}^\infty(G) \to \mathbb{C}$ on the space $\mathcal{C}^\infty(G) = \bigcup_{r=1}^\infty \mathcal{C}(G/N_r)$ of locally constant functions. It has the property $|L(f)| \le \|f\|$ where $\|f\| = \sup_{x\in G} |f(x)|$, and this permits us to extend L to all of $\mathcal{C}(G)$ once we check $\mathcal{C}^\infty(G) \subset \mathcal{C}(G)$ is dense. This density follows from the Heine-Cantor Theorem: Any $f \in \mathcal{C}(G)$ is uniformly continuous so given $\epsilon > 0$ there is a large enough r such that $x^{-1}y \in N_r \Rightarrow |f(x) - f(y)| < \epsilon$. Pick a decomposition into cosets $G = \bigsqcup gN_r$ and let $\tilde{f} \in \mathcal{C}(G/N_r)$ be the function such that $\tilde{f}(g) = f(g)$ for all representatives $g \in G/N_r$. Then $\|f - \tilde{f}\| < \epsilon$.

In summary, for a (first-countable) profinite group G we have constructed a G-invariant positive functional $L : \mathcal{C}(G) \to \mathbb{C}$. The associated Haar measure is given by $\mu(A) = L(\mathrm{Ch}_A)$. Note that $\mu(G) = 1$, so this μ is the probability measure on G.

We finally address the locally profinite case. Choose a profinite open subgroup $G_0 \subset G$ once and for all, and endow it with the probability measure μ_0 we just constructed. Decompose G into cosets $G = \bigsqcup yG_0$. We will always assume G/G_0 is countable. Any Borel set $A \subset G$ then decomposes as a countable disjoint union

$$A = \bigsqcup_{y \in G/G_0} A \cap yG_0 = \bigsqcup_{y \in G/G_0} y(y^{-1}A \cap G_0).$$

If μ is a translation-invariant measure on G extending μ_0 we must have

$$\mu(A) = \sum_{y \in G/G_0} \mu_0(y^{-1}A \cap G_0).$$

Conversely, one easily checks this actually defines a Haar measure μ on G, and $\mu(G_0) = 1$.

Exercise 2.53 Explain why $\mathbb{Q}_p/\mathbb{Z}_p$ and $\mathbb{Q}_p^\times/\mathbb{Z}_p^\times$ are countable, but $\mathbb{Q}_p$ is not.

Exercise 2.54 Let ν be a Haar measure on a finite abelian group G. Show that $\nu = |G|\nu(\{e\}) \cdot \mu$ where μ is the probability measure. (The measure ν with $\nu(\{e\}) = 1$ is called the counting measure.)

2.3.3 Fourier Transforms

Let G be a locally compact abelian group. We now switch to additive notation. We equip G with a Haar measure μ. Thus $\mu(g + A) = \mu(A)$ for Borel sets A.

The Fourier transformation of an integrable function $f \in L^1(G)$ is a function on the dual group $\widehat{G}$ defined by the integral

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}d\mu(x).$$

For a preferred measure $\widehat{\mu}$ on $\widehat{G}$ one can repeat this process and consider the Fourier transformation $\hat{\hat{f}}$ on $\hat{\hat{G}}$, granted $\widehat{f}$ is again integrable. Via Pontryagin duality $G \xrightarrow{\sim} \hat{\hat{G}}$ one identifies $\hat{\hat{f}}$ with a function on G. Fourier inversion says there is a unique choice of $\widehat{\mu}$ such that the relation $\hat{\hat{f}}(x) = f(-x)$ holds almost everywhere on G (outside a set of measure zero):

Theorem 2.55 (The Fourier inversion formula) *Fix G and μ as above. There is a unique dual Haar measure $\widehat{\mu}$ on $\widehat{G}$ such that for all $f \in L^1(G)$ with $\widehat{f} \in L^1(\widehat{G})$ the equality*

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x)d\widehat{\mu}(\chi)$$

holds μ-almost everywhere on G (everywhere if f in addition is continuous).

Proof See [RV99, Sect. 3.3]. □

Exercise 2.56 Check that $\widehat{c\mu} = \frac{1}{c}\widehat{\mu}$ for $c > 0$.

Exercise 2.57 Suppose G is compact (and therefore $\widehat{G}$ is discrete). Show that the probability measure on G is dual to the counting measure on $\widehat{G}$. (`Hint`: Compute the Fourier transformation of $f = \mathrm{Ch}_G$.)

General properties of the Gelfand transformation (applied to the Banach algebra $L^1(G)$) tell us that the Fourier transformation $\widehat{(\cdot)} : L^1(G) \longrightarrow \mathcal{C}_0(\widehat{G})$ lands in the space of continuous functions vanishing at infinity, which means the subset $\{\chi \in \widehat{G} : |\widehat{f}(\chi)| \geq \epsilon\}$ is compact $\forall \epsilon > 0$ (provided f is integrable). See the last paragraph of [Ped89, Ex. 4.2.8, p. 140] for example. This is the Riemann-Lebesgue Lemma in this context.

Exercise 2.58 Show that $\widehat{f}$ is a bounded *continuous* function on $\widehat{G}$ for all $f \in L^1(G)$. (`Hint`: Let $\epsilon > 0$. By regularity pick a compact subset $C \subset G$ such that $\int_{G-C} |f(x)|d\mu(x) < \epsilon$. Consider $W(C, U)$ for small enough neighborhoods $1 \in U \subset \mathbb{C}^1$. Now estimate $|\widehat{f}(\chi_1) - \widehat{f}(\chi_2)|$ assuming $\chi_1^{-1}\chi_2 \in W(C, U)$.)

We now assume G is self-dual and pick an isomorphism $B : G \xrightarrow{\sim} \widehat{G}$ of topological groups. This excludes infinite compact/discrete groups. Our main examples are K_v and $\mathbb{A}_K$, where B depends on the choice of a non-trivial additive character ψ. We use the same notation B for the associated perfect pairing $G \times G \longrightarrow \mathbb{C}^1$ given by $(x, y) \mapsto B(x)(y)$.

A measure μ on G has a pushforward measure $B_*\mu$ on $\widehat{G}$ defined by $(B_*\mu)(A) = \mu(B^{-1}(A))$. We say that μ is self-dual (relative to B) if $B_*\mu = \widehat{\mu}$.

Exercise 2.59 Suppose G admits a self-duality $B : G \xrightarrow{\sim} \widehat{G}$. Prove that there exists a literally unique (not just up to a constant) Haar measure μ on G which is self-dual relative to B. (`Hint`: Fix a Haar measure ν on G and write $B_*\nu = \beta\widehat{\nu}$ for some $\beta > 0$. Find the $c > 0$ for which $B_*(c\nu) = \widehat{c\nu}$.)

Via B we may view $\widehat{f}$ as a function on G:

$$\widehat{f}(y) = \int_G f(x)\overline{B(y, x)}d\mu(x).$$

(Here the characters of G are parametrized as $\chi_y = B(y, -)$ as $y \in G$ varies.) Taking μ to be the self-dual Haar measure on G, with respect to B, the Fourier inversion formula translates into

$$f(x) = \int_G \widehat{f}(y) B(y, x) d\mu(y)$$

a.e., provided f and $\widehat{f}$ are integrable.

2.3.4 The General Poisson Summation Formula

Our setup will be the following. For now G denotes an arbitrary locally compact abelian group endowed with a choice of Haar measure μ. A new player enters: Suppose $\Gamma \subset G$ is a discrete subgroup such that G/Γ is compact (we say Γ is cocompact). Note that $\widehat{G/\Gamma}$ is then a discrete subgroup of $\widehat{G}$, consisting of all characters χ satisfying $\chi|_\Gamma = 1$. Think of $\mathbb{Z} \subset \mathbb{R}$ or our main example $K \hookrightarrow \mathbb{A}_K$.

The compact group G/Γ carries the quotient measure $\dot{\mu}$ determined by the formula

$$\int_{G/\Gamma} \left(\sum_{\gamma \in \Gamma} f(x + \gamma) \right) d\dot{\mu}(x) = \int_G f(x) d\mu(x)$$

valid at least for $f \in \mathcal{C}_c(G)$.

Theorem 2.60 (Poisson summation) *For all $f \in \mathcal{C}_c(G)$ with $\hat{f}$ integrable we have the identity*

$$\sum\nolimits_{\gamma \in \Gamma} f(\gamma) = \frac{1}{\dot{\mu}(G/\Gamma)} \sum\nolimits_{\chi \in \widehat{G/\Gamma}} \widehat{f}(\chi).$$

Proof Introduce the averaging function $F(x) = \sum_{\gamma \in \Gamma} f(x + \gamma)$ belonging to $\mathcal{C}(G/\Gamma)$. Consider its Fourier transformation

$$\begin{aligned}
\widehat{F}(\chi) &= \int_{G/\Gamma} \left(\sum_{\gamma \in \Gamma} f(x + \gamma) \right) \overline{\chi(x)} d\dot{\mu}(x) \\
&= \int_{G/\Gamma} \left(\sum_{\gamma \in \Gamma} f(x + \gamma) \overline{\chi(x + \gamma)} \right) d\dot{\mu}(x) \\
&= \int_G f(x) \overline{\chi(x)} d\mu(x) \\
&= \widehat{f}(\chi)
\end{aligned}$$

for $\chi \in \widehat{G/\Gamma}$. We used $\chi(\gamma) = 1$ for $\gamma \in \Gamma$ in the second step above. Therefore, by the Fourier inversion formula $\forall x \in G/\Gamma$ we have

$$F(x) = \sum\nolimits_{\chi \in \widehat{G/\Gamma}} \widehat{f}(\chi)\chi(x)$$

provided the counting measure on $\widehat{G/\Gamma}$ is dual to $\dot{\mu}$. In other words $\dot{\mu}$ must be the probability measure on G/Γ. Substituting $x = 0$ gives the result $F(0) = \sum_{\gamma\in\Gamma} f(\gamma) = \sum_{\chi\in\widehat{G/\Gamma}} \widehat{f}(\chi)$. If $\dot{\mu}$ is not the probability measure one must normalize the right-hand side as in the Theorem. □

The Poisson summation formula is more of a modus operandi. The hypothesis that $f \in \mathcal{C}_c(G)$ is too restrictive, and eventually we will apply the formula to (generalizations of) Schwartz functions. As long as the test function f has rapid decay this guarantees all sums in the proof converge. The problem is to formulate a suitable decay condition for an abstract G, but we will not pursue this.

We revisit the self-dual situation, and again assume G admits an isomorphism $B : G \xrightarrow{\sim} \widehat{G}$. Introduce the notation

$$\Gamma^{\perp} = \{y \in G : B(y, \Gamma) = 1\}$$

for the "orthogonal complement" of Γ in G. By definition there is an isomorphism $\Gamma^{\perp} \xrightarrow{\sim} \widehat{G/\Gamma}$ induced by restricting B, and $G/\Gamma^{\perp} \xrightarrow{\sim} \widehat{\Gamma}$. In particular $\Gamma^{\perp}$ is also a discrete co-compact subgroup of G.

Exercise 2.61 Let $G = \mathbb{R}$ and $B(x, y) = e^{-2\pi i x y}$. Consider the subgroup $\Gamma = N\mathbb{Z}$ for some $N \in \mathbb{R}_{>0}$. Find the $N^{\perp} \in \mathbb{R}_{>0}$ such that $\Gamma^{\perp} = N^{\perp}\mathbb{Z}$. For which N is $\Gamma^{\perp} = \Gamma$?

In terms of $\widehat{f}(y) = \int_G f(x)\overline{B(y, x)}d\mu(x)$ the Poisson summation formula may now be restated as follows.

Corollary 2.62 *We adopt the preceding notation and assumptions. Then for all $f \in \mathcal{C}_c(G)$ with $\hat{f}$ integrable we have*

$$\sum\nolimits_{\gamma\in\Gamma} f(\gamma) = \frac{1}{\dot{\mu}(G/\Gamma)} \sum\nolimits_{y\in\Gamma^{\perp}} \widehat{f}(y).$$

Proof This follows immediately from the isomorphism $\Gamma^{\perp} \xrightarrow{\sim} \widehat{G/\Gamma}$. □

Exercise 2.63 Suppose $\Gamma = \Gamma^{\perp}$. Show that μ is self-dual with respect to B if and only if $\dot{\mu}(G/\Gamma) = 1$. (`Hint`: Apply the Poisson summation formula twice to a suitable f and its $\widehat{f}$.)

Here is our motivating example:

Example 2.64 We take $G = \mathbb{A}_K$ and $\Gamma = K$ in the above discussion. As noted earlier the choice of a non-trivial character $\psi : \mathbb{A}_K/K \to \mathbb{C}^1$ gives a self-duality $B_{\psi} : \mathbb{A}_K \xrightarrow{\sim} \widehat{\mathbb{A}_K}$ via the pairing $(x, y) \mapsto \psi(xy)$. Moreover B_{ψ} restricts to an isomorphism $K \xrightarrow{\sim} \widehat{\mathbb{A}_K/K}$. In other words $K^{\perp} = K$ regardless of the choice of ψ–as long as $\psi|_K = 1$ of course.

Pick μ to be the self-dual measure on $\mathbb{A}_K$, relative to any ψ. In other words the measure for which $\dot{\mu}(\mathbb{A}_K/K) = 1$. In terms of $\widehat{f}(y) = \int_{\mathbb{A}_K} f(x)\overline{\psi(xy)}d\mu(x)$ the Poisson summation formula says that

$$\sum_{\gamma\in K} f(\gamma) = \sum_{\gamma\in K} \widehat{f}(\gamma)$$

for all test functions f of rapid decay (a notion to be made precise later when we discuss Schwartz-Bruhat functions on $\mathbb{A}_K$). This simple formula will essentially give us the functional equation for Hecke L-functions.

2.4 The Local Theory

2.4.1 Measures on Local Fields

For a while we will be working purely locally, and we switch to local notation. Thus K denotes a local field of characteristic 0, with normalized absolute value $|\cdot|$. If K is non-archimedean $\mathcal{O}$ denotes its valuation ring, $\mathfrak{p} = (\varpi)$ is the maximal ideal thereof, and $k = \mathcal{O}/\mathfrak{p} \simeq \mathbb{F}_q$ is the residue field.

Since $(K, +)$ is locally compact it admits a Haar measure μ, unique modulo $\mathbb{R}^\times_{>0}$. The following important observation justifies our normalization of $|\cdot|$.

Lemma 2.65 *$\mu(aA) = |a|\mu(A)$ for all Borel sets $A \subset K$ and $a \in K^\times$.*

Proof Fix an $a \in K^\times$. Then $A \mapsto \mu(aA)$ is a Haar measure on K, and therefore there exists a constant $c_a \in \mathbb{R}^\times_{>0}$ such that $\mu(aA) = c_a\mu(A)$ for all A. When $K = \mathbb{R}$ and $A = [0, 1]$ this shows $c_a = |a|$. When $K = \mathbb{C}$ and A is the unit disc centered at the origin this shows $c_a = |a| = |a|^2_{\mathbb{C}}$ (the radius is being scaled by $|a|_{\mathbb{C}}$ so the area of the disc is being scaled by $|a|^2_{\mathbb{C}}$).

Finally, if $K/\mathbb{Q}_p$ is a finite extension we partition $\mathcal{O}$ into cosets $\mathcal{O} = \bigsqcup_x x + \varpi\mathcal{O}$ where x runs over a set of representatives for $k = \mathcal{O}/\mathfrak{p}$. Then, by the translation-invariance of μ, we have

$$\mu(\mathcal{O}) = \sum_x \mu(x + \varpi\mathcal{O}) = \sum_x \mu(\varpi\mathcal{O}) = q\mu(\varpi\mathcal{O}).$$

In other words $\mu(\varpi\mathcal{O}) = q^{-1}\mu(\mathcal{O}) = |\varpi|\mu(\mathcal{O})$. Clearly $a \mapsto c_a$ is multiplicative and trivial on $\mathcal{O}^\times$. As we have just verified $c_\varpi = |\varpi|$. Therefore $c_a = |a|$ for all nonzero $a \in K$. □

We give another interpretation of this result. For $a \in K^\times$ consider the map $m_a : K \xrightarrow{\sim} K$ sending $x \mapsto ax$. In terms of the pushforward measure the previous Lemma says that $m_{a*}\mu = |a|^{-1}\mu$.

The choice of a Haar measure μ on the additive group K determines a Haar measure on the multiplicative group $K^\times$ as follows.

$$\mu^\times(A) = \int_A \frac{d\mu(x)}{|x|}$$

for Borel sets $A \subset K^\times$. (The invariance of $\mu^\times$ follows from Lemma 2.65.) We will sometimes summarize the definition as $d\mu^\times(x) = \frac{d\mu(x)}{|x|}$.

Example 2.66 In this example we assume K is non-archimedean. Our goal is to express $\mu^\times(\mathcal{O}^\times)$ in terms of $\mu(\mathcal{O})$. We decompose $\mathcal{O} - \{0\} = \bigsqcup_{r=0}^\infty \varpi^r \mathcal{O}^\times$. Removing the point 0 does not change the measure since $\mu(\{0\}) = 0$, and we obtain

$$\mu(\mathcal{O}) = \mu(\mathcal{O} - \{0\}) = \sum_{r=0}^\infty \mu(\varpi^r \mathcal{O}^\times) = \sum_{r=0}^\infty |\varpi|^r \mu(\mathcal{O}^\times) = \mu(\mathcal{O}^\times) \sum_{r=0}^\infty q^{-r} = \mu(\mathcal{O}^\times) \frac{1}{1 - q^{-1}}.$$

On the other hand $\mu^\times(\mathcal{O}^\times) = \int_{\mathcal{O}^\times} \frac{d\mu(x)}{|x|} = \int_{\mathcal{O}^\times} d\mu(x) = \mu(\mathcal{O}^\times)$. We conclude that

$$\mu^\times(\mathcal{O}^\times) = (1 - q^{-1})\mu(\mathcal{O}).$$

In particular the normalization $\mu(\mathcal{O}) = 1$ does *not* imply $\mu^\times(\mathcal{O}^\times) = 1$.

We embark on a discussion of Fourier theory for K. To get the ball rolling we fix a non-trivial continuous character $\psi : K \longrightarrow \mathbb{C}^1$ which gives a self-duality $B_\psi : K \xrightarrow{\sim} \widehat{K}$ sending $a \mapsto \psi_a$. The associated bilinear pairing $K \times K \longrightarrow \mathbb{C}^1$ takes $(a, b) \mapsto \psi(ab)$.

Take μ to be the Haar measure on K which is self-dual with respect to B_ψ. The Fourier transformation of $f \in L^1(K)$ is given by the integral

$$\widehat{f}(y) = \int_K f(x)\psi(xy)d\mu(x).$$

(The alert reader will notice a discrepancy with the conventions in Sect. 2.3.3. This is intentional. With our earlier conventions we should really have a factor $\overline{\psi(xy)}$ in the integral, so in essence we have replaced ψ by $\bar{\psi} = \psi_{-1}$. This does not change the measure μ as the reader will verify below in Exercise 2.67.)

If $\widehat{f}$ is again integrable the Fourier inversion formula holds:

$$f(x) = \int_K \widehat{f}(y)\overline{\psi(xy)}d\mu(y).$$

For this to really be computationally useful we need to understand the self-dual measure better. The following exercise reduces this to understanding the self-dual measure relative to the standard character ψ_{std}.

Exercise 2.67 Let $a \in K^\times$. Observe that $B_{\psi_a} = B_\psi \circ m_a$. Use this to show μ is self-dual relative to B_ψ if and only if $\sqrt{|a|}\mu$ is self-dual relative to B_{ψ_a}.

Let $K = \mathbb{R}$. Recall that $\psi_{\text{std}} = e^{-2\pi i(\cdot)}$. In this case the Lebesgue measure is self-dual relative to $B_{\psi_{\text{std}}}$ as we know from Fourier analysis on $\mathbb{R}$.

Let $K = \mathbb{C}$. In this case $\psi_{\text{std}} = e^{-2\pi i \text{Tr}_{\mathbb{C}/\mathbb{R}}(\cdot)} = e^{-4\pi i \text{Re}(\cdot)}$. Here the *double* Lebesgue measure on $\mathbb{C}$ is self-dual relative to $B_{\psi_{\text{std}}}$. This may come as a surprise so we work out the details. We will use the fact that the Gaussian $e^{-\pi\|x\|^2}$ on $\mathbb{R}^n$ is its own Fourier transform, relative to the Lebesgue measure on $\mathbb{R}^n$. We saw this for $n = 1$ and the general case easily reduces to the one-dimensional case. Let λ denote Lebesgue measure on $\mathbb{C}$ and let us compute the Fourier transformation of a "complex" Gaussian $f(z) = e^{-\pi c z\bar{z}}$ with respect to 2λ. Here $c > 0$ is any constant, which gives us some flexibility. Let $w = u + iv$.

$$\begin{aligned}\widehat{f}(w) &= \int_{\mathbb{C}} f(z)e^{-4\pi i \text{Re}(zw)} 2d\lambda(z) \\ &= \int_{\mathbb{R}^2} f(x, y)e^{-2\pi i (x,y)\bullet(2u,-2v)} 2dxdy \\ &= \frac{2}{c} e^{-\frac{\pi}{c}\|(2u,-2v)\|^2} \\ &= \frac{2}{c} e^{-4\pi w\bar{w}/c}.\end{aligned}$$

(In the third step we used that $e^{-\pi c\|x\|^2}$ on $\mathbb{R}^n$ has Fourier transform $c^{-n/2}e^{-\frac{\pi}{c}\|x\|^2}$ relative to Lebesgue, as follows from example 1.24.) For $c = 2$ we infer that $e^{-2\pi z\bar{z}}$ is its own Fourier transformation, but relative to 2λ. This allows us to deduce 2λ must be the self-dual measure on $\mathbb{C}$, relative to ψ_{std}.

Now let $K/\mathbb{Q}_p$ be a p-adic field. Here $\psi_{\text{std}} = \psi_{\mathbb{Q}_p} \circ \text{Tr}_{K/\mathbb{Q}_p}$. To pin down the self-dual measure μ we calculate $\mu(\mathcal{O})$. In the process we compute some widely used non-archimedean Fourier transforms:

Lemma 2.68 (Uncertainty principle) *Let ψ be any non-trivial character of K, and μ any Haar measure. Then for arbitrary $a \in K$ and $r \in \mathbb{Z}$ we have*

$$\widehat{Ch}_{a+\mathfrak{p}^r} = q^{-r}\mu(\mathcal{O})\psi_a Ch_{\mathfrak{p}^{cond(\psi)-r}}.$$

In particular $\widehat{Ch}_{\mathcal{O}} = \mu(\mathcal{O})Ch_{\mathfrak{p}^{cond(\psi)}}$. Moreover, the measure μ is self-dual with respect to ψ precisely when

$$\mu(\mathcal{O}) = q^{\frac{1}{2}cond(\psi)}.$$

Proof By definition

$$\begin{aligned}\widehat{\text{Ch}}_{a+\mathfrak{p}^r}(y) &= \int_{a+\mathfrak{p}^r} \psi(xy)d\mu(x) \\ &= \int_{\mathfrak{p}^r} \psi((a + x')y)d\mu(x')\end{aligned}$$

$$= \psi_a(y) \int_{\mathfrak{p}^r} \psi(x'y) d\mu(x').$$

If $y \in \mathfrak{p}^{\mathrm{cond}(\psi)-r}$ we are integrating 1 over $\mathfrak{p}^r$ and thus

$$\widehat{\mathrm{Ch}}_{a+\mathfrak{p}^r}(y) = \psi_a(y)\mu(\mathfrak{p}^r) = q^{-r}\mu(\mathcal{O})\psi_a(y).$$

On the other hand, if $y \notin \mathfrak{p}^{\mathrm{cond}(\psi)-r}$ we are integrating the non-trivial character ψ_y on $\mathfrak{p}^r$. This of course comes out to zero.

Applying the formula twice yields

$$\widehat{\widehat{\mathrm{Ch}}}_{\mathcal{O}} = \mu(\mathcal{O})\widehat{\mathrm{Ch}}_{\mathfrak{p}^{\mathrm{cond}(\psi)}} = \mu(\mathcal{O})^2 q^{-\mathrm{cond}(\psi)} \mathrm{Ch}_{\mathcal{O}}.$$

Therefore μ is self-dual with respect to ψ exactly when $\mu(\mathcal{O})^2 q^{-\mathrm{cond}(\psi)} = 1$. □

Fourier inversion immediately gives the Fourier transform of a truncated additive character. This will be very useful later on. Let us record it here:

Lemma 2.69 *Let ψ be any non-trivial character of K, and μ any Haar measure. Then for all $a \in K$ and $r \in \mathbb{Z}$ we have*

$$\widehat{\psi_a \mathrm{Ch}_{\mathfrak{p}^{-r}}} = q^r \mu(\mathcal{O}) \mathrm{Ch}_{-a+\mathfrak{p}^{\mathrm{cond}(\psi)+r}}.$$

Proof Let $s = r - \mathrm{cond}(\psi)$ in Lemma 2.68 and take the Fourier transform. When μ is self-dual, Fourier inversion applies:

$$\mathrm{Ch}_{-a+\mathfrak{p}^{s+\mathrm{cond}(\psi)}} = \widehat{\widehat{\mathrm{Ch}}}_{a+\mathfrak{p}^{s+\mathrm{cond}(\psi)}} = q^{-(s+\mathrm{cond}(\psi))}\mu(\mathcal{O})\widehat{\psi_a \mathrm{Ch}_{\mathfrak{p}^{-s}}}$$

Now insert $\mu(\mathcal{O}) = q^{\frac{1}{2}\mathrm{cond}(\psi)}$ and move symbols around. For a general μ the result follows by scaling. □

Recall that $\mathfrak{p}^{\mathrm{cond}(\psi_{\mathrm{std}})} = \partial^{-1}_{K/\mathbb{Q}_p}$ is the inverse different, and therefore $q^{\mathrm{cond}(\psi_{\mathrm{std}})} = N(\partial_{K/\mathbb{Q}_p})^{-1}$. In the case of the standard character, μ is therefore self-dual if and only if $\mu(\mathcal{O}) = \sqrt{N(\partial_{K/\mathbb{Q}_p})}^{-1}$.

2.4.2 Local Zeta Integrals

We have already defined Schwartz functions on $\mathbb{R}$. The definition carries over to $\mathbb{R}^n$ almost verbatim. The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of $\mathcal{C}^\infty$-functions $f : \mathbb{R}^n \to \mathbb{C}$ of rapid decay, which means

$$\|f\|_{\alpha,\beta} = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial_\beta f(x)| < \infty$$

for all $\alpha, \beta \in \mathbb{Z}^n_{\geq 0}$. Here $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ and $\partial_\beta = \left(\frac{\partial}{\partial x_1}\right)^{\beta_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\beta_n}$. Via the identification $\mathbb{C} \simeq \mathbb{R}^2$ this defines the Schwartz space $\mathcal{S}(\mathbb{C})$ of functions on the complex plane.

For a p-adic field K the definition of Schwartz functions is much simpler. One simply takes $\mathcal{S}(K)$ to be the space $\mathcal{C}_c^\infty(K)$ of all compactly supported locally constant functions $f : K \to \mathbb{C}$. For instance $\mathrm{Ch}_{a+\mathfrak{p}^r}$ belongs to $\mathcal{S}(K)$ for all $a \in K$ and $r \in \mathbb{Z}$.

Exercise 2.70 Prove that the characteristic functions $\mathrm{Ch}_{a+\mathfrak{p}^r}$ span $\mathcal{S}(K)$ as $a \in K$ and $r \in \mathbb{Z}$ vary. (`Hint`: Pick $R \gg 0$ such that $\mathrm{supp}(f) \subset \mathfrak{p}^{-R}$. Now consider the coset decomposition $\mathfrak{p}^{-R} = \bigsqcup a + \mathfrak{p}^r$ for large enough r.)

Combining this exercise with Lemma 2.68 shows that $\widehat{(\cdot)}$ is an automorphism of $\mathcal{S}(K)$ also in the non-archimedean case.

We now enter the heart of Tate's thesis and define zeta integrals. They provide a uniform explanation of the shape of our local L-factors $L(\chi)$ which we defined in a somewhat ad hoc manner. K is any local field of characteristic 0, and $\chi : K^\times \to \mathbb{C}^\times$ is a continuous character.

Lemma 2.71 *Let $f \in \mathcal{S}(K)$ be an arbitrary Schwartz function, and fix a Haar measure $\mu^\times$ on $K^\times$. If $\chi : K^\times \to \mathbb{C}^\times$ has exponent $a_\chi > 0$ the local zeta integral*

$$Z(f, \chi) = \int_{K^\times} f(x)\chi(x) d\mu^\times(x)$$

is absolutely convergent.

Proof Let $a = a_\chi > 0$ to simplify notation. For a Haar measure μ on K we must show that

$$\int_{K^\times} |f(x)| \cdot |x|^a \frac{d\mu(x)}{|x|} < \infty.$$

This is easy for $K = \mathbb{R}$ and $K = \mathbb{C}$. Indeed, since f has rapid decay there is no problem at infinity, and at $x = 0$ (where f is defined) the integral compares to $\int_0^1 x^{a-1} dx < \infty$.

For non-archimedean K we decompose the multiplicative group into shells $K^\times = \bigsqcup_{r \in \mathbb{Z}} \varpi^r \mathcal{O}^\times$. The above integral then splits up as a sum of terms

$$\int_{\varpi^r \mathcal{O}^\times} |f(x)| \cdot |x|^{a-1} d\mu(x) = q^{-ar} \int_{\mathcal{O}^\times} |f(\varpi^r u)| d\mu(u).$$

Since $\mathrm{supp}(f) \subset \mathfrak{p}^{-R}$ for some large enough R the above rth term vanishes for $r < -R$. At the other extreme, since f is defined at 0 (and locally constant) $f(\varpi^r u) = f(0)$ for $u \in \mathcal{O}$ and $r \gg 0$. In this range the rth term is $q^{-ar}|f(0)|\mu(\mathcal{O}^\times)$. It remains to note $\sum_{r \gg 0} q^{-ar}$ is a finite geometric sum since $a > 0$. □

One of the fundamental ideas behind Tate's approach is to compare $Z(f, \chi)$ with $Z(\widehat{f}, \chi^\vee)$ where we recall the shifted dual character $\chi^\vee(x) = \chi(x)^{-1}|x|$. When

$0 < a_\chi < 1$ both integrals converge absolutely. It turns out their ratio is completely independent of f, and this is the key to defining the local ϵ-factors.

2.4.3 Archimedean Computations

We first take $K = \mathbb{R}$ and the character $\chi(x) = \chi_{\mathrm{e},s}(x) = \mathrm{sign}(x)^{\mathrm{e}}|x|^s$ where $\mathrm{e} \in \{0, 1\}$ and $s \in \mathbb{C}$. Our preferred Schwartz function will be the Gaussian $f_0(x) = e^{-\pi x^2}$ or its variant $f_1(x) = xe^{-\pi x^2}$ according to whether χ is even or odd.

Lemma 2.72 *For $e \in \{0, 1\}$ let $f_e(x) = x^e e^{-\pi x^2}$. Then $Z(f_e, \chi_{e,s}) = L_{\mathbb{R}}(s + e)$ for $Re(s) > 0$. (Here we take the measure $\frac{dx}{|x|}$ on $\mathbb{R}^\times$ where dx is Lebesgue.)*

Proof This is a straightforward calculation. The reader should be vigilant with the different meanings of $e = \exp(1)$ and $\mathrm{e} \in \{0, 1\}$ here:

$$\begin{aligned}
Z(f_{\mathrm{e}}, \chi_{\mathrm{e},s}) &= \int_{\mathbb{R}^\times} x^{\mathrm{e}} e^{-\pi x^2} \mathrm{sign}(x)^{\mathrm{e}} |x|^s \frac{dx}{|x|} \\
&= \int_{\mathbb{R}^\times} e^{-\pi x^2} |x|^{s+\mathrm{e}} \frac{dx}{|x|} \\
&= 2\int_0^\infty e^{-\pi x^2} x^{s+\mathrm{e}} \frac{dx}{x} \\
&= 2\int_0^\infty e^{-t} \left(\frac{t}{\pi}\right)^{\frac{s+\mathrm{e}}{2}} \frac{dx}{x} \\
&= \int_0^\infty e^{-t} \left(\frac{t}{\pi}\right)^{\frac{s+\mathrm{e}}{2}} \frac{dt}{t} \\
&= \pi^{-\frac{s+\mathrm{e}}{2}} \int_0^\infty e^{-t} t^{\frac{s+\mathrm{e}}{2}-1} dt \\
&= \pi^{-\frac{s+\mathrm{e}}{2}} \Gamma\left(\frac{s+\mathrm{e}}{2}\right) \\
&= L_{\mathbb{R}}(s + \mathrm{e}).
\end{aligned}$$

We used $x^{\mathrm{e}} = \mathrm{sign}(x)^{\mathrm{e}}|x|^{\mathrm{e}}$ and substituted $t = \pi x^2$ so $\frac{dt}{t} = 2\frac{dx}{x}$. □

Since $f_0(x) = e^{-\pi x^2}$ is its own Fourier transformation (relative to ψ_{std}) it follows by differentiation that $f_1(x) = xe^{-\pi x^2}$ has Fourier transform $\widehat{f_1} = i^{-1} f_1$. We summarize this as $\widehat{f_{\mathrm{e}}} = i^{-\mathrm{e}} f_{\mathrm{e}}$.

Noting that $\chi^\vee_{\mathrm{e},s} = \chi_{\mathrm{e},1-s}$ we deduce $Z(\widehat{f_{\mathrm{e}}}, \chi^\vee_{\mathrm{e},s}) = i^{-\mathrm{e}} L_{\mathbb{R}}(1 - s + \mathrm{e})$ for $\mathrm{Re}(s) < 1$. In particular the ratio of zeta integrals is

$$\frac{Z(\widehat{f_{\mathrm{e}}}, \chi^\vee_{\mathrm{e},s})}{Z(f_{\mathrm{e}}, \chi_{\mathrm{e},s})} = i^{-\mathrm{e}} \frac{L_{\mathbb{R}}(1 - s + \mathrm{e})}{L_{\mathbb{R}}(s + \mathrm{e})}$$

provided $0 < \mathrm{Re}(s) < 1$. The right-hand side meromorphically continues the ratio to all of $\mathbb{C}$. We will later refer to this as the local γ-factor.

Remark 2.73 The $i^{-\mathrm{e}}$ above is the "same" factor appearing in the root number of a primitive Dirichlet character.

Next we deal with $K = \mathbb{C}$. We take the character $\chi(x) = \chi_{n,s}(x) = \big(\frac{x}{|x|_{\mathbb{C}}}\big)^n |x|^s$ with $n \in \mathbb{Z}$ and $s \in \mathbb{C}$. We treat the cases $n \geq 0$ and $n \leq 0$ separately. This seems natural in light of the definition of the local L-factor $L(\chi_{n,s}) = L_{\mathbb{C}}(s + \frac{|n|}{2})$ which only sees the absolute value of n.

Lemma 2.74 *For $n \in \mathbb{Z}_{\geq 0}$ let $f_n(z) = \bar{z}^n e^{-2\pi z\bar{z}}$. Then $Z(f_n, \chi_{n,s}) = \pi L_{\mathbb{C}}(s + \frac{n}{2})$ for $Re(s) > 0$. (Here $\mathbb{C}^\times$ carries the measure $\frac{dz}{|z|}$ where $dz = 2dxdy$ is <u>double</u> Lebesgue.)*

Proof Unpacking definitions and switching to polar coordinates we find

$$\begin{aligned}
Z(f_n, \chi_{n,s}) &= \int_{\mathbb{C}^\times} \bar{z}^n e^{-2\pi z\bar{z}} \big(\frac{z}{|z|_{\mathbb{C}}}\big)^n |z|^s \frac{dz}{|z|} \\
&= \int_0^\infty \int_0^{2\pi} r^n e^{-in\theta} e^{-2\pi r^2} e^{in\theta} r^{2s} \frac{2rdrd\theta}{r^2} \\
&= 2\pi \int_0^\infty e^{-2\pi r^2} r^{n+2s} \frac{2dr}{r} \\
&= 2\pi \int_0^\infty e^{-t} \big(\frac{t}{2\pi}\big)^{s+\frac{n}{2}} \frac{dt}{t} \\
&= 2\pi (2\pi)^{-(s+\frac{n}{2})} \int_0^\infty e^{-t} t^{s+\frac{n}{2}-1} dt \\
&= 2\pi (2\pi)^{-(s+\frac{n}{2})} \Gamma(s + \frac{n}{2}) \\
&= \pi L_{\mathbb{C}}(s + \frac{n}{2}).
\end{aligned}$$

We changed variables $t = 2\pi r^2$ so that $\frac{dt}{t} = \frac{2dr}{r}$. □

Exercise 2.75 For $n \in \mathbb{Z}_{\leq 0}$ let $f_n(z) = z^{|n|} e^{-2\pi z\bar{z}}$. Mimic the above calculation and verify the identity $Z(f_n, \chi_{n,s}) = \pi L_{\mathbb{C}}(s + \frac{|n|}{2})$ for $\mathrm{Re}(s) > 0$. (With the same double Lebesgue measure $\frac{dz}{|z|}$ on $\mathbb{C}^\times$.)

We found that $f_0(z) = e^{-2\pi z\bar{z}}$ equals its own Fourier transformation (relative to ψ_{std} and the double Lebesgue measure). Applying the Wirtinger operators $\frac{\partial}{\partial z} = \frac{1}{2}\big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\big)$ and $\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\big)$ a number of times (induction) implies that more generally $\widehat{f_n} = i^{-n} f_{-n}$ for $n \geq 0$. By Fourier inversion this extends even further to $\widehat{f_n} = i^{-|n|} f_{-n}$ for all $n \in \mathbb{Z}$.

Noting that $\chi_{n,s}^\vee = \chi_{-n,1-s}$ we deduce $Z(\widehat{f_n}, \chi_{n,s}^\vee) = i^{-|n|} \pi L_{\mathbb{C}}(1 - s + \frac{|n|}{2})$ for $\mathrm{Re}(s) < 1$. In this case the ratio of zeta integrals is

$$\frac{Z(\widehat{f_n}, \chi_{n,s}^\vee)}{Z(f_n, \chi_{n,s})} = i^{-|n|} \frac{L_{\mathbb{C}}(1-s+\frac{|n|}{2})}{L_{\mathbb{C}}(s+\frac{|n|}{2})}$$

again provided $0 < \mathrm{Re}(s) < 1$ so both zeta integrals converge. This gives the local γ-factor in the complex case once we define those.

2.4.4 Unramified Computations

Now assume K is non-archimedean and let $\chi : K^\times \to \mathbb{C}^\times$ be a continuous character. For the most part we will assume χ is unramified in this section, and in the subsequent section consider ramified χ.

Our favorite Schwartz functions are the characteristic functions of balls $\mathrm{Ch}_{a+\mathfrak{p}^r}$, which as we know span $\mathcal{S}(K)$. We set out to compute their zeta integrals $Z(\mathrm{Ch}_{a+\mathfrak{p}^r}, \chi)$ (relative to some choice of measure $\mu^\times$ on the multiplicative group). Initially we take the balls to be centered at the origin ($a = 0$).

Lemma 2.76 *Let χ be a continuous character of $K^\times$ with exponent $a_\chi > 0$. Then*

$$Z(Ch_{\mathfrak{p}^r}, \chi) = \begin{cases} \mu^\times(\mathcal{O}^\times)\chi(\varpi)^r L(\chi) & \text{if } \chi \text{ is unramified} \\ 0 & \text{if } \chi \text{ is ramified.} \end{cases}$$

Proof Using the shell decomposition $\mathfrak{p}^r - \{0\} = \bigsqcup_{s=r}^\infty \varpi^s \mathcal{O}^\times$ into $\mathcal{O}^\times$-orbits we get

$$\begin{aligned} Z(\mathrm{Ch}_{\mathfrak{p}^r}, \chi) &= \int_{\mathfrak{p}^r - \{0\}} \chi(x) d\mu^\times(x) \\ &= \sum_{s=r}^\infty \int_{\varpi^s \mathcal{O}^\times} \chi(x) d\mu^\times(x) \\ &= \sum_{s=r}^\infty \int_{\mathcal{O}^\times} \chi(\varpi^s u) d\mu^\times(u) \\ &= \sum_{s=r}^\infty \chi(\varpi)^s \int_{\mathcal{O}^\times} \chi(u) d\mu^\times(u). \end{aligned}$$

If χ is ramified $\int_{\mathcal{O}^\times} \chi(u) d\mu^\times(u) = 0$. If χ is unramified this integral is $\mu^\times(\mathcal{O}^\times)$ and the zeta integral becomes a geometric sum $\mu^\times(\mathcal{O}^\times) \sum_{s=r}^\infty \chi(\varpi)^s = \mu^\times(\mathcal{O}^\times)\chi(\varpi)^r \frac{1}{1-\chi(\varpi)}$ provided $|\chi(\varpi)| < 1$. □

In particular for $r = 0$ we find that $Z(\mathrm{Ch}_{\mathcal{O}}, \chi) = \mu^\times(\mathcal{O}^\times) L(\chi)$ for unramified χ.

We also need the zeta integral $Z(\widehat{\mathrm{Ch}}_{\mathfrak{p}^r}, \chi^\vee)$. From Lemma 2.68 with $a = 0$ we know what the Fourier transformation is, namely

$$\widehat{\mathrm{Ch}}_{\mathfrak{p}^r} = q^{-r}\mu(\mathcal{O})\mathrm{Ch}_{\mathfrak{p}^{\mathrm{cond}(\psi)-r}}.$$

Therefore

$$\begin{aligned} Z(\widehat{\mathrm{Ch}}_{\mathfrak{p}^r}, \chi^\vee) &= q^{-r}\mu(\mathcal{O})Z(\mathrm{Ch}_{\mathfrak{p}^{\mathrm{cond}(\psi)-r}}, \chi^\vee) \\ &= q^{-r}\mu(\mathcal{O})\mu^\times(\mathcal{O}^\times)\chi^\vee(\varpi)^{\mathrm{cond}(\psi)-r}L(\chi^\vee) \\ &= q^{-r}\mu(\mathcal{O})^2(1-q^{-1})\chi^\vee(\varpi)^{\mathrm{cond}(\psi)-r}L(\chi^\vee) \\ &= q^{-r}\mu(\mathcal{O})^2(1-q^{-1})q^{r-\mathrm{cond}(\psi)}\chi(\varpi)^{r-\mathrm{cond}(\psi)}L(\chi^\vee) \\ &= \mu(\mathcal{O})^2(1-q^{-1})q^{-\mathrm{cond}(\psi)}\chi(\varpi)^{r-\mathrm{cond}(\psi)}L(\chi^\vee) \end{aligned}$$

if χ is unramified with $a_\chi < 1$. When $0 < a_\chi < 1$ both zeta integrals converge and we look at their ratio. After some cancellation we arrive at the formula

$$\frac{Z(\widehat{\mathrm{Ch}}_{\mathfrak{p}^r}, \chi^\vee)}{Z(\mathrm{Ch}_{\mathfrak{p}^r}, \chi)} = \mu(\mathcal{O})\chi^\vee(\varpi)^{\mathrm{cond}(\psi)}\frac{L(\chi^\vee)}{L(\chi)}.$$

Note that the right-hand side is independent of r.

Exercise 2.77 Suppose $\psi = \psi_{\mathrm{std}}$ is the standard character, and μ is self-dual relative to ψ. Write the unramified character as $\chi = |\cdot|^s$ for some $s \in \mathbb{C}$. Show that

$$\mu(\mathcal{O})\chi^\vee(\varpi)^{\mathrm{cond}(\psi)} = N(\partial_{K/\mathbb{Q}_p})^{\frac{1}{2}-s}.$$

2.4.5 *Ramified Computations*

In this section we are primarily concerned with zeta integrals for ramified characters $\chi : K^\times \to \mathbb{C}^\times$. Thus $\chi|_{\mathcal{O}^\times} \neq 1$ whereas $\chi|_{1+\mathfrak{p}^r} = 1$ for $r \geq \mathrm{cond}(\chi) > 0$ so in that range $\chi|_{\mathcal{O}^\times}$ factors through a non-trivial character of $(\mathcal{O}/\mathfrak{p}^r)^\times$. We always assume χ has positive exponent a_χ so the zeta integrals converge. Recall that by definition $L(\chi) = 1$.

We saw in the previous section that $Z(\mathrm{Ch}_{\mathfrak{p}^r}, \chi) = 0$ for all $r \in \mathbb{Z}$, so to get interesting zeta integrals we need to move the center from the origin and consider Schwartz functions $\mathrm{Ch}_{a+\mathfrak{p}^r}$ with $a \neq 0$.

Pick an $a \neq 0$ and choose $r \gg 0$ such that $a + \mathfrak{p}^r \subset K^\times$. By possibly increasing r we even assume χ is constant on $a + \mathfrak{p}^r$. Under these assumptions

$$\begin{aligned} Z(\mathrm{Ch}_{a+\mathfrak{p}^r}, \chi) &= \int_{a+\mathfrak{p}^r} \chi(x)d\mu^\times(x) \\ &= \mu^\times(a+\mathfrak{p}^r)\chi(a) \end{aligned}$$

$$\begin{aligned}
&= \mu^\times(1+a^{-1}\mathfrak{p}^r)\chi(a)\\
&= \mu(1+a^{-1}\mathfrak{p}^r)\chi(a)\\
&= \mu(a^{-1}\mathfrak{p}^r)\chi(a)\\
&= q^{-r}\mu(\mathcal{O})\frac{\chi(a)}{|a|}
\end{aligned}$$

where $d\mu^\times(x) = \frac{d\mu(x)}{|x|}$. The case $a = \pm 1$ will play an important role so we emphasize it here:

$$Z(\mathrm{Ch}_{\pm 1+\mathfrak{p}^r}, \chi) = q^{-r}\mu(\mathcal{O})\chi(\pm 1)$$

for $r \gg 0$. In this computation we have not used that χ is ramified, and the following two exercises apply to both ramified and unramified characters:

Exercise 2.78 Fix a unitary character $\chi_0 : K^\times \to \mathbb{C}^\times$ and let $\chi_s = \chi_0|\cdot|^s$ for varying $s \in \mathbb{C}$. Let $f \in \mathcal{S}(K)$ be an arbitrary Schwartz function and consider $Z(f, \chi_s)$ as a function of s with $\mathrm{Re}(s) > 0$. Deduce from our computations so far that

$$\frac{Z(f,\chi_s)}{L(\chi_s)} \in \mathbb{C}[q^{\pm s}].$$

Here $\mathbb{C}[q^{\pm s}]$ denotes the ring of polynomial functions in q^{+s} and q^{-s}. In particular $\frac{Z(f,\chi_s)}{L(\chi_s)}$ is entire.

Exercise 2.79 In continuation of the previous exercise show the following: There exists a Schwartz function f_0 which only depends on χ_0 (not s) such that $\frac{Z(f_0,\chi_s)}{L(\chi_s)} = 1$ for all s.

We summarize the results of these exercises by saying $L(\chi_s)$ is a GCD of all the zeta integrals $Z(f, \chi_s)$ where $f \in \mathcal{S}(K)$ varies. This interpretation of $L(\chi_s)$ is instrumental when defining L-factors for other groups such as GL_n.

Now that we computed $Z(\mathrm{Ch}_{a+\mathfrak{p}^r}, \chi)$ we need to get our hands on $Z(\widehat{\mathrm{Ch}}_{a+\mathfrak{p}^r}, \chi^\vee)$. We will first do this for $a = 1$ which is sufficient in order to compute the ratio of zeta integrals in general. We recall from Lemma 2.68 that $\widehat{\mathrm{Ch}}_{1+\mathfrak{p}^r} = q^{-r}\mu(\mathcal{O})\psi\mathrm{Ch}_{\mathfrak{p}^{\mathrm{cond}(\psi)-r}}$ so our task is to compute $Z(\psi\mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee)$ where we have introduced the parameter $s = r - \mathrm{cond}(\psi)$. Preliminarily we find that

$$\begin{aligned}
Z(\psi\mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee) &= \int_{\mathfrak{p}^{-s}-\{0\}} \psi(x)\chi^\vee(x)d\mu^\times(x)\\
&= \sum_{n=-s}^{\infty}\int_{\varpi^n\mathcal{O}^\times} \psi(x)\chi^\vee(x)d\mu^\times(x)\\
&= \sum_{n=-s}^{\infty}\int_{\mathcal{O}^\times} \psi(\varpi^n u)\chi^\vee(\varpi^n u)d\mu^\times(u)\\
&= \sum_{n=-s}^{\infty}\chi^\vee(\varpi)^n\int_{\mathcal{O}^\times} \psi_{\varpi^n}(u)\chi^\vee(u)d\mu^\times(u)
\end{aligned}$$

$$= \sum_{n=-s}^{\infty} \chi^{\vee}(\varpi)^n \tau(\chi^{\vee}, \psi_{\varpi^n})$$

where $\tau(\chi^{\vee}, \psi_{\varpi^n})$ is a local Gauss sum:

Definition 2.80 For a pair of continuous characters $\chi : \mathcal{O}^{\times} \to \mathbb{C}^1$ and $\psi : \mathcal{O} \to \mathbb{C}^1$ we introduce the local Gauss sum

$$\tau(\chi, \psi) = \int_{\mathcal{O}^{\times}} \chi(u)\psi(u) d\mu^{\times}(u)$$

relative to a choice of Haar measure $\mu^{\times}$ on the units.

Note that $\tau(\chi, \psi)$ may be written as an actual finite sum: Pick an $r \gg 0$ for which $\chi|_{1+\mathfrak{p}^r} = 1$ and $\psi|_{\mathfrak{p}^r} = 1$. Then

$$\tau(\chi, \psi) = q^{-r}\mu(\mathcal{O}) \cdot \sum\nolimits_{u \in (\mathcal{O}/\mathfrak{p}^r)^{\times}} \chi(u)\psi(u).$$

(We have used the fact that $\mu^{\times}(1 + \mathfrak{p}^r) = q^{-r}\mu(\mathcal{O})$ and $\psi(uv) = \psi(u)\psi(u(v - 1)) = \psi(u)$ for $u \in \mathcal{O}^{\times}$ and $v \in 1 + \mathfrak{p}^r$.) This makes it more reminiscent of a classical Gauss sum.

Example 2.81 Let $\mathcal{O} = \mathbb{Z}_p$ and take $\psi : \mathbb{Z}_p \to \mathbb{C}^1$ to be the standard character given by inflating the exponential function $e^{2\pi i(\cdot)/p}$ on $\mathbb{Z}/p\mathbb{Z}$. Let us take $\chi : \mathbb{Z}_p^{\times} \to \mathbb{C}^1$ to be the inflation of a Dirichlet character $(\mathbb{Z}/p\mathbb{Z})^{\times} \to \mathbb{C}^1$ also denoted by χ. In this situation we may take $r = 1$, and if we normalize μ such that $\mu(\mathbb{Z}_p) = p$ we see that

$$\tau(\chi, \psi) = \sum\nolimits_{u \in (\mathbb{Z}/p\mathbb{Z})^{\times}} \chi(u)e^{2\pi i u/p} = \tau(\chi)$$

is on the nose a classical Gauss sum.

In the next section we will study properties of local Gauss sum at length. One outcome will be that $\tau(\chi^{\vee}, \psi_{\varpi^n})$ is nonzero for a unique n which depends on the conductors of χ and ψ. That means the sum equalling $Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^{\vee})$ only has one nonzero term, so this zeta integral is essentially a Gauss sum. We will return to its computation below.

2.4.6 Local Gauss Sums

The Gauss sum of a primitive Dirichlet character modulo N has absolute value $\sqrt{N}$. We want an analogue of this result for the local Gauss sums $\tau(\chi, \psi)$ introduced in the last section. Recall that $\chi : \mathcal{O}^{\times} \to \mathbb{C}^1$ and $\psi : \mathcal{O} \to \mathbb{C}^1$ is any pair of multiplicative/additive continuous characters.

Proposition 2.82 *Suppose χ is ramified. Then $\tau(\chi,\psi)\neq 0$ if and only if $cond(\chi) = cond(\psi)$, in which case*

$$|\tau(\chi,\psi)| = \mu(\mathcal{O})q^{-\frac{1}{2}cond(\psi)}.$$

In other words $|\tau(\chi,\psi)| = 1$ when μ is the ψ-self-dual measure on K, and $d\mu^\times(u) = \frac{d\mu(u)}{|u|} = d\mu(u)$.

Proof First we observe that

$$\begin{aligned}
|\tau(\chi,\psi)|^2 &= \int_{\mathcal{O}^\times}\int_{\mathcal{O}^\times} \chi(uv^{-1})\psi(u-v)d\mu^\times(u)d\mu^\times(v)\\
&= \int_{\mathcal{O}^\times}\int_{\mathcal{O}^\times} \chi(uv^{-1})\psi((uv^{-1}-1)v)d\mu^\times(u)d\mu^\times(v)\\
&= \int_{\mathcal{O}^\times}\int_{\mathcal{O}^\times} \chi(x)\psi((x-1)v)d\mu^\times(x)d\mu^\times(v)\\
&= \int_{\mathcal{O}^\times}\int_{\mathcal{O}^\times} \chi(x)\psi((x-1)v)d\mu^\times(v)d\mu^\times(x)\\
&= \int_{\mathcal{O}^\times} \chi(x)\left(\int_{\mathcal{O}^\times} \psi_{x-1}(v)d\mu^\times(v)\right)d\mu^\times(x)\\
&= \int_{\mathcal{O}^\times} \chi(x)h(x)d\mu^\times(x)
\end{aligned}$$

where we have introduced the auxiliary function $h(x) = \int_{\mathcal{O}^\times}\psi_{x-1}(v)d\mu^\times(v)$. In the third step we substituted $x = uv^{-1}$ for a fixed $v\in\mathcal{O}^\times$ in the inner integral. We need to evaluate $h(x)$. First of all $d\mu^\times(v) = \frac{d\mu(v)}{|v|}$ and $|v| = 1$ on $\mathcal{O}^\times$ so we can replace the multiplicative measure by the additive measure and split up $h(x)$ as a difference

$$h(x) = \int_{\mathcal{O}^\times}\psi_{x-1}(v)d\mu(v) = \int_{\mathcal{O}}\psi_{x-1}(v)d\mu(v) - \int_{\mathfrak{p}}\psi_{x-1}(v)d\mu(v).$$

The two terms are easily evaluated. For instance $\psi_{x-1}|_{\mathcal{O}} = 1$ if and only if $(x-1)\mathcal{O}\subset\mathfrak{p}^{\mathrm{cond}(\psi)}$. In other words when $\mathrm{ord}(x-1)\geq\mathrm{cond}(\psi)$. Therefore

$$\int_{\mathcal{O}}\psi_{x-1}(v)d\mu(v) = \begin{cases}\mu(\mathcal{O}) & \text{if } \mathrm{ord}(x-1)\geq\mathrm{cond}(\psi)\\ 0 & \text{if } \mathrm{ord}(x-1)<\mathrm{cond}(\psi).\end{cases}$$

Similarly

$$\int_{\mathfrak{p}}\psi_{x-1}(v)d\mu(v) = \begin{cases}\mu(\mathfrak{p}) & \text{if } \mathrm{ord}(x-1)+1\geq\mathrm{cond}(\psi)\\ 0 & \text{if } \mathrm{ord}(x-1)+1<\mathrm{cond}(\psi).\end{cases}$$

Combining these allows us to conclude

$$h(x) = \begin{cases} \mu(\mathcal{O}) - \mu(\mathfrak{p}) & \text{if } \mathrm{ord}(x-1) \geq \mathrm{cond}(\psi) \\ -\mu(\mathfrak{p}) & \text{if } \mathrm{ord}(x-1) = \mathrm{cond}(\psi) - 1 \\ 0 & \text{if } \mathrm{ord}(x-1) < \mathrm{cond}(\psi) - 1. \end{cases}$$

Inserting this into the above formula for $|\tau(\chi,\psi)|^2$ leads to the following expression in which we let $c = \mathrm{cond}(\psi)$ for legibility. Note that we may assume $c \geq 1$ here. Otherwise the inequality $\mathrm{ord}(x-1) \geq c$ is automatic for all $x \in \mathcal{O}^\times$ and since $\int_{\mathcal{O}^\times} \chi(x) d\mu^\times(x) = 0$ for ramified χ we find that $\tau(\chi,\psi) = 0$ when $c \leq 0$. When $c = 1$ we interpret $1 + \mathfrak{p}^{c-1}$ as being the unit group $\mathcal{O}^\times$.

$$\begin{aligned} |\tau(\chi,\psi)|^2 &= \int_{1+\mathfrak{p}^c} \chi(x)(\mu(\mathcal{O}) - \mu(\mathfrak{p})) d\mu^\times(x) + \int_{(1+\mathfrak{p}^{c-1})-(1+\mathfrak{p}^c)} \chi(x)(-\mu(\mathfrak{p})) d\mu^\times(x) \\ &= \mu(\mathcal{O}) \int_{1+\mathfrak{p}^c} \chi(x) d\mu^\times(x) - \mu(\mathfrak{p}) \int_{1+\mathfrak{p}^{c-1}} \chi(x) d\mu^\times(x). \end{aligned}$$

At this point we divide the argument into three cases according to the ramification of χ.

- If $\chi|_{1+\mathfrak{p}^c} \neq 1$ both integrals vanish and $\tau(\chi,\psi) = 0$.
- If $\chi|_{1+\mathfrak{p}^c} = 1$ but $\chi|_{1+\mathfrak{p}^{c-1}} \neq 1$ only the second integral vanishes and we get

$$|\tau(\chi,\psi)|^2 = \mu(\mathcal{O})\mu^\times(1+\mathfrak{p}^c) = \mu(\mathcal{O})^2 q^{-c}.$$

- If $\chi|_{1+\mathfrak{p}^{c-1}} = 1$ both integrals are nonzero, but they cancel out:

$$\begin{aligned} |\tau(\chi,\psi)|^2 &= \mu(\mathcal{O})\mu^\times(1+\mathfrak{p}^c) - \mu(\mathfrak{p})\mu^\times(1+\mathfrak{p}^{c-1}) \\ &= \mu(\mathcal{O})^2 q^{-c} - \mu(\mathcal{O}) q^{-1} \mu(\mathcal{O}) q^{-(c-1)} \\ &= 0. \end{aligned}$$

  Note that the condition $\chi|_{1+\mathfrak{p}^{c-1}} = 1$ forces $c > 1$ since we are assuming χ is ramified.

The three cases correspond to $c < \mathrm{cond}(\chi)$, $c = \mathrm{cond}(\chi)$, and $c > \mathrm{cond}(\chi)$ respectively. Altogether this proves the Proposition. □

We can now resume our calculation of $Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee)$.

Corollary 2.83 *Suppose χ is ramified, and let $n = cond(\psi) - cond(\chi)$. Then for all $s \geq -n$ we have*

$$Z(\psi Ch_{\mathfrak{p}^{-s}}, \chi^\vee) = \chi^\vee(\varpi)^n \tau(\chi^\vee, \psi_{\varpi^n})$$

independently of s. For $s < -n$ the zeta integral vanishes.

Proof In our sum $Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee) = \sum_{n=-s}^{\infty} (\cdots)$ the nth term only contributes when

$$\mathrm{cond}(\chi) = \mathrm{cond}(\chi^\vee) \stackrel{?}{=} \mathrm{cond}(\psi_{\varpi^n}) = \mathrm{cond}(\psi) - n.$$

In other words for $n = \mathrm{cond}(\psi) - \mathrm{cond}(\chi)$. If this quantity is $< -s$ we have $Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee) = 0$. Otherwise

$$Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee) = \chi^\vee(\varpi)^n \tau(\chi^\vee, \psi_{\varpi^n})$$

as claimed. The right-hand side is visibly independent of s. □

Remark 2.84 In particular we have an explicit formula for the absolute value of the zeta integral:

$$|Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee)| = \mu(\mathcal{O}) q^{-n a_{\chi^\vee}} q^{-\frac{1}{2}\mathrm{cond}(\chi)}$$

so the zeta integral is nonzero for $s \gg 0$. This non-vanishing is in fact one of the main byproducts of all our Gauss sum calculations, and it will play a critical role shortly.

The quantity we were originally after was the ratio of the zeta integrals:

$$\frac{Z(\widehat{\mathrm{Ch}}_{1+\mathfrak{p}^r}, \chi^\vee)}{Z(\mathrm{Ch}_{1+\mathfrak{p}^r}, \chi)} = \frac{q^{-r}\mu(\mathcal{O}) Z(\psi \mathrm{Ch}_{\mathfrak{p}^{-s}}, \chi^\vee)}{q^{-r}\mu(\mathcal{O})\chi(+1)} = \chi^\vee(\varpi)^n \tau(\chi^\vee, \psi_{\varpi^n})$$

for $r \gg 0$. Recall that χ is ramified and $n = \mathrm{cond}(\psi) - \mathrm{cond}(\chi)$ in this formula. One should imagine an invisible factor $\frac{L(\chi^\vee)}{L(\chi)} = 1$ occurring on the right-hand side of the zeta ratio formula, by analogy with the unramified case.

Exercise 2.85 We keep the notation above. Pick an arbitrary $a \in K^\times$ such that $\mathrm{cond}(\chi) = \mathrm{cond}(\psi_a)$ (for example one could take $a = \varpi^n$ so a differs from ϖ^n by a unit). Show that

$$\chi^\vee(\varpi)^n \tau(\chi^\vee, \psi_{\varpi^n}) = \chi^\vee(a)\tau(\chi^\vee, \psi_a).$$

Exercise 2.86 In continuation of the previous exercise, suppose μ is ψ-self-dual and the character χ has exponent $a_\chi = \frac{1}{2}$. Verify that $|\chi^\vee(a)\tau(\chi^\vee, \psi_a)| = 1$.

2.4.7 The γ-Factors and ϵ-Factors

In this section K denotes an arbitrary local field of characteristic zero (possibly archimedean) and $\chi : K^\times \to \mathbb{C}^\times$ is a continuous character with $0 < a_\chi < 1$ at first. Our explicit calculations suggest the ratio $\frac{Z(\widehat{f}, \chi^\vee)}{Z(f, \chi)}$ might be independent of $f \in \mathcal{S}(K)$ provided $Z(f, \chi) \neq 0$ of course. The next Lemma affirms this.

Lemma 2.87 *Let $\chi : K^\times \longrightarrow \mathbb{C}^\times$ be a continuous character with $0 < a_\chi < 1$. Then $\forall f, g \in \mathcal{S}(K)$ we have the identity*

$$Z(f,\chi)Z(\widehat{g},\chi^\vee) = Z(\widehat{f},\chi^\vee)Z(g,\chi).$$

Proof First, the substitution $t = yx^{-1}$ for a fixed x and a simple application of Fubini's Theorem gives:

$$\begin{aligned}
Z(f,\chi)Z(\widehat{g},\chi^\vee) &= \int_{K^\times}\int_{K^\times} f(x)\chi(x)\widehat{g}(y)\chi^\vee(y)d\mu^\times(x)d\mu^\times(y)\\
&= \int_{K^\times}\int_{K^\times} f(x)\chi(x)\widehat{g}(y)\chi(y^{-1})|y|d\mu^\times(y)d\mu^\times(x)\\
&= \int_{K^\times}\int_{K^\times} f(x)\widehat{g}(tx)\chi(t^{-1})|tx|d\mu^\times(t)d\mu^\times(x)\\
&= \int_{K^\times}\left(\int_{K^\times} f(x)\widehat{g}(tx)|x|d\mu^\times(x)\right)\chi^\vee(t)d\mu^\times(t)\\
&= \int_{K^\times}\langle f,g\rangle(t)\chi^\vee(t)d\mu^\times(t)
\end{aligned}$$

where we have introduced

$$\begin{aligned}
\langle f,g\rangle(t) &= \int_{K^\times} f(x)\widehat{g}(tx)|x|d\mu^\times(x)\\
&= \int_{K^\times} f(x)\widehat{g}(tx)d\mu(x)\\
&= \int_{K} f(x)\widehat{g}(tx)d\mu(x).
\end{aligned}$$

(In the last step we added in $x = 0$ where both f and g are defined.) It remains to check this symbol is symmetric, that is $\langle f,g\rangle = \langle g,f\rangle$. This follows by inserting the definition of $\widehat{g}$. We give the details:

$$\begin{aligned}
\langle f,g\rangle(t) &= \int_K f(x)\left(\int_K g(y)\psi(txy)d\mu(y)\right)d\mu(x)\\
&= \int_K\int_K f(x)g(y)\psi(txy)d\mu(x)d\mu(y).
\end{aligned}$$

This double integral is visibly invariant under interchanging f and g. □

This is indeed quite powerful. As we have seen, the ratio $\frac{Z(\widehat{f},\chi^\vee)}{Z(f,\chi)}$ can be computed explicitly for simple Schwartz functions f, such as f_e, f_n, and $\mathrm{Ch}_{a+\mathfrak{p}^r}$. Then for more complicated Schwartz functions f we deduce a highly non-trivial relation between the two zeta integrals $Z(\widehat{f},\chi^\vee)$ and $Z(f,\chi)$.

Definition 2.88 Let $\chi : K^\times \longrightarrow \mathbb{C}^\times$ be a continuous character with $0 < a_\chi < 1$, and $\psi : K \to \mathbb{C}^1$ a non-trivial additive character (used to define the Fourier transformation). The γ-factor is the ratio

$$\gamma(\chi, \psi, \mu) = \frac{Z(\widehat{f}, \chi^\vee)}{Z(f, \chi)}$$

for any choice of $f \in \mathcal{S}(K)$ for which $Z(f, \chi) \neq 0$. The dependence on ψ and μ is through $\widehat{f}$. We often take μ to be the ψ-self-dual measure, in which case we omit μ from the notation and just write $\gamma(\chi, \psi)$.

In all our preceding brute force calculations the γ-factor $\gamma(\chi, \psi, \mu)$ has the general shape

$$\gamma(\chi, \psi, \mu) = \epsilon(\chi, \psi, \mu) \cdot \frac{L(\chi^\vee)}{L(\chi)}$$

where the ϵ-factor $\epsilon(\chi, \psi, \mu)$ has a nice description. We summarize our findings below. Again, when μ is ψ-self-dual we omit it from the notation $\epsilon(\chi, \psi, \mu)$.

- When $K = \mathbb{R}$,
$$\epsilon(\chi_{\mathrm{e},s}, \psi_{\mathrm{std}}) = i^{-\mathrm{e}}.$$
- When $K = \mathbb{C}$,
$$\epsilon(\chi_{n,s}, \psi_{\mathrm{std}}) = i^{-|n|}.$$
- When K is non-archimedean and χ is unramified,
$$\epsilon(\chi, \psi) = \left(q^{\frac{1}{2}} \chi^\vee(\varpi)\right)^{\mathrm{cond}(\psi)}$$
- When K is non-archimedean and χ is ramified,
$$\epsilon(\chi, \psi) = \chi^\vee(a)\tau(\chi^\vee, \psi_a)$$
for any choice of $a \in K^\times$ such that $\mathrm{cond}(\chi) = \mathrm{cond}(\psi_a)$. The Gauss sum is taken with respect to the ψ-self-dual measure.

Soon we will let our character χ vary in a family of twists $\chi_s = \chi_0|\cdot|^s$ for a fixed unitary character χ_0. In this case we will adopt the notation $\epsilon(s, \chi_0, \psi) = \epsilon(\chi_s, \psi)$. In this notation $\epsilon(s, \mathrm{sign}^{\mathrm{e}}, \psi_{\mathrm{std}}) = i^{-\mathrm{e}}$ in the real case, and $\epsilon(s, \mathrm{phase}^n, \psi_{\mathrm{std}}) = i^{-|n|}$ in the complex case where $\mathrm{phase}(z) = \frac{z}{|z|_{\mathbb{C}}}$. In the non-archimedean case $\epsilon(s, \chi_0, \psi_{\mathrm{std}})$ has the following form:

Corollary 2.89 *Let K be a p-adic field and $\chi_0 : K^\times \to \mathbb{C}^1$ a unitary continuous character. Then*

$$\epsilon(s, \chi_0, \psi_{std}) = \epsilon(\frac{1}{2}, \chi_0, \psi_{std}) N(\partial_{K/\mathbb{Q}_p} \mathcal{N}_{\chi_0})^{\frac{1}{2}-s}$$

where the local root number $\epsilon(\frac{1}{2}, \chi_0, \psi_{std}) \in \mathbb{C}^1$ *is given by*

$$\epsilon(\frac{1}{2}, \chi_0, \psi_{std}) = \begin{cases} \chi_0(\partial_{K/\mathbb{Q}_p}) & \text{if } \chi_0 \text{ is unramified} \\ \overline{\chi_0(a)}|a|^{\frac{1}{2}}\tau(\bar{\chi}_0, \psi_a) & \text{if } \chi_0 \text{ is ramified.} \end{cases}$$

Here $\chi_0(\partial_{K/\mathbb{Q}_p}) = \chi_0(\varpi^t)$ *if* $\partial_{K/\mathbb{Q}_p} = (\varpi^t)$, *and we have introduced* $\mathcal{N}_{\chi_0} = \mathfrak{p}^{cond(\chi_0)}$. *Moreover, as above* $a \in K^\times$ *is any element with* $ord(a) = cond(\psi_{std}) - cond(\chi_0)$.

Proof Taking norms in $\mathfrak{p}^{\mathrm{cond}(\psi_{\mathrm{std}})} = \partial_{K/\mathbb{Q}_p}^{-1}$ shows $q^{-\mathrm{cond}(\psi_{\mathrm{std}})} = N(\partial_{K/\mathbb{Q}_p})$. Therefore, assuming χ_0 is unramified and just writing $\psi = \psi_{\mathrm{std}}$, we get

$$\begin{aligned} \epsilon(s, \chi_0, \psi) &= \left(q^{\frac{1}{2}}\chi_0(\varpi)^{-1}q^{s-1}\right)^{\mathrm{cond}(\psi)} \\ &= \chi_0(\varpi^{-\mathrm{cond}(\psi)})q^{-\mathrm{cond}(\psi)(\frac{1}{2}-s)} \\ &= \chi_0(\partial_{K/\mathbb{Q}_p})N(\partial_{K/\mathbb{Q}_p})^{\frac{1}{2}-s}. \end{aligned}$$

When χ_0 is ramified we pick an $a \in K^\times$ with $\mathrm{ord}(a) = \mathrm{cond}(\psi) - \mathrm{cond}(\chi_0)$. Then

$$\begin{aligned} \epsilon(s, \chi_0, \psi) &= \chi_s^\vee(a)\tau(\chi_s^\vee, \psi_a) \\ &= \overline{\chi_0(a)}|a|^{1-s}\tau(\bar{\chi}_0, \psi_a) \\ &= \epsilon(\frac{1}{2}, \chi_0, \psi)|a|^{\frac{1}{2}-s} \\ &= \epsilon(\frac{1}{2}, \chi_0, \psi)q^{-\mathrm{cond}(\psi)(\frac{1}{2}-s)}q^{+\mathrm{cond}(\chi_0)(\frac{1}{2}-s)} \\ &= \epsilon(\frac{1}{2}, \chi_0, \psi)N(\partial_{K/\mathbb{Q}_p}\mathcal{N}_{\chi_0})^{\frac{1}{2}-s} \end{aligned}$$

Note that the Gauss sum is an integral over $\mathcal{O}^\times$ so $\tau(\chi, \psi)$ only depends on χ through its restriction $\chi|_{\mathcal{O}^\times}$. In particular $\tau(\chi_s, \psi) = \tau(\chi_0, \psi)$ for all $s \in \mathbb{C}$. □

Exercise 2.90 Check that $\overline{\epsilon(\frac{1}{2}, \chi_0, \psi_{\mathrm{std}})} = \chi_0(-1)\epsilon(\frac{1}{2}, \bar{\chi}_0, \psi_{\mathrm{std}})$ for all K and all unitary χ_0.

We note in passing that the local ϵ-factors have the general shape $\epsilon(s, \chi_0, \psi) = Wq^{m(\frac{1}{2}-s)}$ for some $W \in \mathbb{C}^1$ and $m \in \mathbb{Z}_{\geq 0}$. By convention $q = 1$ for $K = \mathbb{R}, \mathbb{C}$. In particular $\epsilon(s, \chi_0, \psi)$ is an invertible function on $\mathbb{C}$, in the sense that it is an entire function without zeroes. More precisely $\epsilon(s, \chi_0, \psi)$ is a unit of $\mathbb{C}[q^{\pm s}]$ in the non-archimedean case. We also stress that $\epsilon(s, \chi_0, \psi_{\mathrm{std}}) = 1$ when χ_0 and $K/\mathbb{Q}_p$ are both unramified.

The next example relates our computations to the root number of a Dirichlet character.

Example 2.91 Here we take $K = \mathbb{Q}_p$ and let $\chi_0 : \mathbb{Q}_p^\times \to \mathbb{C}^1$ be a unitary character defined as follows. Let $\chi_0(p) = 1$ and on $\mathbb{Z}_p^\times$ we take χ_0 to be the inflation of a non-trivial Dirichlet character $(\mathbb{Z}/p\mathbb{Z})^\times \to \mathbb{C}^1$ denoted by χ_0^{Dir}. In this case $\partial_{K/\mathbb{Q}_p} = (1)$ and $\mathcal{N}_{\chi_0} = (p)$ so

$$\epsilon(s, \chi_0, \psi_{\mathrm{std}}) = \epsilon(\frac{1}{2}, \chi_0, \psi_{\mathrm{std}}) p^{\frac{1}{2}-s}.$$

Since ψ_{std} is unramified we may take our $a = p^{-1}$ and the ψ_{std}-self-dual measure μ is determined by $\mu(\mathbb{Z}_p) = 1$. Therefore, by Example 2.81 we end up with

$$\begin{aligned}
\epsilon(\frac{1}{2}, \chi_0, \psi_{\mathrm{std}}) &= \overline{\chi_0(p^{-1})} |p^{-1}|^{\frac{1}{2}} \tau(\bar{\chi}_0, \psi_{p^{-1}}) \\
&= \chi_0(p) p^{\frac{1}{2}} \tau(\bar{\chi}_0, \psi_{p^{-1}}) \\
&= p^{\frac{1}{2}} p^{-1} \mu(\mathbb{Z}_p) \tau(\bar{\chi}_0^{\mathrm{Dir}}) \\
&= \frac{\tau(\bar{\chi}_0^{\mathrm{Dir}})}{\sqrt{p}}.
\end{aligned}$$

Up to the "archimedean" factor $i^{-\mathrm{e}}$ this is exactly the classical root number of $\bar{\chi}_0^{\mathrm{Dir}}$. The reader may want to generalize this example to characters χ_0^{Dir} which are primitive modulo a p-power.

2.4.8 The Main Local Theorem

Let K be any local field of characteristic 0 and fix a unitary continuous character $\chi_0 : K^\times \to \mathbb{C}^1$. For $s \in \mathbb{C}$ consider the twist $\chi_s = \chi_0 |\cdot|^s$. Given a Schwartz function $f \in \mathcal{S}(K)$ we view the zeta integral $Z(f, \chi_s)$ as a holomorphic function in the right-half plane $\mathrm{Re}(s) > 0$ (the domain of absolute convergence). We already observed in Exercise 2.78 that $\frac{Z(f,\chi_s)}{L(\chi_s)}$ extends to an entire function in the non-archimedean case (and the ratio equals 1 for a clever choice of Schwartz function f_0). This is true in general.

Theorem 2.92 (The main local Theorem) *We adopt the notation just introduced. The following holds:*

(1) $Z(f, \chi_s)$ has meromorphic continuation to $\mathbb{C}$ for all $f \in \mathcal{S}(K)$;
(2) The ratio $\frac{Z(f,\chi_s)}{L(\chi_s)}$ is an entire function for all $f \in \mathcal{S}(K)$;
(3) There exists an $f_0 \in \mathcal{S}(K)$ such that $\frac{Z(f_0,\chi_s)}{L(\chi_s)} = 1$ for all s;
(4) For arbitrary $f \in \mathcal{S}(K)$ we have the functional equation

$$\frac{Z(\widehat{f}, \chi_s^\vee)}{L(\chi_s^\vee)} = \epsilon(s, \chi_0, \psi) \frac{Z(f, \chi_s)}{L(\chi_s)}.$$

(Here $\chi_s^\vee(x) = (\chi_s)^\vee(x) = \overline{\chi_0(x)}|x|^{1-s}$.*)*

Proof We have already done most of the hard work. $Z(f, \chi_s)$ is holomorphic for $\mathrm{Re}(s) > 0$ and $Z(\widehat{f}, \chi_s^\vee)$ is holomorphic for $\mathrm{Re}(s) < 1$. Furthermore, from our explicit calculation of $\gamma(s, \chi_0, \psi)$ it readily extends to a meromorphic function on $\mathbb{C}$. On the other hand, by the very definition of the γ-factors we have the identity $Z(\widehat{f}, \chi_s^\vee) = \gamma(s, \chi_0, \psi)Z(f, \chi_s)$ which allows us to extend $Z(f, \chi_s)$ to a meromorphic function on all of $\mathbb{C}$. Also, the definition of ϵ-factors immediately translates into the functional equation

$$\frac{Z(\widehat{f}, \chi_s^\vee)}{L(\chi_s^\vee)} = \epsilon(s, \chi_0, \psi)\frac{Z(f, \chi_s)}{L(\chi_s)}.$$

Here $\frac{Z(f,\chi_s)}{L(\chi_s)}$ is holomorphic for $\mathrm{Re}(s) > 0$ and $\frac{Z(\widehat{f},\chi_s^\vee)}{L(\chi_s^\vee)}$ is holomorphic for $\mathrm{Re}(s) < 1$ (since the L-factors have no zeroes in these regions). Our long computations show that $\epsilon(s, \chi_0, \psi) = Wq^{m(\frac{1}{2}-s)}$ has no zeroes anywhere, and therefore the functional equation lets us extend $\frac{Z(f,\chi_s)}{L(\chi_s)}$ to an entire function.

Finally, we already gave examples of Schwartz functions f_0 for which $Z(f_0, \chi_s) = L(\chi_s)$ for all s. When χ_0 is unramified take $f_0 = c\mathrm{Ch}_{\mathcal{O}}$. When χ_0 is ramified take $f_0 = c'\mathrm{Ch}_{1+\mathfrak{p}^r}$ for a large enough r. Here $c, c' > 0$ are suitable constants. For $K = \mathbb{R}$ take $f_0 = f_{\mathrm{e}}$ and for $K = \mathbb{C}$ take $f_0 = \frac{1}{\pi} f_n$ (we apologize for the ambiguous notation used here). □

Thus the L-factor $L(\chi_s)$ may be interpreted as a greatest common divisor of all zeta integrals $Z(f, \chi_s)$ as f varies. Here divisibility pertains to the field of meromorphic functions $\mathrm{Frac}(\mathcal{H}(\mathbb{C}))$. (If f, g are meromorphic functions on $\mathbb{C}$ we say that g divides f if $\frac{f}{g}$ is entire; in other words when $\mathrm{ord}_{s=s_0}(f) \geq \mathrm{ord}_{s=s_0}(g)$ for all $s_0 \in \mathbb{C}$.)

2.5 The Global Theory

2.5.1 Measures on the Adeles

Let $K/\mathbb{Q}$ denote a number field, with completions K_v for $v \in V$. Suppose we have a selection of Haar measures $\{\mu_v\}_{v\in V}$ on all completions $(K_v, +)$. We assume $\mu_v(\mathcal{O}_v) = 1$ for all but at most finitely many $v \in V_f$. Then there is a unique Haar measure μ on $(\mathbb{A}_K, +)$ such that on all boxes $U = \prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v$ we have

$$\mu(U) = \prod_{v\in S} \mu_v(U_v) \times \prod_{v\notin S} \mu_v(\mathcal{O}_v).$$

Note that this is a finite product. We will denote this measure by $\mu = \otimes_{v\in V}\mu_v$. However, we should emphasize that the individual factors μ_v are of course not determined

by μ (at two places $v \neq w$ one can replace the measures by $c\mu_v$ and $c^{-1}\mu_w$ without changing μ).

There is a set of functions on $\mathbb{A}_K$ which play an important role in what follows: Given a collection of integrable functions $\{f_v\}_{v\in V}$ on the various K_v, with the constraint that $f_v = \mathrm{Ch}_{\mathcal{O}_v}$ for all but at most finitely many $v \in V_f$, there is an associated function $f = \otimes_{v\in V} f_v$ on $\mathbb{A}_K$ defined by the finite product

$$f(x) = \prod_{v\in V} f_v(x_v)$$

for $x = (x_v)_{v\in V} \in \mathbb{A}_K$. We name such f factorizable functions, and most functions of interest will be linear combinations of factorizable functions. One can show that $f \in L^1(\mathbb{A}_K)$ and

$$\int_{\mathbb{A}_K} f(x)d\mu(x) = \prod_{v\in V} \int_{K_v} f_v(x_v)d\mu_v(x_v).$$

This generalizes the formula for $\mu(U)$ above. We refer to [RV99, p. 185] for more details on the construction of $\mu = \otimes_{v\in V}\mu_v$ and [RV99, Propositions 5 and 6] for its properties. Needless to say, this way of factoring integrals over $\mathbb{A}_K$ is crucial when relating the local and global theories.

We march on to measures on $\mathbb{I}_K$.

Lemma 2.93 *$\mu(aA) = |a|\mu(A)$ for all $a \in \mathbb{I}_K$ and Borel sets $A \subset \mathbb{A}_K$.*

Proof It suffices to show this for boxes $U = \prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v$. Here $aU = \prod_{v\in S} a_v U_v \times \prod_{v\notin S} a_v\mathcal{O}_v$, so by the local analogue of the Lemma

$$\mu(aU) = \prod_{v\in S} |a_v|_v\mu_v(U_v) \times \prod_{v\notin S} |a_v|_v\mu_v(\mathcal{O}_v) = |a|\mu(U).$$

We recall the idele norm is defined as $|a| = \prod_{v\in V} |a_v|_v$. □

This lets us associate a Haar measure $\mu^\times$ on $\mathbb{I}_K$ in the usual fashion; $\mu^\times(A) = \int_A \frac{d\mu(x)}{|x|}$ for $A \subset \mathbb{I}_K$. However, there is a small wrinkle here: This measure $\mu^\times$ is *not* the product of the local measures $\mu_v^\times$. Indeed their product does not make sense! The problem is that

$$\mu_v^\times(\mathcal{O}_v^\times) = (1 - q_v^{-1})\mu_v(\mathcal{O}_v)$$

for $v \in V_f$, and this is not 1 for all but finitely many v. For this reason we often renormalize $\mu_v^\times$ at the finite places and instead consider

$$\mu_v^\star = \frac{1}{1 - q_v^{-1}}\mu_v^\times.$$

The normalization factor guarantees the product $\mu^\star = \otimes_{v\in V}\mu_v^\star$ is well-defined, analogously to what we did for the adeles. (At $v \in V_\infty$ the measure $\mu_v^\star$ is chosen at random.)

2.5.2 *Fourier Theory*

We have already touched upon the Fourier transformation for $\mathbb{A}_K$. It depend on the choice of a non-trivial additive character $\psi : \mathbb{A}_K/K \to \mathbb{C}^1$. The latter gives a self-duality $\mathbb{A}_K \xrightarrow{\sim} \widehat{\mathbb{A}_K}$ by sending $a \mapsto \psi_a = \psi(a(\cdot))$. As remarked in Example 2.64, μ is self-dual (with respect to any choice of ψ) exactly when $\dot{\mu}(\mathbb{A}_K/K) = 1$.

With these choices, an integrable function f on $\mathbb{A}_K$ has Fourier transformation

$$\widehat{f}(y) = \int_{\mathbb{A}_K} f(x)\psi(xy)d\mu(x).$$

If $\widehat{f}$ is again integrable, the Fourier inversion formula lets us recover f from $\widehat{f}$ as

$$f(x) = \int_{\mathbb{A}_K} \widehat{f}(y)\overline{\psi(xy)}d\mu(x).$$

(We have deliberately veered from the conventions in the aforementioned Example 2.64, by changing the roles of ψ and $\bar{\psi} = \psi_{-1}$.) Note that the above integrals factor as a product of local integrals when f is factorizable. For instance, for $f = \otimes_{v\in V} f_v$ we have

$$\widehat{f}(y) = \prod_{v\in V}\int_{K_v} f_v(x_v)\psi_v(x_v y_v)d\mu_v(x_v) = \prod_{v\in V}\widehat{f_v}(y_v).$$

In other words $\widehat{f} = \otimes_{v\in V}\widehat{f_v}$. (This makes sense since $\widehat{\mathrm{Ch}_{\mathcal{O}_v}} = \mu_v(\mathcal{O}_v)\mathrm{Ch}_{\mathfrak{p}_v^{\mathrm{cond}(\psi_v)}} = \mathrm{Ch}_{\mathcal{O}_v}$ for all but finitely many $v \in V_f$.)

Here is the global definition of a Schwartz-Bruhat function on $\mathbb{A}_K$.

Definition 2.94 The Schwartz space $\mathcal{S}(\mathbb{A}_K)$ is the span of all factorizable functions $f = \otimes_{v\in V} f_v$ where $f_v \in \mathcal{S}(K_v)$ for all v, and $f_v = \mathrm{Ch}_{\mathcal{O}_v}$ for all but at most finitely many $v \in V_f$.

From what we just observed it is clear that $\widehat{(\cdot)}$ is an automorphism of $\mathcal{S}(\mathbb{A}_K)$.

In Example 2.64 we highlighted the Poisson summation formula in this setting, which says that for all functions f of "rapid decay" we have the identity

$$\sum_{\gamma\in K} f(\gamma) = \sum_{\gamma\in K}\widehat{f}(\gamma).$$

We never defined what it means for f to have rapid decay, but being a Schwartz-Bruhat function ensures both sides of the Poisson summation formula converge. We will not give the somewhat long and technical proof of convergence here, but only refer the reader to [RV99, p. 261].

The following reformulation is vital for establishing the functional equation for Hecke L-functions.

Proposition 2.95 (The Riemann-Roch formula) *For all $f \in \mathcal{S}(\mathbb{A}_K)$ and $a \in \mathbb{I}_K$ we have the identity*

$$\sum_{\gamma \in K} f(a\gamma) = \frac{1}{|a|} \sum_{\gamma \in K} \widehat{f}(a^{-1}\gamma).$$

(Here $\widehat{f}$ is taken relative to the measure μ for which $\dot{\mu}(\mathbb{A}_K/K) = 1$.)

Proof Apply the Poisson summation formula to $h(x) = f(ax)$, which has Fourier transformation

$$\widehat{h}(y) = \int_{\mathbb{A}_K} f(ax)\psi(xy)d\mu(x) = \int_{\mathbb{A}_K} f(x')\psi(a^{-1}x'y)\frac{1}{|a|}d\mu(x') = \frac{1}{|a|}\widehat{f}(a^{-1}y).$$

(We substituted $x' = ax$ and used $d\mu(x') = |a|d\mu(x)$.) □

Taking $a = 1$ gives back the Poisson summation formula as a special case.

Remark 2.96 The Riemann-Roch formula derives its name from the observation that for the function field $K/\mathbb{F}_p(t)$ of an algebraic curve it recovers the classical Riemann-Roch formula $\ell(D) - \ell(\mathcal{K} - D) = \deg(D) - g + 1$ by taking $f_v = \mathrm{Ch}_{\mathcal{O}_v}$ for all v. The intrigued reader should try this as an exercise or study [RV99, p. 268] where this is discussed in detail.

2.5.3 Global Zeta Integrals

We now introduce the key players in Tate's theory.

Lemma 2.97 *Let $f \in \mathcal{S}(\mathbb{A}_K)$ be an arbitrary Schwartz function, and fix a Haar measure $\mu^\star$ on $\mathbb{I}_K$. Let $\chi : C_K \to \mathbb{C}^\times$ be a Hecke character. If χ has exponent $a_\chi > 1$ the global zeta integral*

$$Z(f, \chi) = \int_{\mathbb{I}_K} f(x)\chi(x)d\mu^\star(x)$$

is absolutely convergent.

Proof We may assume $f = \otimes_{v\in V} f_v$ and $\mu^\star = \otimes_{v\in V}\mu_v^\star$. Since $a_{\chi_v} = a_\chi$ is positive for all v the local zeta integrals $Z(f_v, \chi_v)$ are absolutely convergent. It remains to show the product

$$Z(f, \chi) = \prod_{v\in V} Z(f_v, \chi_v)$$

converges absolutely. Pick a finite set of places $S \supset V_\infty$ such that $f_v = \mathrm{Ch}_{\mathcal{O}_v}$, $\mu_v^\star(\mathcal{O}_v^\times) = 1$, and χ_v is unramified for all $v \notin S$. For such v we have $Z(f_v, \chi_v) = L(\chi_v)$. We are left with verifying $\prod_{v\notin S} \frac{1}{1-\chi_v(\varpi_v)}$ is absolutely convergent, but we already took care of this when we introduced Hecke L-functions. □

Note that local zeta integrals converge for exponent > 0 but global zeta integrals only converge for exponent > 1.

The previous proof already shows a close relationship between global zeta integrals and Hecke L-functions. Suppose we fix a unitary Hecke character $\chi_0 : C_K \to \mathbb{C}^1$ and let $\chi_s = \chi_0|\cdot|^s$. We found a finite set of places S such that

$$Z(f, \chi_s) = \left(\prod_{v\in S} \frac{Z(f_v, \chi_{v,s})}{L(\chi_{v,s})}\right) \cdot L^{\mathrm{aut}}(s, \chi_0)$$

for $\mathrm{Re}(s) > 1$, and we know from the local theory that the finite product $\prod_{v\in S} \frac{Z(f_v,\chi_{v,s})}{L(\chi_{v,s})}$ is an entire function. We can even choose specific functions $\{f_{v,0}\}_{v\in S}$ for which this product is 1. If we introduce the function $f_0 = (\bigotimes_{v\in S} f_{v,0}) \otimes (\bigotimes_{v\notin S} \mathrm{Ch}_{\mathcal{O}_v})$ the L-function $L^{\mathrm{aut}}(s, \chi_0)$ is on the nose a zeta integral:

$$Z(f_0, \chi_s) = L^{\mathrm{aut}}(s, \chi_0).$$

Our pressing goal is obviously to extend $Z(f, \chi_s)$ to a meromorphic function on $\mathbb{C}$, understand its poles, and establish a functional equation relating $Z(f, \chi_s)$ to $Z(\widehat{f}, \chi_s^\vee)$. In fact the two zeta integrals happen to always be equal–in other words all global γ-factors are trivial.

Before stating the main global Theorem precisely we take a few initial steps towards its proof. Following our general strategy we first find an integral representation of $Z(f, \chi)$ as some sort of Mellin transform of the analogue of a theta function. (We drop the subscript s in χ_s for now to improve readability.)

Once and for all we choose a splitting $\sigma : \mathbb{R}_{>0}^\times \to \mathbb{I}_K$ of the idele norm. For $t > 0$ and $x \in \mathbb{I}_K$ we will henceforth adopt the notation $tx = \sigma(t)x$. Thus $(t, x) \mapsto tx$ gives an isomorphism $\mathbb{R}_{>0}^\times \times \mathbb{I}_K^1 \xrightarrow{\sim} \mathbb{I}_K$ of topological groups. Under this identification $\mu^\star$ corresponds to a product $\frac{dt}{t} \times \nu^\star$ for a uniquely determined Haar measure $\nu^\star$ on $\mathbb{I}_K^1$ (here dt is the Lebesgue measure on $\mathbb{R}$).

For $f \in \mathcal{S}(\mathbb{A}_K)$ and a Hecke character χ of exponent $a_\chi > 1$ we may then break up the zeta integral as

$$Z(f,\chi) = \int_0^\infty Z_t(f,\chi)\frac{dt}{t}$$

where

$$Z_t(f,\chi) = \int_{\mathbb{I}_K^1} f(tx)\chi(tx)d\nu^\star(x).$$

We find it instructive to think of the first integral as a Mellin transform, and to interpret $Z_t(f,\chi)$ as an analogue of the θ-function $\theta_\chi(it)t^{\frac{s+\mathfrak{e}}{2}}$ from our discussion of Dirichlet L-functions.

The second step of our general strategy is to employ Poisson summation and establish a transformation property of $Z_t(f,\chi)$. Here it is.

Lemma 2.98 *Let $f \in \mathcal{S}(\mathbb{A}_K)$ and consider a Hecke character χ (of arbitrary exponent). For each fixed $t > 0$ the integral defining $Z_t(f,\chi)$ is absolutely convergent, and we have*

$$Z_t(f,\chi) = Z_{t^{-1}}(\widehat{f},\chi^\vee) + E$$

where

$$E = E_t(f,\chi) = \widehat{f}(0)\int_{C_K^1}\chi^\vee(t^{-1}x)d\dot{\nu}^\star(x) - f(0)\int_{C_K^1}\chi(tx)d\dot{\nu}^\star(x)$$

(Here $\dot{\nu}^\star$ denotes the quotient measure of $\nu^\star$ on the norm one idele class group $C_K^1 = \mathbb{I}_K^1/K^\times$, and $\widehat{f}$ is relative to the self-dual measure.)

Proof Since $\chi(\gamma) = 1$ for $\gamma \in K^\times$ the definition of $\dot{\nu}^\star$ leads to

$$\begin{aligned} Z_t(f,\chi) &= \int_{C_K^1}\sum_{\gamma\in K^\times} f(t\gamma x)\chi(t\gamma x)d\dot{\nu}^\star(x) \\ &= \int_{C_K^1}\Big(\sum_{\gamma\in K^\times} f(t\gamma x)\Big)\chi(tx)d\dot{\nu}^\star(x) \\ &= \int_{C_K^1}\Big(\sum_{\gamma\in K} f(t\gamma x)\Big)\chi(tx)d\dot{\nu}^\star(x) - f(0)\int_{C_K^1}\chi(tx)d\dot{\nu}^\star(x). \end{aligned}$$

The point of taking out the $\gamma = 0$ term is we are now in a position to use the Riemann-Roch formula:

$$\sum_{\gamma\in K} f(t\gamma x) = \frac{1}{|tx|}\sum_{\gamma\in K}\widehat{f}(\gamma t^{-1}x^{-1}).$$

Inserting this above we conclude that

$$\begin{aligned} Z_t(f,\chi) + f(0)\int_{C_K^1}\chi(tx)d\dot{\nu}^\star(x) &= \int_{C_K^1}\left(\sum_{\gamma\in K} f(t\gamma x)\right)\chi(tx)d\dot{\nu}^\star(x) \\ &= \int_{C_K^1}\left(\frac{1}{|tx|}\sum_{\gamma\in K}\widehat{f}(\gamma t^{-1}x^{-1})\right)\chi(tx)d\dot{\nu}^\star(x) \\ &= \int_{C_K^1}\left(\sum_{\gamma\in K}\widehat{f}(\gamma t^{-1}y)\right)|t^{-1}y|\chi(ty^{-1})d\dot{\nu}^\star(y) \\ &= \int_{C_K^1}\left(\sum_{\gamma\in K}\widehat{f}(\gamma t^{-1}y)\right)\chi^\vee(t^{-1}y)d\dot{\nu}^\star(y) \\ &= Z_{t^{-1}}(\widehat{f},\chi^\vee) + \widehat{f}(0)\int_{C_K^1}\chi^\vee(t^{-1}x)d\dot{\nu}^\star(x). \end{aligned}$$

(In the last step we invoked our initial computation with t, f, χ replaced by $t^{-1}, \widehat{f}, \chi^\vee$.) We made the change of variables $y = x^{-1}$. In general the Haar measure on a locally compact abelian group is invariant under inversion. Finally we remark that all integrals above converge since f is a Schwartz function and C_K^1 is compact. □

Exercise 2.99 Check that $E_{t^{-1}}(\widehat{f},\chi^\vee) = -E_t(f,\chi)$ in the notation of the previous Lemma.

Exercise 2.100 Show that $E_t(f,\chi) = 0$ unless $\chi = |\cdot|^s$ for some $s \in \mathbb{C}$. (`Hint`: $\int_{C_K^1}\chi(x)d\dot{\nu}^\star(x) = 0$ when $\chi|_{C_K^1} \neq 1$.)

2.5.4 The Main Global Theorem

Here is the holy grail of Tate's thesis.

Theorem 2.101 (The main global Theorem) *Let $\chi_0 : C_K \longrightarrow \mathbb{C}^1$ be a unitary Hecke character and let $\chi_s = \chi_0|\cdot|^s$ for $s \in \mathbb{C}$. Then the following holds:*

(1) $Z(f,\chi_s)$ has meromorphic continuation to all of $\mathbb{C}$.
(2) $Z(f,\chi_s)$ is an entire function unless $\chi_0 = |\cdot|^{-i\tau}$ for some $\tau \in \mathbb{R}$, in which case it has (at most) simple poles at $s = i\tau$ and $s = 1 + i\tau$ with residues

(a) $res_{s=i\tau}Z(f,\chi_s) = -vol(C_K^1)f(0)$
(b) $res_{s=1+i\tau}Z(f,\chi_s) = +vol(C_K^1)\widehat{f}(0)$

respectively.

(3) For all $f \in \mathcal{S}(\mathbb{A}_K)$ we have the functional equation

$$Z(f, \chi_s) = Z(\widehat{f}, \chi_s^\vee).$$

(Here $\widehat{f}$ is with respect to the self-dual measure on $\mathbb{A}_K$, and $vol(C_K^1) = \dot{\nu}^\star(C_K^1)$ where $\frac{dt}{t} \times \nu^\star$ corresponds to the measure $\mu^\star$ used to define the zeta integrals.)

Proof The third step of our general strategy is to split the integral $Z(f, \chi_s) = \int_0^\infty Z_t(f, \chi_s)\frac{dt}{t}$ into a good part $\int_1^\infty$ and a bad part $\int_0^1$. The integral $\int_1^\infty Z_t(f, \chi_s)\frac{dt}{t}$ makes sense for any s and defines an entire function (to see this write it as $\int_1^\infty \int_{C_K^1} \tilde{f}(tx)d\dot{\nu}^\star(x)\frac{dt}{t}$ and observe that $t \mapsto \int_{C_K^1} \tilde{f}(tx)d\dot{\nu}^\star(x) = \chi_0(t)|t|^s \int_{C_K^1} f(tx)\chi_0(x)d\dot{\nu}^\star(x)$ decays rapidly as $t \to \infty$).

Lastly we express the bad integral $\int_0^1 Z_t(f, \chi_s)\frac{dt}{t}$ in terms of the good integral. Assume $\mathrm{Re}(s) > 1$ so the integral converges. Then by the transformation property of $Z_t(f, \chi_s)$ we find that

$$\begin{aligned}\int_0^1 Z_t(f, \chi_s)\frac{dt}{t} &= \int_0^1 Z_{t^{-1}}(\widehat{f}, \chi_s^\vee)\frac{dt}{t} + \int_0^1 E_t(f, \chi_s)\frac{dt}{t} \\ &= \int_1^\infty Z_t(\widehat{f}, \chi_s^\vee)\frac{dt}{t} + \mathcal{E}(f, \chi_s)\end{aligned}$$

where

$$\mathcal{E}(f, \chi_s) = \int_0^1 E_t(f, \chi_s)\frac{dt}{t} = \int_0^1 \left(\widehat{f}(0)\int_{C_K^1} \chi_s^\vee(t^{-1}x)d\dot{\nu}^\star(x) - f(0)\int_{C_K^1} \chi_s(tx)d\dot{\nu}^\star(x)\right)\frac{dt}{t}.$$

Clearly $\mathcal{E}(f, \chi_s) = 0$ when $\chi_0|_{C_K^1} \neq 1$. On the other hand, when $\chi_0|_{C_K^1} = 1$ the Hecke character χ_0 factors through the idele norm and we may write $\chi_0 = |\cdot|^{-i\tau}$ for some $\tau \in \mathbb{R}$ (the sign in the exponent will be a convenient convention). In this case $\chi_s = |\cdot|^{s-i\tau}$ so for $\mathrm{Re}(s) > 1$ we get

$$\begin{aligned}\mathcal{E}(f, \chi_s) &= \int_0^1 \left(\widehat{f}(0)\int_{C_K^1} |t^{-1}x|^{1-s+i\tau}d\dot{\nu}^\star(x) - f(0)\int_{C_K^1} |tx|^{s-i\tau}d\dot{\nu}^\star(x)\right)\frac{dt}{t} \\ &= \int_0^1 \left(\widehat{f}(0)t^{s-1-i\tau}\int_{C_K^1} 1d\dot{\nu}^\star(x) - f(0)t^{s-i\tau}\int_{C_K^1} 1d\dot{\nu}^\star(x)\right)\frac{dt}{t} \\ &= \mathrm{vol}(C_K^1)\cdot\left(\widehat{f}(0)\int_0^1 t^{s-1-i\tau}\frac{dt}{t} - f(0)\int_0^1 t^{s-i\tau}\frac{dt}{t}\right) \\ &= \mathrm{vol}(C_K^1)\cdot\left(\widehat{f}(0)\frac{1}{s-(1+i\tau)} - f(0)\frac{1}{s-i\tau}\right) \\ &= \mathrm{vol}(C_K^1)\widehat{f}(0)\frac{1}{s-(1+i\tau)} - \mathrm{vol}(C_K^1)f(0)\frac{1}{s-i\tau}.\end{aligned}$$

Altogether we find that

$$Z(f,\chi_s)=\int_1^\infty\left(Z_t(f,\chi_s)+Z_t(\widehat{f},\chi_s^\vee)\right)\frac{dt}{t}+\mathcal{E}(f,\chi_s)$$

extends to a meromorphic function on all of $\mathbb{C}$. It is entire unless $\chi_0=|\cdot|^{-i\tau}$ for some $\tau\in\mathbb{R}$, in which case it has (at most) simple poles at $s=i\tau$ and $s=1+i\tau$ with residues $-\mathrm{vol}(C_K^1)f(0)$ and $+\mathrm{vol}(C_K^1)\widehat{f}(0)$ respectively. (Of course, $i\tau$ is not a pole when $f(0)=0$ and $1+i\tau$ is not a pole if $\widehat{f}(0)=0$.)

The functional equation follows rather easily by observing that $Z_t(\hat{\hat{f}},\chi_s)=Z_t(f,\chi_s)$ and $\mathcal{E}(\widehat{f},\chi_s^\vee)=\mathcal{E}(f,\chi_s)$. The first equality follows from Fourier inversion and the fact that $\chi(-1)=1$ since $\chi|_{K^\times}=1$. The second equality follows from integrating the equation in Exercise 2.99. □

2.5.5 Hecke L-Functions at Last

The next result is what motivated the whole story. By combining the local and global theories we almost immediately derive the analytic continuation of $L^{\mathrm{aut}}(s,\chi_0)$ and its functional equation. Recall that

$$L^{\mathrm{aut}}(s,\chi_0)=\prod_{v\in V}L(s,\chi_{0,v})$$

where $L(s,\chi_{0,v})=L(\chi_{0,v}|\cdot|_v^s)$. We are assuming χ_0 is unitary, which ensures the above product converges for $\mathrm{Re}(s)>1$. Note that the product ranges over all places–finite as well as infinite v. The functional equation involves the global ϵ-factor:

Definition 2.102 Let $\chi_0:C_K\to\mathbb{C}^1$ be a unitary Hecke character. Its ϵ-factor is defined to be the (finite) product

$$\epsilon(s,\chi_0)=\prod_{v\in V}\epsilon(s,\chi_{0,v},\psi_v).$$

Here $\psi=\otimes_{v\in V}\psi_v$ is any choice of non-trivial unitary character of $\mathbb{A}_K/K$. We will soon see the product is independent of ψ and we therefore drop it from the notation and just write $\epsilon(s,\chi_0)$.

We will make $\epsilon(s,\chi_0)$ more explicit below, but let us first establish the functional equation for Hecke L-functions (which in particular shows $\epsilon(s,\chi_0)$ is independent of ψ).

Theorem 2.103 (Hecke, Tate) *Let $\chi_0:C_K\to\mathbb{C}^1$ be a unitary Hecke character. Then $L^{aut}(s,\chi_0)$ has meromorphic continuation to the whole complex plane; it is*

entire unless $\chi_0 = |\cdot|^{-i\tau}$ *for some* $\tau \in \mathbb{R}$*, in which case it has simple poles at* $s = i\tau$ *and* $s = 1 + i\tau$*. Furthermore, we have the functional equation*

$$L^{aut}(s, \chi_0) = \epsilon(s, \chi_0) L^{aut}(1 - s, \bar{\chi}_0).$$

Proof We have already found a concrete Schwartz function $f_0 = (\bigotimes_{v \in S} f_{v,0}) \otimes (\bigotimes_{v \notin S} \mathrm{Ch}_{\mathcal{O}_v})$ for which

$$Z(f_0, \chi_s) = L^{\mathrm{aut}}(s, \chi_0).$$

From the main global Theorem we immediately get the meromorphic continuation of $L^{\mathrm{aut}}(s, \chi_0)$ plus the fact that it is entire unless $\chi_0 = |\cdot|^{-i\tau}$ for some $\tau \in \mathbb{R}$. To see that $s = i\tau$ and $s = 1 + i\tau$ actually are poles in that case we only have to observe that $f_0(0)$ and $\widehat{f_0}(0)$ are nonzero. This is easily read off from our explicit choices of functions $f_{0,v}$ (they are Gaussians or characteristic functions of $\mathcal{O}_v$ since $|\cdot|^{-i\tau}$ is everywhere unramified and $\mathrm{e} = 0$ and $n = 0$ at the real/complex places respectively). We will return to the explicit calculation of $f_0(0)$ and $\widehat{f_0}(0)$ in due time.

For arbitrary Schwartz functions $f \in \mathcal{S}(\mathbb{A}_K)$ we also observed that for any large enough finite set of places $S \supset V_\infty$ we have

$$\frac{Z(f, \chi_s)}{L^{\mathrm{aut}}(s, \chi_0)} = \prod_{v \in S} \frac{Z(f_v, \chi_{v,s})}{L(\chi_{v,s})}.$$

By the main local Theorem the product on the right is an entire function. Moreover, using this for $\widehat{f}$ and $\chi_s^\vee$ yields

$$\begin{aligned}
\frac{Z(\widehat{f}, \chi_s^\vee)}{L^{\mathrm{aut}}(1 - s, \bar{\chi}_0)} &= \prod_{v \in S} \frac{Z(\widehat{f_v}, \chi_{v,s}^\vee)}{L(\chi_{v,s}^\vee)} \\
&= \prod_{v \in S} \epsilon(s, \chi_{0,v}, \psi_v) \frac{Z(f_v, \chi_{v,s})}{L(\chi_{v,s})} \\
&= \epsilon(s, \chi_0) \frac{Z(f, \chi_s)}{L^{\mathrm{aut}}(s, \chi_0)}
\end{aligned}$$

Finally, by the main global Theorem $Z(f, \chi_s) = Z(\widehat{f}, \chi_s^\vee)$. Cancelling out that common factor instantly gives the desired functional equation for the Hecke L-functions. □

As a byproduct of the proof $L^{\mathrm{aut}}(s, \chi_0)$ is a greatest common divisor of all zeta integrals $Z(f, \chi_s)$, just as in the local situation.

Gathering our computations of the local root numbers gives the following explicit formula for $\epsilon(s, \chi_0)$.

Proposition 2.104 *Let* $\chi_0 : C_K \to \mathbb{C}^1$ *be a unitary Hecke character. Define its conductor to be the natural number* $N_{\chi_0} = \prod_{v \in V_f} N(\mathfrak{p}_v)^{cond(\chi_{0,v})}$*. Then*

$$\epsilon(s,\chi_0)=\epsilon(\frac{1}{2},\chi_0)\big(|D_K|N_{\chi_0}\big)^{\frac{1}{2}-s}$$

where the global root number $\epsilon(\frac{1}{2},\chi_0)\in\mathbb{C}^1$ *is a product of*

- $\epsilon(\frac{1}{2},\chi_0)_{arc}=i^{-m}$ *with* $m=\sum_{v\in V_{\mathbb{R}}}e_v+\sum_{v\in V_{\mathbb{C}}}|n_v|\in\mathbb{Z}_{\geq 0}$,
- $\epsilon(\frac{1}{2},\chi_0)_{unr}=\prod_{v:\chi_{0,v}unr}\chi_{0,v}(\partial_{K_v/\mathbb{Q}_p})$,
- $\epsilon(\frac{1}{2},\chi_0)_{ram}=\prod_{v:\chi_{0,v}ram}\overline{\chi_{0,v}(a_v)}|a_v|_v^{\frac{1}{2}}\tau(\bar{\chi}_{0,v},\psi_{std,a_v})$.

(At the ramified places $a_v\in K_v^\times$ *is any element with* $ord_v(a_v)=cond(\psi_{std})-cond(\chi_{0,v})$*.)*

Proof Since $\epsilon(s,\chi_0)$ is independent of ψ we may take $\psi=\psi_{\text{std}}$. For all $v\in V_f$ we have

$$\epsilon(s,\chi_{0,v},\psi_v)=\epsilon(\frac{1}{2},\chi_{0,v},\psi_v)N(\partial_{K_v/\mathbb{Q}_p}\mathcal{N}_{\chi_{0,v}})^{\frac{1}{2}-s}$$

where $\mathcal{N}_{\chi_{0,v}}=\mathfrak{p}_v^{\text{cond}(\chi_{0,v})}$. For $v\in V_{\mathbb{R}}$ we have $\epsilon(s,\chi_{0,v},\psi_v)=i^{-\text{e}_v}$, and for $v\in V_{\mathbb{C}}$ we have $\epsilon(s,\chi_{0,v},\psi_v)=i^{-|n_v|}$ in the hopefully obvious notation. Both are independent of s. Taking the product over all $v\in V$ gives us

$$\epsilon(s,\chi_0)=\epsilon(\frac{1}{2},\chi_0)(|D_K|N_{\chi_0})^{\frac{1}{2}-s}$$

where the global root number factors as

$$\epsilon(\frac{1}{2},\chi_0)=\prod_{v\in V}\epsilon(\frac{1}{2},\chi_{0,v},\psi_v).$$

We have used the fact that the discriminant is the norm of the different, which can be computed locally:

$$\prod_{v\in V_f}N(\partial_{K_v/\mathbb{Q}_p})=N(\partial_{K/\mathbb{Q}})=|D_K|.$$

See [RV99, p. 338] and [Wei95, Chap. 8]. The explicit formulas for the archimedean, unramified, and ramified contributions to $\epsilon(\frac{1}{2},\chi_0)$ are just rephrasing the results of our local calculations. □

In broad strokes $\epsilon(\frac{1}{2},\chi_0)$ is a product of factors i^{-1} from the archimedean places, factors $\chi_{0,v}(\varpi_v)$ from the places where $\chi_{0,v}$ is unramified but $K_v/\mathbb{Q}_p$ is ramified, and factors (fudge)(Gauss sum of $\bar{\chi}_{0,v}$) at the ramified places.

Exercise 2.105 Show that $\overline{\epsilon(\frac{1}{2}, \chi_0)} = \epsilon(\frac{1}{2}, \bar{\chi}_0)$ for all unitary χ_0. (`Hint`: Use the result of Exercise 2.90.)

It is reassuring that Theorem 2.103 and Proposition 2.104 are consistent with Theorem 5 on [Wei95, p. 133] and Proposition 14 on [Wei95, p. 132] respectively.

2.5.6 *Dedekind Revisited*

When $\chi_0 = |\cdot|^{-i\tau}$ the L-function $L^{\text{aut}}(s, \chi_0) = \zeta_K^{\text{aut}}(s - i\tau)$ is essentially a shift of the Dedekind zeta function, and there is no loss of generality by taking $\chi_0 = 1$.

From Theorems 2.101 and 2.103 we immediately deduce that $\zeta_K^{\text{aut}}(s)$ has meromorphic continuation to $\mathbb{C}$, with simple poles at $s = 0$ and $s = 1$ having residues

(a) $\text{res}_{s=0}\zeta_K^{\text{aut}}(s) = -\text{vol}(C_K^1) f_0(0)$
(b) $\text{res}_{s=1}\zeta_K^{\text{aut}}(s) = +\text{vol}(C_K^1) \widehat{f_0}(0)$

respectively. Here f_0 is the Schwartz function from the proof of Theorem 2.103, with the property that $Z(f_0, |\cdot|^s) = \zeta_K^{\text{aut}}(s)$. Moreover, taking $\chi_0 = 1$ in Proposition 2.104 shows $\epsilon(s, 1) = |D_K|^{\frac{1}{2}-s}$. We conclude that $\zeta_K^{\text{aut}}(s)$ satisfies the functional equation

$$\zeta_K^{\text{aut}}(s) = |D_K|^{\frac{1}{2}-s} \zeta_K^{\text{aut}}(1 - s)$$

as we claimed prior to Corollary 1.89. In order to prove the analytic class number formula our next task is to compute the above residues explicitly. This splits up into a volume computation and the evaluation of $f_0(0)$ and $\widehat{f_0}(0)$.

Lemma 2.106 $f_0(0) = \pi^{-r_2} \mu_f^\star(\widehat{\mathcal{O}}_K^\times)^{-1}$ *and* $\widehat{f_0}(0) = \sqrt{|D_K|}^{-1} \pi^{-r_2} \mu_f^\star(\widehat{\mathcal{O}}_K^\times)^{-1}$ *(with the convention that we use the usual archimedean measures* $\frac{dx}{|x|}$ *and* $\frac{dz}{|z|}$ *in zeta integrals).*

Proof Instead of f_0 (which depends on the measure $\mu^\star$ used in the zeta integral) we prefer to work with the following concrete variant:

$$F_0 = (e^{-\pi x^2} \text{ at real places}) \otimes (e^{-2\pi z\bar{z}} \text{ at complex places}) \otimes (\text{Ch}_{\mathcal{O}_v} \text{ at finite places}).$$

Calling upon our local zeta integral calculations we find that

$$Z(F_0, |\cdot|^s) = \pi^{r_2} \mu_f^\star(\widehat{\mathcal{O}}_K^\times) \zeta_K^{\text{aut}}(s).$$

Here we are using the measure $\frac{dx}{|x|}$ at the real places and $\frac{dz}{|z|}$ at the complex places where $dz = 2dxdy$ is double Lebesgue. Note here that $\mu_f^\star(\widehat{\mathcal{O}}_K^\times) = \prod_{v \in V_f} \mu_v^\star(\mathcal{O}_v^\times)$ and the factor π^{r_2} is traced back to the identity $Z(e^{-2\pi z\bar{z}}, |\cdot|^s) = \pi L_{\mathbb{C}}(s)$. In other words we found that $F_0 = \pi^{r_2} \mu_f^\star(\widehat{\mathcal{O}}_K^\times) f_0$. Obviously $F_0(0) = 1$ which gives the formula for $f_0(0)$.

On the other hand we know the Gaussians $e^{-\pi x^2}$ and $e^{-2\pi z\bar{z}}$ are fixed under Fourier transformation (with respect to the ψ_{std}-self-dual measures). Furthermore, for finite places v we recall that

$$\widehat{\text{Ch}}_{\mathcal{O}_v} = \sqrt{N(\partial_{K_v/\mathbb{Q}_p})}^{-1}\,\text{Ch}_{\partial^{-1}_{K_v/\mathbb{Q}_p}}$$

again relative to the ψ_{std}-self-dual measure on K_v. We infer that

$$\widehat{F}_0 = \sqrt{|D_K|}^{-1} \cdot (e^{-\pi x^2} \text{ at real places}) \otimes (e^{-2\pi z\bar{z}} \text{ at complex places}) \otimes (\text{Ch}_{\partial^{-1}_{K_v/\mathbb{Q}_p}} \text{ at finite places}).$$

Once again we used the formula $\prod_{v\in V_f} N(\partial_{K_v/\mathbb{Q}_p}) = |D_K|$. At the origin we have $\widehat{F}_0(0) = \sqrt{|D_K|}^{-1}$. □

In terms of the completed Dedekind zeta function $\xi_K(s) = |D_K|^{\frac{s}{2}}\zeta_K^{\text{aut}}(s)$ the residue formulas can be recast as

(a) $\text{res}_{s=0}\xi_K(s) = -\text{vol}(C_K^1)\pi^{-r_2}$
(b) $\text{res}_{s=1}\xi_K(s) = +\text{vol}(C_K^1)\pi^{-r_2}$

under the assumption that we normalize our measures at the finite places such that $\mu_f^\star(\widehat{\mathcal{O}}_K^\times) = 1$. Note how the two $|D_K|^{\frac{1}{2}}$ cancel out at $s = 1$. It remains to compute $\text{vol}(C_K^1)$ with respect to this normalization of measures.

2.5.7 A Volume Computation

We normalize our measure $\mu^\star = \otimes_{v\in V}\mu_v^\star$ on the ideles such that $\mu_f^\star(\widehat{\mathcal{O}}_K^\times) = 1$, and at the real/complex places we take $\frac{dx}{|x|}$ and $\frac{dz}{|z|}$ as usual. (Here dz is double Lebesgue.)

Theorem 2.107 *$vol(C_K^1) = \frac{2^{r_1}(2\pi)^{r_2}}{w}hR$.*

Proof Recall that $\text{vol}(C_K^1) = \dot{\nu}^\star(C_K^1)$ where $\frac{dt}{t}\times\nu^\star$ corresponds to $\mu^\star$ under $\mathbb{R}_{>0}^\times \times \mathbb{I}_K^1 \xrightarrow{\sim} \mathbb{I}_K$. We start with the four-term exact sequence

$$0 \longrightarrow \mathcal{O}_K^\times \longrightarrow K_\infty^1 \times \widehat{\mathcal{O}}_K^\times \longrightarrow C_K^1 \longrightarrow \text{Cl}_K \longrightarrow 0.$$

from the proof of Theorem 2.20. This shows $\frac{K_\infty^1\times\widehat{\mathcal{O}}_K^\times}{\mathcal{O}_K^\times}$ is of index h in C_K^1 and we are left with showing

$$\text{vol}\Big(\frac{K_\infty^1 \times \widehat{\mathcal{O}}_K^\times}{\mathcal{O}_K^\times}\Big) \stackrel{?}{=} \frac{2^{r_1}(2\pi)^{r_2}}{w}R.$$

Re-examining the proof of Theorem 2.20 we find the short exact sequence

$$0 \longrightarrow \frac{T \times \widehat{\mathcal{O}}_K^\times}{\mu_K} \longrightarrow \frac{K_\infty^1 \times \widehat{\mathcal{O}}_K^\times}{\mathcal{O}_K^\times} \longrightarrow \frac{H}{L} \longrightarrow 0.$$

Recall that $T = \{\pm 1\}^{r_1} \times (\mathbb{C}^1)^{r_2}$ is the kernel of the regulator map $K_\infty^\times \to \mathbb{R}^r$. The quotient $\frac{T \times \widehat{\mathcal{O}}_K^\times}{\mu_K}$ is endowed with the quotient of the product measure where

- $\{\pm 1\}$ has the counting measure;
- $\mathbb{C}^1$ has the angle measure (giving the circle perimeter 2π);
- $\widehat{\mathcal{O}}_K^\times$ has the probability measure $\mu_f^\star$;
- μ_K has the counting measure.

Thus $\mathrm{vol}\big(\frac{T \times \widehat{\mathcal{O}}_K^\times}{\mu_K}\big) = \frac{2^{r_1}(2\pi)^{r_2}}{w}$. The measure on H/L is precisely the quotient of $\dot{\nu}^\star$ on $\frac{K_\infty^1 \times \widehat{\mathcal{O}}_K^\times}{\mathcal{O}_K^\times}$ by the measure on $\frac{T \times \widehat{\mathcal{O}}_K^\times}{\mu_K}$ we just defined, and by definition $R = \mathrm{vol}(H/L)$. For more details on the comparison between this definition of R and the classical one we refer the reader to the proof of Proposition 9 on [Wei95, p. 95]. □

This completes the proof of Theorem 1.84.

Chapter 3
Hecke Characters Versus Galois Characters



Weil singled out a particularly interesting family of Hecke characters in [Wei56] which arise naturally in arithmetic geometry. By definition, characters χ of type (A) take values in the algebraic numbers. The subfamily of characters of type (A_0) is defined by imposing an 'algebraicity' condition on the components of χ at the infinite places, namely that $\chi_\infty = \otimes_{v\in V_\infty}\chi_v$ agrees with the restriction of an algebraic character (in a sense to be made precise). Thus Hecke characters of type (A_0) are also known as *algebraic* Hecke characters.

First one promotes the multiplicative group $K^\times$ to an algebraic torus T_K, defined over $\mathbb{Q}$. For example, $\mathbb{Q}(i)$ can be identified with the subring of matrices of the form $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$ with rational entries, which gives us a way to think of $\mathbb{Q}(i)^\times$ as a subgroup of $\mathrm{GL}_2(\mathbb{Q})$. If we allow the matrix entries to vary in $\overline{\mathbb{Q}}$, the corresponding subgroup of $\mathrm{GL}_2(\overline{\mathbb{Q}})$ is isomorphic to the two-dimensional torus $\overline{\mathbb{Q}}^\times \times \overline{\mathbb{Q}}^\times$. In the lingo of linear algebraic groups, we have defined a non-split torus in GL_2 whose group of $\mathbb{Q}$-points is isomorphic to $\mathbb{Q}(i)^\times$. Similarly, for any number field K of degree d over $\mathbb{Q}$, there is a torus T_K in GL_d having $\mathbb{Q}$-points identifiable with $K^\times$. An *infinity type* is now an algebraic character $\xi : T_K \to \mathrm{GL}_1$ defined over $\mathbb{Q}$, and χ is said to be algebraic of type ξ provided χ_∞ and ξ coincide on the subgroup $T_K(\mathbb{R})^\circ$. For the trivial type $\xi = 1$ this definition captures all finite order characters, such as Dirichlet characters. The idele norm $|\cdot|$ is another example of an algebraic Hecke character. Melding together these examples, $\chi|\cdot|^a$ is algebraic for χ of finite order and $a \in \mathbb{Z}$. In fact this gives a complete list of all algebraic characters for totally real fields (such as $\mathbb{Q}$). However, for CM fields (take cyclotomic fields or imaginary quadratic fields if you like) there exists more intriguing examples of algebraic Hecke characters.

CM stands for complex multiplication, taken in the setting of elliptic curves (or more generally abelian varieties). The starter may want to think of an elliptic curve E as being given by a Weierstrass equation $y^2 = x^3 + Ax + B$, but compactified by adding an extra point at infinity. It has the structure of an abelian group where the 'extra point' is the neutral element. If you pick an elliptic curve E at random it will most likely not admit any endomorphisms other than the obvious ones in the image of the homomorphism $\mathbb{Z} \to \mathrm{End}(E)$. However, there are sparse examples where

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$\mathrm{End}(E)$ is larger than $\mathbb{Z}$, and those are precisely the elliptic curves with complex multiplication. One example is the curve given by the equation $y^2 = x^3 - x$, where sending $(x, y) \mapsto (-x, iy)$ gives a 'non-trivial' endomorphism. In this example it turns out that $\mathrm{End}(E) \simeq \mathbb{Z}[i]$ and we say that E has complex multiplication (CM) by $\mathbb{Q}(i)$. There is a beautiful theory of complex multiplication which expresses the L-function $L(s, E)$ of a CM curve E in terms of a Hecke L-function $L(s, \chi)$, where χ is an algebraic Hecke character of a CM field (both depending on E of course). We emphasize that for such curves E one can therefore derive the analytic properties of $L(s, E)$ from the theory of Hecke and Tate.

The modest goal of this Chapter is to outline how one attaches a Galois character $r_\chi : \mathrm{Gal}(\overline{K}/K) \to \overline{\mathbb{Q}}_p^\times$ to an algebraic Hecke character $\chi : C_K \to \mathbb{C}^\times$, where p can be any prime. The generalization to GL_n is the expectation that with any algebraic cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_K)$, one can associate a representation $r_\pi : \mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}_n(\overline{\mathbb{Q}}_p)$ such that, at any place v, the restriction $r_\pi|_{\mathrm{Gal}(\overline{K}_v/K_v)}$ gives rise to π_v via the local Langlands correspondence. This question is at the heart of the Langlands program. For totally real fields, and CM fields, there has been extraordinary progress over the past three decades (at least for *regular* algebraic π, in other words π such that $\pi_\infty \otimes$ (an algebraic representation) has non-trivial Lie algebra cohomology) but for a general number field K the question very much remains open at the time of writing. We will revisit the $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$ case in later chapters, where the π's in question correspond to certain modular forms.

The construction $\chi \rightsquigarrow r_\chi$ is really a two-step process. First one converts the complex character χ into a p-adic character χ^p. This step is the main subject of Sect. 3.1. The outcome is a continuous character $\chi^p : C_K \to \overline{\mathbb{Q}}_p^\times$, which actually takes values in $\overline{\mathbb{Z}}_p^\times$. The second step is to show χ^p factors through the Artin reciprocity map $C_K \to \mathrm{Gal}(\overline{K}/K)^{\mathrm{ab}}$ from class field theory. Some previous exposure to global class field theory is definitely useful here, but Sect. 3.2 serves as a user guide. The reader should be able to appreciate and employ the key results without knowing the fine details of the theory.

The Galois characters r_χ arising from this construction all have a very special shape at the places above p. For such a place v, the restriction $r_\chi|_{\mathrm{Gal}(\overline{K}_v/K_v)}$ is of the form $\gamma \mapsto \prod_\sigma \sigma(\mathrm{Art}_{K_v}^{-1}(\gamma))^{a_\sigma}$ on an open subgroup of inertia. Here Art_{K_v} denotes the local Artin reciprocity map, σ runs over all embeddings $K_v \hookrightarrow \overline{\mathbb{Q}}_p$, and the integers a_σ can be gleaned from the infinity type ξ. Using the jargon of p-adic Hodge theory, r_χ is potentially crystalline at all places above p (with Hodge-Tate weights a_σ). Besides that, r_χ is unramified (i.e. trivial on inertia) for all but at most finitely many places away from p, but this property turns out to hold automatically for *every* continuous character $r : \mathrm{Gal}(\overline{K}/K) \to \overline{\mathbb{Q}}_p^\times$ (see Proposition 3.27). The analogue of the last statement is false for representations, as shown in [Ram00].

Amazingly these local properties completely characterize the Galois characters of the form r_χ, and this sets up a one-to-one correspondence between algebraic Hecke characters, on the one hand, and these 'geometric' Galois characters on the other. This result is outside the purview of this book. We refer the reader to the excellent survey [Far11] for more details.

One of the pillars of this subject is the Fontaine-Mazur conjecture, first stated explicitly in [FM95, Conjecture 1, p. 194]. It predicts a vast generalization of the bijection in the previous paragraph, with characters replaced by representations. A continuous representation $r : \mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}_n(\overline{\mathbb{Q}}_p)$ is called 'geometric' if it is unramified almost everywhere and potentially semistable at all places dividing p. We treat p-adic Hodge theory as a black box here. For $n > 1$ the notion of being potentially *semistable* is more general than being potentially crystalline. It allows for non-trivial monodromy, which is a certain nilpotent matrix in $M_n(\overline{\mathbb{Q}}_p)$. In any case, at least as long as r is also irreducible, the conjecture forecasts that r necessarily 'comes from algebraic geometry'. For the experts, this means r is a subquotient of the étale cohomology $H^i_{\text{ét}}(X_{\overline{K}}, \overline{\mathbb{Q}}_p(j))$ of some smooth and proper variety $X_{/K}$. Combining this with the Langlands philosophy, one therefore anticipates that there should be a bijection $\pi \rightsquigarrow r_\pi$ between the set of algebraic cuspidal automorphic representations π of $\mathrm{GL}_n(\mathbb{A}_K)$ and the set of irreducible geometric representations r as above (up to isomorphism). There has been remarkable progress on the Fontaine-Mazur conjecture for *odd* representations $r : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\overline{\mathbb{Q}}_p)$ as an application of the so-called p-adic local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_p)$. As a sample result, [Em11, Theorem 1.2.4, p. 4] asserts that r corresponds to a modular form of weight at least two if r is geometric with distinct Hodge-Tate weights (and the mod p reduction $\overline{r}$ satisfies some mild conditions).

Section 3.4 is admittedly a bit misplaced because its main result (Theorem 3.29) applies to *all* unitary Hecke characters without any algebraicity constraint. Thus Sect. 3.4 does not fit the general narrative of the text and can be omitted on a first reading. We wanted to include Theorem 3.29 somewhere, as it generalizes the earlier non-vanishing theorems on the line $\mathrm{Re}(s) = 1$ (for zeta and L-functions over $\mathbb{Q}$) to arbitrary number fields. The proof is a clever application of Landau's Lemma 1.135 (not due to the author of course). Theorem 3.29 is just the tip of the iceberg. A famous result of Jacquet and Shalika gives the analogue for cuspidal automorphic representations of $\mathrm{GL}_n(\mathbb{A}_K)$. This can be viewed as a wide-ranging extension of both the prime number theorem and Dirichlet's theorem on primes in arithmetic progressions.

3.1 Algebraic Hecke Characters

3.1.1 The Torus Defined by a Number Field

Let $K/\mathbb{Q}$ be a number field. To start things off we will describe how the multiplicative group $K^\times$ can be thought of as the $\mathbb{Q}$-points of an algebraic torus T_K. We are not assuming the reader is familiar with the theory of algebraic groups; by and large T_K will be interpreted simply as a functor from commutative $\mathbb{Q}$-algebras to (multiplicative) abelian groups.

Definition 3.1 With a commutative $\mathbb{Q}$-algebra R we associate the group $T_K(R) = (K \otimes_{\mathbb{Q}} R)^\times$.

Over $\bar{\mathbb{Q}}$ there is a canonical isomorphism

$$T_K(\bar{\mathbb{Q}}) = (K \otimes_{\mathbb{Q}} \bar{\mathbb{Q}})^\times \xrightarrow{\sim} (\bar{\mathbb{Q}}^\times)^{\mathrm{Hom}(K,\bar{\mathbb{Q}})}$$

(One way to see this is to think of K as a quotient $\mathbb{Q}[x]/(f)$ for some irreducible polynomial f. Then the embeddings in $\mathrm{Hom}(K, \bar{\mathbb{Q}})$ correspond to the roots of f.) Note that $(\bar{\mathbb{Q}}^\times)^{\mathrm{Hom}(K,\bar{\mathbb{Q}})}$ consists of tuples $(t_\sigma)_{\sigma\in\mathrm{Hom}(K,\bar{\mathbb{Q}})}$ with $t_\sigma \in \bar{\mathbb{Q}}^\times$ but we are not putting an order on $\mathrm{Hom}(K, \bar{\mathbb{Q}})$. The exact same argument works with $\bar{\mathbb{Q}}$ replaced by the Galois closure $\widetilde{K} \subset \bar{\mathbb{Q}}$ of K. In the language of algebraic groups this says T_K is a $[K : \mathbb{Q}]$-dimensional algebraic torus defined over $\mathbb{Q}$ which splits over $\widetilde{K}$.

Exercise 3.2 Explain the isomorphisms

- $T_K(\mathbb{Q}) \simeq K^\times$
- $T_K(\mathbb{R}) \simeq (\mathbb{R}^\times)^{r_1} \times (\mathbb{C}^\times)^{r_2} = K_\infty^\times$
- $T_K(\mathbb{Q}_p) \simeq \prod_{v|p} K_v^\times$.

Remark 3.3 Once and for all we pick embeddings $\iota_\infty : \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $\iota_p : \bar{\mathbb{Q}} \hookrightarrow \bar{\mathbb{Q}}_p$. They induce bijections

$$\mathrm{Hom}(K, \bar{\mathbb{Q}}) \xrightarrow{\sim} \mathrm{Hom}(K, \mathbb{C}), \qquad \mathrm{Hom}(K, \bar{\mathbb{Q}}) \xrightarrow{\sim} \mathrm{Hom}(K, \bar{\mathbb{Q}}_p).$$

Moreover, $\mathrm{Hom}(K, \bar{\mathbb{Q}}_p)$ has a natural partition: $\bar{\mathbb{Q}}_p$ carries the absolute value extending the one on $\mathbb{Q}_p$. Via an embedding $K \hookrightarrow \bar{\mathbb{Q}}_p$ it restricts to an absolute value on K, and a fortiori a place $v|p$ along with a continuous (equivalently $\mathbb{Q}_p$-linear) embedding $K_v \hookrightarrow \bar{\mathbb{Q}}_p$. This gives a bijection

$$\mathrm{Hom}(K, \bar{\mathbb{Q}}_p) \xrightarrow{\sim} \bigsqcup_{v|p} \mathrm{Hom}_{\mathrm{cont}}(K_v, \bar{\mathbb{Q}}_p).$$

(Why do the two sides have the same cardinality? The local degrees $[K_v : \mathbb{Q}_p]$ add up to the global degree $[K : \mathbb{Q}]$.) This will be important later when we set up a dictionary between weights at ∞ and so-called Hodge-Tate weights at p.

By an infinity type we mean an algebraic character $\xi : T_K(\bar{\mathbb{Q}}) \longrightarrow \bar{\mathbb{Q}}^\times$. Under the identification with tuples ξ is of the form

$$\xi : (t_\sigma)_{\sigma\in\mathrm{Hom}(K,\bar{\mathbb{Q}})} \longmapsto \prod_{\sigma\in\mathrm{Hom}(K,\bar{\mathbb{Q}})} t_\sigma^{a_\sigma}$$

for a collection of integers $a_\sigma \in \mathbb{Z}$. The exact same recipe gives a complexified character $\xi_\infty : T_K(\mathbb{C}) \longrightarrow \mathbb{C}^\times$ and a p-adified character $\xi_p : T_K(\bar{\mathbb{Q}}_p) \longrightarrow \bar{\mathbb{Q}}_p^\times$, where

we have tacitly used ι_∞ and ι_p to go between the various index sets of embeddings. The ensuing characters fit in the commutative diagram

$$\begin{array}{ccccc} T_K(\mathbb{C}) & \xleftarrow{\iota_\infty} & T_K(\bar{\mathbb{Q}}) & \xrightarrow{\iota_p} & T_K(\bar{\mathbb{Q}}_p) \\ \downarrow{\scriptstyle \xi_\infty} & & \downarrow{\scriptstyle \xi} & & \downarrow{\scriptstyle \xi_p} \\ \mathbb{C}^\times & \xleftarrow{\iota_\infty} & \bar{\mathbb{Q}}^\times & \xrightarrow{\iota_p} & \bar{\mathbb{Q}}_p^\times. \end{array}$$

In the same fashion we have a $[K_v : \mathbb{Q}_p]$-dimensional algebraic torus T_{K_v} defined over $\mathbb{Q}_p$ such that

$$T_K(\bar{\mathbb{Q}}_p) \xrightarrow{\sim} \prod_{v|p} T_{K_v}(\bar{\mathbb{Q}}_p), \qquad T_{K_v}(\bar{\mathbb{Q}}_p) \xrightarrow{\sim} (\bar{\mathbb{Q}}_p^\times)^{\mathrm{Hom}_{\mathrm{cont}}(K_v, \bar{\mathbb{Q}}_p)}.$$

Correspondingly we factor our p-adified infinity type as $\xi_p = \otimes_{v|p}\xi_v$ where $\xi_v : T_{K_v}(\bar{\mathbb{Q}}_p) \longrightarrow \bar{\mathbb{Q}}_p^\times$.

3.1.2 The Algebraicity Condition

First note that $T_K(\mathbb{R})$ has 2^{r_1} connected components since $\mathbb{R}^\times$ has two. In fact $\pi_0(T_K(\mathbb{R})) \simeq (\mathbb{Z}/2\mathbb{Z})^{r_1}$. We denote the identity component by $T_K(\mathbb{R})^\circ$ or sometimes $(K_\infty^\times)^\circ$. Of course there is an isomorphism $T_K(\mathbb{R})^\circ \simeq (\mathbb{R}_{>0}^\times)^{r_1} \times (\mathbb{C}^\times)^{r_2}$ of topological groups.

Exercise 3.4 Explain why $\pi_1(T_K(\mathbb{R})^\circ) \simeq \mathbb{Z}^{r_2}$.

We will narrow our focus and impose a condition on the infinity component $\chi_\infty = \otimes_{v\in V_\infty}\chi_v$ of our Hecke characters $\chi : C_K \to \mathbb{C}^\times$. This condition captures all the arithmetically interesting Hecke characters appearing in nature. For instance, one can associate a Hecke character χ_E of this kind to an elliptic curve E/K with CM ("complex multiplication") and $L(s, E)$ is essentially the same as $L(s, \chi_E)$. (If E has CM by $\mathcal{O}_M$ and $M \subset K$ the Hasse-Weil L-function factors as $L(s, \chi_E)L(s, \bar{\chi}_E)$. See [Sil94, p. 175] for more details.)

Definition 3.5 A Hecke character $\chi : C_K \to \mathbb{C}^\times$ is said to be algebraic of infinity type ξ if $\chi_\infty = \xi_\infty$ on the identity component $T_K(\mathbb{R})^\circ$. (Here we restrict the character ξ_∞ to $T_K(\mathbb{R}) \subset T_K(\mathbb{C})$.)

Example 3.6 If $\xi = 1$ (in other words all $a_\sigma = 0$) a Hecke character χ is algebraic of that type precisely when χ has finite order. Why? Certainly any finite order character χ_∞ must be trivial on the connected group $T_K(\mathbb{R})^\circ$. Conversely, if $\chi_\infty|_{T_K(\mathbb{R})^\circ} = 1$ then $\chi = 1$ on some finite index subgroup $K^\times(K_\infty^\times)^\circ H$ of $\mathbb{I}_K$. Here $H = 1 + N\widehat{\mathcal{O}}_K$ for a large enough N. (Recall that $K^\times K_\infty^\times \widehat{\mathcal{O}}_K^\times$ has index h in $\mathbb{I}_K$.)

This means in particular that any two algebraic Hecke characters of the same (possibly non-trivial) infinity type ξ coincide up to a finite order twist, and vice versa (any finite order twist of an algebraic Hecke character of type ξ is again of type ξ).

Example 3.7 The idele norm $|\cdot| : C_K \to \mathbb{R}^\times_{>0}$ is an algebraic Hecke character of type ξ_1 (the one with all $a_\sigma = 1$). One often writes $N_{K/\mathbb{Q}}$ instead of ξ_1, for obvious reasons (on $\mathbb{Q}$-points ξ_1 restricts to the norm map $K^\times \to \mathbb{Q}^\times$).

Remark 3.8 One may ask why we do not require that $\chi_\infty = \xi_\infty$ on all of $T_K(\mathbb{R})$, but only on the identity component? The former is too confining. For example, if $\chi : C_\mathbb{Q} \to \mathbb{C}^\times$ comes from Dirichlet character, then χ is algebraic of type $\xi = 1$ but $\chi_\infty = \mathrm{sign}^\mathrm{e}$ where $\mathrm{e} \in \{0, 1\}$ is the parity of the Dirichlet character.

Let us unpack the definition a bit. First of all, for any Hecke character χ all the factors χ_v have the same exponent. Namely $a_{\chi_v} = a_\chi$ for all $v \in V$. Now suppose χ is algebraic of type ξ.

- If $v \in V_\mathbb{R}$ corresponds to the embedding $\sigma : K \hookrightarrow \mathbb{C}$, then $\chi_v(x) = x^{a_\sigma}$ for positive x. Therefore a_χ must be an integer, and $a_\sigma = a_\chi$ for all real embeddings σ.
- If $v \in V_\mathbb{C}$ corresponds to a conjugate pair of non-real embeddings $\tau \neq \bar\tau : K \hookrightarrow \mathbb{C}$, then $\chi_v(z) = z^{a_\tau}\bar z^{a_{\bar\tau}} = \left(\frac{z}{|z|_\mathbb{C}}\right)^{a_\tau - a_{\bar\tau}} |z|^{\frac{1}{2}(a_\tau + a_{\bar\tau})}$ for all z. Consequently $\frac{1}{2}(a_\tau + a_{\bar\tau}) = a_\chi$ for all τ. In particular this forces $a_\chi \in \frac{1}{2}\mathbb{Z}$.

One often write $a_\chi = \frac{w}{2}$ where $w \in \mathbb{Z}$ is the so-called motivic weight of χ.

For $K = \mathbb{Q}$ all algebraic Hecke characters essentially arise from Dirichlet characters. More generally, for totally real fields we have the classification below.

Lemma 3.9 *For a totally real field K the algebraic Hecke characters are precisely those of the form $\chi = \chi_0|\cdot|^a$ where $a \in \mathbb{Z}$ and $\chi_0 : C_K \to \mathbb{C}^\times$ is a finite order character. The only possible infinity types are $N^a_{K/\mathbb{Q}}$ where $a \in \mathbb{Z}$.*

Proof This is just summarizing what we have already remarked. We know $|\cdot|^a$ is algebraic of type $N^a_{K/\mathbb{Q}}$, and therefore so is $\chi_0|\cdot|^a$ for any finite order χ_0. Conversely, if χ is algebraic of type ξ we know a_χ is an integer and $a_\sigma = a_\chi$ for all σ. Thus $\xi = N^{a_\chi}_{K/\mathbb{Q}}$. It follows that $\chi|\cdot|^{-a_\chi}$ must be of finite order. □

Algebraic Hecke characters are more substantial for CM fields K (a totally imaginary quadratic extension of a totally real subfield $K^+ \subset K$). For example $K = \mathbb{Q}(\zeta_N)$ or an imaginary quadratic field. In that case the only constraint on the infinity type is that the sums $a_\tau + a_{\bar\tau} = w$ are independent of $\tau : K \hookrightarrow \mathbb{C}$.

3.1.3 The p-Adic Hecke Character

Our eventual goal is to attach a Galois character to an algebraic Hecke character χ, using class field theory. In fact we will do this for each choice of p (and embeddings

ι_∞ and ι_p as above). This will result in a whole compatible system of p-adic Galois characters $r_\chi : \mathrm{Gal}(\bar{K}/K) \longrightarrow \bar{\mathbb{Q}}_p^\times$, and we wish to describe the Galois characters arising in this fashion by local conditions.

Prior to invoking class field theory we first translate χ into a p-adic character, in other words a continuous (now for the p-adic topology!) character $\chi^p : \mathbb{I}_K/K^\times \longrightarrow \bar{\mathbb{Q}}_p^\times$. It depends on ι_∞ and ι_p but we suppress that from the notation. The construction of χ^p proceeds in two steps. The first step is to descend χ to a character $\check{\chi} : \mathbb{I}_K \to \bar{\mathbb{Q}}^\times$ via ι_∞, and the second step is then to base change $\check{\chi}$ to a p-adic character χ^p via ι_p.

We will initially assume $\chi_\infty = \xi_\infty$ holds on all of $T_K(\mathbb{R})$, and then point out what modifications one has to make if equality only holds on $T_K(\mathbb{R})^\circ$.

The very first step is to restrict χ to the finite ideles $\mathbb{I}_{K,f} = \prod'_{v\in V_f} K_v^\times$ consisting of all $x = (x_v)_{v\in V}$ in $\mathbb{I}_K$ with $x_v = 1$ for all $v \in V_\infty$. We denote the restriction by $\chi_f = \otimes_{v\in V_f}\chi_v$. There is still a diagonal embedding $K^\times \hookrightarrow \mathbb{I}_{K,f}$, but now $K^\times$ is usually not a discrete subgroup although $\mathbb{I}_{K,f}/K^\times$ is compact (there is an obvious continuous surjective map $C_K^1 \twoheadrightarrow \mathbb{I}_{K,f}/K^\times$).

Exercise 3.10 Show that the image of $K^\times \hookrightarrow \mathbb{I}_{K,f}$ is discrete if and only if $\mathcal{O}_K^\times$ is finite, in other words when $K = \mathbb{Q}$ or K is an imaginary quadratic field. (`Hint`: If $\mathcal{O}_K^\times$ is infinite, then for every nonzero ideal $I \subset \mathcal{O}_K$ there is a $u \in \mathcal{O}_K^\times$ such that $u \neq 1$ but $u \equiv 1 \pmod{I}$. Use this to find a sequence $u_1, u_2, u_3, \ldots$ in $K^\times - \{1\}$ converging to $1 \in \mathbb{I}_{K,f}$.)

Note that χ_f is typically not trivial on $K^\times \hookrightarrow \mathbb{I}_{K,f}$. Indeed, letting $\gamma_f \in \mathbb{I}_{K,f}$ denote the diagonal embedding of $\gamma \in K^\times$ we have

$$\chi_f(\gamma_f) = \chi_\infty(\gamma_\infty)^{-1} = \xi_\infty(\gamma_\infty)^{-1} = \iota_\infty\big(\xi(\gamma)^{-1}\big).$$

The first equality is a reformulation of $\chi(\gamma) = 1$, and the second uses our initial hypothesis that $\chi_\infty = \xi_\infty$ on all of $T_K(\mathbb{R})$. Observe that $\xi(\gamma) \in \widetilde{K}^\times$ for all $\gamma \in K^\times$ since T_K splits over the Galois closure $\widetilde{K} \subset \bar{\mathbb{Q}}$.

Lemma 3.11 *Under the above assumptions there is a number field $L \subset \bar{\mathbb{Q}}$ such that $\chi_f(\mathbb{I}_{K,f}) \subset \iota_\infty(L^\times)$.*

Proof There is an open subgroup $H \subset \widehat{\mathcal{O}}_K^\times$ such that $\chi_f(H) = \{1\}$. The subgroup $K^\times H$ has finite index in $\mathbb{I}_{K,f}$ since $K^\times\widehat{\mathcal{O}}_K^\times$ has index h. Let $m = [\mathbb{I}_{K,f} : K^\times H]$ and pick an arbitrary $x \in \mathbb{I}_{K,f}$. Since $x^m \in K^\times H$ the preceding calculation of $\chi_f(\gamma_f)$ shows $\chi_f(x^m) \in \iota_\infty(\widetilde{K}^\times)$. Adjoining all mth roots of elements from $\widetilde{K}$ gives a number field L by Kummer theory. □

Definition 3.12 We let $\check{\chi} = \iota_\infty^{-1} \circ \chi_f$ denote the character $\mathbb{I}_{K,f} \to \bar{\mathbb{Q}}^\times$ such that $\chi_f = \iota_\infty \circ \check{\chi}$. (Note that $\check{\chi}$ takes values in $L^\times$ for some number field $L \subset \bar{\mathbb{Q}}$.)

As a result of the above calculation for $\chi_f(\gamma_f)$ the character $\check{\chi}$ satisfies the transformation property

$$\check{\chi}(\gamma_f x) = \xi(\gamma)^{-1}\check{\chi}(x)$$

for all $\gamma \in K^\times$ and $x \in \mathbb{I}_{K,f}$. Furthermore $\check{\chi}|_H = 1$ for some open subgroup $H \subset \widehat{\mathcal{O}}_K^\times$.

This completes the descent step. Next we base change $\check{\chi}$ to a p-adic character. First consider $\iota_p \circ \check{\chi}$. This character $\mathbb{I}_{K,f} \to \bar{\mathbb{Q}}_p^\times$ has the transformation property

$$(\iota_p \circ \check{\chi})(\gamma_f x) = \xi_p(\gamma_p)^{-1}(\iota_p \circ \check{\chi})(x),$$

where $\gamma_p \in T_K(\mathbb{Q}_p)$ denotes the image of γ under the embedding $K^\times \hookrightarrow \prod_{v|p} K_v^\times$.

Definition 3.13 The p-adic character associated with χ is given by $\chi^p(x) = \xi_p(x_p)(\iota_p \circ \check{\chi})(x)$ where we have used the notation $x_p = (x_v)_{v|p} \in \prod_{v|p} K_v^\times$ for a finite idele x.

The front factor $\xi_p(x_p)$ in some sense restores the automorphicity. To wit, the transformation property of $\iota_p \circ \check{\chi}$ translates into $\chi^p|_{K^\times} = 1$. In other words χ^p factors through the compact quotient $\mathbb{I}_{K,f}/K^\times$, and in particular χ^p is p-adically unitary–meaning χ^p takes values in the units $\bar{\mathbb{Z}}_p^\times$.

In fact χ_p is not only continuous for the p-adic topology, it enjoys much stronger properties. To explain that we first factor $\mathbb{I}_{K,f} = \prod_{v|p} K_v^\times \times \mathbb{I}_{K,f}^p$ where $\mathbb{I}_{K,f}^p$ is the group of ideles away from p. We may pick the open subgroup H to be a product $H = H_p H^p$ where $H_p \subset \prod_{v|p} \mathcal{O}_v^\times$ and $H^p \subset \mathbb{I}_{K,f}^p$ are open subgroups. Sometimes H^p is referred to as the tame level of χ, and H_p is called the wild level. With this notation we remark that

- $\chi^p = 1$ on H^p ("smooth away from p");
- $\chi^p = \xi_p$ on H_p ("locally algebraic at p").

In vague terms χ^p is locally given by a polynomial function in the variables $(x_v)_{v|p}$, and it is therefore certainly continuous.

Up to this point we have been working under the hypothesis that $\chi_\infty = \xi_\infty$ on all of $T_K(\mathbb{R})$. When this only holds on $T_K(\mathbb{R})^\circ$ the above construction needs to be modified. Now $\check{\chi}$ is a character on the full group of ideles $\mathbb{I}_K$ given by

$$\check{\chi}(x) = \chi_\infty(x_\infty)\xi_\infty(x_\infty)^{-1}(\iota_\infty^{-1} \circ \chi_f)(x_f).$$

This expression lies in $\bar{\mathbb{Q}}^\times$ since the factor $\chi_\infty(x_\infty)\xi_\infty(x_\infty)^{-1}$ just introduces a sign (recall $\pi_0(T_K(\mathbb{R}))$ is an elementary abelian 2-group). It is easily checked that $\check{\chi}(\gamma x) = \xi(\gamma)^{-1}\check{\chi}(x)$ for $\gamma \in K^\times$ and $x \in \mathbb{I}_K$, not just for finite ideles. The p-adic character is defined by the same formula $\chi^p(x) = \xi_p(x_p)(\iota_p \circ \check{\chi})(x)$, but for $x \in \mathbb{I}_K$. This gives a continuous character $\chi^p : C_K \to \bar{\mathbb{Z}}_p^\times$. We summarize its properties below.

Construction 3.14 Let $\chi : C_K \to \mathbb{C}^\times$ be an algebraic Hecke character of infinity type ξ. Then for each prime p there is continuous character $\chi^p : C_K \to \bar{\mathbb{Z}}_p^\times$ with the following properties:

(1) $\chi^p = 1$ on an open subgroup of $\mathbb{I}_{K,f}^p$
(2) $\chi^p = \xi_p$ on an open subgroup of $\prod_{v|p} K_v^\times$

(3) $\chi^p = \frac{\chi_\infty}{\xi_\infty}$ on $K_\infty^\times$.

It is given explicitly by the formula

$$\chi^p(x) = \chi_\infty(x_\infty)\xi_\infty(x_\infty)^{-1} \cdot \xi_p(x_p) \cdot (\iota_p \circ \iota_\infty^{-1} \circ \chi_f)(x_f).$$

for all $x \in \mathbb{I}_K$.

Exercise 3.15 Let $\chi = |\cdot|$ be the idele norm on C_K. Describe the associated p-adic character χ^p. For instance, for a fixed $v|p$ and $x \in K_v^\times$ show that $\chi^p(x) = N_{K_v/\mathbb{Q}_p}(x)^\times$ where $y^\times = y|y|_p \in \mathbb{Z}_p^\times$ denotes the unit factor of a p-adic number $y \in \mathbb{Q}_p^\times$.

3.2 Global Class Field Theory in a Nutshell

3.2.1 The Artin Reciprocity Map

The basic idea behind the Artin map is pretty simple. Suppose S is a finite set of maximal ideals of $\mathcal{O}_K$, and E/K is a finite abelian extension unramified outside S. Sending $\mathfrak{p} \mapsto \mathrm{Frob}_\mathfrak{p}$ defines a homomorphism $\mathrm{Id}_K^S \to \mathrm{Gal}(E/K)$ which is surjective by the Chebotarev density theorem. Here $\mathrm{Id}_K^S = \bigoplus_{\mathfrak{p}\notin S} \mathbb{Z}\mathfrak{p}$ is the group of nonzero fractional ideals coprime to S, and $\mathrm{Frob}_\mathfrak{p} \in \mathrm{Gal}(E/K)$ is the element inducing the Frobenius automorphism $x \mapsto x^{N(\mathfrak{p})}$ on the residue field of any prime of E above $\mathfrak{p}$. (Since $\mathrm{Gal}(E/K)$ is abelian $\mathrm{Frob}_\mathfrak{p}$ only depends on $\mathfrak{p}$.) The glaring problem is to describe the kernel of the projection $\mathrm{Id}_K^S \twoheadrightarrow \mathrm{Gal}(E/K)$.

Exercise 3.16 Let S_E denote the set of maximal ideals $\mathfrak{P} \subset \mathcal{O}_E$ dividing some $\mathfrak{p} \in S$. Define a norm map $N_{E/K} : \mathrm{Id}_E^{S_E} \to \mathrm{Id}_K^S$ by $N_{E/K}(\mathfrak{P}) = \mathfrak{p}^{f(\mathfrak{P}|\mathfrak{p})}$ when $\mathfrak{P}|\mathfrak{p}$ and $\mathfrak{p} \notin S$. Explain why

$$N_{E/K}(\mathrm{Id}_E^{S_E}) \subset \ker\big(\mathrm{Id}_K^S \longrightarrow \mathrm{Gal}(E/K)\big).$$

The kernel becomes much more transparent if we pass to ideles instead of ideals. The map $\mathbb{I}_K \twoheadrightarrow \mathrm{Id}_K$ restricts to a map $\mathbb{I}_K^S \twoheadrightarrow \mathrm{Id}_K^S$ taking $x \mapsto (x) = \prod_{v\in V_f - S} \mathfrak{p}_v^{\mathrm{ord}_v(x_v)}$. Here $\mathbb{I}_K^S$ is the group of ideles x with $x_v = 1$ for all $v \in S$ (via $v \leftrightarrow \mathfrak{p}_v$ we confuse S with a set of finite places). Composition results in a map $\mathbb{I}_K^S \twoheadrightarrow \mathrm{Gal}(E/K)$ sending $\varpi_v \mapsto \mathrm{Frob}_v$ for $v \notin S$. In other words it restricts to the local Artin map $K_v^\times \twoheadrightarrow \mathrm{Gal}(E_w/K_v)$ for all $w|v$ provided $v \notin S$. Ideally we would like a map $\mathbb{I}_K \twoheadrightarrow \mathrm{Gal}(E/K)$ which restricts to the local Artin map at *all* places $v \in V$ (not just the unramified ones). The global Artin map makes our dreams come true. We will not define it precisely, but only list some of its most important properties.

Notation 3.17 The norm map $N_{E/K} : E^\times \to K^\times$ extends to a homomorphism $N_{E/K} : \mathbb{I}_E \to \mathbb{I}_K$ by mapping the tuple $x = (x_w)_{w\in V_E} \in \mathbb{I}_E$ to $N_{E/K}(x) =$

$(\prod_{w|v} N_{E_w/K_v}(x_w))_{v\in V}$. This norm map on ideles descends to a norm map between idele class groups $N_{E/K} : C_E \to C_K$.

Theorem 3.18 (Artin's reciprocity law) *Let $K/\mathbb{Q}$ be a number field. There exists a continuous surjective homomorphism*

$$Art_K : C_K \longrightarrow Gal(\bar{K}/K)^{ab}$$

which induces an isomorphism

$$Art_{E/K} : C_K/N_{E/K}(C_E) \xrightarrow{\sim} Gal(E/K)$$

for all finite abelian extensions E/K (inside $\bar{K}$).

Proof Omitted. We refer the reader to [Wei95, p. 276]. □

Remark 3.19 In the previous Theorem $K \hookrightarrow \bar{K}$ is a fixed algebraic closure and $\mathrm{Gal}(\bar{K}/K)$ is the absolute Galois group. It carries the Krull topology which makes it a profinite group. Thus a neighborhood basis at the identity consists of the open normal finite index subgroups $\mathrm{Gal}(\bar{K}/E)$ where E ranges over *all* finite Galois extensions of K inside $\bar{K}$. In other words $\mathrm{Gal}(\bar{K}/K) \xrightarrow{\sim} \varprojlim_E \mathrm{Gal}(E/K)$ as topological groups. Intuitively an $\alpha \in \mathrm{Gal}(\bar{K}/K)$ is "close" to the identity if α fixes a "large" finite Galois extension E/K.

The abelianization $\mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}}$ is the quotient of $\mathrm{Gal}(\bar{K}/K)$ by the *closure* of the commutator subgroup. Alternatively $\mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}} \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K)$ where K^{ab} is the union of all finite abelian extensions of K in $\bar{K}$. One of the goals of class field theory is to describe K^{ab} explicitly. For instance, the celebrated Kronecker-Weber theorem says that $\mathbb{Q}^{\mathrm{ab}}$ is generated by all roots of unity.

Note that the Artin map of a number field is *surjective*, see [Tat79, Sect. 1.4.4, p. 6]. This is not true for local fields in general. If v is a non-archimedean place, Art_{K_v} only has dense image in $\mathrm{Gal}(\overline{K_v}/K_v)^{\mathrm{ab}}$ given by the abelianization of the Weil group $W_{K_v}^{\mathrm{ab}}$. The Weil group W_{K_v} is the subgroup of automorphisms inducing *integral* powers of Frobenius on the residue field. Analogously, think of $\mathbb{Z}$ which sits as a dense subgroup of the absolute Galois group of a finite field $\widehat{\mathbb{Z}}$. Every number field K does admit a Weil group W_K, i.e. a group satisfying the conditions (W_1) through (W_4) in [Tat79, Definition 1.1], but all known constructions of W_K are somewhat indirect. We refer the reader to [Tat79, Sect. 1.2] where W_K is constructed as a projective limit of groups $W_{E/K}$ corresponding to certain classes in $H^2(\mathrm{Gal}(E/K), C_E)$.

The reciprocity law comes with a counterpart:

Theorem 3.20 (The existence Theorem) *As E varies over finite abelian extensions of K, the norm groups $N_{E/K}(C_E)$ are exactly the open subgroups of finite index in C_K. More precisely $E \mapsto N_{E/K}(C_E)$ defines a bijection between the two sets.*

Proof Again we guide the reader to Corollary 2 on [Wei95, p. 276]. □

So what is the kernel of Art_K itself? Clearly it contains the identity component C_K°, and miraculously that is the whole kernel! In other words Art_K induces an isomorphism $\pi_0(C_K) \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K)$. See [AT09, Chap. 9] where the structure of C_K° is also determined. It is surprisingly simple:

$$C_K^\circ = \text{image of } \overline{K^\times (K_\infty^\times)^\circ} \text{ under } \mathbb{I}_K \twoheadrightarrow C_K.$$

We summarize this in the short exact sequence below.

$$0 \longrightarrow \overline{K^\times (K_\infty^\times)^\circ} \longrightarrow \mathbb{I}_K \longrightarrow \mathrm{Gal}(K^{\mathrm{ab}}/K) \longrightarrow 0.$$

This is the essential apparatus enabling us to associate Galois characters to Hecke characters, and the other way around. Before doing so we circle back to local-global compatibility.

For each $v \in V$ choose an algebraic closure $K_v \hookrightarrow \overline{K_v}$ and an embedding $\bar{K} \hookrightarrow \overline{K_v}$ which fits in a commutative diagram

$$\begin{array}{ccc} \bar{K} & \longrightarrow & \overline{K_v} \\ \uparrow & & \uparrow \\ K & \longrightarrow & K_v. \end{array}$$

This gives rise to a continuous injective homomorphism $\mathrm{Gal}(\overline{K_v}/K_v) \hookrightarrow \mathrm{Gal}(\bar{K}/K)$. Changing $\bar{K} \hookrightarrow \overline{K_v}$ gives a conjugate subgroup. (This is analogous to a decomposition subgroup at a finite v.) Passing to abelianizations yields a diagram

$$\begin{array}{ccc} C_K & \xrightarrow{\mathrm{Art}_K} & \mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}} \\ \uparrow & & \uparrow \\ K_v^\times & \xrightarrow{\mathrm{Art}_{K_v}} & \mathrm{Gal}(\overline{K_v}/K_v)^{\mathrm{ab}} \end{array}$$

where Art_{K_v} is the local Artin map. It is a marvelous result that this diagram indeed commutes.

Remark 3.21 When varying the base field K, the map Art_K has additional functorial properties which we will only mention in passing. For example, the norm map $N_{E/K} : C_E \to C_K$ corresponds to the (abelianization of the) inclusion $\mathrm{Gal}(\bar{E}/E) \hookrightarrow \mathrm{Gal}(\bar{K}/K)$. The natural inclusion $C_K \hookrightarrow C_E$ corresponds to a more complicated map $V : \mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}} \longrightarrow \mathrm{Gal}(\bar{E}/E)^{\mathrm{ab}}$ known as the transfer map (or Verlagerung) which only exists between the abelianizations, not the Galois groups themselves. The pertaining diagrams can be found on [RV99, p. 225].

3.2.2 The Langlands Correspondence for GL_1

The dual of the Artin map gives a bijection

$$\mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\bar{K}/K), \mathbb{C}^\times) \xrightarrow{1:1} \{\text{finite order Hecke characters } C_K \to \mathbb{C}^\times\}.$$

Why? The continuous complex characters of $\mathbb{I}_K/\overline{K^\times(K_\infty^\times)^\circ}$ are precisely the Hecke characters $\chi : C_K \to \mathbb{C}^\times$ trivial on $(K_\infty^\times)^\circ$. In other words, algebraic Hecke characters of infinity type $\xi = 1$. We already saw in Example 3.6 that those have finite order and vice versa.

Consequently, complex characters of $\mathrm{Gal}(\bar{K}/K)$ do not account for algebraic Hecke characters of infinity type $\xi \neq 1$. The complex topology forces every character $\mathrm{Gal}(\bar{K}/K) \to \mathbb{C}^\times$ to have finite order. In lieu of $\mathbb{C}$ we could take any topological ring R and again get a bijection

$$\mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\bar{K}/K), R^\times) \xrightarrow{1:1} \mathrm{Hom}_{\mathrm{cont}}(\mathbb{I}_K/\overline{K^\times(K_\infty^\times)^\circ}, R^\times).$$

Still, on the right-hand side we have continuous characters $C_K \to R^\times$ which are trivial on $(K_\infty^\times)^\circ$. We will fix a prime p and consider the coefficient rings $R = \bar{\mathbb{Q}}_p$ and $R = \bar{\mathbb{Z}}_p$. This gives much more flexibility on the Galois side.

The output of the construction $\chi \rightsquigarrow \chi^p$ is trivial on $(K_\infty^\times)^\circ$ so we can feed χ^p into the above bijection and associate a p-adic Galois character with χ.

$$r_\chi : \mathrm{Gal}(\bar{K}/K) \longrightarrow \bar{\mathbb{Q}}_p^\times, \qquad r_\chi = \chi^p \circ \mathrm{Art}_K^{-1}.$$

For each $v \in V$ the restriction $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ is independent of the choice of $\bar{K} \hookrightarrow \overline{K_v}$, and via local class field theory it corresponds to χ_v when $v \nmid p$. We make this precise below.

Construction 3.22 Let $\chi : C_K \longrightarrow \mathbb{C}^\times$ be an algebraic Hecke character of infinity type ξ. Fix a prime p plus embeddings $\iota_\infty : \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $\iota_p : \bar{\mathbb{Q}} \hookrightarrow \bar{\mathbb{Q}}_p$ as above. Then there exists a continuous p-adic Galois character

$$r_\chi : \mathrm{Gal}(\bar{K}/K) \longrightarrow \bar{\mathbb{Q}}_p^\times$$

with the following properties:

(1) For $v \in V_f$ <u>not</u> above p we have

$$r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)} \circ \mathrm{Art}_{K_v} = \iota_p \circ \iota_\infty^{-1} \circ \chi_v.$$

(2) For $v \in V_f$ above p we have

$$r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)} \circ \mathrm{Art}_{K_v} = \xi_v \otimes (\iota_p \circ \iota_\infty^{-1} \circ \chi_v)$$

where $\xi_v : K_v^\times \longrightarrow \bar{\mathbb{Q}}_p^\times$ is the vth factor of ξ_p introduced at the end of Sect. 3.1.1.

(3) For $v \in V_{\mathbb{R}}$ corresponding to $\sigma : K \hookrightarrow \mathbb{R}$ we have

$$r_\chi(c_\sigma) = \frac{\chi_v}{\xi_v}(-1)$$

for any complex conjugation $c_\sigma \in \mathrm{Gal}(\bar{K}/K)$ arising from a lift of σ to $\bar{K} \hookrightarrow \mathbb{C}$.

(4) For $v \in V_{\mathbb{C}}$ we have $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)} = 1$.

This all follows from local-global compatibility using the explicit formula for χ^p.

Remark 3.23 At $v|p$ the restriction $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ corresponds to the locally algebraic character $\xi_v \otimes (\iota_p \circ \iota_\infty^{-1} \circ \chi_v)$ via local class field theory. More precisely, if $\chi_v|_{H_v} = 1$ for a small enough open subgroup $H_v \subset \mathcal{O}_v^\times$, then on H_v the locally algebraic character is given by ξ_v.

Composition with Art_{K_v} gives a bijection

$$\mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\overline{K_v}/K_v), \bar{\mathbb{Q}}_p^\times) \xrightarrow{1:1} \mathrm{Hom}_{\mathrm{cont}}(K_v^\times, \bar{\mathbb{Z}}_p^\times)$$

between the p-adic characters of $\mathrm{Gal}(\overline{K_v}/K_v)$ and the p-adically unitary characters of $K_v^\times$. In more detail

$$\begin{aligned}\mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\overline{K_v}/K_v), \bar{\mathbb{Q}}_p^\times) &= \mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\overline{K_v}/K_v), \bar{\mathbb{Z}}_p^\times) \\ &\xrightarrow{1:1} \mathrm{Hom}_{\mathrm{cont}}(\widehat{K_v^\times}, \bar{\mathbb{Z}}_p^\times) \\ &= \mathrm{Hom}_{\mathrm{cont}}(K_v^\times, \bar{\mathbb{Z}}_p^\times).\end{aligned}$$

This is the p-adic local Langlands correspondence for GL_1.

Exercise 3.24 Let $\chi = |\cdot|$ be the idele norm on C_K. Show that r_χ is the *inverse* of the p-adic cyclotomic character

$$r_{\mathrm{cyc}} : \mathrm{Gal}(\bar{K}/K) \longrightarrow \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q}) \xrightarrow{\sim} \mathbb{Z}_p^\times$$

defined by $\alpha(\zeta_{p^r}) = \zeta_{p^r}^{r_{\mathrm{cyc}}(\alpha)}$ for all $r > 0$. (`Hint`: For $K = \mathbb{Q}$ note that $(r_{\mathrm{cyc}} \circ \mathrm{Art}_{\mathbb{Q}})(\ell) = \ell = \chi^p(\ell)^{-1}$.)

3.3 Geometric Galois Characters

3.3.1 Local Properties of p-Adic Galois Characters

Recall that the inertia subgroup at $v \in V_f$ is defined by the short exact sequence

$$0 \longrightarrow I_v \longrightarrow \mathrm{Gal}(\overline{K_v}/K_v) \longrightarrow \mathrm{Gal}(\overline{k_v}/k_v) \longrightarrow 0.$$

The Frobenius element $\mathrm{Frob}_v \in \mathrm{Gal}(\overline{K_v}/K_v)/I_v$ induces the automorphism $x \mapsto x^{q_v}$ on the residue field of $\overline{K_v}$. If R is a topological ring, a continuous character $r_v : \mathrm{Gal}(\overline{K_v}/K_v) \to R^\times$ is said to be unramified if $r_v|_{I_v} = 1$. For unramified r_v we can unambiguously make sense of the element $r_v(\mathrm{Frob}_v) \in R^\times$.

Our choice of Frob_v is called the arithmetic Frobenius. Some authors prefer the geometric Frobenius which induces the inverse Frobenius automorphism $x \mapsto x^{1/q_v}$ on the residue field. In many situations the latter is the more natural choice (cf. Exercise 3.24).

The local Artin map restricts to an isomorphism $\mathcal{O}_v^\times \xrightarrow{\sim} I_v^{\mathrm{ab}}$. Thus r_v is unramified if and only if $r_v \circ \mathrm{Art}_{K_v}$ is unramified, in which case $r_v(\mathrm{Frob}_v) = (r_v \circ \mathrm{Art}_{K_v})(\varpi_v)$.

We apply this to the Galois character r_χ attached to an algebraic Hecke character χ. We deduce that $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ is unramified if and only if χ_v is unramified, provided $v \nmid p$. In particular r_χ is unramified at all but at most finitely places $v \in V_f$, and at such v we have $r_\chi(\mathrm{Frob}_v) = (\iota_p \circ \iota_\infty^{-1})(\chi_v(\varpi_v))$. In addition

$$|(\iota_\infty \circ \iota_p^{-1})(r_\chi(\mathrm{Frob}_v))|_{\mathbb{C}} = |\chi_v(\varpi_v)|_{\mathbb{C}} = |\varpi_v|_v^{a_\chi} = q_v^{-\frac{w}{2}}.$$

Thus $r_\chi(\mathrm{Frob}_v)$ is an example of what is called a Weil number.

At the places $v|p$ the story is more intricate. When $\xi_v = 1$ we can draw the same conclusions as in the previous paragraph, but when $\xi_v \neq 1$ the restriction $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ is never unramified. If χ_v is unramified we can only say that $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)} = \xi_v \circ \mathrm{Art}_{K_v}^{-1}$ on the whole inertia group I_v (as opposed to an open subgroup thereof). This naturally leads to the following definition from p-adic Hodge theory.

Definition 3.25 Let $v|p$ and fix a tuple of integers $(a_\sigma) \in \mathbb{Z}^{\mathrm{Hom}_{\mathrm{cont}}(K_v, \bar{\mathbb{Q}}_p)}$. A continuous Galois character

$$r_v : \mathrm{Gal}(\overline{K_v}/K_v) \longrightarrow \bar{\mathbb{Q}}_p^\times$$

is said to be crystalline and have Hodge-Tate weights (a_σ) if its restriction to inertia is given by

$$r_v(\gamma) = \prod\nolimits_{\sigma \in \mathrm{Hom}_{\mathrm{cont}}(K_v, \bar{\mathbb{Q}}_p)} \sigma\big(\mathrm{Art}_{K_v}^{-1}(\gamma)\big)^{a_\sigma}$$

for all $\gamma \in I_v$. We say r_v is potentially crystalline if the above identity only holds when γ belongs to some open subgroup of I_v.

In this terminology $r_\chi|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ is potentially crystalline, and its Hodge-Tate weights can be read off from ξ_v. Moreover, it is crystalline if and only if χ_v is unramified.

Exercise 3.26 Show that $\mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\overline{K_v}/K_v), \bar{\mathbb{Q}}_p^\times) \xrightarrow{1:1} \mathrm{Hom}_{\mathrm{cont}}(K_v^\times, \bar{\mathbb{Z}}_p^\times)$ restricts to a bijection between the following two sets:

- Potentially crystalline characters $\mathrm{Gal}(\overline{K_v}/K_v) \longrightarrow \bar{\mathbb{Q}}_p^\times$;
- Locally algebraic characters $K_v^\times \longrightarrow \bar{\mathbb{Q}}_p^\times$ which are p-adically unitary.

3.3.2 The Geometricity Condition

We now consider a global Galois character $r : \mathrm{Gal}(\bar{K}/K) \to \bar{\mathbb{Q}}_p^\times$. In [FM95] Fontaine and Mazur introduced the notion of a geometric Galois representation. For a character r this means

- $r|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ is unramified for all but at most finitely many places v;
- $r|_{\mathrm{Gal}(\overline{K_v}/K_v)}$ is potentially crystalline for all $v|p$.

We first observe that (for characters) the first condition is superfluous.

Proposition 3.27 *Any continuous character $r : Gal(\bar{K}/K) \to \bar{\mathbb{Q}}_p^\times$ is unramified outside a finite set of places, and r factors through $\mathcal{O}_L^\times$ for some finite extension $L/\mathbb{Q}_p$.*

Proof As a preliminary we note that r in fact takes values in $\mathcal{O}_L^\times$ for some finite extension $L/\mathbb{Q}_p$. This follows from the Baire category theorem ([Ped89, Proposition 2.2.2, p. 52]) which we briefly recall: Suppose $F_1, F_2, \ldots$ is a sequence of closed subsets of a locally compact Hausdorff space. If all the F_i have empty interior then so does their union $\bigcup_{i=1}^\infty F_i$. If G is a profinite group and $r : G \longrightarrow \bar{\mathbb{Q}}_p^\times$ is a continuous character we consider the countable collection of closed subsets $r^{-1}(L^\times)$. (There are only finitely many extensions $L/\mathbb{Q}_p$ of a given degree and $L \subset \bar{\mathbb{Q}}_p$ is closed since it is complete.) Clearly $G = \bigcup_L r^{-1}(L^\times)$ since $\bar{\mathbb{Q}}_p = \bigcup_L L$. Obviously G is compact and Hausdorff, so Baire's theorem tells us $r^{-1}(L^\times)$ has non-empty interior for some L. It is a group so all its elements must be interior points, in particular the identity. This means there is an open $N \lhd G$ of finite index such that $N \subset r^{-1}(L^\times)$. Thus $r(g)^{[G:N]} \in L^\times$ for all $g \in G$. By passing to a finite extension of L if necessary we can therefore accommodate all the values $r(g)$. Finally observe that $r(G)$ is a compact subgroup of $L^\times$ and hence lies in the maximal compact subgroup $\mathcal{O}_L^\times$.

We return to ramification. Consider the higher units $1 + \mathfrak{m}_L^n$ for $n > \frac{1}{p-1} e(L/\mathbb{Q}_p)$. In this range exp and log are mutually inverse isomorphisms of topological groups $1 + \mathfrak{m}_L^n \xrightarrow{\sim} \mathfrak{m}^n$. We infer that $1 + \mathfrak{m}_L^n$ is a torsion-free pro-p group (non-canonically isomorphic to $\mathbb{Z}_p^{[L:\mathbb{Q}_p]}$). By continuity of $r \circ \mathrm{Art}_K$ there exists an open neighborhood $1 \in U = \prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v^\times \subset \mathbb{I}_K$ such that $(r \circ \mathrm{Art}_K)(U) \subset 1 + \mathfrak{m}_L^n$. Here we may assume S contains no places above p. In particular $(r \circ \mathrm{Art}_K)(\mathcal{O}_v^\times) \subset 1 + \mathfrak{m}_L^n$ for $v \notin S$. This forces $(r \circ \mathrm{Art}_K)(\mathcal{O}_v^\times) = \{1\}$ for $v \notin S$. Why? Suppose $v|\ell$. Then $1 + \mathfrak{p}_v$ is pro-ℓ and therefore does not admit a non-trivial homomorphism to a pro-p group since $p \neq \ell$. In other words $(r \circ \mathrm{Art}_K)|_{\mathcal{O}_v^\times}$ factors through the finite quotient $\mathcal{O}_v^\times / 1 + \mathfrak{p}_v$. At last, since $1 + \mathfrak{m}_L^n$ has no elements of finite order > 1 we must have $(r \circ \mathrm{Art}_K)|_{\mathcal{O}_v^\times} = 1$. We conclude that $r|_{I_v} = 1$ for $v \notin S$. □

We have seen that r_χ is geometric for every algebraic Hecke character χ. Amazingly the converse holds. To put it another way the construction $\chi \rightsquigarrow r_\chi$ gives a bijection between the two sets:

- Geometric Galois characters $\mathrm{Gal}(\bar{K}/K) \longrightarrow \bar{\mathbb{Q}}_p^\times$;

- Algebraic Hecke characters $C_K \longrightarrow \mathbb{C}^\times$.

The proof of the converse is beyond the scope of these notes. We refer to the reader to [Ser89] for an exposition of some of the methods going into the proof.

Remark 3.28 The definition of crystalline representations is quite involved. In fact, in higher dimension p-adic Hodge theory gives a whole hierarchy of representations (crystalline $\Rightarrow$ semistable $\Rightarrow$ de Rham). The geometric representations $r : \mathrm{Gal}(\bar{K}/K) \longrightarrow \mathrm{GL}_n(\bar{\mathbb{Q}}_p)$ are by definition those that are (1) unramified outside a finite set, and (2) de Rham at all places $v|p$. It is expected that such r appear as subquotients of $H^i_{\text{ét}}(X_{\bar{K}}, \bar{\mathbb{Q}}_p(j))$ for some variety $X_{/K}$, provided r is irreducible. See [FM95, Conjecture 1].

When r is potentially abelian, which means $r|_{\mathrm{Gal}(\bar{E}/E)}$ factors through $\mathrm{Gal}(\bar{E}/E)^{\mathrm{ab}}$ for some finite extension E/K, one can show that r is in some sense built from Tate modules of abelian varieties $A_{/K}$ which are potentially of CM type. We direct the reader to [FM95, p. 201] for the precise formulation. For the uninitiated we note that abelian varieties are higher-dimensional generalizations of elliptic curves. They are projective varieties $A_{/K}$ with a group law, and the p-adic Tate module $T_p(A) = \varprojlim_n A(\bar{K})[p^n]$ is free of rank $2 \dim A$ over $\mathbb{Z}_p$. One should think of $V_p(A) = T_p(A) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ as the dual of $H^1_{\text{ét}}(A_{\bar{K}}, \mathbb{Q}_p)$. We say A is of CM type if $\mathrm{End}(A) \otimes \mathbb{Q}$ is a CM field of degree $2 \dim A$ over $\mathbb{Q}$.

3.4 Non-vanishing of Hecke L-Functions for Real Part One

We saw in Theorem 1.15 that $\zeta(s) \neq 0$ for $\mathrm{Re}(s) = 1$, and by Theorem 1.49 we know $L(1, \chi) \neq 0$ for all Dirichlet characters χ. These result were instrumental in proving the prime number theorem and the infinitude of primes in arithmetic progressions, respectively. In this section we subsume these non-vanishing results in a more general assertion valid for arbitrary (unitary) Hecke characters of any number field.

Theorem 3.29 *Let $\chi_0 : C_K \to \mathbb{C}^1$ be a unitary Hecke character. Then $L(s, \chi_0)$ has no zeroes with real part equal to one.*

Proof First note that $L(s + it, \chi_0) = L(s, \chi_{it})$ for $t \in \mathbb{R}$. It therefore suffices to show $s = 1$ is not a zero of $L(s, \chi_0)$ for any unitary χ_0. For the sake of contradiction, assume $L(1, \chi_0) = 0$ for some χ_0. Then the same holds for the complex conjugate character, $L(1, \overline{\chi}_0) = 0$. (Indeed $L(s, \overline{\chi}_0) = \overline{L(\overline{s}, \chi_0)}$ for $\mathrm{Re}(s) > 1$ by taking complex conjugates in the Euler product defining the L-function of χ_0. Since both sides are meromorphic functions on $\mathbb{C}$, the identity holds everywhere.)

The trick is to consider the auxiliary function

$$\mathbb{L}(s) := \zeta_K(s)^2 L(s, \chi_0) L(s, \overline{\chi}_0).$$

The only possible poles of $\mathbb{L}(s)$ are $s = 1$ and $s = 1 \pm i\tau$ when $\chi_0 = |\cdot|^{-i\tau}$ for some $\tau \in \mathbb{R}$.

Certainly for $\mathrm{Re}(s) > 1$ we may write $\mathbb{L}(s) = \exp\big(D(s)\big)$ with

$$D(s) := \sum_{v \in V_f} \sum_{r=1}^{\infty} \left(2 + 2\mathrm{Re}\big(\chi_{0,v}(\varpi_v)^r\big)\right) \frac{1}{r} q_v^{-rs}.$$

This can be expressed as a Dirichlet series with *non-negative coefficients* since $\chi_{0,v}(\varpi_v)^r \in \mathbb{C}^1$ obviously has real part ≥ -1. Below we will apply Landau's Lemma 1.135 to $D(s)$.

We observe that $\mathrm{ord}_{s=1}\mathbb{L}(s) = 0$. To see this, note that at least $s = 1$ is not a pole since

$$\mathrm{ord}_{s=1}\mathbb{L}(s) = 2\mathrm{ord}_{s=1}\zeta_K(s) + \mathrm{ord}_{s=1}L(s, \chi_0) + \mathrm{ord}_{s=1}L(s, \overline{\chi}_0) \geq -2 + 1 + 1 = 0$$

by our assumption that $L(1, \chi_0) = 0 = L(1, \overline{\chi}_0)$. Moreover, $\mathbb{L}(1) \neq 0$. Indeed $D(\sigma) \geq 0$ for $\sigma > 1$, and therefore $\mathbb{L}(\sigma) \geq 1$ for $\sigma > 1$. Letting $\sigma \to 1^+$ shows that $\mathbb{L}(1) \geq 1$. In particular $\mathbb{L}(1)$ is nonzero.

Consequently $\mathbb{L}(s)$ has a logarithm in a sufficiently small open disc centered at $s = 1$. This logarithm differs from $D(s)$ by a constant, and hence $D(s)$ has analytic continuation to this disc. By Lemma 1.135 (which applies to $D(s)$ since its coefficients are ≥ 0) there is a $t' < 1$ such that the series defining $D(s)$ converges absolutely for $\mathrm{Re}(s) > t'$. In other words $D(s)$ has abscissa of convergence $t^\sharp < 1$.

We claim that in fact $t^\sharp = -\infty$; in other words that the series $D(s)$ is absolutely convergence for all s. Suppose $t^\sharp \neq -\infty$. As we have just seen, $t^\sharp < 1$ and thus $t^\sharp$ is not a pole of $\mathbb{L}(s)$. In fact we must have $\mathbb{L}(t^\sharp) = 0$, for otherwise (by the argument in the previous paragraph) $D(s)$ would have analytic continuation to a small enough open disc centered at $t^\sharp$. However, by the remarks after the proof of Lemma 1.135 we know $t^\sharp$ is a pole for $D(s)$.

We recall that $\mathrm{ord}_{s=-2}\zeta_K(s) = r_1 + r_2 \geq 1$ by 1.89. As $L(s, \chi_0)$ and $L(s, \overline{\chi}_0)$ do not have a pole at $s = -2$ we deduce that $\mathbb{L}(-2) = 0$. On the other hand $D(-2)$ (which we now know converges) is clearly ≥ 0, and thus $\mathbb{L}(-2) = \exp D(-2) \geq 1$. This is a contradiction. □

As mentioned at the very end of Sect. 1.3.3, the non-vanishing of $\zeta_K(s)$ on the line $\mathrm{Re}(s) = 1$ is essentially Landau's prime ideal theorem about the asymptotic behavior of $\#\{\mathfrak{p} : N(\mathfrak{p}) \leq x\}$.

We note that there is a vast generalization of Theorem 3.29, due to Jacquet and Shalika, stating that the L-function $L(s, \pi)$ of a so-called cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_K)$ does not vanish on the line $\mathrm{Re}(s) = 1$. Theorem 3.29 is the $n = 1$ case.

Chapter 4
Artin L-Functions



Emil Artin first defined the L-function of a complex Galois representation in [Art23]. Satz 3 on page 105 of that paper gives the *meromorphic* continuation of these L-functions, and in a few paragraphs after the theorem Artin writes:

> Völlig neue Methoden aber wird wohl der Nachweis verlangen, daß es sich bei unseren L-Reihen um ganze Funktionen (abgesehen vom Hauptcharakter) handelt.

This roughly translates into "Completely new methods will probably be required to prove that our L-series are entire functions (apart from the principal character)". The development of the methods Artin called for has kept number theorists busy over the past century.

If $G = \mathrm{Gal}(E/K)$ is the Galois group of a finite Galois extension of number fields E/K, we consider the finite-dimensional complex representations $r : G \to \mathrm{GL}(V)$. Oftentimes we view r as a representation of the absolute Galois group $\mathrm{Gal}(\overline{K}/K)$ by composing it with the restriction map $\mathrm{Gal}(\overline{K}/K) \twoheadrightarrow G$. Vice versa, any continuous representation $\mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}(V)$ factors through a finite quotient of the absolute Galois group. In other words, the complex topology on V does not really play a role and we may as well think of V as a discrete vector space over $\mathbb{C}$.

The Artin L-function $L(s, r)$ is defined by an Euler product for $\mathrm{Re}(s) > 1$. For each finite place v of K, the vth Euler factor is essentially built out of the characteristic polynomial of a Frobenius element at v. Thus, except for finitely many v, the (inverse) Euler factor is of degree $n = \dim_{\mathbb{C}} V$ in q_v^{-s}.

In the one-dimensional case, every Galois character $r : G^{\mathrm{ab}} \to \mathbb{C}^\times$ corresponds to a Hecke character $\chi = r \circ \mathrm{Art}_{E/K}$ via class field theory, and the Artin L-function $L(s, r)$ equals the Hecke L-function $L(s, \chi)$. So for $n = 1$ the analytic properties of $L(s, r)$ follow immediately from the Hecke-Tate theory. The overall goal of this Chapter is to write *any* Artin L-function as an alternating product of Hecke L-functions over various extensions of K, i.e. to express $L(s, r)$ as a finite product $\prod_\chi L(s, \chi)^{(-1)^{a_\chi}}$. In this product χ denotes a Hecke character of some intermediate field L_χ/K, which depends on χ, and $a_\chi \in \{0, 1\}$. This will allow us to deduce the meromorphic continuation of Artin L-functions, and their functional equations, again

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from the theory of Hecke and Tate. We have very little control over the exponents $(-1)^{a_\chi}$ in general and this is the reason why one cannot use Artin's method to rule out poles of $L(s,r)$ when $r \neq 1_G$.

In Sect. 4.3 we establish two fundamental invariance properties of Artin L-functions. The first one basically says $L(s,r)$ is independent of the top field E. More precisely, composing r with $\mathrm{Gal}(\widetilde{E}/K) \twoheadrightarrow \mathrm{Gal}(E/K)$ gives rise to the same L-function for any finite Galois extension $\widetilde{E}/K$ containing E as an intermediate field. We say that Artin L-functions are invariant under *inflation*. This property permits us to unambiguously attach an L-function to a representation of $\mathrm{Gal}(\overline{K}/K)$. The second property, on the other hand, is paramount. It says that $L(s,r)$ is invariant under *induction* in the following sense: Let L be an intermediate field of E/K corresponding to $H := \mathrm{Gal}(E/L)$. A representation $\rho : H \to \mathrm{GL}(W)$ induces a representation $r := \mathrm{Ind}_H^G \rho$ of G, and it has the *same* Artin L-function as ρ. This is remarkable, as we are now comparing an Euler product over places of K to one over places of a different field L. For example, if $\rho : H \to \mathbb{C}^\times$ is one-dimensional, $L(s, \mathrm{Ind}_H^G \rho)$ is nothing but a Hecke L-function in disguise. In the extreme case $H = \{e\}$ the induced representation $r_{\mathrm{reg}} = \mathrm{Ind}_e^G 1$ is the 'regular' representation on the space of *all* functions $f : G \to \mathbb{C}$. While $L(s, r_{\mathrm{reg}}) = \zeta_E(s)$, Frobenius reciprocity tells us any irreducible representation r of G appears in r_{reg} with multiplicity $\dim r$. Note how this gives the factorization formula $\zeta_E(s) = \prod_r L(s,r)^{\dim r}$ with great ease. One of the factors in this product is $\zeta_K(s)$, corresponding to the principal character $r = 1_G$. Thus the ratio $\frac{\zeta_E(s)}{\zeta_K(s)}$ is a product of Artin L-functions believed to be entire. In Sect. 4.4 we give a direct proof that this ratio of Dedekind zeta functions is entire for *Galois* extensions E/K. This is a result known as the Aramata-Brauer theorem. The Dedekind conjecture, which is still open, states that $\frac{\zeta_E(s)}{\zeta_K(s)}$ is entire for *any* finite extension of number fields E/K. This can be deduced from the Artin conjecture by tweaking the factorization formula slightly.

Section 4.4 is purely group-theoretic. For *any* finite group G, and any representation r thereof, the theme is to express the character $\chi_r = \mathrm{trace}(r)$ as a linear combination of characters of induced representations. The main result is the Brauer induction theorem (Theorem 4.24 and its Corollaries 4.25 and 4.31) which is indispensable when establishing the functional equation for Artin L-functions. Despite its importance, this theorem is not usually included in an undergraduate algebra course, and for the sake of completeness we include a full proof of Brauer's result. The latter is preceded by Artin's induction theorem (see Theorem 4.18 and Corollary 4.19) which essentially says χ_r can be written as a $\mathbb{Q}$-linear combination $\sum_{\text{cyclic } H \leq G} a_H \mathrm{Ind}_H^G \chi_H$ of characters induced from cyclic subgroups. Brauer's theorem says that one can improve this to a linear combination with *integer* coefficients if one takes a slightly bigger family of subgroups H–the so-called elementary subgroups. A group is said to be *p-elementary* if it is isomorphic to a product $C \times P$, where P is a p-group and C is cyclic of order prime to p. As p varies this gives the family of all elementary groups. For us the most significant feature of elementary groups is the fact that they are *nilpotent*, and consequently every irreducible representation is induced from a character of a subgroup (where 'character' here means an actual homomorphism into $\mathbb{C}^\times$). In other words nilpotent groups are *monomial* groups. To orient the reader,

nilpotent $\Rightarrow$ monomial $\Rightarrow$ solvable. Altogether this enables us to write any Artin L-function as a ratio of products of Hecke L-functions.

In Sect. 4.5 we complete $L(s,r)$ by introducing Euler factors at the infinite places $v \in V_\infty = V_\mathbb{R} \cup V_\mathbb{C}$. When v is a complex place we let $L_v(s,r) = L_\mathbb{C}(s)^n$ where $n = \dim_\mathbb{C} V$. When v is the real place of K corresponding to an embedding $\sigma : K \hookrightarrow \mathbb{R}$ there are two possibilities: Either the extensions of σ to E all take real values, or they come in complex conjugate pairs of non-real embeddings $E \hookrightarrow \mathbb{C}$. In the first case, where v stays real in E, we let $L_v(s,r) = L_\mathbb{R}(s)^n$. The second case where v splits in E is more interesting. Here we have a corresponding complex conjugation in G, and $n^\pm$ denote the dimensions of its (± 1)-eigenspaces in V. In terms of these the vth Euler factor is defined to be $L_\mathbb{R}(s)^{n^+} L_\mathbb{R}(s+1)^{n^-}$. These definitions are tailored to make $L_\infty(s,r) = \prod_{v \in V_\infty} L_v(s,r)$ satisfy the same invariance properties under inflation and induction as Artin L-functions.

The completed L-function $\widehat{L}(s,r)$ in the functional equation (Theorem 4.53) involves not only $L_\infty(s,r)$ but also the factor $\mathcal{N}_r^{\frac{s}{2}}$. The natural number $\mathcal{N}_r$ enjoys similar invariance properties under inflation and restriction, which ultimately lets us reduce the proof of the functional equation to Hecke-Tate theory. By definition, $\mathcal{N}_r = |D_K|^n N(\mathfrak{f}(r))$ with $\mathfrak{f}(r)$ denoting the *Artin conductor*. This is a nonzero ideal in $\mathcal{O}_K$ defined locally by giving the exponents in its factorization into prime ideals $\prod_{v \in V_f} \mathfrak{p}_v^{\mathfrak{f}_v(r)}$. The number $\mathfrak{f}_v(r)$ is defined to be the sum $\sum_{i \geq 0} \frac{|G_i|}{|G_0|} \operatorname{codim} V^{G_i}$ where G_i denotes the ith ramification group at v. Clearly $\mathfrak{f}_v(r)$ is a non-negative rational number, and $\mathfrak{f}_v(r) = 0$ for all but at most finitely many places v. However, it is not at all clear that in fact $\mathfrak{f}_v(r)$ is an *integer*–so that $\mathfrak{f}(r)$ is well-defined. Again the Brauer induction theorem comes to the rescue. It brings the question down to the one-dimensional case, which is essentially the Hasse-Arf theorem for abelian extensions of local fields (the jumps in the upper numbering of the ramification groups are integers). This is all worked out in Sect. 4.6.

In the last Sect. 4.9 we apply this machinery to prove the Chebotarev density theorem [Tsc26] over $\mathbb{Q}$, which is a wide-ranging generalization of Dirichlet's theorem on the uniform distribution of primes in arithmetic progressions. With $G = \operatorname{Gal}(E/\mathbb{Q})$ as above, and $C \subset G$ a conjugacy class, Chebotarev's theorem asserts that the set of primes p with Frobenius in C has density $\frac{|C|}{|G|}$. For example, if $C = \{e\}$ this shows the set of split primes $\operatorname{Spl}_{E/\mathbb{Q}}$ has density $\frac{1}{[E:\mathbb{Q}]}$. We recover Dirichlet's theorem in the special case where $E = \mathbb{Q}(\zeta_N)$ and $C = \{[a]\}$ contains a single residue class $[a] = a + N\mathbb{Z}$.

4.1 Complex Galois Representations

Instead of characters we now consider representations of $\operatorname{Gal}(\bar{K}/K)$ on complex vector spaces. More precisely, if V is a finite-dimensional $\mathbb{C}$-vector space we endow $\operatorname{GL}(V)$ with the complex topology (choose a basis for V and identify $\operatorname{GL}(V)$ with

$\mathrm{GL}_n(\mathbb{C})$–the resulting topology is independent of the choice of basis) and consider a continuous homomorphism

$$r : \mathrm{Gal}(\bar{K}/K) \longrightarrow \mathrm{GL}(V).$$

The absolute Galois group carries the Krull topology.

We will associate an L-function to r and establish its basic analytic properties. First we observe that the continuity of r amounts to r having finite image, by the following generalization of Lemma 2.25 to representations:

Lemma 4.1 *Let G be a profinite group. Then any continuous homomorphism $r : G \to GL_n(\mathbb{C})$ factors through a finite quotient of G.*

Proof This is the no small subgroups argument. There is an open neighborhood $1 \in \Omega \subset \mathrm{GL}_n(\mathbb{C})$ not containing any subgroups of $\mathrm{GL}_n(\mathbb{C})$ other than $\{1\}$. Why? The exponential map for matrices restricts to a diffeomorphism $B_\epsilon \xrightarrow{\sim} \exp(B_\epsilon)$ on a small enough open ball $0 \in B_\epsilon \subset M_n(\mathbb{C})$ for the matrix norm $\|\cdot\|$. We claim $\Omega = \exp(B_{\epsilon/2})$ works. For suppose $H \subset \exp(B_{\epsilon/2})$ is a subgroup of $\mathrm{GL}_n(\mathbb{C})$ and pick an $x \in H$. Write $x = \exp(X)$ with $X \in B_{\epsilon/2}$. Then $x^2 = \exp(2X)$ and $2X \in B_\epsilon$. On the other hand $x^2 \in H$ and it therefore has the form $x^2 = \exp(X')$ for a unique $X' \in B_{\epsilon/2}$. We must have $X' = 2X$ since $\exp|_{B_\epsilon}$ is injective. Thus $\|X\| = \frac{1}{2}\|X'\| < \frac{\epsilon}{4}$. Continuing in this fashion shows that $X \in B_{\epsilon/2^m}$ for all $m \in \mathbb{Z}_{>0}$. We conclude $X = 0$, and hence $x = 1$. □

In other words, the complex topology on V is somewhat immaterial and we may as well take V to be discrete when we talk about continuous representations of $\mathrm{Gal}(\bar{K}/K)$ on V.

There is no loss of generality by considering representations of $\mathrm{Gal}(E/K)$ for some finite Galois extension of number fields E/K. For instance one can take the fixed field $E = \bar{K}^{\ker(r)}$ and think of r as a faithful representation $\mathrm{Gal}(E/K) \hookrightarrow \mathrm{GL}(V)$.

4.2 Definition of the L-Series

We start with a finite Galois extension of number fields E/K with Galois group $G = \mathrm{Gal}(E/K)$. Let $r : G \longrightarrow \mathrm{GL}(V)$ be an n-dimensional complex representation. To define the L-function $L(s, r)$ we must first introduce some notation.

Let v be a finite place of K. We pick a place $w|v$ of E with decomposition group G_w and inertia group I_w. Let $\mathrm{Frob}_w \in G_w/I_w$ be the Frobenius element inducing the map $x \mapsto x^{q_v}$ on the residue field of w.

Since I_w is normal in G_w, the subspace of I_w-invariant vectors V^{I_w} is preserved by G_w and we get a representation of G_w/I_w on V^{I_w} defined by r. In particular $r(\mathrm{Frob}_w) \in \mathrm{GL}(V^{I_w})$ makes sense.

Remark 4.2 Any other place $w'|v$ is of the form $w' = gw$ for some $g \in G$. Thus $G_{w'} = gG_w g^{-1}$ and $I_{w'} = gI_w g^{-1}$. Conjugation by g gives an isomorphism $G_w/I_w \xrightarrow{\sim} G_{w'}/I_{w'}$ mapping $\mathrm{Frob}_w \mapsto \mathrm{Frob}_{w'}$. With this notation $r(g)$ restricts to an isomorphism $V^{I_w} \xrightarrow{\sim} V^{I_{w'}}$ which intertwines the action of $r(\mathrm{Frob}_w)$ on the source with that of $r(\mathrm{Frob}_{w'})$ on the target. In particular the characteristic polynomial of $r(\mathrm{Frob}_w)$ depends only on v–not on the choice of $w|v$.

Definition 4.3 In the above setup the Artin L-function of r is defined by the Euler product

$$L(s, r) = \prod_{v \in V_f} \det_{V^{I_w}} (1 - r(\mathrm{Frob}_w) q_v^{-s})^{-1}.$$

We say r is unramified at v if $V^{I_w} = V$ for some (or equivalently all) $w|v$. This obviously holds if v is unramified in E, and in particular for all but finitely many v.

Proposition 4.4 *The infinite product defining $L(s, r)$ is absolutely convergent for $Re(s) > 1$, uniformly in $Re(s) > 1 + \epsilon$ for all $\epsilon \geq 0$.*

Proof Let S be the finite set of places v where r is ramified. It suffices to show the product over $v \notin S$ has the claimed convergence properties. Let $\zeta_{w,1}, \ldots, \zeta_{w,n}$ denote the eigenvalues of $r(\mathrm{Frob}_w) \in \mathrm{GL}(V)$ listed with algebraic multiplicity. All these $\zeta_{w,i}$ are roots of unity of order dividing $|G|$, and in particular $|\zeta_{w,i}| = 1$. The absolute convergence of

$$\prod_{v \notin S} \frac{1}{(1 - \zeta_{w,1} q_v^{-s}) \cdots (1 - \zeta_{w,n} q_v^{-s})}$$

is equivalent to the absolute convergence of

$$\sum_{v \notin S} \ln\Big(\frac{1}{1 - \zeta_{w,1} q_v^{-s}}\Big) + \cdots + \ln\Big(\frac{1}{1 - \zeta_{w,n} q_v^{-s}}\Big) = \sum_{v \notin S} \sum_{i=1}^{n} \sum_{r=1}^{\infty} \frac{\zeta_{w,i}^r q_v^{-rs}}{r}.$$

The same for uniform convergence. Now the argument reduces to the case of the Dedekind zeta function $\zeta_K(s)$, see Sect. 1.3. □

Exercise 4.5 For $i = 1, 2$ let $r_i : G \longrightarrow \mathrm{GL}(V_i)$ be two representations. Verify the following.

(a) $r_1 \simeq r_2 \Longrightarrow L(s, r_1) = L(s, r_2)$;
(b) $L(s, r_1 \oplus r_2) = L(s, r_1) L(s, r_2)$.

4.2.1 Examples

For the trivial character $r = 1_G$ we recover the Dedekind zeta function:

$$L(s, 1_G) = \zeta_K(s).$$

More generally let $r : G \longrightarrow \mathbb{C}^\times$ be any character. It factors through G^{ab}, so by passing to a subfield of E we may as well assume E/K is an abelian extension to begin with. Composition with the Artin map then gives a Hecke character $\chi = r \circ \mathrm{Art}_{E/K}$. By class field theory $\mathrm{Art}_{E/K}(\mathcal{O}_v^\times) = \{1\}$ when v is unramified in E/K, in which case $\mathrm{Art}_{E/K}(\varpi_v) = \mathrm{Frob}_w$ for all $w|v$. For ramified v we have $\mathrm{Art}_{E/K}(\mathcal{O}_v^\times) = I_w$. Altogether these remarks yield the equality

$$L(s, r) = L(s, \chi).$$

The analytic properties of Hecke L-functions thus transfer to Artin L-functions for characters.

Generalizing this to n-dimensional representations r is at the heart of the Langlands philosophy. The substitute for χ should be an automorphic representation of $\mathrm{GL}_n(\mathbb{A}_K)$. For $n = 2$ and $K = \mathbb{Q}$ a fundamental theorem of Deligne and Serre associates an r to a modular form of weight one.

4.3 Invariance Properties

We keep the notation introduced above. Thus $G = \mathrm{Gal}(E/K)$ is the Galois group of a finite Galois extension of number fields E/K, and $r : G \to \mathrm{GL}(V)$ is an n-dimensional complex representation.

In this section we consider an intermediate field $K \subset L \subset E$ and let $H = \mathrm{Gal}(E/L)$ be the corresponding subgroup of G. Thus L is the fixed field E^H by Galois theory.

4.3.1 Invariance Under Inflation

Here we assume L/K is Galois. In other words H is a normal subgroup of G, with quotient group $G/H \simeq \mathrm{Gal}(L/K)$.

Suppose r factors through G/H. That is, there is a (necessarily unique) representation $r' : G/H \to \mathrm{GL}(V)$ which inflates to r. This means r arises from r' by composition with the natural projection:

$$G \xrightarrow{\mathrm{pr}_H} G/H \xrightarrow{r'} \mathrm{GL}(V).$$

In this case r and r' have the same Artin L-function.

Proposition 4.6 *We have the equality $L(s,r) = L(s,r')$ where $r = r' \circ pr_H$ is the inflation as above.*

Proof For any $v \in V_f$ we compare the Euler factors at v. Fix a place $w|v$ of E as above and let $\nu = w|_L$ be the place of L below w. Clearly $\mathrm{pr}_H(G_w) = (G/H)_\nu$, and similarly for inertia. (It may be helpful to think of G_w as $\mathrm{Gal}(E_w/K_v)$ etc. to see this.) In particular $V^{I_w} = V^{I_\nu}$. Finally, noting that the induced map on quotients $G_w/I_w \to (G/H)_\nu/I_\nu$ sends $\mathrm{Frob}_w \mapsto \mathrm{Frob}_\nu$ concludes the proof. □

This shows the top field E is less important, and allows us to associate an Artin L-function to any continuous representation r of $\mathrm{Gal}(\bar{K}/K)$ by factoring r through some finite quotient $\mathrm{Gal}(E/K)$. The resulting L-function is independent of the choice of E, as we have just shown.

4.3.2 Invariance Under Induction

Here we relinquish the assumption that L/K is Galois. So H may not be normal in G.

We take r to be induced from H. Start with a representation $\rho : H \to \mathrm{GL}(W)$ and let $r = \mathrm{Ind}_H^G \rho$. Thus r is the G-representation on the space of functions

$$V = \{f : G \longrightarrow W \text{ such that } f(hx) = \rho(h) f(x) \text{ for all } h \in H \text{ and } x \in G\}$$

given by right translations $(r(g)f)(x) = f(xg)$. The uninitiated reader should check this indeed defines a $\mathbb{C}$-linear left G-action on V. Frobenius reciprocity is the fact that the induction functor Ind_H^G is both left and right adjoint to the restriction functor Res_H^G (for finite groups G in general).

Exercise 4.7 Let $\{x_1, \ldots, x_N\}$ be a choice of right coset representatives. That is $G = \bigsqcup_{i=1}^N H x_i$. Check that the evaluation map $f \mapsto (f(x_i))_{i=1}^N$ gives an isomorphism of $\mathbb{C}$-vector spaces $V \xrightarrow{\sim} W^N$ and deduce the dimension formula

$$\dim_{\mathbb{C}} \mathrm{Ind}_H^G \rho = [G : H] \cdot \dim_{\mathbb{C}} \rho.$$

The following notation will be used below:

Definition 4.8 For $\sigma \in G$ we define a representation $\sigma_* \rho$ of $\sigma^{-1} H \sigma$ on the same vector space W by $(\sigma_* \rho)(\sigma^{-1} h \sigma) = \rho(h)$. In other words $\sigma_* \rho$ is the composition with conjugation by σ:

$$\sigma^{-1} H \sigma \xrightarrow{\sim} H \xrightarrow{\rho} \mathrm{GL}(W).$$

When we endow W with this $\sigma^{-1} H \sigma$-action we will write $\sigma_* W$.

Exercise 4.9 (*Mackey decomposition*) Let Γ be a subgroup of G and pick a set of representatives Σ for the double cosets $H\backslash G/\Gamma$. Show that sending f to the tuple of functions $(f_\sigma)_{\sigma\in\Sigma}$, where $f_\sigma(\gamma) = f(\sigma\gamma)$, gives an isomorphism of Γ-representations

$$\mathrm{Ind}_H^G \rho \xrightarrow{\sim} \bigoplus\nolimits_{\sigma\in\Sigma} \mathrm{Ind}_{\Gamma\cap\sigma^{-1}H\sigma}^{\Gamma} \sigma_*\rho.$$

(We have omitted various restrictions on both sides for legibility. The left-hand side is really $\mathrm{Res}_\Gamma^G \mathrm{Ind}_H^G \rho$.) Prove that $f \mapsto (f(\sigma))_{\sigma\in\Sigma}$ gives an isomorphism of vector spaces $(\mathrm{Ind}_H^G\rho)^\Gamma \xrightarrow{\sim} \bigoplus_{\sigma\in\Sigma} \rho^{H\cap\sigma\Gamma\sigma^{-1}}$.

A key feature of Artin L-functions is their insensitivity to induction:

Proposition 4.10 *We have the equality $L(s, r) = L(s, \rho)$ where $r = Ind_H^G \rho$ as above.*

Proof Fix a $v \in V_f$ and let $\nu_1 \ldots, \nu_t$ be the places of L above v. For each ν_i we select a place $w_i|\nu_i$ of E. Note that $H \cap G_{w_i}$ and $H \cap I_{w_i}$ are the decomposition and inertia groups relative to the top layer E/L. Via the inclusion (of finite cyclic groups)

$$(H \cap G_{w_i})/(H \cap I_{w_i}) \hookrightarrow G_{w_i}/I_{w_i} \tag{4.1}$$

we identify $\mathrm{Frob}_{w_i}^{f_i}$ with the Frobenius of w_i in the layer E/L, where f_i is the inertia degree of $\nu_i|v$. That is $q_{\nu_i} = q_v^{f_i}$. In this notation it suffices to show the polynomial identity

$$\det\nolimits_{V^{I_{w_1}}}(1 - r(\mathrm{Frob}_{w_1})T) = \prod_{i=1}^{t} \det\nolimits_{W^{H\cap I_{w_i}}}(1 - \rho(\mathrm{Frob}_{w_i}^{f_i})T^{f_i}) \tag{4.2}$$

in $\mathbb{C}[T]$ and evaluate at $T = q_v^{-s}$. Since G acts transitively on the set of places of E above v, which includes $w_1, \ldots, w_t$, we can write $w_i = \sigma_i w_1$ for some $\sigma_i \in G$. In the notation of 4.8 we have

$$W^{H\cap I_{w_i}} = (\sigma_{i*}W)^{I_{w_1}\cap\sigma_i^{-1}H\sigma_i} \qquad \rho(\mathrm{Frob}_{w_i}^{f_i}) = (\sigma_{i*}\rho)(\mathrm{Frob}_{w_1}^{f_i})$$

for all $i = 1, \ldots, t$. Note that $\sigma \mapsto \sigma w_1|_L$ gives a bijection between $H\backslash G/G_{w_1}$ and $\{\nu_1, \ldots, \nu_t\}$ by basic splitting theory. In other words $\{\sigma_1, \ldots, \sigma_t\}$ is a set of representatives for $H\backslash G/G_{w_1}$. From the Mackey decomposition 4.9 we get an isomorphism of G_{w_1}-representations

$$V \xrightarrow{\sim} \bigoplus_{i=1}^{t} \mathrm{Ind}_{G_{w_1}\cap\sigma_i^{-1}H\sigma_i}^{G_{w_1}} \sigma_{i*}W = \bigoplus_{i=1}^{t} V_i$$

where V_i is the ith summand. We have thus reduced the proof of (4.2) to verifying the equality

$$\det_{V_i^{I_{w_1}}}(1 - r(\mathrm{Frob}_{w_1})T) = \det_{(\sigma_{i*}W)^{I_{w_1}\cap\sigma_i^{-1}H\sigma_i}}(1 - (\sigma_{i*}\rho)(\mathrm{Frob}_{w_1}^{f_i})T^{f_i})$$

for all i. Note that f_i is the index of the inclusion in (4.1). This index changes to f_1 if we replace H by $\sigma_i^{-1}H\sigma_i$, as is easily checked. Therefore it suffices to consider $i = 1$, and it remains to show that

$$\det_{V_1^{I_{w_1}}}(1 - r(\mathrm{Frob}_{w_1})T) = \det_{W^{I_{w_1}\cap H}}(1 - \rho(\mathrm{Frob}_{w_1}^{f_1})T^{f_1}) \tag{4.3}$$

where now $V_1 = \mathrm{Ind}_{G_{w_1}\cap H}^{G_{w_1}} W$. Note that (with $i = 1$ say) the inclusion (4.1) has cokernel

$$G_{w_1}/I_{w_1}(G_{w_1} \cap H),$$

which is cyclic of order f_1, generated by the image of Frob_{w_1}. Let $\varphi \in G_{w_1}$ be a lift of Frob_{w_1}. Since $I_{w_1} \lhd G_{w_1}$ this shows the first f_1 powers of φ give a set of double coset representatives

$$\{1, \varphi, \varphi^2, \ldots, \varphi^{f_1-1}\} \overset{1:1}{\longleftrightarrow} (G_{w_1} \cap H)\backslash G_{w_1}/I_{w_1}.$$

Now, again by 4.9 (the last part) the map $f \mapsto (f(1), f(\varphi), \ldots, f(\varphi^{f_1-1}))$ gives an isomorphism ($I_{w_1} \lhd G_{w_1}$ is invariant under conjugation by φ-powers):

$$V_1^{I_{w_1}} \xrightarrow{\sim} \left(W^{I_{w_1}\cap H}\right)^{\oplus f_1}.$$

The action of Frob_{w_1} on $V_1^{I_{w_1}}$ corresponds to the following operation on the right-hand side: An f_1-tuple of vectors $(\xi_0, \xi_1, \ldots, \xi_{f_1-2}, \xi_{f_1-1})$ is sent to $(\xi_1, \xi_2, \ldots, \xi_{f_1-1}, \rho(\mathrm{Frob}_{w_1}^{f_1})\xi_0)$. Pick a basis for $W^{I_{w_1}\cap H}$ and let A be the matrix of $\rho(\mathrm{Frob}_{w_1}^{f_1})$ relative to this basis. Then the matrix of $r(\mathrm{Frob}_{w_1})$ has the following block form if $f_1 = 4$ say:

$$\begin{pmatrix} & & & A \\ I & & & \\ & I & & \\ & & I & \end{pmatrix}.$$

We conclude that

$$\det_{V_1^{I_{w_1}}}(1 - r(\mathrm{Frob}_{w_1})T) = \det\begin{pmatrix} I & & & -AT \\ -IT & I & & \\ & -IT & I & \\ & & -IT & I \end{pmatrix} = \det(I - AT^4)$$

which is (4.3). (The last step above follows from straightforward column operations: Add T times the fourth column to the third. Then add T times the (new) third column

to the second, and so on. In the end this gives a block upper-triangular matrix with a diagonal block $I - AT^4$ in the upper left corner, followed by identity matrices on the diagonal.) □

4.3.3 An Application to Dedekind Zeta Functions

As an important application of Proposition 4.10 we can now extend the factorization formula in Theorem 1.102 to non-abelian Galois extensions E/K. As above we let $G = \mathrm{Gal}(E/K)$. The finite set of (isomorphism classes of) irreducible G-representations will be denoted by $\mathrm{Irr}_{\mathbb{C}}(G)$. When we take out the trivial representation 1_G we write $\mathrm{Irr}^*_{\mathbb{C}}(G)$.

Corollary 4.11 $\zeta_E(s) = \prod_{r\in Irr_{\mathbb{C}}(G)} L(s,r)^{\dim_{\mathbb{C}} r} = \zeta_K(s) \cdot \prod_{r\in Irr^*_{\mathbb{C}}(G)} L(s,r)^{\dim_{\mathbb{C}} r}$.

Proof We apply Proposition 4.10 to the regular representation $r_{\mathrm{reg}} = \mathrm{Ind}_e^G(1)$. This is the space of all functions $\mathcal{C}(G,\mathbb{C})$ with G-action $(gf)(x) = f(xg)$. We infer that $L(s, r_{\mathrm{reg}}) \overset{4.10}{=\!=} L(s,1) = \zeta_E(s)$. On the other hand, r_{reg} decomposes as a direct sum of irreducible representations in which any irreducible r has multiplicity

$$\dim_{\mathbb{C}} \mathrm{Hom}_G(r, r_{\mathrm{reg}}) = \dim_{\mathbb{C}} \mathrm{Hom}_G(r, \mathrm{Ind}_e^G(1)) = \dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}}(r,\mathbb{C}) = \dim_{\mathbb{C}} r^* = \dim_{\mathbb{C}} r$$

by Frobenius reciprocity (and Schur's Lemma). That is, $r_{\mathrm{reg}} \simeq \bigoplus_{r\in \mathrm{Irr}_{\mathbb{C}}(G)} r^{\oplus \dim_{\mathbb{C}} r}$. Therefore, since Artin L-functions convert direct sums into products, we obtain the factorization formula. □

One of the biggest unsolved problems in number theory is the following.

Conjecture 4.12 (The Artin conjecture) $L(s,r)$ *extends to an entire function for all* $r \in Irr^*_{\mathbb{C}}(G)$.

As a consequence of the factorization formula 4.11, Artin's conjecture implies Dedekind's conjecture (cf. 1.105) that $\frac{\zeta_E(s)}{\zeta_K(s)}$ is entire–for *Galois* extensions E/K, but the Galois case is known unconditionally by Aramata-Brauer. See Corollary 4.20 below.

For a general finite extension of number fields E/K, consider a Galois closure $\widetilde{E}/K$ with Galois group $G = \mathrm{Gal}(\widetilde{E}/K)$ and let $H = \mathrm{Gal}(\widetilde{E}/E)$. Instead of the regular representation we work with the induced representation $\mathrm{Ind}_H^G(1_H)$. We know $L(s, \mathrm{Ind}_H^G(1_H)) = \zeta_E(s)$ by 4.10. Furthermore, $r \in \mathrm{Irr}_{\mathbb{C}}(G)$ appears in $\mathrm{Ind}_H^G(1_H)$ with multiplicity

$$\dim_{\mathbb{C}} \mathrm{Hom}_G(r, \mathrm{Ind}_H^G(1_H)) = \dim_{\mathbb{C}} \mathrm{Hom}_H(r|_H, 1_H) = \dim_{\mathbb{C}}(r^*)^H.$$

In particular $r = 1_G$ has multiplicity one (realized as the subspace of constant functions $H\backslash G \longrightarrow \mathbb{C}$). We thus arrive at the more general factorization formula

$$\frac{\zeta_E(s)}{\zeta_K(s)} = \prod\nolimits_{r\in \mathrm{Irr}^*_{\mathbb{C}}(G)} L(s,r)^{\dim_{\mathbb{C}}(r^*)^H}.$$

The Artin conjecture predicts all the L-functions appearing on the right-hand side have analytic continuation to entire functions. This shows Artin $\Longrightarrow$ Dedekind.

Remark 4.13 A central conjecture of Langlands asserts that $L(s,r)$ equals the L-function of a so-called automorphic representation of $\mathrm{GL}_n(\mathbb{A}_K)$, where $n = \dim_{\mathbb{C}} r$. These automorphic L-functions vastly generalize Hecke L-functions and are known to have analytic continuation (and a functional equation plus other desirable properties). Consequently, Langlands $\Longrightarrow$ Artin.

4.4 The Brauer Induction Theorem

In this section we discuss a fundamental result of Brauer which will allow us to show $L(s,r)$ at least has *meromorphic* continuation to the whole complex plane.

This is within the realm of representation theory of finite groups, not Galois theory. Thus G now denotes an arbitrary finite group. Recall that the character of a representation $r : G \to \mathrm{GL}(V)$ is the class function $\chi_r(g) = \mathrm{tr} r(g)$. For example, $\chi_r(e) = \dim_{\mathbb{C}} V$ is the dimension (or degree) of r.

Example 4.14 $\chi_{r_{\mathrm{reg}}}(g) = |G| \cdot \delta_e(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e. \end{cases}$ (The trace of a permutation matrix is the number of fixed points.)

Example 4.15 The dual (or contragredient) representation has character $\chi_{r^*}(g) = \chi_r(g^{-1})$.

Example 4.16 $\chi_{r_1 \otimes r_2}(g) = \chi_{r_1}(g) \cdot \chi_{r_2}(g)$. (We leave the verification to the reader.)

We recall the formula for an induced character.

Lemma 4.17 *Let $r = Ind_H^G \rho$ be an induced representation. Then for all $g \in G$ we have the formula*

$$\chi_r(g) = (Ind_H^G \chi_\rho)(g) = \frac{1}{|H|} \cdot \sum\nolimits_{s\in G, sgs^{-1}\in H} \chi_\rho(sgs^{-1}) = \sum\nolimits_{s\in \Sigma, sgs^{-1}\in H} \chi_\rho(sgs^{-1})$$

for any set of representatives Σ for $H\backslash G$. In particular χ_r vanishes outside $\bigcup_{s\in G} s^{-1}Hs$.

Proof Let W be the space of ρ. For $s \in G$ and $w \in W$ let $f_{s,w}$ denote the function in $V = \mathrm{Ind}_H^G W$ with $\mathrm{supp}(f_{s,w}) \subset Hs$ and $f_{s,w}(s) = w$. One easily checks the relations $g f_{s,w} = f_{sg^{-1},w}$ and $f_{hs,w} = f_{s,h^{-1}w}$ for $g \in G$ and $h \in H$. Pick a set of

representatives $\Sigma = \{s_1, \ldots, s_\ell\}$ for $H\backslash G$ and a basis $\{w_1, \ldots, w_m\}$ for W. This gives basis for V,

$$f_{s_1,w_1}, \ldots, f_{s_\ell,w_1}, \ldots, f_{s_1,w_m}, \ldots, f_{s_\ell,w_m},$$

relative to which $r(g)$ is block-diagonal (with m blocks of size $\ell \times \ell$). For $j = 1, \ldots, m$ the trace of the jth block only gets a contribution from the $s_i \in \Sigma$ satisfying $Hs_ig^{-1} = Hs_i$. For such s_i note that

$$gf_{s_i,w_j} = f_{s_ig^{-1},w_j} = f_{(s_ig^{-1}s_i^{-1})s_i,w_j} = f_{s_i,(s_igs_i^{-1})w_j}.$$

The jth trace comes out to

$$\sum\nolimits_{s_i \in \Sigma, s_igs_i^{-1} \in H} \rho(s_igs_i^{-1})_{jj}.$$

Summing over j gives the result. □

The formula in Lemma 4.17 gives a $\mathbb{C}$-linear map Ind_H^G from the class functions on H to class functions on G. This even works integrally, as we now explain. The irreducible characters $\chi_{r_1}, \ldots, \chi_{r_d}$ for G form an orthonormal basis for the space of class functions on G. They generate the free abelian (Grothendieck) group of *virtual* characters

$$R(G) = \mathbb{Z}\chi_{r_1} \oplus \cdots \oplus \mathbb{Z}\chi_{r_d},$$

which can also be thought of as consisting of all differences $\chi - \chi'$ of characters χ, χ'. The actual characters form a submonoid which we will denote by $R^+(G)$ (the $\mathbb{Z}_{\geq 0}$-linear combinations of the χ_{r_i}). Note that $R(G) \otimes_{\mathbb{Z}} \mathbb{C}$ is the space of class functions on G.

From what we have said, it is now clear that Ind_H^G defines a homomorphism

$$\begin{aligned} \mathrm{Ind}_H^G : R(H) &\longrightarrow R(G), \\ \chi \longmapsto g \mapsto (\mathrm{Ind}_H^G\chi)(g) &:= \frac{1}{|H|} \cdot \sum\nolimits_{s \in G, sgs^{-1} \in H} \chi(sgs^{-1}) \end{aligned}$$

which takes $R^+(H) \longrightarrow R^+(G)$ by Lemma 4.17.

4.4.1 Artin's Theorem

Below we will be interested in various families of subgroups $\mathcal{F}$, and the goal is to control the cokernel of the homomorphism

$$\begin{aligned}\mathrm{Ind}_{\mathcal{F}}^{G} : \bigoplus\nolimits_{H\in\mathcal{F}} R(H) &\longrightarrow R(G)\\ (\chi_H)_{H\in\mathcal{F}} &\longmapsto \sum\nolimits_{H\in\mathcal{F}} \mathrm{Ind}_H^G \chi_H.\end{aligned}$$

Any function in the image of $\mathrm{Ind}_{\mathcal{F}}^{G}$ vanishes outside $\bigcup_{H\in\mathcal{F}}\bigcup_{s\in G} s^{-1}Hs$. In particular, if $\mathrm{Ind}_{\mathcal{F}}^{G}$ has finite cokernel any class function on G must vanish outside this union (of G-conjugacy classes). Therefore the union must be all of G. This condition turns out to also be sufficient.

Theorem 4.18 (Artin.) *$\mathrm{Ind}_{\mathcal{F}}^{G}$ has finite cokernel if (and only if) $G = \bigcup_{H\in\mathcal{F}}\bigcup_{s\in G} s^{-1}Hs$.*

Proof The family $\mathcal{F}_{\mathrm{cyc}}$ of all cyclic subgroups clearly satisfies the hypothesis $G = \bigcup_{H\in\mathcal{F}_{\mathrm{cyc}}}\bigcup_{s\in G} s^{-1}Hs$, and by the transitivity of induction it suffices to show that $|G|\cdot\mathrm{cok}(\mathrm{Ind}_{\mathcal{F}_{\mathrm{cyc}}}^{G}) = 0$. (Any cyclic subgroup has a conjugate contained in some subgroup from $\mathcal{F}$. Therefore the image of $\mathrm{Ind}_{\mathcal{F}_{\mathrm{cyc}}}^{G}$ is contained in that of $\mathrm{Ind}_{\mathcal{F}}^{G}$.)

For any finite cyclic group H we introduce the auxiliary function $\theta_H : H \longrightarrow \mathbb{Z}_{\geq 0}$ given by

$$\theta_H(x) = |H|\cdot \mathrm{char}_{\{\text{generators}\}}(x) = \begin{cases} |H| & \text{if } H = \langle x\rangle \\ 0 & \text{if } H \neq \langle x\rangle \end{cases}.$$

As a preliminary observation, we note that $\sum_{H\in\mathcal{F}_{\mathrm{cyc}}} \mathrm{Ind}_H^G\theta_H$ is the constant function $g \mapsto |G|$. Indeed

$$\begin{aligned}\sum\nolimits_{H\in\mathcal{F}_{\mathrm{cyc}}} (\mathrm{Ind}_H^G\theta_H)(g) &= \sum\nolimits_{H\in\mathcal{F}_{\mathrm{cyc}}} \frac{1}{|H|}\cdot \sum\nolimits_{s\in G, sgs^{-1}\in H} \theta_H(sgs^{-1})\\ &= \sum\nolimits_{s\in G}\sum\nolimits_{H\in\mathcal{F}_{\mathrm{cyc}}, sgs^{-1}\in H} \frac{1}{|H|}\cdot\theta_H(sgs^{-1})\\ &= \sum\nolimits_{s\in G}\sum\nolimits_{H\in\mathcal{F}_{\mathrm{cyc}}, \langle sgs^{-1}\rangle = H} 1\\ &= \sum\nolimits_{s\in G} 1\\ &= |G|.\end{aligned}$$

In particular we deduce that $\theta_H \in R(H)$ by induction on $|H|$:

$$\theta_H + \sum\nolimits_{\text{cyclic } H'<H} \mathrm{Ind}_{H'}^{H}\theta_{H'} = \mathrm{const}_{|H|} = |H|\cdot\chi_{1_H} \in R(H).$$

Consequently $\mathrm{const}_{|G|} = |G|\cdot\chi_{1_G}$ lies in the image of $\mathrm{Ind}_{\mathcal{F}_{\mathrm{cyc}}}^{G}$, which is an ideal of $R(G)$ by the projection formula

$$U\otimes_{\mathbb{C}} \mathrm{Ind}_H^G W \xrightarrow{\sim} \mathrm{Ind}_H^G(U|_H\otimes_{\mathbb{C}} W).$$

We conclude that $|G|$ annihilates the cokernel of $\mathrm{Ind}_{\mathcal{F}_{\mathrm{cyc}}}^{G}$, as claimed. □

The previous proof immediately gives the following.

Corollary 4.19 *Any virtual character $\chi \in R(G)$ can be expressed as a sum*

$$\chi = \frac{1}{|G|} \cdot \sum\nolimits_{cyclic\ H \leq G} Ind_H^G \chi_H$$

for suitable $\chi_H \in R(H)$. (Thus χ is a $\mathbb{Q}$-linear combination of characters induced from characters of cyclic subgroups.)

Brauer's Theorem will allow us to express χ as a $\mathbb{Z}$-linear combination of induced characters, but from the larger family of so-called *elementary* subgroups–to be introduced in the next section.

First, as an application we can now prove the Aramata-Brauer Theorem alluded to earlier right after Conjecture 1.105.

Corollary 4.20 (Aramata-Brauer) $\frac{\zeta_E(s)}{\zeta_K(s)}$ *is entire for any Galois extension of number fields E/K.*

Proof For a finite cyclic group H we introduce the function $\lambda_H = \varphi(|H|)\chi_{H,\mathrm{reg}} - \theta_H$. (Here φ is Euler's function, $\chi_{H,\mathrm{reg}} = |H|\delta_e$ is the regular character of H, and θ_H is the virtual character introduced in the proof of Theorem 4.18.) From the proof of Theorem 4.18 we infer that

$$\begin{aligned}\sum\nolimits_{\text{cyclic } H \leq G} \mathrm{Ind}_H^G \lambda_H &= \sum\nolimits_{\text{cyclic } H \leq G} \varphi(|H|)\mathrm{Ind}_H^G \chi_{H,\mathrm{reg}} - \sum\nolimits_{\text{cyclic } H \leq G} \mathrm{Ind}_H^G \theta_H \\ &= \sum\nolimits_{\text{cyclic } H \leq G} \varphi(|H|)\chi_{G,\mathrm{reg}} - |G|\chi_{1_G} \\ &= |G| \cdot (\chi_{G,\mathrm{reg}} - \chi_{1_G}).\end{aligned}$$

Note that we may write G as a disjoint union $G = \bigcup_{\text{cyclic } H \leq G} \{\text{generators of } H\}$. Comparing cardinalities gives the formula $|G| = \sum_{\text{cyclic } H \leq G} \varphi(|H|)$ used above.

Next we argue that λ_H is in fact a *character* of H. This essentially follows from the next two properties:

- $\langle \lambda_H, \chi_{1_H} \rangle = 0$;
- $\lambda_H(h) \leq 0$ for all $h \neq e$ (and $\lambda_H(e) \geq 0$).

Both are easy to check. Our task is now to show that $\langle \lambda_H, \chi \rangle \geq 0$ for all $\chi \in R^+(H)$. This is simple:

$$\begin{aligned}|H| \cdot \langle \lambda_H, \chi \rangle &= \sum_{h \in H} \lambda_H(h)\chi(h^{-1}) \\ &= \sum_{h \in H} \mathrm{Re}\big(\lambda_H(h)\chi(h^{-1})\big) \\ &= \lambda_H(e)\chi(e) + \sum_{h \in H \setminus \{e\}} \lambda_H(h)\mathrm{Re}\big(\chi(h^{-1})\big)\end{aligned}$$

$$\begin{aligned} &\geq \sum_{h \in H} \lambda_H(h)\chi(e) \\ &= \chi(e)|H| \cdot \langle \lambda_H, \chi_{1_H} \rangle \\ &= 0 \end{aligned}$$

using the estimate

$$\mathrm{Re}\chi(h^{-1}) \leq |\mathrm{Re}\chi(h^{-1})| \leq |\chi(h^{-1})| \leq \chi(e).$$

(The last inequality follows by writing $\chi = \sum_j \chi_j$ for certain homomorphisms $\chi_j : H \longrightarrow \mathbb{C}^\times$ and noting χ_j is unitary.) This proves that $\lambda_H \in R^+(H)$ by letting χ vary over the irreducible characters.

Finally, we apply this to $G = \mathrm{Gal}(E/K)$. From Proposition 4.10 we get

$$\left(\frac{\zeta_E(s)}{\zeta_K(s)}\right)^{|G|} = \left(\frac{L(s, \chi_{G,\mathrm{reg}})}{L(s, 1_G)}\right)^{|G|} = \prod\nolimits_{\text{cyclic } H \leq G} L(s, \lambda_H),$$

and $L(s, \lambda_H)$ is an entire function since χ_{1_H} does not appear in λ_H. Therefore $\zeta_K(s)$ divides $\zeta_E(s)$. □

4.4.2 Brauer's Theorem

For the applications to Artin L-functions we have in mind, it will be crucial for us to express any $\chi \in R(G)$ as a $\mathbb{Z}$-linear (not just $\mathbb{Q}$-linear) combination of induced characters. Brauer's refinement of Artin's Theorem (really Corollary 4.25) achieves this by enlarging $\mathcal{F}_{\mathrm{cyc}}$ to a family of certain nilpotent groups. (Recall that H being *nilpotent* means it admits a central series:

$$\{e\} = H_0 \subset H_1 \subset H_2 \subset \cdots \subset H_N = H$$

such that $H_i \lhd H$ and $H_{i+1}/H_i \subset Z(H/H_i)$ for all i. Obviously nilpotent $\Rightarrow$ solvable.)

Definition 4.21 Let p be a prime. We say a finite group H is *p-elementary* if $H \simeq C \times P$ where C is cyclic of order prime-to-p, and P is a p-group.

We say H is *elementary* if it is p-elementary for some prime p. (Such H are nilpotent.)

Note that a cyclic group is p-elementary for all p.

Exercise 4.22 Show that any subgroup of a p-elementary group is again p-elementary. (`Hint`: If A, B are groups with coprime cardinalities, then any subgroup of $A \times B$ is of the form $A' \times B'$ for for subgroups A' and B' of A and B respectively.)

Example 4.23 Start with an element $g \in G$ of order prime-to-p. We associate p-elementary subgroups with g as follows. Let $C = \langle g \rangle$ and take P to be a Sylow p-subgroup of the centralizer $Z_G(g)$. Then $H = C \times P$ is a p-elementary subgroup of $Z_G(g)$–uniquely determined by g up to $Z_G(g)$-conjugation.

Our imminent objective is to bound the cokernel of $\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G$ where $\mathcal{F}_{p\text{-el}}$ is the family of all p-elementary subgroups of G.

Theorem 4.24 (Brauer) *The prime-to-p part of $|G|$ annihilates $cok(Ind_{\mathcal{F}_{p\text{-el}}}^G)$.*

This is the namesake result of this section. Before proceeding to its proof, we record the following offshoot which will be our main tool in the meromorphic continuation of Artin L-functions.

Corollary 4.25 *Any virtual character $\chi \in R(G)$ can be expressed as a sum*

$$\chi = \sum\nolimits_{\text{elementary } H \leq G} Ind_H^G \chi_H$$

for suitable $\chi_H \in R(H)$. (Thus χ is a $\mathbb{Z}$-linear combination of characters induced from characters of elementary subgroups.)

Proof We are claiming $\mathrm{Ind}_{\mathcal{F}_{\mathrm{el}}}^G$ is surjective, where $\mathcal{F}_{\mathrm{el}} = \bigcup_p \mathcal{F}_{p\text{-el}}$. Obviously the image of $\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G$ lies in the image of $\mathrm{Ind}_{\mathcal{F}_{\mathrm{el}}}^G$, so by Theorem 4.24 the cokernel of $\mathrm{Ind}_{\mathcal{F}_{\mathrm{el}}}^G$ has order prime-to-p for all p. In other words $\mathrm{cok}(\mathrm{Ind}_{\mathcal{F}_{\mathrm{el}}}^G)$ must be trivial. □

We start by laying some groundwork preparing us for the proof of 4.24. Let $\mathcal{O}$ be the ring of integers in the cyclotomic field $\mathbb{Q}(\zeta_{|G|})$. Note that $\mathcal{O}$ contains the values of all characters of G (a matrix of order dividing $|G|$ has its eigenvalues, and hence its trace, in $\mathcal{O}$). In particular we may view $R(G) \otimes_{\mathbb{Z}} \mathcal{O}$ as a subring of the ring of class functions $G \longrightarrow \mathcal{O}$. Applying the functor $(-) \otimes_{\mathbb{Z}} \mathcal{O}$ gives the exact sequence

$$\bigoplus\nolimits_{H \in \mathcal{F}_{p\text{-el}}} R(H) \otimes_{\mathbb{Z}} \mathcal{O} \longrightarrow R(G) \otimes_{\mathbb{Z}} \mathcal{O} \longrightarrow \mathrm{cok}(\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G) \otimes_{\mathbb{Z}} \mathcal{O} \longrightarrow 0$$

Since $\mathcal{O}/\mathbb{Z}$ is flat (even finite free) it suffices to show the prime-to-part of $|G|$ annihilates

$$\mathrm{cok}\left(\bigoplus\nolimits_{H \in \mathcal{F}_{p\text{-el}}} R(H) \otimes_{\mathbb{Z}} \mathcal{O} \longrightarrow R(G) \otimes_{\mathbb{Z}} \mathcal{O} \right) \simeq \mathrm{cok}(\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G) \otimes_{\mathbb{Z}} \mathcal{O}.$$

The map in parentheses will be denoted $\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G \otimes_{\mathbb{Z}} \mathcal{O}$ in what follows.

We cleave the proof of 4.24 into a series of Lemmas. The first has the same flavor as Corollary 4.19.

Lemma 4.26 *Any $\mathcal{O}$-valued class function $f : G \longrightarrow \mathbb{C}$ can be expressed as a sum*

$$f = \frac{1}{|G|} \cdot \sum\nolimits_{\text{cyclic } H \leq G} a_H Ind_H^G \chi_H$$

for suitable $\chi_H \in R(H)$ and coefficients $a_H \in \mathcal{O}$. (Thus $|G|f$ is an $\mathcal{O}$-linear combination of characters induced from characters of cyclic subgroups.)

Proof Recall the functions $\theta_H \in R(H)$ introduced in the proof of Theorem 4.18, where we found that $\sum_{H\in\mathcal{F}_{\text{cyc}}} \text{Ind}_H^G \theta_H$ is the constant function sending everything to $|G|$. By the projection formula we can express any class function f as

$$f = \frac{1}{|G|} \cdot \sum\nolimits_{H\in\mathcal{F}_{\text{cyc}}} f \cdot \text{Ind}_H^G \theta_H = \frac{1}{|G|} \cdot \sum\nolimits_{H\in\mathcal{F}_{\text{cyc}}} \text{Ind}_H^G (f|_H \cdot \theta_H).$$

At this point we just need to check $f|_H \cdot \theta_H \in R(H) \otimes_{\mathbb{Z}} \mathcal{O}$, but this is clear. Indeed, for any homomorphism $\chi : H \longrightarrow \mathbb{C}^\times$ we have $\langle f|_H \cdot \theta_H, \chi\rangle \in \mathcal{O}$ since f is $\mathcal{O}$-valued and θ_H takes values in $|H|\mathbb{Z}$. □

Any $g \in G$ factors uniquely as $g = g_p g_{p'} = g_{p'} g_p$ where g_p has p-power order, $g_{p'}$ has prime-to-p order, and both are powers of g. (To see this, decompose $\langle g\rangle$ as a product of cyclic subgroups of prime-power orders.)

Lemma 4.27 *Suppose $\chi \in R(G) \otimes_{\mathbb{Z}} \mathcal{O}$ is $\mathbb{Z}$-valued (when viewed as a class function). Then*

$$\chi(g) \equiv \chi(g_{p'}) \pmod{p}$$

for all $g \in G$.

Proof By restricting χ to $\langle g\rangle$ we may assume $G = \langle g\rangle$ to begin with. Expand $\chi = \sum_i a_i\chi_i$ where $a_i \in \mathcal{O}$ and the $\chi_i : G \longrightarrow \mathbb{C}^\times$ are degree one characters. Let q be the order of the p-part g_p. So $g^q = g_{p'}^q$. Then, working modulo $p\mathcal{O}$, we have congruences

$$\chi(g)^q \equiv \sum_i a_i^q \chi_i(g)^q = \sum_i a_i^q \chi_i(g^q) = \sum_i a_i^q \chi_i(g_{p'}^q) = \sum_i a_i^q \chi_i(g_{p'})^q \equiv \chi(g_{p'})^q.$$

Since $\mathbb{Z} \cap p\mathcal{O} = p\mathbb{Z}$, and χ is assumed to be $\mathbb{Z}$-valued, we are done by Fermat's little theorem. □

The next result holds the clue.

Lemma 4.28 *Let $g \in G$ be an element of prime-to-p order, and consider a p-elementary subgroup H associated with g as in Example 4.23. Then there exists a $\mathbb{Z}$-valued $\psi \in R(H) \otimes_{\mathbb{Z}} \mathcal{O}$ whose induction $\Psi = \text{Ind}_H^G \psi$ satisfies the following two properties:*

(1) $\Psi(g) \not\equiv 0 \pmod{p}$;
(2) $\Psi(y) = 0$ for all p'-elements y which are <u>not</u> conjugate to g.

Proof Employ the notation $H = C \times P$ introduced in Example 4.23. Start with the function on $C = \langle g\rangle$ given by $\psi_C = |C|\delta_g$. It can be written as $\psi_C = \sum_\chi \chi(g^{-1})\chi$ (summing over all degree one characters χ of C) and therefore visibly belongs to $R(C) \otimes_{\mathbb{Z}} \mathcal{O}$. We take $\psi \in R(H) \otimes_{\mathbb{Z}} \mathcal{O}$ to be the composition of ψ_C with the

projection $H \to C$. Clearly ψ is $\mathbb{Z}$-valued since ψ_C is. We verify that $\Psi = \mathrm{Ind}_H^G \psi$ satisfies (1) and (2)–starting with (2): Suppose $y \in G$ has prime-to-p order and the sth term of the sum

$$\Psi(y) = \frac{1}{|H|} \cdot \sum\nolimits_{s \in G, sys^{-1} \in H} \psi(sys^{-1})$$

is nonzero. Since sys^{-1} has order prime-to-p it necessarily lies in C, so that $\psi(sys^{-1}) = \psi_C(sys^{-1})$. We are assuming this is nonzero, which forces $sys^{-1} = g$. Thus y must be conjugate to g if such an s exists. This proves (2). For part (1) observe, for the same reasons as in the proof of (2), that

$$\Psi(g) = \frac{1}{|H|} \cdot \sum\nolimits_{s \in G, sgs^{-1} \in H} \psi(sgs^{-1}) = \frac{1}{|H|} \cdot \sum\nolimits_{s \in G, sgs^{-1} = g} |C| = [Z_G(g) : P]$$

is prime-to-p since $P \subset Z_G(g)$ is a Sylow p-subgroup. □

Proof of Theorem 4.24: Let $(g_i)_{i \in I}$ be a set of representatives for the conjugacy classes of p'-elements in G. Let H_i be a p-elementary subgroup associated with g_i, as in 4.23. In Lemma 4.28 we found a $\mathbb{Z}$-valued $\psi_i \in R(H_i) \otimes_{\mathbb{Z}} \mathcal{O}$ such that $\Psi_i = \mathrm{Ind}_{H_i}^G \psi_i$ satisfies:

(1) $\Psi_i(g_i) \not\equiv 0 \pmod{p}$;
(2) $\Psi_i(g_j) = 0$ for all $j \neq i$.

Introduce the $\mathbb{Z}$-valued function $\Psi_+ = \sum_{i \in I} \Psi_i$ belonging to the image of $\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G \otimes_{\mathbb{Z}} \mathcal{O}$. We note that

$$\Psi_+(g) \not\equiv 0 \pmod{p} \tag{4.4}$$

for all $g \in G$. Indeed, write $g = g_p g_{p'}$ with $g_{p'}$ conjugate to g_j, say. Then by Lemma 4.27 we see that

$$\Psi_+(g) \equiv \Psi_+(g_{p'}) = \Psi_+(g_j) = \sum_{i \in I} \Psi_i(g_j) = \Psi_j(g_j) \not\equiv 0 \pmod{p}.$$

Write $|G| = p^n \ell$ with $p \nmid \ell$. From what we noted in (4.4) we deduce the congruence

$$\Psi_+(g)^{\varphi(p^n)} \equiv 1 \pmod{p^n}.$$

In other words the function $g \mapsto \ell(\Psi_+(g)^{\varphi(p^n)} - 1)$ takes values in $|G|\mathbb{Z}$. By Lemma 4.26 it can be written as an $\mathcal{O}$-linear combination

$$\ell(\Psi_+^{\varphi(p^n)} - \chi_{1_G}) = \sum\nolimits_{\text{cyclic } H \leq G} a_H \mathrm{Ind}_H^G \chi_H.$$

Since cyclic groups are p-elementary the right-hand side is in the image of $\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^G \otimes_{\mathbb{Z}} \mathcal{O}$, which is an ideal containing Ψ_+. We conclude that the constant function $\ell \cdot \chi_{1_G}$

lies in this image as well. In other words ℓ annihilates the cokernel of $\mathrm{Ind}_{\mathcal{F}_{p\text{-el}}}^{G} \otimes_{\mathbb{Z}} \mathcal{O}$ which implies Theorem 4.24. Done! □

Before wrapping up this section we need to understand the irreducible characters of an elementary group. We will make use of the following general property.

Proposition 4.29 *Let $r : G \longrightarrow GL(V)$ be an irreducible representation, and let $N \lhd G$. Suppose $r|_N$ is non-isotypic. Then there exists an $H < G$, containing N, and a (necessarily irreducible) representation $\rho : H \longrightarrow GL(W)$ such that $r \simeq Ind_H^G \rho$.*

Proof Let $V = \bigoplus_{i \in I} V_i$ be the decomposition of $r|_N$ into isotypic N-representations. For each $g \in G$, the automorphism $r(g)$ permutes the N-invariant subspaces of V since $N \lhd G$. In particular $r(g)$ permutes the V_i–and G acts transitively on $\{V_i\}_{i \in I}$ since r is irreducible. Pick one of these isotypic subspaces, say V_j. We stress that $V_j \neq V$ by our assumption that $r|_N$ is not isotypic. Therefore the stabilizer

$$H := \{g \in G : gV_j = V_j\}$$

is a *proper* subgroup of G, which obviously contains N. We may take ρ to be the representation of H on V_j obtained by restricting r. More precisely, $\rho(g) = r(g)|_{V_j}$ for $g \in H$. Sending $f \mapsto \sum_{g \in H \backslash G} g^{-1} f(g)$ then gives a G-equivariant linear map

$$\mathrm{Ind}_H^G V_j \longrightarrow V$$

which is injective because $V = \bigoplus_{g \in H \backslash G} g^{-1} V_j$. It is therefore an isomorphism since r is irreducible. □

For a nilpotent group, the irreducible representations are *monomial*. That is, they all arise by inducing a degree one character of a subgroup. This applies in particular to elementary groups.

Proposition 4.30 *Let $r : G \longrightarrow GL(V)$ be an irreducible representation of a nilpotent group G. Then there exists a subgroup H and a homomorphism $\rho : H \longrightarrow \mathbb{C}^\times$ such that $r \simeq Ind_H^G \rho$.*

Proof We proceed by induction on $|G|$. The base case is trivial, so assume $|G| > 1$ and that the statement is true for all nilpotent groups of smaller cardinality. We may assume r is faithful. (Otherwise $K = \ker(r) \neq \{e\}$ so by induction $r \simeq \mathrm{Ind}_{H/K}^{G/K} \rho$ for some subgroup $H \subset G$ containing K, and some character $\rho : H/K \longrightarrow \mathbb{C}^\times$. One checks that in this case $r \simeq \mathrm{Ind}_H^G \rho$.) We may obviously also assume G is non-abelian.

Now pick an abelian normal subgroup $N \lhd G$ such that $N > Z(G)$. (For instance, take N to be a subgroup such that $N/Z(G)$ is a non-trivial cyclic subgroup of $Z(G/Z(G))$. The latter is non-trivial since G is assumed to be non-abelian.)

As r is faithful, $r(N)$ is not a central subgroup of $r(G)$. In particular there must be an $n \in N$ such that $r(n)$ is not scaling by a constant. Since N is abelian we

deduce that $r|_N$ is non-isotypic. Proposition 4.29 gives the existence of an $H' < G$ along with a representation $\rho' : H' \longrightarrow \mathrm{GL}(W)$ such that $r \simeq \mathrm{Ind}_{H'}^G \rho'$. In turn, by induction $\rho' \simeq \mathrm{Ind}_H^{H'} \rho$ for some degree one character $\rho : H \longrightarrow \mathbb{C}^\times$. Transitivity yields $r \simeq \mathrm{Ind}_{H'}^G \mathrm{Ind}_H^{H'} \rho \simeq \mathrm{Ind}_H^G \rho$ as desired. □

Corollary 4.25 and Proposition 4.30 combined gives the following crucial result:

Corollary 4.31 *Any virtual character $\chi \in R(G)$ can be expressed as a $\mathbb{Z}$-linear combination of monomial characters:*

$$\chi = \sum\nolimits_{\text{elementary } H \leq G} a_H \mathrm{Ind}_H^G \chi_H$$

for suitable χ_H which are finite sums of homomorphisms $H \longrightarrow \mathbb{C}^\times$, and coefficients $a_H \in \mathbb{Z}$.

Unfortunately, if $\chi \in R^+(G)$ is a character, one *cannot* always arrange for all the coefficients a_H to be in $\mathbb{Z}_{\geq 0}$. For instance, an irreducible non-monomial character does not have this form for obvious reasons. This is essentially why the Artin conjecture is still open in general.

A group G is called *monomial* if all its irreducible characters are monomial (induced from a degree one character of a subgroup). Proposition 4.30 shows that nilpotent groups are monomial. Actually, the proof shows the slightly stronger statement that supersolvable groups are monomial. Furthermore, a result of Taketa shows that a monomial group is necessarily solvable (see [Isa94, Corollary 5.13, p. 67]). This gives a hierarchy of groups:

$$\text{elementary} \Longrightarrow \text{nilpotent} \Longrightarrow \text{supersolvable} \Longrightarrow \text{monomial} \Longrightarrow \text{solvable}$$

where all implications are strict. For example, S_4 is monomial but not supersolvable, and $\mathrm{SL}_2(\mathbb{F}_3)$ can be shown to be solvable but not monomial. Both groups have order 24. Moreover, S_3 is an example of a supersolvable group which is not nilpotent (it has trivial center).

4.5 The Euler Factors at the Infinite Places

We return to number fields. Thus G again denotes the Galois group of a finite Galois extension of number fields E/K, and $r : G \longrightarrow \mathrm{GL}(V)$ is an n-dimensional complex representation with Artin L-function $L(s, r)$.

Theorem 4.32 *$L(s, r)$ admits meromorphic continuation to the whole complex plane.*

Proof By Corollary 4.31 we can write $\chi_r = \sum_{H \leq G} a_H \mathrm{Ind}_H^G \chi_H$ as in that statement. Proposition 4.10 lets us express $L(s, r)$ as an alternating product of Hecke L-functions

$$L(s,r) = \prod\nolimits_{H\leq G} L(s, \mathrm{Ind}_H^G \chi_H)^{a_H} = \prod\nolimits_{H\leq G} L(s,\chi_H)^{a_H}.$$

We stress that some of the a_H could very well be negative. The right-hand side is visibly a meromorphic function. □

Just like for Hecke L-functions, the functional equation for $L(s,r)$ takes a much more transparent form for a modified version $L^{\mathrm{aut}}(s,r)$ which takes the infinite places V_∞ into account.

Recall that $V_\infty = V_{\mathbb{R}} \cup V_{\mathbb{C}}$ where the real places $v \in V_{\mathbb{R}}$ correspond to real embeddings $\sigma : K \hookrightarrow \mathbb{R}$, and the complex places $v \in V_{\mathbb{C}}$ correspond to pairs $\{\tau, \bar{\tau}\}$ of non-real embeddings $K \hookrightarrow \mathbb{C}$.

Let $v \in V_\infty$, and pick a place $w|v$ of E. The Galois group G acts transitively on the set of such w, and we let G_w be the stabilizer of our choice w. There are three possible scenarios:

(1) v is complex (and then w is necessarily complex as well);
(2) v is real and w is real;
(3) v is real and w is complex.

Here $G_w = \{e\}$ in case (1) and (2) and $G_w = \langle c_w \rangle = \{e, c_w\}$ in case (3).

Definition 4.33 Suppose v is real and $w|v$ is complex. Let $V = V^+ \oplus V^-$ be the decomposition into (± 1)-eigenspaces for $r(c_w)$. We introduce their dimensions

$$n^\pm = \dim_{\mathbb{C}} V^\pm = \frac{1}{2}(n \pm \chi_r(c_w)).$$

The Euler factor at $v \in V_\infty$ is defined case-by-case as follows:

$$L_v(s,r) = \begin{cases} L_{\mathbb{C}}(s)^n & \text{if } v \text{ and } w \text{ are complex} \\ L_{\mathbb{R}}(s)^n & \text{if } v \text{ and } w \text{ are real} \\ L_{\mathbb{R}}(s)^{n^+} L_{\mathbb{R}}(s+1)^{n^-} & \text{if } v \text{ is real and } w \text{ is complex.} \end{cases}$$

(For the convenience of the reader we recall that $L_{\mathbb{R}}(s) = \pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$ and $L_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$.)

We continue "Examples".

Example 4.34 Suppose $G = \mathrm{Gal}(E/K)$ is abelian, and $r : G \longrightarrow \mathbb{C}^\times$ is a degree one character with associated Hecke character $\chi = r \circ \mathrm{Art}_{E/K}$. By local-global compatibility at $v \in V_\infty$ we have

$$\mathrm{Art}_{E/K}(\underbrace{-1}_{\text{at } v}, \underbrace{1,1,1,\ldots}_{\text{away from } v}) = c_w$$

for all $w|v$ in case (3). So $r(c_w) = \chi_v(-1) = (-1)^{\mathrm{e}_v}$ shows $L_v(s,r) = L_{\mathbb{R}}(s + \mathrm{e}_v) = L(s,\chi_v)$ in this case. In cases (1) and (2) we have $E_w = K_v$–so $K_v^\times$ is clearly

in $N_{E/K}(C_E)$, and therefore $\chi_v = 1$. For trivial reasons we have $L_v(s, r) = L(s, \chi_v)$ as well in these cases.

The definition of the Euler factors at the infinite places is tailored in such a way that the invariance properties in Sect. 4.3 carry over to the functions

$$L_\infty(s, r) = \prod_{v \in V_\infty} L_v(s, r).$$

Here is the precise statement.

Proposition 4.35 *Let $K \subset L \subset E$ be an intermediate field and let $H = Gal(E/L)$.*

(a) Assume $H \lhd G$. Then

$$L_\infty(s, r) = L_\infty(s, r')$$

where $r = r' \circ pr_H$ is the inflation of $r' : G/H \to GL(V)$;

(b) For any (not necessarily normal) subgroup H we have the equality

$$L_\infty(s, r) = L_\infty(s, \rho)$$

where $r = Ind_H^G \rho$ is the induction of $\rho : H \to GL(W)$.

Proof The proof of (a) is fairly straightforward. For a fixed $v \in V_\infty$ for K we must check that $L_v(s, r) = L_v(s, r')$. Pick a place $w|v$ of E and let $\nu = w|_L$. If v is complex, so are w and ν, and both sides equal $L_{\mathbb{C}}(s)^n$. If v and w are real, so is ν, and both sides equal $L_{\mathbb{R}}(s)^n$. The only case which requires a moments thought is when v is real and w is complex, where $L_v(s, r) = L_{\mathbb{R}}(s)^{n^+} L_{\mathbb{R}}(s+1)^{n^-}$. This divides into two sub-cases–according to whether ν is real or complex:

(i) v is real, w is complex, and ν is real:
Here $L_v(s, r') = L_{\mathbb{R}}(s)^n$ and it suffices to note that $n^+ = n$ since $c_w \in H$ acts trivially on V.

(ii) v is real, w is complex, and ν is complex:
Here $L_v(s, r) = L_{\mathbb{R}}(s)^{m^+} L_{\mathbb{R}}(s+1)^{m^-}$ where $m^\pm$ denote the dimensions of the (± 1)-eigenspaces for $r'(c_\nu)$. Clearly $c_\nu = \mathrm{pr}_H(c_w)$ so $m^\pm = n^\pm$.

The proof of (b) is slightly more complicated. Again we fix a place $v \in V_\infty$ for K and consider all places $\nu|v$ of L above it. We need to verify the equality

$$L_v(s, r) = \prod_{\nu|v} L_\nu(s, \rho).$$

If v is complex, so are all the $\nu|v$, and both sides equal $L_{\mathbb{C}}(s)^n$. (There are exactly $[L : K]$ places $\nu|v$, and $n = [G : H] \dim_{\mathbb{C}} \rho$.) If v is real, and some (equivalently all) $w|v$ are real, an identical argument shows both sides equal $L_{\mathbb{R}}(s)^n$. The more difficult case is when v is real and some (equivalently all) $w|v$ are complex. We fix a $w|v$ once and for all. Then the G-action gives a bijection

$$\begin{aligned} H\backslash G/G_w &\xrightarrow{1:1} \{\nu|v\} \\ \sigma &\longmapsto \sigma w|_L. \end{aligned}$$

Note that $\nu = \sigma w|_L$ is a real place if and only if $c_{\sigma w} (= \sigma c_w \sigma^{-1})$ belongs to H. To see this, suppose w corresponds to the pair of embeddings $\{\tau, \bar{\tau}\}$ and consider the diagram

$$\begin{array}{ccccccc} L & \xrightarrow{\text{incl}} & E & \xrightarrow{\sigma^{-1}} & E & \xrightarrow{\tau} & \mathbb{C} \\ \| & & \downarrow{\scriptstyle c_{\sigma w}} & & \downarrow{\scriptstyle c_w} & & \downarrow{\scriptstyle \overline{(\cdot)}} \\ L & \xrightarrow{\text{incl}} & E & \xrightarrow{\sigma^{-1}} & E & \xrightarrow{\tau} & \mathbb{C} \end{array}$$

The horizontal embedding $L \hookrightarrow \mathbb{C}$ in this diagram represents ν. The diagram commutes if and only if ν is real, which in turn happens precisely when $c_{\sigma w}$ restricts to the identity on L (look at the leftmost square). Note also that $H\sigma G_w = H\sigma \iff \sigma G_w \sigma^{-1} \subset H$

We conclude that the real places $\nu|v$ have the following parametrization in terms of our choice w:

$$\begin{aligned} H\backslash\{\sigma \in G : \sigma G_w \sigma^{-1} \subset H\} &\xrightarrow{1:1} \{\text{real } \nu|v\} \\ \sigma &\longmapsto \sigma w|_L. \end{aligned}$$

In particular,

$$\chi_r(c_v) = \frac{1}{|H|} \cdot \sum\nolimits_{\sigma\in G, \sigma c_w \sigma^{-1} \in H} \chi_\rho(\sigma c_w \sigma^{-1}) = \sum\nolimits_{\text{real } \nu|v} \chi_\rho(c_\nu).$$

(Here c_v denotes any choice of c_w with $w|v$, and similarly for c_ν.) Before winding up the argument we introduce some auxiliary notation. Let $m = \dim_{\mathbb{C}} \rho$, and let $m_\nu^\pm$ denote the dimensions of the (± 1)-eigenspaces of $\rho(c_\nu)$. We conclude from the duplication formula $L_{\mathbb{R}}(s) L_{\mathbb{R}}(s+1) = L_{\mathbb{C}}(s)$ that

$$\begin{aligned} \prod_{\nu|v} L_\nu(s, \rho) &= \prod\nolimits_{\text{complex } \nu|v} L_{\mathbb{C}}(s)^m \cdot \prod\nolimits_{\text{real } \nu|v} L_{\mathbb{R}}(s)^{m_\nu^+} L_{\mathbb{R}}(s+1)^{m_\nu^-} \\ &= L_{\mathbb{R}}(s)^{\tilde{n}^+} L_{\mathbb{R}}(s+1)^{\tilde{n}^-} \end{aligned}$$

where we have introduced exponents

$$\begin{aligned} \tilde{n}^\pm &= m \cdot \#\{\text{complex } \nu|v\} + \sum\nolimits_{\text{real } \nu|v} m_\nu^\pm \\ &= m \cdot \#\{\text{complex } \nu|v\} + \sum\nolimits_{\text{real } \nu|v} \frac{1}{2}(m \pm \chi_\rho(c_\nu)) \\ &= \frac{1}{2}\left(2\,m \cdot \#\{\text{complex } \nu|v\} + m \cdot \#\{\text{real } \nu|v\} \pm \sum\nolimits_{\text{real } \nu|v} \chi_\rho(c_\nu) \right) \end{aligned}$$

$$= \frac{1}{2}\Big(m[L:K] \pm \chi_r(c_v)\Big) = \frac{1}{2}(n \pm \chi_r(c_v)) = n^{\pm}.$$

This gives the result. □

Consequently the modified Artin L-function $L^{\mathrm{aut}}(s,r) = L_\infty(s,r)L(s,r)$ also satisfies the analogous invariance properties under inflation and induction. As in the proof of Theorem 4.32 this allows us to express $L^{\mathrm{aut}}(s,r)$ as an alternating product of (normalized) Hecke L-functions $L^{\mathrm{aut}}(s,\chi_H)$.

4.6 The Artin Conductor

The functional equation for $L(s,r)$ entails a numerical invariant of r which is the (norm of the) so-called Artin conductor. This is an ideal of $\mathcal{O}_K$ which in some sense measures the ramification of r. The conductor is defined as a finite product $\mathfrak{f}(r) = \prod_{v\in V_f} \mathfrak{p}_v^{\mathfrak{f}_v(r)}$ with exponents $\mathfrak{f}_v(r) \in \mathbb{Z}_{\geq 0}$ defined purely locally. We thus switch setup for a moment, and work over a local field.

4.6.1 The Local Artin Conductor

In this section E/K denotes a finite Galois extension of local fields with Galois group G, and $r : G \longrightarrow \mathrm{GL}(V)$ is an n-dimensional complex representation.

Recall the ith ramification subgroup G_i is defined as follows:

$$G_i = \{\sigma \in G : \mathrm{ord}_E(\sigma(x) - x) \geq i+1, \forall x \in \mathcal{O}_E\}. \tag{4.5}$$

Upon replacing i by its roof we may take i to be an integer, in which case G_i is the kernel of the natural homomorphism $G \to \mathrm{Aut}_{\mathcal{O}_K\text{-alg}}(\mathcal{O}_E/\mathfrak{p}_E^{i+1})$. In particular $G_i \lhd G$. For small i we get familiar groups:

$$G_i = \begin{cases} G & \text{if } i \leq -1 \\ I & \text{if } -1 < i \leq 0 \\ P & \text{if } 0 < i \leq 1. \end{cases}$$

(Here I is the inertia subgroup, and $P \lhd I$ is its Sylow p-subgroup–known as "wild inertia".)

Once and for all we pick a θ such that $\mathcal{O}_E = \mathcal{O}_K[\theta]$. Then θ also generates $\mathcal{O}_E/\mathfrak{p}_E^{i+1}$ as an $\mathcal{O}_K$-algebra. Therefore, to check that $\sigma \in G_i$ it is enough to verify the inequality $\mathrm{ord}_E(\sigma(\theta) - \theta) \geq i+1$. In particular $G_i = \{e\}$ for $i \gg 0$.

To measure the ramification of r we consider the subspace of G_i-invariants V^{G_i} as i varies. Note that G/G_i acts on V^{G_i} via r. This gives a filtration of V by G-invariant subspaces

$$V^G = V^{G_{-1}} \subset V^{G_0} \subset V^{G_1} \subset V^{G_2} \subset \cdots \subset V^{G_i} = V, \quad i \gg 0.$$

How deep do we have to go in the sequence $(G_i)_{i \geq -1}$ to get nonzero G_i-invariants? When $V^{G_0} = V$ we say V is unramified. When $V^{G_1} = V$ we say V is tamely ramified. Otherwise V is called wildly ramified. The Artin conductor is a number built from all the dimensions $\dim_{\mathbb{C}} V^{G_i}$ as $i \geq 0$ varies:

Definition 4.36 $\mathfrak{f}(r) = \sum_{i=0}^{\infty} \frac{|G_i|}{|G_0|} \cdot \operatorname{codim}_{\mathbb{C}} V^{G_i}$.

The *codimension* of a subspace $W \subset V$ is $\dim_{\mathbb{C}} V - \dim_{\mathbb{C}} W$.

Note that the sum defining $\mathfrak{f}(r)$ is finite since $\operatorname{codim}_{\mathbb{C}} V^{G_i} = 0$ for all $i \gg 0$. It is clear that $\mathfrak{f}(r) = 0$ if and only if $G_0 = I$ acts trivially on all of V; that is when V is unramified. It is also plain that in general $\mathfrak{f}(r) \in \mathbb{Q}_{\geq 0}$. What is not at all obvious is the fact that $\mathfrak{f}(r)$ is an integer! This is the main result of this section, which allows us to make sense of the ideal $\mathfrak{p}_K^{\mathfrak{f}(r)}$.

Theorem 4.37 $\mathfrak{f}(r) \in \mathbb{Z}_{\geq 0}$.

To set the stage for the proof we introduce the function

$$\begin{aligned} i_G : G &\longrightarrow \mathbb{Z}_{\geq 0} \cup \{\infty\} \\ \sigma &\longmapsto \operatorname{ord}_E\big(\sigma(\theta) - \theta\big). \end{aligned}$$

(By convention $i_G(e) = \infty$.) In light of our earlier remarks it is clear that $i_G(\sigma) \geq i + 1 \iff \sigma \in G_i$. From this observation it is easy to deduce the following properties:

- $i_G(\sigma) = i + 1$ if $\sigma \in G_i$ but $\sigma \notin G_{i+1}$;
- i_G is independent of the choice of θ;
- $i_G(\tau\sigma\tau^{-1}) = i_G(\sigma)$ for all $\sigma, \tau \in G$.

Exercise 4.38 Let $H \subset G$ be a subgroup. Show that $i_G(\sigma) = i_H(\sigma)$ for all $\sigma \in H$.

We introduce a variant of i_G, essentially by changing its value at the identity to get a zero average. Here $f = f_{E/K}$ denotes the inertia degree of the extension E/K, which will prove to be a convenient normalizing factor, inserted to make part (a) of Lemma 4.42 below hold:

Definition 4.39 $a_G(\sigma) = \begin{cases} -f i_G(\sigma) & \text{if } \sigma \neq e \\ \sum_{\tau \neq e} f i_G(\tau) & \text{if } \sigma = e. \end{cases}$

This gives an $f\mathbb{Z}$-valued class function $a_G : G \longrightarrow \mathbb{C}$ with average value $\frac{1}{|G|} \sum_{\sigma \in G} a_G(\sigma) = 0$. In other words, $\langle a_G, \chi_{1_G} \rangle = 0$. Furthermore, $a_G(\sigma) \leq 0$ for all $\sigma \neq e$. Using ideas from the proof of Corollary 4.20 one can deduce that $\operatorname{Re}\langle a_G, \chi \rangle \geq 0$ for all $\chi \in R^+(G)$ (cf. [Ser96, Exercise 9.1, p. 69]). We will not use this. Instead we will directly show that $a_G \in R^+(G)$ and $\langle a_G, \chi_r \rangle = \mathfrak{f}(r)$.

Remark 4.40 Although a_G is the character of a representation, there seems to be no natural way to realize this representation–other than just taking the sum $\bigoplus_i \mathfrak{f}(r_i)r_i$. One can realize it over $\mathbb{Q}_\ell$ when ℓ is different from the residual characteristic p of K. It is not always realizable over $\mathbb{R}$, $\mathbb{Q}$, or $\mathbb{Q}_p$.

To start off, we establish the connection between a_G and $\mathfrak{f}$.

Lemma 4.41 *For any class function $\gamma : G \longrightarrow \mathbb{C}$ we have*

$$\langle \gamma, a_G \rangle = \sum_{i=0}^{\infty} \frac{|G_i|}{|G_0|} \cdot (\gamma(e) - Av_{G_i}(\gamma)),$$

where $Av_{G_i}(\gamma) = \frac{1}{|G_i|} \sum_{\sigma \in G_i} \gamma(\sigma)$. In particular $\langle \chi_r, a_G \rangle = \mathfrak{f}(r)$.

Proof We will in fact show the exact formula for a_G below:

$$a_G = \sum_{i=0}^{\infty} \frac{|G_i|}{|G_0|} \cdot \mathrm{Ind}_{G_i}^{G}(\chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}}). \tag{4.6}$$

Applying $\langle \gamma, - \rangle$ gives the result since $\langle \gamma|_{G_i}, \chi_{G_i,\mathrm{reg}} \rangle = \gamma(e)$ and $\langle \gamma|_{G_i}, \chi_{1_{G_i}} \rangle = \mathrm{Av}_{G_i}(\gamma)$. Also, for $\gamma = \chi_r$ note that $\mathrm{Av}_{G_i}(\chi_r) = \langle \chi_r|_{G_i}, \chi_{1_{G_i}} \rangle = \dim_{\mathbb{C}} V^{G_i}$ (and of course $\chi_r(e) = \dim_{\mathbb{C}} V$).

To show (4.6) we first observe that, since G_i is a normal subgroup of G,

$$\mathrm{Ind}_{G_i}^{G}(\chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}})(g) = \frac{1}{|G_i|} \cdot \sum_{s \in G, sgs^{-1} \in G_i} (\chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}})(sgs^{-1}) = \begin{cases} \frac{|G|}{|G_i|}(|G_i| - 1) & \text{if } g = e, \\ -\frac{|G|}{|G_i|} & \text{if } g \in G_i - \{e\}, \\ 0 & \text{if } g \notin G_i. \end{cases}$$

Now suppose $g \neq e$ and pick j such that $g \in G_j$ but $g \notin G_{j+1}$. In other words $i_G(g) = j + 1$. Then

$$\begin{aligned} \sum_{i=0}^{\infty} \frac{|G_i|}{|G_0|} \cdot \mathrm{Ind}_{G_i}^{G}(\chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}})(g) &= \sum_{i=0}^{j} \frac{|G_i|}{|G_0|} \cdot \mathrm{Ind}_{G_i}^{G}(\chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}})(g) \\ &= -\sum_{i=0}^{j} \frac{|G_i|}{|G_0|} \cdot \frac{|G|}{|G_i|} \\ &= -(j+1) f_{E/K} \\ &= a_G(g). \end{aligned}$$

One deduces the equality for $g = e$ using that both sides of (4.6) are orthogonal to χ_{1_G}: We have already observed that $\langle a_G, \chi_{1_G} \rangle = 0$, and

$$\langle \mathrm{Ind}_{G_i}^{G}(\chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}}), \chi_{1_G} \rangle = \langle \chi_{G_i,\mathrm{reg}} - \chi_{1_{G_i}}, \chi_{1_{G_i}} \rangle = 0$$

since 1_{G_i} appears with multiplicity one in the regular representation of G_i (as the subspace of constant functions). □

The overall idea behind the proof of Theorem 4.37 is to reduce it to the degree one case using the Brauer induction theorem. The degree one case then boils down to the Hasse-Arf theorem. To carry this out, we first need to understand how the function a_G behaves with respect to inflation and induction.

Lemma 4.42 *Let $K \subset L \subset E$ be an intermediate field corresponding to the subgroup $H = Gal(E/L)$.*

(a) Suppose $H \lhd G$. Then, for all $\tau \in G$,

$$a_{G/H}(\tau H) = \frac{1}{|H|} \sum\nolimits_{\sigma \in \tau H} a_G(\sigma).$$

(b) Without restrictions on H we have, for all $\sigma \in H$,

$$a_G(\sigma) = ord_K(D_{L/K})\chi_{H,reg}(\sigma) + f_{L/K} a_H(\sigma).$$

(Here $D_{L/K}$ denotes the discriminant of the bottom layer L/K.)

Proof Using the property $f_{E/K} = f_{E/L} f_{L/K}$, part (a) is easily checked to be equivalent to the identity

$$i_{G/H}(\tau H) = \frac{1}{e_{E/L}} \sum\nolimits_{\sigma \in \tau H} i_G(\sigma) \tag{4.7}$$

where $e_{E/L}$ is the ramification index of the top layer. To verify this formula, write $\mathcal{O}_L = \mathcal{O}_K[\beta]$. We may assume $\tau \notin H$ (otherwise both sides are infinite). Then $i_{G/H}(\tau H) = \frac{1}{e_{E/L}} \mathrm{ord}_E(\tau(\beta) - \beta)$. In other words we have show that

$$\tau(\beta) - \beta = u \cdot \prod\nolimits_{\sigma \in \tau H} (\sigma(\theta) - \theta)$$

for some $u \in \mathcal{O}_E^\times$. Let $f \in \mathcal{O}_L[T]$ be the minimal polynomial of θ over L. Applying τ to its coefficients yields $f^\tau(T) = \prod_{\sigma \in \tau H}(T - \sigma(\theta))$. In particular $f^\tau(\theta) = \pm \prod_{\sigma \in \tau H}(\sigma(\theta) - \theta)$. Note that $\tau(\beta) \neq \beta$ so all coefficients of $f^\tau - f$ are $\mathcal{O}_L$-multiples of $\tau(\beta) - \beta$. Substituting $T = \theta$ shows $\tau(\beta) - \beta$ at least divides $\prod_{\sigma \in \tau H}(\sigma(\theta) - \theta)$ in $\mathcal{O}_E$. To show the opposite divisibility write $\beta = g(\theta)$ for some $g \in \mathcal{O}_K[T]$, and write $g - \beta = fq$ for some $q \in \mathcal{O}_L[T]$ (by the Gauss lemma). Applying τ and evaluating at $T = \theta$ now lets us conclude that

$$\beta - \tau(\beta) = g(\theta) - \tau(\beta) = g^\tau(\theta) - \tau(\beta) = f^\tau(\theta) \cdot q^\tau(\theta)$$

indeed is an $\mathcal{O}_E$-multiple of $\prod_{\sigma \in \tau H}(\sigma(\theta) - \theta)$ as desired. (Note that $g = g^\tau$ since $g \in K[T]$.) Altogether this proves Eq. (4.7) and therefore part (a).

Moving on to part (b) we first note that the formula trivially holds when $\sigma \neq e$. (In this case $\chi_{H,\mathrm{reg}}$ vanishes on σ, and $\mathcal{O}_E = \mathcal{O}_L[\theta]$ so $i_G(\sigma) = i_H(\sigma)$. The claim comes down to $f_{E/K} = f_{E/L} f_{L/K}$.) The real content of part (b) is the relation

$$a_G(e) = \mathrm{ord}_K(D_{L/K})|H| + f_{L/K} a_H(e).$$

Ultimately this amounts to the basic transitivity property of discriminants. Indeed, once we verify the formula $a_G(e) = \mathrm{ord}_K(D_{E/K})$, the claimed relation is equivalent to

$$D_{E/K} = D_{L/K}^{[E:L]} N_{L/K}(D_{E/L}).$$

The latter is a standard fact from algebraic number theory. See [Neu99, Corollary 2.10, p. 202] for example. To go between the two interpretations we have used the fact that $f_{L/K}\mathrm{ord}_L = \mathrm{ord}_K \circ N_{L/K}$, which is readily verified.

We have dwindled the proof down to showing $a_G(e) = \mathrm{ord}_K(D_{E/K})$. Since $D_{E/K}$ is the norm of the different $\mathcal{D}_{E/K}$, see [Neu99, Theorem 2.9, p. 201], it suffices to show $a_G(e) = f_{E/K}\mathrm{ord}_E(\mathcal{D}_{E/K})$. Let $m \in \mathcal{O}_K[T]$ be the minimal polynomial of θ over K. Then $\mathcal{D}_{E/K} = (m'(\theta))$ by [Neu99, Proposition 2.4, p. 197]. Note that $m(T) = \prod_{\sigma \in G}(T - \sigma(\theta))$. Differentiating and substituting $T = \theta$ shows $m'(\theta) = \prod_{\sigma \neq e}(\theta - \sigma(\theta))$. Therefore

$$\mathrm{ord}_E(\mathcal{D}_{E/K}) = \mathrm{ord}_E m'(\theta) = \sum_{\sigma \neq e} \mathrm{ord}_E(\theta - \sigma(\theta)) = \sum_{\sigma \neq e} i_G(\sigma) = \frac{1}{f_{E/K}} a_G(e)$$

as desired. □

This has the following more or less immediate adjoint reformulation.

Lemma 4.43 *In the notation of Lemma 4.42 the following identities hold for any two class functions α on G/H, and β on H:*

(a) $\langle \alpha \circ pr_H, a_G \rangle = \langle \alpha, a_{G/H} \rangle$;
(b) $\langle Ind_H^G \beta, a_G \rangle = ord_K(D_{L/K})\beta(e) + f_{L/K}\langle \beta, a_H \rangle$.

Proof More generally one has the following two adjunction properties for all class functions γ on G, where we let $\gamma_\sharp(\tau H) = \frac{1}{|H|}\sum_{\sigma \in \tau H}\gamma(\sigma)$ denote the corresponding class function on G/H,

- $\langle \alpha \circ \mathrm{pr}_H, \gamma \rangle = \langle \alpha, \gamma_\sharp \rangle$;
- $\langle \mathrm{Ind}_H^G \beta, \gamma \rangle = \langle \beta, \gamma|_H \rangle$.

They are both easily verified, and we leave their verification as an exercise for the reader. Since $a_{G/H} = (a_G)_\sharp$ the first property implies part (a). The formula for $a_G|_H$ in part (b) of 4.42 winds up the proof. □

As a start, let us deduce Theorem 4.37 for degree one characters from the Hasse-Arf theorem. For the convenience of the reader, we recall the statement of Hasse-Arf. We need the Herbrand function

$$\phi_{E/K}(x) = \int_0^x \frac{dt}{[G_0 : G_t]}.$$

(The definition of G_t for non-integral t is (4.5) with t instead of i.). By convention $\phi_{E/K}(x) = x$ for $-1 \leq x < 0$. This gives a continuous, piecewise linear, increasing, concave bijection from $[-1, \infty)$ to itself.

Theorem 4.44 (Hasse-Arf) *If E/K is an abelian extension, then $G_j > G_{j+1} \Longrightarrow \phi_{E/K}(j) \in \mathbb{Z}$.*

This says the jumps in the upper numbering of the ramification groups are integers. See [Neu99, Proposition 6.3, p. 355] for a proof.

Lemma 4.45 *$\mathfrak{f}(r) \in \mathbb{Z}_{\geq 0}$ for any homomorphism $r : G \longrightarrow \mathbb{C}^\times$. Furthermore,*

$$\mathfrak{f}(r) = \phi_{E/K}(j) + 1$$

where $j \in \mathbb{Z}$ is chosen such that $r|_{G_j} \neq 1$ but $r|_{G_{j+1}} = 1$. (For $r = 1$ we let $j = -1$.)

Proof Let $H = \ker(r)$. Then r factors through the cyclic quotient G/H as $r = r' \circ \mathrm{pr}_H$. By part (a) of Lemma 4.43 we find that $\mathfrak{f}(r) = \langle r, a_G \rangle = \langle r', a_{G/H} \rangle = \mathfrak{f}(r')$. We may, and will, therefore assume E/K is abelian to begin with.

Obviously $\mathfrak{f}(1) = 0$. If $r \neq 1$, consider the $j \in \mathbb{Z}$ such that $r|_{G_j} \neq 1$ but $r|_{G_{j+1}} = 1$. Then

$$\mathfrak{f}(r) = \sum_{i=0}^{j} \frac{|G_i|}{|G_0|},$$

with this sum being interpreted as being 0 if $j = -1$ (that is, when r is unramified). On the other hand

$$\phi_{E/K}(j) = \sum_{s=0}^{j-1} \int_s^{s+1} \frac{dt}{[G_0 : G_t]} = \frac{1}{[G_0 : G_1]} + \frac{1}{[G_0 : G_2]} + \cdots + \frac{1}{[G_0 : G_j]} = \mathfrak{f}(r) - 1,$$

the -1 coming from the $i = 0$ term. (Note that $G_t = G_{s+1}$ for $s < t \leq s + 1$.) By our choice of j we clearly have $G_j \neq G_{j+1}$, so $\mathfrak{f}(r) = \phi_{E/K}(j) + 1$ is an integer by the Hasse-Arf theorem. □

This result has a natural reformulation:

Example 4.46 Suppose $G = \mathrm{Gal}(E/K)$ is abelian and let $r : G \longrightarrow \mathbb{C}^\times$ be a homomorphism. Consider the associated character $\chi = r \circ \mathrm{Art}_{E/K}$ of $K^\times$. Recall that we defined the conductor of χ as

$$\mathrm{cond}(\chi) = \min\{n \geq 0 : \chi|_{1+\mathfrak{p}^n} = 1\} = \max\{m \geq 0 : \chi|_{1+\mathfrak{p}^m} \neq 1\} + 1.$$

Since $\mathrm{Art}_{E/K}$ maps $1 + \mathfrak{p}^m$ onto $G^n = G_{\phi_{E/K}^{-1}(m)}$ (by [Neu99, Theorem 6.2, p. 354] for instance) Lemma 4.45 translates into $\mathrm{cond}(\chi) = \mathfrak{f}(r)$.

Finally we have the prerequisites to reach the goal of this section.

Proof of Theorem 4.37: By the Brauer induction theorem (Corollary 4.31 to be precise) we can express the character χ_r as a $\mathbb{Z}$-linear combination $\chi_r = \sum_{H\leq G} c_H \mathrm{Ind}_H^G \chi_H$ for certain degree one characters χ_H. Then

$$\begin{aligned}
\mathfrak{f}(r) &\overset{4.41}{=\!=} \langle \chi_r, a_G\rangle \\
&= \sum_{H\leq G} c_H \langle \mathrm{Ind}_H^G \chi_H, a_G\rangle \\
&\overset{4.43}{=\!=} \sum_{H\leq G} c_H \big(\mathrm{ord}_K(D_{E^H/K})\chi_H(e) + f_{E^H/K}\langle \chi_H, a_H\rangle\big) \\
&\overset{4.41}{=\!=} \sum_{H\leq G} c_H \mathrm{ord}_K(D_{E^H/K}) + \sum_{H\leq G} c_H f_{E^H/K}\mathfrak{f}(\chi_H).
\end{aligned}$$

As $\mathfrak{f}(\chi_H)$ is an integer by Lemma 4.45, so is $\mathfrak{f}(r)$. The fact that $\mathfrak{f}(r)$ is non-negative follows from (4.6) which shows that at least $|G_0|a_G$ is the character of a representation. (Note that the regular representation of any $H \leq G$ breaks up as $r_{H,\mathrm{reg}} = 1_H \oplus r_{H,\mathrm{aug}}$ where $r_{H,\mathrm{aug}}$ is the "augmentation" representation on the subspace of $\mathcal{C}(H,\mathbb{C})$ consisting of functions f with average $\frac{1}{|H|}\sum_{h\in H} f(h) = 0$.)

This finishes the proof of Theorem 4.37, and we now have a well-defined integral ideal $\mathfrak{p}_K^{\mathfrak{f}(r)}$ associated with r–the local Artin conductor.

4.6.2 The Global Artin Conductor

We return to the global setup. So E/K again denotes a finite Galois extension of number fields with Galois group G, and $r : G \longrightarrow \mathrm{GL}(V)$ is an n-dimensional complex representation.

For every $v \in V_f$ pick a place $w|v$ of E. Restrict r to $G_w \simeq \mathrm{Gal}(E_w/K_v)$ and consider the local Artin conductor $\mathfrak{f}(r|_{G_w})$. This is independent of the choice of w. (Any other $w'|v$ is of the form $w' = gw$ for some $g \in G$, and the ramification groups are conjugate, $G_{w',i} = gG_{w,i}g^{-1}$. In particular $v \mapsto gv$ gives an isomorphism $V^{G_{w,i}} \xrightarrow{\sim} V^{G_{w',i}}$ for all i.) We let $\mathfrak{f}_v(r) = \mathfrak{f}(r|_{G_w})$. Note that $\mathfrak{f}_v(r) = 0$ if v is unramified in E, so the product below is finite:

Definition 4.47 $\mathfrak{f}(r) = \prod_{v\in V_f} \mathfrak{p}_v^{\mathfrak{f}_v(r)}$. (This is a nonzero ideal of $\mathcal{O}_K$.)

This is the Artin conductor of r. We need the global analogue of Lemma 4.43, describing how $\mathfrak{f}$ behaves under inflation and induction:

Lemma 4.48 *Let $K \subset L \subset E$ be an intermediate field corresponding to the subgroup $H = Gal(E/L)$.*

(a) *Suppose* $H \lhd G$. *Then* $\mathfrak{f}(r) = \mathfrak{f}(r')$ *if* $r = r' \circ pr_H$ *is the inflation of* $r' : G/H \longrightarrow GL(V)$;
(b) *Without restrictions on* H *we have*

$$\mathfrak{f}(r) = D_{L/K}^{\dim \rho} \cdot N_{L/K}\big(\mathfrak{f}(\rho)\big)$$

if $r = Ind_H^G \rho$.

Proof Part (a) follows almost immediately from its local counterpart in Lemma 4.43: Fix a $w|v$ as above, and let $\nu = w|_L$ be the place of L below it. Then $r|_{G_w}$ is the inflation of $r'|_{\mathrm{Gal}(L_\nu/K_v)}$. From Lemmas 4.43 and 4.41 we deduce that $\mathfrak{f}(r|_{G_w}) = \mathfrak{f}(r'|_{\mathrm{Gal}(L_\nu/K_v)})$. Hence $\mathfrak{f}_v(r) = \mathfrak{f}_v(r')$ for all $v \in V_f$.

Part (b) also reduces to Lemma 4.43, but this requires more work: Fix a $v \in V_f$ and a $w|v$ of E. We need to show:

$$\mathfrak{f}_v(r) = \mathrm{ord}_v(D_{L/K}) \dim \rho + \sum_{\nu|v} f_{L_\nu/K_v} \mathfrak{f}_\nu(\rho).$$

By the Mackey decomposition (Exercise 4.9) we have an isomorphism of G_w-representations

$$r|_{G_w} = (\mathrm{Ind}_H^G \rho)|_{G_w} \xrightarrow{\sim} \bigoplus_{\sigma \in \Sigma} \mathrm{Ind}_{G_w \cap \sigma^{-1} H \sigma}^{G_w} \sigma_* \rho$$

where Σ is a set of representatives for $H \backslash G / G_w$. Note that $\chi_{\sigma_*\rho}(g) = \chi_\rho(\sigma g \sigma^{-1}) =: \chi_\rho^\sigma(g)$. As we have used multiple times previously, $\sigma \mapsto \sigma w|_L$ gives a bijection between Σ and the set of places $\nu|v$ of L. If $\nu \leftrightarrow \sigma$ under this correspondence, we have

$$G_w \cap \sigma^{-1} H \sigma = \sigma^{-1}(G_{\sigma w} \cap H)\sigma = \sigma^{-1} H_{\sigma w} \sigma,$$

and $H_{\sigma w} \simeq \mathrm{Gal}(E_{\sigma w}/L_\nu)$. In particular,

$$\mathrm{Ind}_{G_w \cap \sigma^{-1} H \sigma}^{G_w} \sigma_* \rho = \mathrm{Ind}_{\sigma^{-1} H_{\sigma w} \sigma}^{G_w} \sigma_* \rho = \mathrm{Ind}_{\sigma^{-1} H_{\sigma w} \sigma}^{\sigma^{-1} G_{\sigma w} \sigma} \sigma_* \rho \xrightarrow{\sim} \sigma_* \mathrm{Ind}_{H_{\sigma w}}^{G_{\sigma w}} \rho.$$

By part (b) of Lemma 4.43 we deduce that

$$\begin{aligned}
\mathfrak{f}_v(r) &= \langle \chi_r|_{G_w}, a_{G_w} \rangle \\
&= \sum_{\sigma \in \Sigma} \langle \big(\mathrm{Ind}_{H_{\sigma w}}^{G_{\sigma w}} \chi_\rho\big)^\sigma, a_{G_w} \rangle \\
&= \sum_{\sigma \in \Sigma} \langle \mathrm{Ind}_{H_{\sigma w}}^{G_{\sigma w}} \chi_\rho, a_{G_{\sigma w}} \rangle \\
&= \sum_{\sigma \in \Sigma} \Big(\mathrm{ord}_{K_v}(D_{E^{H_{\sigma w}}/K_v}) \chi_\rho(e) + f_{E^{H_{\sigma w}}/K_v} \langle \chi_\rho|_{H_{\sigma w}}, a_{H_{\sigma w}} \rangle \Big) \\
&= \sum_{\nu|v} \Big(\mathrm{ord}_{K_v}(D_{L_\nu/K_v}) \dim \rho + f_{L_\nu/K_v} \mathfrak{f}_\nu(\rho) \Big)
\end{aligned}$$

$$= \operatorname{ord}_v(D_{L/K}) \dim \rho + \sum_{\nu|v} f_{L_\nu/K_v} \mathfrak{f}_\nu(\rho).$$

We note that $a_{G_w}(\sigma^{-1} g \sigma) = a_{G_{\sigma w}}(g)$, and similarly with i_{G_w} instead of a_{G_w}, since $G_{\sigma w,i} = \sigma G_{w,i} \sigma^{-1}$. This was used in the third step. In the last step we used the standard fact that $D_{L/K}$ factors as a product of all the local discriminants D_{L_ν/K_v}. See [Neu99, Corollary 2.11, p. 202] for example. □

Example 4.49 Taking $\rho = 1_H$ we can express $D_{L/K}$ as an Artin conductor:

$$D_{L/K} = \mathfrak{f}(\operatorname{Ind}_H^G 1_H).$$

When $H = \{e\}$–in other words when $L = E$–the induced representation is the regular representation. Since $r_{G,\mathrm{reg}} = \bigoplus_{r \text{ irred.}} r^{\oplus \dim r}$ by Frobenius reciprocity, this gives the conductor-discriminant formula for (possibly *non*-abelian) Galois extensions E/K:

$$D_{E/K} = \prod\nolimits_{r \text{ irred.}} \mathfrak{f}(r)^{\dim r}.$$

This is a wide-ranging generalization of Theorem 1.108.

4.7 The Functional Equation

The relationship between $L(s, r)$ and $L(1 - s, r^*)$ involves the following natural number:

Definition 4.50 $\mathcal{N}_r = |D_K|^{\dim r} \cdot N\big(\mathfrak{f}(r)\big)$.

Lemma 4.51 *Adopt the notation from Lemma 4.48.*

(a) Suppose $H \lhd G$ and $r = r' \circ pr_H$. Then $\mathcal{N}_r = \mathcal{N}_{r'}$.
(b) Without restrictions on H we have

$$\mathcal{N}_r = \mathcal{N}_\rho$$

if $r = Ind_H^G \rho$.

Proof Both parts are immediately deduced from Lemma 4.48. Here are the details for part (b):

$$\begin{aligned}
\mathcal{N}_r &= |D_K|^{[G:H] \dim \rho} \cdot N\big(D_{L/K}^{\dim \rho} \cdot N_{L/K}\big(\mathfrak{f}(\rho)\big)\big) \\
&= \big(|D_K|^{[L:K]} N(D_{L/K})\big)^{\dim \rho} \cdot N\big(N_{L/K}\big(\mathfrak{f}(\rho)\big)\big) \\
&= |D_L|^{\dim \rho} \cdot N(\mathfrak{f}(\rho)) \\
&= \mathcal{N}_\rho,
\end{aligned}$$

using the transitivity property of discriminants [Neu99, Corollary 2.10, p. 202]. Note also that $N(I)$ is the positive generator of the ideal $N_{K/\mathbb{Q}}(I)$, and $|D_K|$ is that of $D_{K/\mathbb{Q}}$. □

As in Sect. 1.2.5 we state the functional equation in terms of the *completed* Artin L-function:

$$\widehat{L}(s,r) := \mathcal{N}_r^{\frac{s}{2}} \cdot L_\infty(s,r) \cdot L(s,r)$$

(Note that there is a small discrepancy with the notation in 1.2.5. The completed Dirichlet L-function there involves a constant factor $N^{\frac{s}{2}}$ as well.)

Example 4.52 We continue our running Hecke character example. So, suppose again $G = \mathrm{Gal}(E/K)$ is abelian and let $r : G \longrightarrow \mathbb{C}^\times$ be a degree one character with associated Hecke character $\chi = r \circ \mathrm{Art}_{E/K}$. As we saw in Example 4.46 we have $\mathrm{cond}(\chi_v) = \mathfrak{f}_v(r)$ for all finite places v. Hence

$$\mathcal{N}_r = |D_K| \cdot N(\mathfrak{f}(r)) = |D_K| \cdot N(\prod\nolimits_{v \in V_f} \mathfrak{p}_v^{\mathfrak{f}_v(r)}) = |D_K| \cdot \prod\nolimits_{v \in V_f} N(\mathfrak{p}_v)^{\mathrm{cond}(\chi_v)} = |D_K| N_\chi.$$

(See Proposition 2.104 where the Hecke conductor N_χ was introduced.) In particular, invoking Example 4.34, we find that

$$\widehat{L}(s,r) = (|D_K| N_\chi)^{\frac{s}{2}} L^{\mathrm{aut}}(s,\chi).$$

Theorem 2.103 and Proposition 2.104 translate into $\widehat{L}(s,r)$ having meromorphic continuation and satisfying the functional equation

$$\widehat{L}(s,r) = \epsilon(\frac{1}{2},\chi)\widehat{L}(1-s,r^*).$$

Furthermore, $\widehat{L}(s,r)$ is entire unless $r = 1_G$ is the trivial character. (To aid comparison with 2.103 observe that $|\cdot|^{-i\tau}$ does not have finite order unless $\tau = 0$.)

The Brauer induction theorem enables us to deduce meromorphic continuation for $\widehat{L}(s,r)$ in general from the previous degree one example. This is the result we have been pursuing throughout this chapter.

Theorem 4.53 *Let E/K be a finite Galois extension of number fields, with Galois group G, and let $r : G \longrightarrow GL(V)$ be a finite-dimensional complex representation. The completed Artin L-function $\widehat{L}(s,r)$ has meromorphic continuation to $\mathbb{C}$, and there is a constant $W(r) \in \mathbb{C}^1$ (the "root number") such that*

$$\widehat{L}(1-s,r^*) = W(r)\widehat{L}(s,r).$$

Proof The meromorphic continuation was already observed in Theorem 4.32, and we adopt the notation $\chi_r = \sum_{H\leq G} a_H \mathrm{Ind}_H^G \chi_H$ from its proof. Now that we know $\mathcal{N}_r$ and $L_\infty(s,r)$ are both insensitive to induction (by 4.51 and 4.35 respectively) so is $\widehat{L}(s,r)$ and we can just add hats in the proof of 4.32:

$$\widehat{L}(s,r) = \prod\nolimits_{H\leq G} \widehat{L}(s, \mathrm{Ind}_H^G \chi_H)^{a_H} = \prod\nolimits_{H\leq G} \widehat{L}(s,\chi_H)^{a_H} = \overline{W(r)} \cdot \prod\nolimits_{H\leq G} \widehat{L}(1-s, \overline{\chi_H})^{a_H}$$
$$= \overline{W(r)} \cdot \widehat{L}(1-s, r^*)$$

where $\overline{W(r)} = \prod_{H\leq G} \epsilon(\frac{1}{2}, \chi_H \circ \mathrm{Art}_{E/E^H})^{a_H} \in \mathbb{C}^1$ consequently only depends on r (and not the coefficients a_H in the choice of an expansion of χ_r as a $\mathbb{Z}$-linear combination of monomial characters). □

Again we emphasize that because the coefficients a_H in the previous proof can be negative we have no control over the poles of $\widehat{L}(s,r)$ (unless G is a monomial group–in which case Artin's conjecture 4.12 holds by the remarks in Example 4.52).

Exercise 4.54 Deduce that the root number $W(\cdot)$ is invariant under inflation and induction. Extend Corollary 1.112 to non-abelian extensions by showing that $\prod_{r\in \mathrm{Irr}_{\mathbb{C}}(G)} W(r)^{\dim_{\mathbb{C}}(r)} = 1$.

4.8 Trivial Zeroes

We first emphasize that $L(s,r)$ has no zeroes in the domain of absolute convergence $\mathrm{Re}(s) > 1$. Indeed, in this region $L(s,r)$ is the exponential of

$$\ln L(s,r) = \sum_{v\in V_f} \sum_{m=1}^{\infty} \frac{q_v^{-ms}}{m} \cdot \mathrm{tr}_{V^{I_w}} r(\mathrm{Frob}_w^m).$$

(To see this, recall the formula $\exp \mathrm{tr} A = \det \exp A$ for square complex matrices A. This is obvious for diagonalizable A–which form a dense subset. Why? Any A is triangularizable over $\mathbb{C}$. Now perturb some of the diagonal entries by an $\varepsilon > 0$ to make them distinct.)

Since the other factors of $\widehat{L}(s,r)$ have no zeroes whatsoever, $\widehat{L}(s,r)$ has no zeroes in $\mathrm{Re}(s) > 1$. Consider the point $s = 1-k$ for some integer $k \geq 1$. At first we take $k > 1$ so that k is in the domain of convergence. Then, from the functional equation we deduce

$$0 = \mathrm{ord}_{s=1-k} \widehat{L}(s,r) = \mathrm{ord}_{s=1-k} L(s,r) + \sum_{v\in V_\infty} \mathrm{ord}_{s=1-k} L_v(s,r).$$

In other words,

$$\mathrm{ord}_{s=1-k} L(s,r) = -\sum_{v\in V_\infty} \mathrm{ord}_{s=1-k} L_v(s,r) = \begin{cases} \frac{n}{2}[K:\mathbb{Q}] + \frac{1}{2}\sum_{v\in V_\mathbb{R}} \chi_r(c_w) & \text{if } k \text{ is odd,} \\ \frac{n}{2}[K:\mathbb{Q}] - \frac{1}{2}\sum_{v\in V_\mathbb{R}} \chi_r(c_w) & \text{if } k \text{ is even.} \end{cases} \tag{4.8}$$

For example, when k is *odd* the calculation goes as follows. Recall that $n = \dim_\mathbb{C} r$. (Also, we let $c_w = e$ if v and $w|v$ are both real.)

$$\begin{aligned} -\sum_{v\in V_\infty} \mathrm{ord}_{s=1-k} L_v(s,r) &= \sum_{v\in V_\mathbb{C}} -n\mathrm{ord}_{s=1-k}\Gamma(s) + \sum_{v\in V_\mathbb{R}} -\frac{1}{2}(n+\chi_r(c_w))\mathrm{ord}_{s=1-k}\Gamma(\frac{s}{2}) \\ &= n|V_\mathbb{C}| + \frac{1}{2}(n|V_\mathbb{R}| + \sum_{v\in V_\mathbb{R}} \chi_r(c_w)) \\ &= \frac{n}{2}[K:\mathbb{Q}] + \frac{1}{2}\sum_{v\in V_\mathbb{R}} \chi_r(c_w). \end{aligned}$$

As a quick sanity check we recover Corollary 1.89 when $r = 1_G$. We leave the details as an exercise for the reader.

Exercise 4.55 Let $r : \mathrm{Gal}(E/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\mathbb{C})$ be an *odd* representation (which means complex conjugation has determinant -1). Show that $L(s,r)$ has simple zeroes at all the negative integers.

When $k = 1$ the same argument goes through for irreducible $r \neq 1_G$ once we verify that $\mathrm{ord}_{s=1} L(s,r) = 0$. That is, $L(s,r)$ has no pole nor a zero at $s = 1$. More generally, for *any* (possibly reducible) representation r the pole order at $s = 1$ is given by the dimension of the G-invariants:

Proposition 4.56 $-ord_{s=1} L(s,r) = \dim_\mathbb{C} r^G$.

Proof The strategy is to first reduce to the case where $\dim_\mathbb{C} r = 1$, and then do the degree one case using the factorization formula 4.11.

So suppose $-\mathrm{ord}_{s=1} L(s,\psi) = \begin{cases} 1 & \psi = 1_G, \\ 0 & \psi \neq 1_G \end{cases}$ for characters $\psi : G \longrightarrow \mathbb{C}^\times$, but with G being random. As in the proof of 4.32 we write $\chi_r = \sum_{H\leq G} a_H \mathrm{Ind}_H^G \chi_H$. Then, by our hypothesis applied to H and χ_H, we find that

$$-\mathrm{ord}_{s=1} L(s,r) = -\sum_{H\leq G} a_H \mathrm{ord}_{s=1} L(s,\chi_H) = \sum\nolimits_{H\leq G:\chi_H = 1_H} a_H.$$

On the other hand,

$$\dim_\mathbb{C} r^G = \langle \chi_r, \chi_{1_G}\rangle = \sum_{H\leq G} a_H \langle \mathrm{Ind}_H^G \chi_H, \chi_{1_G}\rangle = \sum_{H\leq G} a_H \langle \chi_H, 1_H\rangle = \sum\nolimits_{H\leq G:\chi_H = 1_H} a_H.$$

It remains to check off the degree one case; in other words that $L(1,\psi) \neq 0$ for $\psi \neq 1_G$. This follows from 4.11, which exhibits $L(1,\psi)$ as a factor of $\frac{\mathcal{M}_E}{\mathcal{M}_K} > 0$. (Note that ψ factors through an abelian quotient of G, so we may as well assume G

is abelian from the get-go. This ensures all the other L-functions in the product 4.11 are entire functions.) □

The numbers appearing on the right-hand side of (4.8) have a very interesting K-theoretic interpretation. As Gross proved in [Gro05] they coincide with $\dim_{\mathbb{C}}(K_{2k-1}(\mathcal{O}_K) \otimes_{\mathbb{Z}} V)^G$. The special case $r = 1_G$ is a seventies result known as the Quillen-Borel theorem.

4.9 Chebotarev's Density Theorem

We now have a proof of the Chebotarev density theorem at our fingertips. Since we only defined density for sets of prime numbers A, we will assume $K = \mathbb{Q}$ for simplicity. Thus let $E/\mathbb{Q}$ be a finite Galois extension with Galois group G. If p is unramified in E, the Frobenius elements $\mathrm{Frob}_w \in G$ are all conjugate as $w|p$ varies. Thus we may assign a conjugacy class Frob_p in G to any unramified prime p. These classes $\{\mathrm{Frob}_p\}$ are equi-distributed as p runs over such primes:

Theorem 4.57 (Chebotarev) *The set of primes $A = \{p$ unramified in E with $Frob_p = C\}$ has analytic density $\frac{|C|}{|G|}$ for any conjugacy class $C \subseteq G$.*

Proof Let S be the set of prime numbers which ramify in E. Any representation $r : G \longrightarrow \mathrm{GL}(V)$ is then clearly unramified outside S, and we let $L^S(s, r)$ be the S-depleted Artin L-function. It is the exponential of

$$\begin{aligned}\ln L^S(s,r) &= \sum_{p\notin S}\sum_{m=1}^{\infty}\frac{p^{-ms}}{m}\cdot\chi_r(\mathrm{Frob}_w^m)\\ &= \sum_{p\notin S} p^{-s}\chi_r(\mathrm{Frob}_p) + \sum_{p\notin S}\sum_{m=2}^{\infty}\frac{p^{-ms}}{m}\cdot\chi_r(\mathrm{Frob}_w^m)\\ &= \sum_{p\notin S} p^{-s}\chi_r(\mathrm{Frob}_p) + \mathcal{E}^S(s,r)\end{aligned}$$

for $\mathrm{Re}(s) > 1$. The inner sum defining $\mathcal{E}^S(s, r)$ starts from $m = 2$, and therefore the double sum converges absolutely for $\mathrm{Re}(s) > \frac{1}{2}$ by the usual argument (as in the proof of Corollary 1.93, but with minor modifications). In particular $\mathcal{E}^S(s, r)$ stays bounded as $s \to 1$.

Now recall the orthogonality relations for G, [Ser96, Proposition 7, p. 20]. For any two conjugacy classes $C, C' \subset G$ we have

$$\sum_{r\in\mathrm{Irr}_{\mathbb{C}}(G)}\overline{\chi_r}(C)\chi_r(C') = \begin{cases}\frac{|G|}{|C|} & \text{if } C = C'\\ 0 & \text{if } C \neq C'.\end{cases}$$

For a fixed conjugacy class $C \subset G$ we deduce that

$$\sum_{r\in \mathrm{Irr}_{\mathbb{C}}(G)} \overline{\chi_r}(C) \ln L^S(s,r) = \sum_{p\notin S} p^{-s}\Big(\sum_{r\in \mathrm{Irr}_{\mathbb{C}}(G)} \overline{\chi_r}(C)\chi_r(\mathrm{Frob}_p)\Big) + \sum_{r\in \mathrm{Irr}_{\mathbb{C}}(G)} \overline{\chi_r}(C)\mathcal{E}^S(s,r)$$
$$= \frac{|G|}{|C|}\cdot \sum_{p\in A} p^{-s} + \mathcal{E}^{S,C}(s)$$

with $\mathcal{E}^{S,C}(s)$ bounded near $s = 1$. On the left-hand side, the term for $r = 1_G$ is the logarithm of the S-depleted Riemann zeta function $\zeta^S(s)$–which has a simple pole at $s = 1$. In particular $\ln \zeta^S(s) \sim \ln\left(\frac{1}{s-1}\right)$ as $s \to 1^+$. For $r \neq 1_G$, Proposition 4.56 shows $L^S(1,r) \neq 0$ (the depleted Euler factors are all nonzero). Thus $\ln L^S(s,r)$ remains bounded as $s \to 1$ (for more details see the last two paragraphs of the proof of Lemma 1.54). Altogether this shows $\sum_{p\in A} p^{-s} \sim \frac{|C|}{|G|}\cdot \ln\left(\frac{1}{s-1}\right)$ as $s \to 1^+$. □

This immediately gives a different proof of Corollary 1.93 by taking $C = \{e\}$ to be the conjugacy class of the identity in G. The condition $\mathrm{Frob}_p = e$ is equivalent to $p \in \mathrm{Spl}_{E/\mathbb{Q}}$, and we conclude that $A = \mathrm{Spl}_{E/\mathbb{Q}}$ has analytic density $\frac{1}{[E:\mathbb{Q}]}$.

Chapter 5
A Streamlined Introduction to Modular Forms



Eichler is often quoted as saying "There are five elementary operations in arithmetic: addition, subtraction, multiplication, division, and ... modular forms". If he said that, it was probably said tongue-in-cheek, but it is hard to overestimate the importance of modular forms in number theory–and also in physics (string theory, conformal field theory, black holes, quantum field theory, etc.). This chapter is meant to serve as a somewhat fast-paced introduction to the subject, and a reference guide to the many excellent sources already available in the literature, such as [Bum97, DI95, DS05, Gel75, Hid93, Shi94] just to name a few.

A well-behaved periodic function on $\mathbb{R}$ is given by a Fourier series $\sum_{n=-\infty}^{\infty} a_n e^{2\pi i n x}$. A modular form has a Fourier expansion $\sum_{n=0}^{\infty} a_n e^{2\pi i n z}$ where z is now a complex variable in the upper-half plane $\mathcal{H}$. Since the sum now only ranges over non-negative integers n, and $\mathrm{Im}(z) > 0$, it converges at least if a_n has at most polynomial growth as $n \to \infty$. However, a modular form has a plethora of symmetries in addition to periodicity: The group $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathcal{H}$ by linear fractional transformations, also known as Möbius transformations, given by the formula $\gamma z = \frac{az+b}{cz+d}$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The real line only has one non-trivial discrete subgroup up to scaling, namely $\mathbb{Z}$. In comparison $\mathrm{SL}_2(\mathbb{R})$ has many interesting discrete subgroups besides $\mathrm{SL}_2(\mathbb{Z})$. For example, one can impose congruence conditions on the matrix entries and get groups like $\Gamma_0(N)$ where $N|c$. Another standard choice is the even smaller group $\Gamma_1(N)$ consisting of matrices $\gamma \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right) \bmod N$. Take Γ to be your favorite example. A modular form for Γ is a holomorphic function $f : \mathcal{H} \to \mathbb{C}$ of moderate growth which is not quite Γ-invariant, but rather it satisfies a transformation property of the form $f(\gamma z) = (cz + d)^k f(z)$ for some integer $k \geq 1$ called the weight. Even this lenient Γ-invariance property is very restrictive and cuts out a finite-dimensional vector space $M_k(\Gamma)$, which often contains nothing but the zero function. The subspace of *cusp* forms $S_k(\Gamma)$ consists of all modular forms which decay exponentially as $\mathrm{Im}(z) \to \infty$, and more generally as z approaches any of the so-called cusps for Γ–a finite set of points on the real axis (in addition to a point at infinity).

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The group $\mathrm{SL}_2(\mathbb{Z})$ only has the one cusp at infinity, and the space of cusp forms has codimension one. Therefore $M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z}))$ decomposes as a direct sum $S_{2\kappa}(\mathrm{SL}_2(\mathbb{Z})) \oplus \mathbb{C}E_\kappa$ where E_κ is a very special modular form known as an *Eisenstein* series, with Fourier expansion $1 - \frac{4\kappa}{B_{2\kappa}} \sum_{n=1}^{\infty} \sigma_{2\kappa-1}(n) e^{2\pi i n z}$. There are no nonzero modular forms for $\mathrm{SL}_2(\mathbb{Z})$ of *odd* weight, as can be seen by taking $\gamma = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. By combining Eisenstein series of various weights we can generate more examples of modular forms. For example $E_2^i E_3^j$ is a modular form for $\mathrm{SL}_2(\mathbb{Z})$ of weight $4i + 6j$. Since the two weight 12 forms E_2^3 and E_3^2 both have constant term one, their difference $E_2^3 - E_3^2$ must be a *cusp* form. Up to a normalization factor this is the modular discriminant function Δ studied by Ramanujan around 1916. It turns out that $S_{12}(\mathrm{SL}_2(\mathbb{Z}))$ is in fact spanned by Δ. Moreover, not only is Δ not identically zero but actually it has no zeroes whatsoever. We prove this in Sect. 5.6 as an application of the Weierstrass $\wp$-function.

Another source of modular forms is quadratic forms. If V is a finite-dimensional vector space over $\mathbb{R}$, endowed with an inner product, certain self-dual lattices $L \subset V$ give rise to modular forms θ_L for $\mathrm{SL}_2(\mathbb{Z})$. The nth Fourier coefficient $r_L(n)$ is the number of solutions $x \in L$ to the equation $\langle x, x \rangle = 2n$. If we choose a basis for V this amounts to counting the number of ways one can represent $2n$ by the quadratic form defined by L. This is a classical problem in number theory. For instance, Lagrange's Four Squares Theorem (1770) is the epic result that any positive integer can be expressed as the sum of four squares. By comparing θ_L to E_κ one can deduce explicit formulas such as $r_L(n) = 240\sigma_3(n)$ when V is 8-dimensional, and also $r_L(n) = 480\sigma_7(n)$ when V is 16-dimensional (due to the fact that $\mathrm{SL}_2(\mathbb{Z})$ has no nonzero cusp forms of weight 4 and 8 respectively).

We are mainly interested in the space $S_k(\Gamma_1(N))$, or more precisely forms f in here with a *nebentype*, which means there is a Dirichlet character χ modulo N such that $f(\gamma z) = \chi(d)(cz + d)^k f(z)$ for all matrices $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$. This defines a direct summand $S_k(\Gamma_0(N), \chi)$ which is trivial unless χ and k have the same parity.

The L-series of a cusp form f as above is defined to be the Dirichlet series $L(s, f) = \sum_{n=1}^{\infty} a_n n^{-s}$ encoding the Fourier coefficients a_n of f. This series converges for $\mathrm{Re}(s)$ sufficiently large relative to the weight. Using the by-now-standard plot, via Mellin transforms, it is fairly straightforward to show that $L(s, f)$ has analytic continuation and a functional equation relating it to $L(k - s, f^W)$ for some other cusp form f^W with nebentype $\overline{\chi}$. This is done in Sect. 5.8 where we also discuss the *twisted* L-series $L(s, f, \psi)$ with respect to a second Dirichlet character ψ.

To really qualify as an L-function we require $L(s, f)$ to have an Euler product, which is not the case for all f. We want a degree two Euler product of the form $\prod_p (1 - a_p p^{-s} + \chi(p) p^{k-1-2s})^{-1}$ which for $k = 2$ resembles the definition of the L-function of an elliptic curve over $\mathbb{Q}$. This forces recursive relations among the Fourier coefficients. For example we must have $a_{mn} = a_m a_n$ when m, n are coprime. Furthermore, assuming $a_1 = 1$, we should have $a_{p^r} = a_p a_{p^{r-1}} - \chi(p) p^{k-1} a_{p^{r-2}}$ for primes p and $r \geq 2$. Ramanujan noted, without proof, that the Fourier coefficients $\tau(n)$ of the Δ-function (suitably normalized) satisfy all these relations with $k = 12$, and Mordell supplied a proof shortly after. The reason behind this marvel is that

$S_{12}(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional, and hence Δ is automatically an eigenvector for all the so-called *Hecke* operators $(T_n)_{n\geq 1}$. In general, on each of the spaces $S_k(\Gamma_0(N), \chi)$ there is a sequence of operators $(T_n)_{n\geq 1}$ satisfying the analogous recursion relations. Moreover there is a Petersson inner product, with respect to which T_n is 'usually' normal (i.e. it commutes with its adjoint). From linear algebra we know this family of commuting diagonalizable operators can be *simultaneously* diagonalized. In other words $S_k(\Gamma_0(N), \chi)$ has a basis of common eigenvectors for all the normal T_n. When f is an eigenvector for *all* the T_n we call it a Hecke *eigenform*, and its eigenvalues turn out to essentially be the same as the Fourier coefficients. In particular $L(s, f)$ has an Euler product for such f.

If truth be told, to clean up some of the imprecision in the previous paragraph we should require f to be a *newform*. This notion is analogous to the primitivity of a Dirichlet character, in that f should not come from a modular form of a smaller level $M|N$ in an obvious way. For example $S_k(\Gamma_1(M))$ is clearly a subspace of $S_k(\Gamma_1(N))$, but there are also other embeddings $i_u : S_k(\Gamma_1(M)) \hookrightarrow S_k(\Gamma_1(N))$ by basically changing the variable $z \mapsto uz$ for each divisor $u|\frac{N}{M}$. Forms arising in this fashion are called oldforms, and by definition the space of newforms is the Petersson orthogonal complement. Recall that a Dirichlet L-function $L(s, \chi)$ only has a nice functional equation, and no 'missing' Euler factors, when χ is primitive. So it should not come as a surprise that a primitivity assumption on f is needed for $L(s, f)$ to enjoy analogous properties.

A key property of a newform is that the first Fourier coefficient a_1 is always nonzero, so by scaling we may assume $a_1 = 1$. Under this assumption the Hecke eigenvalues coincide with the Fourier coefficients on the nose. This gives a finite set of exemplary eigenforms $\mathrm{Prim}_k(\Gamma_0(N), \chi)$ analogous to the set of primitive Dirichlet characters modulo N. Furthermore, by sending f to its complex conjugate $f^\vee$ (with Fourier coefficients $\overline{a_n}$) we get a one-to-one correspondence with $\mathrm{Prim}_k(\Gamma_0(N), \overline{\chi})$. Note that the modular form f^W mentioned in the sixth paragraph is *not* necessarily normalized. However, the first Fourier coefficient $a_1(f^W)$ at least has absolute value one, and $f^W = a_1(f^W) f^\vee$ essentially by the strong multiplicity one theorem (a primitive form is determined by its T_p-eigenvalues for $p \nmid N$, see Proposition 5.93). All this culminates in Corollary 5.96 which shows that $N^{\frac{s}{2}} L_{\mathbb{C}}(s) L(s, f)$ extends to an entire function, and relates it to the counterpart with $f \rightsquigarrow f^\vee$ and $s \rightsquigarrow k - s$. The root number $W(f)$ appearing there is no more than the inverse of $i^k a_1(f^W)$.

In Sect. 5.9 we discuss the Rankin-Selberg method, but mainly for $\mathrm{SL}_2(\mathbb{Z})$. This will play a pivotal role in Chap. 6 where it is being used, perhaps surprisingly, to show the irreducibility of Galois representations attached to modular forms (see Proposition 6.9 for $k = 1$ and 6.35 for $k \geq 2$). The method also takes a central role in the proof of the Deligne-Serre theorem (see Lemma 6.52). What is being used in all these applications is Proposition 5.110, or rather the $\Gamma_1(N)$-version Proposition 6.8, which asserts that $\sum_{p\nmid N} |a_p|^2 p^{-s}$ is bounded above by $\ln(\frac{1}{s-k}) + O(1)$ as $s \to k^+$. (In fact we have equality here if we admit Deligne's bound $|a_p| \leq 2p^{\frac{k-1}{2}}$, but the relatively elementary fact that we have an upper bound suffices for our purposes.)

Rankin and Selberg independently discovered the method in [Ran39, Sel40], and it has been developed massively beyond GL(2) since then. In the simplest case the input is two weight k cusp forms f, g for $\mathrm{SL}_2(\mathbb{Z})$, to fix ideas, with Fourier coefficients $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ respectively. The goal is to understand the Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n b_n}{n^s}$. The heart of the method is to represent this series (up to well-understood factors and the shift $s \rightsquigarrow s+k-1$) as an integral of the form $\int_{\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}} f(z)\overline{g(z)}E(z,s)y^k \frac{dxdy}{y^2}$ for a certain *non*-holomorphic Eisenstein series $E(z,s)$. The latter is an eigenfunction for the Laplacian $-y^2(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2})$, it has analytic continuation to $\mathbb{C}$, and it is invariant under $s \mapsto 1-s$. The last two properties therefore trickle down to analogous properties of the Rankin-Selberg L-function $L(s, f\times g)$. A distinctive feature is that it has a simple pole at $s=k$ with residue $\frac{1}{2}\|f\|^2$ when $f=g$, and this is the secret to bounding $\sum_p |a_p|^2 p^{-s}$ near $s=k$.

In all honesty the Rankin-Selberg L-function $L(s, f\times g)$ is not exactly given by the series $\sum_{n=1}^{\infty} \frac{a_n b_n}{n^s}$, but this sum times $\zeta(2s-2k+2)$. The reason for this correction factor is Theorem 5.107 which expresses the L-function as a degree four Euler product, provided f, g are both primitive forms so that $L(s, f)$ and $L(s, g)$ have Euler products.

In Sect. 5.7 we expound on the automorphic interpretation of modular forms, for at least two reasons: The practical reason is this point of view gives a natural proof of the adjunction property of the Petersson inner product (see Proposition 5.59). The broader reason is that automorphic forms and automorphic representations give the right framework when one develops these ideas further for other groups such as $\mathrm{GL}(n)$, symplectic groups, orthogonal groups, unitary groups, and exceptional groups. In fact we will use this language tangentially in our sketch of the proof of the Langlands-Tunnell theorem in Sect. 6.2.3.

For the reader who has never heard of automorphic forms, here is the basic idea. A modular form $f \in S_k(\Gamma)$ defines a function ϕ_f on $\mathrm{SL}_2(\mathbb{R})$ morally by mapping $g \mapsto f(gi)$. This identifies $S_k(\Gamma)$ with a subspace of nice functions in the Hilbert space $L^2(\Gamma\backslash \mathrm{SL}_2(\mathbb{R}))$. Now, for groups Γ of arithmetic interest, there is a way to identify the quotient $\Gamma\backslash \mathrm{SL}_2(\mathbb{R})$ with the adelic double coset space $Z_\infty^+ \mathrm{GL}_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{A})/U$ where $U \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is a compact open subgroup corresponding to Γ, and Z_∞^+ is the neutral component of the center of $\mathrm{GL}_2(\mathbb{R})$. This boils down to the strong approximation theorem about $\mathrm{SL}_2(\mathbb{Q})$ being dense in $\mathrm{SL}_2(\mathbb{A}^S)$ for every non-empty finite set of places S, which we prove in Sect. 5.7.4. The upshot is we can now think of $S_k(\Gamma)$ as a subspace of $L^2(Z_\infty^+ \mathrm{GL}_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{A}))$. This L^2-space carries a natural action of $\mathrm{GL}_2(\mathbb{A})$ by right translations, and the automorphic representations are basically the irreducible subquotients hereof. This gives a robust foundation for Hecke operators, as Γ and U vary. For instance it is completely obvious that the $\mathrm{GL}_2(\mathbb{Q}_p)$-action commutes with the $\mathrm{GL}_2(\mathbb{Q}_q)$-action for primes $p \neq q$. The definition of a newform also has a simple interpretation in terms of a more general notion of conductors. At the end of the day, any $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ corresponds

to an automorphic representation π of $\mathrm{GL}_2(\mathbb{A})$ which factors as a 'restricted' tensor product $\pi = \otimes_{p\leq\infty}\pi_p$ of irreducible representations π_p of the Lie group $\mathrm{GL}_2(\mathbb{Q}_p)$. This is in perfect accordance with what we saw for Hecke characters.

5.1 Linear Fractional Transformations

Our starting point is the complex upper half-plane $\mathcal{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$ with the usual Euclidean topology. We identify $\mathcal{H}$ with an open subset of $\mathbb{R}^2$ by writing $z = x + iy$ with $y > 0$.

It is sometimes useful to consider the partition of $\mathbb{C} - \mathbb{R}$ into the upper half-plane and the lower half-plane. In that case we will denote the former by $\mathcal{H}^+$ (instead of just $\mathcal{H}$) and the latter by $\mathcal{H}^-$.

For $z \in \mathbb{C} - \mathbb{R}$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{R})$ we let

$$\gamma z = \frac{az + b}{cz + d}.$$

These maps $z \mapsto \gamma z$ are called linear fractional transformations (or Möbius transformations).

Example 5.1 The matrix $\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$ defines the horizontal translation map $z \mapsto z + b$. The matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ gives the map $z \mapsto -\frac{1}{z}$. Note that for $|z| = 1$ the second map is $x + iy \mapsto -x + iy$ (reflection across the imaginary axis).

Exercise 5.2 Adopting the above notation, verify the following basic facts:

(a) $\mathrm{Im}(\gamma z) = \frac{\det(\gamma)}{|cz+d|^2}\mathrm{Im}(z)$;

(b) The group $\mathrm{GL}_2(\mathbb{R})^+ = \{\gamma : \det(\gamma) > 0\}$ acts transitively on both $\mathcal{H}^+$ and $\mathcal{H}^-$ via linear fractional transformations. (`Hint`: For transitivity let $\left(\begin{smallmatrix} y & x \\ 0 & 1 \end{smallmatrix}\right)$ act on $\pm i$.);

(c) The stabilizer of i in $\mathrm{GL}_2(\mathbb{R})^+$ equals $Z_\infty^+\mathrm{SO}_2(\mathbb{R})$. (Here $Z_\infty^+ = \{\left(\begin{smallmatrix} a & 0 \\ 0 & a \end{smallmatrix}\right) : a > 0\}$ is the identity component of the center and $\mathrm{SO}_2(\mathbb{R})$ denotes the group of orthogonal 2×2 matrices with determinant one, i.e. the rotation matrices.);

(d) The map $\gamma \mapsto \gamma i$ gives a bijection

$$\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R}) \simeq \mathrm{GL}_2(\mathbb{R})^+/Z_\infty^+\mathrm{SO}_2(\mathbb{R}) \xrightarrow{\sim} \mathcal{H}^+.$$

(e) The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ induces the negation map $z \mapsto -z$ which identifies $\mathcal{H}^+$ with $\mathcal{H}^-$.

It is often convenient to think of $\mathbb{C}$ as sitting inside the projective line $\mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$. Here we identify $z \in \mathbb{C}$ with the line $\mathbb{C}\left(\begin{smallmatrix} z \\ 1 \end{smallmatrix}\right)$ spanned by the vector $\left(\begin{smallmatrix} z \\ 1 \end{smallmatrix}\right)$ in $\mathbb{C}^2$. The point at infinity is the "missing" line $\infty = \mathbb{C}\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$. The group $\mathrm{GL}_2(\mathbb{C})$ acts transitively on $\mathbb{P}^1(\mathbb{C})$ by permuting the lines in $\mathbb{C}^2$:

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot \mathbb{C}\left(\begin{smallmatrix} z \\ z' \end{smallmatrix}\right) = \mathbb{C}\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\left(\begin{smallmatrix} z \\ z' \end{smallmatrix}\right) = \mathbb{C}\left(\begin{smallmatrix} az+bz' \\ cz+dz' \end{smallmatrix}\right).$$

For instance $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot \infty = \mathbb{C}\left(\begin{smallmatrix} a \\ c \end{smallmatrix}\right)$, from which it follows that the stabilizer of ∞ in $\mathrm{GL}_2(\mathbb{C})$ is the subgroup of all upper triangular matrices.

Instead of the full $\mathrm{GL}_2(\mathbb{C})$ we are more interested in the orbits in $\mathbb{P}^1(\mathbb{C})$ under the action of $\mathrm{GL}_2(\mathbb{R})^+$. Note that $\gamma \in \mathrm{GL}_2(\mathbb{R})$ takes the line $\mathbb{C}\left(\begin{smallmatrix} z \\ 1 \end{smallmatrix}\right)$ given by a $z \in \mathbb{C} - \mathbb{R}$ to the line $\mathbb{C}\left(\begin{smallmatrix} \gamma z \\ 1 \end{smallmatrix}\right)$. With this in mind, a moments thought now shows that

$$\mathbb{P}^1(\mathbb{C}) = \mathbb{P}^1(\mathbb{R}) \cup \mathcal{H}^+ \cup \mathcal{H}^-$$

is the partition of $\mathbb{P}^1(\mathbb{C})$ into $\mathrm{GL}_2(\mathbb{R})^+$-orbits. This point of view becomes particularly useful when we introduce cusps, which are certain orbits in $\mathbb{P}^1(\mathbb{Q})$.

Exercise 5.3 Show that the Cayley transform $z \mapsto \frac{z-i}{z+i}$ identifies $\mathcal{H}$ with the complex open unit disk centered at the origin. (`Hint`: Note that $\mathcal{H}$ can be thought of as the set of points $z \in \mathbb{C}$ which are closer to i than $-i$, in other words such that $|z - i| < |z + i|$.)

5.2 Discrete Subgroups

A subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{R})$ is *discrete* if the induced topology on Γ is discrete, i.e. all subsets are open. Equivalently all singletons $\{\gamma\}$ are open. Since $\mathrm{SL}_2(\mathbb{R})$ is a topological group it suffices to check this for the identity matrix $\gamma = I$. Indeed, if I has an open neighborhood $U \subset \mathrm{SL}_2(\mathbb{R})$ such that $\Gamma \cap U = \{I\}$ then $\Gamma \cap \gamma U = \{\gamma\}$ for all $\gamma \in \Gamma$.

Lemma 5.4 *Discrete subgroups are closed.*

Proof Let Γ and U be as above. Since the map $f : (x, y) \mapsto x^{-1}y$ is continuous there is an open subset $V \ni I$ of $\mathrm{SL}_2(\mathbb{R})$ such that $f(V \times V) \subset U$. Now suppose $(\gamma_n)_{n\geq 1}$ is a sequence in Γ converging to some $g \in \mathrm{SL}_2(\mathbb{R})$. Then $\gamma_n \in gV$ for $n \geq n_0$ say. Consequently $\gamma_m^{-1}\gamma_n \in f(V \times V) \subset U$ for $m, n \geq n_0$. Thus $\gamma_m^{-1}\gamma_n = I$ for large enough m and n, i.e. the sequence is eventually constant. In particular $g = \gamma_n$ for $n \gg 0$, and therefore the limit g lies in Γ. □

Two discrete subgroups Γ and Γ' are said to be *commensurable* (and we write $\Gamma \sim \Gamma'$) if $\Gamma \cap \Gamma'$ has finite index in both Γ and Γ'. This is an equivalence relation. (To see transitivity, assuming $\Gamma \sim \Gamma'$ and $\Gamma' \sim \Gamma''$ check that in fact $\Gamma \cap \Gamma' \cap \Gamma''$ has finite index in Γ, Γ', and Γ''.)

Exercise 5.5 Define the commensurator of Γ to be $\mathrm{Comm}(\Gamma) = \{g \in \mathrm{SL}_2(\mathbb{R}) : \Gamma \sim g\Gamma g^{-1}\}$. Verify this is a subgroup (containing the normalizer of Γ). Show that the commensurator of the modular group $\mathrm{SL}_2(\mathbb{Z})$ contains[1] $\mathrm{SL}_2(\mathbb{Q})$.

[1] The SL_n-version of this is [Shi94, Lemma 3.10, p. 55].

Number theorists are mostly concerned with *arithmetic* groups. By definition a subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{Q})$ is arithmetic if $\Gamma \sim \mathrm{SL}_2(\mathbb{Z})$. For instance, finite index subgroups $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ are arithmetic. A natural way to produce examples is to start with a natural number N along with a subgroup $H \subset \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ and take Γ to be the inverse image $\Gamma_H = \mathrm{pr}_N^{-1}(H)$ under the projection homomorphism

$$\mathrm{pr}_N : \mathrm{SL}_2(\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

which reduces all matrix entries modulo N. It turns out, essentially by the Chinese remainder theorem, that pr_N is surjective. We invite the reader to try this as an exercise, or look up [Shi94, Lemma 1.38, p. 20] where the SL_n-analogue is proved. As a result pr_N gives a bijection (an isomorphism of groups if H is normal):

$$\mathrm{SL}_2(\mathbb{Z})/\Gamma_H \xrightarrow{\sim} \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})/H.$$

This gives a way to calculate the index of Γ_H for interesting choices of H, see [Shi94, Sect. 1.6] for sample computations. Here are some standard choices, in increasing order, for a fixed N:

- $H = \{I\}$ gives the full level N principal congruence subgroup

$$\begin{aligned}\Gamma(N) &= \ker\left(\mathrm{SL}_2(\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})\right)\\ &= \{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}) : \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \equiv \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) \bmod N\}.\end{aligned}$$

- $H = \{\left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)\}$ gives

$$\Gamma_1(N) = \{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}) : \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right) \bmod N\}.$$

- $H = \{\left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)\}$ gives

$$\Gamma_0(N) = \{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}) : \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod N\}.$$

- $H = \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ gives $\mathrm{SL}_2(\mathbb{Z})$.

Note that Γ_H contains $\Gamma(N)$ for any choice of subgroup $H \subset \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. As H varies these Γ_H comprise the *congruence* subgroups, i.e. subgroups $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ containing $\Gamma(N)$ for some N.

We alert the reader that it is *not* true that every finite index subgroup of $\mathrm{SL}_2(\mathbb{Z})$ must be a congruence subgroup. One can exhibit counterexamples in $\Gamma(2)$ by using the fact that there is a surjection $\Gamma(2) \twoheadrightarrow \mathbb{Z}$ and consider $\ker(\Gamma(2) \twoheadrightarrow \mathbb{Z}/m\mathbb{Z})$ for large m. By contrast, amazingly the congruence subgroup problem has a positive answer for $\mathrm{SL}_n(\mathbb{Z})$ when $n \geq 3$. This was shown by BassâŁ“Mennicke-MilnorâŁ“Serre using K-theoretic methods.

Exercise 5.6 Check that $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mapsto d \bmod N$ defines a surjective homomorphism $\Gamma_0(N) \twoheadrightarrow (\mathbb{Z}/N\mathbb{Z})^\times$ with kernel $\Gamma_1(N)$. Conclude with an isomorphism $\Gamma_0(N)/\Gamma_1(N) \xrightarrow{\sim} (\mathbb{Z}/N\mathbb{Z})^\times$.

5.3 Complex Modular Curves

We now restrict the $\mathrm{SL}_2(\mathbb{R})$-action on $\mathcal{H}$ to a finite index subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ and we wish to endow the space of orbits $\Gamma\backslash\mathcal{H}$ with a geometric structure. It carries the quotient topology, i.e. a subset of $\Gamma\backslash\mathcal{H}$ is open when its inverse image in $\mathcal{H}$ is open, but we would like to promote $\Gamma\backslash\mathcal{H}$ to a one-dimensional complex manifold–a so-called Riemann surface. This is complicated by the fact that in general $\Gamma/\Gamma \cap \{\pm I\}$ does not act freely on $\mathcal{H}$–the isotropy groups $\mathrm{Stab}_\Gamma(z)$ may be bigger than $\Gamma \cap \{\pm I\}$ for certain special points $z \in \mathcal{H}$. (Note that $-I$ acts trivially on all of $\mathcal{H}$.)

Lemma 5.7 *The isotropy group $\mathrm{Stab}_\Gamma(z)$ is a finite cyclic group for all $z \in \mathcal{H}$.*

Proof All the isotropy groups are $\mathrm{SL}_2(\mathbb{R})$-conjugate so we may as well take $z = i$. In this case we are looking at $\mathrm{Stab}_\Gamma(i) = \Gamma \cap \mathrm{SO}_2(\mathbb{R})$. This is a finite group since Γ is discrete and $\mathrm{SO}_2(\mathbb{R})$ is compact. The homomorphism $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right) \mapsto e^{i\theta}$ identifies $\mathrm{SO}_2(\mathbb{R})$ with the circle group $\mathbb{C}^1$. Since a finite subgroup of the multiplicative group of a field (in this case $\mathbb{C}$) is cyclic we are done. □

Here are examples of what can go wrong when Γ is too large.

Example 5.8 $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{Stab}_{\mathrm{SL}_2(\mathbb{Z})}(i)$ and $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right) \in \mathrm{Stab}_{\mathrm{SL}_2(\mathbb{Z})}(\omega)$ where $\omega = e^{2\pi i/3} = \frac{1}{2}(-1 + i\sqrt{3})$. The matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has order two in $\mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$ whereas $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ has order three. The points i and ω are examples of *elliptic* points for $\mathrm{SL}_2(\mathbb{Z})$, i.e. points with non-trivial stabilizers modulo $\{\pm I\}$.

For elliptic points z it is more difficult to define a chart of $\Gamma\backslash\mathcal{H}$ around Γz. For the most part we will ignore this issue here by assuming Γ is sufficiently small. We refer the interested reader to [DS05, Sect. 2.2] for a more comprehensive account.

We next establish the key property of the action $\Gamma \times \mathcal{H} \longrightarrow \mathcal{H}$.

Lemma 5.9 *Every point $z \in \mathcal{H}$ admits an open neighborhood $W = W_z$ such that for all $\gamma \in \Gamma$ we have*

$$W \cap \gamma W \neq \varnothing \Longrightarrow \gamma z = z$$

(and of course conversely). If Γ has no elliptic points in $\mathcal{H}$ the group $\Gamma/\Gamma \cap \{\pm I\}$ acts properly discontinuously on $\mathcal{H}$.

Proof (Based on the proof of [Shi94, Proposition 1.7].) We first show that for any two compact subsets $A, B \subset \mathcal{H}$ there are only finitely many $\gamma \in \Gamma$ for which $A \cap \gamma B \neq \varnothing$. Consider the map $h : \mathrm{SL}_2(\mathbb{R}) \twoheadrightarrow \mathcal{H}$ defined by $h(g) = gi$. If we identify $\mathcal{H}$ with $\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R})$ this is nothing but the projection map $\mathrm{SL}_2(\mathbb{R}) \twoheadrightarrow \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R})$.

In particular h is open (it takes open sets to open sets) and proper (the inverse image of a compact subset is compact). To see h is indeed proper let us check that $h^{-1}(A)$ is compact. As $\mathrm{SL}_2(\mathbb{R})$ is locally compact we may cover it by open subsets $(\Omega_i)_{i\in I}$ with compact closures $\overline{\Omega_i}$. Then $(h(\Omega_i))_{i\in I}$ is an open covering of $\mathcal{H}$. Choose a finite subset $J \subset I$ such that $(h(\Omega_j))_{j\in J}$ is a covering of A. Then $h^{-1}(A)$ is a closed subset of the compact set $\bigcup_{j\in J} \overline{\Omega_i}\mathrm{SO}_2(\mathbb{R})$ and therefore $h^{-1}(A)$ itself is compact. Of course $h^{-1}(B)$ is compact as well for the same reason.

For simplicity we introduce the notation $\mathbb{A} := h^{-1}(A)$ and $\mathbb{B} := h^{-1}(B)$. Now suppose $A \cap \gamma B \neq \varnothing$ for some $\gamma \in \Gamma$. This implies $\mathbb{A} \cap \gamma\mathbb{B} \neq \varnothing$ by the surjectivity of h, and therefore γ must lie in the compact set $\mathbb{A}\mathbb{B}^{-1}$. Since Γ is discrete there can only be finitely many such γ, which proves our first claim.

Now let $V \ni z$ denote an arbitrary open neighborhood of z with compact closure $\overline{V}$. Applying our previous observation to $A = B = \overline{V}$ we infer that $\{\gamma \in \Gamma : V \cap \gamma V \neq \varnothing\}$ is a finite set (where we have intentionally omitted the bar). We list its elements as $\{\gamma_1, \ldots, \gamma_r\}$ where the first s members $\{\gamma_1, \ldots, \gamma_s\}$ all fix z, and the remaining elements $\{\gamma_{s+1}, \ldots, \gamma_r\}$ do not fix z. For each $i > s$ we may thus pick open neighborhoods $V_i \ni z$ and $W_i \ni \gamma_i z$ such that $V_i \cap W_i = \varnothing$ (by the Hausdorff property). We claim

$$W := V \cap \bigcap_{i>s} (V_i \cap \gamma_i^{-1} W_i)$$

works. Clearly W is an open neighborhood of z. Suppose $\gamma \in \Gamma$ satisfies $W \cap \gamma W \neq \varnothing$. Since $W \subset V$ this γ must be on the list $\{\gamma_1, \ldots, \gamma_r\}$, say $\gamma = \gamma_j$. Our task is to show $j \leq s$, so that $\gamma_j z = z$. For the sake of contradiction, assume $j > s$. Select a $w \in W \cap \gamma_j W$. Note that $W \subset V_j$ and $W \subset \gamma_j^{-1} W_j$. Consequently $w \in V_j \cap W_j$, which contradicts the disjointness of V_j and W_j. We conclude that $\gamma z = z$ as desired.

As a result, if z is a non-elliptic point, i.e. $\mathrm{Stab}_\Gamma(z) \subset \{\pm I\}$, such a γ must be $\pm I$. This says exactly that the action of $\Gamma/\Gamma \cap \{\pm I\}$ on $\mathcal{H}$ is properly discontinuous if all points z are non-elliptic. □

Exercise 5.10 Adapt some of the ideas from the proof of Lemma 5.9 to show that $\Gamma\backslash\mathcal{H}$ is Hausdorff. This amounts to the following: For any two Γ-inequivalent points $z, z' \in \mathcal{H}$ find open neighborhoods $W \ni z$ and $W' \ni z'$ such that $W \cap \gamma W' = \varnothing$ for all $\gamma \in \Gamma$.

The importance of the proper discontinuity of the action (in the absence of elliptic points) is that the natural projection $\mathrm{pr}_\Gamma : \mathcal{H} \twoheadrightarrow \Gamma\backslash\mathcal{H}$ is a covering map. This means pr_Γ is surjective, open, and most importantly a local homeomorphism. Thus every $z \in \mathcal{H}$ has an open neighborhood W such that pr_Γ restricts to a homeomorphism $W \xrightarrow{\sim} \mathrm{pr}_\Gamma(W)$. (The injectivity follows immediately from Lemma 5.9, and since pr_Γ is open its restriction to W is a homeomorphism.)

Since $\mathcal{H}$ is simply connected, $\mathrm{pr}_\Gamma : \mathcal{H} \twoheadrightarrow \Gamma\backslash\mathcal{H}$ is in fact the universal cover of $\Gamma\backslash\mathcal{H}$. The group of deck transformations can be shown to be isomorphic to $\Gamma/\Gamma \cap \{\pm I\}$.

As W varies the charts $(\mathrm{pr}_\Gamma(W), \mathrm{pr}_\Gamma|_W^{-1})$ constitute an atlas for $\Gamma\backslash\mathcal{H}$, which thus acquires the structure of a Riemann surface. We will often denote it by $Y(\Gamma)$ when we wish to emphasize the complex structure.

The manifold $Y(\Gamma)$ is usually non-compact for the groups Γ we have in mind. For example, if Γ has finite index in $\mathrm{SL}_2(\mathbb{Z})$, a fundamental domain for Γ is a finite union of $\mathrm{SL}_2(\mathbb{Z})$-translates of a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$. The latter can be taken to be

$$\mathcal{F} = \{z \in \mathcal{H} : |z| \geq 1,\ |\mathrm{Re}(z)| \leq \frac{1}{2}\}.$$

See [Ser73, Theorem 1, p. 78] for a proof thereof (and a picture–or draw your own). Note how the elliptic points i and ω sit on the boundary. Obviously $\mathcal{F}$ is not compact since $\mathrm{Im}(z)$ becomes arbitrarily large for $z \in \mathcal{F}$. One can formally add a point ∞ and compactify $\mathcal{F}$. For a general Γ, a fundamental domain may not only go off to ∞ but also get arbitrarily close to a finite set of rational numbers (on the real axis separating $\mathcal{H}^+$ and $\mathcal{H}^-$). These finitely many missing limit points are the *cusps* of Γ, and we should formally add all of them to compactify $Y(\Gamma)$.

Here is the precise definition of cusps.

Definition 5.11 The set of cusps of an arithmetic group Γ is the set of Γ-orbits in $\mathbb{P}^1(\mathbb{Q})$. In other words, $\mathrm{Cusp}_\Gamma := \Gamma\backslash\mathbb{P}^1(\mathbb{Q})$. We let ∞ denote the Γ-orbit of the line $\mathbb{Q}\binom{1}{0}$.

Lemma 5.12 *Cusp_Γ is finite.*

Proof We first note that $\mathrm{Cusp}_{\mathrm{SL}_2(\mathbb{Z})} = \{\infty\}$; in other words $\mathrm{SL}_2(\mathbb{Z})$ only has the one cusp at infinity. To verify this, note that every element of $\mathbb{P}^1(\mathbb{Q})$ can be represented as $\mathbb{Q}\binom{a}{c}$ where $a, c \in \mathbb{Z}$ are coprime (we are allowing one of them to be zero). Find $b, d \in \mathbb{Z}$ such that $ad - bc = 1$. Then $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ belongs to $\mathrm{SL}_2(\mathbb{Z})$ and takes ∞ to $\mathbb{Q}\binom{a}{c}$. This shows that $\mathrm{SL}_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$.

If Γ has finite index in $\mathrm{SL}_2(\mathbb{Z})$, with right coset representatives $\{g_1, \ldots, g_N\}$, then $\mathbb{Q}\binom{a}{c}$ is in the Γ-orbit of $g_i\infty$ for some $i = 1, \ldots, N$.

The case of a general arithmetic group follows from this by observing that $\mathrm{Cusp}_{\Gamma'}$ maps onto Cusp_Γ for a subgroup $\Gamma' \subset \Gamma$. So, if $\mathrm{Cusp}_{\Gamma'}$ is finite then so is Cusp_Γ. Apply this to $\Gamma' = \Gamma \cap \mathrm{SL}_2(\mathbb{Z})$. □

As a first step towards compactifying $Y(\Gamma)$ we consider the upper half-plane adorned with the extra point ∞ and all the rational numbers,

$$\mathcal{H}^* = \mathcal{H} \cup \mathbb{P}^1(\mathbb{Q}).$$

Here we view $\mathcal{H}$ and $\mathbb{P}^1(\mathbb{Q})$ as subsets of $\mathbb{P}^1(\mathbb{C})$ and take their union there. The group $\mathrm{SL}_2(\mathbb{Q})$ acts on $\mathcal{H}^*$ and $\mathbb{P}^1(\mathbb{Q})$ is one of the orbits.

The first order of business is to put a topology on $\mathcal{H}^*$. We definitely want $\mathcal{H} \hookrightarrow \mathcal{H}^*$ to be a topological embedding, i.e. a subset $W \subset \mathcal{H}$ should be open in $\mathcal{H}^*$ precisely when it is open in the Euclidean topology on $\mathcal{H}$. We need to specify the open neighborhoods of points in $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{\infty\}$. We surely want

$$W_{\infty,T} = \{z : \mathrm{Im}(z) > T\} \cup \{\infty\}$$

to be an open neighborhood of ∞ for all $T > 0$. Now consider a $\rho \in \mathbb{Q}$, viewed as an element of $\mathbb{P}^1(\mathbb{Q})$. Pick a $g \in \mathrm{SL}_2(\mathbb{Z})$–not just in $\mathrm{SL}_2(\mathbb{Q})$–such that $\rho = g\infty$. Then let

$$W_{\rho,T} = gW_{\infty,T} = \{gz : \mathrm{Im}(z) > T\} \cup \{\rho\}.$$

This set is independent of the choice of g. For suppose $\rho = g\infty = g'\infty$. Then $g^{-1}g'$ fixes ∞ and is therefore of the form $\pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$ since g and g' have entries in $\mathbb{Z}$. As $\pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$ act on $\mathcal{H}$ by $z \mapsto z + b$ and $\mathrm{Im}(z + b) = \mathrm{Im}(z)$ we conclude that $gW_{\infty,T} = g'W_{\infty,T}$.

The sets $W_{\rho,T}$ form a neighborhood basis for ρ in $\mathcal{H}^*$ as $T > 0$ varies. This determines the topology on $\mathcal{H}^*$. A subset $W \subset \mathcal{H}^*$ is open if for every $z \in W$ there is an open neighborhood $W_z \subset W$. (We refer to [DS05, Fig. 2.5, p. 59] for an illustration of the neighborhoods $W_{\rho,T}$.)

We stress that $\mathcal{H}^*$ is *not* compact. In fact $\mathcal{H}^*$ is not even locally compact since $W_{\infty,T}$ does not have compact closure. One way to see this is to consider the horizontal sequence $z_n = n + 2Ti$, with $n = 1, 2, 3, \ldots$ say, which does not have a convergent subsequence.

For an arithmetic group Γ as above we can now introduce the crucial quotient space

$$X(\Gamma) = \Gamma\backslash\mathcal{H}^* = Y(\Gamma) \cup \mathrm{Cusp}_\Gamma.$$

The proof that $X(\Gamma)$ is Hausdorff is more involved than one might think. We will not include the proof here, but we direct the reader to [Shi94, Theorem 1.28, p. 12] where this is shown in detail. The compactness is easier:

Proposition 5.13 *The space $X(\Gamma)$ is a compactification of $Y(\Gamma)$. More precisely, $X(\Gamma)$ is compact, and the inclusion $Y(\Gamma) \hookrightarrow X(\Gamma)$ is a topological embedding identifying $Y(\Gamma)$ with a dense open subset of $X(\Gamma)$.*

Proof Points are closed in $X(\Gamma)$, by the Hausdorff property for instance, so $Y(\Gamma) = X(\Gamma) - \mathrm{Cusp}_\Gamma$ is open. We extend pr_Γ to a map $\mathrm{pr}_\Gamma : \mathcal{H}^* \twoheadrightarrow \Gamma\backslash\mathcal{H}^*$. Any open subset of $X(\Gamma)$ is of the form $\mathrm{pr}_\Gamma(W)$ for an open set $W \subset \mathcal{H}^*$. Since the neighborhoods $W_{\rho,T}$ all contain points from $\mathcal{H}$, so does W (assuming it is non-empty). Therefore $\mathrm{pr}_\Gamma(W)$ intersects $Y(\Gamma)$ non-trivially, which shows the latter is dense.

Using that $\mathcal{H} \hookrightarrow \mathcal{H}^*$ is a topological embedding, chasing open sets in the diagram

$$\begin{array}{ccc} \mathcal{H} & \hookrightarrow & \mathcal{H}^* \\ \downarrow{\scriptstyle \mathrm{pr}_\Gamma} & & \downarrow{\scriptstyle \mathrm{pr}_\Gamma} \\ Y(\Gamma) & \hookrightarrow & X(\Gamma). \end{array}$$

shows $Y(\Gamma) \hookrightarrow X(\Gamma)$ is a topological embedding as well.

For the compactness of $X(\Gamma)$ we can replace Γ by a subgroup if necessary. We may thus assume Γ is a finite index subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\mathcal{F}$ be the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ defined above. Decompose $\mathrm{SL}_2(\mathbb{Z})$ into cosets $\Gamma g_1 \cup \cdots \cup \Gamma g_N$,

and let $\mathcal{F}_\Gamma := g_1\mathcal{F} \cup \cdots \cup g_N\mathcal{F}$. Note that $\mathcal{H} = \bigcup_{\gamma\in\Gamma} \gamma\mathcal{F}_\Gamma$, in other words the restriction $\mathrm{pr}_\Gamma : \mathcal{F}_\Gamma \to Y(\Gamma)$ is surjective. Similarly, introduce $\mathcal{F}^* = \mathcal{F} \cup \{\infty\}$ and let $\mathcal{F}_\Gamma^* := g_1\mathcal{F}^* \cup \cdots \cup g_N\mathcal{F}^*$. Then $\mathrm{pr}_\Gamma : \mathcal{F}_\Gamma^* \to X(\Gamma)$ is surjective. It now suffices to note that $\mathcal{F}^*$ (and hence $\mathcal{F}_\Gamma^*$) is a compact subset of $\mathcal{H}^*$. This is rather immediate: An open covering of $\mathcal{F}^*$ must contain a neighborhood of ∞, which without loss of generality we may take to be of the form $W_{\infty,T}$. It remains to observe that $\{z \in \mathcal{F} : \mathrm{Im}(z) \le T\}$ is a compact subset of $\mathcal{H}$ by Heine-Borel. □

In order to enhance $X(\Gamma)$ to a compact Riemann surface we need complex charts around the cusps. The key to this is the following cuspidal analogue of Lemma 5.9, which we will not pause to prove:

Lemma 5.14 *Every $\rho \in \mathbb{P}^1(\mathbb{Q})$ admits an open neighborhood $W = W_\rho$ in $\mathcal{H}^*$ such that for all $\gamma \in \Gamma$,*

$$W \cap \gamma W \neq \varnothing \Longrightarrow \gamma\rho = \rho.$$

Proof We omit the proof here. The reader is encouraged to peruse [Shi94, Lemma 1.26, p. 11]. □

For such a $W \ni \rho$ the projection pr_Γ restricts to a bijection $\mathrm{Stab}_\Gamma(\rho)\backslash W \xrightarrow{\sim} \mathrm{pr}_\Gamma(W)$, and we need to identify the source with an open subset of $\mathbb{C}$. One can move ρ to ∞ by choosing a $g \in \mathrm{SL}_2(\mathbb{Z})$ with $g\infty = \rho$. Then $g^{-1}\mathrm{Stab}_\Gamma(\rho)g = \mathrm{Stab}_{g^{-1}\Gamma g}(\infty)$, so that $z \mapsto gz$ defines a homeomorphism

$$\mathrm{Stab}_{g^{-1}\Gamma g}(\infty)\backslash g^{-1}W \xrightarrow{\sim} \mathrm{Stab}_\Gamma(\rho)\backslash W.$$

Because of this it is enough to treat the point ∞. We will also assume Γ is a finite index subgroup of $\mathrm{SL}_2(\mathbb{Z})$. In this case $\mathrm{Stab}_\Gamma(\infty) = \Gamma \cap \{\pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) : b \in \mathbb{Z}\}$.

Exercise 5.15 Explain why $\Gamma\{\pm I\} \cap \{\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) : b \in \mathbb{Z}\}$ is not the trivial group. Take h to be the smallest positive integer such that $\left(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}\right) \in \Gamma\{\pm I\}$. We call h the Γ-*width* of ∞. Deduce that

$$\mathrm{Stab}_\Gamma(\infty)\{\pm I\} = \left\langle \left(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}\right) \right\rangle \{\pm I\} = \{\pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) : b \in h\mathbb{Z}\}.$$

Find h for each of the three congruence subgroups $\Gamma(N)$, $\Gamma_1(N)$, and $\Gamma_0(N)$.

We view $z \mapsto e^{2\pi i z/h}$ as a holomorphic function mapping $\{z : \mathrm{Im}(z) > T\}$ onto the punctured open disk of radius $e^{-2\pi T/h}$ centered at the origin, i.e. $B_{e^{-2\pi T/h}}(0) - \{0\}$ where $B_r(0) = \{z : |z| < r\}$. We extend this map to $W_{\infty,T}$ by sending the extra point ∞ to the lacking point at the origin:

$$\begin{aligned} W_{\infty,T} &\longrightarrow B_{e^{-2\pi T/h}}(0) \\ z &\longmapsto e^{2\pi i z/h} \\ \infty &\longmapsto 0. \end{aligned}$$

Two points $z, z' \in W_{\infty,T}$ have the same image in $B_{e^{-2\pi T/h}}(0)$ precisely when $z - z' \in h\mathbb{Z}$, i.e. when z, z' lie in the same orbit under the action of $\mathrm{Stab}_\Gamma(\infty)$ (by definition of the width h in Exercise 5.15). This results in a homeomorphism

$$\mathrm{Stab}_\Gamma(\infty)\backslash W_{\infty,T} \xrightarrow{\sim} B_{e^{-2\pi T/h}}(0).$$

For $\rho = \infty$ we may take W in Lemma 5.14 to be of the form $W_{\infty,T}$ for $T \gg 0$. We identify $\mathrm{pr}_\Gamma(W_{\infty,T})$ with the disc $B_{e^{-2\pi T/h}}(0)$ and this gives a chart of $X(\Gamma)$ around ∞. At $\rho = g\infty$ we get the chart

$$B_{e^{-2\pi T/h(\rho)}}(0) \xleftarrow{\sim} \mathrm{Stab}_{g^{-1}\Gamma g}(\infty)\backslash W_{\infty,T} \xrightarrow{\sim} \mathrm{Stab}_\Gamma(\rho)\backslash W_{\rho,T} \xrightarrow{\sim} \mathrm{pr}_\Gamma(W_{\rho,T})$$

for $T \gg 0$. Here the radius of the disk depends on $h(\rho)$ which denotes the $g^{-1}\Gamma g$-width of ∞ (this is easily checked to be independent of the choice of $g \in \mathrm{SL}_2(\mathbb{Z})$ such that $g\infty = \rho$). We call $h(\rho)$ the Γ-width of ρ. It only depends on the cusp $\Gamma\rho$ defined by $\rho \in \mathbb{P}^1(\mathbb{Q})$.

Exercise 5.16 Suppose $\mathrm{pr}_\Gamma(W_{\infty,T})$ intersects $\mathrm{pr}_\Gamma(W_z)$ for some $W_z \ni z$ as in Lemma 5.9. Identify their intersection with open subsets $\mathcal{I} \subset B_{e^{-2\pi T/h}}(0)$ and $\mathcal{J} \subset W_z$ using the charts we have defined. Verify the transition maps $\mathcal{I} \to \mathcal{J}$ and $\mathcal{J} \to \mathcal{I}$ are (mutually inverse) holomorphic functions.

We endow $X(\Gamma)$ with the maximal atlas containing all the charts we have defined. It thus becomes a compact Riemann surface.

5.4 The Moduli Interpretation

We do not assume the reader is familiar with the general theory of elliptic curves. However, we would be amiss if we did not at least hint at how points of $Y(\Gamma)$ parametrize isomorphism classes of elliptic curves over $\mathbb{C}$ with extra structure depending on Γ. In other words, how $Y(\Gamma)$ can be viewed as a moduli space.

An elliptic curve over $\mathbb{C}$ is just a complex torus $\mathbb{C}/\Lambda$, with base point $0 + \Lambda$. Here $\Lambda \subset \mathbb{C}$ is a lattice, i.e. a discrete subgroup of the form $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ for an $\mathbb{R}$-basis $\{\omega_1, \omega_2\}$ in $\mathbb{C}$. Note that $\mathbb{C}/\Lambda$ is an abelian group which is topologically a torus $S^1 \times S^1$. (To see this, glue opposite sides of the fundamental parallelogram $\{a_1\omega_1 + a_2\omega_2 : 0 \le a_1, a_2 < 1\}$.)

Suppose $\Lambda' \subset \mathbb{C}$ is another lattice and let $c \in \mathbb{C}$ be a scalar such that $c\Lambda \subset \Lambda'$. Multiplication by c gives a holomorphic homomorphism $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$. All the holomorphic maps $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$, which preserve the base points, arise in this fashion by [Sil09, Theorem 4.1, p. 159]. In particular, $\mathbb{C}/\Lambda$ is isomorphic to $\mathbb{C}/\Lambda'$ if and only if Λ and Λ' are homothetic, i.e. $\Lambda' = c\Lambda$ for some $c \in \mathbb{C}^\times$. We write $\Lambda \sim \Lambda'$.

For $\tau \in \mathcal{H}$ we introduce the lattice $\Lambda_\tau := \mathbb{Z} \oplus \mathbb{Z}\tau$ and the associated elliptic curve $E_\tau := \mathbb{C}/\Lambda_\tau$. Any $\mathbb{C}/\Lambda$ as above is isomorphic to E_τ with $\tau = \pm\frac{\omega_2}{\omega_1}$ (choose the sign such that $\mathrm{Im}(\pm\frac{\omega_2}{\omega_1}) > 0$).

Lemma 5.17 *Let $\tau, \tau' \in \mathcal{H}$. Then $E_\tau \simeq E_{\tau'}$ if and only if $\tau' = \gamma\tau$ for some $\gamma \in SL_2(\mathbb{Z})$. Said differently, the map*

$$\begin{aligned} SL_2(\mathbb{Z})\backslash\mathcal{H} &\longrightarrow \{\textit{elliptic curves over } \mathbb{C}\}/\simeq \\ SL_2(\mathbb{Z})\tau &\longmapsto [E_\tau]\ (= \textit{the isomorphism class of } E_\tau) \end{aligned}$$

is a bijection.

Proof First suppose $\tau' = \gamma\tau$ for some $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then the corresponding lattices are homothetic:

$$\Lambda_{\tau'} = \mathbb{Z} \oplus \mathbb{Z}\tau' = \mathbb{Z} \oplus \mathbb{Z}\frac{a\tau + b}{c\tau + d} \sim \mathbb{Z}(c\tau + d) \oplus \mathbb{Z}(a\tau + b) = \mathbb{Z} \oplus \mathbb{Z}\tau = \Lambda_\tau.$$

We note that two $\mathbb{R}$-bases $\{\omega_1, \omega_2\}$ and $\{\omega_1', \omega_2'\}$ generate the *same* lattice in $\mathbb{C}$ precisely when there is a matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$ such that

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\left(\begin{smallmatrix} \omega_2 \\ \omega_1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} \omega_2' \\ \omega_1' \end{smallmatrix}\right).$$

Conversely, we now assume $\Lambda_{\tau'} = s\Lambda_\tau$ for some $s \in \mathbb{C}^\times$. As just remarked, this yields a matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$ with the property

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\left(\begin{smallmatrix} s\tau \\ s \end{smallmatrix}\right) = \left(\begin{smallmatrix} \tau' \\ 1 \end{smallmatrix}\right).$$

Consequently $\tau' = s(a\tau + b) = \frac{a\tau+b}{c\tau+d}$. Since τ and τ' both lie in $\mathcal{H}$ the last identity forces $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ to have positive determinant, i.e. $ad - bc = 1$. This shows τ and τ' are necessarily $\mathrm{SL}_2(\mathbb{Z})$-conjugate. □

The group E_τ has a preferred element $P_\tau := \frac{1}{N} + \Lambda_\tau$ of order N. We let

$$C_\tau := \langle P_\tau \rangle = \{\frac{m}{N} + \Lambda_\tau : 0 \le m < N\}$$

denote the cyclic subgroup of E_τ generated by P_τ. Consider another point $\tau' = \gamma\tau$ for some $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. From the proof of Lemma 5.17 we know that $\Lambda_\tau = (c\tau + d)\Lambda_{\tau'}$. Multiplication by $c\tau + d$ therefore gives an isomorphism $E_{\tau'} \xrightarrow{\sim} E_\tau$ which takes $P_{\tau'}$ to

$$\frac{c\tau + d}{N} + \Lambda_\tau = \left(\frac{c}{N}\tau + \frac{d}{N}\right) + \Lambda_\tau.$$

This clearly belongs to C_τ if $N|c$, in other words if $\gamma \in \Gamma_0(N)$. Since we know $P_{\tau'}$ (and therefore also $\frac{c\tau+d}{N} + \Lambda_\tau$) has order N we conclude that $\langle \frac{c\tau+d}{N} + \Lambda_\tau \rangle = C_\tau$

provided $N|c$. To summarize, if $\gamma \in \Gamma_0(N)$, multiplication by $c\tau + d$ gives an isomorphism $E_{\tau'} \xrightarrow{\sim} E_\tau$ which restricts to an isomorphism $C_{\tau'} \xrightarrow{\sim} C_\tau$. Furthermore, the same calculation shows that in fact $P_{\tau'} \mapsto P_\tau$ when $\gamma \in \Gamma_1(N)$.

This leads to the following generalization of Lemma 5.17.

Proposition 5.18 *The two maps*

$$\begin{aligned} \Gamma_0(N)\backslash\mathcal{H} &\longrightarrow \{\textit{elliptic curves over } \mathbb{C} \textit{ with a cyclic subgroup of order } N\}/\sim \\ \Gamma_0(N)\tau &\longmapsto [E_\tau, C_\tau] \ (= \textit{the equivalence class of the pair } (E_\tau, C_\tau)) \end{aligned}$$

and

$$\begin{aligned} \Gamma_1(N)\backslash\mathcal{H} &\longrightarrow \{\textit{elliptic curves over } \mathbb{C} \textit{ with a point of order } N\}/\sim \\ \Gamma_1(N)\tau &\longmapsto [E_\tau, P_\tau] \ (= \textit{the equivalence class of the pair } (E_\tau, P_\tau)) \end{aligned}$$

are bijective.

Proof We start by showing injectivity of the first map. So suppose $\tau, \tau' \in \mathcal{H}$ and there is an isomorphism $E_{\tau'} \xrightarrow{\sim} E_\tau$ which restricts to an isomorphism $C_{\tau'} \xrightarrow{\sim} C_\tau$. In the notation of the proof of Lemma 5.17 we know that $\tau' = \gamma\tau$ for some $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and the isomorphisms are given by multiplication by $\frac{1}{s} = c\tau + d$. Thus $\frac{c\tau+d}{N} + \Lambda_\tau \in C_\tau$, i.e. $\frac{c}{N}\tau + \frac{d-m}{N} \in \Lambda_\tau$ for some $m \in \mathbb{Z}$. This happens precisely when $N|c$. Minor modifications of this argument also shows the injectivity of the second map. In that case we are analyzing when $\frac{c\tau+d}{N} + \Lambda_\tau = P_\tau$, i.e. when $\frac{c}{N}\tau + \frac{d-1}{N} \in \Lambda_\tau$. This clearly holds exactly when $\gamma \in \Gamma_1(N)$.

For the surjectivity, start with a pair (E_τ, C) where C is some cyclic subgroup of order N. We need a $\tau' \in \mathcal{H}$ such that $(E_{\tau'}, C_{\tau'}) \sim (E_\tau, C)$. More precisely, we are seeking a $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ such that $\frac{c\tau+d}{N} + \Lambda_\tau \in C$. (This indeed implies that $C_{\tau'}$ maps onto C since both subgroups have order N.) Pick a generator $\frac{n}{N}\tau + \frac{m}{N} + \Lambda_\tau \in C$. Here n, m are necessarily coprime. Any γ of the form $\left(\begin{smallmatrix} * & * \\ n & m \end{smallmatrix}\right)$ works.

This also shows the second map is surjective. Indeed, starting with (E_τ, P) where $P = \frac{n}{N}\tau + \frac{m}{N} + \Lambda_\tau$ has order N, we found a $\gamma = \left(\begin{smallmatrix} * & * \\ n & m \end{smallmatrix}\right)$ for which $(E_{\gamma\tau}, P_{\gamma\tau}) \sim (E_\tau, P)$. □

The situation for the principal congruence subgroup $\Gamma(N)$ is slightly more involved. The moduli interpretation of $\Gamma(N)\backslash\mathcal{H}$ employs the Weil pairing, which we now introduce. Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ as above. The subgroup of N-torsion points

$$E[N] = \frac{1}{N}\Lambda/\Lambda = (\mathbb{Z}/N\mathbb{Z})\Big(\frac{\omega_1}{N} + \Lambda\Big) \oplus (\mathbb{Z}/N\mathbb{Z})\Big(\frac{\omega_2}{N} + \Lambda\Big)$$

is a free $\mathbb{Z}/N\mathbb{Z}$-module of rank two with basis $\{\frac{\omega_1}{N} + \Lambda, \frac{\omega_2}{N} + \Lambda\}$. Once and for all, choose a primitive N^{th} root of unity $\zeta_N \in \mu_N(\mathbb{C})$. To fix ideas, the reader may

want to take $\zeta_N = e^{2\pi i/N}$. The corresponding *Weil* pairing is the unique alternating $\mathbb{Z}/N\mathbb{Z}$-bilinear form

$$\langle -, - \rangle : E[N] \times E[N] \longrightarrow \mu_N(\mathbb{C})$$

such that $\langle \frac{\omega_1}{N} + \Lambda, \frac{\omega_2}{N} + \Lambda \rangle = \zeta_N$. Thus the general formula is

$$\langle \frac{a_1\omega_1 + a_2\omega_2}{N} + \Lambda, \frac{b_1\omega_1 + b_2\omega_2}{N} + \Lambda \rangle = \zeta_N^{a_1b_2 - a_2b_1}$$

for integers a_1, a_2, b_1, b_2. In particular $\langle -, - \rangle$ is a non-degenerate pairing (if $P \in E[N]$ satisfies $\langle P, P' \rangle = 1$ for all $P' \in E[N]$, then $P = 0$). For a comparison with other definitions of the Weil pairing we refer to the last paragraph of [Sil09, p. 352].

In the context of $E_\tau = \mathbb{C}/\Lambda_\tau$ this amount to the following. The N-torsion points $E_\tau[N]$ has $\mathbb{Z}/N\mathbb{Z}$-basis $\{P_\tau, Q_\tau\}$ where $P_\tau = \frac{1}{N} + \Lambda_\tau$ (as before) and $Q_\tau = \frac{\tau}{N} + \Lambda_\tau$. The Weil pairing is determined by the relation $\langle P_\tau, Q_\tau \rangle = \zeta_N$. If we replace τ by $\tau' = \gamma\tau$ for some $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma(N)$, as we have now seen multiple times, the multiplication by $c\tau + d$ map gives an isomorphism $E_{\tau'} \xrightarrow{\sim} E_\tau$ which takes $P_{\tau'}$ to P_τ. Moreover, it sends $Q_{\tau'} = \frac{\tau'}{N} + \Lambda_{\tau'}$ to the point

$$\frac{(c\tau + d)\tau'}{N} + \Lambda_\tau = \frac{a\tau + b}{N} + \Lambda_\tau = Q_\tau$$

since we are now imposing the additional congruences $a \equiv 1$ and $b \equiv 0$ modulo N on γ. All combined, this gives a map

$$\begin{aligned} \Gamma(N)\backslash\mathcal{H} &\longrightarrow \{\text{elliptic curves over } \mathbb{C} \text{ with two points } P, Q \text{ of order } N \text{ with Weil pairing } \zeta_N\}/\sim \\ \Gamma(N)\tau &\longmapsto [E_\tau, P_\tau, Q_\tau] \ (= \text{the equivalence class of the triple } (E_\tau, P_\tau, Q_\tau)). \end{aligned}$$

Its bijectivity follows along the same lines as the proof of Proposition 5.18. The surjectivity requires the condition $\langle P, Q \rangle = \zeta_N$. Note also that this condition implies $\{P, Q\}$ is a $\mathbb{Z}/N\mathbb{Z}$-basis for the group of N-torsion points.

Exercise 5.19 Prove that the previous map $\Gamma(N)\tau \mapsto [E_\tau, P_\tau, Q_\tau]$ is a bijection. (`Hint`: To show surjectivity start with (E_τ, P, Q) where $\langle P, Q \rangle = \zeta_N$. Write $P = \frac{n}{N}\tau + \frac{m}{N} + \Lambda_\tau$ and $Q = \frac{v}{N}\tau + \frac{u}{N} + \Lambda_\tau$. Then $mv - nu \equiv 1$ modulo N. Pick a $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $\gamma \equiv \left(\begin{smallmatrix} v & u \\ n & m \end{smallmatrix}\right)$ modulo N and consider $\tau' = \gamma\tau$. Check that $(E_{\tau'}, P_{\tau'}, Q_{\tau'})$ is equivalent to (E_τ, P, Q).)

Altogether this gives the following tower of modular curves.

$$\begin{array}{ccc}
\Gamma(N)\backslash\mathcal{H} & & (E_\tau, P_\tau, Q_\tau) \\
\downarrow & & \downarrow \\
\Gamma_1(N)\backslash\mathcal{H} & & (E_\tau, P_\tau) \\
\downarrow & & \downarrow \\
\Gamma_0(N)\backslash\mathcal{H} & & (E_\tau, \langle P_\tau\rangle) \\
\downarrow & & \downarrow \\
\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} & & E_\tau.
\end{array}$$

In this diagram, the first and the third maps are obtained by simply forgetting the extra structure Q_τ and $C_\tau = \langle P_\tau\rangle$ respectively. The second map remembers the group generated by P_τ, but not the generator itself.

One can go a little further and look at $\Gamma_H\backslash\mathcal{H}$ for any subgroup $H \subset \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. If E is an elliptic curve over $\mathbb{C}$, a basis $\{P, Q\}$ for $E[N]$ as above amounts to an isomorphism of $\mathbb{Z}/N\mathbb{Z}$-modules

$$\begin{aligned}
\alpha_{P,Q} : \mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z} &\xrightarrow{\sim} E[N] \\
e_1 &\longmapsto Q \\
e_2 &\longmapsto P
\end{aligned}$$

under which the Weil pairing corresponds to the alternating $\mathbb{Z}/N\mathbb{Z}$-bilinear form on $\mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z}$ with $\langle e_2, e_1\rangle = \zeta_N$. Here $e_1 = (1, 0)$ and $e_2 = (0, 1)$ denote the standard basis vectors. Note our convention that e_1 maps to the second point Q, and e_2 maps to the first point P. This will be convenient below.

We consider the set of pairs (E, α), where E is an elliptic curve over $\mathbb{C}$, and $\alpha : \mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z} \xrightarrow{\sim} E[N]$ is an isomorphism intertwining the alternating pairings on the source and target, as described in the previous paragraph. On this set we define an equivalence relation $\sim_H$ by declaring that $(E, \alpha) \sim_H (E', \alpha')$ if there is an isomorphism $E' \xrightarrow{\sim} E$ which, when restricted to the N-torsion points, makes the diagram

$$\begin{array}{ccc}
\mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z} & \xrightarrow{\alpha'} & E'[N] \\
{\scriptstyle (x,y)\mapsto(x,y)\cdot\eta}\downarrow & & \downarrow{\scriptstyle \simeq} \\
\mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z} & \xrightarrow{\alpha} & E[N].
\end{array}$$

commute for some $\eta \in H$ (which is necessarily unique). For instance, if $\tau' = \gamma\tau$ with $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_H$, then the multiplication by $c\tau + d$ isomorphism $E_{\tau'} \xrightarrow{\sim} E_\tau$ fits in the commutative diagram

$$\begin{array}{ccc} \mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z} & \xrightarrow{\alpha'} & E_{\tau'}[N] \\ {\scriptstyle (x,y)\mapsto (x,y)\cdot \mathrm{pr}_N(\gamma)}\downarrow & & \downarrow{\scriptstyle \simeq} \\ \mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z} & \xrightarrow{\alpha} & E_{\tau}[N]. \end{array}$$

where we have used the abbreviations $\alpha' = \alpha_{P_{\tau'},Q_{\tau'}}$ and $\alpha = \alpha_{P_\tau,Q_\tau}$ for legibility. (Remember that $\alpha(e_1) = Q_\tau$ and $\alpha(e_2) = P_\tau$.)

Proposition 5.20 *The map*

$$\begin{aligned} &\Gamma_H\backslash\mathcal{H} \longrightarrow \{(E,\alpha)\}/\sim_H \\ &\quad \Gamma_H\tau \longmapsto [E_\tau, \alpha_{P_\tau,Q_\tau}]_H \ (= \textit{the } \sim_H \textit{ equivalence class of the pair } (E_\tau, \alpha_{P_\tau,Q_\tau})) \end{aligned}$$

is a bijection.

Proof For the surjectivity we may assume $\Gamma_H = \Gamma(N)$. This case was proved in Exercise 5.19. For the injectivity, suppose $(E_\tau, \alpha_{P_\tau,Q_\tau}) \sim_H (E_{\tau'}, \alpha_{P_{\tau'},Q_{\tau'}})$. Then $\tau' = \gamma\tau$ for some $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and the multiplication by $c\tau + d$ isomorphism $E_{\tau'} \xrightarrow{\sim} E_\tau$ corresponds to (right) multiplication by $\mathrm{pr}_N(\gamma)$ on $\mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z}$. By definition of $\sim_H$ we require that $\mathrm{pr}_N(\gamma) \in H$. In other words $\gamma \in \Gamma_H$. □

Exercise 5.21 Let $H = \{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) : x \in \mathbb{Z}/N\mathbb{Z}\}$. For two points $\tau, \tau' \in \mathcal{H}$, verify that

$$(E_\tau, \alpha_{P_\tau,Q_\tau}) \sim_H (E_{\tau'}, \alpha_{P_{\tau'},Q_{\tau'}}) \Longleftrightarrow (E_\tau, P_\tau) \sim (E_{\tau'}, P_{\tau'}).$$

Formulate and show a similar assertion for the group of all upper triangular matrices $\{(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix})\} \subset \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$.

Remark 5.22 There is also a moduli interpretation of the compactification $X(\Gamma_H)$ which involves *generalized* elliptic curves. It would lead us too far afield to discuss those in detail here (and it would require heavier use of algebraic geometry) but over $\mathbb{C}$ they are either elliptic curves or so-called Néron polygons. A Néron n-gon is obtained from n projective lines $\ell_1 \simeq \mathbb{P}^1, \ldots, \ell_n \simeq \mathbb{P}^1$ glued as follows: The point $0 \in \ell_i$ is identified with the point $\infty \in \ell_{i+1}$ on the next line in the sequence (with $i = 1, \ldots, n$ taken modulo n). The cusps $\mathrm{Cusp}_{\Gamma_H} \subset X(\Gamma_H)$ parametrize Néron polygons with extra structure.

5.5 The Definition of Modular Forms

Modular forms are global sections of line bundles on $X = X(\Gamma)$. A line bundle is a locally free $\mathcal{O}_X$-module of rank one ($\mathcal{O}_X$ is the sheaf of holomorphic functions). A fundamental result, often attributed to Grauert and Röhrl, asserts that on a *non*-compact Riemann surface every holomorphic vector bundle is trivial. The reader who

is uncomfortable with bundles on Riemann surfaces is encouraged to skip ahead to Definition 5.24 below–our remarks leading up to it are motivational only.

As is well-known, $H^0(X, \mathcal{O}_X) = \mathbb{C}$. This follows from the open mapping theorem (a non-constant holomorphic function $X \to \mathbb{C}$ takes open sets to open sets, but X maps onto a compact set). Instead of $\mathcal{O}_X$ we consider the sheaf Ω^1_X of holomorphic 1-forms. It comes with a universal derivation $d_X : \mathcal{O}_X \to \Omega^1_X$. A key feature is that 1-forms can be pulled back via a morphism $\alpha : X' \to X$, as follows. The pushforward functor α_* from sheaves on X' to sheaves on X gives a diagram

$$\begin{array}{ccc} \mathcal{O}_X & \longrightarrow & \alpha_*\mathcal{O}_{X'} \\ {\scriptstyle d_X}\downarrow & & \downarrow{\scriptstyle \alpha_*(d_{X'})} \\ \Omega^1_X & \dashrightarrow & \alpha_*\Omega^1_{X'}. \end{array}$$

For open subsets $W \subset X$ the dashed arrow gives a map $H^0(W, \Omega^1_X) \longrightarrow H^0(\alpha^{-1}(W), \Omega^1_{X'})$. In coordinates $f(w)dw$ pulls back to $f(\alpha(z))\alpha'(z)dz$. In particular one can restrict 1-forms along any inclusion $X' \hookrightarrow X$.

We can describe the global sections $H^0(X, \Omega^1_X)$ by choosing an open covering $X = \bigcup_{i\in I} W_i$ and giving sections in $H^0(W_i, \Omega^1_X)$, one for each i, which are compatible when restricted to the intersection $W_i \cap W_j$ whenever this is non-empty. This is precisely the sheaf property. Typically the W_i will be charts, and therefore non-compact, in which case $\Omega^1_X|_{W_i}$ is trivial and elements of $H^0(W_i, \Omega^1_X)$ can be expressed as ψdw where $\psi \in H^0(W_i, \mathcal{O}_X)$ and w is a coordinate function on W_i.

Here we contemplate the open covering of $X = X(\Gamma)$ consisting of the open modular curve $Y = Y(\Gamma)$ and the open neighborhoods $\mathrm{Stab}_\Gamma(\rho)\backslash W_{\rho,T} \xrightarrow{\sim} \mathrm{pr}(W_{\rho,T})$ at all the cusps ρ (for a fixed large enough T say) which we identify with open disks in the usual fashion. We keep track of the various restriction maps for this covering in the diagram below,

$$\begin{array}{ccccc} & & H^0(X, \Omega^1_X) & & \\ & \swarrow & \downarrow & \searrow & \\ H^0(\mathrm{pr}(W_{\infty,T}), \Omega^1_X) & & H^0(Y, \Omega^1_X) & & H^0(\mathrm{pr}(W_{\rho,T}), \Omega^1_X) \\ \downarrow & \swarrow & & \searrow & \downarrow \\ H^0(\mathrm{pr}(W_{\infty,T}) \cap Y, \Omega^1_X) & & & & H^0(\mathrm{pr}(W_{\rho,T}) \cap Y, \Omega^1_X). \end{array}$$

The three spaces in the middle row can be described rather explicitly. The compatibility on the intersections forces holomorphy conditions at the cusps, as we now proceed to explain.

We first describe $H^0(Y, \Omega^1_X)$ by pulling back 1-forms to $\mathcal{H}$ via the projection map $\mathrm{pr} = \mathrm{pr}_\Gamma : \mathcal{H} \to Y$. Each $\gamma \in \Gamma$ defines a transformation $z \mapsto \gamma z$ which results in a Γ-action on $H^0(\mathcal{H}, \Omega^1_{\mathcal{H}})$ via pullbacks. Since pr commutes with γ we get a map $H^0(Y, \Omega^1_X) \longrightarrow H^0(\mathcal{H}, \Omega^1_{\mathcal{H}})^\Gamma$. It is easy to see this is bijective, by using pr is

a covering map (the Γ-invariance ensures we can glue sections on $W_z \xrightarrow{\sim} \mathrm{pr}(W_z)$ uniquely to a section on Y). Now, a Γ-invariant element of $H^0(\mathcal{H}, \Omega^1_{\mathcal{H}})$ has the form $f(z)dz$ where $f : \mathcal{H} \to \mathbb{C}$ is a holomorphic function such that

$$f(z)dz = f(\gamma z)d(\gamma z) = f(\gamma z)(cz+d)^{-2}dz$$

for all $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$. (Indeed an easy calculation shows $z \mapsto \gamma z = \frac{az+b}{cz+d}$ has derivative $(cz+d)^{-2}$ as γ has determinant one.) To summarize, $H^0(Y, \Omega^1_X)$ can be identified with the space of holomorphic functions $f : \mathcal{H} \to \mathbb{C}$ with the following *weight two* transformation property for all $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$,

$$f(\gamma z) = (cz+d)^2 f(z).$$

Next, at a $\rho \in \mathbb{P}^1(\mathbb{Q})$ we have the chart $B_{e^{-2\pi T/h(\rho)}}(0) \xrightarrow{\sim} \mathrm{pr}(W_{\rho,T})$ which enables us to identify sections of Ω^1_X on $\mathrm{pr}(W_{\rho,T})$ with holomorphic functions on an open disk centered at the origin. As such, they have Taylor expansions. For example, for $\rho = \infty$ elements of $H^0(\mathrm{pr}(W_{\infty,T}), \Omega^1_X)$ have the form $\tilde{f}(q)dq$, where q is the coordinate on the disc, and $\tilde{f}(q) = \sum_{n=0}^{\infty} c_n q^n$. We recall that the identification with a disk comes from the map $z \mapsto e^{2\pi i z/h}$ in this case, so the pullback of $\tilde{f}(q)dq$ to $\mathcal{H}$ is the 1-form

$$\tilde{f}(e^{2\pi i z/h})(2\pi i/h)e^{2\pi i z/h}dz = (2\pi i/h)\Big(\sum_{n=0}^{\infty} c_n e^{2\pi i(n+1)z/h}\Big)dz.$$

At $\rho = \infty$ we thus require f to have a Fourier expansion starting from $n = 1$,

$$f(z) = \sum_{n=1}^{\infty} a_n e^{2\pi i n z/h}.$$

At a $\rho \neq \infty$ we write $\rho = g\infty$ with $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. This g is used to identify $\mathrm{pr}(W_{\rho,T})$ with a disk via the transformation $z \mapsto gz$. The pullback of $f(z)dz$ along this map should now be holomorphic at infinity. This pullback is given by

$$f(gz)d(gz) = f(gz)(cz+d)^{-2}dz = (f|g)(z)dz, \quad (f|g)(z) := f(gz)(cz+d)^{-2}.$$

In other words $f|g$ should now have a Fourier expansion $\sum_{n=1}^{\infty} a_n(\rho)e^{2\pi i n z/h(\rho)}$ for all $g \in \mathrm{SL}_2(\mathbb{Z})$. Such functions f are precisely the weight two cusp forms for Γ. In the notation to be introduced in Definition 5.25 we have thus identified $H^0(X, \Omega^1_X)$ with $S_2(\Gamma)$.

Replacing Ω^1_X by $(\Omega^1_X)^{\otimes\kappa}$ in the previous discussion does not quite give modular forms of weight 2κ, but close. The transformation property becomes $f(\gamma z) = (cz+d)^{2\kappa} f(z)$ but now $\tilde{f}$ is allowed to have a pole at the origin, $\tilde{f}(q) = \sum_{n=-\kappa}^{\infty} c_n q^n$. (There are further complications in the presence of elliptic points but we are avoiding those issues by assuming Γ is small enough.) Thus the space $M_{2\kappa}(\Gamma)$, to be introduced

in Definition 5.24, can be identified with the global sections of a certain *twist* of $(\Omega^1_X)^{\otimes \kappa}$.

The reader may want to consult [DS05, Sect. 3.3] and [Shi94, Sect. 2.4] for more details.

Definition 5.23 On the space $\mathrm{Hol}(\mathcal{H})$ of holomorphic functions $f : \mathcal{H} \to \mathbb{C}$ we define a right action of $\mathrm{GL}_2(\mathbb{R})^+$ by

$$(f|_k g)(z) := \det(g)^{\frac{k}{2}} j(g,z)^{-k} f(gz)$$

where $j(g,z) := cz + d$ for $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{R})^+$ is the *automorphy* factor. Here the *weight* k can be any integer.

The reader should check this indeed defines a right action $\mathrm{Hol}(\mathcal{H}) \times \mathrm{GL}_2(\mathbb{R})^+ \longrightarrow \mathrm{Hol}(\mathcal{H})$, which essentially boils down to verifying the cocycle condition

$$j(gg', z) = j(g, g'z) j(g', z)$$

for $g, g' \in \mathrm{GL}_2(\mathbb{R})^+$. The reason we include the factor $\det(g)^{\frac{k}{2}}$ in the definition of $(f|_k g)(z)$ is it ensures $Z^+_\infty = \{\left(\begin{smallmatrix} a & 0 \\ 0 & a \end{smallmatrix}\right) : a > 0\}$ acts trivially on $\mathrm{Hol}(\mathcal{H})$. Note also that $f|_k \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right) = (-1)^k f$.

Definition 5.24 A weight k modular form for Γ is an $f \in \mathrm{Hol}(\mathcal{H})$ such that

(a) $f|_k \gamma = f$ for all $\gamma \in \Gamma$; unwinding the definitions gives the weight k transformation property

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z)$$

for all $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$.

(b) $(f|_k g)(z)$ remains bounded, for all $g \in \mathrm{SL}_2(\mathbb{Z})$, as $\mathrm{Im}(z) \to \infty$. This amounts to assuming each translate $f|_k g$ has a Fourier expansion

$$(f|_k g)(z) = \sum_{n=0}^{\infty} a_n(\rho) e^{2\pi i n z / h(\rho)}$$

as discussed above. When $g = I$ we talk about *the* Fourier expansion of f (at ∞).

These functions obviously comprise a complex vector space $M_k(\Gamma) = \{\text{weight } k \text{ modular forms for } \Gamma\}$.

In some sense, the modular forms genuinely associated with $\mathrm{GL}_2(\mathbb{R})$ are those which vanish at all the cusps:

Definition 5.25 We say $f \in M_k(\Gamma)$ is a *cusp* form if $(f|_k g)(z) \to 0$ as $\mathrm{Im}(z) \to \infty$ for all $g \in \mathrm{SL}_2(\mathbb{Z})$. In other words, the Fourier expansion

$$(f|_k g)(z) = \sum_{n=1}^{\infty} a_n(\rho) e^{2\pi i n z/h(\rho)}$$

has zero constant term (the sum starts from $n = 1$). They clearly form a subspace $S_k(\Gamma)$. Here S stands for *spitzen* which is the German word for cusps.

Remark 5.26 When $-I \in \Gamma$ the space $M_k(\Gamma)$ is trivial unless $k = 2\kappa$ is even, in which case $M_k(\Gamma)$ can be identified with the global sections of a twist of $(\Omega^1_{X(\Gamma)})^{\otimes\kappa}$ as explained earlier. When $-I \notin \Gamma$ there is still a way to interpret $M_k(\Gamma)$ as the global sections of a line bundle $\omega^{\otimes k}_{X(\Gamma)}$ for all (possibly odd) k. We will not go into the definition of $\omega_{X(\Gamma)}$ here, but only emphasize that there is a Kodaira-Spencer isomorphism between $\omega^{\otimes 2}_{X(\Gamma)}$ and the twist of $\Omega^1_{X(\Gamma)}$ by the divisor $C := \mathrm{Cusp}_\Gamma$. Thus

$$M_k(\Gamma) \simeq H^0(X(\Gamma), \omega^{\otimes k}_{X(\Gamma)}), \qquad S_k(\Gamma) \simeq H^0(X(\Gamma), \omega^{\otimes k}_{X(\Gamma)}(-C)).$$

This geometric interpretation gives a way to compute $\dim_{\mathbb{C}} M_k(\Gamma)$ with the help of the Riemann-Roch formula. See [DS05, Sect. 3.5–3.6] and the notes [Mil, pp. 50–53] for detailed derivations of these dimension formulas.

The graded $\mathbb{C}$-algebra $\bigoplus_{k\geq 0} H^0(X(\Gamma), \omega^{\otimes k}_{X(\Gamma)})$ plays an essential role when upgrading $X(\Gamma)$ to a complex algebraic curve via the Proj construction.

Note that a cusp form necessarily decays exponentially at all the cusps. More precisely, $(f|_k g)(z)$ has magnitude $O(e^{-2\pi y/h(\rho)})$ as $y \to \infty$. This observation brings about the next crucial estimate for the Fourier coefficients of a cusp form.

Proposition 5.27 *Let $f(z) = \sum_{n=1}^{\infty} a_n e^{2\pi i n z/h}$ be an element of $S_k(\Gamma)$. Then $a_n = O(n^{\frac{k}{2}})$ as $n \to \infty$.*

Proof To simplify the notation we will assume ∞ has width $h = 1$. The function $z \mapsto |f(z)|\mathrm{Im}(z)^{\frac{k}{2}}$ is Γ-invariant and has exponential decay at all the cusps. In particular it must be bounded on $\mathcal{H}$. (It is enough to check boundedness on a fundamental domain for Γ.) Pick a constant A such that $|f(z)| \leq A\mathrm{Im}(z)^{-\frac{k}{2}}$ for all $z \in \mathcal{H}$. Fix an arbitrary $y > 0$ and consider the periodic function $x \mapsto f(x + iy)$ with Fourier expansion

$$f(x + iy) = \sum_{n=1}^{\infty} a_n e^{-2\pi n y} e^{2\pi i n x}.$$

It has Fourier coefficients

$$a_n e^{-2\pi n y} = \int_0^1 f(x + iy) e^{-2\pi i n x} dx.$$

Therefore,

$$|a_n|e^{-2\pi ny} \leq \int_0^1 |f(x+iy)|dx \leq Ay^{-\frac{k}{2}}.$$

Here y is arbitrary. Taking $y = \frac{1}{n}$ for a given n shows $|a_n| \leq Ae^{2\pi}n^{\frac{k}{2}}$ as desired. □

A deep result of Deligne improves this to $a_n = O(n^{\frac{k-1}{2}+\epsilon})$ for all $\epsilon > 0$.

5.5.1 *Modular Forms with a Nebentype*

In number theory one studies $S_k(\Gamma)$ for congruence subgroups Γ. Perhaps the most interesting example is $S_k(\Gamma_1(N))$–which of course contains $S_k(\Gamma_0(N))$ as a subspace. Note that $S_k(\Gamma_0(N)) = \{0\}$ for odd k, whereas $S_k(\Gamma_1(N))$ is non-trivial at least for large enough N. Customarily one refers to N as the level.

For each Dirichlet character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$ we introduce the subspace $S_k(\Gamma_0(N), \chi)$ consisting of those $f \in S_k(\Gamma_1(N))$ with the transformation property

$$f(\frac{az+b}{cz+d}) = \chi(d)(cz+d)^k f(z)$$

for all $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$. A more succinct way of writing this is $f|_k\gamma = \chi(\gamma)f$ with $\gamma \in \Gamma_0(N)$, where we inflate χ to a character of $\Gamma_0(N)$ via the isomorphism $\Gamma_0(N)/\Gamma_1(N) \xrightarrow{\sim} (\mathbb{Z}/N\mathbb{Z})^\times$ from Exercise 5.6. We say f has nebentype (or nebentypus) χ. Neben means next to in German. By taking $\gamma = -I \in \Gamma_0(N)$ we get the following parity constraint:

$$S_k(\Gamma_0(N), \chi) \neq \{0\} \Longrightarrow \chi(-1) = (-1)^k.$$

In other words χ and k must have the same parity if $S_k(\Gamma_0(N), \chi)$ is non-trivial.

Definition 5.28 Let $d \in \mathbb{Z}$ be coprime to N. Choose any matrix in $\Gamma_0(N)$ of the form $\left(\begin{smallmatrix} * & * \\ * & d \end{smallmatrix}\right)$ and denote it by $\langle d \rangle$. This symbol is well-defined modulo $\Gamma_1(N)$ and we therefore get a linear map

$$\begin{aligned} S_k(\Gamma_1(N)) &\longrightarrow S_k(\Gamma_1(N)) \\ f &\longmapsto f|_k\langle d \rangle \end{aligned}$$

called a *diamond* operator. Note that $\Gamma_1(N)$ is a normal subgroup of $\Gamma_0(N)$ and consequently $f|_k\langle d \rangle$ indeed belongs to $S_k(\Gamma_1(N))$.

With this notation f lies in $S_k(\Gamma_0(N), \chi)$ exactly when $f|_k\langle d \rangle = \chi(d)f$ for all d coprime to N.

Proposition 5.29 $S_k(\Gamma_1(N)) = \bigoplus_\chi S_k(\Gamma_0(N), \chi)$.

Proof Consider the operators $f \mapsto f|_k\langle d\rangle$ on $S_k(\Gamma_1(N))$. From the theory of Riemann surfaces we know the latter is a finite-dimensional vector space over $\mathbb{C}$. Since the operators have finite order, their minimal polynomials must be separable (since they divide $T^{\phi(N)} - 1$ which has no repeated roots). As they commute, they are simultaneously diagonalizable. A common eigenvector is the same as a modular form with a nebentype. □

We close this section with an important exercise, which shows how one can twist a modular form by a Dirichlet character. This will be useful later on when we discuss the analogous operation on L-series. The solution involves some unpleasant matrix calculations which are better left to the reader.

Exercise 5.30 Suppose $f \in S_k(\Gamma_0(N), \chi)$ has Fourier expansion $f = \sum_{n=1}^{\infty} a_n q^n$ where $q = e^{2\pi i z}$. Let $\psi : (\mathbb{Z}/r\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ be a primitive Dirichlet character with conductor r coprime to N. Show that the ψ-twisted Fourier expansion

$$f \times \psi := \sum_{n=1}^{\infty} \psi(n) a_n q^n$$

gives an element of $S_k(\Gamma_0(Nr^2), \chi\psi^2)$. (`Hint`: Using standard properties of Gauss sums, verify the formula

$$\tau(\overline{\psi}) \cdot f \times \psi = \sum_{0 \le u < r} \overline{\psi(u)} \cdot f|_k \begin{pmatrix} 1 & \frac{u}{r} \\ 0 & 1 \end{pmatrix}.$$

For $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(Nr^2)$ find explicit integers a', b', d' such that

$$\begin{pmatrix} 1 & \frac{u}{r} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a' & b' \\ c & d' \end{pmatrix} \begin{pmatrix} 1 & \frac{d^2 u}{r} \\ 0 & 1 \end{pmatrix}.)$$

Remark 5.31 There is a more precise statement in [Shi94, Proposition 3.64, p. 92] which does not require the assumption $\mathrm{GCD}(r, N) = 1$. If χ has conductor $s|N$, a close inspection of the solution to the previous exercise reveals that in fact $f \times \psi$ belongs to $S_k(\Gamma_0(M), \chi\psi^2)$ where $M = \mathrm{LCM}(N, r^2, rs)$.

5.6 Examples of Modular Forms

5.6.1 *Eisenstein Series for $SL_2(\mathbb{Z})$*

Given a lattice $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 \subset \mathbb{C}$ we form the infinite series

$$G_\kappa(\Lambda) = \sum\nolimits_{0 \neq \lambda \in \Lambda} \lambda^{-2\kappa} = \sum\nolimits_{(0,0) \neq (m,n) \in \mathbb{Z}^2} (m\omega_1 + n\omega_2)^{-2\kappa}.$$

This is absolutely convergent for all integers $\kappa \geq 2$. One way to see this is to first observe the following estimates for the number of lattice points in the disk of radius R (centered at the origin):

$$\frac{\pi}{\mathrm{vol}(\mathbb{C}/\Lambda)}(R - \mathrm{diam}(\mathbb{C}/\Lambda))^2 \leq \#\{\lambda \in \Lambda : |\lambda| \leq R\} \leq \frac{\pi}{\mathrm{vol}(\mathbb{C}/\Lambda)}(R + \mathrm{diam}(\mathbb{C}/\Lambda))^2.$$

Here $\mathrm{vol}(\mathbb{C}/\Lambda)$ and $\mathrm{diam}(\mathbb{C}/\Lambda)$ denote the area and the diameter of the fundamental parallelogram respectively. The two inequalities are easily established by covering the radius R disk with translates of the parallelogram $\{a_1\omega_1 + a_2\omega_2 : 0 \leq a_1, a_2 < 1\}$. Thus[2]

$$\#\{\lambda \in \Lambda : |\lambda| \leq R\} = \frac{\pi}{\mathrm{vol}(\mathbb{C}/\Lambda)}R^2 + O(R).$$

In particular, as $R \to \infty$, the number of lattice points in an annulus obeys the asymptotics

$$\#\{\lambda \in \Lambda : R < |\lambda| \leq R + 1\} = O(R)$$

where the implied constant only depends on $\mathrm{vol}(\mathbb{C}/\Lambda)$ and $\mathrm{diam}(\mathbb{C}/\Lambda)$. Returning to the sum $G_\kappa(\Lambda)$ we now deduce that

$$\begin{aligned}
\sum\nolimits_{0\neq\lambda\in\Lambda} |\lambda|^{-2\kappa} &= \sum_{n=0}^{\infty} \sum\nolimits_{\lambda\in\Lambda : n<|\lambda|\leq n+1} |\lambda|^{-2\kappa} \\
&< \sum_{n=0}^{\infty} \sum\nolimits_{\lambda\in\Lambda : n<|\lambda|\leq n+1} n^{-2\kappa} \\
&= \sum_{n=0}^{\infty} \#\{\lambda \in \Lambda : n < |\lambda| \leq n+1\} \cdot n^{-2\kappa} \\
&\leq \text{constant} \cdot \sum_{n=0}^{\infty} n^{-(2\kappa-1)} < \infty
\end{aligned}$$

provided $\kappa > 1$.

If we scale the lattice Λ by $s \in \mathbb{C}^\times$ and insert $s\Lambda$ instead, we immediately find that

$$G_\kappa(s\Lambda) = \sum\nolimits_{0\neq\lambda\in\Lambda} (s\lambda)^{-2\kappa} = s^{-2\kappa} G_\kappa(\Lambda).$$

Specializing to lattices of the form $\Lambda_z = \mathbb{Z} \oplus \mathbb{Z}z$, with $z \in \mathcal{H}$, gives the weight 2κ transformation property of the function $G_\kappa(z) := G_\kappa(\Lambda_z)$.

[2] The famous Gauss circle problem asks whether the error $O(R)$ can be improved to $O(R^{\frac{1}{2}+\epsilon})$ for the lattice $\mathbb{Z} \oplus \mathbb{Z}i$.

Proposition 5.32 *$G_\kappa(z) = \sum_{(0,0)\neq(m,n)\in\mathbb{Z}^2}(mz+n)^{-2\kappa}$ defines an element of $M_{2\kappa}(SL_2(\mathbb{Z}))$ for all $\kappa \in \mathbb{Z}_{\geq 2}$. It has the Fourier expansion*

$$G_\kappa(z) = 2\zeta(2\kappa) + \frac{2(2\pi i)^{2\kappa}}{(2\kappa-1)!}\cdot\sum_{n=1}^{\infty}\sigma_{2\kappa-1}(n)e^{2\pi i n z}$$

where $\sigma_{2\kappa-1}(n) = \sum_{d|n} d^{2\kappa-1}$.

Proof Revisiting the above arguments shows the sum defining $G_\kappa(z)$ is uniformly convergent in compact subsets (since $\mathrm{vol}(\mathbb{C}/\Lambda_z)$ and $\mathrm{diam}(\mathbb{C}/\Lambda_z)$ are bounded when z varies in a compact subset). Therefore G_κ is a holomorphic function on $\mathcal{H}$. Also, for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ we have

$$G_\kappa(\gamma z) = G_\kappa(\Lambda_{\gamma z}) = G_\kappa((cz+d)^{-1}\Lambda_z) = (cz+d)^{2\kappa}G_\kappa(\Lambda_z) = (cz+d)^{2\kappa}G_\kappa(z)$$

since $\Lambda_{\gamma z} = (cz+d)^{-1}\Lambda_z$. Since $\mathrm{SL}_2(\mathbb{Z})$ only has the cusp ∞, to show G_κ is a modular form we must check $G_\kappa(z)$ is bounded as $\mathrm{Im}(z)\to\infty$. This follows from the Fourier expansion, to be proved next.

We first establish the following key identity for all $k \in \mathbb{Z}_{\geq 2}$,

$$\sum_{n\in\mathbb{Z}}\frac{1}{(z+n)^k} = (-1)^k\frac{(2\pi i)^k}{(k-1)!}\cdot\sum_{\ell=1}^{\infty}\ell^{k-1}q^\ell. \tag{5.1}$$

To prove this we start with the Weierstrass factorization formula for sine ([Rud87, Theorem 15.10, p. 303]):

$$\sin(\pi z) = \pi z\prod_{n=1}^{\infty}\left(1-(\frac{z}{n})^2\right).$$

Taking logarithmic derivatives on both sides results in the contangent series

$$H(z) := \pi\cot(\pi z) = \frac{1}{z} + \sum_{n=1}^{\infty}\left(\frac{1}{z+n}+\frac{1}{z-n}\right).$$

(Morally the right-hand side is the divergent series $\sum_{n\in\mathbb{Z}}\frac{1}{z+n}$. Regrouping the terms as above guarantees convergence.) Successive term-wise differentiation gives the formula

$$H^{(k-1)}(z) = (-1)^{k-1}(k-1)!\sum_{n\in\mathbb{Z}}\frac{1}{(z+n)^k}$$

for all $k \in \mathbb{Z}_{\geq 2}$. On the other hand, $H(z)$ has the Fourier expansion (with $q = e^{2\pi i z}$):

$$H(z) = \pi \frac{(e^{\pi i z} + e^{-\pi i z})/2}{(e^{\pi i z} - e^{-\pi i z})/2i} = \pi i \frac{q+1}{q-1} = \pi i + \frac{2\pi i}{q-1} = \pi i - 2\pi i \sum_{\ell=0}^{\infty} q^{\ell}.$$

Differentiating the q-expansion $k-1$ times yields another expression

$$H^{(k-1)}(z) = -(2\pi i)^k \sum_{\ell=1}^{\infty} \ell^{k-1} q^{\ell}$$

valid for $k \in \mathbb{Z}_{\geq 2}$. Comparing the two expressions for $H^{(k-1)}(z)$ immediately shows Eq. (5.1).

We now apply (5.1) to the Eisenstein series.

$$\begin{aligned} G_\kappa(z) &= \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} (mz+n)^{-2\kappa} \\ &= 2\zeta(2\kappa) + 2\sum_{m=1}^{\infty}\sum_{n\in\mathbb{Z}} (mz+n)^{-2\kappa} \\ &= 2\zeta(2\kappa) + \frac{2(2\pi i)^{2\kappa}}{(2\kappa-1)!}\sum_{m=1}^{\infty}\sum_{\ell=1}^{\infty} \ell^{2\kappa-1} q^{m\ell} \\ &= 2\zeta(2\kappa) + \frac{2(2\pi i)^{2\kappa}}{(2\kappa-1)!}\sum_{r=1}^{\infty}\Big(\sum_{\ell|r} \ell^{2\kappa-1}\Big) q^{r} \\ &= 2\zeta(2\kappa) + \frac{2(2\pi i)^{2\kappa}}{(2\kappa-1)!}\sum_{r=1}^{\infty} \sigma_{2\kappa-1}(r) q^{r} \end{aligned}$$

as desired. □

Combining this with the special value formula $\zeta(2\kappa) = (-1)^{\kappa+1}\frac{B_{2\kappa}(2\pi)^{2\kappa}}{2(2\kappa)!}$ from Corollary 1.35 easily gives the following reformulation.

Corollary 5.33 $E_\kappa(z) := \frac{1}{2\zeta(2\kappa)} G_\kappa(z) = 1 - \frac{4\kappa}{B_{2\kappa}} \cdot \sum_{n=1}^{\infty} \sigma_{2\kappa-1}(n) e^{2\pi i n z}$.

Proof Our task is to check that

$$\frac{2(2\pi i)^{2\kappa}}{(2\kappa-1)!} \cdot \frac{1}{2\zeta(2\kappa)} = \frac{2(2\pi i)^{2\kappa}}{(2\kappa-1)!} \cdot \frac{2(2\kappa)!}{(-1)^{\kappa+1} 2 B_{2\kappa} (2\pi)^{2\kappa}} = -\frac{4\kappa}{B_{2\kappa}}.$$

(We leave the gritty details of the last step to the reader.) □

We remind the reader that, by our convention in Sect. 1.1.9, $B_{2\kappa}$ can be negative. This differs from [Ser73, Sect. 4.1] for example, and we advise the reader to be cautious.

Note that the normalized Eisenstein series $E_\kappa \in M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z}))$ has constant term one, and all its other Fourier coefficients are rational numbers. We often confuse

$q = e^{2\pi i z}$ with an indeterminate and identify E_κ with a formal power series in $\mathbb{Q}[\![q]\!]$. For certain primes p we even have good p-integrality properties, as we now expound.

The denominator of $B_{2\kappa}$ (written in the lowest terms) is completely understood by the celebrated von Staudt-Clausen theorem from elementary number theory:

Theorem 5.34 (Von Staudt-Clausen) $B_{2\kappa} + \sum_{p:(p-1)|2\kappa} \frac{1}{p} \in \mathbb{Z}$ *for all* $\kappa \geq 1$.

Proof This is [HW79, Theorem 11.8, p. 91]–with a slick proof in [HW79, Sect. 7.10]. To aid comparison note that our $B_{2\kappa}$ equals $\beta_{2\kappa}$ in the notation of [HW79]. (Their B_κ satisfies $\beta_{2\kappa} = (-1)^{k-1} B_\kappa$.) □

The denominator of $B_{2\kappa}$ is therefore the product of all primes p for which $p-1$ divides 2κ. In particular the denominator is always a square-free multiple of 6. For example,

- $B_2 = \frac{1}{6} = \frac{1}{2\cdot 3}$ and $2 \equiv 0$ modulo $p-1$ for $p \in \{2, 3\}$;
- $B_4 = -\frac{1}{30} = -\frac{1}{2\cdot 3\cdot 5}$ and $4 \equiv 0$ modulo $p-1$ for $p \in \{2, 3, 5\}$;
- $B_6 = \frac{1}{42} = \frac{1}{2\cdot 3\cdot 7}$ and $6 \equiv 0$ modulo $p-1$ for $p \in \{2, 3, 7\}$;
- $B_8 = -\frac{1}{30} = -\frac{1}{2\cdot 3\cdot 5}$ and $8 \equiv 0$ modulo $p-1$ for $p \in \{2, 3, 5\}$;
- $B_{10} = \frac{5}{66} = \frac{5}{2\cdot 3\cdot 11}$ and $10 \equiv 0$ modulo $p-1$ for $p \in \{2, 3, 11\}$;
- $B_{12} = -\frac{691}{2730} = -\frac{691}{2\cdot 3\cdot 5\cdot 7\cdot 13}$ and $12 \equiv 0$ modulo $p-1$ for $p \in \{2, 3, 5, 7, 13\}$.

We apply this to E_κ, viewed as a power series in $\mathbb{Q}[\![q]\!]$. For which p does E_κ belong to $\mathbb{Z}_{(p)}[\![q]\!]$? (We remind the reader that $\mathbb{Z}_{(p)}$ is the localization of $\mathbb{Z}$ at (p), i.e. the subring of fractions $\frac{a}{b}$ with $p \nmid b$.)

Proposition 5.35 *Suppose* $\kappa \geq 2$ *and let* p *be a prime number such that* $2\kappa \equiv 0$ *modulo* $p-1$. *Then* $E_\kappa \in \mathbb{Z}_{(p)}[\![q]\!]$. *Furthermore,*

$$E_\kappa \equiv 1 \text{ modulo } p\mathbb{Z}_{(p)}[\![q]\!].$$

Proof Let v denote the p-adic valuation on $\mathbb{Q}$, i.e. $v(p^r \frac{a}{b}) = r$ if $p \nmid ab$. It has the property that

$$v(x_1 + \cdots + x_N) \geq \min\{v(x_1), \ldots, v(x_N)\},$$

and this inequality is an equality when the minimum is achieved exactly once (meaning there is a j such that $v(x_j) < v(x_i)$ for all $i \neq j$). By Theorem 5.34 we may write

$$B_{2\kappa} = i + \frac{1}{p} + \sum_{\ell \neq p:(\ell-1)|2\kappa} \frac{1}{\ell}$$

with $i \in \mathbb{Z}$. As $v(i) \geq 0$ and $v(\frac{1}{\ell}) = 0$ we infer that $v(B_{2\kappa}) = v(\frac{1}{p}) = -1$ by the initial observation ($\frac{1}{p}$ is the only term in the sum with a negative p-adic valuation). We apply this to the nth Fourier coefficient $c_n := -\frac{4\kappa}{B_{2\kappa}} \sigma_{2\kappa-1}(n)$ of E_κ, simply using that -4κ and $\sigma_{2\kappa-1}(n)$ are integers,

$$v(c_n) = v\big(-\frac{4\kappa}{B_{2\kappa}}\sigma_{2\kappa-1}(n)\big) \geq v(\frac{1}{B_{2\kappa}}) = 1.$$

This not only shows that c_n is p-integral, but indeed that $c_n \in p\mathbb{Z}_{(p)}$ for $n \geq 1$ as claimed. □

This result plays a central role when associating a Galois representation to a modular form f of weight one. It gives a congruence modulo p between f and fE_κ (for κ such that $p-1$ divides 2κ). The point of this is fE_κ is now a modular form of weight $1+2\kappa$ and the same level as f. Increasing the weight this way gives access to Galois representations arising from the arithmetic geometry of modular curves.

Here is the first handful of Eisenstein series written out explicitly:

- $E_2(z) = 1 + 240 \cdot \sum_{n=1}^{\infty} \sigma_3(n)q^n$;
- $E_3(z) = 1 - 504 \cdot \sum_{n=1}^{\infty} \sigma_5(n)q^n$;
- $E_4(z) = 1 + 480 \cdot \sum_{n=1}^{\infty} \sigma_7(n)q^n$;
- $E_5(z) = 1 - 264 \cdot \sum_{n=1}^{\infty} \sigma_9(n)q^n$;
- $E_6(z) = 1 + \frac{65520}{691} \cdot \sum_{n=1}^{\infty} \sigma_{11}(n)q^n$.

Note that $65520 = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13$. In particular E_6 is p-integral for all $p \neq 691$. Also, $E_6 \equiv 1$ modulo $p\mathbb{Z}_{(p)}$ for $p \in \{2, 3, 5, 7, 13\}$. We also note that 691 is the first irregular prime to appear in the numerator of a Bernoulli number.

We close this section with a few facts proved in [Ser73, Sect. 3.2]. First of all, since $\mathrm{SL}_2(\mathbb{Z})$ only has the cusp ∞ it is obvious that

$$\begin{aligned} M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z})) &= S_{2\kappa}(\mathrm{SL}_2(\mathbb{Z})) \oplus \mathbb{C}E_\kappa \\ f &= (f - f(\infty)E_\kappa) + f(\infty)E_\kappa, \end{aligned}$$

where $f(\infty)$ denotes the constant term of the Fourier expansion of f. By part (ii) of [Ser73, Theorem 4, p. 88] we have no nonzero cusp forms for $\mathrm{SL}_2(\mathbb{Z})$ of weight less than 12, so

$$M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}E_\kappa, \quad \kappa = 2, 3, 4, 5.$$

Moreover, by [Ser73, Corollary 2, p. 89] the algebra of modular forms $\bigoplus_{\kappa\geq 0} M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z}))$ is generated by E_2 and E_3. More precisely, the ring homomorphism

$$\begin{aligned} \mathbb{C}[X, Y] &\longrightarrow \bigoplus_{\kappa\geq 0} M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z})) \\ X &\longmapsto E_2 \\ Y &\longmapsto E_3 \end{aligned}$$

is an isomorphism. Since $E_2^m E_3^n$ has weight $4m + 6n$ this essentially says that $M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z}))$ has a basis consisting of the monomials $E_2^m E_3^n$ with $m, n \in \mathbb{Z}_{\geq 0}$ such that $2m + 3n = \kappa$. For $\kappa = 4, 5$ this forces the relations

$$E_4 = E_2^2, \quad E_5 = E_2 E_3.$$

(Each is true up to a constant for dimension reasons. The constant must be one by evaluating both sides at ∞.)

Exercise 5.36 Reformulate the relation $E_4 = E_2^2$ as a recursive formula expressing σ_7 in terms of σ_3. Similarly, rephrase the relation $E_5 = E_2 E_3$ as an expression for σ_9 in terms of σ_3 and σ_5.

5.6.2 The Modular Discriminant

The first cusp form for $\mathrm{SL}_2(\mathbb{Z})$ is the modular discriminant

$$\Delta(z) := (2\pi)^{12} \cdot \frac{E_2(z)^3 - E_3(z)^2}{1728}.$$

Since $E_2^3, E_3^2 \in M_{12}(\mathrm{SL}_2(\mathbb{Z}))$ both have constant term one, it is clear that $\Delta \in S_{12}(\mathrm{SL}_2(\mathbb{Z}))$. Including the extra fudge factors $(2\pi)^{12}$ and $1728 = 12^3 = 2^6 \cdot 3^3$ turns out to be a convenient normalization. Since E_2 and E_3 are algebraically independent, it is clear that Δ is not identically zero. What is not at all obvious is the fact that $\Delta(z)$ is nonzero for all $z \in \mathcal{H}$. Perhaps the best way to see this is via elliptic curves and the Weierstrass $\wp$-function.

We first rewrite $\Delta(z)$ in terms of $g_2(z) := 60G_2(z)$ and $g_3(z) := 140G_3(z)$.

Lemma 5.37 $\Delta(z) = g_2(z)^3 - 27g_3(z)^2$.

Proof Note that $E_2(z) = \frac{1}{2\zeta(4)} G_2(z) = \frac{45}{\pi^4} G_2(z) = \frac{3}{4}\pi^{-4} g_2(z)$ and similarly $E_3(z) = \frac{27}{8}\pi^{-6} g_3(z)$. So,

$$E_2(z)^3 - E_3(z)^2 = \pi^{-12}\big((\frac{3}{4})^3 g_2(z)^3 - (\frac{27}{8})^2 g_3(z)^2\big) = (2\pi)^{-12} \cdot 1728 \cdot \big(g_2(z)^3 - 27 g_3(z)^2\big).$$

The reader may want to spell out the gory details of the last step. □

Conceptually this says $16\Delta(z)$ is the discriminant of the cubic polynomial $4X^3 - g_2(z)X - g_3(z)$. In particular, $\Delta(z) \neq 0$ precisely when this cubic has no roots with multiplicity greater than one. This happens exactly when the projective curve in $\mathbb{P}^2$ defined by the equation $Y^2 = 4X^3 - g_2(z)X - g_3(z)$ has no singularities. (Of course, the curve in question is really defined by the associated homogeneous equation $Y^2Z = 4X^3 - g_2(z)XZ^2 - g_3(z)Z^3$–see [Sil09, Chap. 3].) We analyze the curve by relating it to the complex torus $\mathbb{C}/\Lambda_z$, by employing the Weierstrass $\wp$-function and its derivative.

Instead of Λ_z we prefer to work with an arbitrary lattice $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 \subset \mathbb{C}$ and the complex torus $\mathbb{C}/\Lambda$. In order to embed $\mathbb{C}/\Lambda$ into $\mathbb{P}^2$ we need Λ-periodic meromorphic functions on $\mathbb{C}$. Poles are inevitable: An entire Λ-periodic function is bounded and therefore constant by Liouville's theorem.

Definition 5.38 To a lattice $\Lambda \subset \mathbb{C}$ we attach the following meromorphic function on $\mathbb{C}$,

$$\wp_\Lambda(w) = \frac{1}{w^2} + \sum_{0 \neq \lambda \in \Lambda} \left(\frac{1}{(w-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

This series converges absolutely for all $w \in \mathbb{C} - \Lambda$, uniformly on compact subsets. For the benefit of the reader we include the estimates here. Suppose $|w| \leq R$ and $|\lambda| > 2R$. Then

$$\begin{aligned}
\left| \frac{1}{(w-\lambda)^2} - \frac{1}{\lambda^2} \right| &= \left| \frac{2w\lambda - w^2}{(w-\lambda)^2 \lambda^2} \right| \\
&< \frac{|2w\lambda - w^2|}{|\frac{1}{2}\lambda|^2 |\lambda|^2} \\
&= \frac{4|2w\lambda - w^2|/|\lambda|}{|\lambda|^3} \\
&\leq \frac{4(2R|\lambda| + R^2)/|\lambda|}{|\lambda|^3} \\
&= \frac{4R(2 + R/|\lambda|)}{|\lambda|^3} \\
&< \frac{4R(2 + \frac{1}{2})}{|\lambda|^3} \\
&= \frac{10R}{|\lambda|^3}.
\end{aligned}$$

In the second step we used the triangle inequality $|w - \lambda| \geq |\lambda| - |w| > \frac{1}{2}|\lambda|$. The way we proved the convergence of $G_\kappa(\Lambda)$ now carries over: For a fixed $w \in \mathbb{C} - \Lambda$ take $R = |w|$ above. This results in

$$\begin{aligned}
\sum\nolimits_{\lambda \in \Lambda : |\lambda| > 2|w|} \left| \frac{1}{(w-\lambda)^2} - \frac{1}{\lambda^2} \right| &= \sum\nolimits_{n \geq 2|w|} \sum\nolimits_{\lambda \in \Lambda : n < |\lambda| \leq n+1} \left| \frac{1}{(w-\lambda)^2} - \frac{1}{\lambda^2} \right| \\
&< \sum\nolimits_{n \geq 2|w|} \#\{\lambda \in \Lambda : n < |\lambda| \leq n+1\} \cdot \frac{10|w|}{n^3} \\
&< \text{constant} \cdot \sum\nolimits_{n \geq 2|w|} n^{-2} < \infty.
\end{aligned}$$

Minor modifications of this argument show the series converges uniformly when w varies in a compact subset of $\mathbb{C} - \Lambda$. Therefore at least $\wp_\Lambda$ is holomorphic here. On

the other hand, the defining series shows each $\lambda \in \Lambda$ is a double pole of $\wp_\Lambda$ (with residue 0).

Lemma 5.39 *$\wp_\Lambda$ is even and Λ-periodic.*

Proof Clearly $\wp_\Lambda(-w) = \wp_\Lambda(w)$ by the substitution $\lambda \mapsto -\lambda$ in the defining sum. We can differentiate term-wise (by the uniform convergence in compact subsets) so that

$$\wp'_\Lambda(w) = -\sum_{\lambda\in\Lambda} \frac{2}{(w-\lambda)^3}.$$

Evidently the derivative $\wp'_\Lambda$ is Λ-periodic, so for each $\lambda \in \Lambda$ there is a constant c_λ such that $\wp_\Lambda(w+\lambda) = \wp_\Lambda(w) + c_\lambda$ for all $w \in \mathbb{C} - \Lambda$. Here we may assume $\lambda \in \{\omega_1, \omega_2\}$ to ensure $\frac{1}{2}\lambda \notin \Lambda$. Taking $w = -\frac{1}{2}\lambda$ gives rise to the identity $\wp_\Lambda(\frac{1}{2}\lambda) = \wp_\Lambda(-\frac{1}{2}) + c_\lambda$, which shows $c_\lambda = 0$ since $\wp_\Lambda$ is even. □

Exercise 5.40 This gives the Laurent expansion of $\wp_\Lambda$ at the origin. Show that for all $w \in \mathbb{C} - \Lambda$,

$$\wp_\Lambda(w) = \frac{1}{w^2} + \sum_{n=1}^{\infty}(2n+1)G_{n+1}(\Lambda)w^{2n}.$$

(`Hint`: Start by writing $\frac{1}{(w-\lambda)^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}\left[\frac{1}{(1-\frac{w}{\lambda})^2} - 1\right] = \sum_{r=1}^{\infty}(r+1)w^r/\lambda^{r+2}$ for $\left|\frac{w}{\lambda}\right| < 1$. Now observe that the sum $\sum_{0\neq\lambda\in\Lambda}\frac{1}{\lambda^{r+2}}$ vanishes for odd r, using the substitution $\lambda \mapsto -\lambda$.)

The Weierstrass $\wp$-function of Λ satisfies the following fundamental differential equation.

Proposition 5.41 $\wp'_\Lambda(w)^2 = 4\wp_\Lambda(w)^3 - g_2(\Lambda)\wp_\Lambda(w) - g_3(\Lambda)$.

Proof We first emphasize that $g_2(\Lambda) := 60G_2(\Lambda)$ and $g_3(\Lambda) := 140G_3(\Lambda)$. By Exercise 5.40,

- $\wp'_\Lambda(w)^2 = \frac{4}{w^6} - 24G_2(\Lambda)\frac{1}{w^2} - 80G_3(\Lambda) + O(w)$;
- $\wp_\Lambda(w) = \frac{1}{w^2} + 3G_2(\Lambda)w^2 + 5G_3(\Lambda)w^4 + O(w^6)$;
- $\wp_\Lambda(w)^3 = \frac{1}{w^6} + 9G_2(\Lambda)\frac{1}{w^2} + 15G_3(\Lambda) + O(w)$.

From these asymptotics as $w \to 0$ we derive that

$$D(w) := \wp'_\Lambda(w)^2 - 4\wp_\Lambda(w)^3 + g_2(\Lambda)\wp_\Lambda(w) + g_3(\Lambda) = O(w).$$

(We encourage the reader to check the last equality to see how the singular parts and the constant terms cancel out.) Thus D is a Λ-periodic function with a zero at the origin. Hence all the points of Λ are zeroes of D. In particular D is an *entire* Λ-periodic function and therefore constant by Liouville's theorem. We conclude that D is identically zero, which is exactly the differential equation we sought. □

We continue with a couple of other key features of $\wp_\Lambda$.

Proposition 5.42 *The function $\wp_\Lambda : \mathbb{C} - \Lambda \longrightarrow \mathbb{C}$ is surjective. Furthermore, for $w, w' \in \mathbb{C} - \Lambda$,*

$$\wp_\Lambda(w) = \wp_\Lambda(w') \Longleftrightarrow w \equiv \pm w' \text{ modulo } \Lambda.$$

Proof Let $c \in \mathbb{C}$ be arbitrary. We will show the function $f(w) := \wp_\Lambda(w) - c$ has a zero in $\mathbb{C} - \Lambda$. For $\delta \in \mathbb{C}$ we introduce the parallelogram

$$P_\delta := \delta + \{a_1\omega_1 + a_2\omega_2 : 0 \leq a_1, a_2 \leq 1\}.$$

We choose a δ such that its boundary ∂P_δ contains no points from Λ (i.e., no poles of $\wp_\Lambda$). In particular there is a unique point from Λ in the interior $\overset{\circ}{P}_\delta$, say λ_0. If the reader prefers, take $\delta = -\frac{1}{2}(\omega_1 + \omega_2)$, in which case P_δ is centered at the origin and $\lambda_0 = 0$.

By the argument principle in complex analysis, $\frac{1}{2\pi i} \oint_{\partial P_\delta} \frac{f'(w)}{f(w)} dw = \mathcal{Z} - \mathcal{P}$ where $\mathcal{Z}$ is the number of zeroes of f in $\overset{\circ}{P}_\delta$, and $\mathcal{P}$ is the number of poles of f in $\overset{\circ}{P}_\delta$ (here the zeroes and poles are counted with multiplicities). This is a simple application of the residue theorem, noting that $\frac{f'}{f}$ has a simple pole at w^* with residue $\text{ord}_{w=w^*} f(w)$ when w^* is a zero or a pole of f.

Our specific f is Λ-periodic (by Lemma 5.39) so the contour integrals of $\frac{f'}{f}$ along opposite sides of P_δ cancel out, and as a result $\frac{1}{2\pi i} \oint_{\partial P_\delta} \frac{f'(w)}{f(w)} dw = 0$. In other words $\mathcal{Z} = \mathcal{P}$. We know $\wp_\Lambda$ has a double pole at each lattice point. Thus λ_0 is the only pole of f in $\overset{\circ}{P}_\delta$, and it has multiplicity two. We conclude that $\mathcal{Z} = \mathcal{P} = 2$, so f has a zero in $\overset{\circ}{P}_\delta$. This proves the surjectivity of $\wp_\Lambda$ as a function $\mathbb{C} - \Lambda \longrightarrow \mathbb{C}$.

The implication $\Longleftarrow$ in the statement is exactly Lemma 5.39. To show the other implication $\Longrightarrow$ suppose $\wp_\Lambda(w_1) = \wp_\Lambda(w_2)$ for some $w_1, w_2 \in \mathbb{C} - \Lambda$. We adopt the setup from the proof of surjectivity with $c := \wp_\Lambda(w_2)$. Then $f(w_1) = f(w_2) = 0$.

We just found a zero of f in $\overset{\circ}{P}_\delta$, say w_0. As $\mathcal{Z} = 2$ there are two possible scenarios: Either w_0 has multiplicity two, or w_0 is simple and there is another simple zero of f in $\overset{\circ}{P}_\delta$. To examine the situation, observe that $f(-w_0) = 0$ since f is even. Of course $-w_0$ may not lie in $\overset{\circ}{P}_\delta$, but there is a unique $\widetilde{w_0} \in \overset{\circ}{P}_\delta$ such that $\widetilde{w_0} \equiv -w_0$ modulo Λ. Then $f(\widetilde{w_0}) = 0$. We return to the two scenarios:

- If $\widetilde{w_0} \neq w_0$, these are the only two zeroes of f in $\overset{\circ}{P}_\delta$ (and they are both simple). In this case we conclude that either $w_1 \equiv w_0$ or $w_1 \equiv \widetilde{w_0}$ modulo Λ. Similarly for w_2. From this we infer that indeed $w_1 \equiv \pm w_2$ modulo Λ.
- If $\widetilde{w_0} = w_0$, this is the only zero of f in $\overset{\circ}{P}_\delta$. To see this, it suffices to verify w_0 has multiplicity at least two, i.e. that $f'(w_0) = 0$. However, f' is an odd Λ-periodic function, so

$$f'(w_0) = -f'(-w_0) = -f'(\widetilde{w_0}) = -f'(w_0).$$

(In the last step we used the hypothesis $\widetilde{w_0} = w_0$.) In this case we conclude that both w_1 and w_2 must be $\equiv w_0$ modulo Λ, so certainly $w_1 \equiv w_2$ modulo Λ.

This completes the proof. □

As an immediate Corollary, $\wp_\Lambda$ has zeroes. If we pick a zero $w^\diamond$ then all other zeroes belong to $\pm w^\diamond + \Lambda$. In the early eighties Eichler and Zagier gave an explicit formula for $w^\diamond$ in [EZ82]. For the lattice Λ_z their formula is

$$w^\diamond = \frac{1}{2} + \frac{1}{2\pi i}\ln(5 + 2\sqrt{6}) + 144\pi i\sqrt{6}\int_z^{i\infty}(t - z)\Delta(t)E_6(t)^{-3/2}dt.$$

Here is a (less immediate) Corollary which was the motivation behind our $\wp_\Lambda$ detour.

Corollary 5.43 *Let $\omega_3 := \omega_1 + \omega_2$. The polynomial $4X^3 - g_2(\Lambda)X - g_3(\Lambda)$ has three distinct roots*

$$\wp_\Lambda\big(\tfrac{1}{2}\omega_1\big), \quad \wp_\Lambda\big(\tfrac{1}{2}\omega_2\big), \quad \wp_\Lambda\big(\tfrac{1}{2}\omega_3\big).$$

Consequently, $\Delta(z) \neq 0$ for all $z \in \mathcal{H}$.

Proof The listed values of $\wp_\Lambda$ are distinct by Proposition 5.42. For example, $\frac{1}{2}\omega_1 \not\equiv \pm\frac{1}{2}\omega_3$ modulo Λ implies $\wp_\Lambda\big(\frac{1}{2}\omega_1\big) \neq \wp_\Lambda\big(\frac{1}{2}\omega_3\big)$. To check these values are roots of the cubic polynomial note that the derivative $\wp'_\Lambda$ is an odd Λ-periodic function, and therefore

$$\wp'_\Lambda\big(\tfrac{1}{2}\omega_i\big) = -\wp'_\Lambda\big(-\tfrac{1}{2}\omega_i\big) = -\wp'_\Lambda\big(-\tfrac{1}{2}\omega_i + \omega_i\big) = -\wp'_\Lambda\big(\tfrac{1}{2}\omega_i\big) = 0$$

for $i = 1, 2, 3$. The differential equation in Proposition 5.41 now shows that indeed

$$4\wp_\Lambda\big(\tfrac{1}{2}\omega_i\big)^3 - g_2(\Lambda)\wp_\Lambda\big(\tfrac{1}{2}\omega_i\big) - g_3(\Lambda) = \wp'_\Lambda\big(\tfrac{1}{2}\omega_i\big)^2 = 0$$

for $i = 1, 2, 3$. The last claim follows from Lemma 5.37 which identifies $16\Delta(z)$ with the discriminant of the polynomial $4X^3 - g_2(z)X - g_3(z)$. □

From a more geometric point of view, Corollary 5.43 says the curve $C_\Lambda \subset \mathbb{P}^2$ defined by the equation

$$Y^2Z - \big(4X^3 - g_2(\Lambda)XZ^2 - g_3(\Lambda)Z^3\big) = 0$$

has no singularities. Let $\mathbb{A}^2 \subset \mathbb{P}^2$ be the plane where $Z \neq 0$. Note that $C_\Lambda \cap \mathbb{A}^2$ is the set of points $[x : y : 1]$ satisfying $y^2 = 4x^3 - g_2(\Lambda)x - g_3(\Lambda)$ whereas $C_\Lambda \cap \{Z = 0\}$ consists of a single point $[0 : 1 : 0]$. To summarize, the *uniformization* map

$$\begin{aligned} \mathbb{C}/\Lambda &\longrightarrow C_\Lambda \\ w+\Lambda &\longmapsto [\wp_\Lambda(w) : \wp'_\Lambda(w) : 1] \quad (w \notin \Lambda) \\ 0+\Lambda &\longmapsto [0:1:0] \end{aligned}$$

is bijective. We recommend the reader checks this in detail, we only sketch the overall ideas:

- Starting with $[x : y : 1] \in C_\Lambda$ write $x = \wp_\Lambda(w)$ for some $w \in \mathbb{C} - \Lambda$. This forces the relation $y = \pm\wp'_\Lambda(w)$. Now replace w by $-w$ if necessary to arrange that $[x : y : 1] = [\wp_\Lambda(w) : \wp'_\Lambda(w) : 1]$.
- If $\wp_\Lambda(w_1) = \wp_\Lambda(w_2)$ and $\wp'_\Lambda(w_1) = \wp'_\Lambda(w_2)$ we know $w_1 \equiv \pm w_2 \bmod \Lambda$ from the first relation, and the sign must be a plus by the second relation. (Otherwise $\wp'_\Lambda(w_1) = \wp'_\Lambda(w_2) = 0$ shows $\wp_\Lambda(w_1) = \wp_\Lambda(w_2) = \wp_\Lambda\left(\frac{1}{2}\omega_i\right)$ for some i. Thus $w_1 \equiv \pm\frac{1}{2}\omega_i$ and similarly for w_2. Any sign combination leads to $w_1 \equiv w_2 \bmod \Lambda$.)

There is a natural way to view C_Λ as a complex manifold. Doing so, the uniformization map is an analytic isomorphism identifying C_Λ with the complex torus $\mathbb{C}/\Lambda$.

Remark 5.44 In continuation of the remarks at the end of Sect. 5.6.1, we mention without proof that $S_{12}(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}\Delta$. More generally, part (iii) of [Ser73, Theorem 4, p. 88] asserts that the multiplication by Δ map

$$\begin{aligned} M_k(\mathrm{SL}_2(\mathbb{Z})) &\longrightarrow S_{k+12}(\mathrm{SL}_2(\mathbb{Z})) \\ f &\longmapsto f\Delta \end{aligned}$$

is an isomorphism for all even $k \geq 0$.

We also wish to point out that Δ has an elegant product formula, due to Jacobi:

$$(2\pi)^{-12}\Delta(z) = \sum_{n=1}^{\infty} \tau(n) q^n = q \prod_{n=1}^{\infty} (1-q^n)^{24} = q - 24q^2 + 252q^3 - 1472q^4 + \cdots .$$

In particular, the Fourier coefficients $\tau(n)$ of the normalization $(2\pi)^{-12}\Delta(z)$ are integers. The Ramanujan τ-function has a slew of remarkable arithmetic properties. The Lehmer conjecture predicts that $\tau(n)$ is never zero.

The product formula is often written as $(2\pi)^{-12}\Delta(z) = \eta^{24}(z)$ where $\eta(z) := q^{\frac{1}{24}} \prod_{n=1}^{\infty}(1-q^n)$ is the Dedekind η-function. For an elementary proof of the product formula consult [Ser73, Theorem 6, p. 95].

5.6.3 The j-Invariant

This is not quite an example of a modular form, but almost. Since $\Delta(z) \neq 0$ for $z \in \mathcal{H}$ it makes sense to define

$$j(z) := 1728 \cdot \frac{g_2(z)^3}{\Delta(z)}.$$

This is a holomorphic function $j : \mathcal{H} \longrightarrow \mathbb{C}$. Moreover, since both g_2^3 and Δ are weight 12 modular forms for $\mathrm{SL}_2(\mathbb{Z})$, the j-function is $\mathrm{SL}_2(\mathbb{Z})$-invariant and can be thought of as a function

$$j : \mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \longrightarrow \mathbb{C}.$$

It is not hard to show this is a bijection, but we will not include the argument here. We refer the reader to part (c) of [Ser73, Proposition 5, p. 89] and its proof. Recall that the points of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ correspond bijectively to isomorphism classes of elliptic curves over $\mathbb{C}$. Thus a complex elliptic curve E_z is determined up to isomorphism by its j-invariant $j(E_z) = j(z)$.

What prevents j from being a weight zero modular form? It is not holomorphic at ∞. In fact j has a simple pole at ∞ with residue *one* (which explains the factor 1728 in its definition). To see this,

$$g_2(z) = 60G_2(z) = 60 \cdot 2\zeta(4) \cdot E_2(z) = 60 \cdot \frac{\pi^4}{45} \cdot E_2(z) = \frac{4}{3}\pi^4\big(1 + 240 \cdot \sum_{n=1}^{\infty} \sigma_3(n)q^n\big) \Longrightarrow$$
$$g_2(z)^3 = \frac{64}{27}\pi^{12}\big(1 + 720q + O(q^2)\big),$$

and

$$g_3(z) = 140G_3(z) = 140 \cdot 2\zeta(6) \cdot E_3(z) = 280 \cdot \frac{\pi^6}{945} \cdot E_3(z) = \frac{8}{27}\pi^6\big(1 - 504 \cdot \sum_{n=1}^{\infty} \sigma_5(n)q^n\big) \Longrightarrow$$
$$g_3(z)^2 = \frac{64}{729}\pi^{12}\big(1 - 1008q + O(q^2)\big).$$

Therefore $\Delta(z) = g_2(z)^3 - 27g_3(z)^2 = 1728 \cdot \frac{64}{27}\pi^{12}q + O(q^2) = (2\pi)^{12}q + O(q^2)$ and finally

$$j(z) = 1728 \cdot \frac{\frac{64}{27}\pi^{12}\big(1 + 720q + \cdots\big)}{(2\pi)^{12}q + \cdots} = \frac{1}{q} + O(1).$$

With more work, the first few terms of the Laurent expansion of j at ∞ can be shown to be

$$j(z) = \frac{1}{q} + 744 + 196884q + 21493760q^2 + 864299970q^3 + 20245856256q^4 + \cdots.$$

The coefficients of the q-series are positive integers, which are mysteriously linked to representations of the monster group through moonshine theory.

Exercise 5.45 Show that $j(i) = 1728$ and $j(\omega) = 0$. (`Hint`: Scale the series defining $G_3(i)$ and $G_2(\omega)$ by suitable powers of i and ω respectively to deduce both Eisenstein special values are zero.)

The two points on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$ given by (the orbits of) i and ω are examples of so-called Heegner points. By definition those are projections of points $z \in \mathcal{H}$ which belong to an imaginary quadratic subfield of $\mathbb{C}$. For such z the elliptic curve $E_z = \mathbb{C}/\Lambda_z$ has complex multiplication–which means its endomorphism algebra $\mathrm{End}(E_z)$ is larger than $\mathbb{Z}$. For instance, if z is an algebraic integer, multiplication by z is an endomorphism of E_z since $z^2 \in \mathbb{Z} \oplus \mathbb{Z}z = \Lambda_z$. At such points z the j-invariant is an algebraic integer. (There are three proofs of integrality in [Sil94, Chap. 2, Sect. 6]–see Theorem 6.1 there.)

This is only the beginning of a much longer story: Fix an imaginary quadratic subfield $K \subset \mathbb{C}$ and let E be a complex elliptic curve with $\mathrm{End}(E) \simeq \mathcal{O}_K$. There are exactly h_K such curves up to isomorphism (essentially because the homothety class of the lattice corresponds to an ideal class). Then

$$K(j(E)) = H_K = \text{the Hilbert class field of } K$$
$$(= \text{the maximal unramified abelian extension of } K).$$

Furthermore, $[\mathbb{Q}(j(E)) : \mathbb{Q}] = h_K$. For a very user-friendly exposition of these results we refer to [Sil09, Appendix C, Sect. 11] and [Sil94, Chap. 2, Sect. 4]. In the special case where $h_K = 1$ there is a unique curve E, up to isomorphism, with $\mathrm{End}(E) \simeq \mathcal{O}_K$. The j-invariant of this curve E is in $\mathbb{Z}$. A famous result of Baker, Heegner, and Stark, says there are exactly nine such fields:

$$K = \mathbb{Q}(\sqrt{-d}), \quad d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}.$$

The corresponding j-invariants are tabulated in [Sil94, p. 483]. Perhaps the most interesting example is the last Heegner number $d = 163$:

$$j\Big(\frac{1+\sqrt{-163}}{2}\Big) = -640320^3.$$

Note that $q = -e^{-\pi\sqrt{163}}$ when $z = \frac{1+\sqrt{-163}}{2}$. This $q \sim 10^{-18}$ is extremely small. Therefore, from the above q-series we read off that

$$-640320^3 = j\Big(\frac{1+\sqrt{-163}}{2}\Big) = -e^{\pi\sqrt{163}} + 744 + \text{extremely small}.$$

Thus the transcendental number $e^{\pi\sqrt{163}}$ is extremely close to being an integer! In fact

$$e^{\pi\sqrt{163}} = 262537412640768743.99999999992500725971981....$$

This is known as Ramanujan's constant. Bizarrely enough, its nearest integer $640320^3 + 744$ has a rather large prime factor. The prime factorization is $2^3 \cdot 3 \cdot 10939058860032031$.

5.6.4 Modular Forms Associated with Quadratic Forms

We first spell out the main results of Sect. 2.3.4 in the case where $V \simeq \mathbb{R}^N$ is an N-dimensional $\mathbb{R}$-vector space and $L \subset V$ is a discrete cocompact subgroup. Equivalently, L is a lattice in V, i.e. $L = \mathbb{Z}b_1 \oplus \cdots \oplus \mathbb{Z}b_N$ for an $\mathbb{R}$-basis $\{b_1, \ldots, b_N\}$ of V.

Once and for all we choose an inner product $\langle -, - \rangle$ on V (a symmetric, positive definite, bilinear form). This lets us identify the Pontryagin dual $\widehat{V}$ with V via the pairing

$$\begin{aligned} B : V \times V &\longrightarrow \mathbb{C}^1 \\ (x, y) &\longmapsto e^{2\pi i \langle x, y \rangle}. \end{aligned}$$

The corresponding dual lattice of L is

$$L^\perp = \{x \in V : \langle x, y \rangle \in \mathbb{Z},\ \forall y \in L\}.$$

We always endow V with the Lebesgue measure μ, and consider the quotient measure $\dot{\mu}$ on V/L. (Saying μ is Lebesgue means a unit box, with respect to any orthonormal basis, has measure one.) The resulting Fourier transformation $\widehat{f}$ of an integrable function $f \in L^1(V)$ is given by the formula

$$\widehat{f}(y) = \int_V f(x) e^{-2\pi i \langle x, y \rangle} d\mu(x).$$

A key tool in this section will be the Poisson summation formula in this setup:

Corollary 5.46 (Poisson's formula for V) *For every Schwartz function $f \in \mathcal{S}(V)$ we have the identity*

$$\sum\nolimits_{\ell \in L} f(\ell) = \frac{1}{\dot{\mu}(V/L)} \sum\nolimits_{y \in L^\perp} \widehat{f}(y).$$

Proof This was proved more generally for any locally compact abelian group in Sect. 2.3.4. □

When we established the functional equation for the Riemann zeta function, we used Poisson's formula for $\mathbb{R}$ to first obtain a transformation property of the function

$\theta(z) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 z}$. Here our primary interest is the following generalization of the θ-series,

$$\theta_L(z) = \sum_{\ell\in L} e^{\pi i\langle \ell,\ell\rangle z}$$

with $z \in \mathcal{H}$. For certain lattices L this will be shown to be a modular form. The next transformation property tells us how θ_L and $\theta_{L^\perp}$ are related via the linear fractional transformation $z \mapsto -\frac{1}{z}$ defined by the matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$.

Lemma 5.47 *$\theta_{L^\perp}(\frac{i}{t}) = \dot{\mu}(V/L)t^{\frac{N}{2}}\theta_L(it)$ for all $t > 0$.*

Proof When t is positive, $\theta_L(it) = \sum_{\ell\in L} f(\ell)$ with $f(x) = e^{-\pi t\langle x,x\rangle}$. To compute $\widehat{f}$ choose an orthonormal basis $\{v_1, \ldots, v_N\}$ for V and write $x = x_1v_1 + \cdots + x_Nv_N$. Then $f(x) = e^{-\pi t(x_1^2+\cdots+x_N^2)}$ and, by reducing to the $N = 1$ case, we have

$$\begin{aligned}
\widehat{f}(y) = \widehat{f}(y_1v_1 + \cdots + y_Nv_N) &= \int_{\mathbb{R}^n} e^{-\pi t(x_1^2+\cdots+x_N^2)} e^{-2\pi i(x_1y_1+\cdots+x_Ny_N)} dx_1\cdots dx_N \\
&= \prod_{j=1}^{N} \int_{\mathbb{R}} e^{-\pi t x_j^2} e^{-2\pi i x_j y_j} dx_j \\
&= \prod_{j=1}^{N} \frac{1}{\sqrt{t}} e^{-\pi y_j^2/t} \\
&= t^{-N/2} e^{-\pi\langle y,y\rangle/t}
\end{aligned}$$

as we normalize μ such that it corresponds to the Lebesgue measure on $\mathbb{R}^N$. On the right-hand side of Poisson's formula we now have the sum $\sum_{y\in L^\perp}\widehat{f}(y) = t^{-N/2}\theta_{L^\perp}(i/t)$. This gives the result. □

When $4|N$ we can restate the transformation property as follows:

$$\theta_{L^\perp}\Big(-\frac{1}{z}\Big) = \dot{\mu}(V/L)(iz)^{\frac{N}{2}}\theta_L(z)$$

for all $z \in \mathcal{H}$. (Indeed both sides are holomorphic functions on $\mathcal{H}$, and since $(-1)^{\frac{N}{2}} = 1$ they agree on the imaginary ray $i\mathbb{R}_{>0}$ as Lemma 5.47 shows.)

We now impose somewhat serious restrictions on what kind of lattices we allow from now on.

Assumption 5.48 Let V and $\langle -, -\rangle$ be as above. We assume $L \subset V$ is a lattice satisfying:

(i) $L = L^\perp$ (explicitly this means $x \in L \iff \langle x, L\rangle \subset \mathbb{Z}$);
(ii) $\langle \ell, \ell\rangle \in 2\mathbb{Z}$ for all $\ell \in L$.

Remark 5.49 As above, write $L = \mathbb{Z}b_1 \oplus \cdots \oplus \mathbb{Z}b_N$ for an $\mathbb{R}$-basis $\{b_1, \ldots, b_N\}$ of V. Then the choice of an inner product $\langle -, - \rangle$ is completely determined by the associated symmetric and positive definite $N \times N$-matrix

$$M := \big(\langle b_i, b_j \rangle\big)_{i,j=1,\ldots,N}.$$

In terms of this matrix, condition (i) translates into $M \in \mathrm{SL}_N(\mathbb{Z})$ (remember M is positive definite so $\det(M) > 0$). Knowing M is symmetric with integer entries, condition (ii) now translates into the diagonal entries $\langle b_i, b_i \rangle$ being even numbers.

The *quadratic form* defined by M is the homogeneous quadratic N-variable polynomial

$$Q_M(X_1, \ldots, X_N) = \sum_{i,j=1}^{N} \langle b_i, b_j \rangle X_i X_j \quad \in \mathbb{Z}[X_1, \ldots, X_N].$$

Employing this notation, the theta series can be rewritten as $\theta_L(z) = \sum_{(x_1,\ldots,x_N)\in\mathbb{Z}^N} e^{\pi i Q_M(x_1,\ldots,x_N)z}$.

Condition (i) implies $\dot{\mu}(V/L) = 1$ by applying Lemma 5.47 twice. Hence $\theta_L|_{\frac{N}{2}} \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = i^{\frac{N}{2}} \theta_L$ provided $4|N$. Condition (ii) ensures that $\theta_L|_{\frac{N}{2}} \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = \theta_L$. Also, of course $\theta_L|_{\frac{N}{2}} \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right) = \theta_L$ when $4|N$.

Proposition 5.50 *If there exists a lattice $L \subset V$ satisfying assumptions (i) and (ii) in 5.48 then $8|N$, where $N = \dim_{\mathbb{R}} V$. Moreover, θ_L is a modular form for $SL_2(\mathbb{Z})$ of weight $\frac{N}{2}$. It has Fourier expansion*

$$\theta_L(z) = \sum_{n=0}^{\infty} r_L(n) e^{2\pi i n z} = 1 + \sum_{n=1}^{\infty} r_L(n) e^{2\pi i n z}$$

where

$$r_L(n) = \#\{\ell \in L : \langle \ell, \ell \rangle = 2n\} = \#\{(x_1, \ldots, x_N) \in \mathbb{Z}^N : Q_M(x_1, \ldots, x_N) = 2n\}.$$

Proof As a preliminary, we recall the standard presentation of $\mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$. The quotient is generated by (the images of) $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ subject to the relations

$$S^2 = (ST)^3 = I$$

in $\mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$. This is [Ser73, Theorem 2, p. 78] and the remark immediately after its proof.

First we show how $N \equiv 4 \pmod 8$ leads to a contradiction. Equivalently, $i^{\frac{N}{2}} = -1$ so that $\theta_L|_{\frac{N}{2}} S = -\theta_L$. Thus also $\theta_L|_{\frac{N}{2}} ST = -\theta_L$, from which we deduce $\theta_L|_{\frac{N}{2}} (ST)^3 = -\theta_L$. However, $(ST)^3 \in \{\pm I\}$ acts trivially on θ_L. This is a contradiction since clearly θ_L is nonzero (indeed $\theta_L(it) > 0$ for $t > 0$).

If $N \equiv 1, 3, 5, 7 \pmod 8$ consider the lattice $L^{\oplus 4} \subset V^{\oplus 4}$ which satisfies (i) and (ii) since L does. Since $\dim_{\mathbb{R}}(V^{\oplus 4}) \equiv 4 \pmod 8$ the previous paragraph gives a contradiction. Similarly, if $N \equiv 2, 6 \pmod 8$ consider the lattice $L^{\oplus 2} \subset V^{\oplus 2}$ and use the fact that $\dim_{\mathbb{R}}(V^{\oplus 2}) \equiv 4 \pmod 8$ in this case. Either way this gives a contradiction unless $N \equiv 0 \pmod 8$.

Since $8|N$, i.e. $i^{\frac{N}{2}} = 1$, we now know $\theta_L|_{\frac{N}{2}} S = \theta_L$ and $\theta_L|_{\frac{N}{2}} T = \theta_L$. Because S and T generate $\mathrm{SL}_2(\mathbb{Z})$ modulo $\{\pm I\}$ (which also acts trivially on θ_L) we conclude that $\theta_L|_{\frac{N}{2}}\gamma = \theta_L$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. What remains is to check that $\theta_L(z)$ stays bounded as $\mathrm{Im}(z) \to \infty$, but clearly $\theta_L(z) \to 1$ since $\langle \ell, \ell \rangle > 0$ for $\ell \in L \backslash \{0\}$. So, indeed $\theta_L \in M_{\frac{N}{2}}(\mathrm{SL}_2(\mathbb{Z}))$.

The Fourier expansion is easily obtained by rearranging terms in the theta series,

$$\theta_L(z) = \sum_{n=0}^{\infty} \sum_{\ell \in L : \langle \ell, \ell \rangle = 2n} e^{2\pi i n z} = \sum_{n=0}^{\infty} \#\{\ell \in L : \langle \ell, \ell \rangle = 2n\} e^{2\pi i n z} = \sum_{n=0}^{\infty} r_L(n) e^{2\pi i n z}.$$

The final interpretation of $r_L(n)$ in terms of Q_M is obvious from Remark 5.49. □

Since θ_L and the Eisenstein series $E_{\frac{N}{4}}$ both have constant term one, their difference

$$\theta_L(z) - E_{\frac{N}{4}}(z) = \sum_{n=1}^{\infty} \big(r_L(n) + \frac{N}{B_{\frac{N}{2}}} \sigma_{\frac{N}{2}-1}(n)\big) e^{2\pi i n z}$$

is a weight $\frac{N}{2}$ *cusp* form for $\mathrm{SL}_2(\mathbb{Z})$. (We note that $8|N$ implies $\frac{N}{4} \in \mathbb{Z}_{\geq 2}$ so that $E_{\frac{N}{4}}$ is defined. For the same reason, $\frac{N}{2} \in 4\mathbb{Z}$ implies $B_{\frac{N}{2}} < 0$.) Whenever $S_{\frac{N}{2}}(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$ we get a truly remarkable explicit formula which expresses $r_L(n)$ in terms of $\sigma_{\frac{N}{2}-1}(n)$ and a Bernoulli number:

$$r_L(n) = \frac{N}{|B_{\frac{N}{2}}|} \sigma_{\frac{N}{2}-1}(n)$$

for all $n \geq 1$. It is not clear a priori that the right-hand side is even an integer for such N. Here are the first few $N \in 8\mathbb{Z}_{>0}$ for which this works:

- For $N = 8$, $r_L(n) = 240\sigma_3(n)$;
- For $N = 16$, $r_L(n) = 480\sigma_7(n)$.

For $N = 24$, all we can say is $\theta_L - E_6$ is a multiple of Δ. Since the Fourier coefficients of $(2\pi)^{-12}\Delta$ are the integers $\tau(n)$ we prefer to express $\theta_L - E_6$ as a multiple of the normalization $(2\pi)^{-12}\Delta$,

$$\theta_L - E_6 = \beta_L (2\pi)^{-12} \Delta, \quad \beta_L \in \mathbb{C}.$$

In other words, for all $n \geq 1$ we have the equality

$$r_L(n) = \frac{65520}{691}\sigma_{11}(n) + \beta_L \tau(n).$$

Recall that $B_{12} = -\frac{691}{2730}$. For $n = 1$ this yields $\beta_L = r_L(1) - \frac{65520}{691}$. In particular $\beta_L \in \mathbb{Q}^\times$ due to the fact that $r_L(1)$ is an integer but $\frac{65520}{691}$ is not.

Exercise 5.51 Show the inequalities $n^{2\kappa-1} \leq \sigma_{2\kappa-1}(n) \leq \zeta(2\kappa-1)n^{2\kappa-1}$ for all $n \geq 1$ and $\kappa > 1$. Deduce that the Fourier coefficients of E_κ have order of magnitude $O(n^{2\kappa-1})$ as $n \to \infty$.

For cusp forms the order of magnitude of the Fourier coefficients is significantly smaller, as we saw in Proposition 5.27. In general, for N such that $S_{\frac{N}{2}}(\mathrm{SL}_2(\mathbb{Z})) \neq \{0\}$ we thus get the crude asymptotics

$$r_L(n) = \frac{N}{|B_{\frac{N}{2}}|}\sigma_{\frac{N}{2}-1}(n) + O(n^{\frac{N}{4}}).$$

The leading term here has magnitude $O(n^{\frac{N}{2}-1})$.

5.7 The Automorphic Point of View

5.7.1 The NAK Decomposition and Invariant Measures

In $\mathrm{SL}_2(\mathbb{R})$ we have three natural subgroups

- $N = \{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) : x \in \mathbb{R}\}$;
- $A = \{\left(\begin{smallmatrix} \sqrt{y} & 0 \\ 0 & 1/\sqrt{y} \end{smallmatrix}\right) : y > 0\}$;
- $K = \mathrm{SO}_2(\mathbb{R}) = \{\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right) : \theta \in \mathbb{R}\}$.

They give a decomposition $\mathrm{SL}_2(\mathbb{R}) = NAK$. More precisely, the multiplication map $N \times A \times K \longrightarrow \mathrm{SL}_2(\mathbb{R})$ is a homeomorphism. To see this, note that every $g \in \mathrm{SL}_2(\mathbb{R})$ can be factored uniquely as

$$g = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} \sqrt{y} & 0 \\ 0 & 1/\sqrt{y} \end{smallmatrix}\right)\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$$

with $x \in \mathbb{R}$, $y > 0$, and $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. Why? Write $gi = x + iy = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} \sqrt{y} & 0 \\ 0 & 1/\sqrt{y} \end{smallmatrix}\right) i$ and recall that K is the isotropy group at i.

Exercise 5.52 Deduce from $\mathrm{SL}_2(\mathbb{R}) = NAK$ that K is a *maximal* compact subgroup of $\mathrm{SL}_2(\mathbb{R})$.

In these coordinates a Haar measure on $\mathrm{SL}_2(\mathbb{R})$ is $d\theta\frac{dxdy}{y^2}$ where $d\theta$ is the Lebesgue quotient measure on the circle group $\mathbb{R}/2\pi\mathbb{Z} \simeq K$. See [Lan85, p. 43]. The other part $\frac{dxdy}{y^2}$ is the invariant measure on $\mathcal{H}$.

Exercise 5.53 Verify that the measure $\frac{dxdy}{y^2}$ has the following invariance property. For all $g \in \mathrm{GL}_2(\mathbb{R})^+$ and any integrable function h on a measurable subset $F \subset \mathcal{H}$ we have the identity

$$\int_{gF} h(z)\frac{dxdy}{y^2} = \int_F h(gz)\frac{dxdy}{y^2}.$$

(`Hint`: Compute the Jacobian determinant of the change of variables $z \mapsto gz$ using the Cauchy-Riemann equations.)

Since $\mathrm{SL}_2(\mathbb{R})$ is unimodular we may endow $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$ with the quotient measure (which is $\mathrm{SL}_2(\mathbb{R})$-invariant). Our next goal is to identify $S_k(\Gamma)$ with a subspace of the Hilbert space $L^2(\Gamma\backslash\mathrm{SL}_2(\mathbb{R}))$.

5.7.2 *From Modular Forms to Automorphic Forms*

We start with an arbitrary function $f : \mathcal{H} \longrightarrow \mathbb{C}$ and a $k \in \mathbb{Z}$ which will remain fixed in the entire discussion. For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{R})^+$ we let

$$\phi_f(g) := (f|_k g)(i) = \det(g)^{\frac{k}{2}} j(g,i)^{-k} f(gi) = \det(g)^{\frac{k}{2}}(ci+d)^{-k} f(gi).$$

This function $\phi_f : \mathrm{GL}_2(\mathbb{R})^+ \longrightarrow \mathbb{C}$ is clearly invariant under $Z_\infty^+ = \{\left(\begin{smallmatrix} a & 0 \\ 0 & a \end{smallmatrix}\right) : a > 0\}$. Matrix multiplication gives an isomorphism of topological groups $Z_\infty^+ \times \mathrm{SL}_2(\mathbb{R}) \xrightarrow{\sim} \mathrm{GL}_2(\mathbb{R})^+$ and we interchangeably view ϕ_f as a function on $\mathrm{SL}_2(\mathbb{R})$. In general ϕ_f transforms as follows under right K-translations,

$$\phi_f(g\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)) = e^{-ik\theta}\phi_f(g).$$

This is straightforward to check by unwinding the definition of ϕ_f. Note that $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right) \mapsto e^{-ik\theta}$ gives a character of K (and every character has this form if we allow k to vary over all integers).

Let Γ be a discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$. We will always assume $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$ has finite measure.

- f has the weight k transformation property for Γ, i.e. $f|_k\gamma = f$ for all $\gamma \in \Gamma$, if and only if

  $$\phi_f(\gamma g) = \phi_f(g)$$

  for $\gamma \in \Gamma$. (We leave the details to the reader. For the "if" part use that $\mathrm{SL}_2(\mathbb{R})$ acts transitively on $\mathcal{H}$, so one only needs to check that $(f|_k\gamma)(gi) = f(gi)$ for all

$g \in \mathrm{SL}_2(\mathbb{R})$.). In other words, for such f the corresponding function ϕ_f factors through $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$.

- Suppose $(f|_k g)(z) = \sum_{n=0}^{\infty} a_n(\rho) e^{2\pi i n z/h(\rho)}$ with $g \in \mathrm{SL}_2(\mathbb{Z})$ fixed (and $\rho = g\infty$). We can recover the constant term $a_0(\rho)$ from ϕ_f by the formula

$$\begin{aligned}\int_0^1 \phi_f\big(g\left(\begin{smallmatrix}1 & xh(\rho)\\ 0 & 1\end{smallmatrix}\right)\big)dx &= \int_0^1 (f|_k g)(xh(\rho)+i)dx\\ &= \int_0^1 \sum_{n=0}^{\infty} a_n(\rho) e^{-2\pi n/h(\rho)} e^{2\pi i n x} dx\\ &= \sum_{n=0}^{\infty} a_n(\rho) e^{-2\pi n/h(\rho)} \int_0^1 e^{2\pi i n x} dx\\ &= a_0(\rho).\end{aligned}$$

  We conclude that f is cuspidal if and only if these integrals vanish for all $g \in \mathrm{SL}_2(\mathbb{Z})$.
- The holomorphicity of f is reflected by ϕ_f being an eigenfunction for the Laplace-Beltrami operator

$$\Delta := -y^2\big(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\big) - y\frac{\partial^2}{\partial x \partial \theta}$$

  on $\mathcal{C}^\infty(\mathrm{SL}_2(\mathbb{R}))$. In terms of the NAK factorization of g above, by using the cocycle condition of the automorphy factor j, we find that $j(g,i) = \frac{1}{\sqrt{y}} e^{i\theta}$. Hence $\phi_f(g) = e^{-ik\theta} y^{\frac{k}{2}} f(x+iy)$. An easy but tedious calculation, which we omit here, then gives the equation

$$\Delta\phi_f = -\frac{k}{2}\big(\frac{k}{2} - 1\big)\phi_f.$$

  The holomorphy of f enters the computation via the Cauchy-Riemann equations $\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y} = 0$, which in particular implies f is harmonic, i.e. $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$.

Exercise 5.54 Do the calculation and verify the relation $\Delta\phi_f = -\frac{k}{2}\big(\frac{k}{2} - 1\big)\phi_f$.

Remark 5.55 The Laplace-Beltrami operator Δ is, up to a constant, the differential operator defined by the Casimir element $\Omega \in U(\mathfrak{sl}_2(\mathbb{R}))$ which generates the center as an algebra. For our purposes the exact formula for Ω is not important, but we give it anyway. The Lie algebra $\mathfrak{sl}_2(\mathbb{R})$ of trace zero matrices has the following standard presentation. It is spanned by

$$X = \left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right), \quad Y = \left(\begin{smallmatrix}0 & 0\\ 1 & 0\end{smallmatrix}\right), \quad H = \left(\begin{smallmatrix}1 & 0\\ 0 & -1\end{smallmatrix}\right)$$

and these basis vectors satisfy the commutator relations

$$[X,Y] = H, \quad [H,Y] = -2Y, \quad [H,X] = 2X.$$

The Casimir element is $\Omega = XY + YX + \frac{1}{2}H^2$ (where XY, and so on, of course means multiplication in the universal enveloping algebra–not matrix multiplication). The main point here is that Ω belongs to the center of $U(\mathfrak{sl}_2(\mathbb{R}))$ which can easily be checked using the commutator relations. For example,

$$\begin{aligned}\Omega H &= XYH + YXH + \frac{1}{2}H^3 \\ &= X(HY + 2Y) + Y(HX - 2X) + \frac{1}{2}H^3 \\ &= (HX - 2X)Y + 2XY + (HY + 2Y)X - 2YX + \frac{1}{2}H^3 \\ &= HXY + HYX + \frac{1}{2}H^3 \\ &= H\Omega.\end{aligned}$$

We also remind the reader that a $\xi \in \mathfrak{sl}_2(\mathbb{R})$ defines an operator on smooth functions by exponentiating. Namely, for $\phi \in \mathcal{C}^\infty(\mathrm{SL}_2(\mathbb{R}))$ we let

$$(\xi\phi)(g) = \frac{d}{dt}\phi(g\exp(t\xi))|_{t=0}.$$

The formulas for the action of $\{X, Y, H\}$ in (x, y, θ)-coordinates are a bit complicated. We refer the interested reader to [Bum97, Proposition 2.2.5, p. 155] where it is also shown that Ω gives the operator Δ (up to a constant).

Exercise 5.56 Finish the verification that Ω lies in the center by checking that $\Omega X = X\Omega$ and $\Omega Y = Y\Omega$.

For $f \in S_k(\Gamma)$ we still need to check ϕ_f is square-integrable on $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$. More generally, let us start with two cusp forms $f_1, f_2 \in S_k(\Gamma)$ and relate the L^2-inner product $\langle\phi_{f_1}, \phi_{f_2}\rangle$ to the so-called *Petersson* inner product:

$$\langle f_1, f_2\rangle = \frac{1}{\mathrm{meas}(\Gamma\backslash\mathcal{H})}\int_{\Gamma\backslash\mathcal{H}} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}.$$

Here $\Gamma\backslash\mathcal{H}$ can safely be thought of as the choice of a nice enough fundamental domain $\mathcal{F}_\Gamma \subset \mathcal{H}$. Note that the integrand $f_1(z)\overline{f_2(z)}y^k$ is Γ-invariant, so the integral does not depend on the choice of $\mathcal{F}_\Gamma$ by the invariance of the measure $\frac{dxdy}{y^2}$ established in Exercise 5.53.

Since cusp forms decay exponentially at the cusps, the integral is absolutely convergent–assuming $\mathrm{meas}(\mathcal{F}_\Gamma)$ is finite. (The alert reader will note that it is enough to assume at least *one* of the modular forms, f_1 or f_2, is a cusp form.)

Proposition 5.57 *In the notation of the previous paragraph,* $\langle\phi_{f_1}, \phi_{f_2}\rangle = \langle f_1, f_2\rangle$. *(See the proof for the precise normalization of the* L^2*-inner product on the left-hand side.)*

Proof Let $dg = d\theta \frac{dxdy}{y^2}$ denote our choice of Haar measure on $\mathrm{SL}_2(\mathbb{R})$ and let $d\dot{g}$ be the quotient measure on $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$. We select a subset $\widetilde{\mathcal{F}}_\Gamma \subset \mathrm{SL}_2(\mathbb{R})$ such that $\mathrm{SL}_2(\mathbb{R}) = \bigcup_{\gamma\in\Gamma} \gamma\widetilde{\mathcal{F}}_\Gamma$ and all overlaps in this union have measure zero. We construct an $\widetilde{\mathcal{F}}_\Gamma$ as follows. Start with a fundamental domain $\mathcal{F}_\Gamma \subset \mathcal{H}$ as above, then introduce

$$\begin{aligned}\widetilde{\mathcal{F}}_\Gamma &:= \{g \in \mathrm{SL}_2(\mathbb{R}) : gi \in \mathcal{F}_\Gamma\} \\ &= \{\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}\sqrt{y} & 0\\ 0 & 1/\sqrt{y}\end{smallmatrix}\right)\left(\begin{smallmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{smallmatrix}\right) : x+iy \in \mathcal{F}_\Gamma,\ \theta \in \mathbb{R}\}.\end{aligned}$$

Unwinding the definitions we find that

$$\begin{aligned}\mathrm{meas}(\Gamma\backslash\mathrm{SL}_2(\mathbb{R}))\cdot\langle\phi_{f_1},\phi_{f_2}\rangle &:= \int_{\Gamma\backslash\mathrm{SL}_2(\mathbb{R})} \phi_{f_1}(g)\overline{\phi_{f_2}(g)}d\dot{g} \\ &= \int_{\widetilde{\mathcal{F}}_\Gamma} \phi_{f_1}(g)\overline{\phi_{f_2}(g)}dg \\ &= \int_{\widetilde{\mathcal{F}}_\Gamma} e^{-ik\theta}y^{\frac{k}{2}}f_1(x+iy)\overline{e^{-ik\theta}y^{\frac{k}{2}}f_2(x+iy)}d\theta\frac{dxdy}{y^2} \\ &= \int_{\widetilde{\mathcal{F}}_\Gamma} f_1(x+iy)\overline{f_2(x+iy)}y^k d\theta\frac{dxdy}{y^2} \\ &= \mathrm{meas}(K)\cdot\int_{\mathcal{F}_\Gamma} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2} \\ &= \mathrm{meas}(K)\cdot\mathrm{meas}(\Gamma\backslash\mathcal{H})\cdot\langle f_1, f_2\rangle.\end{aligned}$$

The exact same computation with $k = 0$ and $f_1 = f_2 = 1$ (constant functions) yields

$$\mathrm{meas}(\Gamma\backslash\mathrm{SL}_2(\mathbb{R})) = \mathrm{meas}(K)\cdot\mathrm{meas}(\Gamma\backslash\mathcal{H}).$$

Canceling this common factor on both sides gives the desired equality of inner products. □

Altogether the map $f \mapsto \phi_f$ gives an isometric embedding of $S_k(\Gamma)$ into the space of automorphic forms in $L^2(\Gamma\backslash\mathrm{SL}_2(\mathbb{R}))$. In general, an *automorphic form* on a semisimple Lie group G is a smooth function $\phi : G \to \mathbb{C}$ such that

- ϕ factors through $\Gamma\backslash G$ for some discrete subgroup Γ (for which $\Gamma\backslash G$ has finite measure);
- ϕ is K-finite, where K is a maximal compact subgroup. (This means the space of K-translates $\mathrm{span}_{\mathbb{C}}\{\kappa\phi : \kappa \in K\}$ is finite-dimensional. Here $(\kappa\phi)(g) = \phi(g\kappa)$.);
- ϕ is $Z(\mathfrak{g})$-finite, where $Z(\mathfrak{g})$ denotes the center of the universal enveloping algebra $U(\mathfrak{g})$ of the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$. (Analogously, this means the subspace $\mathrm{span}_{\mathbb{C}}\{\xi\phi : \xi \in Z(\mathfrak{g})\}$ is finite-dimensional.);
- ϕ has moderate growth–in a sense we will not make precise here.

In our case of $G = \mathrm{SL}_2(\mathbb{R})$ and $\phi = \phi_f$ the two finite-dimensional spaces above are in fact one-dimensional (as K acts on ϕ_f via the character $\theta \mapsto e^{-ik\theta}$ and ϕ_f is an eigenfunction for Δ, which generates $Z(\mathfrak{sl}_2(\mathbb{R}))$).

A key feature of $L^2(\Gamma\backslash \mathrm{SL}_2(\mathbb{R}))$ is the action of $\mathrm{SL}_2(\mathbb{R})$ by right translations, i.e. $(g\phi)(x) = \phi(xg)$. By the invariance of the quotient measure, the $\mathrm{SL}_2(\mathbb{R})$-action preserves the L^2-inner product:

$$\langle g\phi_1, g\phi_2\rangle = \langle \phi_1, \phi_2\rangle, \quad \phi_1, \phi_2 \in L^2(\Gamma\backslash \mathrm{SL}_2(\mathbb{R})). \tag{5.2}$$

Thus $L^2(\Gamma\backslash \mathrm{SL}_2(\mathbb{R}))$ is a unitary representation of $\mathrm{SL}_2(\mathbb{R})$. Similarly for any semisimple Lie group G. The use of representation theory is absolutely instrumental in the field of automorphic forms–especially when one goes beyond $\mathrm{SL}_2(\mathbb{R})$ and studies groups of higher rank.

Remark 5.58 In what follows it will occcasionally be more convenient to work with $Z_\infty^+\Gamma\backslash \mathrm{GL}_2(\mathbb{R})^+$ rather than $\Gamma\backslash \mathrm{SL}_2(\mathbb{R})$. Of course, the two quotients are canonically isomorphic, hence so are their L^2-spaces:

$$L^2(Z_\infty^+\Gamma\backslash \mathrm{GL}_2(\mathbb{R})^+) \xrightarrow{\sim} L^2(\Gamma\backslash \mathrm{SL}_2(\mathbb{R})).$$

The point of this is the source now carries a $\mathrm{GL}_2(\mathbb{R})^+$-action (albeit Z_∞^+ acts trivially) and the unitarity (5.2) can be rephrased as

$$\langle g\phi_1, g\phi_2\rangle = \langle \phi_1, \phi_2\rangle, \quad \phi_1, \phi_2 \in L^2(Z_\infty^+\Gamma\backslash \mathrm{GL}_2(\mathbb{R})^+),$$

but this now holds for all $g \in \mathrm{GL}_2(\mathbb{R})^+$.

5.7.3 *Invariance Properties of the Petersson Inner Product*

We start by clearing a potential pitfall. If $\Gamma' \subset \Gamma$ is a smaller finite index subgroup, we clearly have inclusions $M_k(\Gamma) \subset M_k(\Gamma')$ and $S_k(\Gamma) \subset S_k(\Gamma')$. Given $f_1, f_2 \in S_k(\Gamma)$ we may think of them as elements of the larger space $S_k(\Gamma')$ and take their Petersson inner product with respect to Γ'. However, we will check this coincides with the one relative to Γ, so there is no ambiguity. To see this, build a fundamental domain for Γ' out of $\mathcal{F}_\Gamma$ by setting

$$\mathcal{F}_{\Gamma'} := \bigcup_{\sigma \in \Gamma'\backslash\Gamma} \sigma\mathcal{F}_\Gamma$$

(with σ running over a set of right coset representatives for Γ' in Γ). Note that

$$\mathrm{meas}(\mathcal{F}_{\Gamma'}) = [\Gamma : \Gamma'] \cdot \mathrm{meas}(\mathcal{F}_\Gamma).$$

Consequently, for $f_1, f_2 \in S_k(\Gamma)$,

$$\begin{aligned}
\frac{1}{\mathrm{meas}(\Gamma'\backslash\mathcal{H})}\int_{\Gamma'\backslash\mathcal{H}} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2} &= \frac{1}{\mathrm{meas}(\mathcal{F}_{\Gamma'})}\int_{\mathcal{F}_{\Gamma'}} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}\\
&= \frac{1}{\mathrm{meas}(\mathcal{F}_{\Gamma'})}\sum_{\sigma\in\Gamma'\backslash\Gamma}\int_{\sigma\mathcal{F}_\Gamma} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}\\
&= \frac{1}{\mathrm{meas}(\mathcal{F}_{\Gamma'})}\sum_{\sigma\in\Gamma'\backslash\Gamma}\int_{\mathcal{F}_\Gamma} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}\\
&= \frac{1}{\mathrm{meas}(\mathcal{F}_{\Gamma'})}[\Gamma:\Gamma']\int_{\mathcal{F}_\Gamma} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}\\
&= \frac{1}{\mathrm{meas}(\mathcal{F}_\Gamma)}\int_{\mathcal{F}_\Gamma} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}\\
&= \frac{1}{\mathrm{meas}(\Gamma\backslash\mathcal{H})}\int_{\Gamma\backslash\mathcal{H}} f_1(z)\overline{f_2(z)}y^k\frac{dxdy}{y^2}.
\end{aligned}$$

In the third step we used Exercise 5.53 and the Γ-invariance of the expression $f_1(z)\overline{f_2(z)}y^k$. As a result the notation $\langle f_1, f_2\rangle$ is unambiguous–it is independent of the choice of a Γ such that $f_1, f_2 \in S_k(\Gamma)$.

The objective of this section is to convert the unitarity in Remark 5.58 into a property of the Petersson inner product. We begin with a precursory observation. Start with $f \in S_k(\Gamma)$ and let $g \in \mathrm{GL}_2(\mathbb{R})^+$ act on the right. The resulting form $f|_k g$ belongs to $S_k(g^{-1}\Gamma g)$. We will frequently view $f|_k g$ as an element of the possibly larger space $S_k(\Gamma \cap g^{-1}\Gamma g)$ which also contains $S_k(\Gamma)$. This point of view will be in effect in the next result.

Proposition 5.59 *Let $f_1, f_2 \in S_k(\Gamma)$ and $g \in GL_2(\mathbb{R})^+$ be arbitrary. Then*

$$\langle f_1|_k g, f_2\rangle = \langle f_1, f_2|_k g^{-1}\rangle.$$

(On the left-hand side we are pairing two forms in $S_k(\Gamma \cap g^{-1}\Gamma g)$, whereas on the right-hand side we view the two forms as elements of $S_k(\Gamma \cap g\Gamma g^{-1})$.)

Proof First note that $\phi_{f_1|_k g}(x) = (f_1|_k g|_k x)(i) = \phi_{f_1}(gx)$ and similarly $\phi_{f_2|_k g^{-1}}(x) = \phi_{f_2}(g^{-1}x)$. Therefore, by Proposition 5.57,

$$\begin{aligned}
\langle f_1|_k g, f_2\rangle &= \frac{1}{\mathrm{meas}(Z_\infty^+(\Gamma\cap g^{-1}\Gamma g)\backslash\mathrm{GL}_2(\mathbb{R})^+)}\int_{Z_\infty^+(\Gamma\cap g^{-1}\Gamma g)\backslash\mathrm{GL}_2(\mathbb{R})^+}\phi_{f_1}(gx)\overline{\phi_{f_2}(x)}d\dot{x}\\
&= \frac{1}{\mathrm{meas}(Z_\infty^+(\Gamma\cap g^{-1}\Gamma g)\backslash\mathrm{GL}_2(\mathbb{R})^+)}\int_{Z_\infty^+(\Gamma\cap g^{-1}\Gamma g)\backslash\mathrm{GL}_2(\mathbb{R})^+}\phi_{f_1}(gxg^{-1}g)\overline{\phi_{f_2}(x)}d\dot{x}\\
&= \frac{1}{\mathrm{meas}(Z_\infty^+(\Gamma\cap g\Gamma g^{-1})\backslash\mathrm{GL}_2(\mathbb{R})^+)}\int_{Z_\infty^+(\Gamma\cap g\Gamma g^{-1})\backslash\mathrm{GL}_2(\mathbb{R})^+}\phi_{f_1}(yg)\overline{\phi_{f_2}(g^{-1}yg)}d\dot{y}\\
&= \frac{1}{\mathrm{meas}(Z_\infty^+(\Gamma\cap g\Gamma g^{-1})\backslash\mathrm{GL}_2(\mathbb{R})^+)}\int_{Z_\infty^+(\Gamma\cap g\Gamma g^{-1})\backslash\mathrm{GL}_2(\mathbb{R})^+}\phi_{f_1}(y)\overline{\phi_{f_2}(g^{-1}y)}d\dot{y}\\
&= \langle f_1, f_2|_k g^{-1}\rangle.
\end{aligned}$$

The fourth step invoked the invariance of the quotient measure $d\dot{y}$, as discussed in Remark 5.58. The swapping of g and g^{-1} in the third step is justified by the fact that the substitution $x \mapsto gxg^{-1}$ gives a bijection

$$Z_\infty^+(\Gamma \cap g^{-1}\Gamma g)\backslash \mathrm{GL}_2(\mathbb{R})^+ \xrightarrow{\sim} Z_\infty^+(\Gamma \cap g\Gamma g^{-1})\backslash \mathrm{GL}_2(\mathbb{R})^+.$$

This proves the claim. □

5.7.4 *Strong Approximation*

In this section K will again denote a number field, not $\mathrm{SO}_2(\mathbb{R})$. We continue to denote its set of places by V. If $S \subset V$ is a finite subset we employ the notation $K_S = \prod_{v\in S} K_v$ and

$$\mathbb{A}_K^S = \prod_{v\notin S}{}' K_v = \{x = (x_v)_{v\notin S} \in \prod_{v\notin S} K_v : x_v \in \mathcal{O}_v \text{ for all but finitely many } v \notin S\}.$$

The weak approximation theorem ([Cas67, p. 48]) states that the image of the diagonal embedding $K \hookrightarrow K_S$ is dense. Let us unwind what this means: Given a family of elements $x_v \in K_v$ at each $v \in S$, and an $\epsilon > 0$, there exists a $\gamma \in K$ such that $|\gamma - x_v|_v < \epsilon$ for all $v \in S$.

The strong approximation theorem ([Cas67, p. 67]) gives the following strengthening: For $S \neq \varnothing$ the image of $K \hookrightarrow \mathbb{A}_K^S$ is dense. This amounts to the following: Let $S' \subset V$ be a finite subset disjoint from S. Given $x_v \in K_v$ for $v \in S'$, and an $\epsilon > 0$, there exists a $\gamma \in K$ such that

- $|\gamma - x_v|_v < \epsilon$ for all $v \in S'$, and
- $|\gamma|_v \leq 1$ for all $v \notin S \cup S'$.

(This implies weak approximation for S'. Note how the role of S has changed from the previous paragraph.) The case of a general $S \neq \varnothing$ can immediately be deduced from the singleton case $S = \{v_0\}$. We stress that the theorem of course fails for $S = \varnothing$, in which case the image of $K \hookrightarrow \mathbb{A}_K$ is discrete.

We encourage the reader to work out the following useful reformulation of the strong approximation theorem for the additive group $\mathbb{G}_a$ over K.

Exercise 5.60 Identify K_S and K with subsets of $\mathbb{A}_K$ and consider the set $K_S + K$ of all sums. (For K_S take adeles which are zero outside S.) Show that $K_S + K$ is dense in $\mathbb{A}_K$ if and only if the image of $K \hookrightarrow \mathbb{A}_K^S$ is dense.

In contrast, the strong approximation property does *not* hold for the multiplicative group $\mathbb{G}_m = \mathrm{GL}_1$. Taking $S = V_\infty$ to be the set of all archimedean places, we have seen that $K_\infty^\times K^\times \widehat{\mathcal{O}}_K^\times$ has index h_K in $\mathbb{I}_K$. Also, for $K = \mathbb{Q}$ the subgroup $\mathbb{R}^\times \mathbb{Q}^\times (1 +$

$N\widehat{\mathbb{Z}}$) has index greater than one in $\mathbb{I}$ for large enough N. Therefore $K_\infty^\times K^\times$ is not dense in $\mathbb{I}_K$. (Otherwise $x(1 + N\widehat{\mathcal{O}}_K^\times)$ would intersect $K_\infty^\times K^\times$ for all $x \in \mathbb{I}_K$, and all $N \geq 1$.)

For any linear algebraic group G over K (i.e., a subgroup of some GL_n cut out by polynomials in the matrix entries with coefficients in K) one can ask the same questions. In the sixties Kneser proved rather definitive results, thereby characterizing the pairs (G, S) satisfying the strong approximation property.

The weak approximation property holds in great generality. The density of $G(K) \hookrightarrow G(K_S)$ is true for all S under mild restrictions on G. Note that it holds for $\mathbb{G}_m$, for example.

For the reader who is familiar with the lingo of linear algebraic groups, we mention an important special case of Kneser's theorem: If G is a simply connected almost K-simple linear algebraic group over K, then the image of $G(K) \hookrightarrow G(\mathbb{A}_K^S)$ is dense for all finite sets S such that $G(K_S)$ is non-compact. This applies to SL_n, and symplectic groups, but not GL_n. (The latter is reductive, but not semisimple because it has center $\mathbb{G}_m$.)

We give a direct proof of Kneser's theorem in the SL_2-case.

Proposition 5.61 *The image of $SL_2(K) \hookrightarrow SL_2(\mathbb{A}_K^S)$ is dense for all finite sets $S \neq \varnothing$.*

Proof For the duration of this proof let C denote the closure of $\mathrm{SL}_2(K)$. As C is a closed subgroup of $\mathrm{SL}_2(\mathbb{A}_K^S)$ it suffices to show $\mathrm{SL}_2(K_v) \subset C$ for all $v \notin S$. Before we show this, let us elaborate a bit on this initial reduction step.

Let $g = (g_v)_{v \notin S} \in \mathrm{SL}_2(\mathbb{A}_K^S)$. If $S' \subset V$ is a finite set disjoint from S, we define the S'-truncation $g^{S'}$ by

$$g_v^{S'} = \begin{cases} g_v & v \in S' \\ I & v \notin S \cup S'. \end{cases}$$

From the way we topologize $\mathrm{SL}_2(\mathbb{A}_K^S)$, as a restricted direct product, it is clear that $g^{S'} \to g$ as $|S'| \to \infty$. (Check that indeed $g^{-1}g^{S'}$ belongs to any open box around the identity for sufficiently large S'.) As C is closed, it suffices to show $g^{S'} \in C$. As C is a subgroup, and $g^{S'}$ is a *finite* product of elements from $\mathrm{SL}_2(K_v)$ with $v \in S'$, it is enough to show $\mathrm{SL}_2(K_v) \subset C$ for all $v \notin S$.

Over any field F, the elementary matrices $\left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ * & 1 \end{smallmatrix}\right)$ generate $\mathrm{SL}_2(F)$–see [Lan02, Lemma 8.1, p. 537] or try this as an exercise in row and column operations. Now, for $v \notin S$ we have

$$\{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) : x \in K\} \subset C \Longrightarrow \{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) : x \in \mathbb{A}_K^S\} \subset C \Longrightarrow \{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) : x \in K_v\} \subset C$$

by using the density of $K \hookrightarrow \mathbb{A}_K^S$. Similarly for the matrices $\left(\begin{smallmatrix} 1 & 0 \\ * & 1 \end{smallmatrix}\right)$. This gives the result. □

Remark 5.62 The proof generalizes to SL_n, noting that $\mathrm{SL}_n(F)$ is generated by transvections for any field F. A transvection is an elementary matrix corresponding

to row addition, i.e. a matrix of the form $I + tE_{ij}$ where E_{ij} has (i, j)-entry 1 and all other entries equal to zero.

Equivalently, the subset of products $\mathrm{SL}_2(K_S)\mathrm{SL}_2(K)$ is dense in $\mathrm{SL}_2(\mathbb{A}_K)$. In our applications below we only use the special case $K = \mathbb{Q}$ and $S = V_\infty$, namely that $\mathrm{SL}_2(\mathbb{R})\mathrm{SL}_2(\mathbb{Q})$ is dense in $\mathrm{SL}_2(\mathbb{A})$. Thus, for any open subgroup $\mathcal{U} \subset \mathrm{SL}_2(\mathbb{A}_f)$ over the finite adeles $\mathbb{A}_f = \mathbb{A}^{V_\infty}$ we have the equality

$$\mathrm{SL}_2(\mathbb{A}) = \mathrm{SL}_2(\mathbb{Q})\mathrm{SL}_2(\mathbb{R})\mathcal{U}.$$

From this, one easily obtains a GL_2-version: For every open subgroup $U \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$ such that $\det(U) = \widehat{\mathbb{Z}}^\times$ we find that

$$\mathrm{GL}_2(\mathbb{A}) = \mathrm{GL}_2(\mathbb{Q})\mathrm{GL}_2(\mathbb{R})^+U.$$

(For $x \in \mathrm{GL}_2(\mathbb{A})$ we write $\det(x) = \det(\gamma)\det(g)\det(u)$ with $\gamma \in \mathrm{GL}_2(\mathbb{Q})$, $g \in \mathrm{GL}_2(\mathbb{R})^+$, $u \in U$, according to the factorization $\mathbb{Q}^\times \times \mathbb{R}^\times_{>0} \times \widehat{\mathbb{Z}}^\times \xrightarrow{\sim} \mathbb{I}$. Then factor the element $\gamma^{-1}xu^{-1}g^{-1} \in \mathrm{SL}_2(\mathbb{A})$ relative to $\mathcal{U} := U \cap \mathrm{SL}_2(\mathbb{A}_f)$ and note that elements in $\mathrm{GL}_2(\mathbb{R})$ and U commute.)

We associate a discrete subgroup $\Gamma = \Gamma_U \subset \mathrm{SL}_2(\mathbb{R})$ with U as follows. First consider the intersection $\mathrm{GL}_2(\mathbb{Q}) \cap \mathrm{GL}_2(\mathbb{R})^+U$ in $\mathrm{GL}_2(\mathbb{A})$ consisting of matrices $\gamma \in \mathrm{GL}_2(\mathbb{Q})$ with positive determinant such that $\gamma_f \in U$. Note that such a γ necessarily has determinant one since $\mathbb{Q}^\times \cap \mathbb{R}^\times_{>0}\widehat{\mathbb{Z}}^\times = \{1\}$. We take Γ to be the image of this intersection under the embedding $\mathrm{GL}_2(\mathbb{Q}) \hookrightarrow \mathrm{GL}_2(\mathbb{R})$, i.e.

$$\begin{aligned} \mathrm{GL}_2(\mathbb{Q}) \cap \mathrm{GL}_2(\mathbb{R})^+U &\xrightarrow{\sim} \Gamma \\ \gamma &\longmapsto \gamma_\infty. \end{aligned}$$

The corresponding quotient $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$ has a nice adelic double coset description: There are canonical $\mathrm{SL}_2(\mathbb{R})$-equivariant homeomorphisms,

$$\begin{aligned} \Gamma\backslash\mathrm{SL}_2(\mathbb{R}) \xrightarrow{\sim} Z^+_\infty\Gamma\backslash\mathrm{GL}_2(\mathbb{R})^+ &\xrightarrow{\sim} Z^+_\infty\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})/U \\ g &\longmapsto Z^+_\infty\mathrm{GL}_2(\mathbb{Q})gU. \end{aligned}$$

(We leave most details to the reader, but here is the idea. If $g, g' \in \mathrm{GL}_2(\mathbb{R})^+$ define the same double coset, write $g' = z\gamma gu$. Comparing components at ∞ we find that $g' = z\gamma_\infty g$. At the finite places we get $\gamma_f = u^{-1} \in U$. Thus $\gamma_\infty \in \Gamma$.)

Exercise 5.63 Suppose $U \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is an open subgroup containing the scalar matrices $\widehat{\mathbb{Z}}^\times I$. Then there is a natural bijection

$$Z^+_\infty\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})/U \xrightarrow{\sim} Z(\mathbb{A})\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})/U$$

where Z denotes the center of GL_2.

Although we will not use it, we briefly sketch what happens in the number field case. For simplicity we stick to SL_2 and GL_2, but the arguments apply more generally. Let $U \subset \mathrm{GL}_2(\widehat{\mathcal{O}}_K)$ be an open subgroup with $\det(U) = \widehat{\mathcal{O}}_K^\times$. Taking the determinant gives a bijection

$$\mathrm{GL}_2(K)\backslash \mathrm{GL}_2(\mathbb{A}_K)/\mathrm{GL}_2(K_\infty)U \xrightarrow{\sim} \mathbb{I}_K/K^\times K_\infty^\times \widehat{\mathcal{O}}_K^\times \xrightarrow{\sim} \mathrm{Cl}_K.$$

Pick double coset representatives $\{\sigma_1, \ldots, \sigma_h\}$ in $\mathrm{GL}_2(\mathbb{A}_K)$ where $h = h_K$ is the class number. We may, and will, assume that $\sigma_i \in \mathrm{GL}_2(\mathbb{A}_{K,f})$ for all i. So,

$$\mathrm{GL}_2(\mathbb{A}_K) = \bigsqcup_{i=1}^{h} \mathrm{GL}_2(K)\sigma_i \mathrm{GL}_2(K_\infty)U.$$

In this case we have a discrete subgroup $\Gamma_i \subset \mathrm{GL}_2(K_\infty)$ associated with each σ_i, namely the image of the intersection $\mathrm{GL}_2(K) \cap \sigma_i \mathrm{GL}_2(K_\infty)U\sigma_i^{-1}$ under the embedding $\mathrm{GL}_2(K) \hookrightarrow \mathrm{GL}_2(K_\infty)$. Thus

$$\begin{aligned} \mathrm{GL}_2(K) \cap \sigma_i \mathrm{GL}_2(K_\infty)U\sigma_i^{-1} &\xrightarrow{\sim} \Gamma_i \\ \gamma &\longmapsto \gamma_\infty. \end{aligned}$$

This yields a homeomorphism compatible with the $\mathrm{GL}_2(K_\infty)$-actions on the source and target,

$$\begin{aligned} \bigsqcup_{i=1}^{h} \Gamma_i\backslash \mathrm{GL}_2(K_\infty) &\xrightarrow{\sim} \mathrm{GL}_2(K)\backslash \mathrm{GL}_2(\mathbb{A}_K)/U \\ g &\longmapsto \mathrm{GL}_2(K)\sigma_i g U. \end{aligned}$$

The arguments are very similar to the case $K = \mathbb{Q}$ and we omit the details.

5.7.5 *Passing to Adelic Automorphic Forms*

In this section we start with an open subgroup $U \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$ such that $\det(U) = \widehat{\mathbb{Z}}^\times$ and let $\Gamma = \Gamma_U$ be the discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ attached to U. We recall that Γ is the image of the intersection $\mathrm{GL}_2(\mathbb{Q}) \cap \mathrm{GL}_2(\mathbb{R})^+ U$ under $\mathrm{GL}_2(\mathbb{Q}) \hookrightarrow \mathrm{GL}_2(\mathbb{R})$. Observe that Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of finite index since U necessarily has finite index in $\mathrm{GL}_2(\widehat{\mathbb{Z}})$ (by virtue of being an open subgroup of a compact group).

A cusp form $f \in S_k(\Gamma)$ gives the automorphic form $\phi_f(g) = (f|_k g)(i)$. We now view ϕ_f as a function on the double coset space $Z_\infty^+ \mathrm{GL}_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{A})/U$. This gives a function $\Phi_f : \mathrm{GL}_2(\mathbb{A}) \to \mathbb{C}$ which is uniquely determined by the following properties:

- Φ_f is left-invariant under $Z_\infty^+\mathrm{GL}_2(\mathbb{Q})$;
- Φ_f is right-invariant under U;
- $\Phi_f(g) = \phi_f(g) = (f|_k g)(i)$ for $g \in \mathrm{GL}_2(\mathbb{R})^+$.

This Φ_f is an example of an automorphic form on $\mathrm{GL}_2(\mathbb{A})$. We will not delve into the precise meaning of this, but only direct the interested reader to [BJ79, Sect. 4.2, p. 195].

Note that $Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})$ has a $\mathrm{GL}_2(\mathbb{A})$-invariant quotient measure $\dot{\mu}$, relative to which the quotient has finite measure. This follows straightforwardly from a comparison with $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$: The push-forward measure $\dot{\mu}_U$ via

$$Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A}) \longrightarrow Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})/U \simeq \Gamma\backslash\mathrm{SL}_2(\mathbb{R})$$

is an $\mathrm{SL}_2(\mathbb{R})$-invariant measure on $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})$, relative to which the latter quotient has the same measure as $Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})$. Consequently, for $f_1, f_2 \in S_k(\Gamma)$,

$$\langle \Phi_{f_1}, \Phi_{f_2} \rangle := \frac{1}{\dot{\mu}(Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A}))} \int_{Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})} \Phi_{f_1}(x)\overline{\Phi_{f_2}(x)} d\dot{\mu}(x) = \langle \phi_{f_1}, \phi_{f_2} \rangle = \langle f_1, f_2 \rangle.$$

In other words, we have an isometric embedding

$$\begin{aligned} S_k(\Gamma) &\longrightarrow L^2(Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})) \\ f &\longmapsto \Phi_f. \end{aligned}$$

Remark 5.64 What do we gain from doing this? The ambient L^2-space now has a $\mathrm{GL}_2(\mathbb{A})$-action by right translations. Thus, not only does $\mathrm{GL}_2(\mathbb{R})$ act on it, but there are commuting $\mathrm{GL}_2(\mathbb{Q}_p)$-actions for all p. Decomposing $L^2(Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A}))$ into irreducible representations, and analogously with GL_2 replaced by more complicated groups, is the scope of the theory of *automorphic representations* which is at the heart of the Langlands program. The subrepresentation generated by Φ_f is one of the main examples of an automorphic representation of $\mathrm{GL}_2(\mathbb{A})$. Others arise from Maass wave forms, i.e. Γ-invariant $\mathcal{C}^\infty$-functions $f : \mathcal{H} \to \mathbb{C}$ which are eigenfunctions for the hyperbolic Laplacian $-y^2\left(\frac{\partial^2}{\partial x} + \frac{\partial^2}{\partial y}\right)$.

We mostly take U to be one of the following examples:

- $U_0(N) = \{u \in \mathrm{GL}_2(\widehat{\mathbb{Z}}) : u \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod N\widehat{\mathbb{Z}}\}$;
- $U_1(N) = \{u \in \mathrm{GL}_2(\widehat{\mathbb{Z}}) : u \equiv \left(\begin{smallmatrix} * & * \\ 0 & 1 \end{smallmatrix}\right) \bmod N\widehat{\mathbb{Z}}\}$.

They contain the matrices $\left(\begin{smallmatrix} * & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and therefore satisfy the condition $\det(U) = \widehat{\mathbb{Z}}^\times$. The corresponding discrete subgroups are easily checked to be $\Gamma_0(N)$ and $\Gamma_1(N)$ respectively. (Note that $\Gamma_U \subset \mathrm{SL}_2(\mathbb{Z})$ when $U \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$.) Both are examples of factorizable groups, i.e. $U = \prod_p U_p$, with $U_p \subset \mathrm{GL}_2(\mathbb{Z}_p)$ an open subgroup for all p, and with the constraint that $U_p = \mathrm{GL}_2(\mathbb{Z}_p)$ for all but finitely many p. For instance,

$$U_0(N)_p = \{u \in \mathrm{GL}_2(\mathbb{Z}_p) : u \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod N\mathbb{Z}_p\}.$$

Since $N\mathbb{Z}_p = p^{\mathrm{ord}_p(N)}\mathbb{Z}_p$ we find that $U_0(N)_p = \mathrm{GL}_2(\mathbb{Z}_p)$ for all $p \nmid N$.

As a refinement of the map $f \mapsto \Phi_f$, suppose $f \in S_k(\Gamma_0(N), \chi)$ has nebentype $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$. In this case Φ_f is a function on $Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})/U_1(N)$ with a central character, i.e. $Z(\mathbb{A}) \simeq \mathbb{I}$ acts on Φ_f via a Hecke character. Let $\chi_{\mathbb{I}} : C_{\mathbb{Q}} \to \mathbb{C}^\times$ be the Hecke character attached to χ. (Thus $\chi_{\mathbb{I},\infty}$ is trivial on $\mathbb{R}_{>0}^\times$ and $\chi_{\mathbb{I},f}|_{\widehat{\mathbb{Z}}^\times}$ is the composition $\widehat{\mathbb{Z}}^\times \twoheadrightarrow (\mathbb{Z}/N\mathbb{Z})^\times \xrightarrow{\chi} \mathbb{C}^\times$.) With our conventions, the central character of Φ_f is the *inverse* of $\chi_{\mathbb{I}}$.

Proposition 5.65 *In the current setup,* $\left(\begin{smallmatrix} c & 0 \\ 0 & c \end{smallmatrix}\right)\Phi_f = \chi_{\mathbb{I}}(c)^{-1}\Phi_f$ *for all* $c \in \mathbb{I}$.

Proof Both sides of the equality are functions on $Z_\infty^+\mathrm{GL}_2(\mathbb{Q})\backslash\mathrm{GL}_2(\mathbb{A})/U_1(N)$. It suffices to show they agree on $\mathrm{GL}_2(\mathbb{R})^+$. In other words, we need to check that

$$\Phi_f(g\left(\begin{smallmatrix} c & 0 \\ 0 & c \end{smallmatrix}\right)) = \chi_{\mathbb{I}}(c)^{-1}\Phi_f(g)$$

for $g \in \mathrm{GL}_2(\mathbb{R})^+$. This is obvious if $c \in \mathbb{R}_{>0}^\times$ or $c \in \mathbb{Q}^\times$ by the invariance of Φ_f under $Z_\infty^+\mathrm{GL}_2(\mathbb{Q})$. So we may assume $c \in \widehat{\mathbb{Z}}^\times$. Pick a $d \in \mathbb{Z}$, relatively prime to N, such that $c \mapsto d + N\mathbb{Z}$ via the natural projection $\widehat{\mathbb{Z}}^\times \twoheadrightarrow (\mathbb{Z}/N\mathbb{Z})^\times$. Once and for all, choose a matrix $A = \left(\begin{smallmatrix} a & b \\ N & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ which we view as an element of $\mathrm{GL}_2(\mathbb{Q}) \hookrightarrow \mathrm{GL}_2(\mathbb{A})$. Note that $A_f\left(\begin{smallmatrix} c & 0 \\ 0 & c \end{smallmatrix}\right)^{-1} \in U_1(N)$. Also, $A_f g = gA_f$ as $g \in \mathrm{GL}_2(\mathbb{R})^+$. Therefore

$$\Phi_f(g\left(\begin{smallmatrix} c & 0 \\ 0 & c \end{smallmatrix}\right)) = \Phi_f(gA_f) = \Phi_f(A_f A_\infty A_\infty^{-1} g) = \Phi_f(AA_\infty^{-1} g) = \Phi_f(A_\infty^{-1} g) = (f|_k A_\infty^{-1} g) = \chi(d)^{-1}\Phi_f(g)$$

since A_∞ defines the diamond operator $\langle d\rangle$. We are done as $\chi_{\mathbb{I}}(c) = \chi(d)$ by the very definition of $\chi_{\mathbb{I}}$. □

5.8 The *L*-Series of a Modular Form

5.8.1 Definitions and First Properties

Let $f(z) = \sum_{n=1}^\infty a_n e^{2\pi i n z/h}$ be an element of $S_k(\Gamma)$. Then the Fourier coefficients have rate of growth $a_n = O(n^{\frac{k}{2}})$ by Proposition 5.27. Therefore the Dirichlet series

$$L(s, f) = \sum_{n=1}^\infty a_n n^{-s}$$

converges absolutely in the right half plane $\mathrm{Re}(s) > \frac{k}{2} + 1$. (In fact for $\mathrm{Re}(s) > \frac{k+1}{2}$ by Deligne's theorem.) When f is not a cusp form one can of course still adopt the same definition of $L(s, f)$–which is still a sum starting from $n = 1$ (the constant term a_0 of f does not contribute). However, since the Fourier coefficients of weight k Eisenstein series grow like $O(n^{k-1})$ the domain of convergence is much smaller.

Here is a toy example.

Example 5.66 For $\kappa \in \mathbb{Z}_{\geq 2}$ consider $E_\kappa^*(z) := -\frac{B_{2\kappa}}{4\kappa} E_\kappa(z) = -\frac{B_{2\kappa}}{4\kappa} + \sum_{n=1}^{\infty} \sigma_{2\kappa-1}(n) q^n$. Remember this is a weight 2κ modular form for $\mathrm{SL}_2(\mathbb{Z})$. It has L-series, absolutely convergent for $\mathrm{Re}(s) > 2\kappa$, given by

$$L(s, E_\kappa^*) = \sum_{n=1}^{\infty} \sigma_{2\kappa-1}(n) n^{-s} = \zeta(s)\zeta(s - 2\kappa + 1)$$

because $\sigma_{2\kappa-1} = \mathbf{1} * (\cdot)^{2\kappa-1}$. Since the Riemann zeta function has an Euler product, so does $L(s, E_\kappa^*)$–but now the reciprocal factors have degree two:

$$\begin{aligned} L(s, E_\kappa^*) = \zeta(s)\zeta(s - 2\kappa + 1) &= \prod_p \frac{1}{1 - p^{-s}} \cdot \frac{1}{1 - p^{2\kappa-1} p^{-s}} \\ &= \prod_p \frac{1}{1 - (1 + p^{2\kappa-1}) p^{-s} + p^{2\kappa-1} p^{-2s}} \\ &= \prod_p \frac{1}{1 - \sigma_{2\kappa-1}(p) p^{-s} + p^{2\kappa-1} p^{-2s}}. \end{aligned}$$

Equivalently, $\sigma_{2\kappa-1}$ is a multiplicative arithmetic function satisfying the recurrence relation

$$\sigma_{2\kappa-1}(p^r) = \sigma_{2\kappa-1}(p)\sigma_{2\kappa-1}(p^{r-1}) - p^{2\kappa-1}\sigma_{2\kappa-1}(p^{r-2})$$

for $r \geq 2$. (The initial values are of course $\sigma_{2\kappa-1}(1) = 1$ and $\sigma_{2\kappa-1}(p) = 1 + p^{2\kappa-1}$.) It is good practice to deduce the recurrence relation from the Euler product and vice versa. At the same time, as a sanity check, the reader should directly check the recurrence relation by using the formula $\sigma_{2\kappa-1}(p^r) = \frac{p^{(r+1)(2\kappa-1)}-1}{p^{2\kappa-1}-1}$. This is a completely elementary calculation.

From the analytic properties of the Riemann zeta function we infer that $L(s, E_\kappa^*)$ has meromorphic continuation to $\mathbb{C}$, with at *most* simple poles at $s = 1$ and $s = 2\kappa$. However, $s = 1$ is *not* a pole since $\zeta(2 - 2\kappa) = 0$ for $\kappa \in \mathbb{Z}_{\geq 2}$. On the other hand $s = 2\kappa$ *is* a pole since $\zeta(2\kappa) \neq 0$.

There is no nice functional equation of the usual kind, the reason being that $\xi(s)$ is invariant under $s \mapsto 1 - s$ whereas $\xi(s - 2\kappa + 1)$ is invariant under $s \mapsto (4\kappa - 1) - s$.

In light of this example we will exclusively consider cusp forms f for $\Gamma_1(N)$ from now on. The matrix below plays a key role when establishing the basic analytic properties of $L(s, f)$.

$$W = W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{R})^+.$$

It satisfies the following easily verifiable properties:

- $\det(W) = N$;
- $W^2 = \left(\begin{smallmatrix} -N & 0 \\ 0 & -N \end{smallmatrix}\right) = -N \cdot I$;
- W fixes the point $\frac{i}{\sqrt{N}} \in \mathcal{H}$;
- $W \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) W^{-1} = \left(\begin{smallmatrix} d & -c/N \\ -bN & a \end{smallmatrix}\right)$. In particular W normalizes both $\Gamma_1(N)$ and $\Gamma_0(N)$, i.e.

$$W\Gamma_1(N)W^{-1} = \Gamma_1(N), \quad W\Gamma_0(N)W^{-1} = \Gamma_0(N).$$

Consequently $f \mapsto f|_k W$ gives a linear operator on $S_k(\Gamma_1(N))$ which sends $S_k(\Gamma_0(N), \chi)$ to the subspace of forms with complex conjugate nebentype $S_k(\Gamma_0(N), \overline{\chi})$. (Why? In the notation of the fourth bullet point above we have $\chi(a) = \overline{\chi(d)}$ if $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$.)

We introduce a lighter notation for this ubiquitous operation.

Definition 5.67 $f^W(z) = (f|_k W)(z) = N^{-\frac{k}{2}} z^{-k} f(\frac{-1}{Nz})$.

This is an involution when k is even. In general we have $f^{WW} = (-1)^k f$. That being so, W has order dividing four, and $S_k(\Gamma_1(N))$ can be decomposed into eigenspaces for the W-action. The eigenvalues obviously belong to $\langle i \rangle = \{\pm 1, \pm i\}$.

Our first mission is to analytically continue $L(s, f)$ and relate it to $L(k - s, f^W)$.

Theorem 5.68 *Let $f \in S_k(\Gamma_1(N))$. The completed L-series*

$$\widehat{L}(s, f) := N^{\frac{s}{2}} L_{\mathbb{C}}(s) L(s, f) = N^{\frac{s}{2}} 2(2\pi)^{-s} \Gamma(s) L(s, f)$$

extends to an entire function (similarly for f^W) and satisfies the equation

$$\widehat{L}(s, f) = i^k \widehat{L}(k - s, f^W).$$

Proof We start by computing the Mellin transformation of the function $t \mapsto f(it)$ on $\mathbb{R}^\times_{>0}$.

$$\begin{aligned}
\int_0^\infty f(it) t^s \frac{dt}{t} &= \int_0^\infty \sum_{n=1}^\infty a_n e^{-2\pi n t} t^s \frac{dt}{t} \\
&= \sum_{n=1}^\infty \int_0^\infty a_n e^{-2\pi n t} t^s \frac{dt}{t} \\
&= \sum_{n=1}^\infty a_n \int_0^\infty e^{-x} \left(\frac{x}{2\pi n}\right)^s \frac{dx}{x} \\
&= (2\pi)^{-s} \cdot \sum_{n=1}^\infty a_n n^{-s} \int_0^\infty e^{-x} x^s \frac{dx}{x} \\
&= (2\pi)^{-s} \Gamma(s) L(s, f) \\
&= \frac{1}{2} N^{-\frac{s}{2}} \widehat{L}(s, f),
\end{aligned}$$

assuming $\mathrm{Re}(s) > \frac{k}{2} + 1$. This gives the integral representation of $\widehat{L}(s, f)$. We split the integral as a sum of a good integral and a bad integral,

$$I(s, f) := \int_{\frac{1}{\sqrt{N}}}^{\infty} f(it) t^s \frac{dt}{t}, \qquad J(s, f) := \int_0^{\frac{1}{\sqrt{N}}} f(it) t^s \frac{dt}{t}.$$

(The choice of $\frac{1}{\sqrt{N}}$ as the cut-point will become clear momentarily.) Since $f(it) = O(e^{-2\pi t})$ the first integral is absolutely convergent for all $s \in \mathbb{C}$ and represents an entire function. For the second integral we have a problem at $t = 0$ unless $\mathrm{Re}(s)$ is large enough. We use the transformation property

$$f^W(ix) = N^{-\frac{k}{2}} (ix)^{-k} f(\frac{i}{Nx})$$

to express $J(s, f)$ in terms of $I(k - s, f^W)$. Assuming $\mathrm{Re}(s) \gg 0$ the substitution $x = \frac{1}{Nt}$ gives

$$\begin{aligned} J(s, f) &= \int_0^{\frac{1}{\sqrt{N}}} f(it) t^s \frac{dt}{t} \\ &= \int_{\frac{1}{\sqrt{N}}}^{\infty} f(\frac{i}{Nx}) (\frac{1}{Nx})^s \frac{dx}{x} \\ &= \int_{\frac{1}{\sqrt{N}}}^{\infty} N^{\frac{k}{2}} (ix)^k f^W(ix) (Nx)^{-s} \frac{dx}{x} \\ &= i^k N^{\frac{k}{2} - s} \int_{\frac{1}{\sqrt{N}}}^{\infty} f^W(ix) x^{k-s} \frac{dx}{x} \\ &= i^k N^{\frac{k}{2} - s} I(k - s, f^W). \end{aligned}$$

The resulting expression for the completed L-series,

$$\frac{1}{2} \widehat{L}(s, f) = N^{\frac{s}{2}} I(s, f) + i^k N^{\frac{k-s}{2}} I(k - s, f^W),$$

shows $\widehat{L}(s, f)$ admits continuation to an entire function. Moreover,

$$\begin{aligned} i^k \frac{1}{2} \widehat{L}(k - s, f^W) &= i^k N^{\frac{k-s}{2}} I(k - s, f^W) + (-1)^k N^{\frac{s}{2}} I(s, f^{WW}) \\ &= i^k N^{\frac{k-s}{2}} I(k - s, f^W) + N^{\frac{s}{2}} I(s, f) \\ &= \frac{1}{2} \widehat{L}(s, f). \end{aligned}$$

This completes the proof. □

If $f^W = i^m f$ the functional equation can be restated as $\widehat{L}(s, f) = i^{k+m}\widehat{L}(k - s, f)$.

Exercise 5.69 This is a continuation of Exercise 5.30 and we adopt its setup. The ψ-twisted L-series

$$L(s, f, \psi) := L(s, f \times \psi) = \sum_{n=1}^{\infty} \psi(n) a_n n^{-s}$$

is absolutely convergent for $\mathrm{Re}(s) > \frac{k}{2} + 1$ and extends to an entire function by Theorem 5.68 applied to the cusp form $f \times \psi$. Going on, verify the functional equation

$$\widehat{L}(s, f, \psi) := (Nr^2)^{\frac{s}{2}} 2(2\pi)^{-s}\Gamma(s)L(s, f, \psi) ==$$

$$i^k \chi(r)\psi(N)\tau(\psi)^2 r^{-1} \cdot \widehat{L}(k - s, f^W, \overline{\psi}).$$

(`Hint`: This amounts to checking that

$$(f \times \psi)^{W_{Nr^2}} = \chi(r)\psi(N)\tau(\psi)^2 r^{-1} \cdot f^W \times \overline{\psi}.$$

To see this, use the formula for $f \times \psi$ obtained in Exercise 5.30 along with the following trick. For a u coprime to r pick $d, w \in \mathbb{Z}$ such that $dr - Nuw = 1$. Then

$$\begin{pmatrix} 1 & \frac{u}{r} \\ 0 & 1 \end{pmatrix} W_{Nr^2} = r W \begin{pmatrix} r & -w \\ -Nu & d \end{pmatrix} \begin{pmatrix} 1 & \frac{w}{r} \\ 0 & 1 \end{pmatrix}.$$

Inserting this, using elementary properties of Gauss sums, and remembering that f^W has nebentype $\overline{\chi}$, gives the result. For further help consult [Shi94, Proposition 3.65, Theorem 3.66].)

Plainly, we recover Theorem 5.68 by taking $r = 1$ and ψ to be trivial in the previous exercise.

5.8.2 *Euler Products and Hecke Operators*

The Artin L-function of a two-dimensional Galois representation $r : \mathrm{Gal}(E/\mathbb{Q}) \longrightarrow \mathrm{GL}(V)$ is given by a product

$$L(s, r) = \prod_{p \mid \mathcal{N}_r} \det{}_{V^{I_w}}(1 - r(\mathrm{Frob}_w)p^{-s})^{-1} \cdot \prod_{p \nmid \mathcal{N}_r} (1 - \mathrm{tr}\, r(\mathrm{Frob}_w)p^{-s} + \det r(\mathrm{Frob}_w)p^{-2s})^{-1}$$

for $\mathrm{Re}(s) > 1$. Note that the coefficient $\det r(\mathrm{Frob}_w)$ is a root of unity.

The L-function of an elliptic curve $\mathcal{E}/\mathbb{Q}$ is modeled on this. Let $N_{\mathcal{E}}$ denote its conductor. For $p \nmid N_{\mathcal{E}}$, the curve has good reduction and it makes sense to count the number of points over $\mathbb{F}_p$, say $\#\mathcal{E}(\mathbb{F}_p) = p + 1 - a_p$. Here a_p should be thought of as an error term. Indeed $|a_p| \leq 2\sqrt{p}$ for all primes p. The L-function is given by

$$L(s, \mathcal{E}) = \prod_{p|N_{\mathcal{E}}} F_p(p^{-s})^{-1} \cdot \prod_{p\nmid N_{\mathcal{E}}} (1 - a_p p^{-s} + p^{1-2s})^{-1}$$

for $\mathrm{Re}(s) > \frac{3}{2}$, where $F_p(X) = 1 \pm X$ when $\mathcal{E}$ has multiplicative reduction at p, and $F_p(X) = 1$ when $\mathcal{E}$ has additive reduction at p. (The sign $\pm$ in $F_p(X)$ in the first case depends on whether the reduction is non-split or split, respectively.)

These remarks, in addition to the example $L(s, E_\kappa^*)$, raises the question whether $L(s, f)$ has an Euler product of the form

$$L(s, f) = \prod_p (1 - a_p p^{-s} + \chi(p) p^{k-1-2s})^{-1} \tag{5.3}$$

for $\mathrm{Re}(s) \gg 0$ and $f \in S_k(\Gamma_0(N), \chi)$. (We stick to the convention that $\chi(p) = 0$ if $p \mid N$, and a_n now denotes the nth Fourier coefficient of f.) However, this is too much to hope for in general. First of all, in order to write

$$L(s, f) = \sum_{n=1}^{\infty} a_n n^{-s} = \prod_p \sum_{r=0}^{\infty} a_{p^r} p^{-rs}$$

we need the Fourier coefficients to be multiplicative, i.e. $a_{mn} = a_m a_n$ when $\mathrm{GCD}(m, n) = 1$. If this is granted, secondly we need the identity

$$(1 - a_p X + \chi(p) p^{k-1} X^2) \sum_{r=0}^{\infty} a_{p^r} X^r = 1$$

for all p (and $X = p^{-s}$ if the reader prefers). By expanding the left-hand side, this identity immediately translates into to the recurrence relations

- $a_1 = 1$;
- $a_{p^r} = a_p a_{p^{r-1}} - \chi(p) p^{k-1} a_{p^{r-2}}$ for $r \geq 2$.

(We found that the Fourier coefficients $\sigma_{2\kappa-1}(p^r)$ of E_κ^* satisfy these relations with $k = 2\kappa$ and $\chi = 1$.)

Next we will define a sequence of commuting and often diagonalizable operators $T_n \in \mathrm{End}_{\mathbb{C}} S_k(\Gamma_0(N), \chi)$ for $n = 1, 2, 3, \ldots$, the first of which is the identity, which satisfy the relations

- $T_{mn} = T_m T_n$ provided $\mathrm{GCD}(m, n) = 1$;
- $T_{p^r} = T_p T_{p^{r-1}} - \chi(p) p^{k-1} T_{p^{r-2}}$ for $r \geq 2$.

On q-expansions the T_p-operator takes a simple form: If $f = \sum_{n=1}^{\infty} a_n q^n$ then $T_p f = \sum_{n=1}^{\infty} a_n(T_p f) q^n$ where

$$a_n(T_p f) = a_{np} + \chi(p) p^{k-1} a_{n/p} \tag{5.4}$$

with the convention that $a_{n/p} = 0$ if $p \nmid n$. The problem with using this as a *definition* of T_p, is it is not at all clear that $T_p f$ is a modular form.

From linear algebra we know the operators $(T_n)_{n\geq 1}$ are simultaneously diagonalizable *if* they are diagonalizable[3] of course, i.e. $S_k(\Gamma_0(N), \chi)$ has basis of eigenvectors for all the T_n-operators–under the standing hypotheses. Pick a basis vector and call it f. Then $T_n f = \lambda_n f$ for all $n \geq 1$, where λ_n is the eigenvalues of T_n on f. Clearly $\lambda_1 = 1$, $\lambda_{mn} = \lambda_m \lambda_n$ when $\mathrm{GCD}(m, n) = 1$, and

$$\lambda_{p^r} = \lambda_p \lambda_{p^{r-1}} - \chi(p) p^{k-1} \lambda_{p^{r-2}}$$

for $r \geq 2$. On the other hand (5.4) implies that $\lambda_p a_n = a_{np} + \chi(p) p^{k-1} a_{n/p}$. For $n = 1$ this shows $\lambda_p a_1 = a_p$. We usually normalize f such that $a_1 = 1$. Under this assumption $\lambda_p = a_p$, and more generally (as we will see later) the Hecke eigenvalues equal the Fourier coefficients.

The outcome is that for a normalized *eigenform* f, as in the previous paragraph, the L-series $L(s, f)$ can be expressed as an Euler product (5.3) at least for $\mathrm{Re}(s) > \frac{k}{2} + 1$. Moreover, $S_k(\Gamma_0(N), \chi)$ has a basis of eigenforms.

5.8.3 The Definition of Hecke Operators

If $\Gamma, \Gamma' \subset \mathrm{SL}_2(\mathbb{R})$ are arithmetic subgroups, there is a natural way to obtain linear transformations $M_k(\Gamma) \longrightarrow M_k(\Gamma')$ from double cosets $\Gamma h \Gamma'$ with $h \in \mathrm{GL}_2(\mathbb{Q})^+$. There is a bijection

$$\begin{aligned} (\Gamma' \cap h^{-1}\Gamma h)\backslash\Gamma' &\xrightarrow{\sim} \Gamma\backslash\Gamma h\Gamma' \\ \gamma &\longmapsto h\gamma, \end{aligned}$$

and the two sets here are finite since $h \in \mathrm{GL}_2(\mathbb{Q})$ belongs to the commensurator of $\mathrm{SL}_2(\mathbb{Z})$ in $\mathrm{GL}_2(\mathbb{R})$:

$$h^{-1}\Gamma h \sim h^{-1}\mathrm{SL}_2(\mathbb{Z})h \sim \mathrm{SL}_2(\mathbb{Z}) \sim \Gamma'.$$

[3] For a general $N > 1$ one has to replace $S_k(\Gamma_0(N), \chi)$ by the subspace of *newforms* to achieve this.

This gives rise to the double coset transformation

$$\begin{aligned} M_k(\Gamma) &\longrightarrow M_k(\Gamma') \\ f &\longmapsto f|_k\Gamma h\Gamma' := \sum\nolimits_{g\in\Gamma\backslash\Gamma h\Gamma'} f|_k g = \sum\nolimits_{\gamma\in(\Gamma'\cap h^{-1}\Gamma h)\backslash\Gamma'} f|_k h\gamma. \end{aligned}$$

Example 5.70 When $\Gamma = \Gamma' = \Gamma_1(N)$ the diamond operator $f \mapsto f|_k\langle d\rangle$ coincides with the operator defined by the double coset $\Gamma\langle d\rangle\Gamma = \Gamma\langle d\rangle$. (Recall that $\langle d\rangle = \left(\begin{smallmatrix} * & * \\ * & d \end{smallmatrix}\right) \in \Gamma_0(N)$ provided $\mathrm{GCD}(d, N) = 1$, and Γ is a normal subgroup of $\Gamma_0(N)$.)

Remark 5.71 In what follows we interpret $f \mapsto f|_k\langle d\rangle$ as the zero operator when $\mathrm{GCD}(d, N) > 1$.

To fix ideas we let $\Gamma = \Gamma' = \Gamma_1(N)$ throughout this section. The most important example to us is the pth Hecke operator T_p, for each prime p, defined by the matrix $h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$:

$$\begin{aligned} T_p : M_k(\Gamma) &\longrightarrow M_k(\Gamma) \\ f &\longmapsto p^{\frac{k}{2}-1} \cdot f|_k\Gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma. \end{aligned}$$

As will soon become clear, the factor $p^{\frac{k}{2}-1}$ ensures that T_p preserves the integrality of Fourier coefficients (as follows from (5.4) at least if χ is trivial.) To make T_p more explicit, and to establish (5.4), we need the following basic observation:

Lemma 5.72 *For $\Gamma = \Gamma_1(N)$ we have:*

(i) $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma = \{A \in Mat_{2\times 2}(\mathbb{Z}) : A \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & p \end{smallmatrix}\right) \text{ mod } N, \text{ and } \det A = p\}$ *if $p \nmid N$;*

(ii) $\Gamma\backslash\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$ *has a complete set of mutually inequivalent coset representatives consisting of*

$$\left(\begin{smallmatrix} 1 & u \\ 0 & p \end{smallmatrix}\right), \quad u = 0, 1, 2, \ldots, p-1,$$

together with $\langle p\rangle\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ if $p \nmid N$. (When $p|N$ the last representative is omitted.)

Proof In (i) the inclusion $\subset$ is clear. For the other inclusion $\supset$ we use an argument from [DS05, p. 105]. Let $L = \mathbb{Z}^2$–thought of as the lattice of column vectors $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ with $x, y \in \mathbb{Z}$. Let $L_0 \subset L$ be the subgroup of vectors $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ with $y \in N\mathbb{Z}$. Start with an $A \in \mathrm{Mat}_{2\times 2}(\mathbb{Z})$ such that $A \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & p \end{smallmatrix}\right)$ mod N and $\det A = p$. We need to show $A \in \Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)\Gamma$. Clearly $AL_0 \subset L_0$, using only that $A \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$ mod N. Also, the index is $[L_0 : AL_0] = |\det A| = p$ (pick a $\mathbb{Z}$-basis for L_0 and use the Smith normal form for example). Similarly for L. Thus

$$[L : AL_0] = [L : L_0] \cdot [L_0 : AL_0] = Np.$$

By the theory of elementary divisors, there exist a basis $\{v_1, v_2\}$ of L and positive integers $\{d_1, d_2\}$ such that $d_1|d_2$ and $\{d_1v_1, d_2v_2\}$ is a basis of AL_0. We may assume the matrix $V := \left(\begin{smallmatrix} v_1 & v_2 \end{smallmatrix}\right)$ lies in $\mathrm{SL}_2(\mathbb{Z})$ by replacing v_1 with $-v_1$ if necessary. Note that $d_1d_2 = Np$.

Look at the first column Ae_1 of A. On the one hand, $Ae_1 \equiv \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \bmod N$ by assumption. On the other hand, $e_1 \in L_0 \Rightarrow Ae_1 \in AL_0$, so the entries of Ae_1 are divisible by d_1. Together this shows $\mathrm{GCD}(d_1, N) = 1$. Since p is prime, and $d_1d_2 = Np$, we conclude that $d_1 = 1$ and $d_2 = Np$. (We cannot have $d_1 = p$. Otherwise p^2 divides d_1d_2 and therefore $p|N$, but then d_1 is not relatively prime to N.)

To summarize, we have

$$L = \mathbb{Z}v_1 \oplus \mathbb{Z}v_2 \supset AL_0 = \mathbb{Z}v_1 \oplus \mathbb{Z}Npv_2.$$

As $L/AL_0 \simeq \mathbb{Z}/Np\mathbb{Z}$ there is a unique intermediate lattice of index N in L, namely $\mathbb{Z}v_1 \oplus \mathbb{Z}Nv_2$, and a unique intermediate lattice of index p in L, namely $\mathbb{Z}v_1 \oplus \mathbb{Z}pv_2$. From the uniqueness we infer that

$$L_0 = \mathbb{Z}v_1 \oplus \mathbb{Z}Nv_2, \quad AL = \mathbb{Z}v_1 \oplus \mathbb{Z}pv_2.$$

In particular, since $v_1 \in L_0$ we observe that $V = \left(\begin{smallmatrix} v_1 & v_2 \end{smallmatrix}\right)$ in fact belongs to $\Gamma_0(N)$.

Now consider the matrix $U := (V \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right))^{-1} A$. In other words $A = V \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) U$. We claim that $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ also belongs to $\Gamma_0(N)$. Indeed,

- $\det U = 1$;
- $Ae_1 = av_1 + pcv_2 \in AL_0 = \mathbb{Z}v_1 \oplus \mathbb{Z}Npv_2 \Longrightarrow a \in \mathbb{Z},\ c \in N\mathbb{Z}$;
- $Ae_2 = bv_1 + pdv_2 \in AL = \mathbb{Z}v_1 \oplus \mathbb{Z}pv_2 \Longrightarrow b \in \mathbb{Z},\ d \in \mathbb{Z}$.

At this point we have shown that $A \in \Gamma_0(N) \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma_0(N)$. As a last step, in order to replace $\Gamma_0(N)$ with $\Gamma = \Gamma_1(N)$ here, we need to check that

$$\Gamma_0(N) \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma_0(N) = \Gamma_1(N) \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma_0(N). \tag{5.5}$$

(For if $A = \widetilde{V} \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \widetilde{U}$ with $\widetilde{V} \in \Gamma_1(N)$ and $\widetilde{U} \in \Gamma_0(N)$ then $\widetilde{U}$ automatically lies in $\Gamma_1(N)$ by reading the matrix equation modulo N, and using the assumption that $p \nmid N$.)

To check the non-trivial inclusion $\subset$ in (5.5) note that every coset in $\Gamma_1(N)\backslash\Gamma_0(N) \xrightarrow{\sim} (\mathbb{Z}/N\mathbb{Z})^\times$ is represented by a matrix of the form $\left(\begin{smallmatrix} r & s \\ N & t \end{smallmatrix}\right)$. It therefore suffices to verify that

$$\left(\begin{smallmatrix} r & s \\ N & t \end{smallmatrix}\right) \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \in \Gamma_1(N) \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma_0(N).$$

One way to see this is as follows: By the Chinese remainder theorem (which applies since $p \nmid N$) there is a $z \in \mathbb{Z}$ such that $z \equiv 1 \bmod N$ and $z \equiv -r \bmod p$. Pick a matrix $\delta = \left(\begin{smallmatrix} * & * \\ N & z \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, which then necessarily belongs to $\Gamma_1(N)$. A tedious matrix calculation, which we leave to the reader, then shows that

$$\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)^{-1}\delta\left(\begin{smallmatrix}r&s\\N&t\end{smallmatrix}\right)\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\in\Gamma_0(N).$$

This proves part (i).

For part (ii) an instant computation shows $\left(\begin{smallmatrix}1&u\\0&p\end{smallmatrix}\right)\left(\begin{smallmatrix}1&v\\0&p\end{smallmatrix}\right)^{-1}\in\Gamma$ if and only if $u\equiv v \bmod p$. When $p\nmid N$ we ask the reader to check that $\langle p\rangle\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)\left(\begin{smallmatrix}1&u\\0&p\end{smallmatrix}\right)^{-1}\notin\Gamma$. (The upper right entry of $\langle p\rangle$ is not a multiple of p.) Thus the listed matrices at least represent distinct left cosets of Γ, and the matrices are in $\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma$ by (i) say, noting that $\langle p\rangle\equiv\left(\begin{smallmatrix}*&*\\0&p\end{smallmatrix}\right) \bmod N$.

It remains to see that

$$[\Gamma:\Gamma\cap\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)^{-1}\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)]=\begin{cases}p+1 & \text{if } p\nmid N\\ p & \text{if } p\mid N.\end{cases}$$

In other words we need to find the index of

$$\widetilde{\Gamma}:=\Gamma\cap\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)^{-1}\Gamma\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)=\{\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma : p|b\}.$$

in Γ. Let us first assume $p\nmid N$. In this case $\Gamma=R^{-1}(\{\left(\begin{smallmatrix}1&*\\0&1\end{smallmatrix}\right)\}\times \mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}))$ where R is the surjective homomorphism

$$R:\mathrm{SL}_2(\mathbb{Z})\twoheadrightarrow\mathrm{SL}_2(\mathbb{Z}/Np\mathbb{Z})\xrightarrow{\sim}\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})\times\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}).$$

(The surjectivity of the first map can be found in [Shi94, Lemma 1.38, p. 20].) Furthermore,

$$\widetilde{\Gamma}=R^{-1}(\{\left(\begin{smallmatrix}1&*\\0&1\end{smallmatrix}\right)\}\times\{\left(\begin{smallmatrix}*&0*&*\end{smallmatrix}\right)\}).$$

Therefore R induces a bijection

$$\widetilde{\Gamma}\backslash\Gamma\xrightarrow{\sim}\{\left(\begin{smallmatrix}*&0*&*\end{smallmatrix}\right)\}\backslash\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}).$$

The latter is identified with $\mathbb{P}^1(\mathbb{Z}/p\mathbb{Z})$ via the (transitive) $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$-action on the lines in $(\mathbb{Z}/p\mathbb{Z})^{\oplus 2}$ and noting that $\{\left(\begin{smallmatrix}*&0*&*\end{smallmatrix}\right)\}$ is the stabilizer of the line spanned by $e_2=\left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$. Thus indeed $[\Gamma:\widetilde{\Gamma}]=p+1$.

Finally we deal with the case where $p|N$. Here one explicitly checks that $\Gamma=\bigcup_{0\le x<p}\widetilde{\Gamma}\left(\begin{smallmatrix}1&x\\0&1\end{smallmatrix}\right)$ and this is clearly a disjoint union. The assumption $p|N$ comes in as follows: Given $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma$ we can find an x such that $ax\equiv b \bmod p$ since $p\nmid a$. (Indeed $p|N|c$ and $ad-bc=1$.) We conclude that $[\Gamma:\widetilde{\Gamma}]=p$. □

Exercise 5.73 Use the theory of elementary divisors (as in the proof of (i) above) to show that

$$\mathrm{SL}_2(\mathbb{Z})\backslash\mathrm{GL}_2(\mathbb{Q})^+/\mathrm{SL}_2(\mathbb{Z})$$

has a complete set of double coset representatives consisting of matrices $\left(\begin{smallmatrix} d_1 & 0 \\ 0 & d_2 \end{smallmatrix}\right)$ where $d_1, d_2 \in \mathbb{Q}^\times$ and $\frac{d_2}{d_1} \in \mathbb{Z}_{>0}$. (Hint: Starting with a $g \in \mathrm{GL}_2(\mathbb{Q})^+$ pick $M \in \mathbb{Z}_{>0}$ such that $Mg \in \mathrm{Mat}_{2\times 2}(\mathbb{Z})$ and consider the subgroup of $\mathbb{Z}^2$ generated by the columns.)

Exercise 5.74 Show that T_p commutes with the diamond operators $f \mapsto f|_k\langle d\rangle$ on $S_k(\Gamma)$. Deduce that T_p restricts to an operator on $S_k(\Gamma_0(N), \chi)$ for all Dirichlet characters χ modulo N. (Hint: Let $\gamma = \left(\begin{smallmatrix} * & * \\ * & d \end{smallmatrix}\right) \in \Gamma_0(N)$. Check $\gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \gamma^{-1} \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & p \end{smallmatrix}\right)$ mod N and deduce that $\gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \gamma^{-1} \in \Gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma$. Combine this with the fact that γ normalizes Γ.)

We can now prove (5.4). First off, we have

$$T_p f = p^{\frac{k}{2}-1} \cdot \sum_{u=0}^{p-1} f|_k \left(\begin{smallmatrix} 1 & u \\ 0 & p \end{smallmatrix}\right) + p^{\frac{k}{2}-1} \cdot f|_k\langle p\rangle \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right),$$

with the convention that we drop the last term when $p|N$. Next, we express the different parts of this formula in terms of the Fourier coefficients of f:

$$\begin{aligned}
\sum_{u=0}^{p-1} (f|_k \left(\begin{smallmatrix} 1 & u \\ 0 & p \end{smallmatrix}\right))(z) &= \sum_{u=0}^{p-1} p^{-\frac{k}{2}} f\Big(\frac{z+u}{p}\Big) \\
&= p^{-\frac{k}{2}} \cdot \sum_{u=0}^{p-1} \sum_{n=1}^{\infty} a_n e^{2\pi i n(z+u)/p} \\
&= p^{-\frac{k}{2}} \cdot \sum_{n=1}^{\infty} a_n e^{2\pi i n z/p} \Big(\sum_{u=0}^{p-1} e^{2\pi i n u/p} \Big) \\
&= p^{-\frac{k}{2}+1} \cdot \sum_{n'=1}^{\infty} a_{n'p} e^{2\pi i n' z}
\end{aligned}$$

using the geometric sum $\sum_{u=0}^{p-1} e^{2\pi i n u/p} = \begin{cases} 0 & \text{if } p \nmid n \\ p & \text{if } p \mid n \end{cases}$. Also, when $p \nmid N$ the last term comes out to

$$\begin{aligned}
(f|_k\langle p\rangle \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right))(z) &= \chi(p) p^{\frac{k}{2}} f(pz) \\
&= \chi(p) p^{\frac{k}{2}} \cdot \sum_{n=1}^{\infty} a_n e^{2\pi i n p z} \\
&= \chi(p) p^{\frac{k}{2}} \cdot \sum_{n'=1}^{\infty} a_{n'/p} e^{2\pi i n' z} \\
&= p^{-\frac{k}{2}+1} \cdot \chi(p) p^{k-1} \cdot \sum_{n'=1}^{\infty} a_{n'/p} e^{2\pi i n' z}.
\end{aligned}$$

Comparing Fourier coefficients gives the formula $a_n(T_p f) = a_{np} + \chi(p)p^{k-1}a_{n/p}$ in (5.4) at once.

Remark 5.75 Let $\mathbb{Z}[\chi]$ denote the smallest subring of $\mathbb{C}$ containing all the values of χ. This is the ring of integers in a cyclotomic field. If $f \in S_k(\Gamma_0(N), \chi)$ has all its Fourier coefficients in $\mathbb{Z}[\chi]$, then so does $T_p f$ for all p. In this sense T_p preserves integrality.

Exercise 5.76 We adopt the setup from Exercise 5.30. Check that $T_p(f \times \psi) = \psi(p)(T_p f \times \psi)$ for all p. (Remember $f \times \psi$ has nebentype $\chi\psi^2$.) Conclude that if f is an eigenvector for T_p then so is $f \times \psi$.

Exercise 5.77 Prove that $T_p T_q = T_q T_p$ for any two primes p and q. (`Hint`: Use (5.4) twice to get an expression for $a_n(T_p T_q f)$ which is symmetric in p and q.)

We now define T_m for arbitrary $m \in \mathbb{Z}_{>0}$. Starting with prime powers $m = p^r$, we declare that

$$T_{p^r} = T_p T_{p^{r-1}} - \chi(p)p^{k-1}T_{p^{r-2}}$$

for $r \geq 2$. Thus T_{p^r} is a polynomial in T_p with coefficients in $\mathbb{Z}[\chi]$. In particular $(T_{p^r})_{r\geq 0}$ is a commuting family of operators. Moreover, by Exercise 5.77 we even have commutativity when p varies, i.e. $T_{p^r}T_{q^s} = T_{q^s}T_{p^r}$. This allows us to define T_m in general, via the prime factorization $m = p_1^{r_1} \cdots p_t^{r_t}$, as the product

$$T_m := T_{p_1^{r_1}} \cdots T_{p_t^{r_t}}$$

since the order of the factors is irrelevant. With this definition we obviously have the relation $T_{mn} = T_m T_n$ when m and n are relatively prime, and $(T_m)_{m\geq 1}$ is a commuting family of operators.

Here is the generalization of (5.4) which produces the Fourier coefficients of $T_m f$ as a combination of those of f, but we are now allowing m to be an arbitrary positive integer.

Proposition 5.78 *Let $(a_n)_{n\geq 1}$ be the Fourier coefficients of $f \in S_k(\Gamma_0(N), \chi)$. Let $m \in \mathbb{Z}_{>0}$. Then the nth Fourier coefficient of $T_m f$ is given by the formula*

$$a_n(T_m f) = \sum\nolimits_{0<d|GCD(m,n)} \chi(d)d^{k-1}a_{mn/d^2}.$$

As a special case hereof, $a_1(T_m f) = a_m$.

Proof A moment's thought shows the avowed formula boils down to (5.4) when $m = p$ is a prime (do the cases $p \nmid n$ and $p|n$ separately). More generally, for $m = p^r$ with $r \geq 2$ we first note that

$$\begin{aligned} a_n(T_{p^r} f) &= a_n(T_p T_{p^{r-1}} f) - \chi(p) p^{k-1} a_n(T_{p^{r-2}} f) \\ &= a_{np}(T_{p^{r-1}} f) + \chi(p) p^{k-1} a_{n/p}(T_{p^{r-1}} f) - \chi(p) p^{k-1} a_n(T_{p^{r-2}} f). \end{aligned}$$

For the purpose of this proof we introduce a notation for the right-hand side of the claimed formula,

$$A_{n,m} := \sum\nolimits_{0<d|\mathrm{GCD}(m,n)} \chi(d) d^{k-1} a_{mn/d^2}.$$

If we can verify this quantity satisfies the same recurrence relation, namely

$$A_{n,p^r} = A_{np,p^{r-1}} + \chi(p) p^{k-1} A_{n/p,p^{r-1}} - \chi(p) p^{k-1} A_{n,p^{r-2}},$$

we can finish the prime-power case by induction on r. To verify this relation, observe that

$$\begin{aligned} A_{n,p^r} &= \sum\nolimits_{s\le r : p^s|n} \chi(p^s) p^{s(k-1)} a_{np^r/p^{2s}} \\ &= a_{np^r} + \sum\nolimits_{0<s\le r : p^s|n} \chi(p^s) p^{s(k-1)} a_{np^r/p^{2s}} \\ &= a_{np^r} + \chi(p) p^{k-1} \sum\nolimits_{s'\le r-1 : p^{s'+1}|n} \chi(p^{s'}) p^{s'(k-1)} a_{np^{r-2}/p^{2s'}} \\ &= a_{np^r} + \chi(p) p^{k-1} \sum\nolimits_{s'\le r-1 : p^{s'}|\frac{n}{p}} \chi(p^{s'}) p^{s'(k-1)} a_{\frac{n}{p}p^{r-1}/p^{2s'}} \\ &= a_{np^r} + \chi(p) p^{k-1} A_{n/p,p^{r-1}}. \end{aligned}$$

A similar calculation, which we omit here, gives $A_{np,p^{r-1}} = a_{np^r} + \chi(p) p^{k-1} A_{n,p^{r-2}}$ (or just reuse the previous calculation with different parameters). In conjunction these last two relations do the job.

Now suppose $m \in \mathbb{Z}_{>0}$ is arbitrary and it factors as $m = m_1 m_2$ with $\mathrm{GCD}(m_1, m_2) = 1$. If the Proposition holds for both m_1 and m_2 we deduce that

$$\begin{aligned} a_n(T_m f) = a_n(T_{m_1} T_{m_2} f) &= \sum\nolimits_{0<d|\mathrm{GCD}(m_1,n)} \chi(d) d^{k-1} a_{m_1 n/d^2}(T_{m_2} f) \\ &= \sum\nolimits_{0<d|\mathrm{GCD}(m_1,n)} \sum\nolimits_{0<e|\mathrm{GCD}(m_2,m_1 n/d^2)} \chi(de)(de)^{k-1} a_{m_1 m_2 n/(de)^2}. \end{aligned}$$

As d and e vary in their respective sets of common divisors, de runs over the common divisors of m and n. For instance, if $n = ds$, then $e|s$ since $e|\frac{m_1 n}{d^2} = \frac{m_1 s}{d} = \frac{m_1}{d} s$ and $e|m_2$ is coprime to $\frac{m_1}{d}$. Conversely, any $g|\mathrm{GCD}(m,n)$ can be written uniquely as $g = de$ with $d|m_1$ and $e|m_2$. Here $d|g|n$ and $\frac{m_1 n}{d^2 e} = \frac{m_1 n}{dg} = \frac{m_1}{d} \cdot \frac{n}{g}$ is visibly an integer. In conclusion, the double sum representing $a_n(T_m f)$ above simplifies to $A_{n,m}$. Using this observation repeatedly for the prime factorization $m = p_1^{r_1} \cdots p_t^{r_t}$ we have reduced the general case to the prime-power case, which we already dealt with. The $n = 1$ example is now trivial. □

5.8.4 Adjoints and Diagonalizability

In this section we will compute the adjoint of T_p on level N forms when $p \nmid N$. Here the adjoint T_p^* is with respect to the Petersson inner product $\langle -, - \rangle$ on $S_k(\Gamma_1(N))$, i.e.

$$\langle T_p f_1, f_2 \rangle = \langle f_1, T_p^* f_2 \rangle$$

for any two $f_1, f_2 \in S_k(\Gamma_1(N))$. An important consequence of the nature of T_p^* will be that it commutes with T_p. In other words, T_p is a *normal* operator. By the spectral theorem from linear algebra such operators are not just diagonalizable, but *orthogonally* diagonalizable. Thus $S_k(\Gamma_1(N))$ admits an orthonormal basis (ONB) consisting of eigenvectors for T_p. Since the T_p commute, as $p \nmid N$ varies, we can in fact do this simultaneously: There is an ONB of eigenvectors for *all* such T_p at the same time. Remember this has important consequences for the representation of $L(s, f)$ as an Euler product when f is such an eigenvector–or eigenform. However, the situation for $p|N$ is more subtle. We will return to these issues shortly.

We begin with an arbitrary double coset operator $f \mapsto f|_k \Gamma h \Gamma$ on $S_k(\Gamma)$ for some $h \in \mathrm{GL}_2(\mathbb{Q})^+$. By Proposition 5.59 we have

$$\begin{aligned}
\langle f_1|_k \Gamma h \Gamma, f_2 \rangle &= \sum\nolimits_{\gamma \in (\Gamma \cap h^{-1}\Gamma h)\backslash \Gamma} \langle f_1|_k h\gamma, f_2 \rangle \\
&= \sum\nolimits_{\gamma \in (\Gamma \cap h^{-1}\Gamma h)\backslash \Gamma} \langle f_1, f_2|_k \gamma^{-1} h^{-1} \rangle \\
&= \sum\nolimits_{\gamma \in (\Gamma \cap h^{-1}\Gamma h)\backslash \Gamma} \langle f_1, f_2|_k h^{-1} \rangle \\
&= [\Gamma : \Gamma \cap h^{-1}\Gamma h] \cdot \langle f_1, f_2|_k h^{-1} \rangle
\end{aligned}$$

for $f_1, f_2 \in S_k(\Gamma)$. Similarly,

$$\langle f_1, f_2|_k \Gamma h^{-1} \Gamma \rangle = [\Gamma : \Gamma \cap h\Gamma h^{-1}] \cdot \langle f_1|_k h, f_2 \rangle.$$

Thus, again by Proposition 5.59, we get the general adjunction property

$$\frac{1}{[\Gamma : \Gamma \cap h^{-1}\Gamma h]} \cdot \langle f_1|_k \Gamma h \Gamma, f_2 \rangle = \frac{1}{[\Gamma : \Gamma \cap h\Gamma h^{-1}]} \cdot \langle f_1, f_2|_k \Gamma h^{-1} \Gamma \rangle. \quad (5.6)$$

We now zero in on the case $\Gamma = \Gamma_1(N)$ and $h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$ with $p \nmid N$.

Lemma 5.79 *The matrix $h' := p\langle p \rangle h^{-1}$ belongs to $\Gamma h \Gamma$.*

Proof Clearly $\det(h') = p$. If $\langle p \rangle = \left(\begin{smallmatrix} a & b \\ c & p \end{smallmatrix}\right) \in \Gamma_0(N)$ an easy check gives $h' = \left(\begin{smallmatrix} pa & b \\ pc & p \end{smallmatrix}\right)$. This indeed lies in $\mathrm{Mat}_{2\times 2}(\mathbb{Z})$ and $h' \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & p \end{smallmatrix}\right) \bmod N$ since $N|c$ and $ap - bc = 1$. Now appeal to part (i) of Lemma 5.72. □

This leads to the main result of this section:

Proposition 5.80 *$T_p^* = \chi(p)^{-1}T_p$ as operators on $S_k(\Gamma_0(N), \chi)$ where $p \nmid N$. Hence $T_p^*T_p = T_pT_p^*$.*

Proof In the proof of Lemma 5.72 we found that $[\Gamma : \Gamma \cap h^{-1}\Gamma h] = p + 1$. Since $\Gamma h'\Gamma = \Gamma h\Gamma$ we infer that also $[\Gamma : \Gamma \cap h'^{-1}\Gamma h'] = p + 1$. As $\langle p\rangle$ normalizes Γ this is equivalent to $[\Gamma : \Gamma \cap h\Gamma h^{-1}] = p + 1$. Thus, in the present case, the two fudge factors in (5.6) cancel out.

For $f \in S_k(\Gamma)$ we have

$$f|_k\Gamma h^{-1}\Gamma = f|_k\Gamma\langle p\rangle^{-1}p^{-1}h'\Gamma = \big(f|_k\langle p\rangle^{-1}\big)|_k\Gamma h'\Gamma = \big(f|_k\langle p\rangle^{-1}\big)|_k\Gamma h\Gamma.$$

In light of these observations (5.6) yields:

$$\langle T_pf_1, f_2\rangle = \langle f_1, T_p(f_2|_k\langle p\rangle^{-1})\rangle.$$

In other words $T_p^*f = T_p(f|_k\langle p\rangle^{-1}) = \chi(p)^{-1}T_p$ if f has nebentype χ. □

In particular T_p is self-adjoint on $S_k(\Gamma_0(N))$.

Remark 5.81 The situation is more intricate when $p|N$. In this case the Atkin-Lehner operator T_p (soon to be denoted by U_p) is not normal in general. Conjecturally T_p is diagonalizable if $p^3 \nmid N$.

5.8.5 *Oldforms and Newforms*

When $M|N$ a Dirichlet character mod M inflates to a Dirichlet character mod N. Recall that $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$ is primitive if it does not factor through $(\mathbb{Z}/N\mathbb{Z})^\times \twoheadrightarrow (\mathbb{Z}/M\mathbb{Z})^\times$ for any $M|N$ other than $M = N$. In this sense χ genuinely lives on $(\mathbb{Z}/N\mathbb{Z})^\times$ and N is its conductor.

Similarly for modular forms. When $M|N$ we have $\Gamma_1(N) \subset \Gamma_1(M)$ and therefore an inclusion

$$i_1 : S_k(\Gamma_1(M)) \hookrightarrow S_k(\Gamma_1(N)).$$

In fact, for any positive divisor $u|\frac{N}{M}$ we have $\left(\begin{smallmatrix} u & 0 \\ 0 & 1 \end{smallmatrix}\right)\Gamma_1(N)\left(\begin{smallmatrix} u & 0 \\ 0 & 1 \end{smallmatrix}\right)^{-1} \subset \Gamma_1(M)$ and consequently an inclusion

$$\begin{aligned} i_u : S_k(\Gamma_1(M)) &\longrightarrow S_k(\Gamma_1(N)) \\ f &\longmapsto f|_k\left(\begin{smallmatrix} u & 0 \\ 0 & 1 \end{smallmatrix}\right) \end{aligned}$$

which restricts to an inclusion

$$i_u : S_k(\Gamma_0(M), \chi) \hookrightarrow S_k(\Gamma_0(N), \chi \circ \mathrm{pr}_{N,M})$$

for all Dirichlet character χ mod M (with inflation $\chi \circ \mathrm{pr}_{N,M}$). The reader should check that i_u intertwines the diamond operators $\langle d \rangle$ with d coprime to N (and a fortiori also coprime to M).

Exercise 5.82 Suppose $f \in S_k(\Gamma_0(M), \chi)$ has Fourier coeffcients $(a_n)_{n \geq 1}$. Check that $i_u(f)$ has Fourier expansion

$$i_u(f)(z) = \left(f|_k \left(\begin{smallmatrix} u & 0 \\ 0 & 1 \end{smallmatrix}\right)\right)(z) = u^{\frac{k}{2}} f(uz) = \sum_{n=1}^{\infty} u^{\frac{k}{2}} a_{n/u} e^{2\pi i n z}.$$

Use this observation, along with (5.4), to show that i_u intertwines the T_p-operators for $p \nmid N$.

When $p|N$ and $p \nmid M$ we run into a notational conflict. In this case i_u does not always commute with T_p, even for $u = 1$: The T_p-operator on $S_k(\Gamma_0(N), \chi \circ \mathrm{pr}_{N,M})$ does not in general restrict to the T_p-operator on the subspace $S_k(\Gamma_1(M), \chi)$. The issue is that $\chi(p) \neq 0$ but $(\chi \circ \mathrm{pr}_{N,M})(p) = 0$.

As we vary the level N, from now on we will denote the operator T_p by U_p when $p|N$. This is known as the pth Atkin-Lehner operator and it plays a pivotal role in the theory of newforms:

Definition 5.83 For $p|N$ the operator U_p on $S_k(\Gamma_1(N))$ takes $f = \sum_{n=1}^{\infty} a_n q^n$ to $U_p f = \sum_{n=1}^{\infty} a_{np} q^n$.

Exercise 5.84 With notation as in the previous definition, show that the adjoint of U_p satisfies the equation $U_p^*(f)^W = U_p(f^W)$ with $W = W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$. (`Hint`: Note that $W \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) W^{-1} = p \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)^{-1}$ and recall that W normalizes $\Gamma_1(N)$. Now mimic the proof of Proposition 5.80.)

The space of oldforms is spanned by the images of all the embeddings i_u as $M|N$ varies over proper divisors and $u|\frac{N}{M}$:

$$S_k(\Gamma_1(N))^{\text{old}} := \mathrm{im}\left(\sum i_u : \bigoplus_{M|N,\, M \neq N} \bigoplus_{u|\frac{N}{M}} S_k(\Gamma_1(M)) \longrightarrow S_k(\Gamma_1(N)) \right).$$

The space of newforms is the orthogonal complement with respect to the Petersson inner product:

$$S_k(\Gamma_1(N))^{\text{new}} := \left(S_k(\Gamma_1(N))^{\text{old}} \right)^{\perp}.$$

Vaguely the newforms are the ones which genuinely live on $\Gamma_1(N)$, by analogy with primitive Dirichlet characters. Note that the nebentype of an oldform is necessarily not primitive.

Example 5.85 We find it instructive to flesh out the simplest case where $N = q$ is a prime. Here $M = 1$ and $u \in \{1, q\}$. We only have two embeddings i_1 and i_q. The space of oldforms is the image of their sum:

$$\begin{aligned} i_1 + i_q : S_k(\mathrm{SL}_2(\mathbb{Z})) \oplus S_k(\mathrm{SL}_2(\mathbb{Z})) &\longrightarrow S_k(\Gamma_1(q)) \\ (f_1, f_2) &\longmapsto f_1 + f_2|_k \left(\begin{smallmatrix} q & 0 \\ 0 & 1 \end{smallmatrix}\right). \end{aligned}$$

In other words, oldforms are functions $z \mapsto f_1(z) + f_2(qz)$ where f_1 and f_2 are cusp forms on $\mathrm{SL}_2(\mathbb{Z})$. Newforms for $\Gamma_1(q)$ are perpendicular to such functions.

Since i_1 and i_q commute with T_p, for all $p \neq q$, the space $S_k(\Gamma_1(q))^{\mathrm{old}}$ is preserved by T_p. Therefore the oldforms are preserved by $T_p^* = T_p \circ \langle p \rangle^{-1}$ as well, from which we conclude that $S_k(\Gamma_1(q))^{\mathrm{new}}$ is stable under T_p.

The same holds for the Atkin-Lehner operator U_q, but this is less obvious. By computing the nth Fourier coefficient of both compositions, one easily verifies the diagram

$$\begin{array}{ccc} S_k(\mathrm{SL}_2(\mathbb{Z})) \oplus S_k(\mathrm{SL}_2(\mathbb{Z})) & \dashrightarrow & S_k(\mathrm{SL}_2(\mathbb{Z})) \oplus S_k(\mathrm{SL}_2(\mathbb{Z})) \\ \downarrow{\scriptstyle i_1+i_q} & & \downarrow{\scriptstyle i_1+i_q} \\ S_k(\Gamma_1(q)) & \xrightarrow{U_q} & S_k(\Gamma_1(q)) \end{array}$$

commutes, where the dashed arrow is

$$(f_1, f_2) \longmapsto (f_1, f_2) \begin{pmatrix} T_q & -q^{\frac{k}{2}-1} \\ q^{\frac{k}{2}} & 0 \end{pmatrix} = (T_q f_1 + q^{\frac{k}{2}} f_2, -q^{\frac{k}{2}-1} f_1).$$

In particular U_q maps $S_k(\Gamma_1(q))^{\mathrm{old}}$ to itself.

Similarly, straightforward computations show the commutativity of the diagram

$$\begin{array}{ccc} S_k(\mathrm{SL}_2(\mathbb{Z})) \oplus S_k(\mathrm{SL}_2(\mathbb{Z})) & \dashrightarrow & S_k(\mathrm{SL}_2(\mathbb{Z})) \oplus S_k(\mathrm{SL}_2(\mathbb{Z})) \\ \downarrow{\scriptstyle i_1+i_q} & & \downarrow{\scriptstyle i_1+i_q} \\ S_k(\Gamma_1(q)) & \xrightarrow{f \mapsto f^W} & S_k(\Gamma_1(q)) \end{array}$$

where the dashed arrow is now the swap

$$(f_1, f_2) \longmapsto (f_2^{\widetilde{W}}, f_1^{\widetilde{W}}) = (f_2, f_1)$$

with $\widetilde{W} = W_1 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $W = W_q = \left(\begin{smallmatrix} 0 & -1 \\ q & 0 \end{smallmatrix}\right)$. As above we infer that $f \mapsto f^W$ preserves $S_k(\Gamma_1(q))^{\mathrm{old}}$, and consequently so does the adjoint U_q^* by Exercise 5.84. The upshot of all this is that U_q preserves the newforms $S_k(\Gamma_1(q))^{\mathrm{new}}$.

Exercise 5.86 Verify that the two diagrams in the previous example indeed commute.

The next result asserts that, in general, the spaces of oldforms and newforms are stable under T_m for all $m \in \mathbb{Z}_{>0}$.

Proposition 5.87 *The orthogonal decomposition* $S_k(\Gamma_1(N)) = S_k(\Gamma_1(N))^{old} \oplus S_k(\Gamma_1(N))^{new}$ *is preserved by the diamond operators* $\langle d \rangle$ *with* $GCD(d, N) = 1$, *the Hecke operators* T_p *with* $p \nmid N$, *and the Atkin-Lehner operators* U_p *for* $p|N$.

Proof Example 5.85 generalizes as follows. Let $N > 1$ be arbitrary and pick a prime divisor $q|N$. Write $N = qM$ and consider the map

$$i_1 + i_q : S_k(\Gamma_1(M)) \oplus S_k(\Gamma_1(M)) \longrightarrow S_k(\Gamma_1(N)).$$

The dashed arrow in the analogous first diagram is now right-multiplication by $\begin{pmatrix} T_q & -q^{\frac{k}{2}-1}\langle q \rangle \\ q^{\frac{k}{2}} & 0 \end{pmatrix}$. In the second diagram the dashed arrow is $(f_1, f_2) \mapsto (f_2^{\widetilde{W}}, f_1^{\widetilde{W}})$ where now $\widetilde{W} = W_M = \left(\begin{smallmatrix} 0 & -1 \\ M & 0 \end{smallmatrix}\right)$. We conclude that $\mathrm{im}(i_1 + i_q)$ is preserved by U_q and its adjoint.

Now let $p|N$ be prime different from q. In this case we have the commutative diagram

$$\begin{array}{ccc} S_k(\Gamma_1(N/p)) \oplus S_k(\Gamma_1(N/p)) & \dashrightarrow & S_k(\Gamma_1(N/p)) \oplus S_k(\Gamma_1(N/p)) \\ \downarrow{\scriptstyle i_1+i_p} & & \downarrow{\scriptstyle i_1+i_p} \\ S_k(\Gamma_1(N)) & \xrightarrow{\quad U_q \quad} & S_k(\Gamma_1(N)) \end{array}$$

with the dashed morphism being $(f_1, f_2) \mapsto (U_q f_1, U_q f_2)$. To see this, compare the Fourier coefficients of the two compositions. For this reason U_q and U_q^* also preserve $\mathrm{im}(i_1 + i_p)$ for such p.

Note that every map i_u, with $u \neq 1$, factors through $i_p : S_k(\Gamma_1(N/p)) \hookrightarrow S_k(\Gamma_1(N))$ for some $p|u|N$. Therefore the space of oldforms is spanned by the images of these maps i_p, along with i_1. In other words,

$$S_k(\Gamma_1(N))^{\mathrm{old}} = \bigoplus_{p|N} \mathrm{im}(i_1 + i_p).$$

From what we have observed above, it is now clear that U_q and U_q^* preserve $S_k(\Gamma_1(N))^{\mathrm{old}}$ and therefore also $S_k(\Gamma_1(N))^{\mathrm{new}}$ for all $q|N$. The analogous statement for the operators T_p, with $p \nmid N$, follows exactly as in Example 5.85. □

We return to the maps $i_p : S_k(\Gamma_1(N/p)) \hookrightarrow S_k(\Gamma_1(N))$ for primes $p|N$. If $f = \sum_{n=1}^{\infty} a_n q^n \in \mathrm{im}(i_p)$ then clearly $a_n = 0$ for $p \nmid n$. In particular $a_n = 0$ when $\mathrm{GCD}(n, N) = 1$. In other words,

$$f \in \sum_{p|N} \mathrm{im}(i_p) \implies a_n(f) = 0 \text{ for all } n \text{ coprime to } N.$$

The central ingredient in Atkin-Lehner theory is the amazing result that the converse ($\Longleftarrow$) holds:

Theorem 5.88 (Atkin, Lehner) *If the Fourier coefficients $(a_n)_{n\geq 1}$ of a cusp form $f \in S_k(\Gamma_1(N))$ have the property that $a_n = 0$ whenever $GCD(n, N) = 1$, then f belongs to the image of*

$$\sum_{p|N} i_p : \bigoplus_{p|N} S_k(\Gamma_1(N/p)) \longrightarrow S_k(\Gamma_1(N))$$
$$f \longmapsto \sum_{p|N} f|_k \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right).$$

In particular any such f is necessarily an oldform.

To keep the momentum of these notes we will not pause to prove it, our goal is just to give an overview of the theory. The proof is in fact rather delicate. We refer to [DS05, Sect. 5.7] where the authors expose an intriguing argument, due to Carlton, based on the observation that $S_k(\Gamma(N))$ is a representation of $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. See [DS05, Theorem 5.7.1, p. 190] and its proof.

As a first output we highlight the following:

Corollary 5.89 *Let $f \in S_k(\Gamma_0(N), \chi)$ be an eigenvector for T_m, for all $m \in \mathbb{Z}_{>0}$ coprime to N. Assume f is a newform. Then:*

(a) The first Fourier coefficient $a_1(f)$ is nonzero;
(b) f is an eigenvector for T_m for all $m \in \mathbb{Z}_{>0}$.

Proof Suppose $f \in S_k(\Gamma_0(N), \chi)$ is an eigenvector for all T_m with $\mathrm{GCD}(m, N) = 1$; say $T_m f = \lambda_m f$ for all such m. By Proposition 5.78 we know the Fourier coefficients $(a_n)_{n\geq 1}$ of f satisfy

$$a_m = a_1(T_m f) = a_1(\lambda_m f) = \lambda_m a_1.$$

If $a_1 = 0$, we deduce that $a_m = 0$ when $\mathrm{GCD}(m, N) = 1$. Theorem 5.88 tells us f must be an oldform. However, we are assuming f is a newform. This gives the contradiction $f = 0$. This proves part (a).

For the proof of (b) we may assume $a_1 = 1$ by scaling (knowing the first Fourier coefficient is nonzero). Fix an arbitrary $m_0 \in \mathbb{Z}_{>0}$. By Proposition 5.87 we know T_{m_0} preserves $S_k(\Gamma_1(N))^{\mathrm{new}}$. Therefore the auxiliary form $g := T_{m_0} f - a_{m_0} f$ is a newform. Since $(T_m)_{m\geq 1}$ is a commuting family we also know that $T_m g = a_m g$ for all m coprime to N. However,

$$a_1(g) = a_1(T_{m_0} f) - a_{m_0} a_1 = a_{m_0} - a_{m_0} = 0.$$

By part (a) applied to g this scenario can only happen if $g = 0$. We conclude that $T_{m_0} f = a_{m_0} f$. Since m_0 was arbitrary, this proves (b). □

We will often scale a newform f as in the previous Corollary to arrange that $a_1(f) = 1$. When this is satisfied, we call f a *normalized* newform. Under this assumption the Fourier coefficients and the Hecke eigenvalues of f agree on the nose, and $L(s, f)$ has an Euler product. These forms are the crucial building blocks analogous to primitive Dirichlet characters.

Definition 5.90 $\mathrm{Prim}_k(\Gamma_0(N), \chi)$ denotes the set of *primitive* forms, i.e. all $f \in S_k(\Gamma_1(N))^{\mathrm{new}}$ with nebentype χ such that

- f is an eigenvector for T_m for all $m \in \mathbb{Z}_{>0}$;
- $a_1 = 1$.

Equivalently, $T_m f = a_m f$ for all $m \in \mathbb{Z}_{>0}$. (Here $(a_n)_{n\geq 1}$ are the Fourier coefficients of f.)

Example 5.91 For $N = 1$ and $k = 12$ there is a unique primitive form:

$$(2\pi)^{-12}\Delta(z) = \sum_{n=1}^{\infty} \tau(n) q^n == q \prod_{n=1}^{\infty} (1 - q^n)^{24} = q - 24q^2 + 252q^3 - 1472q^4 + \cdots .$$

In particular the coefficients satisfy the identities (conjectured by Ramanujan, proved by Mordell):

- $\tau(mn) = \tau(m)\tau(n)$ provided $\mathrm{GCD}(m, n) = 1$;
- $\tau(p^r) = \tau(p)\tau(p^{r-1}) - p^{11}\tau(p^{r-2})$ for $r \geq 2$.

(Since $S_{12}(\mathrm{SL}_2(\mathbb{Z}))$ is spanned by Δ, the latter is clearly an eigenfunction for all the T_m-operators.)

Remark 5.92 $\mathrm{Prim}_k(\Gamma_0(N), \chi)$ is a basis for $S_k(\Gamma_0(N), \chi)^{\mathrm{new}}$. Why? This space is preserved by all Hecke operators, and T_m is diagonalizable at least when $m \in \mathbb{Z}_{>0}$ is coprime to N. Thus $S_k(\Gamma_0(N), \chi)^{\mathrm{new}}$ has a basis (f_i) of common eigenvectors for T_m with $\mathrm{GCD}(m, N) = 1$. By Corollary 5.89 these basis vectors f_i are *automatically* eigenvectors for the remaining T_m with $\mathrm{GCD}(m, N) > 1$, and we may normalize all the f_i to turn them into primitive forms.

In fact $\mathrm{Prim}_k(\Gamma_0(N), \chi)$ is always an orthogonal basis for $S_k(\Gamma_0(N), \chi)^{\mathrm{new}}$ as a consequence of the next result, which will also be used below for other purposes.

Proposition 5.93 (Strong Multiplicity One) *If $f_1, f_2 \in Prim_k(\Gamma_0(N), \chi)$ have the same T_p-eigenvalues for all $p \nmid N$, then $f_1 = f_2$.*

Proof Their difference $h := f_1 - f_2$ belongs to $S_k(\Gamma_0(N), \chi)^{\mathrm{new}}$ and has Fourier coefficients

$$a_m(h) = a_m(f_1) - a_m(f_2) = 0$$

for all m coprime to N. By Theorem 5.88 we know h must also be an oldform, so that indeed $h = 0$. □

Recall that two eigenvectors of a normal operator with distinct eigenvalues are orthogonal. (To see this note that a normal operator T satisfies $\|Tv\| = \|T^*v\|$. Applying this to $T - \lambda I$ shows $Tv = \lambda v$ if and only if $T^*v = \overline{\lambda}v$. Suppose $Tw = \mu w$ with $\lambda \neq \mu$. Now compute $\langle Tv, w\rangle$ in two ways and infer that $\langle v, w\rangle = 0$.) In our setup, if $f_1, f_2 \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ are distinct, then by Proposition 5.93 they have distinct T_p-eigenvalues for some $p \nmid N$. As T_p is normal for $p \nmid N$ we find that $\langle f_1, f_2\rangle = 0$.

5.8.6 The Dual Modular Form

Oftentimes, exploiting the geometry of the curve $X(\Gamma)$ gives a basis for $S_k(\Gamma)$ consisting of modular forms with Fourier coefficients in $\mathbb{Q}$. In this case the entire group of field automorphisms $\mathrm{Aut}(\mathbb{C})$ acts on $S_k(\Gamma)$. An automorphism σ acts on $f = \sum_{n=1}^{\infty} a_n q^n$ by letting

$$f^\sigma = \sum_{n=1}^{\infty} \sigma(a_n) q^n.$$

(To see f^σ belongs to $S_k(\Gamma)$ expand f in the $\mathbb{Q}$-basis and let σ act on the coefficients.) For complex conjugation, which is the unique *continuous* automorphism of $\mathbb{C}$ other than the identity, there is a more hands-on approach.

Start with the transformation $z \mapsto -\bar{z}$ of $\mathcal{H}$. Geometrically, this is reflection across the imaginary axis. For any $f \in \mathrm{Hol}(\mathcal{H})$ we let $f^\vee(z) := \overline{f(-\bar{z})}$. Clearly $f^\vee$ is again a holomorphic function on $\mathcal{H}$ (look at the Taylor expansion of f) called the dual of f. One easily checks that $q(z) = e^{2\pi i z}$ is self-dual. Therefore, if $f = \sum_{n=1}^{\infty} a_n q^n$, its dual has complex conjugate Fourier coefficients:

$$f^\vee = \sum_{n=1}^{\infty} \overline{a_n} q^n.$$

Suppose $f \in S_k(\Gamma_0(N), \chi)$. We claim that in this case $f^\vee \in S_k(\Gamma_0(N), \overline{\chi})$. To see this, begin with an arbitrary matrix $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$ and first note that

$$-\overline{\gamma z} = -\frac{a\bar{z} + b}{c\bar{z} + d} = \frac{-a(-\bar{z}) + b}{c(-\bar{z}) - d} = \gamma'(-\bar{z})$$

where $\gamma' := \left(\begin{smallmatrix} -a & b \\ c & -d \end{smallmatrix}\right) \in \Gamma_0(N)$. Therefore

$$f^\vee(\gamma z) = \overline{f(-\overline{\gamma z})} = \overline{\chi(-d)(c(-\bar{z}) - d)^k f(-\bar{z})} = \overline{\chi(d)}(cz + d)^k f^\vee(z).$$

We used the tacit assumption that $\chi(-1) = (-1)^k$. Without it the claim is vacuous. The growth condition at the cusps is established in a similar fashion, using the next exercise.

Exercise 5.94 For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{R})^+$ introduce $g' := \left(\begin{smallmatrix} -a & b \\ c & -d \end{smallmatrix}\right)$. Verify the identity

$$(f|_k g)^\vee = (-1)^k \cdot (f^\vee|_k g')$$

for all $f \in \mathrm{Hol}(\mathcal{H})$.

Here is a short list of almost immediate properties. We encourage the reader to fill in the details.

- Taking the dual commutes with T_m for all $m \in \mathbb{Z}_{>0}$;
- The map $f \mapsto f^\vee$ preserves both $S_k(\Gamma_0(N), \chi)^{\mathrm{old}}$ and $S_k(\Gamma_0(N), \chi)^{\mathrm{new}}$;
- Passing to the dual form gives a bijection $\mathrm{Prim}_k(\Gamma_0(N), \chi) \xrightarrow{1:1} \mathrm{Prim}_k(\Gamma_0(N), \overline{\chi})$.

For the second bullet point, the change of variables $(x, y) \mapsto (-x, y)$ shows that $\langle f_1, f_2 \rangle = \langle f_2^\vee, f_1^\vee \rangle$.

For a primitive form $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ we now relate $f^\vee$ to the form $f^W = f|_k W$ appearing in the functional equation for $L(s, f)$. We know that at least $f^W \in S_k(\Gamma_0(N), \overline{\chi})$ is a newform since $(\cdot)^W$ is involutive up to a sign and it preserves oldforms. Furthermore, for $p \nmid N$ we have

$$T_p(f^W) = T_p^*(f)^W = \overline{a_p} f^W$$

where we used the normality of T_p in the last step. In particular f^W is automatically an eigenvector for all the T_m-operators, by Corollary 5.89, and its first Fourier coefficient is nonzero. However, f^W may not be normalized.

Lemma 5.95 *$f^W = a_1(f^W) f^\vee$ and $a_1(f^W)$ has absolute value one.*

Proof Exercise 5.94 with $g = W = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right) = g'$ gives $(f^W)^\vee = (-1)^k \cdot (f^\vee)^W$. On the other hand, by strong multiplicity one (Proposition 5.93) we know $f^W = a_1(f^W) f^\vee$. Therefore

$$\overline{a_1(f^W)} f = \left(a_1(f^W) f^\vee\right)^\vee = (f^W)^\vee = (-1)^k \cdot (f^\vee)^W = (-1)^k \cdot \left(a_1(f^W)^{-1} f^W\right)^W = a_1(f^W)^{-1} f$$

since $f^{WW} = (-1)^k f$. Considering the first Fourier coefficient above shows that indeed $a_1(f^W) \in \mathbb{C}^1$. □

For primitive forms the functional equation for the L-series can now be restated as follows.

Corollary 5.96 *Let $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$. Then the completed L-function $\widehat{L}(s, f) := N^{\frac{s}{2}} L_{\mathbb{C}}(s) L(s, f)$ admits analytic continuation to an entire function satisfying the equation*

$$\widehat{L}(k-s, f^\vee) = W(f)\widehat{L}(s, f)$$

where $W(f) := i^{-k}a_1(f^W)^{-1} \in \mathbb{C}^1$. *(This is called the root number of* f*.)*

Proof By Theorem 5.68 we know $\widehat{L}(s, f)$ extends to an entire function and we have the functional equation

$$\widehat{L}(s, f) = i^k\widehat{L}(k-s, f^W) = i^k a_1(f^W)\widehat{L}(k-s, f^\vee) = \overline{W(f)} \cdot \widehat{L}(k-s, f^\vee)$$

as claimed. □

5.9 The Rankin-Selberg Method

5.9.1 Real Analytic Eisenstein Series

The hyperbolic Laplacian $\Delta := -y^2\left(\frac{\partial^2}{\partial x} + \frac{\partial^2}{\partial y}\right)$ is $\mathrm{GL}_2(\mathbb{R})^+$-invariant in the sense that $\Delta(f \circ g) = \Delta(f) \circ g$ for all smooth functions f on $\mathcal{H}$ and all $g \in \mathrm{GL}_2(\mathbb{R})^+$. (We identify g with a linear fractional transformation of $\mathcal{H}$.)

The function $\mathcal{H} \ni x + iy \mapsto y^s$ is clearly an eigenfunction for Δ with eigenvalue $s(1-s)$. Therefore, so is the function $\mathcal{H} \ni z \mapsto \mathrm{Im}(gz)^s$ for all $g \in \mathrm{GL}_2(\mathbb{R})^+$ by the previous paragraph. If Γ is a finite index subgroup of $\mathrm{SL}_2(\mathbb{Z})$, a natural way to produce a Γ-invariant eigenfunction is to, in a suitable sense, take the average of all the functions $\mathcal{H} \ni z \mapsto \mathrm{Im}(\gamma z)^s$ with $\gamma \in \Gamma$. To guarantee convergence we need to assume $\mathrm{Re}(s)$ is large enough, and we must also mod out by the subgroup

$$\Gamma_\infty := \mathrm{Stab}_\Gamma(\infty) = \Gamma \cap \{\pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) : b \in \mathbb{Z}\}.$$

Indeed $\mathrm{Im}(\gamma z)$ only depends on the coset $\gamma \in \Gamma_\infty\backslash\Gamma$. When $\mathrm{Re}(s) \gg 0$ it then makes sense to consider the infinite series

$$\mathcal{E}(z, s) := \mathcal{E}_s(z) = \sum\nolimits_{\gamma \in \Gamma_\infty\backslash\Gamma} \mathrm{Im}(\gamma z)^s.$$

Ignoring convergence issues for now, this is a Γ-invariant eigenfunction for Δ with eigenvalue $s(1-s)$:

- $\mathcal{E}(\gamma z, s) = \mathcal{E}(z, s)$ for all $\gamma \in \Gamma$;
- $\Delta\mathcal{E}_s = s(1-s)\mathcal{E}_s$.

Note that $\mathcal{E}_s$ is *not* a holomorphic function on $\mathcal{H}$, it is only real-analytic (meaning it is given by a Taylor series in the two variables x and y).

From now on we will focus exclusively on the case $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. Thus $\Gamma_\infty = \{\pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) : b \in \mathbb{Z}\}$. In this situation we have a natural bijection

$$\begin{aligned}\Gamma_\infty\backslash\Gamma &\xrightarrow{1:1} \{(c,d) : c,d \in \mathbb{Z} \text{ coprime}\}/\pm \\ \Gamma_\infty \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) &\longmapsto \text{the equivalence class of } (c,d).\end{aligned}$$

(The injectivity is a one line matrix computation left to the reader.) This allows us to make the sum $\mathcal{E}(z,s)$ more explicit in the $\mathrm{SL}_2(\mathbb{Z})$-case:

$$2\mathcal{E}(z,s) = \sum\nolimits_{(c,d):c,d\in\mathbb{Z} \text{ coprime}} \frac{\mathrm{Im}(z)^s}{|cz+d|^{2s}}.$$

The factor 2 is there because $\pm(c,d)$ contribute the same amount to the sum. In particular, still avoiding convergence questions, we have

$$\begin{aligned}\mathrm{Im}(z)^s \cdot \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} \frac{1}{|mz+n|^{2s}} &= \sum_{N=1}^{\infty} \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2:\mathrm{GCD}(m,n)=N} \frac{\mathrm{Im}(z)^s}{|mz+n|^{2s}} \\ &= \sum_{N=1}^{\infty} \sum\nolimits_{(c,d):c,d\in\mathbb{Z} \text{ coprime}} \frac{\mathrm{Im}(z)^s}{|Ncz+Nd|^{2s}} \\ &= \sum_{N=1}^{\infty} N^{-2s} \cdot 2\mathcal{E}(z,s) \\ &= 2\zeta(2s)\cdot\mathcal{E}(z,s).\end{aligned}$$

For $s=\kappa\in\mathbb{Z}_{\geq 2}$ the initial sum is reminiscent of $G_\kappa(z)$, except for the factor $\mathrm{Im}(z)^s$ and the absolute value in the denominator. Hence we think of $\mathcal{E}(z,s)$ as a non-holomorphic analogue of $E_\kappa(z)$.

The proof of absolute convergence for $G_\kappa(\Lambda)$ now carries over verbatim, showing that

$$\sigma > 1 \implies \sum\nolimits_{0\neq\lambda\in\Lambda_z} |\lambda|^{-2\sigma} < \infty.$$

Consequently, for a fixed $z\in\mathcal{H}$, the Eisenstein series $\mathcal{E}(z,s)$ is absolutely convergent for $\mathrm{Re}(s)>1$ and it represents a holomorphic function in that region. Still keeping z fixed, our next order of business is to extend $s\mapsto\mathcal{E}(z,s)$ to a meromorphic function on $\mathbb{C}$.

Theorem 5.97 *For any fixed $z\in\mathcal{H}$, the function $E(z,s) := \pi^{-s}\Gamma(s)\zeta(2s)\mathcal{E}(z,s)$ has meromorphic continuation to the whole s-plane and satisfies the functional equation*

$$E(z,1-s) = E(z,s).$$

It has exactly the two poles $s=1$ and $s=0$. Both poles are simple, with residues $\pm\frac{1}{2}$ respectively.

Proof We adopt the setup from Sect. 5.6.4 with $V=\mathbb{C}\simeq\mathbb{R}^2$ and $L=\frac{1}{\sqrt{y}}\Lambda_z = \mathbb{Z}\frac{1}{\sqrt{y}}\oplus\mathbb{Z}(\frac{x}{\sqrt{y}}+i\sqrt{y})$ for a fixed $z=x+iy\in\mathcal{H}$. Note that the fundamental parallelogram of L has base $\frac{1}{\sqrt{y}}$ and height $\sqrt{y}$. Thus $\dot{\mu}(V/L)=1$ where μ is the

Lebesgue measure on $\mathbb{C}$. The dot product on $\mathbb{R}^2$ transfers to the inner product $\langle w, w' \rangle = \mathrm{Re}(\bar{w}w')$ on $\mathbb{C}$. The dual lattice $L^\perp$ consists of all $w = a + ib \in \mathbb{C}$ such that

(i) $\mathrm{Re}(\bar{w}\frac{1}{\sqrt{y}}) = \frac{a}{\sqrt{y}} \in \mathbb{Z}$;
(ii) $\mathrm{Re}(\bar{w}(\frac{x}{\sqrt{y}} + i\sqrt{y})) = \frac{ax}{\sqrt{y}} + b\sqrt{y} \in \mathbb{Z}$.

From this one infers that $L^\perp = i^{-1}L$. As a result L and $L^\perp$ have the same theta series, namely

$$\theta_{L^\perp}(w) = \theta_L(w) = \sum_{\ell \in L} e^{\pi i |\ell|^2 w} = \sum\nolimits_{(m,n)\in\mathbb{Z}^2} e^{\pi i |mz+n|^2 w/y}.$$

By Lemma 5.47 it has the transformation property

$$\theta_L\Big(\frac{i}{t}\Big) = t\theta_L(it)$$

for all $t > 0$. We connect the θ-series to the Eisenstein series $\mathcal{E}(z, s)$ by considering the Mellin transformation, assuming $\mathrm{Re}(s) > 1$ of course:

$$\begin{aligned}
\frac{1}{2}\int_0^\infty \Big(\theta_L(it) - 1\Big) t^s \frac{dt}{t} &= \frac{1}{2}\int_0^\infty \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} e^{-\pi|mz+n|^2 t/y} t^s \frac{dt}{t} \\
&= \frac{1}{2}\cdot \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} \int_0^\infty e^{-\pi|mz+n|^2 t/y} t^s \frac{dt}{t} \\
&= \frac{1}{2}\cdot \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} \int_0^\infty e^{-u} \left(\frac{uy}{\pi|mz+n|^2}\right)^s \frac{du}{u} \\
&= \frac{1}{2}\cdot \sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} \pi^{-s} y^s \frac{1}{|mz+n|^{2s}} \int_0^\infty e^{-u} u^s \frac{du}{u} \\
&= \pi^{-s}\Gamma(s)\zeta(2s)\mathcal{E}(z, s) \\
&= E(z, s).
\end{aligned}$$

The usual trick of splitting the integral as $\int_0^1$ plus $\int_1^\infty$, and using the above transformation property for the θ-series, now gives

$$2E(z, s) = I(s) + I(1 - s) + \frac{1}{s - 1} - \frac{1}{s}$$

where $I(s) := \int_1^\infty \big(\theta_L(it) - 1\big) t^s \frac{dt}{t}$. The expression on the right-hand side of the above relation visibly has all the desired properties since $\theta_L(it) - 1$ decays exponentially as $t \to \infty$. □

Note that the factor $\xi(2s)$ which appears in the normalization of the Eisenstein series is invariant under $s \mapsto \frac{1}{2} - s$. Therefore the unnormalized Eisenstein series

$\mathcal{E}(z,s)$ does not itself have any nice invariance properties, and working with the normalized version $E(z,s)$ tends to be more useful.

We found that the nth Fourier coefficient of $G_\kappa(z)$ is essentially $\sigma_{2\kappa-1}(n)$ for $n>0$. Somewhat surprisingly, an analogous result is true for $E(z,s)$ as we will show next.

The Fourier expansion of $E(z,s)$ involves the K-Bessel function. There are different conventions about the notation, see Eq. 10.32.10 in

`dlmf.nist.gov/10.32`

for example, but in number theory the standard normalization is

$$K_s(y) := \frac{1}{2}\int_0^\infty e^{-y(t+t^{-1})/2} t^s \frac{dt}{t}.$$

Here $y>0$, so that the integrand decays exponentially as $t\to\infty$ and $t\to 0^+$, and the integral is defined for all $s\in\mathbb{C}$ and results in an entire function $s\mapsto K_s(y)$. It is not that important to us exactly what $K_s(y)$ is. The key properties we will use are:

- $K_s(y)=K_{-s}(y)$;
- For all $s\in\mathbb{C}$ there is a constant $C_s>0$ such that $|K_s(y)|\le C_s e^{-y/2}$ for all $y>4$.

The first one is clear from the definition of $K_s(y)$ by the change of variables $t'=\frac{1}{t}$. The second property is less obvious. One way to prove it is to observe that

$$A,B>2 \Longrightarrow A>2>\frac{B}{B-1} \Longrightarrow AB>A+B \Longrightarrow e^{-AB}<e^{-A}e^{-B}.$$

Taking $A=y/2>2$ and $B=t+t^{-1}>2$ for positive $t\neq 1$ gives the estimate

$$|K_s(y)|\le \frac{1}{2}\int_0^\infty e^{-y(t+t^{-1})/2} t^{\mathrm{Re}(s)}\frac{dt}{t} \le e^{-y/2}\frac{1}{2}\int_0^\infty e^{-(t+t^{-1})} t^{\mathrm{Re}(s)}\frac{dt}{t} = K_{\mathrm{Re}(s)}(2)e^{-y/2}.$$

In conclusion, $C_s := K_{\mathrm{Re}(s)}(2)$ works.

Proposition 5.98 *The normalized real-analytic Eisenstein series has Fourier expansion*

$$E(z,s)=\xi(2s)y^s+\xi(2s-1)y^{1-s}+\sum_{r\in\mathbb{Z}} 2\sqrt{y}|r|^{s-\frac{1}{2}}K_{s-\frac{1}{2}}(2\pi|r|y)\sigma_{1-2s}(|r|)e^{2\pi irx}$$

for all $s\in\mathbb{C}$. In particular $E(z,s)$ has moderate growth as $Im(z)\to\infty$. More precisely, as $y\to\infty$,

$$E(x+iy,s)=O(y^\alpha)$$

for any $\alpha\ge\max\{Re(s),Re(1-s)\}$ and $s\notin\{0,1\}$.

Proof Clearly $E(z+1,s) = E(z,s)$ by taking $\gamma = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ above. From the proof of Theorem 5.97 it is also easy to deduce that $E(z,s)$ is a smooth function on $\mathcal{H}$. Therefore it has a Fourier series

$$E(z,s) = E(x+iy,s) = \sum_{r\in\mathbb{Z}} a_r(y,s)e^{2\pi irx}, \qquad a_r(y,s) = \int_0^1 E(x+iy,s)e^{-2\pi irx}dx.$$

Inserting the series expansion of $2\zeta(2s)\mathcal{E}(z,s)$ the rth Fourier coefficient is given by

$$\begin{aligned} a_r(y,s) &= \pi^{-s}\Gamma(s)\frac{1}{2}\sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} \int_0^1 \frac{y^s}{|mz+n|^{2s}}e^{-2\pi irx}dx \\ &= \pi^{-s}\Gamma(s)\frac{1}{2}\sum\nolimits_{(0,0)\neq(m,n)\in\mathbb{Z}^2} \int_0^1 \frac{y^s}{((mx+n)^2+(my)^2)^s}e^{-2\pi irx}dx \\ &= \pi^{-s}\Gamma(s)\frac{1}{2}\sum\nolimits_{(m,n)\in\mathbb{Z}^2:m\neq 0} m^{-2s}\int_0^1 \frac{y^s}{((x+n/m)^2+y^2)^s}e^{-2\pi irx}dx \\ &\quad + \pi^{-s}\Gamma(s)\frac{1}{2}\sum_{n\neq 0} n^{-2s}\int_0^1 y^s e^{-2\pi irx}dx. \end{aligned}$$

When $r = 0$ this simplifies to

$$a_0(y,s) = \pi^{-s}\Gamma(s)\zeta(2s)y^s + \pi^{-s}\Gamma(s)\frac{1}{2}\sum\nolimits_{(m,n)\in\mathbb{Z}^2:m\neq 0} m^{-2s}\int_0^1 \frac{y^s}{((x+n/m)^2+y^2)^s}dx.$$

When $r \neq 0$ the expression reduces to

$$a_r(y,s) = \pi^{-s}\Gamma(s)\frac{1}{2}\sum\nolimits_{(m,n)\in\mathbb{Z}^2:m\neq 0} m^{-2s}\int_0^1 \frac{y^s}{((x+n/m)^2+y^2)^s}e^{-2\pi irx}dx.$$

Since $\pm(m,n)$ give identical terms we may write $\sum_{m=1}^{\infty}\sum_{n\in\mathbb{Z}}$ instead of $\frac{1}{2}\sum_{(m,n)\in\mathbb{Z}^2:m\neq 0}$. Then we replace the inner sum over $n\in\mathbb{Z}$ with a sum over the residue classes $\nu + m\mathbb{Z}$. Our task then becomes to evaluate the sums

$$y^s\sum_{m=1}^{\infty}\sum_{\nu=0}^{m-1} m^{-2s}\int_{-\infty}^{\infty}\frac{1}{((x+\nu/m)^2+y^2)^s}e^{-2\pi irx}dx.$$

Note how the integral $\int_0^1$ turned into an integral over the whole real line. We start out by analyzing the indefinite integrals. Invoking the result of Exercise 5.99 right after this proof we find that

$$\begin{aligned}\int_{-\infty}^{\infty}\frac{1}{((x+\nu/m)^2+y^2)^s}e^{-2\pi irx}dx &= \int_{-\infty}^{\infty}\frac{1}{(x'^2+y^2)^s}e^{-2\pi ir(x'-\nu/m)}dx'\\ &= e^{2\pi ir\nu/m}\int_{-\infty}^{\infty}\frac{1}{(x^2+y^2)^s}e^{-2\pi irx}dx\\ &= e^{2\pi ir\nu/m}\pi^s\frac{1}{\Gamma(s)}y^{-s}\cdot\begin{cases}\pi^{\frac{1}{2}-s}\Gamma(s-\frac{1}{2})y^{1-s} & \text{if } r=0\\ 2\sqrt{y}|r|^{s-\frac{1}{2}}K_{s-\frac{1}{2}}(2\pi|r|y) & \text{if } r\neq 0\end{cases}.\end{aligned}$$

Combining everything for $r = 0$ gives the constant (i.e., independent of x) term

$$a_0(y,s) = \pi^{-s}\Gamma(s)\zeta(2s)y^s + \pi^{\frac{1}{2}-s}\Gamma(s-\frac{1}{2})\zeta(2s-1)y^{1-s} = \xi(2s)y^s + \xi(2s-1)y^{1-s}.$$

Note how the two terms on the right-hand side are interchanged under $s \mapsto 1-s$. For $r \neq 0$ we get the formula

$$a_r(y,s) = 2\sqrt{y}|r|^{s-\frac{1}{2}}K_{s-\frac{1}{2}}(2\pi|r|y)\sigma_{1-2s}(|r|),$$

using the fact that $\sum_{\nu=0}^{m-1}e^{2\pi ir\nu/m} = \begin{cases}m & \text{if } m|r\\ 0 & \text{if } m\nmid r\end{cases}$. In more detail we have

$$\sum_{m=1}^{\infty}\sum_{\nu=0}^{m-1}m^{-2s}e^{2\pi ir\nu/m} = \sum_{0<m|r}m^{1-2s} = \sigma_{1-2s}(r).$$

This finishes the proof when $\mathrm{Re}(s)$ is sufficiently large. It remains to note that the Fourier expansion converges for all $s \in \mathbb{C}$ (uniformly in compact subsets) since $|K_{s-\frac{1}{2}}(2\pi|r|y)| \leq C_{s-\frac{1}{2}}e^{-\pi|r|y}$ for large $|r|$ and therefore the rth term decays exponentially as $|r| \to \infty$. □

Exercise 5.99 Assuming $\mathrm{Re}(s)$ is sufficiently large, show that

$$\pi^{-s}\Gamma(s)y^s\int_{-\infty}^{\infty}\frac{1}{(x^2+y^2)^s}e^{-2\pi irx}dx = \begin{cases}\pi^{\frac{1}{2}-s}\Gamma(s-\frac{1}{2})y^{1-s} & \text{if } r=0\\ 2\sqrt{y}|r|^{s-\frac{1}{2}}K_{s-\frac{1}{2}}(2\pi|r|y) & \text{if } r\neq 0\end{cases}.$$

(`Hint`: Express the left-hand side as a double integral by inserting the definition of $\Gamma(s)$. Make a substitution and use the fact that $f_a(t) = e^{-a^2t^2}$ has Fourier transformation $\widehat{f_a}(x) = \frac{\sqrt{\pi}}{a}e^{-\pi^2(x/a)^2}$ for $a > 0$.)

The Fourier expansion of $E(z,s)$ gives a strengthening of Theorem 5.97: It furnishes a very explicit meromorphic continuation from which the poles and their residues are immediately read off. The functional equation for $E(z,s)$ is also a byproduct. To see this, as noted above $K_s(y) = K_{-s}(y)$ and for the σ-functions we have the obvious relation $\sigma_s(r) = r^s\sigma_{-s}(r)$.

5.9.2 Unfolding

The fundamental idea behind the Rankin-Selberg method is to consider the integral

$$\mathrm{RS}_{f,g}(s) := \int_{\Gamma\backslash\mathcal{H}} E(z,s) f(z) \overline{g(z)} \mathrm{Im}(z)^k d\mu(z)$$

for two cusp forms $f, g \in S_k(\Gamma)$. We remind the reader that Γ is shorthand for $\mathrm{SL}_2(\mathbb{Z})$ in this section and $d\mu(z) = \frac{dxdy}{y^2}$ is the invariant measure on $\mathcal{H}$. (The method works in much greater generality, but for the sake of exposition we narrow our scope to the level one case where most features already come into play.)

The integral defining $\mathrm{RS}_{f,g}(s)$ converges for all $s \in \mathbb{C} - \{0, 1\}$ since $f(z)$ and $g(z)$ both grow like $O(e^{-2\pi y})$ and $E(z,s) = O(y^\alpha)$ as $y = \mathrm{Im}(z) \to \infty$. We infer from Theorem 5.97 that $\mathrm{RS}_{f,g}(s)$ is a meromorphic function on $\mathbb{C}$ with at most simple poles at $s = 1$ and $s = 0$. If they are poles, their residues are $\pm\frac{1}{2}\langle f, g\rangle$ respectively. In particular they *are* poles when $f = g \neq 0$ where the residues are $\pm\frac{1}{2}\|f\|^2$. In addition we get the invariance of $\mathrm{RS}_{f,g}(s)$ under $s \mapsto 1 - s$ for free.

As we will see shortly, $\mathrm{RS}_{f,g}(s)$ is closely related to the Dirichlet series $\sum_{n=1}^\infty \frac{a_n \overline{b_n}}{n^s}$ encoding the Fourier coefficients $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ of f and g respectively. This will enable us to extend the L-series to a meromorphic function on $\mathbb{C}$, determine its poles with residues, and establish a functional equation of the usual kind. Therein lies the power of the Rankin-Selberg method.

Definition 5.100 Let $f = \sum_{n=1}^\infty a_n q^n$ and $g = \sum_{n=1}^\infty b_n q^n$ be forms in $S_k(\mathrm{SL}_2(\mathbb{Z}))$. The *Rankin-Selberg* L-function of the pair $\{f, g\}$ is

$$L(s, f \times g) := \zeta(2s - 2k + 2) \sum_{n=1}^\infty \frac{a_n b_n}{n^s}.$$

Remark 5.101 Using the crude $O(n^{\frac{k}{2}})$ estimate for a_n and b_n shows the Dirichlet series $\sum_{n=1}^\infty \frac{a_n b_n}{n^s}$ at least converges absolutely for $\mathrm{Re}(s) > k + 1$. Deligne's bound $O(n^{\frac{k-1}{2}+\epsilon})$ improves this to $\mathrm{Re}(s) > k$. For $f = g^\vee$ we will in fact *deduce* this from the Rankin-Selberg method in Corollary 5.104 below, without appealing to Deligne. The general case is gathered from this special case in Exercise 5.106.

It may seem strange to include the factor $\zeta(2s - 2k + 2)$ in the definition of $L(s, f \times g)$ rather than having the L-function be the actual Dirichlet series. The justification comes in the next section: If f and g are Hecke eigenforms, so that $L(s, f)$ and $L(s, g)$ are both given by Euler products, then $L(s, f \times g)$ has a natural degree four Euler product. The Dirichlet series $\sum_{n=1}^\infty \frac{a_n b_n}{n^s}$ does not.

Theorem 5.102 (Rankin, Selberg) *Let f and g be a pair of forms in $S_k(SL_2(\mathbb{Z}))$ as above. The completed L-function*

$$\widehat{L}(s, f \times g) := (4\pi)^{-s}\Gamma(s)\pi^{-(s-k+1)}\Gamma(s - k + 1)L(s, f \times g)$$

admits meromorphic continuation to all $s \in \mathbb{C}$ and satisfies the functional equation

$$\widehat{L}(s,\, f \times g) = \widehat{L}(2k-1-s,\, f \times g).$$

The only possible poles of $\widehat{L}(s,\, f \times g)$ are at the points $s = k$ and $s = k-1$. They are at most simple, with residues $\pm\frac{1}{2}\langle f, g^\vee\rangle$ respectively.

Proof The defining integral unfolds as follows:

$$\begin{aligned}
\xi(2s)^{-1}\mathrm{RS}_{f,g}(s) &= \int_{\Gamma\backslash\mathcal{H}} \mathcal{E}(z,s) f(z)\overline{g(z)}\mathrm{Im}(z)^k d\mu(z) \\
&= \int_{\Gamma\backslash\mathcal{H}} \sum\nolimits_{\gamma\in\Gamma_\infty\backslash\Gamma} \mathrm{Im}(\gamma z)^s f(z)\overline{g(z)}\mathrm{Im}(z)^k d\mu(z) \\
&= \int_{\Gamma\backslash\mathcal{H}} \sum\nolimits_{\gamma\in\Gamma_\infty\backslash\Gamma} \mathrm{Im}(\gamma z)^s f(\gamma z)\overline{g(\gamma z)}\mathrm{Im}(\gamma z)^k d\mu(z) \\
&= \int_{\Gamma_\infty\backslash\mathcal{H}} f(z)\overline{g(z)}\mathrm{Im}(z)^{s+k} d\mu(z) \\
&= \int_0^\infty \Big[\int_0^1 f(x+iy)\overline{g(x+iy)}dx\Big] y^{s+k-2}dy.
\end{aligned}$$

A crucial point here (which was used in the third step) is of course that the expression $f(z)\overline{g(z)}\mathrm{Im}(z)^k$ is $\mathrm{SL}_2(\mathbb{Z})$-invariant. As a fundamental domain for $\Gamma_\infty = \{\pm\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right) : b \in \mathbb{Z}\}$ we have taken the vertical strip $\{x+iy : x \in (0,1) \text{ and } y > 0\}$. As an intermediate step we evaluate the integral in the $[\cdots]$ brackets:

$$\begin{aligned}
\int_0^1 f(x+iy)\overline{g(x+iy)}dx &= \int_0^1 \sum_{m,n=1}^\infty a_m\overline{b_n} e^{2\pi i(m-n)x} e^{-2\pi(m+n)y} dx \\
&= \sum_{m,n=1}^\infty a_m\overline{b_n} e^{-2\pi(m+n)y}\Big(\int_0^1 e^{2\pi i(m-n)x}dx\Big) \\
&= \sum_{n=1}^\infty a_n\overline{b_n} e^{-4\pi ny}.
\end{aligned}$$

We now return to the initial calculation, assuming $\mathrm{Re}(s)$ is sufficiently large:

$$\begin{aligned}
\xi(2s)^{-1}\mathrm{RS}_{f,g}(s) &= \int_0^\infty \sum_{n=1}^\infty a_n\overline{b_n} e^{-4\pi ny} y^{s+k-2} dy \\
&= \sum_{n=1}^\infty a_n\overline{b_n}\Big(\int_0^\infty e^{-4\pi ny} y^{s+k-1}\frac{dy}{y}\Big) \\
&= (4\pi)^{-(s+k-1)}\Gamma(s+k-1)\cdot\sum_{n=1}^\infty \frac{a_n\overline{b_n}}{n^{s+k-1}}.
\end{aligned}$$

From Theorem 5.97 we deduce that

$$\mathrm{RS}_{f,g}(s) = (4\pi)^{-(s+k-1)}\Gamma(s+k-1)\pi^{-s}\Gamma(s)L(s+k-1, f\times g^\vee)$$

has meromorphic continuation to $\mathbb{C}$, with the only possible poles at $s=0$ and $s=1$, and it is invariant under $s\mapsto 1-s$. After substituting $s'=s+k-1$ this amounts to the integral representation

$$(4\pi)^{-s}\Gamma(s)\pi^{-(s-k+1)}\Gamma(s-k+1)L(s, f\times g^\vee) = \mathrm{RS}_{f,g}(s-k+1)$$

having meromorphic continuation to $\mathbb{C}$, being invariant under $s\mapsto (2k-1)-s$, and with at most simple poles at $s=k$ and $s=k-1$. The respective residues at those two points are $\pm\frac{1}{2}\langle f, g\rangle$.

Stating these assertions with g replaced by $g^\vee$ gives the result. □

Remark 5.103 If $f, g\in \mathrm{Prim}_k(\mathrm{SL}_2(\mathbb{Z}))$ are normalized Hecke eigenforms, we know that $f=g$ if and only if $\langle f, g\rangle\neq 0$. In this case we find that $\widehat{L}(s, f\times g)$ is an entire function exactly when $f\neq g^\vee$. We should emphasize that $g=g^\vee$ in the $\mathrm{SL}_2(\mathbb{Z})$-case since T_p is self-adjoint (and therefore its eigenvalues are real numbers). For a general congruence subgroup it is important to distinguish between g and $g^\vee$, which is why we prefer to keep $g^\vee$ in our statements–with an eye towards generalizations.

As a compelling application of the Rankin-Selberg method (and Landau's Lemma 1.135) we can now prove Deligne's bound $O(n^{\frac{k-1}{2}+\epsilon})$ on *average*–in the following sense:

Corollary 5.104 *Let $f=\sum_{n=1}^\infty a_n q^n$ be in $S_k(SL_2(\mathbb{Z}))$. Then the Dirichlet series $\sum_{n=1}^\infty \frac{|a_n|^2}{n^s}$ converges absolutely for $Re(s)>k$. Thus $\sum_{n\le x}|a_n|^2 = O(x^{k+\epsilon})$ for all $\epsilon>0$.*

Proof The Dirichlet series $D(s):=\sum_{n=1}^\infty \frac{|a_n|^2}{n^s}$ has non-negative coefficients, so Landau's Lemma 1.135 applies. For $\mathrm{Re}(s)\gg 0$ we have

$$D(s) = \zeta(2s-2k+2)^{-1}L(s, f\times f^\vee) = \zeta(2s-2k+2)^{-1}(4\pi)^s\Gamma(s)^{-1}\pi^{s-k+1}\Gamma(s-k+1)^{-1}\widehat{L}(s, f\times f^\vee).$$

By Theorem 5.102 the right-hand side has meromorphic continuation to $\mathbb{C}$ with a pole at $s=k$. Note that the ζ-factor has no pole at $s=k$ and $\Gamma(s)^{-1}$ is entire with no zero at $s=k$. Similarly for the other factors. All other poles of the right-hand side have real part less than k (including those arising from the zeroes of the ζ-function). Landau lets us conclude that $D(s)$ has abscissa of convergence $t^\sharp=k$. From the discussion in Sect. 1.4.3 we get the order of magnitude of the sum function $A(x)=\sum_{n\le x}|a_n|^2$ (by partial summation). □

For the L-function of f itself we have the following byproduct, which gives the abscissa of convergence without reference to Deligne's theorem:

Corollary 5.105 *For f as above, $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$ converges absolutely for $Re(s) > \frac{k+1}{2}$.*

Proof Note that $|a_n| \leq \max\{n^{\delta}, n^{-\delta}|a_n|^2\}$ for all $\delta > 0$ and $n \geq 1$. This is checked case-by-case for the two possible maxima. We have the trivial bound $\sum_{n \leq x} n^{\delta} \leq \sum_{n \leq x} x^{\delta} \leq x^{\delta+1}$. On the other hand, by Corollary 5.104 the series $\sum_{n=1}^{\infty} n^{-\delta}|a_n|^2 n^{-(k-\delta+\epsilon)}$ converges for all $\epsilon > 0$. By the observations in Sect. 1.4.3, $\sum_{n \leq x} n^{-\delta}|a_n|^2$ has order of magnitude $O(x^{k-\delta+\epsilon})$. For any given $\epsilon > 0$ we can arrange that $\max\{\delta + 1, k - \delta + \epsilon\} = \frac{k+1}{2} + \epsilon$ by taking $\delta = \frac{k-1}{2} + \epsilon$. Using this particular δ we infer that

$$\sum_{n \leq x} |a_n| = O(x^{\frac{k+1}{2}+\epsilon}).$$

Referring to Sect. 1.4.3 again, we conclude that $\sum_{n=1}^{\infty} a_n n^{-s}$ is absolutely convergent whenever $\mathrm{Re}(s)$ is greater than $\frac{k+1}{2} + \epsilon$. Since ϵ was arbitrary we are done. □

Exercise 5.106 For any two forms f and g as in Definition 5.100, prove the Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n b_n}{n^s}$ is absolutely convergent for $\mathrm{Re}(s) > k$. (`Hint`: Use the Cauchy-Schwarz inequality to bound $\sum_{n \leq x} |a_n b_n|$ in terms of $\sum_{n \leq x} |a_n|^2$ and $\sum_{n \leq x} |b_n|^2$. Then use results from Sect. 1.4.3.)

5.9.3 The Degree Four Euler Product

Throughout this section we assume $f, g \in \mathrm{Prim}_k(\mathrm{SL}_2(\mathbb{Z}))$ are normalized Hecke eigenforms, so that their L-functions have Euler products:

- $L(s, f) = \prod_p (1 - a_p p^{-s} + p^{k-1-2s})^{-1} = \prod_p \left((1 - \alpha_{1p} p^{-s})(1 - \alpha_{2p} p^{-s})\right)^{-1}$;
- $L(s, g) = \prod_p (1 - b_p p^{-s} + p^{k-1-2s})^{-1} = \prod_p \left((1 - \beta_{1p} p^{-s})(1 - \beta_{2p} p^{-s})\right)^{-1}$.

Here $\{\alpha_{1p}, \alpha_{2p}\}$ are the reciprocal roots of the Hecke polynomial $1 - a_p X + p^{k-1} X^2$. Thus $\alpha_{1p} + \alpha_{2p} = a_p$ and $\alpha_{1p}\alpha_{2p} = p^{k-1}$. Similarly for $\{\beta_{1p}, \beta_{2p}\}$.

Our goal here is to obtain an Euler product for the Rankin-Selberg L-function $L(s, f \times g)$.

Theorem 5.107 $L(s, f \times g) = \prod_p \left((1 - \alpha_{1p}\beta_{1p} p^{-s})(1 - \alpha_{1p}\beta_{2p} p^{-s})(1 - \alpha_{2p}\beta_{1p} p^{-s})(1 - \alpha_{2p}\beta_{2p} p^{-s})\right)^{-1}$.

Proof Since the Fourier coefficients of f and g are multiplicative, for $\mathrm{Re}(s) \gg 0$ we have

$$L(s, f \times g) = \prod_p \frac{1}{1 - p^{-(2s-2k+2)}} \cdot \prod_p \sum_{r=0}^{\infty} a_{p^r} b_{p^r} p^{-rs}.$$

Fixing the prime p and letting $X = p^{-s}$ it therefore suffices to show the identity

$$\frac{1}{1-p^{2k-2}X^2}\cdot\sum_{r=0}^{\infty}a_{p^r}b_{p^r}X^r=\big((1-\alpha_{1p}\beta_{1p}X)(1-\alpha_{1p}\beta_{2p}X)(1-\alpha_{2p}\beta_{1p}X)(1-\alpha_{2p}\beta_{2p}X)\big)^{-1}.$$

This is the next Lemma with $A_r := a_{p^r}$ and $B_r := b_{p^r}$. Note that $\alpha_{1p}\alpha_{2p}\beta_{1p}\beta_{2p} = p^{2k-2}$. □

In the previous proof we made use of:

Lemma 5.108 *Let $(A_r)_{r\geq 0}$ and $(B_r)_{r\geq 0}$ be sequences in $\mathbb{C}$, and suppose their generating functions have the form*

- $\sum_{r=0}^{\infty} A_r X^r = \frac{1}{(1-\alpha_1 X)(1-\alpha_2 X)}$;
- $\sum_{r=0}^{\infty} B_r X^r = \frac{1}{(1-\beta_1 X)(1-\beta_2 X)}$,

for certain pairs of complex numbers $\{\alpha_1, \alpha_2\}$ and $\{\beta_1, \beta_2\}$. (We view the two bullet points as identities in $\mathbb{C}[\![X]\!]$, disregarding convergence.) Then we have an equality

$$\sum_{r=0}^{\infty} A_r B_r X^r = (1-\alpha_1\alpha_2\beta_1\beta_2 X^2)\cdot\frac{1}{(1-\alpha_1\beta_1 X)(1-\alpha_1\beta_2 X)(1-\alpha_2\beta_1 X)(1-\alpha_2\beta_2 X)}$$

of formal power series over $\mathbb{C}$.

Proof We learned the following argument from the proof of [Bum97, Lemma 1.6.1, p. 73]. We first introduce notation for the two rational functions defined by the generating series:

$$A(x) := \frac{1}{(1-\alpha_1 x)(1-\alpha_2 x)}, \qquad B(x) := \frac{1}{(1-\beta_1 x)(1-\beta_2 x)}$$

where we think of x as a complex variable. The poles of A are $\frac{1}{\alpha_i}$ for all $i = 1, 2$ such that $\alpha_i \neq 0$. In particular its Taylor series $\sum_{r=0}^{\infty} A_r x^r$ necessarily has a positive radius of convergence (the minimum of $|\frac{1}{\alpha_i}|$) by complex analysis. Similarly for B.

The trick is to consider the function $w \mapsto F(w) := A(xw)B(w^{-1})$ for a fixed $x \neq 0$ close enough to the origin. The poles of F are:

(1) The nonzero elements of $\{\beta_1, \beta_2\}$. (When $\beta_1 = \beta_2$ is nonzero this gives a double pole.);
(2) The $\frac{1}{x\alpha_i}$ for all $i = 1, 2$ such that $\alpha_i \neq 0$. (When $\alpha_1 = \alpha_2$ is nonzero this gives a double pole.).

Note that $w = 0$ is *not* a pole; it is a removable singularity. For suitable x the two sets of poles in (1) and (2) do not intersect. (For instance, by taking $|x| > 0$ very small we can move the poles $\frac{1}{x\alpha_i}$ as far away from the origin as we like.) Now pick a contour C in the w-plane encircling the poles in $\{\beta_1, \beta_2\}$, and the origin, but warding off the poles $\frac{1}{x\alpha_i}$. We choose C in such a way that $\sum_{r=0}^{\infty} B_r w^{-r}$ converges for $w \in C$. Shrinking $|x|$ further we may also assume $\sum_{r=0}^{\infty} A_r (xw)^r$ converges for $w \in C$. Now, on the one hand

$$\begin{aligned}\frac{1}{2\pi i}\oint_C F(w)\frac{dw}{w} &= \frac{1}{2\pi i}\oint_C \Big(\sum_{m=0}^{\infty} A_m (xw)^m\Big)\Big(\sum_{n=0}^{\infty} B_n w^{-n}\Big)\frac{dw}{w}\\ &= \sum_{m,n=0}^{\infty} A_m B_n x^m \frac{1}{2\pi i}\oint_C w^{m-n-1}dw\\ &= \sum_{n=0}^{\infty} A_n B_n x^n\end{aligned}$$

since the residue of w^{m-n-1} at $w = 0$ is one if $m = n$ and zero otherwise. On the other hand, by the residue theorem $\frac{1}{2\pi i}\oint_C F(w)\frac{dw}{w}$ is the sum of the residues at the poles of $\frac{F(w)}{w}$ surrounded by C:

- $\operatorname{res}_{w=0}\frac{F(w)}{w} = F(0) = A(0)\lim_{w\to 0} B(w^{-1}) = 0$ if $\beta_1 \neq 0$ or $\beta_2 \neq 0$;
- $\operatorname{res}_{w=\beta_1}\frac{F(w)}{w} = A(x\beta_1)\frac{\beta_1}{\beta_1-\beta_2}$ if $\beta_1 \neq \beta_2$;
- $\operatorname{res}_{w=\beta_2}\frac{F(w)}{w} = A(x\beta_2)\frac{\beta_2}{\beta_2-\beta_1}$ if $\beta_1 \neq \beta_2$.

Assuming β_1 and β_2 are distinct, these residues add up to

$$\frac{1}{2\pi i}\oint_C F(w)\frac{dw}{w} = \frac{1}{(1-\alpha_1\beta_1 x)(1-\alpha_2\beta_1 x)}\cdot\frac{\beta_1}{\beta_1-\beta_2} + \frac{1}{(1-\alpha_1\beta_2 x)(1-\alpha_2\beta_2 x)}\cdot\frac{\beta_2}{\beta_2-\beta_1}.$$

We ask the reader to multiply this expression by $(1-\alpha_1\beta_1 x)(1-\alpha_1\beta_2 x)(1-\alpha_2\beta_1 x)(1-\alpha_2\beta_2 x)$ and check that the product indeed comes out to $1-\alpha_1\alpha_2\beta_1\beta_2 x^2$ as claimed. Since x is allowed to vary in a small punctured disk at the origin this gives the equality in $\mathbb{C}[\![X]\!]$ as stated.

The degenerate case $\beta_1 = \beta_2$ follows by extrapolation from the non-degenerate case by continuity in the parameters, or by a direct calculation of the residue $\frac{d}{dw}A(xw)w|_{w=\beta_1}$. □

5.9.4 *The Rankin-Selberg Logarithm and Applications*

We continue with $f, g \in \mathrm{Prim}_k(\mathrm{SL}_2(\mathbb{Z}))$ and the pertaining notation as in the last section. Here we focus on the logarithm of the Rankin-Selberg L-function, i.e.

$$\ln L(s, f\times g) = \sum_p \sum_{r=1}^{\infty}\big((\alpha_{1p}\beta_{1p})^r + (\alpha_{1p}\beta_{2p})^r + (\alpha_{2p}\beta_{1p})^r + (\alpha_{2p}\beta_{2p})^r\big)\frac{p^{-rs}}{r}$$

which converges absolutely for $\mathrm{Re}(s) > k$ (by Exercise 5.106). The Euler product (5.107) is tantamount to $L(s, f\times g)$ equalling the exponential of $\ln L(s, f\times g)$. In particular the Rankin-Selberg L-function does not vanish for $\mathrm{Re}(s) > k$.

Remark 5.109 Think of $\Phi = \Phi_p = \begin{pmatrix} \alpha_{1p} & 0 \\ 0 & \alpha_{2p} \end{pmatrix}$ and $\Psi = \Psi_p = \begin{pmatrix} \beta_{1p} & 0 \\ 0 & \beta_{2p} \end{pmatrix}$ as linear operators on $\mathbb{C}^2$. Then $\Phi \otimes \Psi$ is a linear operator on $\mathbb{C}^2 \otimes \mathbb{C}^2 \simeq \mathbb{C}^4$ with eigenvalues $\{\alpha_{1p}\beta_{1p}, \alpha_{1p}\beta_{2p}, \alpha_{2p}\beta_{1p}, \alpha_{2p}\beta_{2p}\}$. The eigenvectors are $e_i \otimes e_j$ with $i, j \in \{1, 2\}$. The rth coefficient of $\ln L(s, f \times g)$ may thus be interpreted as the trace of the rth power:

$$\mathrm{Tr}(\Phi^r \otimes \Psi^r) = \mathrm{Tr}((\Phi \otimes \Psi)^r) = (\alpha_{1p}\beta_{1p})^r + (\alpha_{1p}\beta_{2p})^r + (\alpha_{2p}\beta_{1p})^r + (\alpha_{2p}\beta_{2p})^r.$$

We are mostly interested in the case $f = g^\vee$ where the above expression is simply

$$(\alpha_{1p}\overline{\alpha_{1p}})^r + (\alpha_{1p}\overline{\alpha_{2p}})^r + (\alpha_{2p}\overline{\alpha_{1p}})^r + (\alpha_{2p}\overline{\alpha_{2p}})^r = (\alpha_{1p}^r + \alpha_{2p}^r)(\overline{\alpha_{1p}}^r + \overline{\alpha_{2p}}^r) = |\mathrm{Tr}(\Phi^r)|^2.$$

In particular the coefficients of $\ln L(s, f \times f^\vee)$ are all non-negative. Moreover, $|a_p|^2$ is the first coefficient since $\mathrm{Tr}(\Phi) = \alpha_{1p} + \alpha_{2p} = a_p$.

Suppose we are in the situation where $f = g^\vee$. Following the by now standard procedure, we break up the logarithmic series $\ln L(s, f \times f^\vee)$ into a main term from $r = 1$ and an error term combining the $r \geq 2$ terms:

$$\ln L(s, f \times f^\vee) = \sum_p |a_p|^2 p^{-s} + \underbrace{\sum_p \sum_{r=2}^{\infty} |\mathrm{Tr}(\Phi_p^r)|^2 \frac{p^{-rs}}{r}}_{\mathcal{E}(s, f \times f^\vee)}.$$

Note that $\mathcal{E}(s, f \times f^\vee) \geq 0$ for all real $s > k$.

Proposition 5.110 $\sum_p |a_p|^2 p^{-s} \leq \ln\left(\frac{1}{s-k}\right) + O(1)$ *as* $s \to k^+$.

Proof Remember that $L(s, f \times f^\vee)$ has a simple pole at $s = k$ by Theorem 5.102. Thus

$$\ln L(s, f \times f^\vee) = \ln\left(\frac{1}{s-k}\right) + O(1)$$

as $s \to k^+$. As noted, the error term $\mathcal{E}(s, f \times f^\vee)$ is non-negative so this proves the inequality. □

The equivalent conditions in the next exercise were conjectured to hold by Petersson. This was proved by Deligne, more generally, as a result of his work on the Weil conjectures.

Exercise 5.111 Let $f \in \mathrm{Prim}_k(\mathrm{SL}_2(\mathbb{Z}))$ and factor its pth Hecke polynomial as

$$1 - a_p X + p^{k-1} X^2 = (1 - \alpha_{1p} X)(1 - \alpha_{2p} X).$$

Here a_p is a *real* number since T_p is self-adjoint. Show the following three conditions are equivalent:

(1) $\alpha_{1p} = \overline{\alpha_{2p}}$;
(2) $|\alpha_{1p}| = |\alpha_{2p}| = p^{\frac{k-1}{2}}$;
(3) $|a_p| \leq 2p^{\frac{k-1}{2}}$.

(`Hint`: Condition (3) implies (1) by noting that $T^2 - a_p T + p^{k-1}$ has discriminant $a_p^2 - 4p^{k-1} \leq 0$.)

Granted the properties in Exercise 5.111, the inequality in Proposition 5.110 can be strengthened to an equality. Indeed, using (2) we get

$$|\mathrm{Tr}(\Phi_p^r)|^2 = |\alpha_{1p}^r + \alpha_{2p}^r|^2 \leq (2p^{r\frac{k-1}{2}})^2 = 4p^{r(k-1)}$$

and on that account the error term satisfies the upper bound

$$\mathcal{E}(s, f \times f^\vee) \leq 4 \sum_p \sum_{r=2}^{\infty} p^{r(k-1)} \frac{p^{-rs}}{r} = 4 \sum_p \sum_{r=2}^{\infty} \frac{p^{-r(s-k+1)}}{r}$$

at least for all $s > k$. The rightmost double sum converges for $\mathrm{Re}(s) - k + 1 > \frac{1}{2}$ and particularly remains bounded as $s \to k$.

Chapter 6
Galois Representations Attached to Weight One Forms



In early January of 1967 Langlands wrote a 17 pages long letter to Weil, which laid out one of the most influential and exhilarating programs in mathematics. The cover page starts off like this:

> In response to your invitation to come and talk I wrote the enclosed letter. After I wrote it I realized there was hardly a statement in it of which I was certain. If you are willing to read it as pure speculation I would appreciate that; if not—I am sure you have a waste basket handy.

At the time of writing (May 2025) one can find the entire letter here:

`https://publications.ias.edu/rpl/paper/43`

The conjectures he put forth were sharpened in the immediately subsequent years. See his famous 'fairy tale' paper [L79] for instance. A small fragment of the theory, which in itself lies extremely deep, is what is often referred to as the 'Langlands Reciprocity Conjecture'. See [Gel84, Conjecture 1, p. 203] for example. It simply states that for any irreducible representation $r : \mathrm{Gal}(E/K) \to \mathrm{GL}_n(\mathbb{C})$ there is a corresponding cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_K)$ whose L-function $L(s, \pi)$ equals the Artin L-function $L(s, r)$. In particular the latter must be entire as Artin conjectured (when r is not the trivial character). Gelbart's fantastic survey paper [Gel84] does a great job explaining how this conjecture naturally spawns the 'Functoriality Conjecture' which is of utmost importance in number theory. We will not dig into functoriality here, but it predicts that there should be a relationship between the automorphic representations of two reductive algebraic groups G and G' whenever there is a homomorphism between the so-called L-groups ${}^L G \to {}^L G'$. We refer the interested reader to [Bor79], or the more leisurely account [Cog03], for the definition of L-groups and a precise version of the functoriality principle. The reader may have heard of base change and automorphic induction, which are particularly important special cases of functoriality. When available they give us a way to transfer (or lift) an automorphic representation to a different group, where

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C. Sorensen, *From Classical L-Functions to Modern Reciprocity Laws*,
Graduate Texts in Mathematics 307, https://doi.org/10.1007/978-3-032-03035-1_6

algebro-geometric techniques are prevalent for example. As we will see sketched later in Sect. 6.2.3, base change methods are also used to prove instances of the reciprocity conjecture.

The representations π corresponding to an r as above are of a very particular kind. For example, when $n = 1$ we associate a Hecke character of *finite* order to r via class field theory, and thus the correspondence fails to take account of the vast majority of algebraic Hecke characters–which nevertheless are attached to ℓ-adic Galois characters as we saw in Chap. 3. We have a similar situation for odd two-dimensional representations r of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Here the sought-after π can be identified with a modular form f. By comparing the functional equation of $L(s, r)$ to that of $L(s, f)$ one observes that f must have weight $k = 1$ for these L-functions to coincide. To sum up, the posited map $r \mapsto f$ misses the modular forms of weight $k \geq 2$ which 'only' match ℓ-adic Galois representations (in analogy with the one-dimensional case).

In this chapter all our representations $r : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{C})$ are assumed to be odd, which means complex conjugation c has determinant $\det r(c) = -1$. In other words $r(c) \sim \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. For such r the Euler factor at infinity is $L_{\mathbb{C}}(s)$, by the duplication formula for the Γ-function, just as for the completed L-function of a modular form. Such an r is categorized according to its *projective* image, i.e. the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ in $\mathrm{PGL}_2(\mathbb{C}) = \mathrm{GL}_2(\mathbb{C})/\mathbb{C}^\times$, which falls into one of the following categories: The projective image is either cyclic, dihedral, or isomorphic to the group of rotational symmetries of a Platonic solid (A_4, S_4, A_5). In the Platonic case we say that r is either tetrahedral, octahedral, or icosahedral, respectively. Usually we also assume r is irreducible which rules out the cyclic case. In the dihedral case $r \simeq \mathrm{Ind}_L^{\mathbb{Q}} \rho$ for some quadratic extension $L/\mathbb{Q}$, plus a character $\rho : \mathrm{Gal}(\overline{\mathbb{Q}}/L) \to \mathbb{C}^\times$, and in this case one can explicitly write down the corresponding modular form f as a sort of θ-series. This is perhaps the simplest example of automorphic induction, where an automorphic representation (in this case a Hecke character) of $\mathrm{GL}_1(\mathbb{A}_L)$ transfers to an automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$.

Around 1980 Langlands dealt with the A_4-case by making heavy use of the theory of base change for GL_2, and shortly after Tunnell handled the S_4-case by invoking a result of Jacquet, Piatetski-Shapiro, and Shalika, on base change along non-normal cubic extensions. In conjunction this gives the Langlands-Tunnell theorem: For every odd and irreducible r as above, with *solvable* projective image, there is an $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ such that $L(s, r) = L(s, f)$. Furthermore $N = \mathcal{N}_r$ and the nebentype χ corresponds to $\det r$ via the Artin map. Thirty years later Khare and Wintenberger took care of the A_5-case in the course of their resolution of Serre's conjecture on odd mod ℓ representations. See part (ii) of [KW09b, Corollary 10.2, p. 502]. Thus the solvability hypothesis is ultimately redundant.

The main objective of this chapter is how to go the *other* way: Starting from a weight one form f the goal is to explain how one can attach a Galois representation r to f with the same L-function. We will follow a famous construction from the mid-1970s due to Deligne and Serre in their paper [DeS74]. See also [Ser77] for an exposition of the proof. The end result is a one-to-one correspondence $r \leftrightarrow f$

between the set of primitive forms $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ and the set of irreducible representations $r : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{C})$ with $\mathcal{N}_r = N$, with $\det r$ corresponding to χ, and such that all the twists $L(s, r \otimes \psi)$ satisfy the Artin conjecture. Of course the last condition should be superfluous, and in fact [KW09b] tells us it is.

The key input in the proof of the Deligne-Serre theorem is its ℓ-adic analogue for forms of weight $k \geq 2$. For $k = 2$ this is in essence a construction from the mid-1950s due to Eichler and Shimura. For a general weight $k \geq 2$, Deligne built these Galois representations around 1970 using étale cohomology of modular curves. It does not make sense to talk about the Artin L-function of an ℓ-adic representation $r : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$, which typically has infinite image. When we say that r is *attached* to a form $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ we mean that $r(\mathrm{Frob}_p)$ has characteristic polynomial $1 - a_p X + \chi(p) p^{k-1} X^2$ for all $p \nmid N\ell$. (The polynomials here are viewed as having coefficients in $\overline{\mathbb{Q}}$ via a choice of embeddings $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$, and part of the statement is that this makes sense, i.e. the polynomials we are comparing in fact do belong to $\overline{\mathbb{Q}}[X]$. One can even replace $\overline{\mathbb{Q}}$ with a finite extension of $\mathbb{Q}$.)

The trick to connect the $k = 1$ world to the $k \geq 2$ world is to multiply a given weight one form by the Eisenstein series E_κ from Proposition 5.35. Recall that, basically by the Von Staudt-Clausen theorem, there is a congruence $E_\kappa \equiv 1$ (modulo ℓ) for all primes ℓ such that $(\ell - 1) | 2\kappa$. Beginning with an $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ the product $f E_\kappa$ lies in $S_{2\kappa+1}(\Gamma_0(N), \chi)$ and $f E_\kappa \equiv f$ (modulo ℓ) for such primes ℓ. Actually this is a bit inverted. In reality we start with a prime ℓ, and then choose a $\kappa \geq 2$ for which 2κ is divisible by $\ell - 1$. For instance $\kappa = 2(\ell - 1)$ will do. The point here is that $2\kappa + 1 \geq 2$ is in the range where one could hope to associate an ℓ-adic representation to $f E_\kappa$. Unfortunately this does not always work. The problem is that $f E_\kappa$ is very rarely a Hecke eigenform. The Deligne-Serre lifting lemma (Lemma 6.43), which is really just an application of the Going Down Theorem from commutative algebra, gives a way around this issue by producing an eigenform $g \in S_{2\kappa+1}(\Gamma_1(N))$ whose Hecke eigenvalues are congruent to those of f modulo ℓ. Applying Deligne's theorem to g, and reducing the resulting ℓ-adic representation modulo ℓ, gives a representation $\overline{r_{f,\ell}} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ associated with f.

The remainder of the proof is rather technical. One first observes that $\overline{r_{f,\ell}}$ can be conjugated into $\mathrm{GL}_2(\mathbb{F}_\ell)$ for a set of primes ℓ with a positive density. The subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$ have been classified, and one can therefore often argue on a case-by-case basis. The salient point is that the size of the image $G_\ell := \mathrm{im}(\overline{r_{f,\ell}})$ is uniformly bounded as ℓ varies. Startlingly, the Rankin-Selberg method sneaks into establishing this uniform bound. When ℓ is large enough $\overline{r_{f,\ell}}$ lifts uniquely to a homomorphism $r_{f,\ell}^{\mathrm{univ}}$ into $\mathrm{GL}_2(\mathcal{O})$, where $\mathcal{O}$ is the valuation ring in a large enough finite extension $K/\mathbb{Q}_\ell$. Concretely $r_{f,\ell}^{\mathrm{univ}}$ can be realized as a projective cover of $\overline{r_{f,\ell}}$ viewed as an $\mathcal{O}[G_\ell]$-module. Finally, via $K \hookrightarrow \overline{\mathbb{Q}}_\ell$ and the choice of an isomorphism $\overline{\mathbb{Q}}_\ell \xrightarrow{\sim} \mathbb{C}$ we may think of $r_{f,\ell}^{\mathrm{univ}}$ as taking values in a finite subgroup of $\mathrm{GL}_2(\mathbb{C})$. This complex representation turns out to have all the desired properties.

A lot has happened in the field in the first quarter of the 21st century. We recommend the ICM papers [Ha02, Tay02, Ha14, Sc18, Cal23] as overviews of some of

these breakthroughs. The introduction to [HT01] is another great place to get a good idea of what the theory looks like in higher dimensions–we suggest understanding [HT01, Theorem C, p. 6] in particular. For a CM field K, this theorem attaches an n-dimensional ℓ-adic representation $r_{\pi,\ell}$ of $\mathrm{Gal}(\overline{K}/K)$ to a cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_K)$ satisfying the following conditions: (i) π_∞ is regular algebraic; (ii) π is conjugate self-dual; (iii) π_v is square integrable for some finite place v. We do not want to delve into the precise meaning of these conditions here. In fact conditions (ii) and (iii) have been removed, see [HLTT] and [Sc15, Theorem 1.0.4, p. 948]. (In [Sc15] one even finds a construction of Galois representations attached to eigensystems appearing in the *mod p* cohomology of the locally symmetric space of GL_n over K, which is allowed to be CM *or* totally real.) Condition (i) is the analogue of "$k \geq 2$". Conjecturally the *regularity* in (i) should be unnecessary as well, and one anticipates there should be an associated representation $r_{\pi,\ell}$ as long as π_∞ is merely algebraic. By analogy this amounts to allowing the "$k = 1$" case. We direct the interested reader to [Sc15, Conjecture 1.0.1] for a more precise statement (of the global Langlands/Fontaine-Mazur conjecture). Relaxing the regularity hypothesis in general seems to be out of reach with current technology (at the time of writing). For Siegel modular forms of genus two, i.e. in the case of $\mathrm{GSp}_4(\mathbb{A}_\mathbb{Q})$, the best result in low weight is still [Tay91, Theorem 2, p. 302] as far as we know. A big step forward, in line with the overall theme of our manuscript, is the complete resolution of the Artin conjecture for totally odd representations $r : \mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}_2(\mathbb{C})$ over *any* totally real field K. See [Sas19, Theorem 2, p. 176] whose proof relies on [Kas13, Pil17], among other references, regarding Hilbert modular forms of parallel weight one.

6.1 The Theorem of Deligne and Serre

There is a striking resemblance between the L-function $L(s, f)$ of a primitive *weight one* form $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ and the Artin L-function $L(s, r)$ of an *odd* two-dimensional Galois representation $r : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}(V)$.

To summarize, for $k = 1$ the L-function of f has an Euler product

$$L(s, f) = \prod_p \left(1 - a_p p^{-s} + \chi(p) p^{-2s}\right)^{-1}$$

for $\mathrm{Re}(s) > 2$ (in fact for $\mathrm{Re}(s) > 1$ if we admit Deligne's theorem or the Rankin-Selberg method for level N forms). Here a_p is the T_p-eigenvalue of f. When $p|N$ this is the same as the U_p-eigenvalue and $\chi(p) = 0$. At these so-called bad primes $p|N$ the corresponding Euler factor is $\frac{1}{1-a_p p^{-s}}$. Note that the nebentype χ is necessarily an odd Dirichlet character since $\chi(-1) = (-1)^k = -1$.

The completed L-function

$$\widehat{L}(s, f) = N^{\frac{s}{2}} L_{\mathbb{C}}(s) L(s, f)$$

extends to an *entire* function and satisfies the functional equation

$$\widehat{L}(1 - s, f^{\vee}) = W(f)\widehat{L}(s, f)$$

for some $W(f) \in \mathbb{C}^1$.

On the other hand, when $\dim_{\mathbb{C}}(r) = 2$ the Artin L-function of r is defined by the Euler product

$$L(s, r) = \prod_{p | \mathcal{N}_r} \det_{V^{I_w}} (1 - r(\mathrm{Frob}_w) p^{-s})^{-1} \cdot \prod_{p \nmid \mathcal{N}_r} (1 - \mathrm{tr}\, r(\mathrm{Frob}_w) p^{-s} + \det r(\mathrm{Frob}_w) p^{-2s})^{-1}$$

for $\mathrm{Re}(s) > 1$. We quickly remind the reader of the notation here: The representation r factors through a finite quotient $G = \mathrm{Gal}(E/\mathbb{Q})$. For each prime p we select a place $w|p$ of E and let $G_w \supset I_w$ denote the decomposition group and inertia group of w. Finally $\mathrm{Frob}_w \in G_w/I_w$ is the Frobenius element inducing the automorphism $x \mapsto x^p$ on the residue field of w.

The completed Artin L-function is

$$\widehat{L}(s, r) = \mathcal{N}_r^{\frac{s}{2}} L_{\infty}(s, r) L(s, r).$$

Saying r is *odd* means the following: Fix an embedding $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and let $c \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be the restriction of complex conjugation. Let $V = V^+ \oplus V^-$ be the decomposition into (± 1)-eigenspaces for $r(c)$ and let $n^{\pm} = \dim_{\mathbb{C}} V^{\pm}$. Oddness means $n^+ = n^- = 1$. In other words $r(c)$ has the matrix form $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ in a suitable basis. Equivalently, $\det r(c) = -1$. Under this assumption we have

$$L_{\infty}(s, r) = L_{\mathbb{R}}(s) L_{\mathbb{R}}(s + 1) = L_{\mathbb{C}}(s).$$

We know $\widehat{L}(s, r)$ at least has *meromorphic* continuation to $\mathbb{C}$ and satisfies the functional equation

$$\widehat{L}(1 - s, r^*) = W(r)\widehat{L}(s, r)$$

where $W(r) \in \mathbb{C}^1$ is the root number.

If we fix $N \geq 1$, and an odd Dirichlet character χ mod N, this suggests there should be a one-to-one correspondence $f \longleftrightarrow r$ between the two sets

- $\mathrm{Prim}_1(\Gamma_0(N), \chi)$;
- Odd two-dimensional (complex) representations r of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with $\mathcal{N}_r = N$ and $\det(r) = \chi$,

mediated by an equality of L-functions. (We have abused notation slightly and identified χ with its inflation to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ via the natural projection to $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \xrightarrow{\sim} (\mathbb{Z}/N\mathbb{Z})^\times$.) It turns out this correspondence is not quite right as stated, but we will make it precise as we go. For one, r needs to be irreducible and of course we should only consider representations up to isomorphism. Note also that writing *odd* in the second bullet point is redundant; it is implied by the condition $\det(r) = \chi$ since χ is assumed to be odd.

Establishing the modularity direction $r \longmapsto f$ would show that $L(s, r)$ is entire, i.e. give a proof of the Artin conjecture for odd two-dimensional r.

The Deligne-Serre theorem [DeS74, Theorem 4.1, p. 513] gives the other direction $f \longmapsto r$.

Theorem 6.1 (Deligne, Serre) *Let χ be an odd Dirichlet character modulo N. Given $f \in Prim_1(\Gamma_0(N), \chi)$ with Fourier coefficients $(a_n)_{n\geq 1}$, there is a finite Galois extension $E/\mathbb{Q}$ and a two-dimensional complex representation*

$$r = r_f : Gal(\overline{\mathbb{Q}}/\mathbb{Q}) \twoheadrightarrow Gal(E/\mathbb{Q}) \longrightarrow GL(V) \simeq GL_2(\mathbb{C})$$

which is unramified at all $p \nmid N$, and for those primes p the following two conditions are satisfied:

$$tr\, r(Frob_w) = a_p, \qquad \det r(Frob_w) = \chi(p)$$

for all places $w|p$ of E. Such an r is necessarily odd and uniquely determined by f up to isomorphism.

The Theorem holds verbatim for forms $f \in S_1(\Gamma_0(N), \chi)$ which are T_p-eigenvectors for all $p \nmid N$, with eigenvalue a_p. In fact this is how [DeS74, Theorem 4.1, p. 513] is stated. The primitivity of f does not play a role in the above version of the Theorem. We employ the notation r_f even for non-primitive f.

One of the primary goals of this section is to give an overview of the proof of Theorem 6.1, including a discussion of related results.

First off, the last sentence of Theorem 6.1 is a direct consequence of the Chebotarev density theorem (Theorem 4.57): Any $g \in G$ is equal to Frob_w for some w above a prime p unramified in E. In fact there are infinitely many primes p to choose from, so we may assume $p \nmid N$. Thus the character of any r as in 6.1 is completely determined:

$$\chi_r(g) := \mathrm{tr}\, r(g) = \mathrm{tr}\, r(\mathrm{Frob}_w) = a_p.$$

In other words r is unique up to isomorphism. Similarly $\det r(g) = \chi(p)$ shows that $\det(r) = \chi$ (when χ is viewed as a Galois character by inflation as above). As χ is odd, $\det r(c) = -1$.

Exercise 6.2 In the setup of Theorem 6.1, deduce (from the existence of r) that $|a_p| \leq 2$ for all $p \nmid N$.

Exercise 6.3 Show that the association $f \mapsto r_f$ is compatible with duality and twisting in the following sense:

- $r_{f^\vee} \simeq r_f^*$;
- $r_{f\times\psi} \simeq r_f \otimes \psi$ (in the setup of Exercises 5.30 and 5.76).

(In the last bullet point we are confusing ψ with a Galois character and $(r_f \otimes \psi)(g) := \psi(g) r_f(g)$.)

6.1.1 Matching the L-Functions and the Conductors

Let $r = r_f$ be the Galois representation attached to a primitive form $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ as in Theorem 6.1. In this section we deduce that r and f have the same L-functions (in fact the same Euler factors at each p) and the same conductors.

Theorem 6.4 *$L(s, r) = L(s, f)$ and $\mathcal{N}_r = N$. (Consequently also $\widehat{L}(s, r) = \widehat{L}(s, f)$ and $W(r) = W(f)$.)*

Proof We commence by taking the ratio of the two functional equations. As r is unramified at all primes $p \nmid N$, such p do not divide $\mathcal{N}_r$, and they give the same Euler factor $L_p(s, r) = L_p(s, f)$. Therefore

$$\left(\frac{N}{\mathcal{N}_r}\right)^{\frac{1-s}{2}} \cdot \prod_{p|N} \frac{L_p(1-s, f^\vee)}{L_p(1-s, r^*)} = \frac{\widehat{L}(1-s, f^\vee)}{\widehat{L}(1-s, r^*)} = \frac{W(f)}{W(r)} \cdot \frac{\widehat{L}(s, f)}{\widehat{L}(s, r)} = \frac{W(f)}{W(r)} \cdot \left(\frac{N}{\mathcal{N}_r}\right)^{\frac{s}{2}} \cdot \prod_{p|N} \frac{L_p(s, f)}{L_p(s, r)}.$$

In Lemma 6.5 below (right after this proof) we take $\omega = \frac{W(f)}{W(r)}$ and $A = \left(\frac{N}{\mathcal{N}_r}\right)^{\frac{1}{2}}$. Moreover, for $p|N$ we let

$$E_p(s) = \frac{L_p(s, f^\vee)}{L_p(s, r^*)}, \qquad F_p(s) = \frac{L_p(s, f)}{L_p(s, r)}.$$

Then part (1) of Lemma 6.5 is satisfied by our previous remarks. It remains to verify part (2) holds as well. It suffices to do this for $F_p(s)$.

We start with $L_p(s, r)$. Pick a $w|p$ and observe that $r(\mathrm{Frob}_w)$ has finite order on V^{I_w}. Therefore its eigenvalues are roots of unity if $V^{I_w} \neq 0$. Consequently $L_p(s, r) = \prod_j (1 - \beta_p^{(j)} p^{-s})^{-1}$ where the $\beta_p^{(j)}$ are roots of unity or zero. So, certainly $|\beta_p^{(j)}| \leq 1 < p^{\frac{1}{2}}$.

For $L_p(s, f) = (1 - a_p p^{-s})^{-1}$ we accept on faith the ad hoc bound $a_n = O(n^{\frac{1}{2}-\epsilon})$. By a result of Rankin, also cited on [DeS74, p. 516], we may even take $\epsilon = \frac{1}{5}$. Recall that $a_p a_n = a_n(U_p f) = a_{np}$ for all $n \geq 1$. In particular $a_{p^r} = a_p^r$ for all r, which has absolute value at most $C p^{r(\frac{1}{2}-\epsilon)}$. Taking r^{th} roots, and then letting $r \to \infty$, results in the inequality $|a_p| \leq p^{\frac{1}{2}-\epsilon} < p^{\frac{1}{2}}$. Hence Lemma 6.5 applies. We conclude that $A = 1$ and $L_p(s, r) = L_p(s, f)$ for all p. □

In the last proof we made a reference to the Lemma below.

Lemma 6.5 ([DeS74, Lemma 4.9, p. 515]) *Let* $E(s) = A^s \prod_p E_p(s)$ *and* $F(s) = A^s \prod_p F_p(s)$ *be two finite products, where* $A > 0$ *and all the (finitely many) factors* $E_p(s)$ *and* $F_p(s)$ *are finite products of the form* $E_p(s) = \prod_i (1 - \alpha_p^{(i)} p^{-s})^{a_i}$ *and* $F_p(s) = \prod_j (1 - \beta_p^{(j)} p^{-s})^{b_j}$ *with exponents* $a_i, b_j \in \{\pm 1\}$. *Assume:*

(1) There is a constant $\omega \in \mathbb{C}^\times$ *such that*

$$E(1 - s) = \omega F(s).$$

(2) $|\alpha_p^{(i)}| < p^{\frac{1}{2}}$ *and* $|\beta_p^{(j)}| < p^{\frac{1}{2}}$ *for all* i, j *and* p.

Then $A = 1$ *and* $E_p(s) = F_p(s) = 1$ *for all* p.

Proof By symmetry it suffices to check that $F_p(s) = 1$ for all p. Suppose there is a p for which $F_p(s) \neq 1$. Consider a j for which $\beta_p^{(j)} \neq 0$ and pick a logarithm $\ln(\beta_p^{(j)})$. Then $s = (\ln(\beta_p^{(j)}) + 2\pi i m)/\ln p$ is either a zero or a pole of $F_p(s)$ for all $m \in \mathbb{Z}$ (depending on whether b_j is positive or negative). Hence $1 - s$ is a pole or a zero of $E(s)$ by (1) and therefore of the form $(\ln(\alpha_p^{(i)}) + 2\pi i n)/\ln p$. However,

$$1 - s = (\ln p - \ln(\beta_p^{(j)}) - 2\pi i m)/\ln p = (\ln(\alpha_p^{(i)}) + 2\pi i n)/\ln p \implies \alpha_p^{(i)} \beta_p^{(j)} = p,$$

which contradicts (2). □

Remark 6.6 In the proof of Theorem 6.4 we utilized the rudimentary bound $a_n = O(n^{\frac{1}{2}-\epsilon})$ to ensure that $|a_p| < p^{\frac{1}{2}}$, in order to apply Lemma 6.5. The latter then implies a_p must be a root of unity or zero (by comparison with the eigenvalues of $r(\mathrm{Frob}_w)$) for $p|N$. Feeding this back into the argument, it is relatively easy to deduce the $k = 1$ case of Deligne's bound, i.e. $a_n = O(n^\epsilon)$. This is the approach in [DeS74, Sect. 9]. See their Corollary 9.2 on page 527.

Exercise 6.7 For $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ as above show that the nth Fourier coefficient satisfies the inequality $|a_n| \leq d_N(n)$, where $d_N(n)$ is the number of positive divisors of n coprime to N. Conclude that $a_n = O(n^\epsilon)$. (`Hint`: Reduce to prime powers $n = p^r$ using that d_N is a multiplicative arithmetic function. Now treat the two cases $p|N$ and $p \nmid N$ separately *using* that the Euler factors match: $L_p(s, r) = L_p(s, f)$.)

The proof of Theorem 6.4 easily adapts to the ψ-twisted setup. Adhering to our standard conventions, ψ denotes a primitive Dirichlet character of conductor N_ψ coprime to N. (We used to denote N_ψ by r but the latter is now reserved for the Galois representation.) We keep $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$. Then $f \times \psi$ belongs to $S_1(\Gamma_0(NN_\psi^2), \chi\psi^2)$ and is a T_p-eigenvector for all p by Exercise 5.76. In particular the L-function is given by the Euler product

$$L(s, f \times \psi) = \prod_p (1 - a_p \psi(p) p^{-s} + (\chi\psi^2)(p) p^{-2s})^{-1}$$

for $\mathrm{Re}(s) \gg 0$. It extends to an entire function, and the completed version

$$\widehat{L}(s, f \times \psi) = (NN_\psi^2)^{\frac{s}{2}} L_{\mathbb{C}}(s) L(s, f \times \psi)$$

satisfies the functional equation

$$\widehat{L}(1-s, f^\vee \times \overline{\psi}) = W(f \times \psi)\widehat{L}(s, f \times \psi), \qquad W(f \times \psi) = \frac{W(f)}{\chi(N_\psi)\psi(N)} \cdot \frac{N_\psi}{\tau(\psi)^2}.$$

By comparison with the Artin L-function $L(s, r \otimes \psi)$ and its functional equation, the proof of Theorem 6.4 yields the equality of L-functions

$$L(s, r \otimes \psi) = L(s, f \times \psi).$$

(We also deduce that $\mathcal{N}_{r\otimes\psi} = NN_\psi^2$ but this can be checked directly from the definition of Artin conductors, knowing that $\mathcal{N}_r = N$.) In particular all these twists $r \otimes \psi$ satisfy the Artin conjecture, i.e. $L(s, r \otimes \psi)$ has no poles if $r \simeq r_f$ and the conductors of f and ψ are relatively prime.

6.1.2 Rankin-Selberg Revisited and Irreducibility

Proposition 5.110 is the $\mathrm{SL}_2(\mathbb{Z})$-case of the following more general result:

Proposition 6.8 *For $f = \sum_{n=1}^\infty a_n q^n \in Prim_k(\Gamma_0(N), \chi)$ (or just a T_p-eigenform in $S_k(\Gamma_0(N), \chi)$ for all $p \nmid N$) we have*

$$\sum_{p \nmid N} |a_p|^2 p^{-s} \leq \ln\Big(\frac{1}{s-k}\Big) + O(1)$$

as $s \to k^+$. This is an equality if we admit the bound $a_n = O(n^{\frac{k-1}{2}+\epsilon})$.

The proof is very similar to the level one case once the basic properties of the Rankin-Selberg L-function $L(s, f \times f^\vee)$ have been established. We give a succinct summary. At the good primes $p \nmid N$ we still have the matrix $\Phi_p = \begin{pmatrix} \alpha_{1p} & 0 \\ 0 & \alpha_{2p} \end{pmatrix}$ and it makes sense to consider the *partial* Euler product

$$\begin{aligned} \mathrm{EP}^N(s) &:= \prod_{p \nmid N} \big((1-\alpha_{1p}\overline{\alpha_{1p}}p^{-s})(1-\alpha_{1p}\overline{\alpha_{2p}}p^{-s})(1-\alpha_{2p}\overline{\alpha_{1p}}p^{-s})(1-\alpha_{2p}\overline{\alpha_{2p}}p^{-s})\big)^{-1} \\ &= \Big(\prod_{p|N}(1-p^{-(2s-2k+2)})(1-|a_p|^2p^{-s})\Big) \cdot L(s, f \times f^\vee). \end{aligned}$$

See [DeS74, Sect. 5]. One has the bound $|a_p| < p^{\frac{k}{2}}$ for $p|N$. In fact [DeS74, Sect. 1.8, p. 509] gives $|a_p|$ explicitly, depending on whether $p^2|N$ and whether χ factors

through $(\mathbb{Z}/\frac{N}{p}\mathbb{Z})^\times$. Therefore $s = k$ is not a zero of the product $\prod_{p|N}(\cdots)$ over the bad primes within the parentheses. Since $L(s, f \times f^\vee)$ can be shown to have a simple pole at $s = k$, so does $\mathrm{EP}^N(s)$. We infer that as $s \to k^+$,

$$\sum_{p\nmid N} |a_p|^2 p^{-s} + \sum_{p\nmid N}\sum_{r=2}^{\infty} |\mathrm{Tr}(\Phi_p^r)|^2 \frac{p^{-rs}}{r} = \ln \mathrm{EP}^N(s) = \ln\Big(\frac{1}{s-k}\Big) + O(1).$$

The convergence of $\sum_{n=1}^{\infty} \frac{|a_n|^2}{n^s}$ when $\mathrm{Re}(s) > k$ follows from Landau's Lemma exactly as in the $\mathrm{SL}_2(\mathbb{Z})$-case. Altogether this gives Proposition 6.8 (granted the analytic properties of $L(s, f \times f^\vee)$ for $N > 1$).

For $k = 1$ we will use this result to show the irreducibility of r_f.

Proposition 6.9 *The Galois representation r_f is irreducible for all $f \in Prim_1(\Gamma_0(N), \chi)$.*

Proof If r_f is reducible it decomposes as a direct sum of two Galois characters, say $r_f \simeq \chi_1 \oplus \chi_2$ where the χ_i are characters of $\mathrm{Gal}(E/\mathbb{Q})$. Since the χ_i factors through an abelian quotient we may assume $E/\mathbb{Q}$ is abelian from the outset. By Kronecker-Weber $E \subset \mathbb{Q}(\zeta_M)$ for some M and we think of the χ_i as being inflations of Dirichlet characters mod M. As each χ_i is unramified at all $p \nmid N$ we can assume $M|N$. Note that $\chi_1\chi_2 = \det(r_f) = \chi$ and $\chi_1(p) + \chi_2(p) = a_p$ for $p \nmid N$. In particular, for $\mathrm{Re}(s) > 1$ we get

$$\sum_{p\nmid N} |a_p|^2 p^{-s} = 2\sum_{p\nmid N} p^{-s} + \sum_{p\nmid N} \chi_1(p)\overline{\chi_2(p)} p^{-s} + \sum_{p\nmid N} \overline{\chi_1(p)}\chi_2(p) p^{-s}.$$

Observe that $\chi_1\overline{\chi_2}$ and $\overline{\chi_1}\chi_2$ are non-trivial characters since $\chi_1 \neq \chi_2$. (Otherwise $\chi = \chi_1\chi_2$ would be an even character.) Therefore, by Lemma 1.54, the above expression is $2\ln(\frac{1}{s-1}) + O(1)$ as $s \to 1^+$. This contradicts Proposition 6.8 because of the factor 2. □

6.1.3 The Converse Theorem

We have seen the representation $r = r_f$ associated with $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ has the following properties: It is a continuous representation $r : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\mathbb{C})$ such that

(1) r is irreducible;
(2) r is odd, i.e. $\det r(c) = -1$;
(3) $L(s, r \otimes \psi)$ is entire for all Galois characters $\psi : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathbb{C}^\times$ with conductor $\mathcal{N}_\psi$ coprime to $\mathcal{N}_r$.

Remark 6.10 Here conditions (1) and (3) are not totally unrelated. Suppose $L(s, r \otimes \psi)$ is entire for *all* characters ψ (now allowing $\mathcal{N}_\psi$ to be arbitrary). Then r must be

irreducible. For if $r \simeq \chi_1 \oplus \chi_2$ we may take $\psi = \chi_1^{-1}$, in which case the L-function of $r \otimes \psi \simeq 1 \oplus \chi_2\psi$ has a pole at $s = 1$. Artin predicts the opposite: (1) implies all $L(s, r \otimes \psi)$ are entire.

We stress that the last condition of course is expected to be vacuous if we believe the Artin conjecture. It is known that these three properties (1)–(3) completely characterize the two-dimensional complex Galois representations arising from weight one forms. This is a consequence of the so-called converse theorem, which we will discuss cursorily below. Summing up, for every odd χ mod N the Deligne-Serre correspondence $f \longmapsto r_f$ gives a bijection

$$\mathrm{Prim}_1(\Gamma_0(N), \chi) \overset{1:1}{\longleftrightarrow}$$

$$\{\text{2-dimensional irreducible } \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})\text{-representations } r \text{ with } \mathcal{N}_r = N \text{ and } \det(r) = \chi \text{ satisfying (3)}\}/\simeq .$$

The correspondence matches the (completed) L-functions, Euler factors, and root numbers defined on both sides. Moreover, it is compatible with taking duals and with twisting by a primitive Dirichlet character ψ with N_ψ coprime to N.

To avoid going off on a tangent we will only give the gist of the converse theorem. We start with the toy example $\mathrm{SL}_2(\mathbb{Z})$ which is particularly simple since it is generated by $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ up to $\pm I$. Start with a Dirichlet series $L(s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}$. For this to converge absolutely for $\mathrm{Re}(s) \gg 0$ we assume $a_n = O(n^C)$ for some $C > 0$. The main insight, which goes back to Hecke and Weil, is that if $\widehat{L}(s) := L_{\mathbb{C}}(s)L(s)$ behaves analytically as if it comes from a modular form–then it does. More precisely, suppose

- $\widehat{L}(s)$ extends to an entire function;
- $\widehat{L}(s) = i^k\widehat{L}(k - s)$ for some even $k \geq 1$;
- $\widehat{L}(s)$ is bounded in vertical strips $\{s : \sigma_1 \leq \mathrm{Re}(s) \leq \sigma_2\}$.

The last condition is somewhat technical and can safely be ignored here. It is there in order to apply the Phragmén-Lindelöf principle and get the decay of $\widehat{L}(\sigma + it)$ as $t \to \infty$. (This plays a role when we move contours of integration and we need to control the error terms.) We will sketch why $f := \sum_{n=1}^{\infty} a_n q^n$ necessarily belongs to $S_k(\mathrm{SL}_2(\mathbb{Z}))$. Clearly the series defining f is convergent for all $z \in \mathcal{H}$ since $a_n = O(n^C)$. Also, the periodicity $f(z + 1) = f(z)$ is obvious from the Fourier expansion. We need to check that $f(\frac{-1}{z}) = z^k f(z)$ and it is enough to do this on the imaginary axis $z = iy$ with $y > 0$.

The usual routine calculation shows that $\widehat{L}(s)$ is essentially the Mellin transformation of $t \mapsto f(it)$, i.e.

$$\int_0^{\infty} f(it)t^s \frac{dt}{t} = \frac{1}{2}\widehat{L}(s)$$

for $\mathrm{Re}(s) \gg 0$. This can be inverted. The Mellin transformation can easily be expressed as a Fourier transformation, and then the Fourier inversion formula takes the shape

$$f(it) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{1}{2} \widehat{L}(s) t^{-s} ds.$$

Here $\sigma \gg 0$ is arbitrary but fixed. The integral $\int_{\sigma - i\infty}^{\sigma + i\infty}$ means the limit of the contour integral $\int_{\sigma - iT}^{\sigma + iT}$ along the vertical line $\mathrm{Re}(s) = \sigma$ as $T \to \infty$. (We recommend the reader derives the Mellin inversion formula as an exercise.)

Now, using the functional equation for $\widehat{L}(s)$ we find that

$$\begin{aligned} f(it) &= i^k \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{1}{2} \widehat{L}(k - s) t^{-s} ds \\ &= i^k t^{-k} \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{1}{2} \widehat{L}(k - s) t^{k - s} ds \\ &= i^k t^{-k} \frac{1}{2\pi i} \int_{k - \sigma - i\infty}^{k - \sigma + i\infty} \frac{1}{2} \widehat{L}(s) t^{s} ds \\ &= i^k t^{-k} \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{1}{2} \widehat{L}(s) t^{s} ds \\ &= i^k t^{-k} f(\frac{i}{t}). \end{aligned}$$

This is the desired result since k is even. In the fourth step we moved the contour of integration, using that $\widehat{L}(s)$ has no poles (and the boundedness assumption to control the contributions from $\mathrm{Im}(s) = T$ as $T \to \infty$). We conclude that $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ and $L(s) = L(s, f)$.

The situation is more complicated for $\Gamma_0(N)$. Here a single functional equation does not always suffice; we need a large enough collection of functional equations for suitable twists. We fix a (not necessarily primitive) Dirichlet character χ modulo N and a $k \geq 1$ compatible with χ, i.e. $\chi(-1) = (-1)^k$. We model the properties $L(s, f \times \psi)$ satisfies for $f \in S_k(\Gamma_0(N), \chi)$.

Start with two complex sequences $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ both of order $O(n^C)$. For every primitive Dirichlet character ψ, of conductor N_ψ, we consider

(a) $L_1(s, \psi) := \sum_{n=1}^{\infty} \frac{\psi(n) a_n}{n^s}$, $\widehat{L}_1(s, \psi) := (N N_\psi^2)^{\frac{s}{2}} L_{\mathbb{C}}(s) L_1(s, \psi)$;

(b) $L_2(s, \overline{\psi}) := \sum_{n=1}^{\infty} \frac{\overline{\psi}(n) b_n}{n^s}$, $\widehat{L}_2(s, \overline{\psi}) := (N N_\psi^2)^{\frac{s}{2}} L_{\mathbb{C}}(s) L_2(s, \overline{\psi})$.

We will not strive for the maximal possible generality here, but only give a sample converse theorem:

Theorem 6.11 (Weil) *Suppose the following holds for all primitive Dirichlet characters ψ with conductor N_ψ coprime to N:*

- *Both $\widehat{L}_1(s,\psi)$ and $\widehat{L}_2(s,\overline{\psi})$ extend to entire functions;*
- *They satisfy the functional equation*

$$\widehat{L}_2(k-s,\overline{\psi}) = \frac{i^{-k}}{\chi(N_\psi)\psi(N)} \cdot \frac{N_\psi}{\tau(\psi)^2} \cdot \widehat{L}_1(s,\psi);$$

- *$\widehat{L}_1(s,\psi)$ and $\widehat{L}_2(s,\overline{\psi})$ are bounded in all vertical strips of finite width.*

Then $f := \sum_{n=1}^{\infty} a_n q^n$ belongs to $S_k(\Gamma_0(N),\chi)$ and $g := \sum_{n=1}^{\infty} b_n q^n = f^W$.

In fact it is enough to impose the three conditions in Theorem 6.11 for N_ψ being either one or a prime number not dividing N. This more general version is [Bum97, Theorem 1.5.1, p. 60]. We see no point in reproducing the detailed proof in [Bum97]. A key point is to first establish the relation between the two functions $f_\psi := \sum_{n=1}^{\infty} \psi(n) a_n q^n$ and $g_{\overline{\psi}} := \sum_{n=1}^{\infty} \overline{\psi(n)} b_n q^n$ on $\mathcal{H}$,

$$f_\psi^{W_{NN_\psi^2}} = \chi(N_\psi)\psi(N) \cdot \frac{\tau(\psi)}{N_\psi^2} \cdot g_{\overline{\psi}}.$$

Compare this to Exercise 5.69. For $\psi = 1$ this simply says that $f^W = g$. The proof of the foregoing relation is virtually identical to the $\mathrm{SL}_2(\mathbb{Z})$-case above, obscured only by additional decorations.

Another central piece of the proof of Theorem 6.11 is the result in the next problem.

Exercise 6.12 Let $h \in \mathrm{Hol}(\mathcal{H})$ and suppose $h|_k R = h$ for some rotation matrix $R \in \mathrm{SO}_2(\mathbb{R})$ of infinite order. Show that $h = 0$. (`Hint`: Identify h with a function on a unit disk via the Cayley transform $\left(\begin{smallmatrix} 1 & -i \\ 1 & i \end{smallmatrix}\right)$. Diagonalize R and find restrictions on the Taylor coefficients at the origin.)

Here is the main application mentioned in the beginning of this section.

Corollary 6.13 *Suppose $r : Gal(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow GL_2(\mathbb{C})$ is a representation satisfying (1)–(3) above, and let $N = \mathcal{N}_r$ and $\chi = \det(r)$. Then there is a unique $f \in Prim_1(\Gamma_0(N),\chi)$ such that $L(s,r) = L(s,f)$.*

Proof (*Sketch*) Define the sequence $(a_n)_{n\geq 1}$ by expanding $L(s,r)$ as a Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$. Then define the other sequence $(b_n)_{n\geq 1}$ by declaring that $\overline{b_n} := i^k W(r) a_n$. For a Dirichlet character ψ as in Theorem 6.11 we then have $L_1(s,\psi) = L(s, r\otimes\psi)$ and $L_2(s,\overline{\psi}) = i^{-k}\overline{W(r)}L(s, r^*\otimes\overline{\psi})$. They both have at least meromorphic continuation and satisfy the Artin functional equation

$$\widehat{L}_2(1-s,\overline{\psi}) = i^{-k}\overline{W(r)}W(r\otimes\psi)\widehat{L}_1(s,\psi).$$

In particular, since $\widehat{L}_1(s,\psi)$ is entire by (3) so is $\widehat{L}_2(s,\overline{\psi})$. The functional equation required in Theorem 6.11 follows from the formula

$$W(r\otimes\psi)=\frac{W(r)}{\chi(N_\psi)\psi(N)}\cdot\frac{N_\psi}{\tau(\psi)^2}.$$

(To verify this, use that $W(\cdot)$ is insensitive to induction and reduce to the case of a character. See also Eq. (3.4.6) in [Tat79, p. 15].) We leave it to the reader to check that the completed Artin L-functions are bounded in vertical strips. We conclude that $f:=\sum_{n=1}^{\infty}a_nq^n\in S_1(\Gamma_0(N),\chi)$. It is a T_p-eigenform for all p since $L(s,f)=L(s,r)$ has an Euler product. It is a newform because otherwise $r\simeq r_f$ would have conductor $<N=\mathcal{N}_r$. Finally, f is normalized since $a_1=1$. (This is true for any Artin L-series as $\frac{1}{1-\zeta p^{-s}}=1+\sum_{r=1}^{\infty}\zeta^r p^{-rs}$.) □

6.1.4 *Examples Emanating from Induced Representations*

Perhaps the simplest examples of representations $r:\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})\longrightarrow\mathrm{GL}_2(\mathbb{C})$ are those induced from a quadratic extension. We fix a finite Galois extension $E/\mathbb{Q}$ and let $G=\mathrm{Gal}(E/\mathbb{Q})$. We think of E as being large, in other words $\mathrm{Gal}(\overline{\mathbb{Q}}/E)$ is a small open subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. The reader may even want to replace E with $\overline{\mathbb{Q}}$ and work with absolute Galois groups instead.

Now take an intermediate field $\mathbb{Q}\subset L\subset E$ such that $L/\mathbb{Q}$ is quadratic and consider the corresponding index two subgroup $H=\mathrm{Gal}(E/L)\leq G$. Note that $H\lhd G$. Start with a character $\rho:H\to\mathbb{C}^\times$ and look at the two-dimensional induced representation of G,

$$r=\mathrm{Ind}_H^G\rho=\mathrm{Ind}_{\mathrm{Gal}(E/L)}^{\mathrm{Gal}(E/\mathbb{Q})}=:\mathrm{Ind}_L^{\mathbb{Q}}\rho.$$

We first analyze when this r satisfies the conditions (1)–(3) in the beginning of the previous section.

– When is r irreducible? For all $\sigma\in G$ we have the character $(\sigma_*\rho)(h)=\rho(\sigma h\sigma^{-1})$ of $H=\sigma^{-1}H\sigma$ appearing in the Mackey decomposition (apply Exercise 4.9 with $\Gamma=H$ normal in G):

$$r|_H=(\mathrm{Ind}_H^G\rho)|_H\simeq\bigoplus\nolimits_{\sigma\in H\backslash G}\sigma_*\rho=\rho\oplus\theta_*\rho$$

for any choice of $\theta\in G$ not in H. Therefore, by Frobenius reciprocity, as a vector space the endomorphism algebra of r is

$$\mathrm{End}_G(r)=\mathrm{Hom}_G(\mathrm{Ind}_H^G\rho,\mathrm{Ind}_H^G\rho)=\mathrm{Hom}_H(\rho,(\mathrm{Ind}_H^G\rho)|_H)\simeq\mathrm{Hom}_H(\rho,\rho)\oplus\mathrm{Hom}_H(\rho,\theta_*\rho).$$

Clearly $\mathrm{Hom}_H(\rho,\rho)=\mathbb{C}$ and $\mathrm{Hom}_H(\rho,\theta_*\rho)=0$ if and only if $\rho\neq\theta_*\rho$. If r is irreducible, $\mathrm{End}_G(r)=\mathbb{C}$ by Schur's Lemma, and therefore $\rho\neq\theta_*\rho$. Conversely, if this last condition holds, r must be irreducible since $\mathrm{End}_G(\chi_1\oplus\chi_2)$ is at least two-dimensional as it contains $\mathrm{End}_G(\chi_1)\oplus\mathrm{End}_G(\chi_2)$. (Here the χ_i are characters of G.) In summary

$$\boxed{\mathrm{Ind}_H^G\rho \text{ is irreducible} \iff \rho\neq\theta_*\rho}$$

Observe that in this case ρ cannot be extended to a character of the (non-abelian) group G.

– When is r odd? We choose an embedding $E\hookrightarrow\mathbb{C}$ and let $c\in G$ be the restriction of complex conjugation. Changing the embedding changes c to a G-conjugate. Recall that r is odd precisely when $r(c)\sim\left(\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right)$. This can be detected with the trace. Indeed r is odd exactly when $\mathrm{tr}\,r(c)=0$. Recall Lemma 4.17 gives an explicit formula for $\chi_r=\mathrm{tr}\,r$. In the case at hand we obtain

$$\chi_r(c)=\mathrm{tr}\,r(c)=\begin{cases}\rho(c)+(\theta_*\rho)(c) & \text{if } c\in H\\ 0 & \text{if } c\notin H\end{cases}.$$

Note that $c\in H$ if and only if L is real quadratic (the embedding $E\hookrightarrow\mathbb{C}$ restricts to a real embedding $L\hookrightarrow\mathbb{R}$). In this case we need $\rho(c)$ and $(\theta_*\rho)(c)$ to have opposite signs. On the other hand, if L is imaginary quadratic, r is odd unconditionally. To summarize,

- If $L/\mathbb{Q}$ is imaginary quadratic, $\mathrm{Ind}_H^G\rho$ is odd;
- If $L/\mathbb{Q}$ is real quadratic, $\mathrm{Ind}_H^G\rho$ is odd $\iff\{\rho(c),(\theta_*\rho)(c)\}=\{\pm1\}$.

Note that in the real quadratic case $\mathrm{Ind}_H^G\rho$ must be irreducible if it is odd.

Exercise 6.14 Suppose $L/\mathbb{Q}$ is real quadratic and H is abelian for simplicity. Deduce that $\mathrm{Ind}_H^G\rho$ is odd if and only if the Hecke character $\rho\circ\mathrm{Art}_{E/L}$ has opposite signatures at the two infinite places of L.

Recall that $\mathcal{N}_r=\mathcal{N}_\rho$ by part (b) of Lemma 4.51. Condition (3) asks whether $L(s,r\otimes\psi)$ is entire for all characters $\psi:\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})\longrightarrow\mathbb{C}^\times$ with $\mathcal{N}_\psi$ coprime to $\mathcal{N}_\rho$. More can be said in our current setup. As long as r is irreducible, $L(s,r\otimes\psi)$ is entire for *all* ψ. To wit, by the projection formula (stated at the end of the proof of Theorem 4.18) and Proposition 4.10 we have

$$L(s,\psi\otimes\mathrm{Ind}_H^G\rho)=L(s,\mathrm{Ind}_H^G(\psi|_H\otimes\rho))=L(s,\psi|_H\otimes\rho).$$

Calling upon Example 4.2.1 the latter equals the L-function of the Hecke character $(\psi|_H\otimes\rho)\circ\mathrm{Art}_{E/L}$. It is entire by Theorem 2.103 since $\psi|_H\otimes\rho\neq1$. (If this was trivial ρ would extend to the character ψ^{-1}, violating the irreducibility criterion $\rho\neq\theta_*\rho$.)

Corollary 6.13 now produces a unique $f\in\mathrm{Prim}_1(\Gamma_0(N),\chi)$ such that $L(s,\rho)=L(s,f)$ where $N:=\mathcal{N}_\rho$ and $\chi:=\det r$. We say f is automorphically induced from

the Hecke character $\rho \circ \mathrm{Art}_{E/L}$. This is one of the first instances of Langlands functoriality–an automorphic form on $\mathrm{GL}_1(\mathbb{A}_L)$ transfers to an automorphic form on $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$ in an L-function preserving manner.

We can write down the Fourier expansion of f: Let $\mathrm{Id}_L = \bigoplus_\mathfrak{p} \mathbb{Z}\mathfrak{p}$ be the group of fractional ideals of L. The Galois character $\rho : H \to \mathbb{C}^\times$ gives a strongly multiplicative function $\chi_\rho : \mathrm{Id}_L \to \mathbb{C}$ by letting $\chi_\rho(\mathfrak{p}) = \rho(\mathrm{Frob}_\mathfrak{p})$ if ρ is unramified at $\mathfrak{p}$, and declaring that $\chi_\rho(\mathfrak{p}) = 0$ otherwise. Then $L(s, \rho)$ is the Dirichlet seres

$$L(s,\rho) = \prod_\mathfrak{p} \frac{1}{1 - \chi_\rho(\mathfrak{p})N(\mathfrak{p})^{-s}} = \sum_{I \neq 0} \chi_\rho(I)N(I)^{-s} = \sum_{n=1}^{\infty} \big(\sum_{I:N(I)=n} \chi_\rho(I)\big)n^{-s},$$

and therefore

$$f = \sum_{n=1}^{\infty} \big(\sum_{I:N(I)=n} \chi_\rho(I)\big)q^n = \sum_{I \neq 0} \chi_\rho(I)q^{N(I)}.$$

In general f happens to be a linear combination of theta series associated with binary quadratic forms.

Remark 6.15 There is a general formula due to Gallagher (see [Gal65]) for the determinant of an induced representation. In the setup of Sect. 4.3.2 it is given by

$$\det r(g) = \mathrm{sign}_{G/H}(g)^{\dim \rho} \cdot \big(\det(\rho) \circ V\big)(g)$$

where $V : G^{\mathrm{ab}} \longrightarrow H^{\mathrm{ab}}$ is Verlagerung and $\mathrm{sign}_{G/H}(g)$ denotes the sign of the permutation $xH \mapsto gxH$ of the cosets G/H. We refer to [RV99, Sect. 6.3, p. 220] for an excellent introduction to the Verlagerung map, also known as the transfer map.

When $[G : H] = 2$ and $\dim \rho = 1$ this formula simplifies to

$$\det r = \omega_{L/\mathbb{Q}} \cdot (\rho \circ V)$$

where $\omega_{L/\mathbb{Q}} : G \longrightarrow \{\pm 1\}$ is the inflation of the isomorphism $\mathrm{Gal}(L/\mathbb{Q}) \xrightarrow{\sim} \{\pm 1\}$. By diagram (ii) in [RV99, p. 225] the Hecke character of $C_\mathbb{Q}$ associated with $\rho \circ V$ via class field theory is the restriction of the Hecke character $\rho \circ \mathrm{Art}_{E/L}$ along the natural embedding $j_{L/\mathbb{Q}} : C_\mathbb{Q} \hookrightarrow C_L$. On the other hand, the Hecke character of $C_\mathbb{Q}$ associated with $\omega_{L/\mathbb{Q}}$ is the one with kernel $N_{L/\mathbb{Q}}(C_L)$. Altogether this gives a description of the nebentype χ of the automorphically induced form f above.

We close this section by making a few remarks on the image of r, or rather the projective image of r. We remind the reader that $\mathrm{PGL}_2(\mathbb{C}) = \mathrm{GL}_2(\mathbb{C})/\mathbb{C}^\times$ where $\mathbb{C}^\times$ is identified with the center $\{\left(\begin{smallmatrix} a & 0 \\ 0 & a \end{smallmatrix}\right) : a \in \mathbb{C}^\times\}$. There is an obvious coordinate-free version of this: If V is a finite-dimensional $\mathbb{C}$-vector space we let $\mathrm{PGL}(V) = \mathrm{GL}(V)/\mathbb{C}^\times$ where $\mathbb{C}^\times$ is now identified with the central subgroup of homotheties $v \mapsto av$. For a representation $r : G \to \mathrm{GL}(V)$ it will be useful to consider the associated *projective* representation, i.e. the homomorphism $r^\diamond : G \to \mathrm{PGL}(V)$ obtained by composing

r with the projection $\mathrm{GL}(V) \twoheadrightarrow \mathrm{PGL}(V)$. For an induced representation $r = \mathrm{Ind}_H^G \rho$ as above, it turns out its projective image $r^\diamond(G)$ is always isomorphic to a *dihedral* group D_n. We say r is of dihedral type. Our notational convention here is that $|D_n| = 2n$. In other words D_n has the presentation

$$D_n = \langle a, b \mid a^2 = b^n = (ab)^2 = e \rangle.$$

So, let $r = \mathrm{Ind}_H^G \rho$ be irreducible. To show $r^\diamond(G) \simeq D_n$ for some n, we first observe that $r(H)$ is abelian, for by the Mackey decomposition $r(h)$ has matrix representation $\left(\begin{smallmatrix} \rho(h) & 0 \\ 0 & (\theta_*\rho)(h) \end{smallmatrix}\right)$ with respect to a suitable basis (independent of $h \in H$). Modulo the group of scalar matrices $\mathbb{C}^\times$ we have

$$\left(\begin{smallmatrix} \rho(h) & 0 \\ 0 & (\theta_*\rho)(h) \end{smallmatrix}\right) \equiv \left(\begin{smallmatrix} 1 & 0 \\ 0 & \rho(h)^{-1}(\theta_*\rho)(h) \end{smallmatrix}\right) \bmod \mathbb{C}^\times.$$

Thus we identify $r^\diamond(H)$ with a finite (and necessarily cyclic) subgroup of $\mathbb{C}^\times$ whose order $n > 1$ is the order of the non-trivial character $(\theta_*\rho)\rho^{-1}$. Pick a generator $B \in r^\diamond(H)$.

Now note that $\theta^2 \in H$ and $r(\theta)^2$ therefore commutes with anything in $r(H)$ and of course with $r(\theta)$. Since these elements generate all of $r(G)$ Schur's Lemma tells us $r(\theta)^2 \in \mathbb{C}^\times$, in other words $A := r^\diamond(\theta)$ has order two (as $r(\theta)$ is not itself a homothety because otherwise r would be reducible). The choice of $\theta \in G - H$ is immaterial, so the same argument shows that $r^\diamond(\theta h)$ has order two for all $h \in H$. In particular $(AB)^2$ is the identity. In total this shows that sending $a \mapsto A$ and $b \mapsto B$ gives the sought-after isomorphism

$$\boxed{D_n \xrightarrow{\sim} r^\diamond(G)}$$

where $n := \mathrm{ord}((\theta_*\rho)\rho^{-1})$ is greater than one since $\rho \neq \theta_*\rho$ when $r = \mathrm{Ind}_H^G \rho$ is irreducible.

The results of this section can be found in [Ser77, Sect. 7, p. 237] which also gives some beautiful numerical examples of forms f expressed in terms of θ-series (in [Ser77, Sect. 7.3]).

6.2 The Theorem of Langlands and Tunnell

Theorem 6.1 is about the direction $f \longmapsto r$. The celebrated Langlands-Tunnell theorem is a partial result in the modularity direction $r \longmapsto f$. In the previous section we introduced the projective representation $r^\diamond : G \to \mathrm{PGL}(V)$ associated with a representation $r : G \to \mathrm{GL}(V)$.

Theorem 6.16 (Langlands, Tunnell) *Suppose we are given an odd and irreducible representation*

$$r : Gal(\overline{\mathbb{Q}}/\mathbb{Q}) \twoheadrightarrow Gal(E/\mathbb{Q}) \longrightarrow GL(V) \simeq GL_2(\mathbb{C})$$

such that the image of $r^{\diamond}$ is a solvable subgroup of $PGL_2(\mathbb{C})$. Let $N := \mathcal{N}_r$ and $\chi := \det r$. Then there exists a unique $f \in Prim_1(\Gamma_0(N), \chi)$ such that $r \simeq r_f$. Equivalently,

$$L(s, r) = L(s, f).$$

($\Longleftrightarrow$ the Artin conjecture holds for r.)

In light of the converse theorem, Theorem 6.16 amounts to the assertion that all the twists $r \otimes \psi$ satisfy the Artin conjecture (at least when $\mathcal{N}_\psi$ and $\mathcal{N}_r$ are coprime) provided r has solvable projective image. The proof of Theorem 6.16 requires base change techniques lying outside the scope of these notes. Our modest goal here is to sharpen the theorem by classifying the finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$ up to isomorphism. In Sect. 6.2.3 we attempt to give the open-minded reader *some* idea of how the proof of (a generalization of) Theorem 6.16 proceeds.

Exercise 6.17 Let r be a two-dimensional representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Show that r is odd if and only if $r^{\diamond}(c)$ is not the identity. Conclude that the projective image of an odd r is not the trivial subgroup.

With Sect. 6.1.4 fresh in mind, we start by explaining the proof of Theorem 6.16 when the image of $r^{\diamond}$ is isomorphic to a dihedral group D_n with $n > 1$. (For $n = 2$ this is the Klein four-group.)

Let $G = \mathrm{Gal}(E/\mathbb{Q})$ and take $r : G \to \mathrm{GL}(V)$ to be a representation as in Theorem 6.16 with dihedral projective image

$$r(G)/r(G) \cap \mathbb{C}^\times \xrightarrow{\sim} r^{\diamond}(G) \xrightarrow{\sim} D_n.$$

Pick a cyclic subgroup $C_n \leq D_n$ of index two. (This is unique when $n > 2$.) Via the above isomorphisms C_n corresponds to a subgroup $C \leq r(G)$ containing $r(G) \cap \mathbb{C}^\times$. Note that this intersection is a finite subgroup of $\mathbb{C}^\times$ and therefore cyclic. The normal (not just subnormal) series

$$\{\mathrm{Id}_V\} \lhd r(G) \cap \mathbb{C}^\times \lhd C \lhd r(G)$$

has cyclic factor groups. Therefore $r(G)$ is supersolvable and thus monomial by the proof of Proposition 4.30. Writing $r(G) \simeq G/N$ we get a subgroup $H \subset G$ containing N and a character $\rho : H/N \to \mathbb{C}^\times$ such that $r \simeq \mathrm{Ind}_{H/N}^{G/N} \rho \simeq \mathrm{Ind}_H^G \rho$. The subgroup H has index $[G : H] = \dim r = 2$ and the results of Sect. 6.1.4 yields an $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ with the same L-function as ρ.

Remark 6.18 Note that a representation r as in Theorem 6.16 can never have a *cyclic* projective image: If $r(G)/r(G) \cap \mathbb{C}^\times$ is cyclic then $r(G)$ must be abelian since $r(G) \cap \mathbb{C}^\times$ is a central subgroup. This would force r to be reducible.

6.2.1 Unitary Groups and the Spin Homomorphism

Let V be a two-dimensional $\mathbb{C}$-vector space with a Hermitian inner product $\langle -, - \rangle$, which we will often suppress from the notation when there is no risk of confusion. The unitary group of V is the group of isometries

$$\mathrm{U}(V) = \{g \in \mathrm{GL}(V) : \langle gv, gw \rangle = \langle v, w \rangle,\ \forall v, w \in V\}.$$

The special unitary group $\mathrm{SU}(V)$ is the kernel of the determinant $\det : \mathrm{U}(V) \twoheadrightarrow \mathbb{C}^1$.

Example 6.19 Upon choosing an orthonormal basis for V we may identify it with $\mathbb{C}^2$ equipped with the standard Hermitian inner product

$$\langle \left(\begin{smallmatrix} v_1 \\ v_2 \end{smallmatrix}\right), \left(\begin{smallmatrix} w_1 \\ w_2 \end{smallmatrix}\right) \rangle = v_1 \overline{w_1} + v_2 \overline{w_2}.$$

For the Hermitian space $\mathbb{C}^2$ we denote the unitary group by $\mathrm{U}_2(\mathbb{C})$ and the special unitary group by $\mathrm{SU}_2(\mathbb{C})$. Thus

$$\mathrm{U}_2(\mathbb{C}) = \{g \in \mathrm{GL}_2(\mathbb{C}) : gg^* = I\}$$

where $g^* = {}^t\bar{g}$ is the conjugate-transpose. Working out the condition $g^* = g^{-1}$ in terms of the matrix entries immediately gives the parametrization

$$\mathrm{SU}_2(\mathbb{C}) = \{\left(\begin{smallmatrix} a & b \\ -\bar{b} & \bar{a} \end{smallmatrix}\right) : |a|^2 + |b|^2 = 1\}.$$

Hence as a manifold $\mathrm{SU}_2(\mathbb{C})$ is the unit 3-sphere S^3 in $\mathbb{R}^4$.

The point of introducing $\mathrm{U}(V)$ here is the following. If $r : G \to \mathrm{GL}(V)$ is a representation of a finite group G we can always endow V with an inner product $\langle -, - \rangle$ such that $r(G) \subset \mathrm{U}(V)$. To see this, start with an arbitrary inner product $[-, -]$ and average it:

$$\langle v, w \rangle := \frac{1}{|G|} \sum_{\sigma \in G} [r(\sigma)v, r(\sigma)w].$$

With respect to this inner product r factors through $\mathrm{U}(V)$ and ergo $r^\diamond$ factors through

$$\mathrm{SU}(V)/\{\pm 1\} \xrightarrow{\sim} \mathrm{U}(V)/\mathbb{C}^1 \hookrightarrow \mathrm{PGL}(V).$$

So, the projective image $r^{\diamond}(G)$ can be identified with a finite subgroup of $\mathrm{SU}_2(\mathbb{C})/\{\pm I\}$. The so-called spin homomorphism, which we will introduce next, identifies $\mathrm{SU}_2(\mathbb{C})/\{\pm I\}$ with the group $\mathrm{SO}_3(\mathbb{R})$ of rotations in $\mathbb{R}^3$. (Recall that a matrix $A \in \mathrm{SL}_3(\mathbb{R})$ lies in $\mathrm{SO}_3(\mathbb{R})$ precisely when ${}^tA = A^{-1}$, or in more geometric terms when $x \mapsto Ax$ is an isometry of $\mathbb{R}^3$.)

Exercise 6.20 Show that any $A \in \mathrm{SO}_3(\mathbb{R})$ has 1 as an eigenvalue, i.e. $\det(A - I) = 0$. Conclude that any A in $\mathrm{SO}_3(\mathbb{R})$ is orthogonally similar to a matrix of the form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

for some angle θ–and vice versa. Interpret this result geometrically as A having an axis of rotation.

The additive analogue of $\mathrm{SU}_2(\mathbb{C})$ is the $\mathbb{R}$-vector space $\mathfrak{g} := \mathfrak{su}_2(\mathbb{C})$ of all complex 2×2-matrices X such that $\mathrm{tr}X = 0$ and $X + X^* = 0$. In other words, the traceless skew-Hermitian matrices. It is easy to check that $\mathfrak{g}$ has a basis consisting of the three quaternionic matrices

$$X_1 := \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad X_2 := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X_3 := \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

We identify $\mathfrak{g}$ with $\mathbb{R}^3$ via these coordinates. The standard dot product on $\mathbb{R}^3$ corresponds to the following inner product on $\mathfrak{g}$, as is immediately checked by a straightforward matrix calculation:

$$\langle X, Y \rangle = -\frac{1}{2}\mathrm{tr}(XY).$$

The group structure on $\mathrm{SU}_2(\mathbb{C})$ is reflected by a Lie algebra structure on $\mathfrak{g}$ given by the commutator Lie bracket $(X, Y) \mapsto XY - YX$. The two are related via the exponential of matrices $\exp : \mathfrak{g} \to \mathrm{SU}_2(\mathbb{C})$. Another crucial fact is that $\mathrm{SU}_2(\mathbb{C})$ acts on $\mathfrak{g}$ via the so-called adjoint representation

$$\begin{aligned} \mathrm{Ad} : \mathrm{SU}_2(\mathbb{C}) &\longrightarrow \mathrm{GL}(\mathfrak{g}) \\ g &\longmapsto \left[X \mapsto gXg^*\right]. \end{aligned}$$

(The reader should check that indeed $gXg^* \in \mathfrak{g}$.) It has kernel $\{\pm I\}$. To see this, observe that $i\mathfrak{g}$ is the space of traceless Hermitian matrices, and therefore $\mathfrak{g} \oplus i\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ is the space of all traceless matrices (any such M decomposes uniquely as the sum of a Hermitian matrix $\frac{1}{2}(M + M^*)$ and a skew-Hermitian matrix $\frac{1}{2}(M - M^*)$ both with trace zero). Hence a $g \in \ker(\mathrm{Ad})$ commutes with any matrix, according to the decomposition $\mathfrak{gl}_2(\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}_2(\mathbb{C})$, and therefore g is a scalar matrix in $\mathrm{SU}_2(\mathbb{C})$, i.e. $g \in \{\pm I\}$.

Example 6.21 We examine the action of $g = e^{tX_1} = \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix}$ for $t \in \mathbb{R}$. Verifying the following three linear relations is an amusing exercise:

- $\mathrm{Ad}(e^{tX_1})X_1 = X_1$;
- $\mathrm{Ad}(e^{tX_1})X_2 = \cos(2t)X_2 + \sin(2t)X_3$;
- $\mathrm{Ad}(e^{tX_1})X_3 = -\sin(2t)X_2 + \cos(2t)X_3$.

This says the matrix representation of $\mathrm{Ad}(e^{tX_1})$ relative to the ordered basis $\{X_1, X_2, X_3\}$ is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 2t & -\sin 2t \\ 0 & \sin 2t & \cos 2t \end{pmatrix}.$$

In other words $\mathrm{Ad}(e^{tX_1})$ is rotation around the axis $\mathbb{R}X_1$ by an angle $2t$ (in the positive direction).

We claim Ad has image exactly $\mathrm{SO}(\mathfrak{g}) \simeq \mathrm{SO}_3(\mathbb{R})$.

Proposition 6.22 *The adjoint representation Ad factors through $SO(\mathfrak{g})$ and induces an isomorphism of topological groups*

$$SU_2(\mathbb{C})/\{\pm I\} \xrightarrow{\sim} SO(\mathfrak{g}).$$

(This is called the spin homomorphism in part (a) of [Art18, Theorem 9.4.1, p. 269].)

Proof First note that at least Ad only takes values in the orthogonal group $\mathrm{O}(\mathfrak{g})$:

$$\langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y\rangle = \langle gXg^*, gYg^*\rangle = -\frac{1}{2}\mathrm{tr}(gXg^*gYg^*) = -\frac{1}{2}\mathrm{tr}(gXYg^{-1}) = -\frac{1}{2}\mathrm{tr}(XY) = \langle X, Y\rangle.$$

Since the determinant maps $\mathrm{O}(\mathfrak{g})$ into $\{\pm 1\}$, and $\mathrm{SU}_2(\mathbb{C}) \simeq S^3$ is connected, $\det \circ \mathrm{Ad}$ must be constant one. Thus Ad factors through $\mathrm{SO}(\mathfrak{g})$. Our task is to show the resulting map is surjective. We do this by comparing dimensions.

Foremost, $\mathrm{SO}_3(\mathbb{R})$ is a three-dimensional connected Lie group since $\mathrm{SO}_3(\mathbb{R})/\mathrm{SO}_2(\mathbb{R}) \simeq S^2$ and $\mathrm{SO}_2(\mathbb{R})$ is the circle group. On the other hand, we have seen that $\mathrm{SU}_2(\mathbb{C}) \simeq S^3$ is a three-dimensional compact Lie group. Since Ad is continuous its image must be a *closed* subgroup of $\mathrm{SO}(\mathfrak{g}) \simeq \mathrm{SO}_3(\mathbb{R})$. By the theory of smooth manifolds the quotient of $\mathrm{SO}(\mathfrak{g})$ by $\mathrm{im}(\mathrm{Ad})$ is a zero-dimensional connected manifold, i.e. a point.

Finally, since $\mathrm{SU}_2(\mathbb{C})$ is compact the continuous bijection $\mathrm{SU}_2(\mathbb{C})/\{\pm I\} \xrightarrow{\sim} \mathrm{SO}(\mathfrak{g})$ is a homeomorphism, and therefore an isomorphism of topological groups. □

We will use the spin isomorphism (Proposition 6.22) to identify $r^{\diamond}(G)$ with a finite subgroup of $\mathrm{SO}_3(\mathbb{R})$, which one can classify in terms of the Platonic solids (tetrahedra, cubes ↔ octahedra, and dodecahedra ↔ icosahedra, where '↔' indicates dual polyhedra).

6.2.2 Rotational Symmetries of Platonic Solids

Our goal in this section is to classify the finite subgroups of $\mathrm{SO}_3(\mathbb{R})$ up to isomorphism. We warm up with the finite subgroups of $\mathrm{O}_2(\mathbb{R})$. The elements hereof are partitioned into the rotations $\mathrm{SO}_2(\mathbb{R})$ and the non-trivial coset consisting of reflections. (An $A \in \mathrm{O}_2(\mathbb{R})$ with $\det A = -1$ necessarily has 1 as an eigenvalue and is therefore orthogonally similar to $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.) For the time being, let $G \leq \mathrm{O}_2(\mathbb{R})$ be a finite subgroup.

- If G is a subgroup of $\mathrm{SO}_2(\mathbb{R}) \simeq \mathbb{C}^1$ it is obviously cyclic.
- Otherwise G is isomorphic to the dihedral group D_n for some $n \geq 1$.

For the second bullet point, we know $G \cap \mathrm{SO}_2(\mathbb{R}) = \langle b \rangle$ is cyclic by the first bullet point. Let $n = |\langle b \rangle|$. Pick a reflection $a \in G$. Then ab is a reflection as well (look at the determinant). Clearly a and b generate G and satisfy the relations $a^2 = b^n = (ab)^2 = e$. (By convention $D_1 = \mathbb{Z}/2\mathbb{Z}$ and $D_2 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.)

Needless to say, cyclic and dihedral groups are realized as subgroups of $\mathrm{O}_2(\mathbb{R})$ by considering rotations and reflections preserving a regular polygon centered at the origin.

The map $A \mapsto \left(\begin{smallmatrix} A & 0 \\ 0 & \det A \end{smallmatrix}\right)$ gives an injective homomorphism $\mathrm{O}_2(\mathbb{R}) \hookrightarrow \mathrm{SO}_3(\mathbb{R})$ which allows us to identify cyclic and dihedral groups with subgroups of $\mathrm{SO}_3(\mathbb{R})$. From a geometric perspective, consider rotational symmetries of a paper-thin polygon in the xy-plane $\mathbb{R}^2 \subset \mathbb{R}^3$. (Note how three-dimensional rotation by π around the x-axis has the effect of reflecting the polygon; similarly for the other axes of symmetry.)

Dual Platonic solids clearly have isomorphic rotational symmetry groups, so in reality we only have three subgroups (A_4, S_4, A_5) of $\mathrm{SO}_3(\mathbb{R})$ arising from regular polyhedra:

- The tetrahedral group $\mathbb{T}$ of order 12. Letting it act on the four vertices of a tetrahedron gives an isomorphism $\mathbb{T} \xrightarrow{\sim} A_4$.
- The octahedral group $\mathbb{O}$ of order 24. Letting it act on the four diagonals of a cube gives an isomorphism $\mathbb{O} \xrightarrow{\sim} S_4$.
- The icosahedral group $\mathbb{I}$ of order 60. Letting it act on the five cubes inscribed in a dodecahedron gives an isomorphism $\mathbb{I} \xrightarrow{\sim} A_5$. (For the icosahedral case, see [Art18, Theorem 7.4.4, p. 199] and its proof.)

We remind the reader that A_5 is a *simple* group (in fact it is the smallest non-abelian simple group). In particular A_5 is *not* a solvable group, in contrast to both A_4 and S_4. For example, S_4 has the normal series

$$\{\mathrm{Id}\} \lhd V_4 \lhd A_4 \lhd S_4$$

with abelian (but not cyclic) factor groups. Here the Klein four-group V_4 is identified with the subgroup of permutations with cycle decomposition $(ab)(cd)$. This collection of groups (C_n, D_n, A_4, S_4, A_5) exhausts all the finite subgroups of $\mathrm{SO}_3(\mathbb{R})$ up to isomorphism:

Theorem 6.23 *Let $G \leq SO_3(\mathbb{R})$ be a finite subgroup. Then G is isomorphic to one of the groups C_n, D_n, A_4, S_4, A_5, for $n \geq 1$. (The only redundancy is $C_2 \simeq D_1$.)*

This is a famous result with plentiful excellent expositions, one of which is [Art18, Theorem 6.12.1, p. 183]. We give the main steps of the proof.

Proposition 6.24 (Burnside's Formula) *Let G be a finite group acting on a finite set S. Then the number of orbits equals the average number of fixed points:*

$$\#\ orbits = |G\backslash S| = \frac{1}{|G|} \cdot \sum_{g \in G} |S^g|$$

where $S^g := \{s \in S : gs = s\}$ is the subset of S fixed by the element g.

Proof Introduce the subset of pairs $\mathfrak{X} = \{(g, s) \in G \times S : gs = s\}$ and compute its cardinality vertically and horizontally. By partitioning $G \times S$ into a union of $\{(g, s) : s \in S\}$, as $g \in G$ varies, we find that

$$|\mathfrak{X}| = \sum_{g \in G} |\{(g, s) : s \in S,\ gs = s\}| = \sum_{g \in G} |\{(g, s) : s \in S^g\}| = \sum_{g \in G} |S^g|.$$

On the other hand, dividing $G \times S$ into subsets $\{(g, s) : g \in G\}$, where $s \in S$ varies, we get

$$|\mathfrak{X}| = \sum_{s \in S} |\{(g, s) : g \in G,\ gs = s\}| = \sum_{s \in S} |\{(g, s) : g \in G_s\}| = \sum_{s \in S} |G_s|.$$

Elements of S in the same G-orbit have conjugate isotropy groups, so $|G_s|$ only depends on the orbit of s. This orbit has cardinality $[G : G_s]$. Partitioning S into orbits $S = Gs_1 \sqcup \cdots \sqcup Gs_N$ allows us to rewrite the previous sum as

$$|\mathfrak{X}| = \sum_{s \in S} |G_s| = \sum_{i=1}^{N} \sum_{s \in Gs_i} |G_s| = \sum_{i=1}^{N} \sum_{s \in Gs_i} |G_{s_i}| = \sum_{i=1}^{N} \sum_{s \in Gs_i} |G_{s_i}| \cdot [G : G_{s_i}] = N|G|.$$

Comparing this to the first formula for $|\mathfrak{X}|$ gives the result. □

We will apply Burnside's formula to the set of *poles* of $G \leq \mathrm{SO}_3(\mathbb{R})$. As an immediate consequence of Exercise 6.20, any non-identity matrix $A \in \mathrm{SO}_3(\mathbb{R})$ has exactly two antipodal fixed points on the unit sphere $S^2 \subset \mathbb{R}^3$; the points where the axis of rotation intersects the sphere. We call these points the poles of A. The pole set of a non-trivial group G is then defined as

$$\mathrm{Pol}_G := \{P \in S^2 : AP = P,\ \text{for some } I \neq A \in G\}.$$

This is obviously a finite set on which G acts. (If P is fixed by A, then gP is fixed by gAg^{-1}.) The multiplicity m_P of a pole $P \in \mathrm{Pol}_G$ is by definition the size of its stabilizer in G. Note that $m_P \geq 2$.

Since a non-identity element $g \in G$ has exactly two fixed points in Pol_G, Burnside's formula yields the equation

$$\#\,\text{orbits} = \frac{1}{|G|} \cdot \big(|\mathrm{Pol}_G| + 2|G| - 2\big). \tag{6.1}$$

This relation gives quite strong constraints on the possible cardinalities of G and Pol_G.

Lemma 6.25 *Let $G \leq SO_3(\mathbb{R})$ be a non-trivial finite subgroup. Then we are in one of the following scenarios:*

	$\lvert G\rvert$	*#orbits*	$\lvert Pol_G\rvert$	*Multiplicities*
A	*n*	*2*	*2*	*n, n*
B	*2m*	*3*	*2m+2*	*2, 2, m*
C	*12*	*3*	*14*	*2, 3, 3*
D	*24*	*3*	*26*	*2, 3, 4*
E	*60*	*3*	*62*	*2, 3, 5*

where $m, n \geq 2$.

Proof Let $\mathrm{Pol}_G = \mathcal{O}_1 \sqcup \cdots \sqcup \mathcal{O}_N$ be the orbit partition, with corresponding multiplicities $m_1, \ldots, m_N$. We label the orbits in such a way that $m_1 \leq \cdots \leq m_N$. Inserting the obvious relation $|\mathrm{Pol}_G| = \sum_{i=1}^{N} \frac{|G|}{m_i}$ into (6.1) gives the key identity

$$2\left(1 - \frac{1}{|G|}\right) = \sum_{i=1}^{N} \left(1 - \frac{1}{m_i}\right). \tag{6.2}$$

Since $|G| > 1$ the left-hand side is in the range $[1, 2)$. Also, each term $1 - \frac{1}{m_i}$ on the right-hand side lies in $[\frac{1}{2}, 1)$ since $m_i > 1$. Thus the right-hand side sum is at least $\frac{N}{2}$ and must have more than one term. We conclude that the number of orbits is either $N = 2$ or $N = 3$.

If $N = 2$ we are in case A of the table. Indeed, in this case (6.2) says that $2 = \frac{|G|}{m_1} + \frac{|G|}{m_2}$, which can only be true if $m_1 = m_2 = |G|$ (since the two terms $\frac{|G|}{m_i} = |\mathcal{O}_i|$ are positive integers).

Suppose $N = 3$ and consider the three multiplicities $m_1 \leq m_2 \leq m_3$. Note that $m_2 \leq 3$, for otherwise $m_3 \geq m_2 \geq 4$ would imply

$$\left(1 - \frac{1}{m_1}\right) + \left(1 - \frac{1}{m_2}\right) + \left(1 - \frac{1}{m_3}\right) \geq \frac{1}{2} + \frac{3}{4} + \frac{3}{4} = 2,$$

but the left-hand side of (6.2) is less than 2. Now the argument ramifies again into two cases, according to whether $m_2 = 2$ or $m_2 = 3$.

- If $m_2 = 2$ we must also have $m_1 = 2$. Under these assumptions (6.2) amounts to

$$2\left(1 - \frac{1}{|G|}\right) = \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{m_3}\right) \Longleftrightarrow |G| = 2m_3.$$

  This is scenario B in the table.
- If $m_2 = 3$ we must have $m_1 = 2$. (Otherwise $m_1 = 3$ and $m_3 \geq 3$ leads to $\sum_{i=1}^{3}\left(1 - \frac{1}{m_i}\right) \geq 2$, again contradicting that (6.2) has left-hand side < 2.) Knowing these multiplicities, Eq. (6.2) now asserts that

$$2\left(1 - \frac{1}{|G|}\right) = \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{m_3}\right) \Longleftrightarrow \frac{2}{|G|} + \frac{1}{6} = \frac{1}{m_3}.$$

  Consequently $m_3 \leq 5$, for otherwise $\frac{2}{|G|} = \frac{1}{m_3} - \frac{1}{6} \leq 0$. The three cases $m_3 = 3, 4, 5$ correspond to scenarios C, D, E in the table respectively.

This finishes the proof. □

Lemma 6.26 *In the five scenarios A–E in Lemma 6.25 the following holds:*

A. $G \simeq C_n$ *is cyclic of order* n.
B. $G \simeq D_m$ *is dihedral of order* $2m$.
C. $G \simeq A_4$.
D. $G \simeq S_4$.
E. $G \simeq A_5$.

Proof In A there are exactly two poles $\{\pm P\}$ each fixed by all $g \in G$. The line through P and $-P$ is fixed by G. Thus $g \in G$ is similar to one of the matrices in Exercise 6.20. This shows G is isomorphic to a finite subgroup of $\mathrm{SO}_2(\mathbb{R})$ and therefore cyclic.

Looking at the third multiplicity in B, there is an orbit consisting of two G-conjugate poles $\{\pm P\}$. Consider the line through $\pm P$ and its orthogonal complement W. After picking an orthonormal basis for W we can represent $g \in G$ by a matrix of the form $\left(\begin{smallmatrix} A & 0 \\ 0 & \det A \end{smallmatrix}\right)$ with $A \in \mathrm{O}_2(\mathbb{R})$. Here the sign $\det A$ is determined by $gP = \pm P$. This identifies G with a finite subgroup of $\mathrm{O}_2(\mathbb{R})$ which is *not* entirely contained in $\mathrm{SO}_2(\mathbb{R})$. Therefore, as noted in the beginning of this section, $G \simeq D_m$ is dihedral.

In C we have at least *two* Sylow 3-subgroups in G. Why? Pick one of the two orbits with four poles, say $\{P_1, P_2, P_3, P_4\}$. Then $|G_{P_i}| = 3$ for all i. If $G_{P_1} = \cdots = G_{P_4}$, any element g hereof would fix all the P_i. Since there are four poles, such a g must fix a pair of non-antipodal poles and therefore be the identity. This is a contradiction.

By Sylow's theorems $|\mathrm{Syl}_3(G)| = 4$ in case C. Each $H \in \mathrm{Syl}_3(G)$ is self-normalizing, i.e. $N_G(H) = H$ because both sides have index 4 in G. In particular G acts faithfully on $\mathrm{Syl}_3(G)$ by conjugation. This gives an injective homomorphism $G \hookrightarrow S_4$, whose image must be the unique index two subgroup A_4.

In case D consider the orbit $\{P_1, \ldots, P_8\}$ with 8 poles. Again $|G_{P_i}| = 3$ for all i, and the G_{P_i} cannot all be the same subgroup (by the exact same argument as in case

C). Using Sylow theory we again find that $|\mathrm{Syl}_3(G)| = 4$. Note also that $\mathrm{ord}(g) \leq 4$ for all $g \in G$ (as g fixes a pole P and $m_P = |G_P|$ lies in $\{2, 3, 4\}$ by the last column of the table).

Still in case D, let G act on $\mathrm{Syl}_3(G)$ by conjugation. We need to show this action is faithful, in other words that the intersection $I := \bigcap_{H \in \mathrm{Syl}_3(G)} N_G(H)$ is trivial. Now $|N_G(H)| = 6$ for all $H \in \mathrm{Syl}_3(G)$, so at least we know $|I| \in \{1, 2, 3, 6\}$. We can immediately rule out that $|I| = 3$. (Otherwise I would be a *normal* Sylow 3-subgroup, but there are four of them–and they are all G-conjugate.) Tweaking this argument a bit also rules out that $|I| = 6$: Otherwise let $K \leq I$ be the *unique* subgroup of I of order 3. Then $gKg^{-1} \leq gIg^{-1} = I$ for all $g \in G$. So, by uniqueness, $gKg^{-1} = K$. Again this gives a normal Sylow 3-subgroup K in G. Contradiction. All that is left is to rule out that $|I| = 2$, and this is where the observation $\mathrm{ord}(g) \leq 4$ comes in: Assume $|I| = 2$. Select an $x \in G$ of order 3 and let $y \in I$ be the non-identity element. Here y is in the *center* of G since $I \lhd G$. Therefore $\mathrm{ord}(xy) = 6 > 4$ since $xy = yx$.

Finally, in case E we need to argue why G is a *simple* group. We start with some preliminary counting:

(1) Consider the size 12 orbit $\{\pm P_1, \ldots, \pm P_6\}$. (Observe that $m_P = m_{-P}$ so the poles with a given multiplicity come in antipodal pairs.) The six stabilizers $G_{P_i} = G_{-P_i}$ are distinct groups with 5 elements. Thus

$$\#\{g \in G : \mathrm{ord}(g) = 5\} = 6(5 - 1) = 24.$$

(2) Similarly, for the size 20 orbit we find that

$$\#\{g \in G : \mathrm{ord}(g) = 3\} = 10(3 - 1) = 20.$$

(3) For the last orbit, the one of size 30, we get

$$\#\{g \in G : \mathrm{ord}(g) = 2\} = 15(2 - 1) = 15.$$

Suppose $Q \lhd G$. We show that $|Q| = 1$ or $|Q| = 60$ by ruling out all the other divisors of 60:

- $|Q| \notin \{3, 6, 12, 15\}$. Otherwise the index of Q belongs to $\{4, 5, 10, 20\}$ and is therefore coprime to 3. By Lagrange's theorem $g^{[G:Q]} \in Q$ for all $g \in G$. If $\mathrm{ord}(g) = 3$ this implies that $g \in Q$. We conclude that Q contains all the 20 elements from part (2). This contradicts $|Q| \leq 15$.
- $|Q| \notin \{5, 10, 20\}$. Otherwise the index of Q belongs to $\{3, 6, 12\}$ and is therefore coprime to 5. Then the 24 elements from part (1) are contained in Q, contradicting $|Q| \leq 20$.
- $|Q| \neq 30$. Otherwise Q has index two and therefore contains all the $24 + 20 = 44 > 30$ elements of order 3 or 5.

- $|Q| \neq 4$. Otherwise Q has odd index and therefore contains all the $15 > 4$ elements of order 2 from part (3).
- $|Q| \neq 2$. Otherwise the non-identity element $y \in Q$ necessarily belongs to the center of G. Picking an $x \in G$ of order 3 we find (as above) that $\mathrm{ord}(xy) = 6$. This is a contradiction (G does not have any elements of order 6).

Altogether this forces Q to be either the trivial subgroup or all of G. It is a standard fact from group theory that a simple group G of order 60 is isomorphic to A_5. We omit the proof. It is a long exercise in Sylow theory. The isomorphism $G \simeq A_5$ comes about by letting G act on $\mathrm{Syl}_2(G)$ by conjugation. (The tricky part is to rule out that $|\mathrm{Syl}_2(G)| = 15$.) See part (b) of [Lan02, Exercise 41, p. 79]. □

This concludes the proof of Theorem 6.23. Circling back to Theorem 6.16, we now know the projective image $r^{\diamond}(G)$ is either dihedral or isomorphic to one of the groups A_4, S_4, A_5 (it is *not* cyclic by Remark 6.18). Moreover, we already established the dihedral case of Theorem 6.16 before Remark 6.18. Since A_5 is not solvable, Theorem 6.16 is really about the two cases $r^{\diamond}(G) \simeq A_4$ (the 'tetrahedral' case) and $r^{\diamond}(G) \simeq S_4$ (the 'octahedral' case). The first case was settled by Langlands in [La80] and Tunnell dealt with the second case soon after in [Tun81].

6.2.3 A Bird's-Eye View of the Proof of Theorem 6.16

With only limited tools at our disposal we can only give a very rough outline of the proof of Theorem 6.16. We refer the more advanced reader to [Gel97] for further details, and a discussion of the role Theorem 6.16 plays in the proof of Fermat's Last Theorem.

A salient point is to be able to work over a number field K rather than just $\mathbb{Q}$. The notion of a primitive form is replaced by a 'cuspidal automorphic representation' π of $\mathrm{GL}_2(\mathbb{A}_K)$. Explaining precisely what that means would take us too far afield, and we ask the reader to treat this notion as a black box. One can think of π as the analogue of a Hecke character, to the effect that $\pi = \otimes_{v \in V} \pi_v$ where π_v is a representation of $\mathrm{GL}_2(K_v)$ which is 'unramified' for all but at most finitely many $v \in V_f$. Being unramified means there are nonzero vectors of π_v fixed by $\mathrm{GL}_2(\mathcal{O}_v)$, and such representations are easy to describe in terms of a pair of unramified characters $\chi_{v,1}, \chi_{v,2} : K_v^{\times} \to \mathbb{C}^{\times}$. There is an associated L-function of the form

$$L(s, \pi) = \prod_{v \in V_f} L(s, \pi_v), \qquad L(s, \pi_v) = \frac{1}{(1 - \chi_{v,1}(\varpi_v) q_v^{-s})(1 - \chi_{v,2}(\varpi_v) q_v^{-s})},$$

where $\pi_v \leftrightarrow (\chi_{v,1}, \chi_{v,2})$ is unramified in the second equality. It is known that $L(s, \pi)$ enjoys the same analytic properties as Hecke L-functions and $L(s, f)$. The notion of an automorphic representation gives a very robust framework even for GL_n and other reductive groups.

The Langlands-Tunnell theorem is more general than Theorem 6.16 in that it applies to any number field K, and it does not require any oddness assumption. (For *even* representations in the $K = \mathbb{Q}$ case their theorem produces Maass forms instead of holomorphic forms.) One can even replace $\mathrm{Gal}(\overline{K}/K)$ with the *Weil* group $W_K \twoheadrightarrow \mathrm{Gal}(\overline{K}/K)$ in the statement below but we refrain from doing so.

Theorem 6.27 (Langlands, Tunnell) *Suppose we are given an irreducible representation*

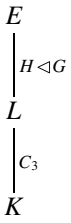
$$r : Gal(\overline{K}/K) \twoheadrightarrow Gal(E/K) \longrightarrow GL(V) \simeq GL_2(\mathbb{C})$$

such that the image of $r^{\diamond}$ is a solvable subgroup of $PGL_2(\mathbb{C})$. Then there exists a unique cuspidal automorphic representation π of $GL_2(\mathbb{A}_K)$, with central character corresponding to $\det r$*, such that*

$$L(s, r) = L(s, \pi).$$

The dihedral case can be checked off by suitably adapting the arguments in Sect. 6.1.4 and before Remark 6.18.

We dive into the *tetrahedral* case. Let r be a representation of $G := \mathrm{Gal}(E/K)$ as in Theorem 6.27 for which $r^{\diamond}(G) \xrightarrow{\sim} A_4$. The inverse image of $V_4 \lhd A_4$ via $G \twoheadrightarrow A_4$ is a normal subgroup $H = \mathrm{Gal}(E/L)$ for some cubic Galois extension L/K.



Now the restriction $r|_H : H \to \mathrm{GL}(V)$ has projective image $V_4 = D_2$ and is therefore of dihedral type. This gives a cuspidal automorphic representation Π of $\mathrm{GL}_2(\mathbb{A}_L)$ such that $L(s, r|_H) = L(s, \Pi)$. Moreover, this Π is invariant under $\mathrm{Gal}(L/K)$ since $r|_H$ is, i.e. $\sigma_*(r|_H) \simeq r|_H$ for all $\sigma \in G$, so $\Pi^\sigma \simeq \Pi$. By the theory of base change, conveniently summarized in [Gel97, p. 194], there exists a cuspidal automorphic representation π of $\mathrm{GL}_2(\mathbb{A}_K)$ such that $\Pi \simeq \mathrm{BC}_{L/K}(\pi)$. In fact there are exactly three such representations π; one can always twist a given π by one of the three characters ω of $C_K/N_{L/K}(C_L)$ and get the same base change.

Let $\chi : C_K \to \mathbb{C}^\times$ be the Hecke character corresponding to $\det r$ via the Artin map. Then Π has central character $\chi \circ N_{L/K}$. The central character χ_π has base change $\chi_\pi \circ N_{L/K} = \chi_\Pi = \chi \circ N_{L/K}$. This shows there is a unique choice of ω for which $\chi_{\pi\otimes\omega} = \chi$, namely $\omega^2 = \chi_\pi^{-1}\chi$. Replacing π with a twist we can assume $\chi_\pi = \chi$ to begin with, and this condition determines the descent π uniquely.

To verify this candidate representation π actually works is of course the hard part of the proof. The rough strategy is to consider r composed with the *adjoint* representation (of $\mathrm{GL}_2(\mathbb{C})$ on the three-dimensional Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ of traceless matrices):

$$\mathrm{Ad} \circ r : G \longrightarrow \mathrm{GL}_2(\mathbb{C}) \xrightarrow{\mathrm{Ad}} \mathrm{GL}_3(\mathbb{C}).$$

This happens to be of the form $\mathrm{Ind}_H^G \theta$ for some character $\theta : H \to \mathbb{C}^\times$. By 'automorphic induction' there exists a cuspidal automorphic representation $\Pi_1 := \mathrm{AI}_{L/K}(\theta \circ \mathrm{Art}_{E/L})$ of $\mathrm{GL}_3(\mathbb{A}_K)$ with L-function $L(s, \theta) = L(s, \mathrm{Ad} \circ r)$. There is another route to an automorphic representation of $\mathrm{GL}_3(\mathbb{A}_K)$ by taking the so-called symmetric square transfer $\Pi_2 := \mathrm{Sym}^2(\pi)$, and the key point is to show $\Pi_1 \simeq \Pi_2$. This is done by showing the GL_3-version of the Rankin-Selberg L-function $L(s, \Pi_1 \times \Pi_2^\vee)$ has a pole at $s = 1$.

The tetrahedral case feeds into the proof of the *octahedral* case $r^\diamondsuit(G) \xrightarrow{\sim} S_4$ as follows. We now reset the notation and let $H = \mathrm{Gal}(E/L)$ denote the inverse image of $A_4 \lhd S_4$ via $G \twoheadrightarrow S_4$. So L/K is now quadratic. The restriction $r|_H$ has projective image A_4, so by the tetrahedral case there is a cuspidal automorphic representation Π of $\mathrm{GL}_2(\mathbb{A}_L)$ such that $L(s, r|_H) = L(s, \Pi)$. Now there are exactly two cuspidal automorphic representations π_1 and π_2 of $\mathrm{GL}_2(\mathbb{A}_K)$ with base change $\mathrm{BC}_{L/K}(\pi_i) = \Pi$ for $i = 1, 2$. However, now $\pi_2 \simeq \pi_1 \otimes \omega$ where ω is a *quadratic* character of C_K, which implies $\chi_{\pi_1} = \chi_{\pi_2}$. It is not clear which π_i should be the candidate for π just by inspecting central characters alone.

Tunnell's idea was to base change to a certain *non-Galois* cubic extension C/K, using a result of Jacquet, Piatetski-Shapiro, and Shalika (which was not available when [La80] was published): There are exactly three Sylow 2-subgroups of S_4, so none of them are normal. They are all isomorphic to D_4. Pick one, and let Q denote its inverse image under the projection $G \twoheadrightarrow S_4$. Then let $C := E^Q$.

Now $r|_Q$ has projective image D_4 and therefore there exists a cuspidal automorphic representation Π' of $\mathrm{GL}_2(\mathbb{A}_C)$ such that $L(s, r|_Q) = L(s, \Pi')$. A key step is to verify that among $\{\pi_1, \pi_2\}$ there is a *unique* π such that $\mathrm{BC}_{C/K}(\pi) \simeq \Pi'$. The point here is that

$$\mathrm{BC}_{CL/C}(\mathrm{BC}_{C/K}(\pi_i)) \simeq \mathrm{BC}_{CL/L}(\mathrm{BC}_{L/K}(\pi_i)) \simeq \mathrm{BC}_{CL/L}(\Pi)$$

for $i = 1, 2$. Since CL/C is quadratic, $\mathrm{BC}_{C/K}(\pi_2) \simeq \mathrm{BC}_{C/K}(\pi_1) \otimes \psi$ for some quadratic Hecke character ψ. From this it is not terribly hard to deduce that $\mathrm{BC}_{C/K}(\pi_i)$ are *distinct* for $i = 1, 2$, so precisely one of them must be Π'. This pins down the candidate representation π.

Showing this π does the job is easier in the octahedral case. One can completely bypass having to consider $\mathrm{Ad} \circ r$. We direct the reader to [Tun81, p. 175] for the slightly tedious calculations. The argument ultimately boils down to the elementary observation that S_4 has no elements of order six (due to the cycles having length at most four).

6.3 Eichler-Shimura Theory and the Theorem of Deligne

For $k \geq 2$ one cannot attach a complex Galois representation to $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ compatible with L-functions. The center of symmetry of $L(s, f)$ is $s = \frac{k}{2}$, whereas for Artin L-functions the center is $s = \frac{1}{2}$.

Taking the cue from algebraic Hecke characters, we should instead search for ℓ-adic Galois representations. For the moment, let $\chi : C_K \longrightarrow \mathbb{C}^\times$ be an algebraic Hecke character (not to be confused with the nebentype) of infinity type ξ. Recall that χ has finite order if and only if $\xi = 1$; this is analogous to the weight one case for modular forms. Construction 3.22 produced a continuous ℓ-adic Galois character

$$r_{\chi,\ell} : \mathrm{Gal}(\bar{K}/K) \longrightarrow \overline{\mathbb{Q}}_\ell^\times$$

such that for $v \in V_f$ not above ℓ we have

$$r_{\chi,\ell}|_{\mathrm{Gal}(\overline{K_v}/K_v)} \circ \mathrm{Art}_{K_v} = \iota_\ell \circ \iota_\infty^{-1} \circ \chi_v.$$

Here ℓ is arbitrary and we have fixed embeddings $\iota_\infty : \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and $\iota_\ell : \bar{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$. (We used p instead of ℓ in 3.22 but this should not lead to confusion.)

Back in the modular forms case, for $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ of weight $k \geq 2$ we should hope for a whole system of compatible ℓ-adic Galois representations

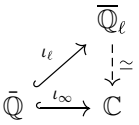
$$r_{f,\ell} : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}(V_\ell) \simeq \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$$

such that at least for primes $p \nmid N\ell$ this is unramified, and any Frobenius element $\mathrm{Frob}_p \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ has characteristic polynomial determined by the T_p-eigenvalue a_p of f:

$$\det(1 - r_{f,\ell}(\mathrm{Frob}_p)X) = 1 - a_p X + \chi(p)p^{k-1}X^2.$$

– How do we interpret this equality? The right-hand side is a polynomial in $\mathbb{C}[X]$ which is independent of ℓ. The left-hand side belongs to $\overline{\mathbb{Q}}_\ell[X]$. It turns out a_p is an *algebraic* number, so via $\iota_\infty : \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$ we can view the right-hand side in $\bar{\mathbb{Q}}[X]$. In succession, via $\iota_\ell : \bar{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ the right-hand side may be thought of as a polynomial with $\overline{\mathbb{Q}}_\ell$-coefficients. This comparison is usually implicit, and we will often suppress the two embeddings ι_∞ and ι_ℓ from the notation.

Remark 6.28 In fact $\overline{\mathbb{Q}}_\ell$ is isomorphic to $\mathbb{C}$ as (abstract) fields–not as topological fields obviously. The isomorphism is hopelessly non-canonical and depends on the choice of transcendence bases over $\mathbb{Q}$ for both fields. Occasionally, choosing an isomorphism $\overline{\mathbb{Q}}_\ell \xrightarrow{\sim} \mathbb{C}$ once and for all serves as a useful gadget. In that situation we insist the embeddings ι_∞ and ι_ℓ are compatible, meaning the diagram



is commutative.

A decisive feature of the weight $k \geq 2$ setting is that we have access to the arithmetic geometry of modular curves. Their (étale) cohomology is a natural habitat for the representations $r_{f,\ell}$. The next big theorem affirms they exist:

Theorem 6.29 (Eichler and Shimura for $k = 2$, Deligne for $k \geq 2$) *Let χ be a Dirichlet character modulo N with $\chi(-1) = (-1)^k$. Given an $f \in Prim_k(\Gamma_0(N), \chi)$ with Fourier coefficients $(a_n)_{n\geq 1}$, for every prime ℓ there is a two-dimensional ℓ-adic representation*

$$r = r_{f,\ell} : Gal(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow GL(V_\ell) \simeq GL_2(\overline{\mathbb{Q}}_\ell)$$

which is unramified at all $p \nmid N\ell$, and for those primes p the following two conditions are satisfied:

$$tr\, r(Frob_p) = a_p, \qquad \det r(Frob_p) = \chi(p)p^{k-1}.$$

The representation $r = r_{f,\ell}$ is irreducible and uniquely determined by f and ℓ up to isomorphism.

Remark 6.30 By the Chebotarev density theorem, $\det r = \chi \cdot \chi_{\text{cyc}}^{k-1}$ where $\chi_{\text{cyc}} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathbb{Z}_\ell^\times$ denotes the ℓ-adic cyclotomic character. In particular $\det r(c) = -1$.

We stress that $r_{f,\ell}$ depends on the choice of embeddings ι_∞ and ι_ℓ although the notation seems to suggest otherwise.

6.3.1 Reduction Modulo ℓ

The Baire category argument in the proof of Proposition 3.27 applies word-for-word to representations. Thus any continuous representation $r : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \text{GL}_2(\overline{\mathbb{Q}}_\ell)$ factors through $\text{GL}_2(L)$ for some finite extension $L/\mathbb{Q}_\ell$. (However, r may be ramified at infinitely many places–in contrast to the GL_1-case in 3.27. In [Ram00] Ramakrishna gives examples of representations $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \twoheadrightarrow \text{GL}_2(\mathbb{Z}_\ell)$ with infinite ramification. Crystallinity can be achieved under GRH.)

By a change of basis we can even devise for the representation to factor through $\text{GL}_2(\mathcal{O}_L)$:

Lemma 6.31 *Let $r : G \longrightarrow GL(V)$ be a finite-dimensional continuous representation of a compact group G on an L-vector space V. Then there exists a G-stable $\mathcal{O}_L$-lattice in V.*

Proof Choose any norm $\|\cdot\|$ on V. For instance, if $\{v_1, \ldots, v_d\}$ is a basis for V let $\|v\| = \max_i |c_i|$ for a vector $v = \sum_i c_i v_i$. Introduce the averaged norm

$$\|v\|_G := \max_{g \in G} \|r(g)v\|.$$

This is finite since r is continuous and G is compact. G acts isometrically with respect to $\|\cdot\|_G$, so the unit ball $V^\circ := \{v \in V : \|v\|_G \leq 1\}$ is at least a G-stable $\mathcal{O}_L$-submodule. It is also a lattice, i.e. the natural map $L \otimes_{\mathcal{O}_L} V^\circ \to V$ is an isomorphism. Surjectivity is clear. For the injectivity suppose $\sum_j a_j \otimes b_j$ lies in the kernel. Pick a nonzero $x \in \mathcal{O}_L$ such that $xa_j \in \mathcal{O}_L$ for all j, such as a sufficiently high power of a uniformizer. Then $\sum_j a_j \otimes b_j = \sum_j x^{-1} \otimes (xa_j)b_j = x^{-1} \otimes \sum_j xa_j b_j = x^{-1} \otimes x \sum_j a_j b_j = 0$. Finally note that $\|v\| \leq \|v\|_G$, so $V^\circ \subset \mathcal{O}_L v_1 \oplus \cdots \oplus \mathcal{O}_L v_d$ is a free $\mathcal{O}_L$-submodule of finite rank necessarily equal to d. □

Exercise 6.32 Prove that $\mathrm{GL}_n(\mathcal{O}_L)$ is a maximal compact subgroup of $\mathrm{GL}_n(L)$. Then show that it is the only one up to conjugation. (`Hint`: A compact $G \leq \mathrm{GL}_n(L)$ can be conjugated into $\mathrm{GL}_n(\mathcal{O}_L)$. If we have an inclusion $\mathrm{GL}_n(\mathcal{O}_L) \leq G$ the two groups must have the same Haar measure.)

In the setup of Lemma 6.45, suppose $V^\circ \subset V$ is a G-stable $\mathcal{O}_L$-lattice. Then we can reduce it modulo $\mathfrak{p}_L$ and get a k_L-vector space

$$\overline{V} := k_L \otimes_{\mathcal{O}_L} V^\circ \simeq V^\circ / \mathfrak{p}_L V^\circ$$

where $k_L = \mathcal{O}_L/\mathfrak{p}_L \simeq \mathbb{F}_{\ell^v}$ is the residue field. This purveys a finite-dimensional G-representation over k_L, which obviously factors through a finite quotient of G. It is often convenient to extend scalars to $\overline{\mathbb{F}}_\ell$ and work with $\overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}$ instead. This is clearly of finite length, i.e. there is a filtration by G-stable subspaces

$$\{0\} = W_0 \subset W_1 \subset W_2 \subset \cdots \subset W_r = \overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}$$

such that W_i / W_{i-1} is irreducible for all $i \geq 1$. The Jordan-Hölder theorem says these irreducible subquotients are independent of the choice of filtration up to isomorphism and rearrangement. Therefore the 'semi-simplification' of $\overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}$ is well-defined up to isomorphism:

$$\left(\overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}\right)^{\mathrm{ss}} := \bigoplus_{i \geq 1} W_i / W_{i-1}.$$

Seriatim, the Brauer-Nesbitt theorem implies that $\left(\overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}\right)^{\mathrm{ss}}$ is independent of the choice of lattice V° (up to isomorphism) and therefore completely determined by V itself.

To add more details, since we are working over the algebraically closed field $\overline{\mathbb{F}}_\ell$, the irreducible characters of a finite group are linearly independent by Wedderburn theory. (The group ring mod the Jacobson radical is Artinian plus semisimple and therefore a product of matrix algebras. See [Sch13, Theorem 3.1.1, p. 88] for a complete proof.) This applies to the above setup by observing that

$$k_L \ni \operatorname{tr} r(g) + \mathfrak{p}_L = \sum_{i \geq 1} \operatorname{tr}\big(g : W_i / W_{i-1} \longrightarrow W_i / W_{i-1}\big)$$

for all $g \in G$. (Represent the G-action on $\overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}$ by block-diagonal matrices and take the trace.) Since the left-hand side is independent of the choice of lattice V° the irreducible summands of $\big(\overline{\mathbb{F}}_\ell \otimes_{k_L} \overline{V}\big)^{\mathrm{ss}}$ (along with their multiplicities) are determined by $\operatorname{tr} r$ alone–and not the lattice.

Remark 6.33 Another argument, which has the benefit of functioning over k_L, is to use the following very general version of the Brauer-Nesbitt theorem: Suppose G is *any* group and k is *any* field. For $i = 1, 2$ let $r_i : G \to \mathrm{GL}_n(k)$ be *semisimple* representations with identical characteristic polynomials:

$$\det(I - r_1(g)X) = \det(I - r_2(g)X)$$

for all $g \in G$. Then $r_1 \simeq r_2$. (When $\operatorname{char}(k) > n$ or $\operatorname{char}(k) = 0$ it is enough that the traces $\operatorname{tr} r_i$ coincide.) This shows the semi-simplification $\overline{V}^{\mathrm{ss}}$ of $\overline{V}$ as a $k_L[G]$-module is independent of V° up to isomorphism. Note that the irreducible summands of $\overline{V}^{\mathrm{ss}}$ may not all be *absolutely* irreducible, i.e. remain irreducible after extending scalars to $\overline{\mathbb{F}}_\ell$.

Feeding $r_{f,\ell}$ into this machine gives a well-defined *semisimple* Galois representation over $\overline{\mathbb{F}}_\ell$.

Corollary 6.34 *Let $f \in Prim_k(\Gamma_0(N), \chi)$ be a cusp form of weight $k \geq 2$ as in Theorem 6.29. Then for every prime ℓ there is an odd and semisimple representation*

$$\overline{r} = \overline{r_{f,\ell}} : Gal(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow GL(\overline{V_\ell}) \simeq GL_2(\overline{\mathbb{F}}_\ell)$$

which is unramified at all $p \nmid N\ell$, and for those primes p the following two conditions hold:

$$tr\, \overline{r}(Frob_p) = a_p + \mathfrak{p}_{\overline{\mathbb{Q}}_\ell}, \qquad \det \overline{r}(Frob_p) = \chi(p)p^{k-1} + \mathfrak{p}_{\overline{\mathbb{Q}}_\ell}.$$

The (semisimple) representation $\overline{r} = \overline{r_{f,\ell}}$ is uniquely determined by f and ℓ up to isomorphism.

In this statement $\mathfrak{p}_{\overline{\mathbb{Q}}_\ell} = \{a \in \overline{\mathbb{Q}}_\ell : |a| < 1\}$ denotes the maximal ideal of $\mathcal{O}_{\overline{\mathbb{Q}}_\ell} = \{a \in \overline{\mathbb{Q}}_\ell : |a| \leq 1\}$. Note that $a_p \in \mathcal{O}_{\overline{\mathbb{Q}}_\ell}$ by Theorem 6.29 since $r_{f,\ell}$ is conjugate to a $\mathrm{GL}_2(\mathcal{O}_{\overline{\mathbb{Q}}_\ell})$-valued representation.

Serre's prominent 'modularity conjecture' is a converse to Corollary 6.34: Changing our perspective, we now start with *some* two-dimensional continuous representation

$$\rho : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$$

assumed to be both odd and irreducible, so it has a chance of arising from a modular form. Very important examples of representations ρ arise from the ℓ-torsion points on elliptic curves over $\mathbb{Q}$. The reader may want to look up the $\ell = 3$ case in [Gel97] which played a crucial role in Wiles' proof of the Shimura-Taniyama-Weil conjecture. In [Ser87, Conjecture 3.2.3, p. 196] Serre conjectured that these two simple hypotheses (ρ is odd and irreducible) implies that ρ is *modular*, i.e. there exists an $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ such that

$$\rho \simeq \overline{r_{f,\ell}}.$$

Serre even gave recipes for an optimal weight $k(\rho)$ and an optimal level $N(\rho)$. The latter is the prime-to-ℓ Artin conductor of ρ. The weight $k(\rho)$ takes a little more effort to describe in a case-by-case manner, see [Ser87, Sects. 2.1–2.4].

Astonishingly, Serre's conjecture was eventually proved around 2008 by Khare and Wintenberger in a series of papers [Kha06, KW09a, KW09b, KW09c]. See also their ICM survey [KW10]. As a consequence of their work, they were able to remove the assumption that $r^\diamond$ has solvable image in Theorem 6.16: Any odd and irreducible representation $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\mathbb{C})$ is associated with a weight one form, even those of icosahedral type. We refer the reader to part (ii) of [KW09b, Corollary 10.2, p. 502].

6.3.2 The Irreducibility and Image of $r_{f,\ell}$

The proof of Proposition 6.9 extends to higher weights k with minor modifications if we assume $a_n = O(n^{\frac{k-1}{2}+\epsilon})$.

Proposition 6.35 *Let f be as in Theorem 6.29. Then $r_{f,\ell}$ is irreducible.*

Proof (*Sketch*) Suppose on the contrary that $r_{f,\ell} \sim \left(\begin{smallmatrix} \eta_1 & * \\ 0 & \eta_2 \end{smallmatrix}\right)$ for two Galois characters $\eta_i : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \overline{\mathbb{Q}}_\ell^\times$. We have not covered p-adic Hodge theory, but hopefully the acceptive reader believes that each η_i is necessarily a *geometric* character in the sense of Sect. 3.3.2. (They are subquotients of a potentially semistable representation.) Essentially by Lemma 3.9, which in particular describes the algebraic Hecke characters over $\mathbb{Q}$, the η_i must have the form $\eta_i = \delta_i \cdot \chi_{\mathrm{cyc}}^{n_i}$ for finite order characters δ_i and $n_i \in \mathbb{Z}$.

The conditions defining $r_{f,\ell}$ translate into the following: The η_i are unramified for all primes $p \nmid N\ell$ and for such p we have

$$\delta_1(p)p^{n_1} + \delta_2(p)p^{n_2} = a_p, \qquad \delta_1(p)\delta_2(p)p^{n_1+n_2} = \chi(p)p^{k-1}.$$

Here $\delta_i(p) := (\iota_\infty \circ \iota_\ell^{-1})(\delta_i(\mathrm{Frob}_p))$ are roots of unity in $\bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$. Taking absolute values in the second equation we find that $n_1 + n_2 = k - 1$ and consequently $\delta_1(p)\delta_2(p) = \chi(p)$. From the first equation,

$$|a_p|^2 = p^{2n_1} + p^{2n_2} + 2\mathrm{Re}\big(\delta_1(p)\overline{\delta_2(p)}\big)p^{k-1}.$$

If $n_1 \neq n_2$ we either have $n_1 > \frac{k-1}{2}$ or $n_2 > \frac{k-1}{2}$. We can replace $\frac{k-1}{2}$ by $\frac{k-1}{2} + \epsilon$ here with an ϵ small enough. Either way we get $|a_p| \sim p^m$ for some $m > \frac{k-1}{2} + \epsilon$ as the prime p goes to infinity (and m is n_1 or n_2). This violates our assumed bound $a_n = O(n^{\frac{k-1}{2}+\epsilon})$ and finishes the proof for even k.

For odd k we still have to get a contradiction when $n_1 = n_2 = \frac{k-1}{2}$. For $\mathrm{Re}(s) > k$ we have

$$\sum_{p\nmid N\ell} |a_p|^2 p^{-s} = 2\sum_{p\nmid N\ell} p^{-s+k-1} + \sum_{p\nmid N\ell} \delta_1(p)\overline{\delta_2(p)}p^{-s+k-1} + \sum_{p\nmid N\ell} \overline{\delta_1(p)}\delta_2(p)p^{-s+k-1}.$$

If $\delta_1 \neq \delta_2$ the last two sums are $O(1)$ as $s \to k^+$ (as explained in the proof of Proposition 6.9). Because of the factor 2 this contradicts the upper bound in Proposition 6.8 (the term from $p = \ell$ is irrelevant). If $\delta_1 = \delta_2$ we have $a_p = 2\delta_1(p)p^{\frac{k-1}{2}}$. Therefore $\sum_{p\nmid N\ell} |a_p|^2 p^{-s} = 4\sum_{p\nmid N\ell} p^{-s+k-1}$ is still too large and Proposition 6.8 again gives a contradiction. □

If E is an elliptic curve defined over $\mathbb{Q}$, its ℓ-adic Tate module $T_\ell(E) = \varprojlim E[\ell^r]$ gives a lattice in a two-dimensional representation $r_{E,\ell} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\mathbb{Q}_\ell)$. If E is 'non-CM' a famous result of Serre says that the image of $r_{E,\ell}$ is an *open* subgroup of $\mathrm{GL}_2(\mathbb{Z}_\ell)$. Moreover, $\mathrm{im}(r_{E,\ell}) = \mathrm{GL}_2(\mathbb{Z}_\ell)$ with only finitely many exceptions ℓ. These results can be subsumed in single statement asserting that the product of *all* these representations $\prod_\ell r_{E,\ell}$ has open image in $\mathrm{GL}_2(\widehat{\mathbb{Z}}) \simeq \prod_\ell \mathrm{GL}_2(\mathbb{Z}_\ell)$.

A similar result is true for $r_{f,\ell}$ but it is more subtle when f does not have rational Fourier coefficients. In general the Fourier coefficients $(a_n)_{n\geq 1}$ of an $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ all lie in a number field:

$$\mathbb{L} = \mathbb{L}_f := \mathbb{Q}(a_n : n \geq 1) = \mathbb{Q}(a_p, \chi(p) : p \text{ prime}).$$

Why? If we accept that $S_k(\Gamma_1(N))$ has a basis of forms with rational Fourier coefficients, $\mathrm{Aut}(\mathbb{C})$ acts by letting $f^\sigma = \sum_{n=1}^\infty \sigma(a_n)q^n$ for any field automorphism σ of $\mathbb{C}$, and the T_n-eigenvalues are at least algebraic numbers. The $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-action on the finite set $\mathrm{Prim}_k(\Gamma_1(N))$ necessarily factors through a finite quotient $\mathrm{Gal}(E/\mathbb{Q})$ for some finite Galois extension $E \supset \mathbb{L}_f$.

Note that $\mathbb{L}_f$ may not be Galois over $\mathbb{Q}$. Indeed $\sigma(\mathbb{L}_f) = \mathbb{L}_{f^\sigma}$ for all $\sigma \in \mathrm{Aut}(\mathbb{C})$. As a preliminary observation, we only need to adjoin a_p and $\chi(p)$ for the *good* primes $p \nmid N$ in order to generate $\mathbb{L}_f$.

Lemma 6.36 $\mathbb{L}_f = \mathbb{Q}(a_p, \chi(p) : p \textit{ prime}, p \nmid N)$.

Proof Suppose $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ fixes a_p and $\chi(p)$ for all $p \nmid N$. Then $f^\sigma \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$ has the same T_p-eigenvalue as f for $p \nmid N$ (and the same nebentype $\chi^\sigma = \chi$). By the strong multiplicity one theorem (5.93) we must have $f^\sigma = f$, i.e. σ is the identity on $\mathbb{L}_f$. □

As a first result on the nature of $\mathbb{L}_f$ we mention:

Proposition 6.37 *Let $f \in \mathrm{Prim}_k(\Gamma_0(N), \chi)$. Then $\mathbb{L}_f$ is a totally real field when χ is trivial, and $\mathbb{L}_f$ is a CM-field when χ is non-trivial. (A 'CM-field' is a totally imaginary quadratic extension of a totally real subfield; see [Shi94, p. 125].)*

Proof For $p \nmid N$ we have $\overline{a_p} = \chi(p)^{-1} a_p$ by Proposition 5.80. When $\chi = 1$ the dual primitive form $f^\vee$ therefore has the same T_p-eigenvalue as f for all $p \nmid N$. Therefore $f^\vee = f$ by the strong multiplicity one theorem (5.93). In other words, f has real Fourier coefficients. For the same reason, so does f^σ for all $\sigma \in \mathrm{Aut}(\mathbb{C})$. This shows $\sigma(a_n) \in \mathbb{R}$ for all embeddings $\sigma : \mathbb{L}_f \hookrightarrow \mathbb{C}$, i.e. $\mathbb{L}_f$ is totally real.

Now assume $\chi \neq 1$. In light of Lemma 6.36, the relation $\overline{a_p} = \chi(p)^{-1} a_p$ for $p \nmid N$ shows that $\mathbb{L}_{f^\vee} = \mathbb{L}_f$. In other words, complex conjugation restricts to an automorphism $c : \mathbb{L}_f \xrightarrow{\sim} \mathbb{L}_f$, which is non-trivial since $\chi \neq 1$. By [Shi94, Proposition 5.11, p. 125] it remains to check that $\overline{\sigma(x)} = \sigma(cx)$ for every $\sigma : \mathbb{L}_f \hookrightarrow \mathbb{C}$. This is easy: For the root of unity $x = \chi(p)$ the relation is obvious (the complex conjugate is the inverse). For $x = a_p$ with $p \nmid N$ the relation follows from comparing $\sigma(\overline{a_p}) = \chi^\sigma(p)^{-1}\sigma(a_p)$ to the identity obtained by replacing f with f^σ above, i.e. $\overline{\sigma(a_p)} = \chi^\sigma(p)^{-1}\sigma(a_p)$. □

We pointed out earlier that $r_{f,\ell}$ depends on the choice of an embedding $\iota_\ell : \bar{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ (and also ι_∞ which we fix once and for all). Since the Fourier coefficients lie in $\mathbb{L}_f$, only the restriction $\mathbb{L}_f \hookrightarrow \overline{\mathbb{Q}}_\ell$ is relevant. This corresponds to a finite place $\lambda | \ell$ together with a continuous embedding of the completion $\mathbb{L}_{f,\lambda} \hookrightarrow \overline{\mathbb{Q}}_\ell$. A sturdy way of packaging the representations $r_{f,\ell}$, as ι_ℓ varies, is to view them as $\mathrm{GL}_2(\mathbb{L}_{f,\lambda})$-valued representations and take their product

$$\widetilde{r_{f,\ell}} : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{Q}_\ell) \simeq \prod_{\lambda|\ell} \mathrm{GL}_2(\mathbb{L}_{f,\lambda}).$$

This is the representation which naturally appears in the (co)homology of modular curves.

Having introduced $\mathbb{L}_f$ and $\widetilde{r_{f,\ell}}$ we now return to the open image questions. Serre and Swinnerton-Dyer considered the case where $N = 1$ and $\mathbb{L}_f = \mathbb{Q}$: First of all they showed that $r_{f,\ell} = \widetilde{r_{f,\ell}}$ always has open image in $\mathrm{GL}_2(\mathbb{Z}_\ell)$. Since $\det r_{f,\ell} = \chi_{\mathrm{cyc}}^{k-1}$ there is the obvious constraint (for $k > 2$) that any $A \in \mathrm{im}(r_{f,\ell})$ has $\det A \in (\mathbb{Z}_\ell^\times)^{k-1}$. This leads to their second result that for all large enough ℓ we have the equality

$$\mathrm{im}(r_{f,\ell}) = \{A \in \mathrm{GL}_2(\mathbb{Z}_\ell) : \det A \in (\mathbb{Z}_\ell^\times)^{k-1}\}.$$

The restriction $\mathbb{L}_f = \mathbb{Q}$ was removed by Ribet [Rib75] and later Momose and Ribet dispensed with the restriction $N = 1$ in [Mom81, Rib85]. The general result is a bit technical. It relies on the observation that f admits *inner twists*, i.e. some $\sigma \in \mathrm{Aut}(\mathbb{C})$ admit a Dirichlet character ψ_σ such that $\sigma(a_p) = \psi_\sigma(p)a_p$ for all but finitely many primes p. For example, if σ is complex conjugation then $\psi_\sigma = \chi^{-1}$. We assume f is 'non-CM' which implies there are no non-trivial self-twists, i.e. there can be at most one character ψ_σ with the above property for each σ. We view ψ_σ as a Galois character interchangeably. Introduce the following notation:

- $\Gamma := \{\sigma \in \mathrm{Aut}(\mathbb{L}_f)$: there exists a ψ_σ with $\sigma(a_p) = \psi_\sigma(p)a_p$ for all but finitely many $p\}$;
- $F := \mathbb{L}_f^\Gamma$;
- $H := \bigcap_{\sigma\in\Gamma} \ker \psi_\sigma$ (viewed as an open subgroup of $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$).

Clearly Γ is a finite group ($\psi_{\sigma\tau} = \psi_\tau^\sigma \psi_\sigma$). It can be shown to be abelian. The general results have to do with the subgroup $\widetilde{r_{f,\ell}}(H)$ of the full image. Momose and Ribet showed that, up to conjugation,

$$\widetilde{r_{f,\ell}}(H) = \{A \in \mathrm{GL}_2(\mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{Z}_\ell) : \det A \in (\mathbb{Z}_\ell^\times)^{k-1}\} \tag{6.3}$$

for all but finitely many ℓ. This is part of a more general result valid for all ℓ: There is a quaternion algebra D over F such that $\widetilde{r_{f,\ell}}(H)$ is an open subgroup of $\{x \in (D \otimes_{\mathbb{Q}} \mathbb{Q}_\ell)^\times : \mathrm{Nrd}_{D/F}(x) \in \mathbb{Q}_\ell^\times\}$. For ℓ outside a finite set we have $D \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \simeq M_2(F \otimes_{\mathbb{Q}} \mathbb{Q}_\ell)$ and [Rib85, Theorem 3.1, p. 191] gives the equality (6.3) above.

6.3.3 *Eichler-Shimura Theory in Broad Strokes*

Eichler and Shimura proved the opposite direction of the Shimura-Taniyama-Weil conjecture: If $f \in \mathrm{Prim}_2(\Gamma_0(N), \chi)$ is a *weight two* form, with rational Fourier coefficients, there is an associated elliptic curve E_f over $\mathbb{Q}$ with L-function $L(s, E_f) = L(s, f)$. In general, when non-rational Fourier coefficients are allowed, one instead defines an abelian variety A_f with

$$L(s, A_f) = \prod_{\sigma\in\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{L}_f,\mathbb{C})} L(s, f^\sigma).$$

Since the left-hand side is given by a degree $2\dim A_f$ Euler product we necessarily have $\dim A_f = [\mathbb{L}_f : \mathbb{Q}]$. The variety A_f is a quotient of the Jacobian of the level $\Gamma_1(N)$ modular curve.

In this section $X = X_1(N)$ denotes the modular curve for $\Gamma_1(N)$. As we spelled out in Section 5.5 we can identify $S_2(\Gamma_1(N))$ with the global sections $H^0(X, \Omega^1_X)$. The dimension of this space is the genus g of the curve X. The homology group $H_1(X, \mathbb{Z}) \simeq \mathbb{Z}^{2g}$ is free on the loops around the g 'holes' of X. Contour integration around these loops gives a bilinear map

$$\begin{aligned} H_1(X,\mathbb{Z}) \times H^0(X,\Omega^1_X) &\longrightarrow \mathbb{C} \\ (\gamma,\omega) &\longmapsto \int_\gamma \omega, \end{aligned}$$

which identifies $H_1(X,\mathbb{Z})$ with a lattice in the dual space $H^0(X,\Omega^1_X)^\vee$. The *Jacobian* of X is the complex torus given by passing to the quotient, i.e.

$$\mathrm{Jac}(X) := H^0(X,\Omega^1_X)^\vee / H_1(X,\mathbb{Z}).$$

By a theorem of Riemann, $\mathrm{Jac}(X)$ can be embedded into a projective space and is therefore an abelian variety over $\mathbb{C}$. We refer the reader to [DS05, Chap. 6] for a more detailed and student-friendly introduction to contour integration, Jacobians, and related material.

Abel's theorem identifies $\mathrm{Jac}(X)$ with the (degree zero) Picard group $\mathrm{Pic}^0(X)$, assuming $g > 0$. This depends on the choice of a base point $x_0 \in X$. The contour integral $\int_{x_0}^{x} \omega$ is only well-defined modulo integrals around loops. This gives a natural map

$$\begin{aligned} X &\longrightarrow \mathrm{Jac}(X) \\ x &\longmapsto \Big(\omega \mapsto \int_{x_0}^{x} \omega\Big). \end{aligned}$$

On the other hand $x \mapsto [x - x_0]$ gives an embedding $X \to \mathrm{Pic}^0(X)$. (Recall that $\mathrm{Pic}^0(X)$ is the group of degree zero divisors $\mathrm{Div}^0(X)$ modulo those arising from $\mathbb{C}(X)$.) Abel's theorem [DS05, Theorem 6.1.2, p. 214] is the result that the above embedding $X \to \mathrm{Jac}(X)$ extends to an isomorphism

$$\begin{aligned} \mathrm{Pic}^0(X) &\xrightarrow{\sim} \mathrm{Jac}(X) \\ \Big[\sum_{x\in X} n_x x\Big] &\longmapsto \Big(\omega \mapsto \sum_{x\in X} n_x \int_{x_0}^{x} \omega\Big). \end{aligned}$$

This suggests how Jacobians should be defined over other base fields (as a moduli space of isomorphism classes of line bundles on a smooth projective algebraic curve).

There is a natural Hecke action on $\mathrm{Jac}(X)$ compatible with the Hecke operators on $S_2(\Gamma_1(N))$. To briefly explain this, as in Sect. 5.8.3 we consider double cosets $\Gamma h \Gamma'$ with $h \in \mathrm{GL}_2(\mathbb{Q})^+$. The geometric actions all arise via pullback and pushforward maps arising from the diagram

$$\begin{array}{ccccc} & & X(\Gamma' \cap h^{-1}\Gamma h) & \xrightarrow{\sim} & X(\Gamma \cap h\Gamma' h^{-1}) \\ & \swarrow^{\pi_1} & & & \searrow^{\pi_2} \\ X(\Gamma') & & & & X(\Gamma). \end{array}$$

The horizontal map h_* is induced by $z \mapsto hz$ and the π_i are the natural projections induced by inclusions. Such diagrams are called Hecke *correspondences*: A point $y \in X(\Gamma')$ 'corresponds' to the set of points $\{\pi_2(h_*x) : x \text{ such that } y = \pi_1(x)\} \subset X(\Gamma)$. One should really take the ramification indices e_x into account and map the *multiset* of points $\{e_x \cdot x : x \in \pi_1^{-1}(y)\}$ to $X(\Gamma)$. The resulting map $y \mapsto \sum_{x \in \pi_1^{-1}(y)} e_x \pi_2(h_*x)$ extends linearly to a map between the divisor groups, ultimately giving a map between Picard groups

$$[\Gamma h \Gamma'] : \mathrm{Pic}^0(X(\Gamma')) \longrightarrow \mathrm{Pic}^0(X(\Gamma)).$$

In turn, via Abel's theorem $[\Gamma h \Gamma']$ corresponds to a map between Jacobians (which can also be defined directly via pullbacks and pushforwards for Jac). The reader who wishes to dig deeper into these definitions should study [DS05, Sect. 6.2].

When $\Gamma = \Gamma' = \Gamma_1(N)$ and h is either $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$ or $\langle d \rangle$ the operators on $\mathrm{Jac}(X)$ will still be denoted by T_p and $\langle d \rangle$ respectively. They can be shown to be compatible with the *duals* of the analogous operators on $S_2(\Gamma_1(N))$. See [DS05, Proposition 6.3.2, p. 229]. (Note that $p^{\frac{k}{2}-1} = 1$ for $k = 2$.)

Introduce the (commutative) Hecke algebra acting faithfully on $S_2(\Gamma_1(N))$ as

$$\mathbb{T} := \mathbb{T}_2(\Gamma_1(N)) = \mathbb{Z}\big[T_n, \langle d \rangle : n \geq 1,\ d \text{ coprime to } N\big] \subset \mathrm{End}_{\mathbb{C}\text{-lin}} S_2(\Gamma_1(N)).$$

This is free of finite rank over $\mathbb{Z}$; it can be identified with an algebra of endomorphisms of $H_1(X, \mathbb{Z})$. Any form $f \in \mathrm{Prim}_2(\Gamma_0(N), \chi)$ defines a Hecke *eigensystem*, i.e. a ring homomorphism

$$\begin{aligned} \theta_f : \mathbb{T} &\longrightarrow \mathbb{C} \\ T_n &\longmapsto a_n \\ \langle d \rangle &\longmapsto \chi(d). \end{aligned}$$

Clearly $\mathrm{im}(\theta_f) = \mathbb{Z}[a_n : n \geq 1]$ is an *order* in the number field $\mathbb{L}_f$ (a subring which is free over $\mathbb{Z}$ of rank equal to the degree). The algebra $\mathbb{T}$ acts on $\mathrm{Jac}(X)$ and we can now define A_f as the quotient

$$A_f := \mathrm{Jac}(X)/\ker(\theta_f)\mathrm{Jac}(X),$$

which obviously carries an action of $\mathbb{T}/\ker(\theta_f) \xrightarrow{\sim} \mathrm{im}(\theta_f)$. How do we view A_f as a complex torus, and what is its dimension? Let

$$\mathbb{C}\mathrm{Orb}(f) := \mathrm{span}_{\mathbb{C}}\{f^\sigma : \sigma \in \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{L}_f, \mathbb{C})\} \subset S_2(\Gamma_1(N))$$

denote the $[\mathbb{L}_f : \mathbb{Q}]$-dimensional subspace spanned by all the Galois-conjugates of f. Restriction of linear forms gives a surjective map $S_2(\Gamma_1(N))^\vee \twoheadrightarrow \mathbb{C}\mathrm{Orb}(f)^\vee$. Let $\mathbb{Z}\mathrm{Lat}(f)$ denote the image of $H_1(X, \mathbb{Z})$. This yields a map

$$A_f \twoheadrightarrow \mathbb{C}\mathrm{Orb}(f)^\vee/\mathbb{Z}\mathrm{Lat}(f)$$

which happens to be an isomorphism ([DS05, Proposition 6.6.4, p. 242]). Furthermore, the target is an $[\mathbb{L}_f : \mathbb{Q}]$-dimensional complex torus. One can go further and produce an isogeny (a surjective homomorphism with finite kernel) from $\mathrm{Jac}(X)$ to a direct sum of tori A_f associated with primitive forms of *all* levels. We refer the reader to [DS05, Theorem 6.6.6, p. 244] for a precise statement.

The big step, which we largely sweep under the rug here, is the reworking of the previous concepts and constructions from an algebro-geometric perspective. To get this off the ground, one must obviously first boost X to a smooth projective genus g curve defined over $\mathbb{Q}$. This alone is a somewhat lengthy story. One approach is via function fields as in [DS05, Chap. 7]. Alternatively, for the reader with a strong background in algebraic geometry [KM85] is the go-to source. We also highly recommend the more introductory account in [DI95, Chap. 8].

With X now defined over $\mathbb{Q}$ let $X_{\bar{\mathbb{Q}}} = X \times_{\mathbb{Q}} \bar{\mathbb{Q}}$. The Jacobian variety $\mathrm{Jac}(X_{\bar{\mathbb{Q}}})$ is now endowed with a $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-action which commutes with the $\mathbb{T}$-action. Playing these two actions off against each other is the name of the game.

The ℓ-adic Tate module of $\mathrm{Jac}(X_{\bar{\mathbb{Q}}})$ gives a $2g$-dimensional $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-representation:

$$T_\ell(\mathrm{Jac}(X_{\bar{\mathbb{Q}}})) := \varprojlim \mathrm{Jac}(X_{\bar{\mathbb{Q}}})[\ell^r] \simeq \mathbb{Z}_\ell^{2g}, \qquad V_\ell(\mathrm{Jac}(X_{\bar{\mathbb{Q}}})) := T_\ell(\mathrm{Jac}(X_{\bar{\mathbb{Q}}})) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \simeq \mathbb{Q}_\ell^{2g}.$$

The really crucial step here is establishing the Eichler-Shimura (congruence) relation, which makes the interplay between T_p and Frob_p precise. Here is one version of it:

Theorem 6.38 (Eichler-Shimura relation) *The $Gal(\bar{\mathbb{Q}}/\mathbb{Q})$-action on each $Jac(X_{\bar{\mathbb{Q}}})[\ell^r]$ is unramified at all primes $p \nmid N\ell$. For any such p we have the relation*

$$T_p = Frob_p + \langle p \rangle p Frob_p^{-1}$$

where both sides are viewed as endomorphisms of $Jac(X_{\bar{\mathbb{Q}}})[\ell^r]$. Similarly for the Tate module $T_\ell(Jac(X_{\bar{\mathbb{Q}}}))$.

Proof See the proofs of [DS05, Theorems 8.7.2, and 9.5.1] □

If we let $r_{X,\ell}$ denote the $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-representation on $V_\ell(\mathrm{Jac}(X_{\bar{\mathbb{Q}}}))$ the theorem says that $r_{X,\ell}(\mathrm{Frob}_p)$ satisfies the equation $x^2 - T_p x + \langle p \rangle p = 0$.

As above $f \in \mathrm{Prim}_2(\Gamma_0(N), \chi)$ defines a quotient A_f of the Jacobian. The actions on $\mathrm{Jac}(X_{\bar{\mathbb{Q}}})$ induce commuting actions on A_f, so again it makes sense to consider the ℓ-adic Tate module

$$T_\ell(A_f) := \varprojlim A_f[\ell^r] \simeq \mathbb{Z}_\ell^{2d}, \qquad V_\ell(A_f) := T_\ell(A_f) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \simeq \mathbb{Q}_\ell^{2d},$$

where $d := [\mathbb{L}_f : \mathbb{Q}]$. The Galois action commutes with the $\mathbb{T}$-action, which factors through $\mathbb{T}/\ker(\theta_f) \xrightarrow{\sim} \mathbb{Z}[a_n : n \geq 1] =: \mathcal{O}_f$. Thus $T_\ell(A_f)$ is an $\mathcal{O}_f \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$-module with a Galois action, and $V_\ell(A_f)$ is a module over $\mathcal{O}_f \otimes_{\mathbb{Z}} \mathbb{Q}_\ell = \mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ with a Galois action. It turns out that $V_\ell(A_f)$ is in fact a *free* $\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$-module of rank

necessarily equal to two. See [DS05, Lemma 9.5.3, p. 390]. Choosing a basis gives a homomorphism

$$\widetilde{r_{f,\ell}} : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{Aut}_{\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{Q}_\ell\text{-lin}}(V_\ell(A_f)) \xrightarrow{\sim} \mathrm{GL}_2(\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{Q}_\ell).$$

It is unramified at primes $p \nmid N\ell$ and for such p the automorphism $\widetilde{r_{f,\ell}}(\mathrm{Frob}_p)$ satisfies the Hecke polynomial equation $x^2 - a_p x + \chi(p)p = 0$.

In the weight two case this gives a geometric realization $V_\ell(A_f)$ of the representation $\widetilde{r_{f,\ell}}$ mentioned after Proposition 6.37. The representation $r_{f,\ell}$ in Theorem 6.29 is obtained from $\widetilde{r_{f,\ell}}$ by projecting onto $\mathrm{GL}_2(\mathbb{L}_{f,\lambda})$ where $\lambda|\ell$ and the embedding $\mathbb{L}_{f,\lambda} \hookrightarrow \overline{\mathbb{Q}}_\ell$ are determined by ι_ℓ in the hopefully obvious way.

Besides [DS05] we recommend [Roh97] and [Kna92, Chap. XI] as great introductions to Eichler-Shimura theory and the pertaining methods.

6.3.4 *The Proof of Deligne's Theorem in Even Broader Strokes*

The universal covering $\mathbb{C}^g$ of a complex torus $\mathbb{C}^g/\Lambda$ has deck transformations Λ. Therefore $H_1(\mathbb{C}^g/\Lambda, \mathbb{Z}) \simeq \Lambda$. In particular, by Poincaré duality (X is a compact oriented 2-manifold) we find that

$$H_1(\mathrm{Jac}(X), \mathbb{Z}) \simeq H_1(X, \mathbb{Z}) \simeq H^1(X, \mathbb{Z}).$$

In étale cohomology the analogous assertion is that $T_\ell(\mathrm{Jac}(X_{\bar{\mathbb{Q}}}))$ is dual to $H^1_{\text{ét}}(X_{\bar{\mathbb{Q}}}, \mathbb{Z}_\ell)$. This suggests that for modular forms of weight $k > 2$ one should look at cohomology with non-constant coefficients.

We adapt the notation from the previous section. Thus $Y \hookrightarrow X$ denote the modular curves of level $\Gamma = \Gamma_1(N)$. For simplicity we assume $N \gg 0$ so that Γ has no elliptic points, i.e. $\mathrm{Stab}_\Gamma(z) = \{I\}$ for all $z \in \mathcal{H}$. Any $\mathbb{Z}[\Gamma]$-module M defines a sheaf $\mathcal{M}$ on $Y = \Gamma\backslash\mathcal{H}$ as follows: Let Γ act diagonally on $\mathcal{H} \times M$ and look at the quotient $\Gamma\backslash(\mathcal{H} \times M)$. For any open subset $U \subset Y$ consider the sections s as in the diagram

$$\Gamma\backslash(\mathcal{H} \times M) \xrightarrow{\text{proj}} \Gamma\backslash\mathcal{H} \xleftarrow{\text{incl}} U, \qquad U \xrightarrow{s} \Gamma\backslash(\mathcal{H} \times M)$$

More precisely, we identify such sections with functions on $\mathrm{pr}_\Gamma^{-1}(U) \subset \mathcal{H}$ and let

$$\mathcal{M}(U) := \{\text{locally constant functions } \mathrm{pr}_\Gamma^{-1}(U) \xrightarrow{s} M \text{ such that } s(\gamma z) = \gamma s(z) \text{ for all } \gamma \in \Gamma\}.$$

Suppose $W \subset \mathcal{H}$ is a connected open subset such that $W \cap \gamma W \neq \varnothing \Longrightarrow \gamma = I$ (see Lemma 5.9) and let $U = \mathrm{pr}_\Gamma(W)$. Since $\mathrm{pr}_\Gamma^{-1}(U) = \bigcup_{\gamma\in\Gamma} \gamma W$ is a disjoint

union, and $s \in \mathcal{M}(U)$ must be constant on each connected component γW, the Γ-equivariance of s shows it is determined by $s|_W$. In particular the restriction $\mathcal{M}|_U \simeq \underline{M}_U$ is a *constant* sheaf. These 'locally constant' sheaves on Y are called *local systems*.

Note that $H^0(Y, \mathcal{M}) \simeq M^\Gamma \simeq H^0(\Gamma, M)$ since $\mathcal{H}$ is connected. This continues to hold for sheaf and group cohomology in higher degrees:

$$H^i(Y, \mathcal{M}) \xrightarrow{\sim} H^i(\Gamma, M).$$

(The map is an edge homomorphism of a spectral sequence which degenerates since $\mathrm{Stab}_\Gamma(z) = \{I\}$. See [Har17, Sect. 2.1.1, p. 48] for instance.)

We get Hecke operators $[\Gamma h \Gamma]$ on $H^i(Y, \mathcal{M})$ essentially by restriction and corestriction in group cohomology (or more geometrically by pullback and pushforward in Hecke correspondences). See [Har17, Sect. 3.1, p. 69] for the precise definition of these double coset operators on cohomology.

There is a group in-between $H^i(Y, \mathcal{M})$ and the cohomology with compact support $H^i_c(Y, \mathcal{M})$ with desirable semisimplicity properties. The *inner* cohomology $H^i_!(Y, \mathcal{M})$ is defined as the image of the compactly supported cohomology, as in the 'fundamental exact sequence' [Har17, Sect. 2.1.4, p. 60]:

$$\cdots \longrightarrow H^i_c(Y, \mathcal{M}) \longrightarrow H^i(Y, \mathcal{M}) \longrightarrow H^i(\partial, \mathcal{M}) \longrightarrow H^{i+1}_c(Y, \mathcal{M}) \longrightarrow \cdots$$
$$\downarrow \quad \nearrow$$
$$H^i_!(Y, \mathcal{M})$$

Here ∂ denotes the boundary of the Borel-Serre compactification of Y. It will not play a significant role in this short survey. We refer the curious reader to [Har17, p. 30] where this compactification is introduced for modular curves.

On the group cohomology side, inner cohomology corresponds to *parabolic* cohomology $H^i_{\mathrm{par}}(\Gamma, M)$. See the appendix to [Hid93, p. 348]. The latter consists of all the classes in $H^i(\Gamma, M)$ which become trivial upon restriction to $H^i(\mathrm{Stab}_\Gamma(\rho), M)$ for all cusps ρ.

We now define a concrete $\mathbb{Z}[\Gamma]$-module M_{k-2} for which $H^i_!(Y, \mathcal{M}_{k-2} \otimes \mathbb{C})$ is very closely related to $S_k(\Gamma)$ via the so-called Eichler-Shimura isomorphism in Theorem 6.39 below.

Start with the tautological $\mathrm{GL}_2(\mathbb{Z})$-representation $\mathbb{Z}^2$ (i.e. integral column vectors with $\mathrm{GL}_2(\mathbb{Z})$ acting by left multiplication). For each $n \geq 0$ we consider the nth symmetric power $\mathrm{Sym}^n \mathbb{Z}^2$, which we think of as the homogeneous polynomials in $\mathbb{Z}[x, y]$ of degree n. This is a free $\mathbb{Z}$-module of rank $n+1$, with basis $\{x^i y^j : i + j = n\}$. The group $\mathrm{GL}_2(\mathbb{Z})$ acts on $\mathrm{Sym}^n \mathbb{Z}^2$ by right translations, i.e.

$$\left(\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) P\right)(x, y) = P\left((x, y)\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\right) = P(ax + cy, bx + dy).$$

This works over any ring R. The exact same recipe gives a representation of $\mathrm{GL}_2(R)$ on $\mathrm{Sym}^n R^2$. It is an example of an *algebraic* representation of GL_2. Another example

is any $\mathbb{Z}$-power of the determinant. Note that the upper-triangular matrix $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ acts on $P_+(x, y) := x^n$ via the character $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \mapsto a^n$. This is called the *highest weight* and it completely determines the (irreducible) algebraic representation up to isomorphism.

We define M_n as the *dual* of the representation $\mathrm{Sym}^n \mathbb{Z}^2$. We think of the dual as a representation on the same $\mathbb{Z}$-module $\langle x^i y^j : i + j = n \rangle$ but the action is now pre-composed with the transpose-inverse. (For GL_2 one usually defines M_n by twisting the dual with $\det^n$ but we will mostly be concerned with SL_2 in what follows. See the conventions in [Hid93, p. 165] where our M_n is denoted by $L(n; \mathbb{Z})$.)

The relation between $S_k(\Gamma)$ and $H^1_{\mathrm{par}}(\Gamma, M_{k-2} \otimes \mathbb{C})$ is based on the observation that $f \in S_k(\Gamma)$ defines an $M_{k-2} \otimes \mathbb{C}$-valued differential form

$$\omega_f := 2\pi i f(z)(x - zy)^{k-2} dz$$

with the property $\gamma^* \omega_f = \gamma \cdot \omega_f$ for all $\gamma \in \Gamma$. This is a straightforward calculation which can be found in [Hid93, p. 167]; we encourage the reader to try this is an exercise. Now for $z \in \mathcal{H}^*$ consider the 'real part' of the contour integral

$$\varphi_{f,z}(\gamma) = \mathrm{Re} \int_z^{\gamma z} \omega_f \in M_{k-2} \otimes \mathbb{R}.$$

This is easily seen to define a cocycle $\varphi_{f,z} : \Gamma \to M_{k-2} \otimes \mathbb{R}$, whose cohomology class $[\varphi_{f,z}] \in H^1(\Gamma, M_{k-2} \otimes \mathbb{R})$ is independent of z and which in fact lies in the parabolic cohomology. Again, we recommend the reader tries to verify these claims as exercises. Hints can be found in [Hid93, Sect. 6.2].

This leads to the instrumental result below, which we repeatedly alluded to in the above discussion.

Theorem 6.39 (Eichler-Shimura isomorphism) *Assume Γ is torsion-free. Then the above map $f \mapsto [\varphi_{f,z}]$ defines an isomorphism of $\mathbb{R}$-vector spaces*

$$S_k(\Gamma) \xrightarrow{\sim} H^1_{par}(\Gamma, M_{k-2} \otimes \mathbb{R}).$$

This isomorphism is compatible with Hecke operators.

Proof See the proof of [Hid93, Theorem 1, p. 168]. The remark about Hecke operators is verified in [Hid93, Sect. 6.3]. □

Viewing $S_k(\Gamma)$ as an $\mathbb{R}$-vector space is a bit odd. Usually one 'complexifies' the map and rephrases Theorem 6.39 as an isomorphism of $\mathbb{C}$-vector spaces (intertwining the Hecke operators):

$$S_k(\Gamma) \oplus \overline{S_k(\Gamma)} \simeq \mathbb{C} \otimes_{\mathbb{R}} S_k(\Gamma) \xrightarrow{\sim} \mathbb{C} \otimes_{\mathbb{R}} H^1_{\mathrm{par}}(\Gamma, M_{k-2} \otimes \mathbb{R}) \simeq H^1_{\mathrm{par}}(\Gamma, M_{k-2} \otimes \mathbb{C}).$$

(Here $\overline{S_k(\Gamma)} = S_k(\Gamma) \otimes_{\mathbb{C},c} \mathbb{C}$ is the group $S_k(\Gamma)$ but with complex conjugate scaling.) For $k = 2$ we note the striking resemblance with the Hodge decomposition for the Kähler manifold X, i.e.

$$H^1(X,\mathbb{C}) = H^{0,1}(X) \oplus H^{1,0}(X) = H^{0,1}(X) \oplus \overline{H^{0,1}(X)} = H^0(X,\Omega^1_X) \oplus \overline{H^0(X,\Omega^1_X)} = S_2(\Gamma) \oplus \overline{S_2(\Gamma)}.$$

The general proof of the surjectivity of $f \mapsto [\varphi_{f,z}]$ exposed in [Hid93, pp. 171–175] is a long technical tour de force of Hodge theory. There is a more highbrow proof in [Har17, Sect. 4.1.3] which gives a version of the Eichler-Shimura isomorphism for any locally symmetric space, see [Har17, Theorem 4.1.2, p. 124].

As in Sect. 6.3.3 we introduce the Hecke algebra, now acting faithfully on weight k forms,

$$\mathbb{T} := \mathbb{T}_k(\Gamma_1(N)) = \mathbb{Z}\big[T_n, \langle d\rangle : n \geq 1,\ d \text{ coprime to } N\big] \subset \mathrm{End}_{\mathbb{C}\text{-lin}} S_k(\Gamma_1(N)).$$

Due to Theorem 6.39 the Hecke-action on $H^1_{\mathrm{par}}(\Gamma, M_{k-2}\otimes\mathbb{C})$ factors through $\mathbb{T}$. Let $f \in \mathrm{Prim}_k(\Gamma_0(N),\chi)$ be a primitive form with Hecke eigensystem $\theta_f : \mathbb{T} \to \mathbb{C}$ and look at the $\ker(\theta_f)$-torsion

$$H^1_{\mathrm{par}}(\Gamma, M_{k-2}\otimes\mathbb{C})[\ker(\theta_f)] \simeq S_k(\Gamma)[\ker(\theta_f)] \oplus \overline{S_k(\Gamma)[\ker(\theta_f)]}.$$

This is a module over $\mathcal{O}_f \otimes_{\mathbb{Z}} \mathbb{C} = \mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{C}$, where we keep letting $\mathcal{O}_f$ denote the subring of $\mathbb{C}$ generated by the Fourier coefficients. As $\theta_{f^\sigma} = \sigma \circ \theta_f$ for $\sigma \in \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{L}_f,\mathbb{C})$ we clearly have at least an inclusion

$$\mathbb{C}\mathrm{Orb}(f) = \mathrm{span}_{\mathbb{C}}\{f^\sigma : \sigma \in \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{L}_f,\mathbb{C})\} \subset S_k(\Gamma)[\ker(\theta_f)].$$

By (a suitable version of) the strong multiplicity one theorem this is an *equality*. See [Roh97, Theorem 3, p. 84] and the argument in [Roh97, p. 91] before Theorem 4. The point is that $\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{C} \simeq \mathbb{C}^{\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{L}_f,\mathbb{C})}$ so its simple modules correspond to the projections onto $\mathbb{C}$ indexed by the embeddings σ. Thus $S_k(\Gamma)[\ker(\theta_f)]$ is spanned by eigenforms g with eigensystem θ_{f^σ} for some σ. Any such g must be a scalar multiple of f^σ by strong multiplicity one ([Roh97, Theorem 3, p. 84]).

Furthermore, $\mathbb{C}\mathrm{Orb}(f) = S_k(\Gamma)[\ker(\theta_f)]$ is free of rank one over $\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{C}$. To see this, observe that a tuple $(z_\tau) \in \mathbb{C}^{\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{L}_f,\mathbb{C})}$ acts by $(z_\tau)\star f^\sigma = z_\sigma f^\sigma$. Therefore $\widetilde{f} := \sum_\sigma f^\sigma$ is a basis. Indeed the map $(z_\tau) \mapsto (z_\tau)\star\widetilde{f}$ is injective since the f^σ are linearly independent. For dimension reasons the map must be an isomorphism $\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} \mathbb{C}\mathrm{Orb}(f)$.

This culminates in the following key offshoot of the Eichler-Shimura isomorphism.

Corollary 6.40 *Assume Γ is torsion-free. Let $\theta_f : \mathbb{T} \to \mathbb{C}$ be the eigensystem of an $f \in \mathrm{Prim}_k(\Gamma_0(N),\chi)$. Then*

$$H^1_!(Y, \mathcal{M}_{k-2}\otimes\mathbb{C})[\ker(\theta_f)] \tag{6.4}$$

is a free $\mathbb{L}_f \otimes_{\mathbb{Q}} \mathbb{C}$-module of rank two.

Now the Galois-action takes the stage. Once and for all we choose an isomorphism of fields $\iota : \overline{\mathbb{Q}}_\ell \xrightarrow{\sim} \mathbb{C}$. Via ι we view θ_f as taking values in $\overline{\mathbb{Q}}_\ell$ and consider the $\mathbb{L}_f \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}_\ell$-module $H^1_!(Y, \mathcal{M}_{k-2} \otimes \overline{\mathbb{Q}}_\ell)[\ker(\theta_f)]$, i.e. the ℓ-adic analogue of the free rank two module in (6.4).

We take for granted that Y can be defined over $\mathbb{Q}$ and let $Y_{\bar{\mathbb{Q}}} = Y \times_{\mathbb{Q}} \bar{\mathbb{Q}}$. The elixir of étale cohomology changes $H^1_!(Y, \mathcal{M}_{k-2} \otimes \overline{\mathbb{Q}}_\ell)$ into an algebro-geometric object "$H^1_{\text{ét},!}(Y_{\bar{\mathbb{Q}}}, \mathcal{M}_{k-2} \otimes \overline{\mathbb{Q}}_\ell)$" endowed with a $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-action commuting with $\mathbb{T}$. For the most part we will treat étale cohomology as a black box, but we recommend [Mil80, MilEt] to the reader who wants to learn the basics.

We first need to address how to make sense of the sheaf $\mathcal{M}_{k-2} \otimes \overline{\mathbb{Q}}_\ell$ in the ℓ-adic world. First, an ℓ-adic sheaf on $Y_{\bar{\mathbb{Q}}}$ (or any Noetherian scheme) is given by an inverse system $\mathcal{F} := \{\mathcal{F}_n\}_{n\geq 1}$, where $\mathcal{F}_n$ is a sheaf of $\mathbb{Z}/\ell^n\mathbb{Z}$-modules on $Y_{\bar{\mathbb{Q}}}$ for the étale topology, and the transition maps induce isomorphisms $\mathcal{F}_{n+1} \otimes_{\mathbb{Z}/\ell^{n+1}\mathbb{Z}} \mathbb{Z}/\ell^n\mathbb{Z} \xrightarrow{\sim} \mathcal{F}_n$. If each $\mathcal{F}_n$ is locally constant we say $\mathcal{F}$ is lisse. This definition is taken from [Stack, Definition 64.18.1]. The cohomology of $\mathcal{F}$ is defined by

- $H^i(Y_{\bar{\mathbb{Q}}}, \mathcal{F}) := \varprojlim_n H^i_{\text{ét}}(Y_{\bar{\mathbb{Q}}}, \mathcal{F}_n)$;
- $H^i(Y_{\bar{\mathbb{Q}}}, \mathcal{F} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell) := H^i(Y_{\bar{\mathbb{Q}}}, \mathcal{F}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

Similarly for cohomology with compact support, see [Stack, Definition 64.18.8].

Because Γ is small enough, Y is a moduli space of elliptic curves with a point of order N. It therefore comes with a *universal* family of elliptic curves (and a 'point' P^{univ})

$$\pi : E^{\text{univ}} \longrightarrow Y$$

such that any pair (E, P) defined over $\mathbb{Q}$ is the fiber product along a unique morphism $\mathrm{Spec}(\mathbb{Q}) \longrightarrow Y$. The complex version $\pi : E^{\text{univ}}(\mathbb{C}) \longrightarrow \Gamma\backslash\mathcal{H}$ has a rather simple description: On $\mathcal{H} \times \mathbb{C}$ we have two actions. Γ acts on the left, via the $\mathcal{H}$-factor, and $\mathbb{Z}^2$ acts on the right by

$$(\tau, z) \bullet \left(\begin{smallmatrix} m \\ n \end{smallmatrix}\right) := (\tau, z + m\tau + n).$$

Then $E^{\text{univ}}(\mathbb{C}) = \Gamma\backslash(\mathcal{H} \times \mathbb{C})/\mathbb{Z}^2$ and π is projection on the first factor. The fiber $\pi^{-1}(\Gamma\tau)$ is naturally identified with the elliptic curve $E_\tau = \mathbb{C}/\Lambda_\tau$, where $\Lambda_\tau = \mathbb{Z} \oplus \mathbb{Z}\tau$. (The 'point' P^{univ} is the section of π taking $\Gamma\tau$ to the equivalence class of $(\tau, \frac{1}{N})$.)

Let us stay in the complex setup for a moment and give a different interpretation of the local system $\mathcal{M}_n$. This description looks more complicated, but it paves the way to the ℓ-adic analogue. Start with the constant sheaf $\mathbb{Z} = \underline{\mathbb{Z}}$ on $E^{\text{univ}}(\mathbb{C})$ and consider the higher direct image $R^1\pi_*\mathbb{Z}$. This is a sheaf on $\Gamma\backslash\mathcal{H}$ whose stalk at $y = \Gamma\tau$ is the cohomology of the fiber

$$(R^1\pi_*\mathbb{Z})_y \simeq H^1(E_\tau, \mathbb{Z}) \simeq H_1(E_\tau, \mathbb{Z}) \simeq \Lambda_\tau \simeq \mathbb{Z}^2.$$

So, at least $\mathcal{M}_n$ and $\mathrm{Sym}^n R^1\pi_*\mathbb{Z}$ have isomorphic stalks at all points. One can strengthen this to an isomorphism of sheaves, using that $R^1\pi_*\mathbb{Z}$ is the sheafification of the presheaf $U \mapsto H^1(\pi^{-1}(U), \mathbb{Z})$.

In the ℓ-adic situation there is a way to make sense of $R^1\pi_*\mathbb{Z}_\ell$ as an ℓ-adic sheaf on $Y_{\bar{\mathbb{Q}}}$. The stalk at a point $y \in Y(\bar{\mathbb{Q}})$ is now

$$(R^1\pi_*\mathbb{Z}_\ell)_y \simeq H^1(E_y, \mathbb{Z}_\ell) \simeq \text{dual of } T_\ell(E_y)$$

where $E_y = E^{\text{univ}} \times_{Y,y} \mathrm{Spec}(\bar{\mathbb{Q}})$ is the elliptic curve over y. See [Stack, Theorem 59.53.1]. Vaguely $R^1\pi_*\mathbb{Z}_\ell$ encapsulates all the ℓ-adic Tate modules in the family E^{univ}.

Finally, the correct way to think about the above space "$H^1_{\text{ét},!}(Y_{\bar{\mathbb{Q}}}, \mathcal{M}_{k-2} \otimes \overline{\mathbb{Q}}_\ell)$" is as follows.

Definition 6.41

$$\begin{aligned}\mathcal{V}_\ell &:= H^1_!(Y_{\bar{\mathbb{Q}}}, \mathrm{Sym}^{k-2} R^1\pi_* \overline{\mathbb{Q}}_\ell) \\ &:= \mathrm{im}\big(H^1_c(Y_{\bar{\mathbb{Q}}}, \mathrm{Sym}^{k-2} R^1\pi_* \overline{\mathbb{Q}}_\ell) \longrightarrow H^1(Y_{\bar{\mathbb{Q}}}, \mathrm{Sym}^{k-2} R^1\pi_* \overline{\mathbb{Q}}_\ell)\big).\end{aligned}$$

By comparison theorems for singular and étale cohomology we know $\mathcal{V}_\ell[\ker(\theta_f)]$ is a free $\mathbb{L}_f \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}_\ell$-module of rank two (with f as in Corollary 6.40). This gives a representation

$$\widetilde{r_{f,\ell}} : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{Aut}_{\mathbb{L}_f \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}_\ell\text{-lin}}\big(\mathcal{V}_\ell[\ker(\theta_f)]\big) \xrightarrow{\sim} \mathrm{GL}_2(\mathbb{L}_f \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}_\ell).$$

To show it has the desired properties we need an analogue of Theorem 6.38.

Theorem 6.42 (Deligne's "Formule de congruence") *The $Gal(\bar{\mathbb{Q}}/\mathbb{Q})$-action on $\mathcal{V}_\ell$ is unramified at all primes $p \nmid N\ell$. For any such p we have the relation*

$$T_p = Frob_p + \langle p \rangle p^{k-1} Frob_p^{-1}$$

where both sides are viewed as $\mathbb{T}$-linear endomorphisms of $\mathcal{V}_\ell$. (Here $Frob_p$ denotes a geometric Frobenius element, i.e. the inverse of the 'usual' arithmetic Frobenius.)

Proof See [Del71, Corollary 3.2] for the assertion about $\mathcal{V}_\ell$ being unramified. See [Del71, Proposition 4.8, Theorem 4.9] for the rest of the statement. (Note that Deligne works with level $\Gamma(N)$ instead of $\Gamma_1(N)$ but this makes no difference here.) □

The proof of Theorem 6.42 gives full play to the scheme-theoretic point of view. The argument analyzes Hecke correspondences modulo p, more precisely on the special fiber of a suitable modular curve. Namely, the modular curve $Y(p)$ of level $\Gamma_1(N) \cap \Gamma_0(p)$, which parametrizes triples (E, P, C) where P is an order N point on the elliptic curve E and C is a subgroup of order p. For $N \geq 5$ the corresponding moduli functor is representable by a regular scheme $\mathcal{Y}(p)_{/\mathbb{Z}[\frac{1}{N}]}$. It is smooth over

$\mathbb{Z}[\frac{1}{Np}]$, with irreducible geometric fibers, but it has 'bad reduction' at p: One can similarly define a smooth model $\mathcal{Y}_{/\mathbb{Z}[\frac{1}{N}]}$ of Y, and the (singular) special fiber $\mathcal{Y}(p)_{\mathbb{F}_p}$ can then be described in terms of two copies $\mathcal{Y}_{\mathbb{F}_p} \sqcup \mathcal{Y}_{\mathbb{F}_p}$ mapped into $\mathcal{Y}(p)_{\mathbb{F}_p}$ via Frobenius and Verschiebung. We refer to [DI95, Sects. 8.3–8.5] for a great survey of this construction.

Theorem 6.42 says exactly that the $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-representation $r_{\mathcal{V}_\ell}$ is unramified for $p \nmid N\ell$ and $r_{\mathcal{V}_\ell}(\mathrm{Frob}_p)$ satisfies the equation $x^2 - T_p x + \langle p \rangle p^{k-1} = 0$ in $\mathrm{End}_{\mathbb{T}}(\mathcal{V}_\ell)$. This of course continues to hold after restriction to $\mathcal{V}_\ell[\ker(\theta_f)]$. Therefore $\widetilde{r_{f,\ell}}(\mathrm{Frob}_p)$ satisfies the equation $x^2 - a_p x + \chi(p)p^{k-1} = 0$ in $M_{2\times 2}(\mathbb{L}_f \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}_\ell)$. Our choice of isomorphism $\iota : \overline{\mathbb{Q}}_\ell \xrightarrow{\sim} \mathbb{C}$ singles out a preferred projection $\mathbb{L}_f \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}_\ell \longrightarrow \overline{\mathbb{Q}}_\ell$. Composing $\widetilde{r_{f,\ell}}$ with this projection gives a representation $r_{f,\ell}$ with the properties in Theorem 6.29. Indeed the minimal polynomial of $r_{f,\ell}(\mathrm{Frob}_p)$ divides $x^2 - a_p x + \chi(p)p^{k-1}$ in $\overline{\mathbb{Q}}_\ell[x]$. If the two polynomials are equal we are done (by the CayleyâŁ“Hamilton theorem). On the other hand, if $r_{f,\ell}(\mathrm{Frob}_p)$ is a *scalar* matrix there is an additional argument which makes use of a certain pairing $\mathcal{V}_\ell \otimes \mathcal{V}_\ell \longrightarrow \mathbb{Q}_\ell$, relative to which Frob_p and $\langle p \rangle p^{k-1}\mathrm{Frob}_p^{-1}$ are adjoint. For this last bit we refer the reader to [Del71, p. 171] for the overall idea, at least for $N = 1$. The point is to consider the *square* of the characteristic polynomial of $r_{\mathcal{V}_\ell}(\mathrm{Frob}_p)$. The reader may also want to look into how the scalar case is dealt with when $k = 2$ in [Roh97, pp. 93–94].

As a supplement to [Del71] there are good notes [Con] by Brian Conrad on this subject, but at the time of writing they are undergoing revisions and can be hard to find.

6.4 The Proof of Deligne-Serre's Theorem 6.1

6.4.1 The Deligne-Serre Lifting Lemma

We start with some preliminary commutative algebra which will be pivotal in the proof of Theorem 6.1.

In this section $\mathcal{O}$ can be any discrete valuation ring (DVR). We are not assuming $\mathcal{O}$ is complete. In our applications $\mathcal{O}$ will be something like $\mathbb{Z}_{(\ell)}$ or more generally the localization of the ring of integers in a number field at a prime ideal. Let $K = \mathrm{Frac}(\mathcal{O})$ and $k = \mathcal{O}/\mathfrak{m}$ be the fraction field and the residue field respectively.

Our general setup is a finite free $\mathcal{O}$-module S along with a commutative $\mathcal{O}$-subalgebra $\mathbb{T} \subset \mathrm{End}_{\mathcal{O}}(S)$. Note that $\mathbb{T}$ is necessarily also finite and free over $\mathcal{O}$. Since $\mathbb{T} \otimes_{\mathcal{O}} K$ is therefore Artinian (indeed finite-dimensional) it is a product of its (finitely many) localizations. See [Lan02, Exercise 9, p. 443] for instance. Each localization is a finite-dimensional local K-algebra whose residue field is a finite extension of K. By replacing K with a finite extension we can often arrange for all the residue fields of $\mathbb{T} \otimes_{\mathcal{O}} K$ to be isomorphic to K via the natural map $K \longrightarrow \mathbb{T} \otimes_{\mathcal{O}} K$ composed with projection. We will *assume* we are already in this situation, which makes the lifting lemma easier to state.

Lemma 6.43 (The "Lifting Lemma" [DeS74, Lemma 6.11, p. 522]) *Assume the residue fields of* $\mathbb{T} \otimes_{\mathcal{O}} K$ *are isomorphic to* K. *Then any* $\mathcal{O}$*-algebra homomorphism* $\overline{\theta} : \mathbb{T} \to k$ *admits a lift to an* $\mathcal{O}$*-algebra homomorphism* $\theta : \mathbb{T} \to \mathcal{O}$ *which appears in* S. *In more detail, there is a* θ *such that the diagram*

$$\begin{array}{ccc} & & \mathcal{O} \\ & \overset{\theta}{\nearrow} & \downarrow \text{mod } \mathfrak{m} \\ \mathbb{T} & \xrightarrow{\overline{\theta}} & k \end{array}$$

commutes and the submodule $S[\ker(\theta)] = \{s \in S : Ts = \theta(T)s,\ \forall T \in \mathbb{T}\}$ *is nonzero.*

Proof Note that $\ker(\overline{\theta}) \subset \mathbb{T}$ is a maximal ideal with $\mathcal{O} \cap \ker(\overline{\theta}) = \mathfrak{m}$. The flat ring extension $\mathcal{O} \hookrightarrow \mathbb{T}$ has the going-down property, see [Stack, Lemma 10.39.19], so there is a prime ideal $\mathfrak{p} \subset \mathbb{T}$ such that the chain $\ker(\overline{\theta}) \supset \mathfrak{p}$ lies above $\mathfrak{m} \supset \{0\}$. The injection $\mathcal{O} \hookrightarrow \mathbb{T}/\mathfrak{p}$ is an isomorphism by the assumption on the residue fields of $\mathbb{T} \otimes_{\mathcal{O}} K$. Indeed $\mathbb{T}/\mathfrak{p}$ is finite and therefore integral over $\mathcal{O}$, and it has fraction field

$$\mathrm{Frac}(\mathbb{T}/\mathfrak{p}) = (\mathbb{T}/\mathfrak{p}) \otimes_{\mathcal{O}} K \simeq (\mathbb{T} \otimes_{\mathcal{O}} K)/(\mathfrak{p} \otimes_{\mathcal{O}} K) \simeq K.$$

By virtue of being a DVR, $\mathcal{O}$ is a normal domain. Therefore we have an isomorphism $\mathcal{O} \xrightarrow{\sim} \mathbb{T}/\mathfrak{p}$, whose inverse is given by an $\mathcal{O}$-algebra homomorphism $\theta : \mathbb{T} \to \mathcal{O}$ with $\ker(\theta) = \mathfrak{p}$.

The requirement that $\ker(\overline{\theta}) \supset \mathfrak{p}$ ensures the diagram below commutes.

$$\begin{array}{ccccc} & & \mathbb{T}/\mathfrak{p} & \xleftarrow{\sim} & \mathcal{O} \\ & \nearrow & \downarrow & & \downarrow \text{mod } \mathfrak{m} \\ \mathbb{T} & \longrightarrow & \mathbb{T}/\ker(\overline{\theta}) & \xrightarrow{\sim} & k \end{array}$$

In other words, θ is a lift of $\overline{\theta}$. It remains to show that $S[\mathfrak{p}] \neq 0$. Observe that $\mathbb{T} \otimes_{\mathcal{O}} K$ acts faithfully on $S \otimes_{\mathcal{O}} K$, i.e. $\mathrm{Ann}_{\mathbb{T} \otimes_{\mathcal{O}} K}(S \otimes_{\mathcal{O}} K) = 0$. In particular the support of $S \otimes_{\mathcal{O}} K$ is all of $\mathrm{Spec}(\mathbb{T} \otimes_{\mathcal{O}} K)$. See [Lan02, Exercise 11, p. 444] for example. Hence $(S \otimes_{\mathcal{O}} K)_{\mathfrak{p} \otimes_{\mathcal{O}} K} \neq 0$ as a special case. By the theory of associated primes, for instance the last part of [Lan02, Proposition 2.10, p. 419], $\mathfrak{p} \otimes_{\mathcal{O}} K$ contains a prime of the form $\mathcal{P} = \mathrm{Ann}_{\mathbb{T} \otimes_{\mathcal{O}} K}(s')$ for some $s' \in S \otimes_{\mathcal{O}} K$. Since $\mathbb{T} \otimes_{\mathcal{O}} K$ has Krull dimension zero we must have $\mathfrak{p} \otimes_{\mathcal{O}} K = \mathcal{P}$, so certainly $\mathfrak{p}$ annihilates s'. Replacing s' with a multiple $s := as'$ for $a \in \mathfrak{m}^i$, $i \gg 0$, gives a nonzero element $s \in S$ annihilated by $\mathfrak{p}$. □

Note that Lemma 6.43 lifts eigensystems but *not* eigenforms: If $\overline{\theta}$ happens to appear in $S/\mathfrak{m}S$, i.e. there is a nonzero $\overline{s} \in S/\mathfrak{m}S$ such that $T\overline{s} = \overline{\theta}(T)\overline{s}$ for all $T \in \mathbb{T}$, then the Lemma does *not* guarantee the existence of an $s \in S$ such that $\overline{s} = s + \mathfrak{m}S$. We only get a lift θ of $\overline{\theta}$ such that $Ts = \theta(T)s$ for *some* $s \neq 0$ with no evident relation to $\overline{s}$ in general.

6.4.2 *Rationality and Integrality Properties of Weight One Forms*

The goal in this section is to establish the basic arithmetic properties of the Fourier coefficients of modular forms of weight $k = 1$. In line with the overall strategy of [DeS74] these properties are deduced from their analogues for modular forms of weight $k \geq 2$ (which follow more or less immediately from the Eichler-Shimura isomorphism 6.39).

The fundamental idea is to increase the weight of f, but not the level, by multiplying f by two particular modular forms for $\mathrm{SL}_2(\mathbb{Z})$:

- The normalized holomorphic Eisenstein series from Corollary 5.33 gives an element of $M_4(\mathrm{SL}_2(\mathbb{Z}))$ with integer Fourier coefficients,

$$E_2(z) = 1 + 240 \cdot \sum_{n=1}^{\infty} \sigma_3(n) q^n.$$

- The normalized modular discriminant from Sect. 5.6.2 gives an element of $S_{12}(\mathrm{SL}_2(\mathbb{Z}))$ with integer Fourier coefficients,

$$\widetilde{\Delta}(z) := (2\pi)^{-12} \Delta(z) = \sum_{n=1}^{\infty} \tau(n) q^n = q \prod_{n=1}^{\infty} (1 - q^n)^{24}.$$

From Corollary 5.43 we know that $\widetilde{\Delta}(z) \neq 0$ for all $z \in \mathcal{H}$. Therefore the multiplication map $f \mapsto f\widetilde{\Delta}$ identifies $S_1(\Gamma_0(N), \chi)$ with a subspace of $S_{13}(\Gamma_0(N), \chi)$, but there is no nice intrinsic description of this subspace. Instead, consider the map

$$\begin{aligned} \Phi = \Phi_\chi : S_1(\Gamma_0(N), \chi) &\longrightarrow S_5(\Gamma_0(N), \chi) \oplus S_{13}(\Gamma_0(N), \chi) \\ f &\longmapsto (fE_2, f\widetilde{\Delta}). \end{aligned}$$

Clearly $\ker(\Phi) = \{0\}$ by looking at the second entry. Also, $\mathrm{im}(\Phi)$ is the subspace of pairs of forms (g, h) such that $g\widetilde{\Delta} = hE_2$. The inverse is $(g, h) \mapsto \frac{h}{\widetilde{\Delta}} = \frac{g}{E_2}$. (Note that $\frac{h}{\widetilde{\Delta}}$ is visibly holomorphic on $\mathcal{H}$, and the alternate expression $\frac{g}{E_2}$ shows this function vanishes at ∞ and therefore at all the cusps.)

The map Φ is *not* Hecke-equivariant (in general) but it still allows us to relate the Fourier coefficients of f to those of $g = fE_2$ and $h = f\widetilde{\Delta}$. Since the Fourier coefficients of E_2 and $\widetilde{\Delta}$ are integers, it is easy to see that $\mathbb{L}_f = \mathbb{L}_g = \mathbb{L}_h$. Recall that $\mathbb{L}_f = \mathbb{Q}(a_n : n \geq 1)$ denotes the subfield of $\mathbb{C}$ generated by the Fourier coefficients $(a_n)_{n \geq 1}$ of f. Similarly for g and h. We even have

$$\mathcal{O}_f = \mathcal{O}_g = \mathcal{O}_h$$

where $\mathcal{O}_f = \mathbb{Z}[a_n : n \geq 1]$ and so on. (View E_2 and $\widetilde{\Delta}$ as formal power series in $\mathbb{Z}[\![q]\!]$. Since E_2 and $q^{-1}\widetilde{\Delta}$ both have constant term one, they are invertible in $\mathbb{Z}[\![q]\!]$. This shows the inclusions $\mathcal{O}_f \subset \mathcal{O}_g$ and $\mathcal{O}_f \subset \mathcal{O}_h$. The opposite inclusions are obvious.)

The result below is essentially [DeS74, Proposition 2.7, p. 512], which they prove using the Tate curve. We follow the proof in [Ser77, Sect. 2.5, p. 202] which elaborates on [DeS74, Remark 2.8, p. 512].

Proposition 6.44 *Let $f = \sum_{n=1}^{\infty} a_n q^n \in S_1(\Gamma_0(N), \chi)$ and $\sigma \in Aut(\mathbb{C})$. Then the following holds:*

(1) $f^\sigma := \sum_{n=1}^{\infty} \sigma(a_n) q^n \in S_1(\Gamma_0(N), \chi^\sigma)$.
(2) If $\mathbb{L}_f \subset \bar{\mathbb{Q}}$ there exists an $M \in \mathbb{Z}_{>0}$ such that Ma_n is an algebraic integer for all $n \geq 1$.
(3) If $f \in Prim_1(\Gamma_0(N), \chi)$ then $\mathbb{L}_f$ is a number field and a_n is an algebraic integer for all $n \geq 1$.

Proof The three claims hold for modular forms of weight $k \geq 2$ (by 6.39 basically). We reduce the $k = 1$ case to the known cases using the map Φ.

Claim (1) follows from a direct inspection of the diagram

$$\begin{array}{ccccc} S_1(\Gamma_0(N), \chi) & \xrightarrow{\Phi_\chi} & \mathrm{im}(\Phi_\chi) & \hookrightarrow & S_5(\Gamma_0(N), \chi) \oplus S_{13}(\Gamma_0(N), \chi) \\ \downarrow{\scriptstyle f \mapsto f^\sigma} & & \downarrow & & \downarrow{\scriptstyle (g,h) \mapsto (g^\sigma, h^\sigma)} \\ S_1(\Gamma_0(N), \chi^\sigma) & \xrightarrow{\Phi_{\chi^\sigma}} & \mathrm{im}(\Phi_{\chi^\sigma}) & \hookrightarrow & S_5(\Gamma_0(N), \chi^\sigma) \oplus S_{13}(\Gamma_0(N), \chi^\sigma) \end{array}$$

Note that (g^σ, h^σ) lies in the image of Φ_{χ^σ} when $(g, h) \in \mathrm{im}(\Phi_\chi)$ since $E_2^\sigma = E_2$ and $\widetilde{\Delta}^\sigma = \widetilde{\Delta}$:

$$g^\sigma \widetilde{\Delta} = (g\widetilde{\Delta})^\sigma = (hE_2)^\sigma = h^\sigma E_2.$$

If $(g, h) = \Phi_\chi(f)$ we deduce that $f^\sigma = (\frac{h}{\widetilde{\Delta}})^\sigma = \frac{h^\sigma}{\widetilde{\Delta}} = \frac{g^\sigma}{E_2}$ indeed belongs to $S_1(\Gamma_0(N), \chi^\sigma)$.

Claim (2) is also rather immediate: If $\mathbb{L}_f$ is contained in $\bar{\mathbb{Q}}$, then obviously so is $\mathbb{L}_g = \mathbb{L}_h$ because it is the same field, and we know there is an $M \in \mathbb{Z}_{>0}$ such that $\mathcal{O}_g = \mathcal{O}_h \subset \frac{1}{M}\bar{\mathbb{Z}}$. In other words $\mathcal{O}_f \subset \frac{1}{M}\bar{\mathbb{Z}}$.

For the final assertion (3) the Hecke-stable lattice $\mathcal{L} \subset S_1(\Gamma_1(N))$ in Lemma 6.45 below at least shows the eigenvalues of T_n on $S_1(\Gamma_1(N))$ are algebraic integers. (Representing T_n in a basis for $\mathcal{L}$ shows its characteristic polynomial is a monic polynomial with $\mathbb{Z}$-coefficients.) Thus $a_n \in \bar{\mathbb{Z}}$ in the setting of (3). It remains to show $\mathbb{L}_f$ is a number field. At least we know $\mathbb{L}_f \subset \bar{\mathbb{Q}}$. Here f is an arbitrary primitive form, so $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\chi))$ acts on $\mathrm{Prim}_1(\Gamma_0(N), \chi)$ by (1) where $\mathbb{Q}(\chi)$ is the cyclotomic subfield of $\mathbb{C}$ generated by the χ-values. As $\mathrm{Prim}_1(\Gamma_0(N), \chi)$ is a finite set, this Galois-action factors through a finite quotient $\mathrm{Gal}(\mathbb{L}/\mathbb{Q}(\chi))$. In particular $a_n \in \bar{\mathbb{Q}}^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{L})} = \mathbb{L}$ for all $n \geq 1$, so $\mathbb{L}_f \subset \mathbb{L}$ has finite degree over $\mathbb{Q}$. □

In the previous proof we referred to the Lemma below, which is interesting in its own right.

Lemma 6.45 *Let $\mathcal{L}$ denote the set of $f \in S_1(\Gamma_1(N))$ such that $f|_1\langle d\rangle$ has integer Fourier coefficients for all $d \in \mathbb{Z}$ coprime to N. Then $\mathcal{L}$ is a finite free $\mathbb{Z}$-submodule and*

$$\mathbb{C} \otimes_{\mathbb{Z}} \mathcal{L} \xrightarrow{\sim} S_1(\Gamma_1(N)).$$

Moreover, $\mathcal{L}$ is preserved by all Hecke and diamond operators.

Proof Again, we admit the corresponding result for forms of weight $k \geq 2$. (The reader may find the case $k = 2$ in [DS05, Corollary 6.5.6, p. 238] particularly instructive.) Let $\mathcal{L}_k$ be the lattice of $F \in S_k(\Gamma_1(N))$ for which $F|_k\langle d\rangle$ has Fourier coefficients in $\mathbb{Z}$ for all d as above.

The map Φ extends to the $\Gamma_1(N)$-situation and gives a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & S_1(\Gamma_1(N)) & \xrightarrow{\Phi} & S_5(\Gamma_1(N)) \oplus S_{13}(\Gamma_1(N)) & \longrightarrow & S_{17}(\Gamma_1(N)) \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \mathcal{L}_1 & \xrightarrow{\Phi} & \mathcal{L}_5 \oplus \mathcal{L}_{13} & \xrightarrow{\cdot\binom{\widetilde{\Delta}}{-E_2}} & \mathcal{L}_{17} \end{array}$$

As is easily verified, for $F \in S_k(\Gamma_1(N))$ and $F' \in S_{k'}(\Gamma_1(N'))$ we have the relation

$$(FF')|_{k+k'}y = (F|_k y)(F'|_{k'} y)$$

for all $y \in \mathrm{GL}_2(\mathbb{R})^+$. Therefore the maps in the top row intertwines the diamond operators. (Note that $E_2|_4\langle d\rangle = E_2$ and similarly for $\widetilde{\Delta}$.) This explains the bottom row. For instance, if $f \in \mathcal{L}_1$ then $(fE_2)|_5\langle d\rangle$ has integer Fourier coefficients, so $fE_2 \in \mathcal{L}_5$. The *exactness* of the bottom row follows by observing that E_2 is invertible in $\mathbb{Z}[\![q]\!]$.

Now clearly $\mathcal{L}_1 \hookrightarrow \mathcal{L}_5 \oplus \mathcal{L}_{13}$ is finite and free over $\mathbb{Z}$. Furthermore, by extending scalars to $\mathbb{C}$ (which is flat over $\mathbb{Z}$) in the bottom row we infer the isomorphism $\mathbb{C} \otimes_{\mathbb{Z}} \mathcal{L} \xrightarrow{\sim} S_1(\Gamma_1(N))$ from the corresponding result for weights 5, 13, 17.

Finally $\mathcal{L}_1$ is preserved by T_p (and of course by $\langle d\rangle$ from the very definition of $\mathcal{L}_1$) since T_p commutes with diamond operators (Exercise 5.74) and preserves integrality by the calculation leading up to 5.74. □

Knowing that $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ has Fourier coefficients in the ring of integers $\mathcal{O}_{\mathbb{L}_f}$, it now makes sense to talk about *congruences*. For example, although fE_2 is *not* an eigenform, it is a cusp form in $S_5(\Gamma_0(N), \chi)$ with Fourier coefficients in $\mathcal{O}_{\mathbb{L}_f}$, and modulo $\ell \in \{2, 3, 5\}$ the nth Fourier coefficients of the two forms are congruent for all $n \geq 1$. We express this property by writing

$$fE_2 \equiv f \pmod{\ell}.$$

Indeed $E_2 \equiv 1 \pmod{\ell}$ for all ℓ dividing $240 = 2^4 \cdot 3 \cdot 5$. Playing the same game with the other holomorphic Eisenstein series

$$E_\kappa(z) = 1 - \frac{4\kappa}{B_{2\kappa}} \cdot \sum_{n=1}^{\infty} \sigma_{2\kappa-1}(n) q^n$$

in $M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z}))$ is at the heart of [DeS74]. Here $\kappa \geq 2$. By the Von Staudt-Clausen Theorem 5.34, more specifically Proposition 5.35, we know that $E_\kappa \equiv 1 \pmod{\ell}$ for primes ℓ satisfying $2\kappa \equiv 0 \pmod{\ell - 1}$.

6.4.3 The Mod ℓ Representation Attached to a Weight One Form

As a first approximation to Theorem 6.1 we extend Corollary 6.34 to weight one forms. We fix an embedding $\iota_\ell : \bar{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ once and for all, but we often suppress it from the notation. The representation $\overline{r_{f,\ell}}$ constructed below will depend on the choice of ι_ℓ. For comparison with the original source, the next result is [DeS74, Theorem 6.7, p. 521]. See also [Ser77, Sect. 4.1].

Proposition 6.46 *Let $f \in Prim_1(\Gamma_0(N), \chi)$. For every prime ℓ there is an odd, semisimple, continuous representation*

$$\overline{r} = \overline{r_{f,\ell}} : Gal(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow GL(\overline{V_\ell}) \simeq GL_2(\overline{\mathbb{F}}_\ell)$$

which is unramified at all $p \nmid N\ell$, and for such primes p the following two conditions hold:

$$tr\,\overline{r}(Frob_p) = a_p + \mathfrak{p}_{\overline{\mathbb{Q}}_\ell}, \qquad \det \overline{r}(Frob_p) = \chi(p) + \mathfrak{p}_{\overline{\mathbb{Q}}_\ell}.$$

The (semisimple) representation $\overline{r} = \overline{r_{f,\ell}}$ is uniquely determined by f and ℓ up to isomorphism.

Proof We are given f and ℓ. Start by choosing any $\kappa \geq 2$ such that $2\kappa \equiv 0 \pmod{\ell - 1}$. For example one can take $\kappa = 2(\ell - 1)$ but we prefer not to work with a specific κ. The Eisenstein series $E_\kappa \in M_{2\kappa}(\mathrm{SL}_2(\mathbb{Z}))$ has Fourier coefficients in $\mathbb{Z}_{(\ell)}$ and $E_\kappa \equiv 1 \pmod{\ell}$ by Proposition 5.35. On the other hand f has Fourier coefficients in the ring of integers of the number field $\mathbb{L}_f$.

Remember that $\mathbb{T}_k(\Gamma_1(N))$ denotes the subring of $\mathrm{End}_{\mathbb{C}\text{-lin}}(S_k(\Gamma_1(N)))$ generated by all the operators T_n and $\langle d \rangle$. There is a subtlety below which compels us to work with the subring $\mathbb{T}_k^\ell(\Gamma_1(N))$ generated only by the T_n with $\ell \nmid n$ (and all $\langle d \rangle$ with d coprime to N).

Choose a finite extension $K/\mathbb{L}_f$ such that all the residue fields of $\mathbb{T}^\ell_{2\kappa+1}(\Gamma_1(N)) \otimes_{\mathbb{Z}} K$ are isomorphic to K (via the canonical inclusions). This is needed shortly when we apply Lemma 6.43.

The embedding ι_ℓ defines a place $\lambda | \ell$ of K by restricting the ℓ-adic absolute value on $\overline{\mathbb{Q}}_\ell$ to $K \subset \bar{\mathbb{Q}}$. We let $\mathcal{O}$ denote the corresponding discrete valuation ring in K (i.e., the localization of $\mathcal{O}_K$ at the prime ideal corresponding to λ). It has fraction

field K and residue field $k = \mathcal{O}/\mathfrak{m}$ (a finite extension of $\mathbb{F}_\ell$). Note that $\mathcal{O}$ contains both $\mathcal{O}_{\mathbb{L}_f}$ and $\mathbb{Z}_{(\ell)}$ so it makes sense to compare f and E_κ over $\mathcal{O}$. Expressly fE_κ belongs to $S_{2\kappa+1}(\Gamma_0(N), \chi)$, it has Fourier coefficients in $\mathcal{O}$, and

$$fE_\kappa \equiv f \pmod{\mathfrak{m}}.$$

Note that fE_κ has weight $2\kappa + 1 \geq 2$ in the range where Theorem 6.29 applies. However, there is no reason for fE_κ to be an eigenform in general, so one should not expect to be able to attach an ℓ-adic Galois representation to it. To get around this roadblock we use the Deligne-Serre Lifting Lemma 6.43.

Let $\mathcal{L}_{2\kappa+1}$ be the lattice from the proof of Lemma 6.45 consisting of $F \in S_{2\kappa+1}(\Gamma_1(N))$ for which $F|_k\langle d\rangle$ has Fourier coefficients in $\mathbb{Z}$ for all d as above. In Lemma 6.43 we let

$$S := \mathcal{L}_{2\kappa+1} \otimes_{\mathbb{Z}} \mathcal{O}, \qquad \mathbb{T} := \mathbb{T}^{\ell}_{2\kappa+1}(\Gamma_1(N)) \otimes_{\mathbb{Z}} \mathcal{O},$$

where we identify $\mathbb{T}$ with an $\mathcal{O}$-subalgebra of $\mathrm{End}_{\mathcal{O}}(S)$. The homomorphism $\overline{\theta}$ comes from the Hecke eigensystem $\theta_f : \mathbb{T}^{\ell}_1(\Gamma_1(N)) \to \mathcal{O}$ by reduction modulo $\mathfrak{m}$. More precisely, at least for all $n \geq 1$ coprime to ℓ we have

$$T_n(fE_\kappa) \equiv T_n f = a_n f \pmod{\mathfrak{m}}$$

by Proposition 5.78 (using that $d'^{(2\kappa+1)-1} = d'^{2\kappa} \equiv 1 \pmod{\ell}$ by Fermat and our assumptions on κ; the sum in 5.78 ranges over d' dividing the index n, so $\ell \nmid d'$ as we are assuming n is coprime to ℓ.) Therefore there is a unique $\mathcal{O}$-algebra homomorphism $\overline{\theta} : \mathbb{T} \to k$ such that

$$\overline{\theta}(T_n) = a_n + \mathfrak{m}, \qquad \overline{\theta}(\langle d\rangle) = \chi(d) + \mathfrak{m}$$

for all n coprime to ℓ, and all d coprime to N. Finally we can invoke Lemma 6.43 and get an $\mathcal{O}$-algebra homomorphism $\theta : \mathbb{T} \to \mathcal{O}$ satisfying the following two properties:

- θ is a lift of $\overline{\theta}$, i.e. for all n and d as above we have

$$\theta(T_n) \equiv a_n \pmod{\mathfrak{m}}, \qquad \theta(\langle d\rangle) \equiv \chi(d) \pmod{\mathfrak{m}}.$$

- θ appears in S, i.e. there exists a nonzero $g \in S$ such that, for all n and d as above,

$$T_n g = \theta(T_n) g, \qquad g|_{2\kappa+1}\langle d\rangle = \theta(\langle d\rangle) g.$$

Since g is an eigenform of weight $2\kappa + 1 \geq 2$, Theorem 6.29 gives a Galois representation $r_{g,\ell}$ such that

$$\det(1 - r_{g,\ell}(\mathrm{Frob}_p)X) = 1 - \theta(T_p)X + \theta(\langle p\rangle)p^{2\kappa}X^2$$

for $p \nmid N\ell$. The reduction mod ℓ, as in Corollary 6.34, gives an odd and semisimple representation $\overline{r_{g,\ell}}$ over $\overline{\mathbb{F}}_\ell$ with all the desired properties (remember that $p^{2\kappa} \equiv 1$ mod ℓ when $p \neq \ell$). So $\overline{r_{f,\ell}} \simeq \overline{r_{g,\ell}}$. □

There is a slight improvement of Proposition 6.46 (and Corollary 6.34) which gives some control over the image of $\overline{r_{f,\ell}}$. Note that the image is finite, and therefore a subgroup of $\mathrm{GL}_2(\mathbb{F}_{\ell^t})$ for some t, since $\mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ is discrete. In fact we can take $\mathbb{F}_{\ell^t}$ to be the residue field $k_{f,\lambda}$ of $\mathbb{L}_{f,\lambda}$, where $\lambda|\ell$ now denotes the place of $\mathbb{L}_f$ defined by the embedding $\iota_\ell|_{\mathbb{L}_f}$. The fact that $\overline{r_{f,\ell}}$ is realizable over $k_{f,\lambda}$ is a consequence of the result that finite fields have Schur index one, as we now explain.

We briefly recall the notion of *Schur index*: Let G be a finite group and take any field k with algebraic closure $\bar{k}$. Let χ_V be the character of a simple $\bar{k}[G]$-module V and introduce the field $k(V) = k(\chi_V(g) : g \in G)$. The Schur index $m_k(V)$ is the minimum of all $m \geq 1$ for which there exists a degree m extension $k'/k(V)$ such that V descends to k', i.e. there exists a $k'[G]$-module V' such that $V \simeq V' \otimes_{k'} \bar{k}$. We refer to [Hup98, Sect. 38] and [Isa94, Chap. 10] for the basics on Schur indices in general.

A famous example of a V with Schur index > 1 occurs for the quaternion group $G = Q_8$ and $k = \mathbb{R}$. Viewing the quaternions as an $\mathbb{R}$-subalgebra $\mathbb{H} \hookrightarrow M_{2\times 2}(\mathbb{C})$ via

$$a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} \longmapsto \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$$

gives an irreducible representation $Q_8 \hookrightarrow \mathrm{GL}_2(\mathbb{C})$ which cannot be realized over $\mathbb{R}$, despite its character clearly being $\mathbb{R}$-valued; $\chi_V(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = 2a$. Therefore this V has $m_{\mathbb{R}}(V) = 2$.

Exercise 6.47 Check that the representation $Q_8 \hookrightarrow \mathrm{GL}_2(\mathbb{C})$ is *not* realizable over $\mathbb{R}$. (`Hint`: Otherwise there would be matrices in $\mathrm{GL}_2(\mathbb{R})$ satisfying the relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$. This would give an embedding $\mathbb{H} \hookrightarrow M_{2\times 2}(\mathbb{R})$ which is necessarily an isomorphism.) Why is it irreducible?

A key result of Brauer states that $m_k(V) = 1$ when $\mathrm{char}(k) > 0$. See [Isa94, Theorems 9.14, and 9.21] and the original [Bra29, B, p. 101]. This essentially boils down to Wedderburn's theorem that a division algebra over a finite field is commutative, in other words that $\mathrm{Br}(k)$ is trivial for finite fields k. By comparison, in the above Q_8-example the base field has Brauer group $\mathrm{Br}(\mathbb{R}) = \frac{1}{2}\mathbb{Z}/\mathbb{Z}$.

We apply this to $\overline{r_{f,\ell}}$, which factors through a finite group G, and $k = k_{f,\lambda}$. Clearly $\mathrm{tr}\,\overline{r_{f,\ell}}(\mathrm{Frob}_p)$ lies in $k_{f,\lambda}$ for all $p \nmid N\ell$, so the character is indeed $k_{f,\lambda}$-valued by the Chebotarev density theorem. If $\overline{r_{f,\ell}}$ is irreducible it therefore descends to $k_{f,\lambda}$. When $\overline{r_{f,\ell}}$ breaks up as a direct sum of two characters there is still a way to run this argument, now using the semisimple version [DeS74, Lemma 6.13, p. 523] of Brauer's result.

Here is how we will use this strengthening: By Corollary 1.93 the set $\mathrm{Spl}_{\mathbb{L}_f/\mathbb{Q}}$ has density $\frac{1}{[\mathbb{L}_f:\mathbb{Q}]}$. In particular $k_{f,\lambda} = \mathbb{F}_\ell$ for a positive density of primes ℓ, and for such ℓ we can therefore conjugate the representation $\overline{r_{f,\ell}}$ into $\mathrm{GL}_2(\mathbb{F}_\ell)$. The next

major step is to show the cardinality of the image $|\mathrm{im}(\overline{r_{f,\ell}})|$ is uniformly bounded for $\ell \in \mathrm{Spl}_{\mathbb{L}_f/\mathbb{Q}}$.

6.4.4 Some Group-Theoretic Technicalities Pertaining to $GL_2(\mathbb{F}_\ell)$

In [DeS74, Sect. 7.1] the authors introduce the following technical condition on a subgroup $G \leq \mathrm{GL}_2(F)$. Here F can be any finite field, for now, but eventually we will take $F = \mathbb{F}_\ell$.

Definition 6.48 For $\eta \in (0, 1)$ and $M \geq 0$ we say that G satisfies condition $C(\eta, M)$ if there exists a *subset* $H \subseteq G$ such that

(1) $|H| \geq (1 - \eta)|G|$ and
(2) $|\{\det(I - hT) : h \in H\}| \leq M$.

Roughly this says the set H should be large relative to $1 - \eta$, which we think of as being close to one, and give at most M distinct characteristic polynomials in $F[T]$.

The key result here is [DeS74, Proposition 7.2, p. 524] which we paraphrase below.

Proposition 6.49 *For $\eta \in (0, \frac{1}{2})$ and $M \geq 0$ there exists a constant $A(\eta, M) > 0$, which is independent of ℓ, such that any semisimple subgroup $G \leq GL_2(\mathbb{F}_\ell)$ satisfying condition $C(\eta, M)$ has cardinality*

$$|G| \leq A(\eta, M).$$

(By G being 'semisimple' we mean the representation $G \hookrightarrow GL_2(\overline{\mathbb{F}}_\ell)$ is semisimple.)

The proof of this proceeds on a case-by-case basis, using the classification of the semisimple subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$. We will only prove a representative case by assuming $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$.

Proof (Partial.) Assume $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$ satisfies $C(\eta, M)$. We rely on the description of all the conjugacy classes in $\mathrm{GL}_2(\mathbb{F}_\ell)$, which can be found in [Lan02, p. 715] for example. We start by reviewing this.

First, the split Cartan subgroup is $T^s = \{\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right) : a, b \in \mathbb{F}_\ell^\times\}$. A non-split Cartan subgroup arises as follows: An $\mathbb{F}_\ell$-basis $\{1, \beta\}$ for the quadratic extension $\mathbb{F}_{\ell^2}$ gives an embedding $\mathbb{F}_{\ell^2}^\times \hookrightarrow \mathrm{GL}_2(\mathbb{F}_\ell)$ mapping $x + y\beta \mapsto \left(\begin{smallmatrix} x & -vy \\ y & x - uy \end{smallmatrix}\right)$ if $\beta^2 + u\beta + v = 0$. (This is the matrix representation of the multiplication by $x + y\beta$ map on $\mathbb{F}_{\ell^2}$ with respect to the basis $\{1, \beta\}$.) This gives a subgroup $T^{ns} = \mathrm{im}(\mathbb{F}_{\ell^2}^\times \hookrightarrow \mathrm{GL}_2(\mathbb{F}_\ell))$ which is well-defined up to conjugation.

Now let $g \in GL_2(\mathbb{F}_\ell)$ and look at the polynomial $\det(I - gT) \in \mathbb{F}_\ell[T]$. This falls into three categories.

- $\det(I - gT) = (1 - aT)(1 - dT)$ for distinct $a, d \in \mathbb{F}_\ell$.
- $\det(I - gT) = (1 - aT)^2$ for some $a \in \mathbb{F}_\ell$.
- $\det(I - gT) = (1 - \beta T)(1 - \bar{\beta} T)$ for some $\beta \in \mathbb{F}_{\ell^2} - \mathbb{F}_\ell$.

The first case happens precisely when g is conjugate to $\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right)$. In the second case g is either equal to $\left(\begin{smallmatrix} a & 0 \\ 0 & a \end{smallmatrix}\right)$ or conjugate to the Jordan normal form $\left(\begin{smallmatrix} a & 1 \\ 0 & a \end{smallmatrix}\right)$. In the third case g lies in a non-split Cartan subgroup.

The table below summarizes Table 12.4 in [Lan02, p. 715].

	The number of classes	The number of elements in each class
$\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right)$ with $a \neq d$ in $\mathbb{F}_\ell^\times$	$\frac{1}{2}(\ell - 1)(\ell - 2)$	$\ell^2 + \ell$
$\left(\begin{smallmatrix} a & 0 \\ 0 & a \end{smallmatrix}\right)$ with $a \in \mathbb{F}_\ell^\times$	$\ell - 1$	1
$\left(\begin{smallmatrix} a & 1 \\ 0 & a \end{smallmatrix}\right)$ with $a \in \mathbb{F}_\ell^\times$	$\ell - 1$	$\ell^2 - 1$
Non-split Cartan	$\frac{1}{2}\ell(\ell - 1)$	$\ell^2 - \ell$

We will only need the data in the rigtht-most column, which come from brute force calculations of the centralizers. For example, in the first row $\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right)$ has centralizer T^s of size $(\ell - 1)^2$. By counting bases for $\mathbb{F}_\ell^2$ we have

$$|\mathrm{GL}_2(\mathbb{F}_\ell)| = (\ell^2 - 1)(\ell^2 - \ell) = \ell(\ell + 1)(\ell - 1)^2$$

so the conjugacy class of $\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right)$ has $[\mathrm{GL}_2(\mathbb{F}_\ell) : T^s] = \ell(\ell + 1)$ elements. The last two rows require more effort. We invite the reader to either fill in the details, as an exercise, or look them up in [Lan02, p. 715].

Consider the map

$$\begin{aligned} \mathrm{char} : \mathrm{GL}_2(\mathbb{F}_\ell) &\longrightarrow \{1 + AT + BT^2 : A, B \in \mathbb{F}_\ell\} \\ g &\longmapsto \det(I - gT). \end{aligned}$$

The inverse image $\mathrm{char}^{-1}(1 + AT + BT^2)$ has cardinality $\ell^2 + \ell$, ℓ^2, or $\ell^2 - \ell$, according to whether the polynomial $1 + AT + BT^2$ has 2, 1, or 0 roots in $\mathbb{F}_\ell$. In particular this cardinality is always at most $\ell^2 + \ell$.

Let $H \subseteq G$ be a subset satisfying (1) and (2) in Definition 6.48. Also, let $r := [G : \mathrm{SL}_2(\mathbb{F}_\ell)]$ so that $|G| = r\ell(\ell^2 - 1)$. Then

$$|H| = \sum\nolimits_{A, B \in \mathbb{F}_\ell} |H \cap \mathrm{char}^{-1}(1 + AT + BT^2)| \leq M(\ell^2 + \ell)$$

since the sum has at most M nonzero terms by (2). On the other hand

$$(1 - \eta) r\ell(\ell^2 - 1) = (1 - \eta)|G| \leq |H|$$

by (1). Combining these inequalities gives the bound $\ell \leq 1 + \frac{M}{(1-\eta)r} \leq 1 + \frac{M}{1-\eta}$ on the prime ℓ, and consequently an upper bound on the size of the group, i.e.

$$|G| \leq |\mathrm{GL}_2(\mathbb{F}_\ell)| = (\ell^2 - 1)(\ell^2 - \ell) < \ell^4 \leq \left(1 + \frac{M}{1-\eta}\right)^4 =: A(\eta, M)$$

as desired. □

The subgroups of $\mathrm{GL}_2(\mathbb{F}_\ell)$ were classified in [Ser72, Sect. 2]. The semisimple ones are categorized as follows:

- $G \supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$;
- $G \subseteq T$ for a Cartan subgroup T (split or non-split);
- $G \subseteq N(T)$ (the normalizer of a Cartan subgroup T) but $G \nsubseteq T$;
- The image of G in $\mathrm{PGL}_2(\mathbb{F}_\ell)$ is isomorphic to A_4, S_4, or A_5.

See [Ser72, Propositions 15, 16]. For example $\mathrm{PGL}_2(\mathbb{F}_3) \simeq S_4$ and $\mathrm{PSL}_2(\mathbb{F}_5) \simeq A_5$. In the bottom three cases, where $G \not\supseteq \mathrm{SL}_2(\mathbb{F}_\ell)$, the proof of Proposition 6.49 is easier but a bit ad hoc. The calculations are in [DeS74, p. 524]. In the third 'normalizer' case one needs the assumption $\eta < \frac{1}{2}$ in order to reduce to the second 'Cartan' case (which is an easy exercise for the reader).

6.4.5 *Uniform Boundedness of the Image Mod ℓ*

We fix an $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ and consider the semisimple representation $\overline{r_{f,\ell}} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$ from Proposition 6.46. We noted that $\overline{r_{f,\ell}}$ can be conjugated into $\mathrm{GL}_2(\mathbb{F}_\ell)$ when $\ell \in \mathrm{Spl}_{\mathbb{L}_f/\mathbb{Q}}$. For such ℓ we let $G_\ell := \mathrm{im}(\overline{r_{f,\ell}})$ be the image. This is a semisimple subgroup of $\mathrm{GL}_2(\mathbb{F}_\ell)$, in the sense of the previous section, determined up to conjugation.

Proposition 6.50 *There exists a constant $A > 0$ such that $|G_\ell| \leq A$ for all $\ell \in Spl_{\mathbb{L}_f/\mathbb{Q}}$.*

This follows immediately from the next Lemma combined with Proposition 6.49. (Take $\eta = \frac{1}{4}$ say and let $A = A(\frac{1}{4}, M_{\frac{1}{4}})$ in the notation to be introduced below.)

Lemma 6.51 *For every $\eta \in (0, 1)$ there exists an $M_\eta \geq 0$ such that G_ℓ satisfies condition $C(\eta, M_\eta)$ for all $\ell \in Spl_{\mathbb{L}_f/\mathbb{Q}}$.*

Proof Select sets X_η and Y_η as in Lemma 6.52 below. Thus X_η has upper density $\leq \eta$ and $\{a_p : p \notin X_\eta\}$ is a finite subset (of $Y_\eta \subset \mathbb{C}$). Let $M_\eta = |\mathcal{M}_\eta|$ where

$$\mathcal{M}_\eta := \{1 - a_p T + \chi(p)T^2 : p \notin X_\eta\}.$$

We verify that G_ℓ satisfies $C(\eta, M_\eta)$ for all $\ell \in \mathrm{Spl}_{\mathbb{L}_f/\mathbb{Q}}$. More precisely, the subset $H_\ell \subset G_\ell$ formed by all the conjugacy classes $\overline{r_{f,\ell}}(\mathrm{Frob}_p)$ with $p \notin X_\eta$ (and $p \nmid N\ell$) satisfies (1) and (2) in Definition 6.48.

To see this, apply the Chebotarev density theorem 4.57 to the Galois extension $E/\mathbb{Q}$ with Galois group $\mathrm{Gal}(E/\mathbb{Q}) \simeq G_\ell$, in other words the fixed field of $\ker(\overline{r_{f,\ell}})$. As $s \to 1^+$ this yields

$$1 - \frac{\sum'_{p\in X_\eta} p^{-s}}{\ln(\frac{1}{s-1})} + o(1) = \frac{\sum'_{p\notin X_\eta} p^{-s}}{\ln(\frac{1}{s-1})} \leq \frac{\sum'_{p:\bar{r}(\mathrm{Frob}_p)\in H_\ell} p^{-s}}{\ln(\frac{1}{s-1})} = \frac{|H_\ell|}{|G_\ell|} + o(1)$$

where the mark in $\sum'$ means we are only summing over primes not dividing $N\ell$. Therefore $\eta \geq 1 - \frac{|H_\ell|}{|G_\ell|}$, which is exactly property (1). Property (2) is simpler: Any element $h \in H_\ell$ is in the class $\overline{r_{f,\ell}}(\mathrm{Frob}_p)$ for some $p \notin X_\eta$, and therefore $\det(I - hT)$ is the reduction mod $\mathfrak{p}_{\overline{\mathbb{Q}}_\ell}$ of one of the M_η many polynomials in the set $\mathcal{M}_\eta$. □

The proof of the previous Lemma made use of yet another application of the Rankin-Selberg method:

Lemma 6.52 *For each $\eta > 0$ there exists a set of prime numbers X_η, and a* **finite** *set of complex numbers Y_η, satisfying the following two properties:*

(1) X_η has 'upper analytic density' at most η, i.e.

$$\limsup_{s\to 1^+}\Big(\sum_{p\in X_\eta} p^{-s}\Big)/\ln\Big(\frac{1}{s-1}\Big) := \inf_{\epsilon>0} \sup_{1<s<1+\epsilon}\Big(\sum_{p\in X_\eta} p^{-s}\Big)/\ln\Big(\frac{1}{s-1}\Big) \leq \eta.$$

(2) $X_\eta \supseteq \{p : a_p \notin Y_\eta\}$.

(Here a_p is the pth Fourier coefficient of the fixed primitive weight one form f.)

Proof Recall that the Fourier coefficients of f all belong to $\mathcal{O}_{\mathbb{L}_f}$. For $c \geq 0$ we introduce the two sets

- $Y(c) = \{a \in \mathcal{O}_{\mathbb{L}_f} : |\sigma(a)|^2 \leq c,\ \forall \sigma : \mathbb{L}_f \hookrightarrow \mathbb{C}\}$;
- $X(c) = \{p \nmid N : a_p \notin Y(c)\}$.

Note that $Y(c) \subset \mathbb{C}$ is finite by Minkowski theory (there are only finitely many lattice points in a ball). Given an $\eta > 0$ it suffices to show that $X(c)$ has upper density $\leq \eta$ for $c \gg 0$. From Proposition 6.44 we know that $f^\sigma := \sum_{n=1}^\infty \sigma(a_n)q^n \in S_1(\Gamma_0(N), \chi^\sigma)$. Applying Proposition 6.8 (a byproduct of the Rankin-Selberg method) to f^σ we find that

$$\sum_{p\nmid N} |\sigma(a_p)|^2 p^{-s} \leq \ln\Big(\frac{1}{s-1}\Big) + O(1)$$

as $s \to 1^+$. Summing over all the embeddings $\sigma : \mathbb{L}_f \hookrightarrow \mathbb{C}$ we thus arrive at the inequality

$$\sum_{p\nmid N}\sum_{\sigma}|\sigma(a_p)|^2p^{-s}\le[\mathbb{L}_f:\mathbb{Q}]\cdot\ln\Big(\frac{1}{s-1}\Big)+O(1)$$

as $s\to 1^+$. If $p\in X(c)$ there exists a σ for which $|\sigma(a_p)|^2>c$, as $a_p\notin Y(c)$. Therefore, for $s>1$,

$$\sum_{p\nmid N}\sum_{\sigma}|\sigma(a_p)|^2p^{-s}\ge\sum_{p\in X(c)}\Big(\sum_{\sigma}|\sigma(a_p)|^2\Big)p^{-s}>c\sum_{p\in X(c)}p^{-s}$$

$$\Longrightarrow\Big(\sum_{p\in X(c)}p^{-s}\Big)/\ln(\frac{1}{s-1})\le\frac{1}{c}[\mathbb{L}_f:\mathbb{Q}]+o(1),$$

from which we infer that $X(c)$ has upper density $\le\frac{1}{c}[\mathbb{L}_f:\mathbb{Q}]\le\eta$ for $c\ge\frac{1}{\eta}[\mathbb{L}_f:\mathbb{Q}]$. In conclusion we may take X_η to be $X(c)$, and Y_η to be $Y(c)$, for large enough c (and including the primes dividing N in X_η if necessary). □

6.4.6 *Lifting Mod ℓ Representations to Characteristic Zero*

In this section we briefly discuss a group-theoretic analogue of the "Lifting Lemma" 6.43. We assume $\mathcal{O}$ is a *complete* DVR, unlike in Lemma 6.43 which holds without the completeness assumption, and adopt the notation $K=\mathrm{Frac}(\mathcal{O})$ and $k=\mathcal{O}/\mathfrak{m}$. We work under the assumptions that $\mathrm{char}(K)=0$ and $\mathrm{char}(k)=\ell>0$.

The goal is to explain how $k[G]$-modules can be lifted canonically to $\mathcal{O}[G]$-modules. Here G is a finite group of order $|G|$ prime to ℓ. Moreover (by analogy with 6.43) we suppose K is sufficiently large in the sense of [Ser96, p. 115], i.e. K contains all the mth roots of unity where m is the least common multiple of the orders of elements from G. This condition guarantees that any simple $K[G]$-module is *absolutely* simple, see [Ser96, Theorem 24, p. 94] (and [Ser96, p. 121] where this fact is mentioned explicitly).

An $\mathcal{O}[G]$-module M can be 'reduced mod $\mathfrak{m}$' giving the $k[G]$-module $M/\mathfrak{m}M$. Conversely, a $k[G]$-module N inflates to an $\mathcal{O}[G]$-module, via the projection $\mathcal{O}[G]\twoheadrightarrow k[G]$, and at least if N is finitely generated we can pass to a projective cover $c:\widetilde{N}\twoheadrightarrow N$. (This means $\widetilde{N}$ is a projective $\mathcal{O}[G]$-module and c is essential, i.e. c is surjective and $c(S)\neq N$ for all submodules $S\subsetneq\widetilde{N}$.) A projective cover exists by [Sch13, Proposition 1.6.10, p. 32] for instance (and it is unique up to isomorphism); there is even a choice of $c:\widetilde{N}\twoheadrightarrow N$ reducing to an isomorphism $\widetilde{N}/\mathfrak{m}\widetilde{N}\xrightarrow{\sim}N/\mathfrak{m}N=N$. See also [Ser96, Sect. 14.3] where projective covers are called projective envelopes.

Here is the analogue of Lemma 6.43 for G-representations.

Proposition 6.53 *Let G be a finite group such that $\ell \nmid |G|$. Assume K is sufficiently large. Then the above maps $[M] \mapsto [M/\mathfrak{m}M]$ and $[N] \mapsto [\widetilde{N}]$ define mutually inverse bijections*

$$\{\textit{isomorphism classes of simple} \mathcal{O}[G]\textit{-modules}\} \longleftrightarrow \{\textit{isomorphism classes of simple } k[G]\textit{-modules}\}.$$

In particular, any representation $\overline{\rho} : G \to GL_n(k)$ admits a lift to a representation $\rho : G \to GL_n(\mathcal{O})$. Moreover, the lift ρ is uniquely determined by $\overline{\rho}$ up to equivalence.

Proof This is a special case of [Ser96, Theorem 38, p. 135] (attributed to Fong-Swan). Alternatively this follows from the bottom diagram in [Sch13, p. 45] which in turn is a consequence of the more general result [Sch13, Proposition 1.7.4, p. 39]. Note that the residue field k is assumed to be algebraically closed in [Sch13, Chap. 2], but this can be relaxed as we assume K is sufficiently large. The one-to-one correspondence extends to all (possibly reducible) representations since $K[G]$ and $k[G]$ are semisimple by Maschke's Theorem (see [Ser96, Exercise 6.1, p. 48]). □

There are far-reaching generalizations of Proposition 6.53, extremely important in modern arithmetic, which give generators and relations for the universal deformation ring of $\overline{\rho} : G \to \mathrm{GL}_n(k)$. We recommend [Gee22] as a user-oriented introduction to deformation theory. Here is the basic setup. First, $K/\mathbb{Q}_\ell$ is a finite extension, and G is allowed to be any profinite group (satisfying Mazur's condition Φ_ℓ). Typically G will be an infinite Galois group. When $\overline{\rho}$ is absolutely irreducible, or at least when $\mathrm{End}_{k[G]}(\overline{\rho}) = k$, there is a *universal* deformation $\rho^{\mathrm{univ}} : G \to \mathrm{GL}_n(R_{\overline{\rho}}^{\mathrm{univ}})$. (See [Gee22, Definition 3.4, p. 86] for the definition of a 'deformation'–which is an equivalence class of lifts.) Thus *any* deformation $\rho : G \to \mathrm{GL}_n(A)$ arises from ρ^{univ} by composition with a corresponding homomorphism $R_{\overline{\rho}}^{\mathrm{univ}} \to A$.

The universal deformation ring $R_{\overline{\rho}}^{\mathrm{univ}}$ is a complete local Noetherian $\mathcal{O}$-algebra with residue field k, and its structure is governed by the cohomology groups $H^i(G, \mathrm{ad}\,\overline{\rho})$ with $i = 1, 2$. Here $\mathrm{ad}\,\overline{\rho}$ denotes the adjoint representation of $\overline{\rho}$, which means we let G act on the space of matrices $M_{n\times n}(k)$ by conjugation. As explained in [Gee22, Sect. 3.10] there is a surjection $\phi : \mathcal{O}[\![x_1, \ldots, x_d]\!] \twoheadrightarrow R_{\overline{\rho}}^{\mathrm{univ}}$, where d is the dimension of the first cohomology group, $d := \dim_k H^1(G, \mathrm{ad}\,\overline{\rho})$. Furthermore [Gee22, Lemma 3.13, p. 87] makes precise how $\ker(\phi)$ is related to the second cohomology group $H^2(G, \mathrm{ad}\,\overline{\rho})$ through an injection

$$\mathrm{Hom}_k\big(\ker(\phi) \otimes_{\mathcal{O}[\![x_1,\ldots,x_d]\!]} k, k\big) \hookrightarrow H^2(G, \mathrm{ad}\,\overline{\rho}).$$

When $H^2(G, \mathrm{ad}\,\overline{\rho}) = 0$ we find that ϕ is an isomorphism $\mathcal{O}[\![x_1, \ldots, x_d]\!] \xrightarrow{\sim} R_{\overline{\rho}}^{\mathrm{univ}}$ (by Nakayama's Lemma) and there is no obstruction to deforming $\overline{\rho}$ to any A by sending the d variables x_i to random values in $\mathfrak{m}_A$. If in addition $H^1(G, \mathrm{ad}\,\overline{\rho}) = 0$ this shows $R_{\overline{\rho}}^{\mathrm{univ}} \simeq \mathcal{O}$, in other words the deformation of $\overline{\rho}$ to A is unique. In particular $\overline{\rho}$ uniquely deforms to $\mathcal{O}$ in this case.

If G is a finite group, restriction-corestriction shows $|G|H^i(G, W) = 0$ for all $i > 0$, where W is any $\mathbb{Z}[G]$-module. If W is an $\mathbb{F}_\ell$-vector space, $\ell H^i(G, W) = 0$

as well. Under the assumption $\ell \nmid |G|$ the cohomology groups $H^i(G, W)$ therefore vanish for $i > 0$. Applying this to $W = \mathrm{ad}\,\overline{\rho}$ gives an alternate proof of Proposition 6.53.

Here is how we will use Proposition 6.53 in the situation of Proposition 6.50, which gives the uniform bound $|G_\ell| \leq A$ for all $\ell \in \mathrm{Spl}_{\mathbb{L}_f/\mathbb{Q}}$. First of all, by only considering primes in the range $\ell > A$ we ensure that $\ell \nmid |G_\ell|$. Secondly, by assuming ℓ splits completely in $\mathbb{L}_f(\zeta_n : n \leq A)$ all its completions are sufficiently large (with respect to G_ℓ). There are infinitely many primes satisfying both conditions, and for such ℓ the representation $\overline{r_{f,\ell}}$ admits a lift $r_{f,\ell}^{\mathrm{univ}}$ as in the (commutative) diagram

$$\begin{array}{ccc} & & \mathrm{GL}_2(\mathcal{O}) \\ & \overset{r_{f,\ell}^{\mathrm{univ}}}{\nearrow} & \downarrow \mathrm{mod}\ \mathfrak{m} \\ \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) & \xrightarrow{\overline{r_{f,\ell}}} & \mathrm{GL}_2(\mathbb{F}_\ell) \end{array}$$

In this diagram $\mathcal{O}$ denotes the valuation ring in the completion K of $\mathbb{L}_f(\zeta_n : n \leq A)$ determined by ι_ℓ. We emphasize that $r_{f,\ell}^{\mathrm{univ}}$ is unique up to equivalence. Also, since $r_{f,\ell}^{\mathrm{univ}}$ factors through a representation of G_ℓ by definition, it is clear that $\mathrm{im}(r_{f,\ell}^{\mathrm{univ}}) \simeq G_\ell$. Essentially for the same reason $r_{f,\ell}^{\mathrm{univ}}$ is unramified for $p \nmid N\ell$, since $\overline{r_{f,\ell}}$ is, and for such p the eigenvalues of $r_{f,\ell}^{\mathrm{univ}}(\mathrm{Frob}_p)$ are roots of unity in K of order $\leq A$. Moreover, its characteristic polynomial satisfies the congruence

$$\det(I - r_{f,\ell}^{\mathrm{univ}}(\mathrm{Frob}_p)T) \equiv 1 - a_p T + \chi(p)T^2$$

modulo $\mathfrak{m}$. The next item of business is to promote this congruence to an equality.

6.4.7 *End of the Proof of Theorem 6.1*

We introduce the number field $\mathbb{E}_f := \mathbb{L}_f(\zeta_n : n \leq A)$ which only depends on $f \in \mathrm{Prim}_1(\Gamma_0(N), \chi)$ (and a choice of uniform bound A). We then consider the *finite* subset of polynomials

$$\mathcal{Y} := \{(1 - \alpha T)(1 - \beta T) : \alpha, \beta \in \mathbb{E}_f^\times \text{ roots of unity of order } \leq A\} \subset \mathcal{O}_{\mathbb{E}_f}[T].$$

For now we let ℓ denote any prime. The chosen embedding $\iota_\ell : \bar{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ determines a place of $\mathbb{E}_f$ above ℓ and we let K denote the corresponding completion; $\mathcal{O}$ is the valuation ring of K and $k = \mathcal{O}/\mathfrak{m}$ is the residue field.

The reduction map $\mathcal{O} \twoheadrightarrow k$ restricts to an isomorphism $\mu_n(\mathcal{O}) \xrightarrow{\sim} \mu_n(k)$ provided $\ell \nmid n$ (by Hensel's Lemma). Therefore the reduction-of-the-coefficients map $\mathcal{O}[T] \twoheadrightarrow k[T]$ restricts to an *injection*

$$\mathcal{Y} \hookrightarrow k[T]$$

if we assume $\ell > A$.

At the end of the previous section we introduced the representation $r_{f,\ell}^{\text{univ}}$ when $\ell \in \text{Spl}_{\mathbb{E}_f/\mathbb{Q}}^{\geq A}$. We found that for $p \nmid N\ell$ the following holds:

- $\det(I - r_{f,\ell}^{\text{univ}}(\text{Frob}_p)T)$ belongs to $\mathcal{Y}$;
- $\det(I - r_{f,\ell}^{\text{univ}}(\text{Frob}_p)T) \equiv 1 - a_p T + \chi(p)T^2 \pmod{\mathfrak{m}}$.

(The eigenvalues of $r_{f,\ell}^{\text{univ}}(\text{Frob}_p)$ have order $\leq A$ so they lie in $\mathcal{O}_{\mathbb{E}_f} \subset \mathcal{O}$.)

We turn the tables by now *fixing* a prime $p \nmid N$ and varying the prime $\ell \in \text{Spl}_{\mathbb{E}_f/\mathbb{Q}}^{\geq A} - \{p\}$. Look at the map

$$\begin{aligned} \text{Spl}_{\mathbb{E}_f/\mathbb{Q}}^{\geq A} - \{p\} &\longrightarrow \mathcal{Y} \\ \ell &\longmapsto \det(I - r_{f,\ell}^{\text{univ}}(\text{Frob}_p)T). \end{aligned}$$

As the source is infinite and $|\mathcal{Y}| < \infty$ there must be a polynomial $(1 - \alpha_p T)(1 - \beta_p T)$ in $\mathcal{Y}$ which is hit by infinitely many primes ℓ. For those ℓ,

$$(1 - \alpha_p T)(1 - \beta_p T) = \det(I - r_{f,\ell}^{\text{univ}}(\text{Frob}_p)T) \equiv 1 - a_p T + \chi(p)T^2 \pmod{\mathfrak{m}}.$$

These infinitely many congruences can only be satisfied if have the equality

$$(1 - \alpha_p T)(1 - \beta_p T) = 1 - a_p T + \chi(p)T^2.$$

In particular the Hecke polynomial $1 - a_p T + \chi(p)T^2$ of f must be an element of $\mathcal{Y}$. Knowing this, the injectivity of $\mathcal{Y} \hookrightarrow \mathbb{F}_\ell[T]$ allows us to conclude that

$$\det(I - r_{f,\ell}^{\text{univ}}(\text{Frob}_p)T) = 1 - a_p T + \chi(p)T^2 \tag{6.5}$$

for **all** $\ell \in \text{Spl}_{\mathbb{E}_f/\mathbb{Q}}^{\geq A} - \{p\}$, using the two bullet points above.

To summarize, for every $\ell \in \text{Spl}_{\mathbb{E}_f/\mathbb{Q}}^{\geq A}$ we have produced a representation $r_{f,\ell}^{\text{univ}} : \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \to \text{GL}_2(\mathcal{O})$, with finite image, which is unramified for all $p \nmid N\ell$. Moreover, the above equality (6.5) holds for such p.

We are now very close to Theorem 6.1, but there is a small wrinkle. Via $K \hookrightarrow \overline{\mathbb{Q}}_\ell \xrightarrow{\sim} \mathbb{C}$ we identify $r_{f,\ell}^{\text{univ}}$ with a continuous representation $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \to \text{GL}_2(\mathbb{C})$ enjoying the desired properties for all $p \nmid N\ell$. The issue is what happens for $p = \ell$ if $\ell \nmid N$.

To resolve this, pick another prime $\ell' \in \text{Spl}_{\mathbb{E}_f/\mathbb{Q}}^{\geq A} - \{\ell\}$ and consider $r_{f,\ell'}^{\text{univ}}$ (again viewed as a complex representation). For $p \nmid N\ell\ell'$ both $r_{f,\ell}^{\text{univ}}$ and $r_{f,\ell'}^{\text{univ}}$ are unramified, and

$$\det(I - r_{f,\ell}^{\text{univ}}(\text{Frob}_p)T) = 1 - a_p T + \chi(p)T^2 = \det(I - r_{f,\ell'}^{\text{univ}}(\text{Frob}_p)T).$$

Together the Brauer-Nesbitt principle (Remark 6.33) and the Chebotarev density theorem imply that at least $r_{f,\ell}^{\mathrm{univ}}$ and $r_{f,\ell'}^{\mathrm{univ}}$ have isomorphic semisimplifications. However, the proof of Proposition 6.9 shows they are both irreducible, so $r_{f,\ell}^{\mathrm{univ}} \simeq r_{f,\ell'}^{\mathrm{univ}}$. As a result $r_{f,\ell}^{\mathrm{univ}}$ is unramified at $p = \ell$, since $r_{f,\ell'}^{\mathrm{univ}}$ is, and $r_{f,\ell}^{\mathrm{univ}}(\mathrm{Frob}_\ell)$ has characteristic polynomial $1 - a_\ell T + \chi(\ell)T^2$ as desired. This proves Theorem 6.1.

Appendix A
Some Background Material

In this appendix we collect material which is typically covered in a standard undergraduate mathematics curriculum, such as Fourier theory on a finite abelian group (the dual group, orthogonality relations, etc.) and the basic theory of Dirichlet series. The latter includes the convolution of arithmetic functions and the Möbius inversion formula.

The reader may not have seen the material on prime-counting in their undergraduate education, although it is completely elementary. Using divisibility properties of factorials and binomial coefficients we show, following [HW79], that $\pi(x) \sim \frac{\psi(x)}{\ln x}$ as $x \to \infty$ where ψ denotes the Chebyshev function $\psi(x) := \sum_{p^r \leq x} \ln p$. Thus the prime number theorem is equivalent to $\psi(x) \sim x$, which is exactly what one deduces from the non-vanishing of the Riemann zeta function on the line $\mathrm{Re}(s) = 1$ using the Wiener-Ikehara theorem (see the discussion after Theorem 1.15 in the main text).

Having this subject matter be easily accessible will hopefully be helpful to the reader.

A.1 Duality for Finite Abelian Groups

Let G be a finite group. A character of G is a homomorphism $\chi : G \longrightarrow \mathbb{C}^\times$. Clearly χ is trivial on all commutators so it factors through the largest abelian quotient G^{ab}. No generality is lost by assuming G is abelian, and we will assume this from now on.

Definition A.1 The dual group of G is the set of all characters $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$ with the composition law $(\chi \cdot \chi')(x) = \chi(x)\chi'(x)$. (The trivial character $1(x) = 1$ is the identity.)

All characters $\chi \in \widehat{G}$ are automatically unitary. That is, they take values in the complex unit circle $\mathbb{C}^1$. More precisely the χ-values are roots of unity of order dividing $|G|$. Indeed

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$$\chi(x)^{|G|} = \chi(x^{|G|}) = \chi(e) = 1.$$

for all $x \in G$ by Lagrange.

Example A.2 Consider $G = \mathbb{Z}/N\mathbb{Z}$. In this case a character $\chi : \mathbb{Z}/N\mathbb{Z} \to \mathbb{C}^\times$ is completely determined by the Nth root of unity $\chi([1])$ and vice versa. Therefore $\chi \mapsto \chi([1])$ gives a canonical isomorphism

$$\widehat{\mathbb{Z}/N\mathbb{Z}} \xrightarrow{\sim} \mu_N.$$

The choice of a primitive Nth root of unity $\zeta_N \in \mathbb{C}^\times$ gives an isomorphism $\mathbb{Z}/N\mathbb{Z} \xrightarrow{\sim} \mu_N$ by sending $[1] \mapsto \zeta_N$. In conjunction this yields a (non-canonical) isomorphism

$$\mathbb{Z}/N\mathbb{Z} \xrightarrow{\sim} \widehat{\mathbb{Z}/N\mathbb{Z}}$$

which takes the generator $[1]$ to the character $\chi_0([n]) = \zeta_N^n$ generating the dual group.

Proposition A.3 *For any finite abelian group G there is a (non-canonical) isomorphism $G \xrightarrow{\sim} \widehat{G}$. In particular $|\widehat{G}| = |G|$.*

Proof By the structure theorem $G \simeq G_1 \times \cdots \times G_t$ for certain cyclic groups $G_i \simeq \mathbb{Z}/N_i\mathbb{Z}$. Drawing on the previous example we find that

$$\widehat{G} \simeq \widehat{G}_1 \times \cdots \times \widehat{G}_t \simeq G_1 \times \cdots \times G_t \simeq G$$

as desired. □

It follows immediately that the biduality map is a canonical isomorphism.

Corollary A.4 *Let G be a finite abelian group. The natural biduality map $G \longrightarrow \hat{\hat{G}}$ is an isomorphism. (This is the map sending $x \mapsto \eta_x$ where $\eta_x(\chi) = \chi(x)$.)*

Proof Clearly the η-map is a homomorphism, and it is injective. Why? The kernel consists of $x \in G$ such that $\chi(x) = 1$ for all $\chi \in \widehat{G}$. This forces $x = e$. Otherwise $\langle x \rangle$ admits a non-trivial character which extends (as we will show shortly) to a character χ of G with the property $\chi(x) \neq 1$. Since we now know $|G| = |\widehat{G}| = |\hat{\widehat{G}}|$ the η-map must be surjective as well. □

We used the following extension result in the previous proof.

Lemma A.5 *Let $H \subset G$ be a subgroup of a finite abelian group G. Then every character of H admits an extension to a character of G.*

Proof We do this by induction on the index $[G : H]$. When $[G : H] = 1$ there is nothing to show. Suppose $[G : H] > 1$ and assume the result holds for subgroups of G with index $< [G : H]$. Pick an $x \in G$ not in H and let $n = \mathrm{ord}(xH) > 1$. We will apply the induction hypothesis to the group $H' = \langle H, x \rangle$.

Let $\chi \in \widehat{H}$ be an arbitrary character (which we want to extend to G). Since $x^n \in H$ it makes sense to consider $\chi(x^n)$. Once and for all pick a $z \in \mathbb{C}$ such that $\chi(x^n) = z^n$. This allows us to extend χ to a character $\widetilde{\chi}$ of H' by declaring that

$$\widetilde{\chi}(hx^r) = \chi(h)z^r$$

for $h \in H$ and $r \in \mathbb{Z}$. (This is well-defined, for if $h_1 x^{r_1} = h_2 x^{r_2}$ we find that $x^{r_1 - r_2} = h_1^{-1} h_2 \in H$ and therefore $n|(r_1 - r_2)$. Consequently $\chi(h_1^{-1}h_2) = \chi(x^{r_1-r_2}) = z^{r_1-r_2}$.) Now by induction $\widetilde{\chi}$ extends all the way to a character of G. □

We may rephrase this Lemma by saying the restriction map $\widehat{G} \to \widehat{H}$ is surjective, and obviously the kernel is the image of $\widehat{G/H}$. We summarize this in the short exact sequence (of finite abelian groups)

$$0 \longrightarrow \widehat{G/H} \longrightarrow \widehat{G} \longrightarrow \widehat{H} \longrightarrow 0.$$

More succinctly $\widehat{(\cdot)}$ is an exact (contravariant) functor. In particular any given character of H has exactly $[G : H]$ extensions to a character of G.

Next we consider the vector space $\mathcal{C}(G)$ of all functions $f : G \longrightarrow \mathbb{C}$. Clearly the set of delta functions $\{\delta_x\}_{x \in G}$ forms a basis, so $\dim_{\mathbb{C}} \mathcal{C}(G) = |G|$. (Here $\delta_x(x) = 1$ and $\delta_x(y) = 0$ for $y \neq x$.) We endow $\mathcal{C}(G)$ with the inner product

$$\langle f, f' \rangle = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{f'(x)}.$$

Obviously $\{\delta_x\}_{x \in G}$ is an orthogonal basis, and $\|\delta_x\| = \frac{1}{\sqrt{|G|}}$ for all $x \in G$. The normalization $\frac{1}{|G|}$ ensures the set of characters is an orthonormal basis:

Proposition A.6 (The orthogonality relations) $\{\chi\}_{\chi \in \widehat{G}}$ *is an orthonormal basis for* $\mathcal{C}(G)$. *More explicitly*

$$\sum_{x \in G} \chi(x)\overline{\chi'(x)} = \begin{cases} |G| & \text{if } \chi = \chi' \\ 0 & \text{if } \chi \neq \chi'. \end{cases}$$

Proof It is rather immediate that $\|\chi\|^2 = \frac{1}{|G|} \sum_{x \in G} |\chi(x)|^2 = 1$ since χ is unitary. Let $\chi \neq \chi'$ be two characters of G. Consider the sum

$$S = \sum_{x \in G} \chi(x)\overline{\chi'(x)} = \sum_{x \in G} (\chi \cdot \chi'^{-1})(x).$$

Pick a $y \in G$ such that $\chi(y) \neq \chi'(y)$. Scaling S by $(\chi \cdot \chi'^{-1})(y) \neq 1$ and making the substitution $x \rightsquigarrow xy$ shows that $S = 0$. In other words $\langle \chi, \chi' \rangle = 0$, and in particular the characters are linearly independent. Since $|\widehat{G}| = |G| = \dim_{\mathbb{C}} \mathcal{C}(G)$ the characters must span the whole space. □

Applying this to the dual group gives us the following obverse.

Corollary A.7 *For $x, y \in G$ we have the relations*

$$\sum_{\chi \in \widehat{G}} \chi(x)\overline{\chi(y)} = \begin{cases} |G| & \text{if } x = y \\ 0 & \text{if } x \neq y. \end{cases}$$

Proof These are the orthogonality relations for $\widehat{G}$ with characters η_x and η_y. □

We expand $f \in \mathcal{C}(G)$ in the character basis as a linear combination $f = \sum_{\chi \in \widehat{G}} a_{f,\chi}\chi$. By orthogonality the coefficients are given by the Fourier transformation

$$a_{f,\chi} = \langle f, \chi \rangle = \widehat{f}(\chi) = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{\chi(x)}.$$

Thus $f(x) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi)\chi(x)$ for all x ("Fourier inversion").

A.2 Arithmetic Functions and Dirichlet Series

Let $\mathbb{N} = \{1, 2, 3, \ldots\}$. An arithmetic function is just a function $f : \mathbb{N} \longrightarrow \mathbb{C}$. We often confuse f with the sequence of complex numbers $a_n = f(n)$ where $n = 1, 2, 3, \ldots$. The vector space $\mathcal{A}$ of all arithmetic functions has a natural algebra structure via convolution:

$$(f * g)(n) = \sum_{d|n} f(d)g(\frac{n}{d})$$

for $f, g \in \mathcal{A}$. The sum runs over all positive divisors of n. This turns $\mathcal{A}$ into a commutative $\mathbb{C}$-algebra, with multiplicative identity

$$1_{\mathcal{A}}(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1. \end{cases}$$

When $f \in \mathcal{A}$ has at most polynomial growth one encodes its values $a_n = f(n)$ in a Dirichlet series.

Definition A.8 $D(s, f) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$.

Note that $D(s, f)$ is absolutely convergent for $\mathrm{Re}(s) > \alpha + 1$ provided $a_n = O(n^{\alpha})$. Viewing convolution as a variant of the Cauchy product it is easy to verify that

$$D(s, f * g) = D(s, f) \cdot D(s, g)$$

for all $f, g \in \mathcal{A}$ (and $s \in \mathbb{C}$ such that all three Dirichlet series converge absolutely).

Definition A.9 An arithmetic function $f : \mathbb{N} \longrightarrow \mathbb{C}$ is said to be multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$ with $\mathrm{GCD}(m, n) = 1$.

We say f is strongly multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$.

Thus a multiplicative function is completely determined by its values on prime-powers $f(p^r)$. A strongly multiplicative function is determined by its values on primes $f(p)$.

For each prime p we introduce $D_p(s, f) = \sum_{r=0}^{\infty} \frac{a_{p^r}}{p^{rs}}$ (when the series is convergent). By the fundamental theorem of arithmetic the Dirichlet series factors as an Euler product

$$D(s, f) = \prod_p D_p(s, f)$$

for all multiplicative $f \in \mathcal{A}$. When f is strongly multiplicative $D_p(s, f)$ is a geometric sum and the Euler product becomes

$$D(s, f) = \prod_p \frac{1}{1 - a_p p^{-s}},$$

provided it is absolutely convergent.

Example A.10 The Euler function $\phi(n) = |(\mathbb{Z}/n\mathbb{Z})^{\times}|$ is multiplicative, and $\phi(p^r) = p^r - p^{r-1}$ for $r > 0$. It is not strongly multiplicative since $\phi(p)^2 = (p-1)^2$ is always less than $\phi(p^2) = p^2 - p$. In this case

$$\begin{aligned} D_p(s, \phi) &= 1 + \sum_{r=1}^{\infty} \phi(p^r) p^{-rs} \\ &= 1 + \sum_{r=1}^{\infty} (p^r - p^{r-1}) p^{-rs} \\ &= \sum_{r=0}^{\infty} p^{r(1-s)} - p^{-s} \sum_{r=0}^{\infty} p^{r(1-s)} \\ &= \frac{1 - p^{-s}}{1 - p^{1-s}} \end{aligned}$$

for $\mathrm{Re}(s) > 1$. In particular $D(s, \phi) = \frac{\zeta(s-1)}{\zeta(s)}$ for such s. We will come back to this example below and give an alternate derivation of this formula.

Example A.11 The Möbius function $\mu : \mathbb{N} \longrightarrow \mathbb{C}$ is defined by $\mu(n) = (-1)^t$ if n factors into t distinct primes $n = p_1 \cdots p_t$. If n is not square-free $\mu(n) = 0$, and by convention $\mu(1) = 1$. The μ-function is multiplicative but not strongly multiplicative since $\mu(p^2) = 0$ and $\mu(p)^2 = 1$. Here

$$D_p(s,\mu) = \sum_{r=0}^{\infty} \mu(p^r) p^{-rs} = 1 - p^{-s},$$

and therefore $D(s,\mu) = \frac{1}{\zeta(s)}$ for $\mathrm{Re}(s) > 1$.

Example A.12 A Dirichlet character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times$ defines a strongly multiplicative arithmetic function $\chi : \mathbb{N} \longrightarrow \mathbb{C}$ by letting $\chi(n) = 0$ when $\mathrm{GCD}(n, N) > 1$.

The convolution of two multiplicative functions is multiplicative.

Proposition A.13 *If $f, g \in \mathcal{A}$ are multiplicative then so is $f * g$.*

Proof Suppose $m, n \in \mathbb{N}$ are coprime. Then any $d|mn$ factors uniquely as $d = ab$ with $a|m$ and $b|n$. So,

$$\begin{aligned}
(f*g)(mn) &= \sum_{d|mn} f(d) g(\frac{mn}{d}) \\
&= \sum_{a|m} \sum_{b|n} f(ab) g(\frac{mn}{ab}) \\
&= \sum_{a|m} \sum_{b|n} f(a) f(b) g(\frac{m}{a}) g(\frac{n}{b}) \\
&= \big(\sum_{a|m} f(a) g(\frac{m}{a})\big)\big(\sum_{b|n} f(b) g(\frac{n}{b})\big) \\
&= (f*g)(m)(f*g)(n).
\end{aligned}$$

In the third step we used that $\mathrm{GCD}(a,b) = 1$ and $\mathrm{GCD}(\frac{m}{a}, \frac{n}{b}) = 1$. □

The analogous result for strongly multiplicative functions is false. Here is a counterexample.

Example A.14 Consider the constant function $\mathbf{1} \in \mathcal{A}$ with $\mathbf{1}(n) = 1$ for all $n \in \mathbb{N}$. We convolve it with $f_s(n) = n^s$ for some $s \in \mathbb{C}$. Note that both $\mathbf{1}$ and f_s are strongly multiplicative. Their convolution is the multiplicative function

$$(f_s * \mathbf{1})(n) = \sum_{d|n} d^s \mathbf{1}(\frac{n}{d}) = \sum_{d|n} d^s = \sigma_s(n).$$

This is a prominent arithmetic function. For instance $\sigma_0(n)$ is the number of positive divisors of n, and $\sigma_1(n)$ is their sum. On prime powers σ_s is given by the geometric sum

$$\sigma_s(p^r) = \sum_{i=0}^{r} p^{is} = \frac{p^{(r+1)s} - 1}{p^s - 1},$$

of course provided $s \notin \frac{2\pi i}{\ln(p)}\mathbb{Z}$. For s in this exceptional set $\sigma_s(p^r) = \sigma_0(p^r) = r+1$. Observe that σ_s is not strongly multiplicative in general, even for $s = 0$. Indeed $\sigma_0(p^2) = 3$ but $\sigma_0(p)^2 = 4$. We conclude that $\mathbf{1}$ is strongly multiplicative, but $\sigma_0 = \mathbf{1} * \mathbf{1}$ is not.

For completeness we evaluate $D(s, \sigma_{s_0})$ for any fixed $s_0 \in \mathbb{C}$. Clearly we have $D(s, f_{s_0}) = \zeta(s - s_0)$ and $D(s, \mathbf{1}) = \zeta(s)$. As $\sigma_{s_0} = f_{s_0} * \mathbf{1}$ we deduce that

$$D(s, \sigma_{s_0}) = \zeta(s)\zeta(s - s_0)$$

for $\mathrm{Re}(s) \gg 0$. One can also see this by working out the Euler product.

In the previous example we encountered the transformation $f \rightsquigarrow f * \mathbf{1}$. This is sometimes called the Möbius transformation. It is given explicitly by the formula

$$(f * \mathbf{1})(n) = \sum_{d|n} f(d).$$

(Remember $\mathbf{1}$ denotes the constant function $n \mapsto 1$ not the identity $1_{\mathcal{A}}$.) We will show that $\mathbf{1} \in \mathcal{A}^\times$ (with inverse μ) so one can recover f from $f * \mathbf{1}$.

Theorem A.15 (The Möbius inversion formula) *Consider an arithmetic function $f \in \mathcal{A}$ with Möbius transformation $g = f * \mathbf{1}$. Then*

$$f(n) = \sum_{d|n} \mu(d) g(\frac{n}{d})$$

for all $n \in \mathbb{N}$.

Proof We just have to show $\mu * \mathbf{1} = 1_{\mathcal{A}}$. Once we know this just convolve both sides of $g = f * \mathbf{1}$ by μ. We know $\mu * \mathbf{1}$ is a multiplicative function since μ and $\mathbf{1}$ are both multiplicative. So is $1_{\mathcal{A}}$. It suffices to check they agree on prime powers. First

$$(\mu * \mathbf{1})(p^r) = \sum_{i=0}^{r} \mu(p^i) = \mu(1) + \mu(p) + 0 + \cdots = 1 - 1 + 0 + \cdots = 0$$

for $r > 0$, and clearly $(\mu * \mathbf{1})(1) = 1$. This agrees with $1_{\mathcal{A}}$. □

Example A.16 Here is a short list of examples of Möbius transformations and convolutions:

- $1_{\mathcal{A}} * \mathbf{1} = \mathbf{1}$
- $\phi * \mathbf{1} = \mathrm{Id}$
- $\mu * \mathbf{1} = 1_{\mathcal{A}}$
- $\mathbf{1} * \mathbf{1} = \sigma_0$
- $f_s * \mathbf{1} = \sigma_s$
- $\phi * \sigma_0 = \sigma_1$.

The second bullet point is a compact way of stating the relation $\sum_{d|n} \phi(d) = n$. (It suffices to check it for $n = p^r$. A more conceptual approach is to count elements of order d in a cyclic group of size n.) Möbius inversion is the formula $\phi(n) = \sum_{d|n} \mu(d)\frac{n}{d}$. This gives an alternative efficient way of getting the expression $D(s, \phi) = \frac{\zeta(s-1)}{\zeta(s)}$ for the Dirichlet series.

Exercise A.17 Consider a field $\mathbb{F}_q$ with q elements and let N_n denote the number of irreducible monic polynomials in $\mathbb{F}_q[t]$ of degree exactly n. By factoring $t^{q^n} - t$ into irreducible polynomials argue that $q^n = \sum_{d|n} dN_d$ and derive an explicit formula for N_n involving the μ-function. Deduce that

$$\pi_{\mathbb{F}_q[t]}(x) = \#\{\text{irreducible monic } f \in \mathbb{F}_q[t] \text{ with } q^{\deg(f)} \leq x\} \sim \frac{x}{\log_q x}$$

as $x \longrightarrow \infty$.

A.3 Counting Primes

One of the overriding goals of number theory is to understand the precise asymptotics of the prime-counting function $\pi(x)$. This is the step function on $\mathbb{R}$ counting the number of primes $p \leq x$. A closer look at Euclid's famous argument that $\pi(x) \to \infty$ reveals the lower bound $\pi(x) \geq \log_2 \log_2 x$. The true asymptotics is $\pi(x) \sim \frac{x}{\ln(x)}$ (meaning the ratio goes to 1 as $x \to \infty$). This is known as the prime number theorem, and one of the principal aims of number theory is to obtain good bounds on the error term.

In this section our modest goal is to reformulate the prime number theorem in terms of the Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$. Here Λ is the von Mangoldt function

$$\Lambda(n) = \begin{cases} \ln(p) & \text{if } n = p^r \text{ with } r > 0 \\ 0 & \text{otherwise .} \end{cases}$$

Note that Λ is not a multiplicative function ($\Lambda(2 \cdot 3) = 0$ but $\Lambda(2) = \ln(2)$ and $\Lambda(3) = \ln(3)$).

Exercise A.18 Show that the von Mangoldt function has Möbius transformation $\Lambda * \mathbf{1} = \ln$. Deduce that $D(s, \Lambda) = -\frac{\zeta'(s)}{\zeta(s)}$ for $\mathrm{Re}(s) > 1$. (`Hint`: Start by showing $\sum_{d|n} \Lambda(d) = \ln(n)$ when $n = p^r$.)

We will relate $\pi(x)$ to $\psi(x)$ through the following variant of the Chebyshev function:

$$\theta(x) = \sum_{p \leq x} \ln(p).$$

Often $\theta(x)$ is referred to as the first Chebyshev function, and $\psi(x)$ is named the second Chebyshev function. We are primarily interested in $\psi(x)$ because it is the summatory function of Λ, and the identity $D(s, \Lambda) = -\frac{\zeta'(s)}{\zeta(s)}$ gives us access to Tauberian results, while $\theta(x)$ will play a more auxiliary role.

Lemma A.19 *We have the following formulas for $\psi(x)$.*

(1) $\psi(x) = \sum_{p \le x} \left[\frac{\ln x}{\ln p}\right] \ln(p)$
(2) $\psi(x) = \sum_{r=1}^{\infty} \theta(x^{1/r})$.

Proof The first formula follows by writing

$$\psi(x) = \sum_{p^r \le x} \ln(p) = \sum_{p \le x} \sum_{r=1}^{[\frac{\ln x}{\ln p}]} \ln(p) = \sum_{p \le x} \Big[\frac{\ln x}{\ln p}\Big] \ln(p).$$

The second follows by interchanging the sums over p and r,

$$\psi(x) = \sum_{r=1}^{\infty} \sum\nolimits_{p \le x^{1/r}} \ln(p) = \sum_{r=1}^{\infty} \theta(x^{1/r}).$$

This is a finite sum. Indeed $\theta(x) = 0$ for $x < 2$ so the sum really only extends from $r = 1$ to $r = \left[\frac{\ln x}{\ln 2}\right]$. □

The next result will eventually tell us $\psi(x) \sim \theta(x)$.

Proposition A.20 $\psi(x) = \theta(x) + O\big(x^{1/2}(\ln x)^2\big)$.

Proof We have the trivial bound $\theta(x) \le x \ln(x)$ for $x \ge 1$, and hence $\theta(x^{1/r}) \le \frac{1}{r} x^{1/r} \ln(x)$ for $x \ge 1$. When $r \ge 2$ we have the estimate $\frac{1}{r} x^{1/r} \ln(x) < x^{1/2} \ln(x)$. Therefore

$$\psi(x) = \sum_{r=1}^{\left[\frac{\ln x}{\ln 2}\right]} \theta(x^{1/r}) = \theta(x) + O(x^{1/2}(\ln x)^2).$$

(The error term has about $\frac{\ln x}{\ln 2}$ terms, each of which is less than $x^{1/2} \ln(x)$.) □

Here is a first approximation to the prime number theorem.

Proposition A.21 *The Chebyshev functions satisfy the following bounds.*

(1) $\theta(x) < \frac{7}{5} x$ *for all* $x \ge 1$.
(2) $\psi(x) > \frac{1}{5} x$ *for all* $x \ge 6$.

Proof We start off with the preliminary observation that $M = \binom{2m+1}{m} < 2^{2m}$ for all $m \in \mathbb{N}$. This follows from the binomial expansion

$$2^{2m+1} = (1+1)^{2m+1} = \sum_{r=0}^{2m+1} \binom{2m+1}{r} = M + M + \cdots > 2M,$$

noting that the $r = m$ and the $r = m+1$ terms both contribute an M to the sum.

As an intermediate step towards (1) we use this estimate of binomial coefficients to deduce the bound

$$\theta(2m+1) - \theta(m+1) < 2\ln(2)m$$

for all $m \in \mathbb{N}$. How? Pick a prime p in the interval $m+1 < p \leq 2m+1$. Note that $p|M$ since p divides the numerator of $M = \frac{1}{m!}(2m+1)(2m)\cdots(m+2)$ but not the denominator. Consequently $\prod_{m+1<p\leq 2m+1} p$ divides M, so that

$$\theta(2m+1) - \theta(m+1) = \sum\nolimits_{m+1<p\leq 2m+1} \ln(p) \leq \ln(M) < 2\ln(2)m.$$

Now to prove part (1) it suffices to show the inequality

$$\theta(n) < 2\ln(2)n$$

for $n \in \mathbb{N}$. Indeed $2\ln(2) < 1.39 < \frac{7}{5}$ and then $\theta(x) = \theta([x]) < \frac{7}{5}[x] \leq \frac{7}{5}x$ for all $x \geq 1$. We proceed by induction on n. For $n = 1, 2$ it is obvious so suppose $n > 2$ and the inequality holds for smaller n.

- When n is even,

$$\theta(n) = \theta(n-1) < 2\ln(2)(n-1) < 2\ln(2)n$$

  since n is not a prime.
- When n is odd, write $n = 2m+1$ for some $m \in \mathbb{N}$. Then

$$\begin{aligned}\theta(n) &= \big(\theta(2m+1) - \theta(m+1)\big) + \theta(m+1)\\ &< 2\ln(2)m + 2\ln(2)(m+1)\\ &= 2\ln(2)\ln(n).\end{aligned}$$

This finishes the proof of part (1).

For part (2) we will employ de Polignac's formula for the p-adic valuation of $n!$. It says that

$$v_p(n!) = \sum_{r=1}^{\infty} \big[\frac{n}{p^r}\big].$$

Why? Among the numbers $\{1, 2, \ldots, n\}$ there are $\left[\frac{n}{p^r}\right]$ multiples of p^r. Therefore exactly $\left[\frac{n}{p^r}\right] - \left[\frac{n}{p^{r+1}}\right]$ of them have p-adic valuation r. We conclude that $v_p(n!)$ is given by the telescoping sum

$$v_p(n!) = \sum_{r=1}^{\infty} r\big(\big[\frac{n}{p^r}\big] - \big[\frac{n}{p^{r+1}}\big]\big) = \sum_{r=1}^{\infty} \big[\frac{n}{p^r}\big].$$

The big hurdle is to show the inequality below for all $n \in \mathbb{N}$:

$$\psi(2n) \geq \ln(2)n.$$

Fix n and introduce $N = \binom{2n}{n} = \frac{(2n)!}{n!^2} = \prod_{p\leq 2n} p^{v_p(N)}$. By de Polignac the exponents are

$$v_p(N) = v_p((2n)!) - 2v_p(n!) = \sum_{r=1}^{\infty} \big(\big[\frac{2n}{p^r}\big] - 2\big[\frac{n}{p^r}\big]\big)$$

where the sum really only goes from $r = 1$ to $r = \big[\frac{\ln(2n)}{\ln(p)}\big]$. The key point is each term is 0 or 1:

$$\big[\frac{2n}{p^r}\big] - 2\big[\frac{n}{p^r}\big] \in \{0, 1\}.$$

Why? We know $\big[\frac{n}{p^r}\big] \leq \frac{n}{p^r} < \big[\frac{n}{p^r}\big] + 1$ and thus $2\big[\frac{n}{p^r}\big] \leq \frac{2n}{p^r} < 2\big[\frac{n}{p^r}\big] + 2$. We infer that $\big[\frac{2n}{p^r}\big] = 2\big[\frac{n}{p^r}\big]$ or $\big[\frac{2n}{p^r}\big] = 2\big[\frac{n}{p^r}\big] + 1$ as claimed.

In particular $v_p(N) \leq \big[\frac{\ln(2n)}{\ln(p)}\big]$ (=number of terms) and therefore

$$\ln(N) = \sum_{p\leq 2n} v_p(N)\ln(p) \leq \sum_{p\leq 2n} \big[\frac{\ln(2n)}{\ln(p)}\big]\ln(p) = \psi(2n).$$

On the other hand

$$N = \frac{(n+1)(n+2)\cdots(2n)}{1\cdot 2\cdots n} = \prod_{s=1}^{n} \frac{n+s}{s} \geq 2^n$$

since $\frac{n+s}{s} \geq 2$ for $s \leq n$. Taking $\ln(\cdot)$ gives the inequality $\psi(2n) \geq \ln(2)n$.

The rest are gritty details. For $x \geq 2$ let $n = [\frac{x}{2}]$. Using $2n \leq x < 2n + 2$ we find that

$$\psi(x) \geq \psi(2n) \geq \ln(2)n > \ln(2)\cdot\frac{x-2}{2} \geq \frac{\ln(2)}{3}x > \frac{1}{5}x$$

provided $x \geq 6$. Note that $\frac{\ln(2)}{3} > 0.23 > \frac{1}{5}$. This finishes the proof of the second half. □

From $\psi(x) > \frac{1}{5}x$ we immediately get that

$$\frac{\theta(x)}{\psi(x)} = 1 + O(x^{-1/2}(\ln x)^2).$$

In particular $\psi(x) \sim \theta(x)$ as $x \longrightarrow \infty$.

Theorem A.22 $\pi(x) \sim \frac{\psi(x)}{\ln x} \sim \frac{\theta(x)}{\ln x}$ *as* $x \longrightarrow \infty$.

Proof It suffices to show $\pi(x) \sim \frac{\theta(x)}{\ln x}$. First note that $\theta(x) = \sum_{p \leq x} \ln(p) \leq \ln(x)\,\pi(x)$. In other words $\frac{\pi(x)\ln(x)}{\theta(x)} \geq 1$. We need to show that for all $\epsilon > 0$ there is a $C > 0$ such that $x > C \Rightarrow \frac{\pi(x)\ln(x)}{\theta(x)} < 1 + \epsilon$.

In preparation we note the following. Suppose $0 < \delta < 1$. Then

$$\begin{aligned}\theta(x) &\geq \sum_{x^{1-\delta} < p \leq x} \ln(p) \\ &> \ln(x^{1-\delta})\big(\pi(x) - \pi(x^{1-\delta})\big) \\ &\geq (1-\delta)\ln(x)\big(\pi(x) - x^{1-\delta}\big).\end{aligned}$$

Here $\pi(x) - \pi(x^{1-\delta})$ is the number of terms in the sum, and each term is at least $\ln(x^{1-\delta})$. We used the trivial bound $\pi(x^{1-\delta}) \leq x^{1-\delta}$ in the third step. Equivalently

$$1 \leq \frac{\pi(x)\ln(x)}{\theta(x)} \leq \frac{x^{1-\delta}\ln(x)}{\theta(x)} + \frac{1}{1-\delta}.$$

Since $\theta(x) \sim \psi(x)$ and $\psi(x) > \frac{1}{5}x$ we find that $\frac{x^{1-\delta}\ln(x)}{\theta(x)} = O(x^{-\delta}\ln x)$ goes to 0 as $x \to \infty$.

Now, given $\epsilon > 0$ first pick a $0 < \delta < 1$ such that $\frac{1}{1-\delta} < 1 + \frac{\epsilon}{2}$. Then choose a $C > 0$ such that $\frac{x^{1-\delta}\ln(x)}{\theta(x)} < \frac{\epsilon}{2}$ for $x > C$. This C works. Indeed $x > C \Rightarrow \frac{\pi(x)\ln(x)}{\theta(x)} < 1 + \epsilon$. □

This achieves the goal of this section.

Corollary A.23 *The prime number theorem is equivalent to* $\psi(x) \sim x$ *as* $x \to \infty$.

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